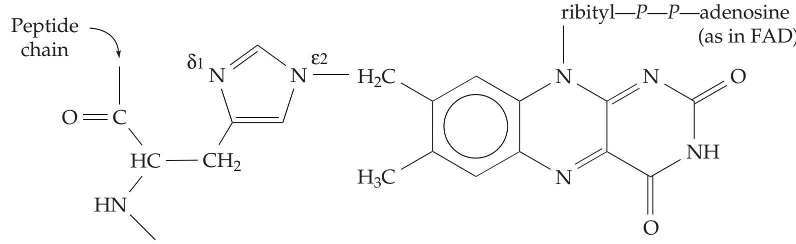
3. More Flavoproteins

Flavoproteins function in virtually every area of metabolism and we have considered only a small fraction of the total number. Here are a few more. Flavin-dependent reductases use hydrogen atoms from NADH or NADPH to reduce many specific substances or classes of compounds. The FAD-containing ferredoxin: $NADP^+$ oxidoreductase catalyzes the reduction of free $NADP^+$ by reduced ferredoxin generated in the chloroplasts of green leaves.[200,201] Similar enzymes, some of which utilize reduced flavodoxins, are found in bacteria.[202,203] The FMN-containing subunit of NADH: ubiquinone oxidoreductase is an essential link in the mitochondrial electron transport chain for oxidation of NADH in plants and animals[183,204,205] and for related processes in bacteria. **Glutamate synthase**, a key enzyme in the nitrogen metabolism of plants and microorganisms, uses electrons from NADPH to reduce 2-oxoglutarate to glutamate in a complex glutamine-dependent process (see Fig. 24-5). The enzyme contains both FMN and FAD and three different iron–sulfur clusters.[205a] Flavin reductases use NADH or NADPH to reduce free riboflavin, FMN, or FAD needed for various purposes[206,206a] including emission of light by luminous bacteria.[207] They provide electrons to many enzymes that react with O_2 such as the cytochromes P450[208,209] and nitric oxide synthase (Chapter 18). An example is adrenodoxin reductase (see chapter banner, p. 764), which passes electrons from NADPH to cytochrome P450 via the small redox protein adrenodoxin. This system functions in steroid biosynthesis as is indicated in Fig. 22-7.[209a,b] Other flavin-dependent reductases have protective functions catalyzing the reduction of ascorbic acid radicals,[210,211] toxic quinones,[212–214] and peroxides.[215–218]

4. Modified Flavin Coenzymes

Mitochondrial succinate dehydrogenase, which catalyzes the reaction of Eq. 15-21, contains a flavin prosthetic group that is covalently attached to a histidine side chain. This modified FAD was isolated and identified as 8α-($N^{\varepsilon 2}$-histidyl)-FAD.[219] The same prosthetic group has also been found in several other dehydrogenases.[220] It was the first identified member of a series of modified FAD or riboflavin 5'-phosphate derivatives that are attached by covalent bonds to the active sites of more than 20 different enzymes.[219]

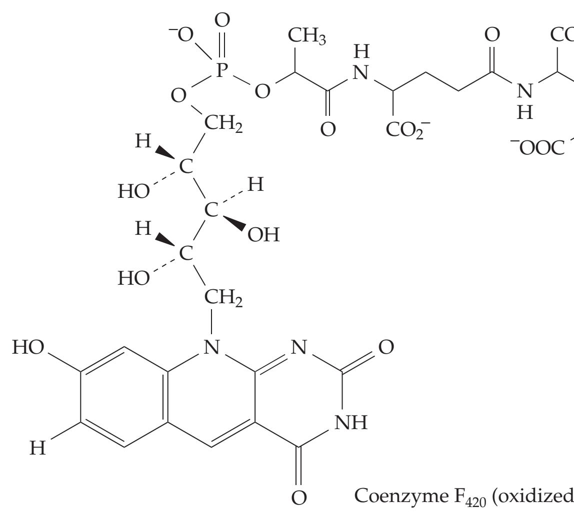
These include 8α-($N^{\varepsilon 2}$-histidyl)-FMN,[221] 8α-($N^{\delta 1}$-histidyl)-FAD,[222] 8α-(*O*-tyrosyl-FAD),[223] and 6-(*S*-cysteinyl)-riboflavin 5'-phosphate, which is found in trimethylamine dehydrogenase (Fig. 15-9).[224] An 8-hydroxy analog of FAD (–OH in place of the 8-CH_3)



Covalently bound modified FAD of succinate dehydrogenase

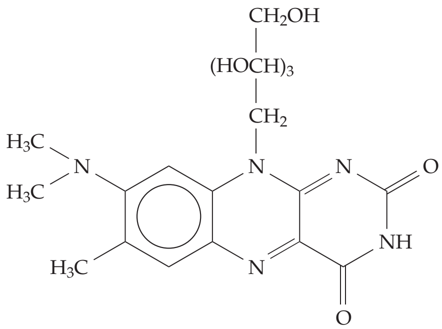
has been isolated from a bacterial electron-transferring flavoprotein.[225] Commercial FAD may contain some riboflavin 5'-pyrophosphate which activates some flavoproteins and inhibits others.[226]

Methanogenic bacteria contain a series of unique coenzymes (Section F) among which is **coenzyme F_{420}**, a 5-deazaflavin substituted by H at position 7 and –OH at position 8 (8-hydroxy-7,8-didemethyl-5-deazariboflavin).[227,228]



Coenzyme F_{420} (oxidized form)

This unique redox catalyst links the oxidation of H_2 or of formate to the reduction of $NADP^+$[229] and also serves as the reductant in the final step of methane biosynthesis (see Section E).[228] It resembles NAD^+ in having a redox potential of about –0.345 volts and the tendency to be only a two-electron donor. More recently free 8-hydroxy-7,8-didemethyl-5-deazaribo-flavin has been identified as an essential light-absorbing chromophore in DNA photolyase of *Methanobacterium*, other bacteria, and eukaryotic algae.[230] **Roseoflavin** is not a coenzyme but an antibiotic from *Streptomyces davawensis*.[231] Many synthetic flavins have been used in studies of mechanisms and for NMR[232] and other forms of spectroscopy.

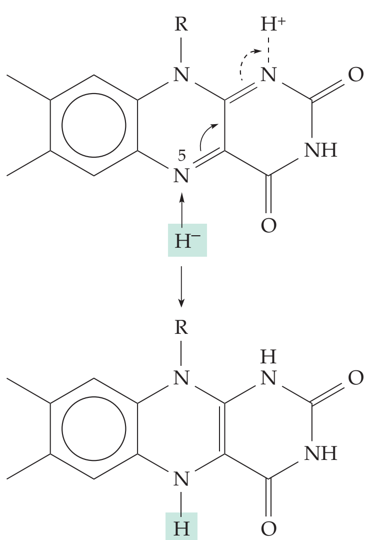


Roseoflavin

5. Mechanisms of Flavin Dehydrogenase Action

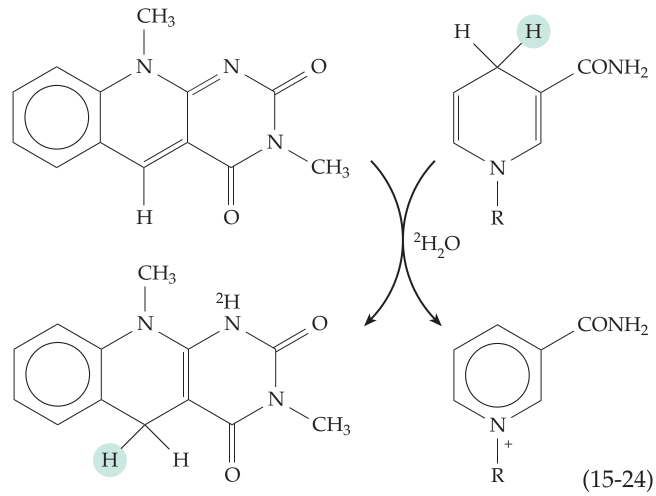
The chemistry of flavins is complex, a fact that is reflected in the uncertainty that has accompanied efforts to understand mechanisms. For flavoproteins at least four mechanistic possibilities must be considered.[153a,233] (a) A reasonable **hydride-transfer** mechanism can be written for flavoprotein dehydrogenases (Eq. 15-23). The hydride ion is donated at N-5 and a proton is accepted at N-1. The oxidation of alcohols, amines, ketones, and reduced pyridine nucleotides can all be visualized in this way. Support for such a mechanism came from study of the nonenzymatic oxidation of NADH by flavins, a reaction that occurs at moderate speed in water at room temperature. A variety of flavins and dihydropyridine derivatives have been studied, and the electronic effects observed for the reaction are compatible with the hydride ion mechanism.[234–236]

According to the mechanism of Eq. 15-23, a hydride ion is transferred directly from a carbon atom in a substrate to the flavin. However, a labeled hydrogen atom transferred to N-5 or N-1 would immediately exchange with the medium, rapid exchange being characteristic of hydrogens attached to nitrogen.



(15-23)

To avoid this problem, Brustlein and Bruice used a **5-deazaflavin** to oxidize NADH nonenzymatically.[237] When this reaction was carried out in ^2H_2O, no ^2H entered the product at C-5, indicating that a hydrogen atom (circled in Eq. 15-24) had been transferred directly from NADH to the C-5 position. Similar direct transfer of hydrogen to C-5 of 5-deazariboflavin 5'-phosphate is catalyzed by flavoproteins such as *N*-methylglutamate synthase[238] and acyl-CoA dehydrogenase.[237–239]



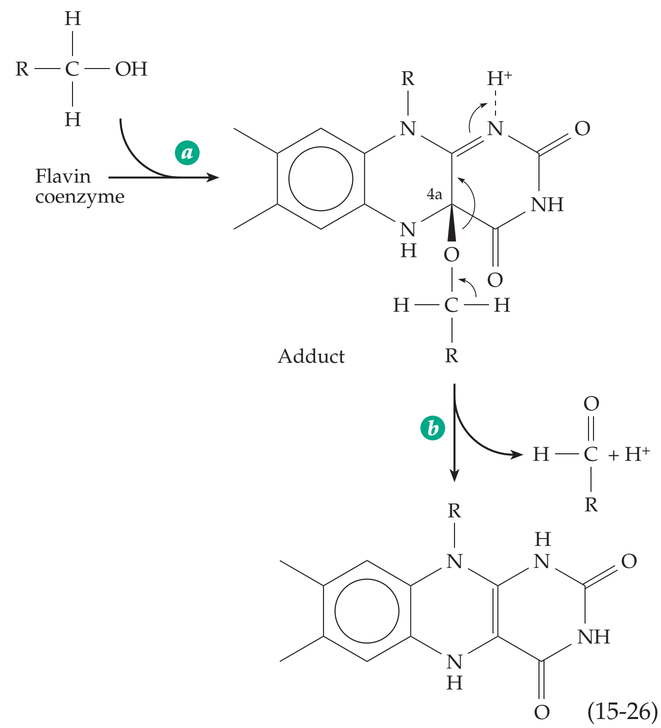
(15-24)

However, these experiments may not have established a mechanism for natural flavoprotein catalysis because the properties of 5-deazaflavins resemble those of NAD^+ more than of flavins.[239] Their oxidation–reduction potentials are low, they do not form stable free radicals, and their reduced forms don't react readily with O_2. Nevertheless, for an acyl-CoA dehydrogenase the rate of reaction of the deazaflavin is almost as fast as that of natural FAD.[238] For these enzymes a hydride ion transfer from the β CH (reaction type D of Table 15-1) is made easy by removal of the α-H of the acyl-CoA to form an enolate anion intermediate.

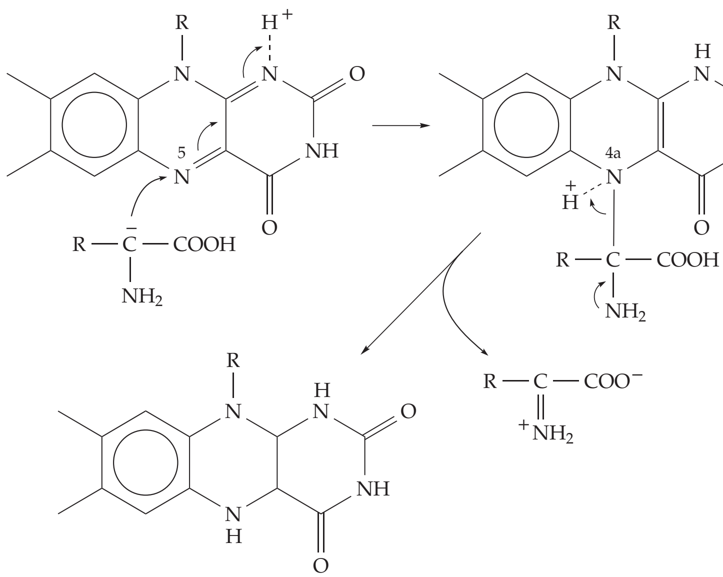
The three-dimensional structure of the medium chain acyl-CoA dehydrogenases with bound substrates and inhibitors is known.[174,175,240] A conserved glutamate side chain is positioned to pull the *pro-R* proton from the α carbon to create the initial enolate anion.[174,175,241] The *pro-R* β C–H lies by N-5 of the flavin ring seemingly ready to donate a hydride ion as in Eq. 15-23. NMR spectroscopy has been carried out with ^{13}C or ^{15}N in each of the atoms of the redox active part of the FAD. The results show directly the effects of strong hydrogen bonding to the protein at N-1, N-3, and N-5 and also suggest that the bound $FADH_2$ is really $FADH^-$ with the negative charge localized on N-1 by strong hydrogen bonding.[149] Many mutants have been made,[242] substrate analogs have been tested,[176,243] kinetic isotope effects have been measured,[242a] and potentiometric titrations have been done.[243] All of the results are compatible with the enolate anion hydride-transfer mechanism. Questions about the acidity of the α-H and the mechanism of its removal to form the enolate anion[242a] are similar to those discussed in Chapter 13, Section B.

A peculiarity of several acyl-CoA dehydrogenases is a bright green color with an absorption maximum at 710 nm. This was found to result from tightly bound coenzyme A persulfide (CoA–S–S⁻).[244,245]

(b) A second possible mechanism of flavin reduction is suggested by the occurrence of addition reactions involving the isoalloxazine ring of flavins. Sulfite adds to flavins by forming an N–S bond at the 5 position and nitroethane, which is readily dissociated to the carbanion $H_3C-CH^--NO_2$, acts as a substrate for D-amino acid oxidase.[246,247] This fact suggested a **carbanion mechanism** according to which normal D-amino acid substrates form carbanions by dissociation of the αH (Eq. 15-25). Ionization would be facilitated by binding of the substrate carboxylate to an adjacent arginine side chain and the carbanion could react at N-5 of the flavin as in Eq. 15-25. Similar mechanisms have been suggested for other flavin enzymes.[248,249]



(15-26)

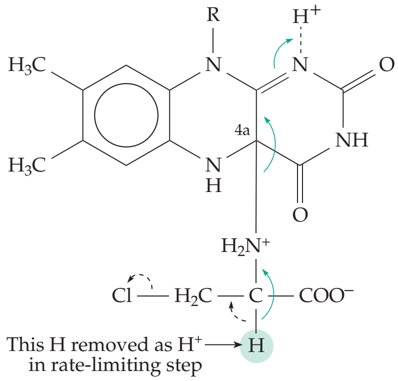


Reduced flavin

(15-25)

(c) The adducts with nitroethane and other compounds[250] pointed to reaction at N-5, but Hamilton[251] suggested that a better position for addition of nucleophiles is carbon 4a, which together with N-5 forms a cyclic Schiff base. He argued that other electrophilic centers in the flavin molecule, such as carbons 2, 4, and 10a, would be unreactive because of their involvement in amide or amidine-type resonance but an amine, alcohol, or other substrate could add to a flavin at position 4a (Eq. 15-26, step *a*). Cleavage of the newly formed C–O bond could then occur by movement of electrons from the alcohol part of the adduct into the flavin as indicated in step *b*. The products of this **4a adduct mechanism** are the reduced flavin and an aldehyde, the same as would be obtained by the hydride ion mechanism. However, in Eq. 15-26, both hydrogens in the original substrate (that on oxygen and that on carbon) have dissociated as protons, the electrons having moved as a pair during the cleavage of the adduct. Hamilton argued that an isolated hydride ion has a large diameter while a proton is small and mobile; for this reason dehydrogenation may often take place by proton transfer mechanisms.

Experimental support for the mechanism of Eq. 15-26 has been obtained using D-chloroalanine as a substrate for D-amino acid oxidase.[252–254] Chloropyruvate is the expected product, but under anaerobic conditions pyruvate was formed. Kinetic data obtained with α-^{2}H and α-^{3}H substrates suggested a common intermediate for formation of both pyruvate and chloro-pyruvate. This intermediate could be an anion formed by loss of H⁺ either from alanine or from a C-4a adduct. The anion could eliminate chloride ion as indicated by the dashed arrows in the following structure. This would lead to formation of pyruvate without reduction of the flavin. Alternatively, the electrons from the carbanion could flow into the flavin (green arrows), reducing it as in Eq. 15-26. A similar mechanism has been suggested for other flavoenzymes.[249,255] Objections to the carbanion mechanism are the expected



very high pK_a for loss of the α-H to form the carbanion[256] and the observed formation of only chloroalanine and no pyruvate in the reverse reaction of chloropyruvate, ammonia, and reduced flavoprotein.

A long-known characteristic of D-amino acid oxidase is its tendency to form charge-transfer complexes with amines, complexes in which a nonbonding electron has been transferred partially to the flavin. Complete electron transfer would yield a flavin radical and a substrate radical which could be intermediates in a **free radical mechanism**, as discussed in the next section.[256]

The three-dimensional structure of the complex of D-amino acid oxidase with the substrate analog benzoate has been determined. The carboxyl group of the inhibitor is bound by an arginine side chain (Fig. 15-11) that probably also holds the amino acid substrate. There is no basic group nearby in the enzyme that could serve to remove the α-H atom in Eq. 15-26 but the position is appropriate for a direct transfer of the hydrogen to the flavin as a hydride ion as in Eq. 15-23.[161,162,257] In spite of all arguments to the contrary the hydride ion mechanism could be correct! However, an adduct mechanism is still possible.

Experimental evidence supports a 4a adduct mechanism for glutathione reductase and related enzymes[191,258,258a] (Fig. 15-12). In this figure the reaction sequence is opposite to that in Eq. 15-26. The enzyme presumably functions as follows. NADPH binds next to the bound FAD and reduces it by transfer of the 4-*pro-S* hydrogen of the NADPH (Fig. 15-12, step *a*). The sulfur atom of Cys 63 is in van der Waals contact with the bound FAD at or near carbon atom 4a. In step *b* the nucleophilic center on atom C-4a of $FADH_2$ attacks a sulfur atom of the disulfide loop between cysteines 58 and 63 in the protein to create a C-4a adduct of a thiolate ion with oxidized FAD and to cleave the –S–S– linkage in the loop. In step *c* the thiol group of cysteine 63 is eliminated, after which the thiol of Cys 58 attacks the nearer sulfur atom of the oxidized glutathione in a nucleophilic displacement (step *d*) to give one reduced glutathione (GSH) and a mixed disulfide of glutathione and the enzyme (G-S-S-Cys 58). The thiolate anion of Cys 63, which is stabilized by interaction with the adjacent flavin ring, then attacks this disulfide (step *e*) to regenerate the internal disulfide and to free the second molecule of reduced glutathione. The imidazole group of the nearby His 467 of the second subunit presumably participates in catalysis as may some other side chains.[191] The disulfide exchange reactions are similar to those discussed in Chapter 12.

A variation is observed for *E. coli* thioredoxin reductase. The reducible disulfide and the NADPH binding site are both on the same side of the flavin rather than on opposite sides as in Fig. 15-12.[190,259] Mercuric reductase also uses NADPH as the reductant transferring the 4*S* hydrogen. The Hg^{2+} presumably binds to a sulfur atom of the reduced disulfide loop and there undergoes reduction. The observed geometry of the active site is correct for this mechanism.

6. Half-Reduced Flavins

A possible mechanism of flavin dehydrogenation consists of consecutive transfer of a hydrogen atom and of an electron with intermediate radicals being formed both on the flavin and on the substrate. Such

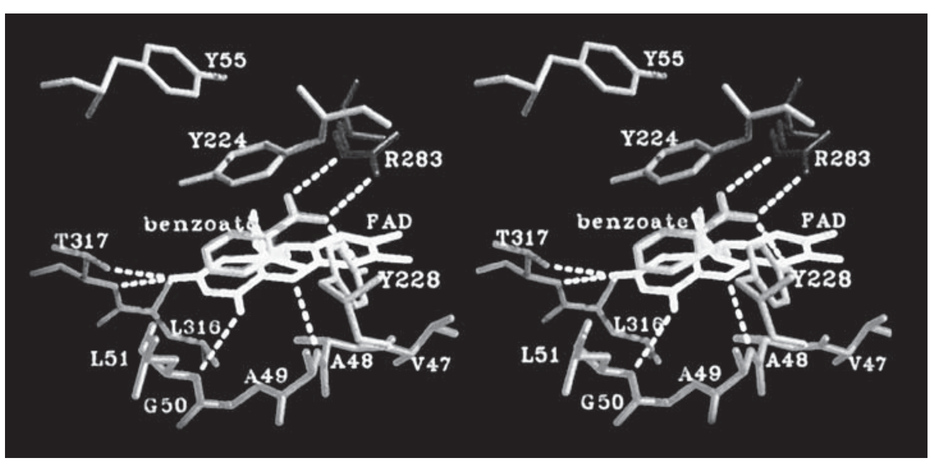


Figure 15-11 Stereoscopic view of the benzoate ion in its complex with D-amino acid oxidase. A pair of hydrogen bonds binds the carboxylate of the ligand to the guanidinium group of R283. Several hydrogen bonds to the flavin ring of the FAD are also indicated. Courtesy of Retsu Miura.[161]

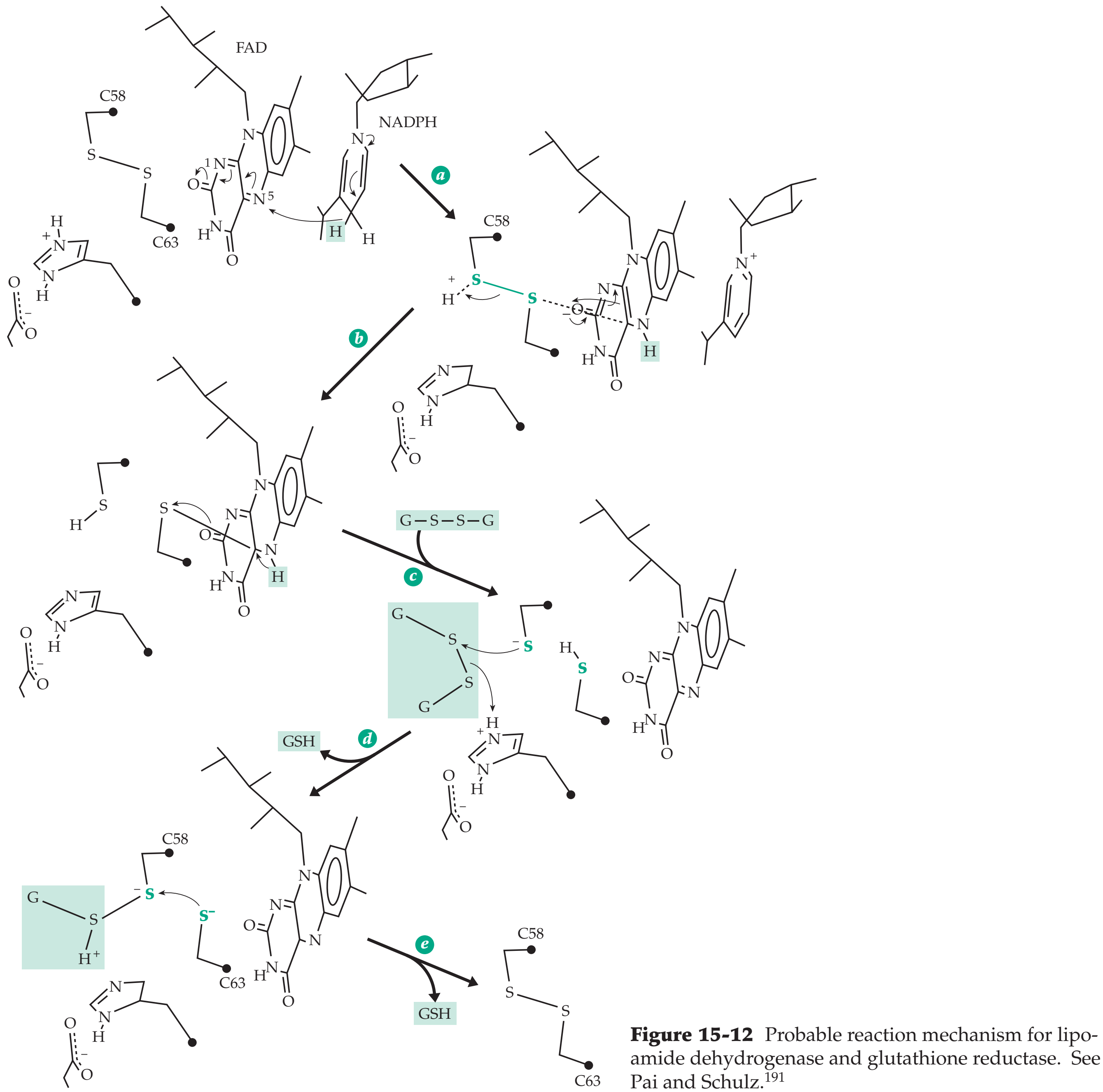


Figure 15-12 Probable reaction mechanism for lipoamide dehydrogenase and glutathione reductase. See Pai and Schulz.[191]

a mechanism takes full advantage of one of the most characteristic properties of flavins, their ability to accept single electrons to form **semiquinone** radicals. If the oxidized form Fl of a flavin is mixed with the reduced form FlH_2, a single hydrogen atom is transferred from FlH_2 to Fl to form two $^\bullet$F1H radicals (Eq. 15-27).

$$Fl + FlH_2 \underset{}{\overset{K_f}{\rightleftharpoons}} 2\ ^{\bullet}FlH \tag{15-27}$$

The equilibrium represented by this equation is independent of pH, but because all three forms of the flavin have different pK_a values (Fig. 15-13) the apparent equilibrium constants relating total concentrations of oxidized, reduced, and radical forms vary with pH.[143,260–262] The fraction of radicals present is greater at low pH and at high pH than at neutrality. For a 3-alkylated flavin the formation constant K_f has been estimated as 2.3×10^{-2} and for riboflavin[260] as 1.5×10^{-2}. From these values and the pK_a values in Fig. 15-13, it is possible to estimate the amount of radical present at any pH.

Neutral flavin radicals have a blue color (the wavelength of the absorption maximum, λ_{max}, is ~560 nm) but either protonation at N-1 or dissociation of a proton from N-5 leads to red cation or anion radicals with λ_{max} at ~477 nm. Both blue and red radicals are

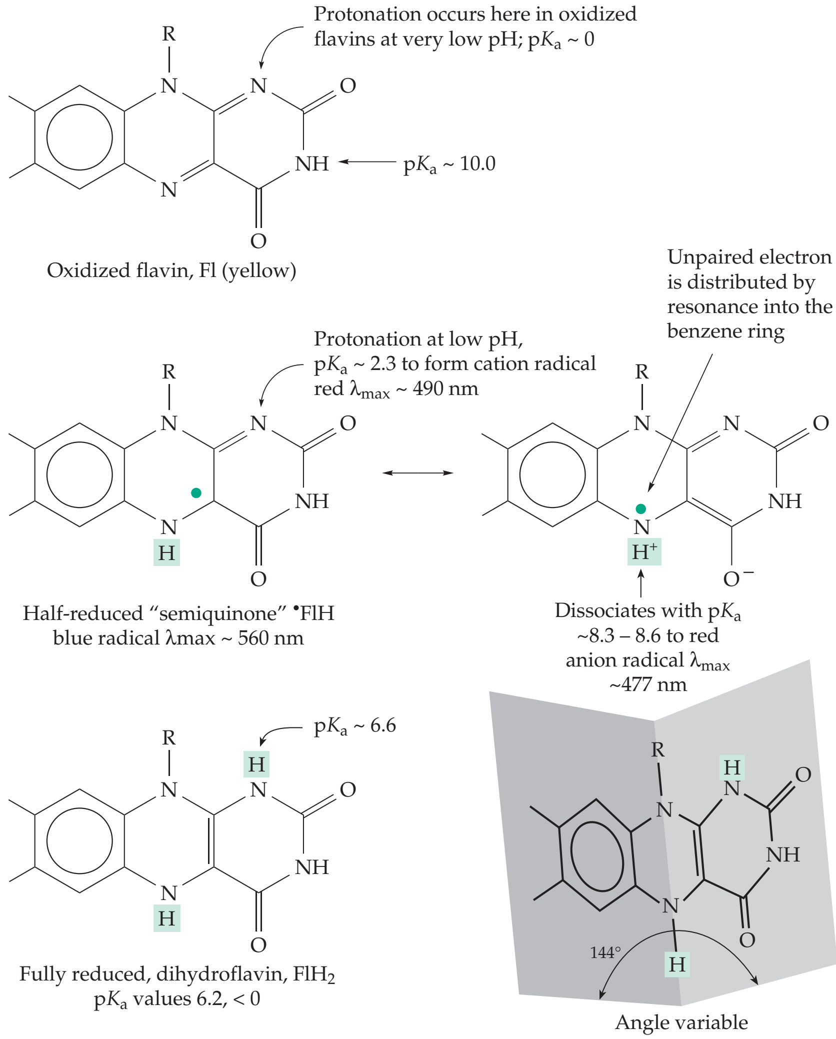
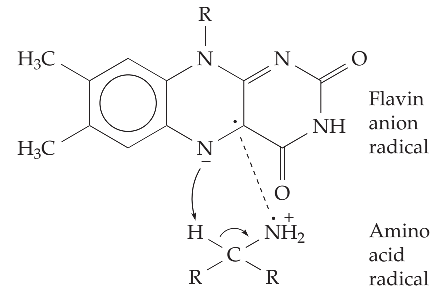


Figure 15-13 Properties of oxidized, half-reduced, and fully reduced flavins. See Müller *et al.*[263,264]

observed in enzymes, with some enzymes favoring one and some the other. Hemmerich suggested that enzymes forming red radicals make a strong hydrogen bond to the proton in the 5 position of the flavin. This increases the basicity of N-1 leading to its protonation and formation of the red cation radicals.

It is possible that many flavoprotein oxygenases and dehydrogenases react via free radicals. For example, instead of the mechanism of Eq. 15-26, an electron could be transferred to the flavin, leaving a radical pair (at right). The crystallographic structure and modeling of the substrate complex support this possibility.[265,265a] In this pair the flavin radical would be more basic than in the fully oxidized form and the amino acid radical would be more acidic than in the uncharged form. A proton transfer as indicated together with coupling of the radical pair would yield the same product as the mechanism of Eq. 15-26.



Another alternative to a radical pair is a hydrogen atom transfer followed by a second electron transfer.[265]

If an enzyme binds a flavin radical much more tightly than the fully oxidized or reduced forms, reduction of the flavoprotein will take place in two one-electron steps. In such proteins the values of $E°'$ for the two steps may be widely separated. The best known examples are the small, low-potential electron-carrying proteins known as **flavodoxins**.[266–269a] These proteins, which carry electrons between pairs of other redox proteins, have a variety of functions in anaerobic and photosynthetic bacteria, cyanobacteria, and green algae. Their functions are similar to those of the **ferredoxins**, iron–sulfur proteins that are considered in Chapter 16. In some bacteria ferredoxin and flavodoxin are interchangeable and the synthesis of flavodoxin is induced if the bacteria become deficient in iron. Flavodoxins all contain riboflavin monophosphate, which functions by cycling between the fully reduced anionic form and a blue semiquinone radical.[270] The two reduction steps, from oxidized flavin to semiquinone and from semiquinone to dihydroflavin, are well separated. For example, the values of $E°'$ (pH 7) for the flavodoxin from *Megasphaera elsdenii* are −0.115 and −0.373 V, while those of the *Azotobacter vinlandii* flavodoxin (azotoflavin) are + 0.050 and – 0.495 V. The latter is the lowest known for any flavoprotein.

Flavodoxins are small proteins with an α/β structure resembling that of the nucleotide binding domain of dehydrogenases. According to ^{31}P NMR data, the phosphate group of the coenzyme bound to flavodoxin is completely ionized,[271] even though it is deeply buried in the protein and is not bound to any positively charged side chain but to the N terminus of an α helix and to four –OH groups of serine and threonine side chains. The flavin ring is partially buried near the surface of the 138-residue protein. An aromatic side chain, from tryptophan or tyrosine, lies against the flavin on the outside of the molecule. Flavodoxins can be crystallized in all three forms: oxidized, semiquinone, and fully reduced. In the crystals the flavin semiquinone, like the oxidized flavin, is nearly planar.

The **DNA photolyase** of *E. coli*, an enzyme that participates in the photochemical repair of damaged DNA (Chapter 23), contains a blue neutral FAD radical with a 580-nm absorption band and an appropriate ESR signal.[230,272] In contrast, the mitochondrial **electron-transferring flavoprotein** (ETF), a 57-kDa αβ dimer containing one molecule of FAD, functions as a single electron carrier cycling between oxidized FAD and the red anionic semiquinone.[273,274] The reduced forms of the acyl-CoA dehydrogenases transfer their electrons one at a time from their FAD to the FAD of two molecules of electron-transferring flavoprotein. Therefore, an intermediate enzyme-bound radical must be present in the FAD of acyl-CoA dehydrogenase at one stage of its catalytic cycle. A related ETF from a methylotrophic bacterium accepts single electrons from reduced trimethylamine dehydrogenase (Fig. 15-9).[275]

Another flavoprotein that makes use of both one- and two-electron transfer reactions is **ferredoxin-NADP$^+$ oxidoreductase** (Eq. 15-28). Its bound FAD accepts electrons one at a time from each of the two

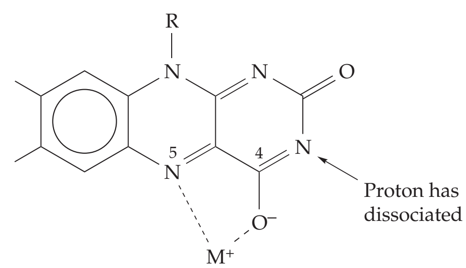
$$2Fd_{red} + NADP^+ + H^+ \rightarrow 2\ Fd_{ox}{}^+ + NADPH \qquad (15\text{-}28)$$

reduced ferredoxins (Fd_{red}) in chloroplasts and then presumably transfers a hydride ion to the NADP$^+$. The enzyme is organized into two structural domains,[201] one of which binds FAD and the other NADP$^+$. Similar single-electron transfers through flavoproteins also occur in many other enzymes. Chorismate mutase, an important enzyme in biosynthesis of aromatic rings (Chapter 25), contains bound FMN. Its function is unclear but involves formation of a neutral flavin radical.[276,277]

7. Metal Complexes of Flavins and Metalloflavoproteins

The presence of metal ions in many flavoproteins suggested a direct association of metal ions and flavins. Although oxidized flavins do not readily bind most metal ions, they form red complexes with Ag^+ and Cu^+ with a loss of a proton from N-3.[278] Flavin semiquinone radicals also form strong red complexes with many metals.[264] If the complexed metal ion can exist in more than one oxidation state, electron transfer between the flavin and a substrate could take place through the metal atom. However, *chelation by flavins in nature has not been observed*. Metalloflavoproteins probably function by having the metal centers close enough to the flavin for electron transfer to occur but not in direct contact. This is the case for a bacterial trimethylamine dehydrogenase in which the FeS cluster is bound about 0.4 nm from the alloxazine ring of riboflavin 5'-phosphate as shown in Fig. 15-9.

Some metalloflavoproteins contain heme groups. The previously mentioned **flavocytochrome *b*$_2$** of yeast is a 230-kDa tetramer, one domain of which carries riboflavin phosphate and another heme. A flavocytochrome from the photosynthetic sulfur bacterium *Chromatium* (cytochrome *c*-552)[279] is a complex of a 21-kDa cytochrome *c* and a 46-kDa flavoprotein containing 8α-(*S*-cysteinyl)-FAD. The 670-kDa **sulfite reductase** of *E. coli* has an $\alpha_8\beta_4$ subunit structure. The eight α chains bind four molecules of FAD and four of riboflavin phosphate, while the β chains bind three or four molecules of **siroheme** (Fig. 16-6) and also contain Fe_4S_4 clusters.[280,281] Many nitrate and some nitrite reductases are flavoproteins which also contain Mo or



Fe prosthetic groups.[282,283] A group of **aldehyde oxidases** and **xanthine dehydrogenases** also contain molybdenum as well as iron (Chapter 16). In every case the metal ions are bound independently of the flavin.[283a]

8. Reactions of Reduced Flavins with Oxygen

Free dihydroriboflavin reacts nonenzymatically in seconds and reduced flavin oxygenases react even faster with molecular oxygen to form hydrogen peroxide (Eq. 15-29).

$$FlH_2 + O_2 \rightarrow Fl + H_2O_2 \qquad (15\text{-}29)$$

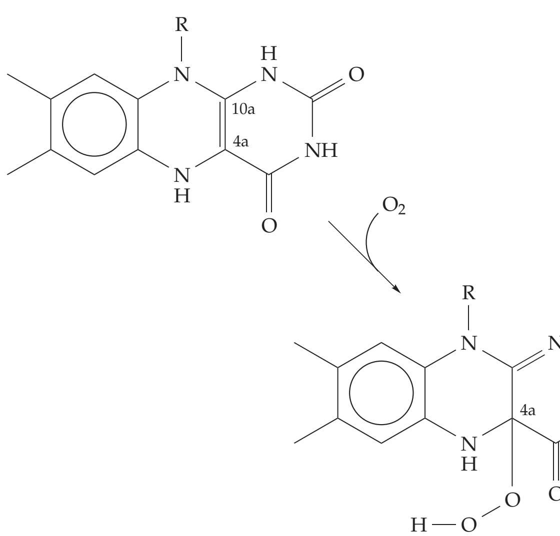
The reaction is more complex than it appears. As soon as a small amount of oxidized flavin is formed, it reacts with reduced flavin to generate flavin radicals $^\bullet$FlH (Eq. 15-27). The latter react rapidly with O_2 each donating an electron to form superoxide anion radicals $^\bullet O_2^-$ (Eq. 15-30a) which can then combine with flavin radicals (Eq. 15-30b).[284]

$$^\bullet FlH + O_2 \rightarrow {}^\bullet O_2^- + H^+ + Fl \qquad (15\text{-}30a)$$

$$^\bullet O_2^- + {}^\bullet FlH + H^+ \rightarrow Fl + H_2O_2 \qquad (15\text{-}30b)$$

During the corresponding reactions of reduced flavoproteins with O_2, intermediates have been detected. For example, spectrophotometric studies of the FAD-containing bacterial *p*-hydroxybenzoate hydroxylase (Chapter 18) revealed the consecutive appearance of three intermediate forms.[285–287] The first, whose absorption maximum is at 380–390 nm, is thought to be an adduct at position 4a (Eq. 15-31). That such a 4a peroxide really forms with the riboflavin phosphate of the light-emitting bacterial **luciferase** (Chapter 23) was demonstrated using coenzyme enriched with ^{13}C at position 4a.[147] A large shift to lower frequency (from 104 to 83 ppm) accompanied formation of the transient adduct. Comparison with reference compounds showed that this change agreed with that predicted.

Other structures for O_2 adducts have also been considered, as has the possibility of rearrangements among these structures.[288] Nevertheless, the products observed from many different flavoprotein reactions can be explained on the basis of a 4a peroxide.[289]



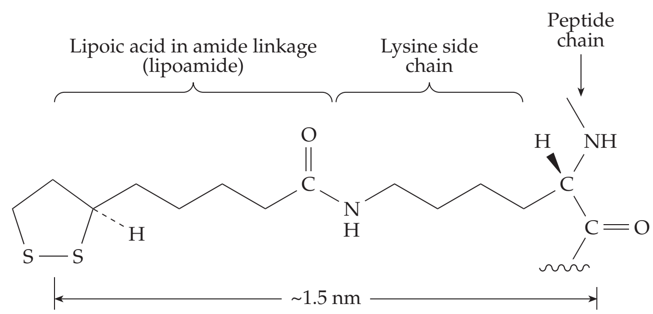
(15-31)

Formation of H_2O_2 by flavin oxidases can occur via elimination of a peroxide anion HOO^- from the adduct of Eq. 15-31 with regeneration of the oxidized flavin. In the active site of a hydroxylase, an OH group can be transferred from the peroxide to a suitable substrate (Eq. 18-42). Although radical mechanisms are likely to be involved, such hydroxylation reactions can also be viewed as transfer of OH^+ to the substrate together with protonation on the inner oxygen atom of the original peroxide to give a 4a –OH adduct. The latter is a covalent hydrate which can be converted to the oxidized flavin by elimination of H_2O. This hydrate is believed to be the third spectral intermediate identified during the action of *p*-hydroxybenzoate hydroxylase.[286,287,290]

C. Lipoic Acid and the Oxidative Decarboxylation of α-Oxoacids

The isolation of lipoic acid in 1951 followed an earlier discovery that the ciliate protozoan *Tetrahymena geleii* required an unknown factor for growth. In independent experiments acetic acid was observed to promote rapid growth of *Lactobacillus casei,* but it could be replaced by an unknown "acetate replacing factor." Another lactic acid bacterium *Streptococcus faecalis* was unable to oxidize pyruvate without addition of "pyruvate oxidation factor." By 1949, all three unknown substances were recognized as identical.[291,291a] After working up the equivalent of 10 tons of water-soluble residue from liver, Lester Reed and his collaborators isolated 30 mg of a fat-soluble acidic material which was named **lipoic acid** (or 6-thioctic acid).[292–294]

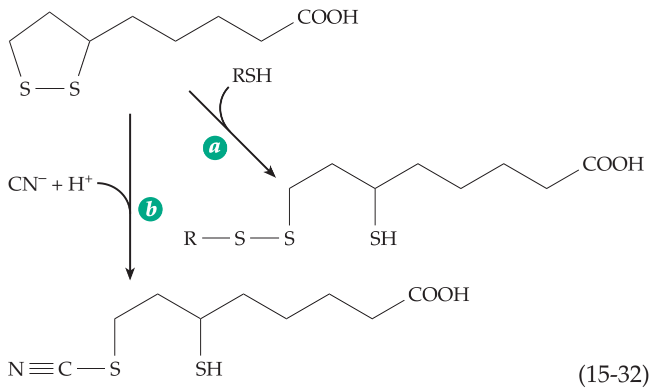
While *Tetrahymena* must have lipoic acid in its diet, we humans can make our own, and it is not considered a vitamin. Lipoic acid is present in tissues in extraordinarily small amounts. Its major function is to participate in the oxidative decarboxylation of α-oxoacids but it also plays an essential role in glycine catabolism in the human body as well as in plants.[295,296] The structure is simple, and the functional group is clearly the cyclic disulfide which swings on the end of a long arm. Like biotin, which is also present in tissues in very small amounts, lipoic acid is bound in covalent amide linkage to lysine side chains in active sites of enzymes:[296a]



1. Chemical Reactions of Lipoic Acid

The most striking chemical property of lipoic acid is the presence of ring strain of ~17–25 kJ mol^{-1} in the cyclic disulfide. Because of this, thiol groups and cyanide ions react readily with oxidized lipoic acid to give mixed disulfides (Eq. 15-32a) and isothiocyanates (Eq. 15-32b), respectively.

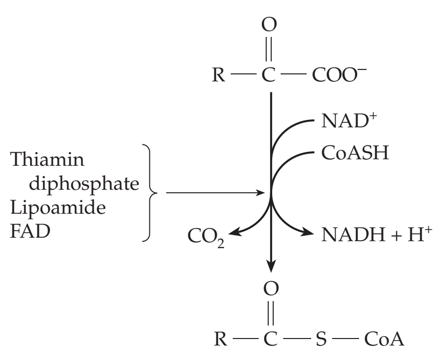
Another result of the ring strain is that the reduction potential $E°'$ (pH 7, 25°C), is –0.30 V, almost the same as that of reduced NAD (–0.32V). Thus, reoxidation of reduced lipoic acid amide by NAD^+ is thermodynamically feasible. Yet another property attributed to the ring strain in lipoic acid is the presence of an absorption maximum at 333 nm.



(15-32)

2. Oxidative Decarboxylation of Alpha-Oxoacids

The oxidative cleavage of an α-oxoacid is a major step in the metabolism of carbohydrates and of amino acids and is also a step in the citric acid cycle. In many bacteria and in eukaryotes the process depends upon both thiamin diphosphate and lipoic acid. The oxoacid anion is cleaved to form CO_2 and the remaining acyl group is combined with coenzyme A (Eq. 15-33). NAD^+ serves as the oxidant. The reaction is catalyzed by a complex of enzymes whose molecular mass varies from ~4 to 10 x 10^6, depending on the species and exact substrate.[297] Separate oxoacid dehydrogenase systems are known for pyruvate,[298–300] 2-oxoglutarate,[301] and the 2-oxoacids with branched side chains derived metabolically from leucine, isoleucine, and



(15-33)

valine.[302,302a] In eukaryotes these enzymes are located in the mitochondria. The **pyruvate** and **2-oxoglutarate dehydrogenase** complexes of *E. coli* and *Azotobacter vinelandii* have been studied most. In both cases there are three major protein components. The first (E_1) is a **decarboxylase** (also referred to as a dehydrogenase) for which the thiamin diphosphate is the dissociable cofactor. The second (E_2) is a lipoic acid amide-containing "core" enzyme which is a **dihydrolipoyl transacylase.** The third (E_3) is the flavoprotein **dihydrolipoyl dehydrogenase**, a member of the glutathione reductase family with a three-dimensional structure and catalytic mechanism similar to those of glutathione reductase (Figs. 15-10, 15-12).[303–305]

Electron microscopy of the core dihydrolipoyl transacylase from *E. coli* reveals a striking octahedral symmetry which has been confirmed by X-ray diffraction.[306–307a] The core from pyruvate dehydrogenase has a mass of ~2390 kDa and contains 24 identical 99.5-kDa E_2 subunits. The 2-oxoglutarate dehydrogenase from *E. coli* has a similar but slightly less symmetric structure. Each core subunit is composed of three domains. A lipoyl group is bound in amide linkage to lysine 42 and protrudes from one end of the domain. A second domain is necessary for binding to subunits E_1 and E_3, while the third major 250-residue domain contains the catalytic acyltransferase center.[308,309] This center closely resembles that of chloramphenicol acetyltransferase (Chapter 12).[310,311] The lipoyl[301,309,312] and catalytic[307] domains of the dihydrolipoyl succinyltransferase from 2-oxoglutarate dehydrogenase resemble those of pyruvate dehydrogenase and also of the branched chain oxoacid dehydrogenase. The three domains of the proteins are joined by long 25- to 30-residue segments rich in alanine, proline, and ionized hydrophilic side chains.[309] This presumably provides flexibility for the lipoyl groups which must move from site to site. The presence of unexpectedly sharp lines in the proton NMR spectrum of the core protein may be a result of this flexibility.[309]

To obtain the X-ray structure of the core protein it was necessary to delete the lipoyl- and $E_1(E_2)$- binding domains. The resulting 24-subunit structure is shown in Fig 15-14A,B.[306] It has been assumed for many years that 12 of the dimeric decarboxylase units (E_1) are bound to the 12 edges of the transacetylase cube, while six (50.6 x 2 kDa) flavoprotein (E_3) dimers bind on the six faces of the cube. The active centers of all three types of subunits are thought to come close together in the regions where the subunits touch, permitting the sequence of catalytic reactions indicated in Fig. 15-15 to take place. Eukaryotic as well as some bacterial pyruvate decarboxylases have a core of 60 subunits in an icosahedral array with 532 symmetry. This can be seen in the image reconstructions of the enzyme from *Saccharomyces cerevisiae* shown in Fig. 15-14C and D. A surprising discovery is that in this yeast enzyme the E_3 units are not on the outside of the 5-fold symmetric faces but protrude into the inner cavity.[299] Each of the 12 E_3 subunits is assisted in binding correctly to the E_2 core by a molecule of the 47-kDa **E_3-binding protein** (BP), also known as protein X.[313,314] Absence of this protein is associated with congenital lactic acidosis.

The unique function of lipoic acid is in the oxidation of the thiamin-bound active aldehyde (Fig. 15-15) in such a way that when the complex with thiamin breaks up, the acyl group formed by the oxidative decarboxylation of the oxoacid is attached to the

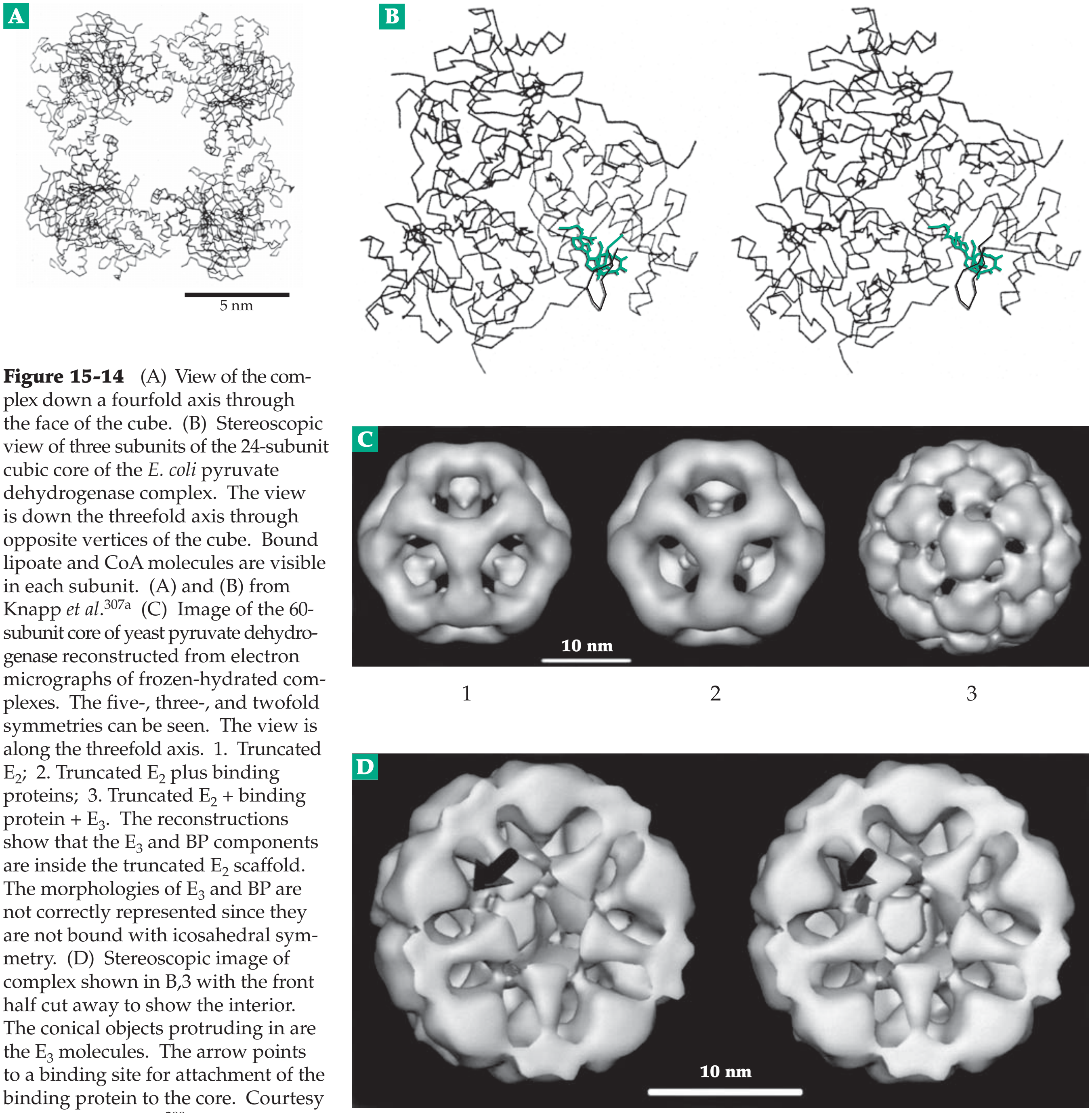


Figure 15-14 (A) View of the complex down a fourfold axis through the face of the cube. (B) Stereoscopic view of three subunits of the 24-subunit cubic core of the *E. coli* pyruvate dehydrogenase complex. The view is down the threefold axis through opposite vertices of the cube. Bound lipoate and CoA molecules are visible in each subunit. (A) and (B) from Knapp *et al*.[307a] (C) Image of the 60-subunit core of yeast pyruvate dehydrogenase reconstructed from electron micrographs of frozen-hydrated complexes. The five-, three-, and twofold symmetries can be seen. The view is along the threefold axis. 1. Truncated E_2; 2. Truncated E_2 plus binding proteins; 3. Truncated E_2 + binding protein + E_3. The reconstructions show that the E_3 and BP components are inside the truncated E_2 scaffold. The morphologies of E_3 and BP are not correctly represented since they are not bound with icosahedral symmetry. (D) Stereoscopic image of complex shown in B,3 with the front half cut away to show the interior. The conical objects protruding in are the E_3 molecules. The arrow points to a binding site for attachment of the binding protein to the core. Courtesy of J. K. Stoops *et al*.[299]

dihydrolipoyl group at the S-8 position.[315] The lipoic acid, which is attached to the flexible lipoyl domains of the core enzyme on a 1.5-nm-long arm, apparently first contacts the thiamin diphosphate site on one of the decarboxylase subunits. Bearing the acyl group, it now swings to the catalytic site on the core enzyme where CoA is bound. The acyl group is transferred to CoA producing a dihydrolipoyl group which then swings to the third subunit where the disulfide loop and a bound FAD of dihydrolipoamide dehydrogenase reoxidize the lipoyl group. The reduced flavin-disulfide enzyme is then oxidized by NAD^+ (Fig. 15-15) by the reverse of the mechanism depicted in Fig. 15-12.

Although the direct reaction of a lipoyl group with the thiamin-bound enamine (active aldehyde) is generally accepted, and is supported by recent studies,[315a] an alternative must be considered.[315] Hexacyanoferrate (III) can replace NAD^+ as an oxidant for pyruvate dehydrogenase and is also able to oxidize nonenzymatically thiamin-bound active acetaldehyde

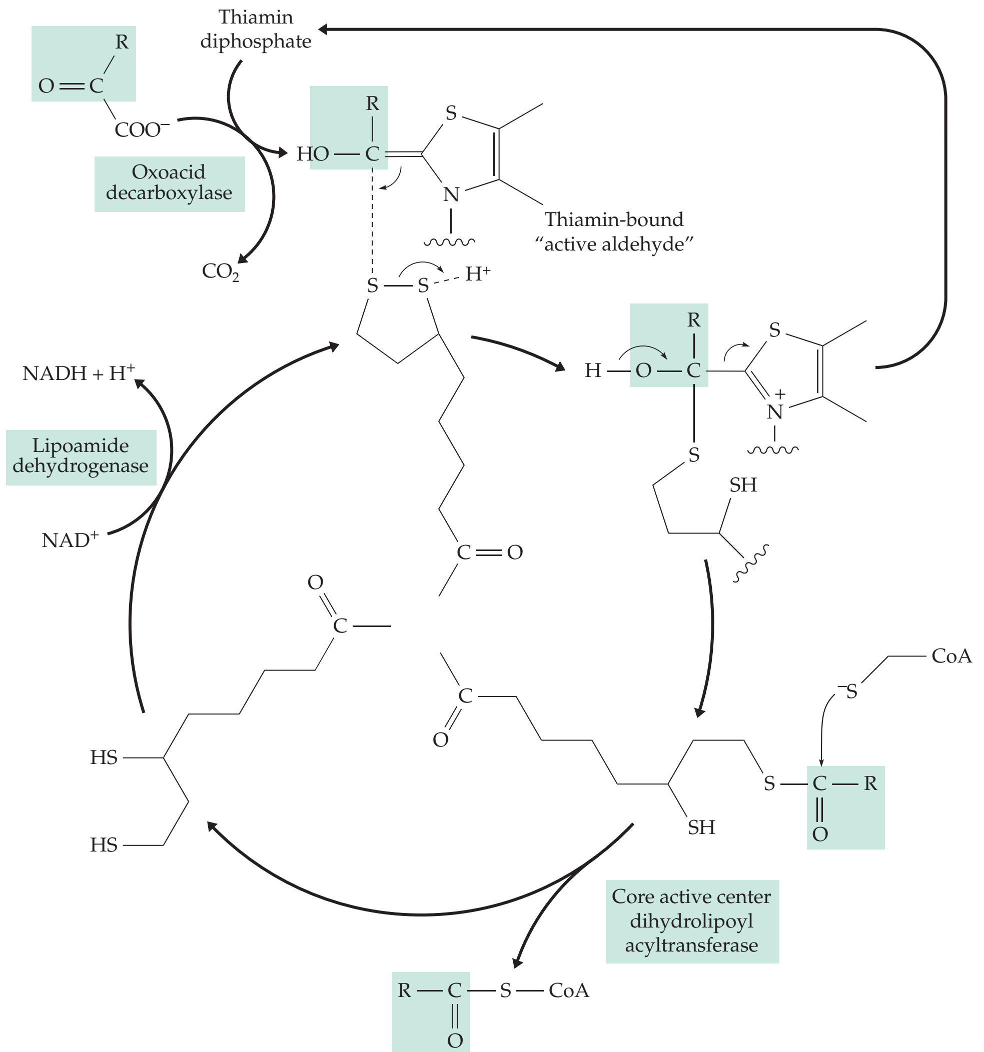


Figure 15-15 Sequence of reactions catalyzed by α-oxoacid dehydrogenases. The substrate and product are shown in boxes, and the path of the oxidized oxoacid is traced by the heavy arrows. The lipoic acid "head" is shown rotating about the point of attachment to a core subunit. However, a whole flexible domain of the core is also thought to move.

to 2-acetylthiamin, a compound in which the acetyl group has a high group transfer potential (Eq. 14-22). Thus, the lipoyl group could first oxidize the active aldehyde to a thiamin-bound acyl derivative, and then in the second step accept the acyl group by a nucleophilic displacement reaction. This mechanism fails to explain the unique role of lipoic acid in oxidative decarboxylation. However, as we will see in the next section, oxidative decarboxylation does not always require lipoic acid and acetylthiamin is probably an intermediate whenever lipoamide is not utilized.

Within many tissues the enzymatic activities of the pyruvate and branched chain oxoacid dehydrogenases complexes are controlled in part by a phosphorylation –dephosphorylation mechanism (see Eq. 17-9). Phosphorylation of the decarboxylase subunit by an ATP-dependent kinase produces an inactive phosphoenzyme. A phosphatase reactivates the dehydrogenase to complete the regulatory cycle (see Eq. 17-9 and associated discussion). The regulation is apparently accomplished, in part, by controlling the affinity of the protein for thiamin diphosphate.[315b] The lipoamide dehydrogenase component of all three dehydrogenase complexes appears to be the same.

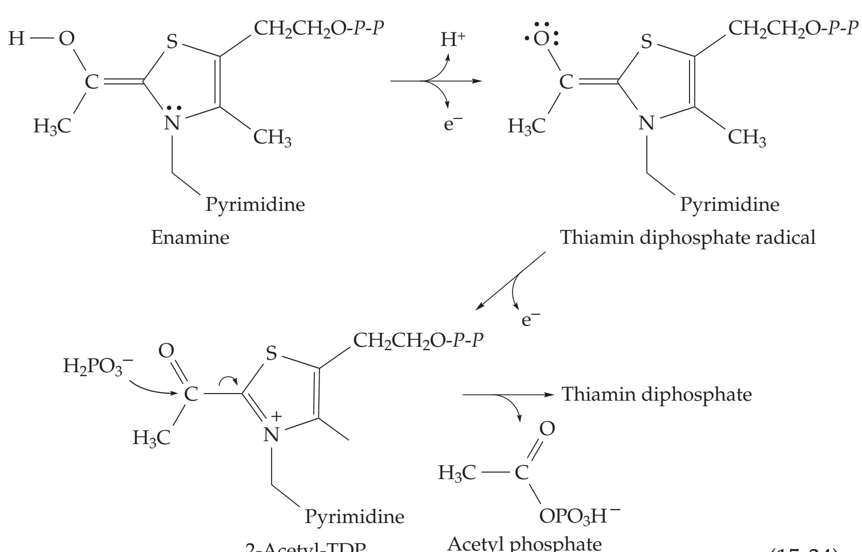
3. Other Functions of Lipoic Acid

In addition to its role in the oxidative decarboxylation of 2-oxoacids, lipoic acid functions in the human body as part of the essential mitochondrial glycine-cleavage system described in Section E. It may also participate in bacterial glycine reductase (Eq. 15-61) and other enzyme systems. Dihydrolipoamide dehydrogenase binds to G4-DNA structures of telomeres and may have a biological role in DNA-binding.[315c] Lipoic acid is being utilized as a nutritional supplement and appears to help in maintaining cellular levels of glutathione.[316]

4. Additional Mechanisms of Oxidative Decarboxylation

Pyruvate is a metabolite of central importance and a variety of mechanisms exist for its cleavage. The pyruvate dehydrogenase complex is adequate for strict aerobes and is very important in aerobic bacteria and in facultative anaerobes such as *E. coli*. Lactic acid bacteria, which lack cytochromes and other heme proteins, are able to carry out limited oxidation with flavoproteins such as **pyruvate oxidase**.[317] However, *strict* anaerobes have to avoid accumulation of reduced pyridine nucleotides and use a nonoxidative cleavage of pyruvate by **pyruvate formate-lyase**. The pyruvate dehydrogenase complex of *our* bodies generates NADH which can be oxidized in mitochondria to provide energy to our cells. However, NADH is a weak reductant. Some cells, such as those of nitrogen-fixing bacteria, require more powerful reductants such as reduced ferredoxin and utilize **pyruvate: ferredoxin oxidoreductase**. *Escherichia coli*, a facultative anaerobe, is adaptable and makes use of all of these types of pyruvate cleavage.[318]

Pyruvate oxidase. The soluble flavoprotein pyruvate oxidase, which was discussed briefly in Chapter 14 (Fig. 14-2, Eq. 14-22), acts together with a membrane-bound electron transport system to convert pyruvate to acetyl phosphate and CO_2.[319] Thiamin diphosphate is needed by this enzyme but lipoic acid is not. The flavin probably dehydrogenates the thiamin-bound intermediate to 2-acetylthiamin as shown in Eq. 15-34. The electron acceptor is the bound FAD and the reaction may occur in two steps as shown with a thiamin diphosphate radical intermediate.[319a] Reaction with inorganic phosphate generates the energy storage metabolite **acetyl phosphate**.



(15-34)

Pyruvate:ferredoxin oxidoreductase. Within clostridia and other strict anaerobes this enzyme catalyzes *reversible* decarboxylation of pyruvate (Eq. 15-35). The oxidant used by clostridia is the low-potential iron–sulfur ferredoxin.[320,320a] Clostridial ferredoxins contain two Fe–S clusters and are therefore two-electron oxidants. Ferredoxin substitutes for NAD^+ in Eq. 15-33 but the Gibbs energy decrease is much less (–16.9 vs – 34.9 kJ / mol. for oxidation by NAD^+).

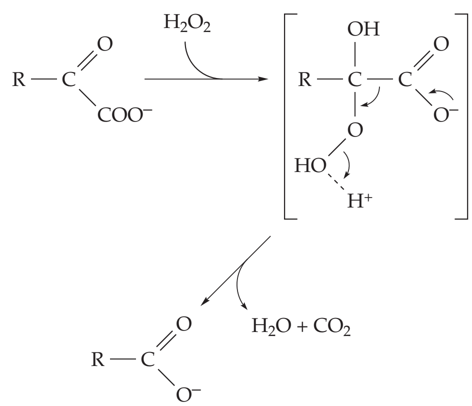
$$\text{Pyruvate}^- + \text{CoA} \xrightarrow[\text{Ferredoxin (ox)} \;\rightarrow\; \text{Ferredoxin (red)}]{} \text{Acetyl-CoA} + CO_2$$

$$\Delta G' \text{ (pH 7)} = -16.9 \text{ kJ mol}^{-1} \qquad (15\text{-}35)$$

The enzyme does not require lipoic acid. It seems likely that a thiamin-bound enamine is oxidized by an iron–sulfide center in the oxidoreductase to 2-acetylthiamin which then reacts with CoA. A free radical intermediate has been detected[318,321] and the proposed sequence for oxidation of the enamine intermediate is that in Eq. 15-34 but with the Fe–S center as the electron acceptor. Like pyruvate oxidase, this enzyme transfers the acetyl group from acetylthiamin to coenzyme A. Cleavage of the resulting acetyl-CoA is used to generate ATP. An indolepyruvate: ferredoxin oxidoreductase has similar properties.[322]

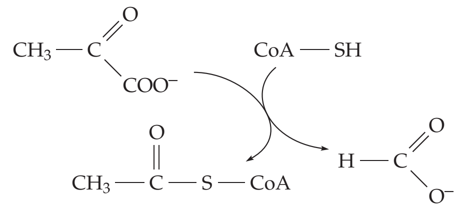
In methanogenic bacteria[320] the low-potential 5-deazaflavin coenzyme F_{420} serves as the reductant in a reversal of Eq. 15-35. A similar enzyme, **2-oxoglutarate synthase**, apparently functions in synthesis of 2-oxoglutarate from succinyl-CoA and CO_2 by photosynthetic bacteria.[323] Either reduced ferredoxin or, in *Azotobacter*, reduced flavodoxin is generated in nitrogen-fixing bacteria (Chapter 24) by cleavage of pyruvate and is used in the N_2 fixation process.

Oxidative decarboxylation by hydrogen peroxide. The nonenzymatic oxidative decarboxylation of α-oxoacids by H_2O_2 is well-known. The first step is the formation of an adduct, an organic peroxide, which breaks up as indicated in Eq. 15-36. An enzyme-catalyzed version of this reaction is promoted by **lactate monooxygenase**, a 360-kDa octameric flavoprotein obtained from *Mycobacterium smegmatis*[324] and a member of the glycolate oxidase family. Under anaerobic conditions, the enzyme produces pyruvate by a simple dehydrogenation. However, the pyruvate dissociates slowly and in the presence of oxygen it forms acetic acid, with one of the oxygen atoms of the carboxyl group coming from O_2.[324,325] Hydrogen peroxide is the usual product formed from oxygen by flavoprotein oxidases, and it seems likely that with lactate monooxygenase the hydrogen peroxide formed immediately oxidizes the pyruvate according to Eq. 15-36. The α-oxoacids formed by amino acid oxidases *in vitro* are also oxidized by accumulating hydrogen peroxide. However, if catalase (Chapter 16) is present it destroys the H_2O_2 and allows the oxoacid to accumulate.



(15-36)

Pyruvate formate-lyase reaction. Anaerobic cleavage of pyruvate to acetyl-CoA and formate (Eq. 15-37) is essential to the energy economy of many cells, including those of *E. coli*. No external oxidant is needed, and the reaction does not require lipoic acid. The reaction has sometimes been called the "phosphoroclastic reaction" because the product acetyl-CoA usually reacts further with inorganic phosphate to form acetyl phosphate (see Fig. 15-16). The latter can then transfer its phospho group to ADP to form ATP.



$\Delta G'$ (pH 7) = −12.9 kJ mol^{-1} (15-37)

The mechanism of the cleavage of the pyruvate in Eq. 15-37 is not obvious. Thiamin diphosphate is not involved, and free CO_2 is not formed. The first identified intermediate is an acetyl-enzyme containing a thioester linkage to a cysteine side chain. This is cleaved by reaction with CoA-SH to give the final product. A clue came when it was found by Knappe and coworkers that the active enzyme, which is rapidly inactivated by oxygen, contains a long-lived free radical.[326] Under anaerobic conditions cells convert the inactive form E_i to the active form E_a by an enzymatic reaction with *S*-adenosylmethionine and reduced flavodoxin Fd(red) as shown in Eq. 15-38.[327–329] A deactivase reverses the process.[330]

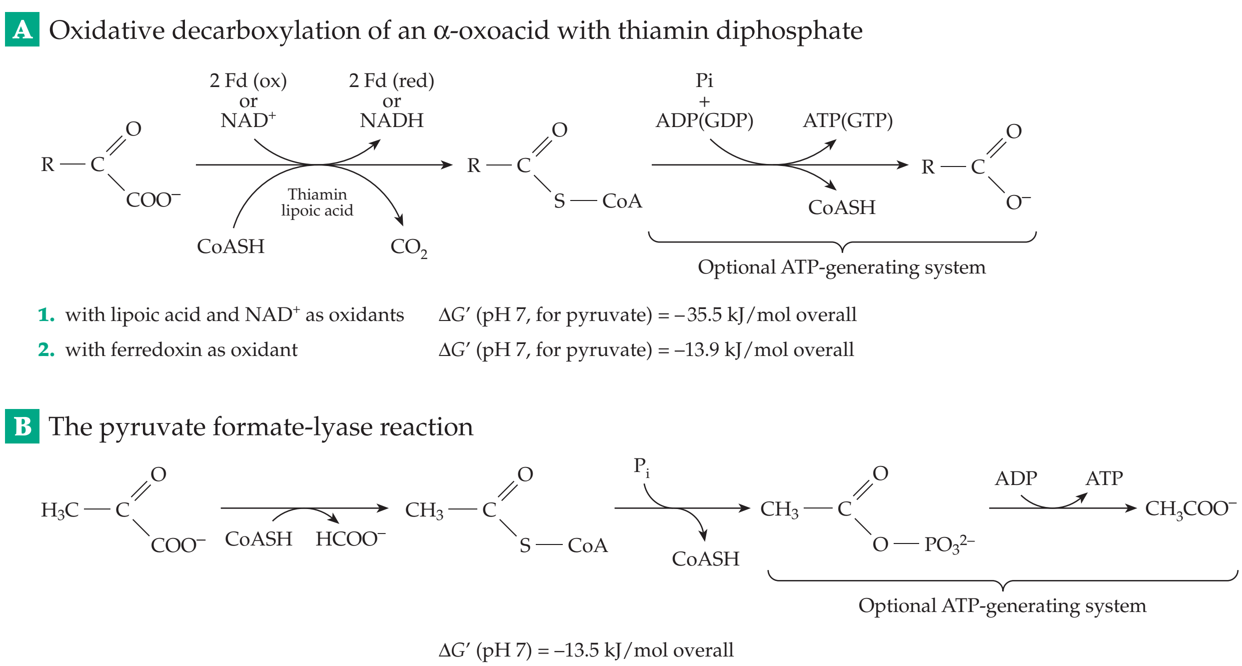
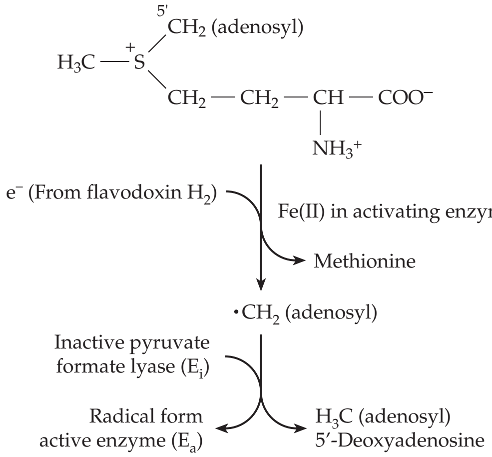
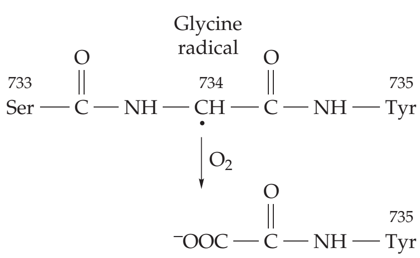


Figure 15-16 Two systems for oxidative decarboxylation of α-oxoacids and for "substrate-level" phosphorylation. The value of $\Delta G'$ = +34.5 kJ mol^{-1} (Table 6-6) was used for the synthesis of ATP^{4-} from ADP^{3-} and HPO_4^{2-} in computing the values of $\Delta G'$ given.



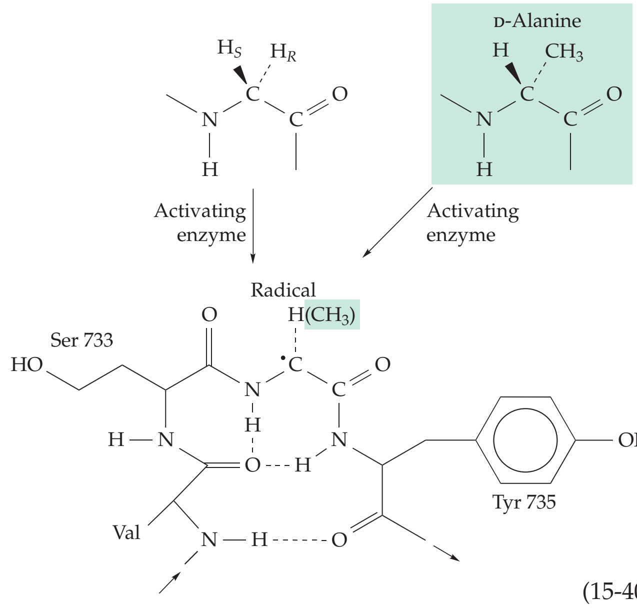
(15-38)

The activating enzyme, which is allosterically activated by pyruvate, is an iron–sulfur (Fe_4S_4) protein (Chapter 16).[331,331a] Formation of the observed radical may proceed via a 5'-deoxyadenosyl radical as has been proposed for lysine 2,3-aminomutase (Eq. 16-42).[332] The activation reaction also resembles the free radical-dependent reactions of vitamin B_{12} which are discussed in Chapter 16. When subjected to O_2 of air at 25°C pyruvate formate-lyase is destroyed with a half-life of ~10 s. The peptide chain is cleaved and sequence analysis and mass spectrometry of the resulting fragments show that the specific sequence Ser-Gly-Tyr at positions 733–755 is cut with formation of an oxalyl group on the N-terminal tyrosine of one fragment (Eq. 15-39). Various ^{13}C-containing amino acids were supplied to growing cells of *E. coli* and were incorporated into the proteins of the bacteria. Pyruvate formate-lyase containing ^{13}C in carbon-2 of glycine gave an EPR spectrum with hyperfine splitting arising from coupling of the unpaired electron of the radical with the adjacent ^{13}C nucleus.[333,333a] This experiment, together with the results described by Eq. 15-39, suggested that the radical is derived from Gly 734.



(15-39)

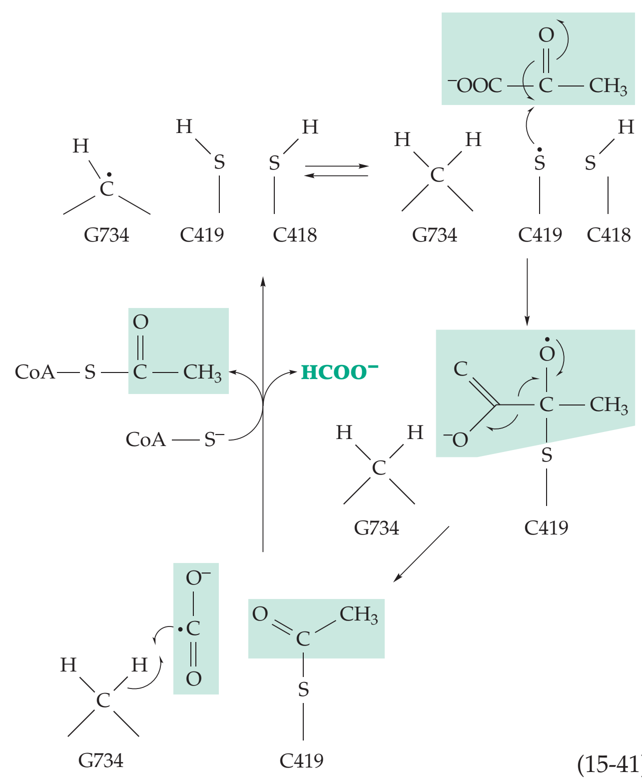
The activating enzyme will also generate radicals from short peptides such as Arg-Val-Ser-Gly-Tyr-Ala-Val, which corresponds to residues 731–737 of the pyruvate formate-lyase active site. If Gly 734 is replaced by L-alanine, no radical is formed, but radical *is* formed if D-alanine is in this position. This suggests that the *pro-S* proton of Gly 734 is removed by the activating



(15-40)

enzyme[329] as illustrated in Eq. 15-40. It has been suggested that the peptide exists in the β-bend conformation shown.

How does the enzyme work? The α-CH proton of the glycyl radical of the original *pro-R* proton undergoes an unexpected exchange with the deuterium of ^2H_2O. The exchange is catalyzed by the thiol group of Cys 419, suggesting that it is close to Gly 734 in the three-dimensional structure.[334] The mutants C419S and C418S are inactive but still allow formation of the Gly 734 radical.[334–335a] The mechanism has been proposed in Eq. 15-41.



(15-41)

5. Cleavage of α-Oxoacids and Substrate-Level Phosphorylation

The α-oxoacid dehydrogenases yield CoA derivatives which may enter biosynthetic reactions. Alternatively, the acyl-CoA compounds may be cleaved with generation of ATP. The pyruvate formate-lyase system also operates as part of an ATP-generating system for anaerobic organisms, for example, in the "mixed acid fermentation" of enterobacteria such as *E. coli* (Chapter 17). These two reactions, which are compared in Fig. 15-16, constitute an important pair of processes both of which accomplish substrate-level phosphorylation. They should be compared with the previously considered examples of substrate level phosphorylation depicted in Eq. 14-23 and Fig. 15-16.

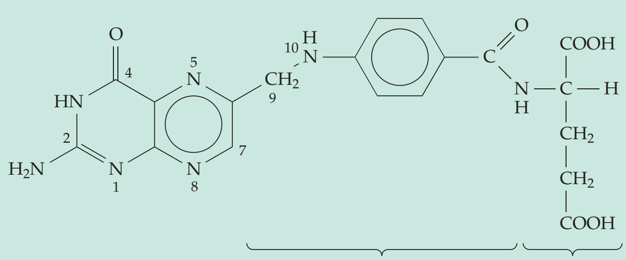
D. Tetrahydrofolic Acid and Other Pterin Coenzymes

In most organisms reduced forms of the vitamin **folic acid** serve as *carriers for one-carbon groups* at three different oxidation levels corresponding to **formic acid**, **formaldehyde**, and the **methyl group** and also facilitate their interconversion.[336,337] Moreover, folic acid is just one of the derivatives of the **pteridine** ring system that enjoy a widespread natural distribution.[338] One of these derivatives functions in the hydroxylation of aromatic amino acids and in nitric oxide synthase and yet another within several molybdenum-containing enzymes. Pteridines also provide coloring to insect wings and eyes and to the skins of amphibians and fish. They appear to act as protective filters in insect eyes and may function as light receptors. An example is a folic acid derivative found in some forms of the DNA photorepair enzyme **DNA photolyase** (Chapter 23).

1. Structure of Pterins

Because of the prevalence of its derivatives, 2-amino-4-hydroxypteridine has been given the trivial name **pterin**. Its structure resembles closely that of guanine, the 5-membered ring of the latter having been expanded to a 6-membered ring. In fact, pterins are derived

BOX 15-D FOLIC ACID (PTEROYLGLUTAMIC ACID)



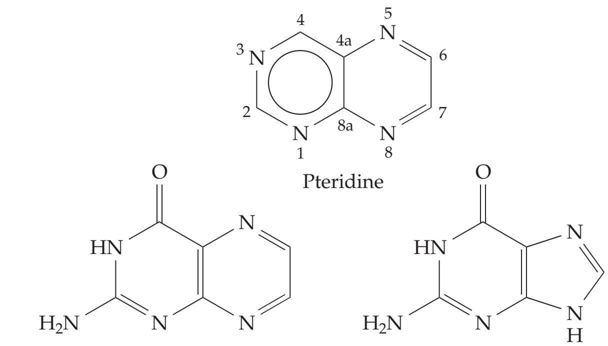
In 1931, Lucy Wills, working in India, observed that patients with **tropical macrocytic anemia**, a disease in which the erythrocytes are enlarged but reduced in numbers, were cured by extracts of yeast or liver. The disease could be mimicked in monkeys fed the local diet, and a similar anemia could be induced in chicks. By 1938 it had also been shown that a factor present in yeast, alfalfa, and other materials was required for the growth of chicks. Isolation of the new vitamin came rapidly after it was recognized that it was also an essential nutrient for *Lactobacillus casei* and *Streptococcus faecalis* R.[a,b] Spinach was a rich source of the new compound, and it was named folic acid (from the same root as the word foliage).

The microbiological activity attributed to folic acid in extracts of natural materials was largely that of di- and triglutamyl derivatives, one of the facts that has led to the description of the history of folic acid as "the most complicated chapter in the story of the vitamin B complex."[a] Metabolic functions for folic acid were suggested by the observations that the requirement for *Streptococcus faecalis* could be replaced by thymine plus serine plus a purine base. Folic acid is required for the biosynthesis of all of these substances. A function in the interconversion of serine and glycine was suggested by the observation that certain mutants of *E. coli* required either serine or glycine for growth. Isotopic labeling experiments established that in the rat as well as in the yeast *Torulopsis* serine and glycine could be interconverted. It was also shown that the amount of interconversion decreased in the folate-deficient rat.

Deficiency of folic acid is a common nutritional problem of worldwide importance.[b] A recommended daily intake is 0.2 mg, but because of the association between low folic acid intake and neural tube defects in infants, women of child-bearing age should have 0.4 mg / day.[c–e]

[a] Wagner, A. F., and Folkers, E. (1964) *Vitamins and Coenzymes*, Wiley (Interscience), New York
[b] Jukes, T. H. (1980) *Trends Biochem. Sci.* **5**, 112–113
[c] Cziezel, A. E., and Dudás, I. (1992) *N. Engl. J. Med.* **327**, 1832–1835
[d] Jukes, T. H. (1997) *Protein Sci.* **6**, 254–256
[e] Rosenquist, T. H., Ratashak, S. A., and Selhub, J. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 15227–15232

biosynthetically from guanine. The two-ring system of pterin is also related structurally and biosynthetically to that of riboflavin (Box 15-B).



Pteridine

Pterin: 2-amino-4-hydroxypteridine

Guanine

Interest in pteridines began with Frederick G. Hopkins, who in 1891 started his investigation of the yellow and white pigments of butterflies. Almost 50 years and a million butterflies later, the structures of the two pigments, **xanthopterin** and **leucopterin** (Fig. 15-17), were established.[339] These pigments are produced in such quantities as to suggest that their synthesis may be a means of deposition of nitrogenous wastes in dry form.

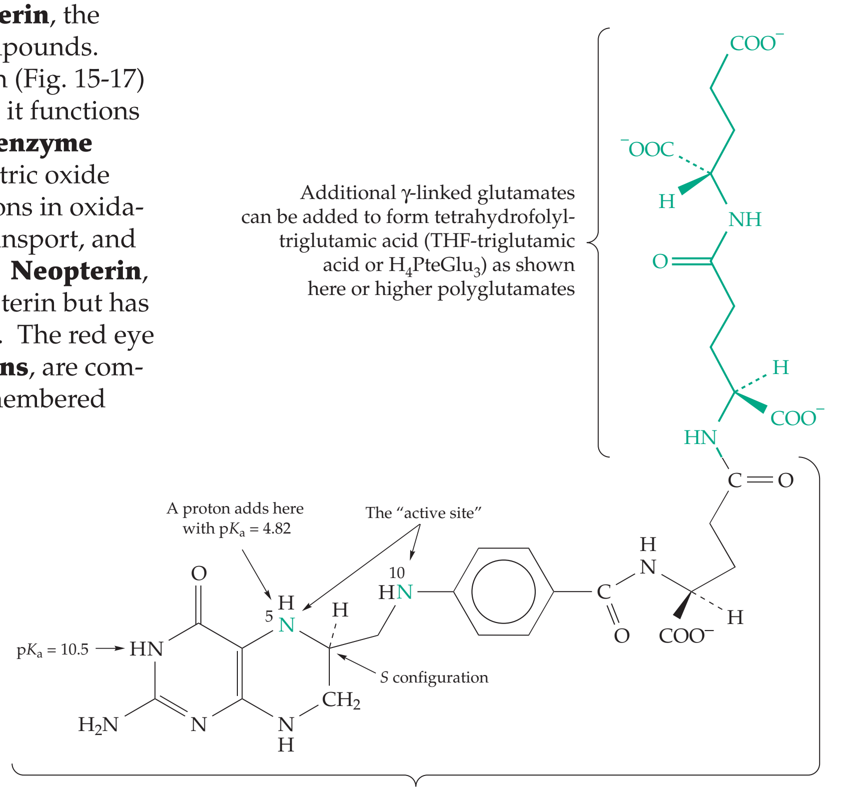
Among the simple pterins isolated from the eyes of *Drosophila*[340] is **sepiapterin** (Fig. 15-17), in which the pyrazine ring has been reduced in the 7,8 position and a short side chain is present at position 6. Reduction of the carbonyl group of sepiapterin with $NaBH_4$ followed by air oxidation produces **biopterin**, the most widely distributed of the pterin compounds. First isolated from human urine, biopterin (Fig. 15-17) is present in liver and other tissues where it functions in a reduced form as a **hydroxylation coenzyme** (see Chapter 18).[338] It is also present in nitric oxide synthase (Chapter 18).[341,342] Other functions in oxidative reactions, in regulation of electron transport, and in photosynthesis have been proposed.[343] **Neopterin**, found in honeybee larvae, resembles biopterin but has a D-*erythro* configuration in the side chain. The red eye pigments of *Drosophila*, called **drosopterins**, are complex dimeric pterins containing fused 7-membered rings (Fig. 15-17).[344,345]

In the pineal gland, as well as in the retina of the eye, light-sensitive pterins may be photochemically cleaved to generate such products as **6-formylpterin**, a compound that could serve as a metabolic regulator.[346] Another pterin acts as a chemical attractant for aggregation of the ameboid cells of *Dictyostelium lacteum*[347](Box 11-C). **Molybdopterin** (Fig. 15-17) is a component of several Mo-containing enzymes discussed in Chapter 16,[348,349] and **methanopterin** is utilized by methanogenic bacteria.[350–353]

2. Folate Coenzymes

The coenzymes responsible for carrying single-carbon units in most organisms are derivatives of **5,6,7,8-tetrahydrofolic acid** (abbreviated H_4PteGlu, H_4folate or THF). However, in methanogenic bacteria the tetrahydro derivative of the structurally unique methanopterin (Fig. 15-17) is the corresponding single-carbon carrier.[353] Naturally occurring tetrahydrofolates contain a chiral center of the *S* configuration.[354,355] They exhibit negative optical rotation at 589 nm. The folate coenzymes are present in extremely low concentrations and the reduced ring is readily oxidized by air.

In addition to the single L-glutamate unit present in tetrahydrofolic acid, the coenzymes occur to the greatest extent as conjugates called **folyl polyglutamates** in which one to eight or more additional molecules of L-glutamic acid have been combined via amide linkages.[356–357a] The first two of the extra glutamates are always joined through the γ (side chain) carboxyl groups but in *E. coli* the rest are joined through their α carboxyls.[358] The distribution of the polyglutamates varies from one organism to the next. Some bacteria contain exclusively the triglutamate derivatives, while in others almost exclusively the tetraglutamate or octaglutamate derivatives predominate.[359] The serum



Tetrahydrofolic acid (THF) or tetrahydropteroylglutamic acid (H_4PteGlu)

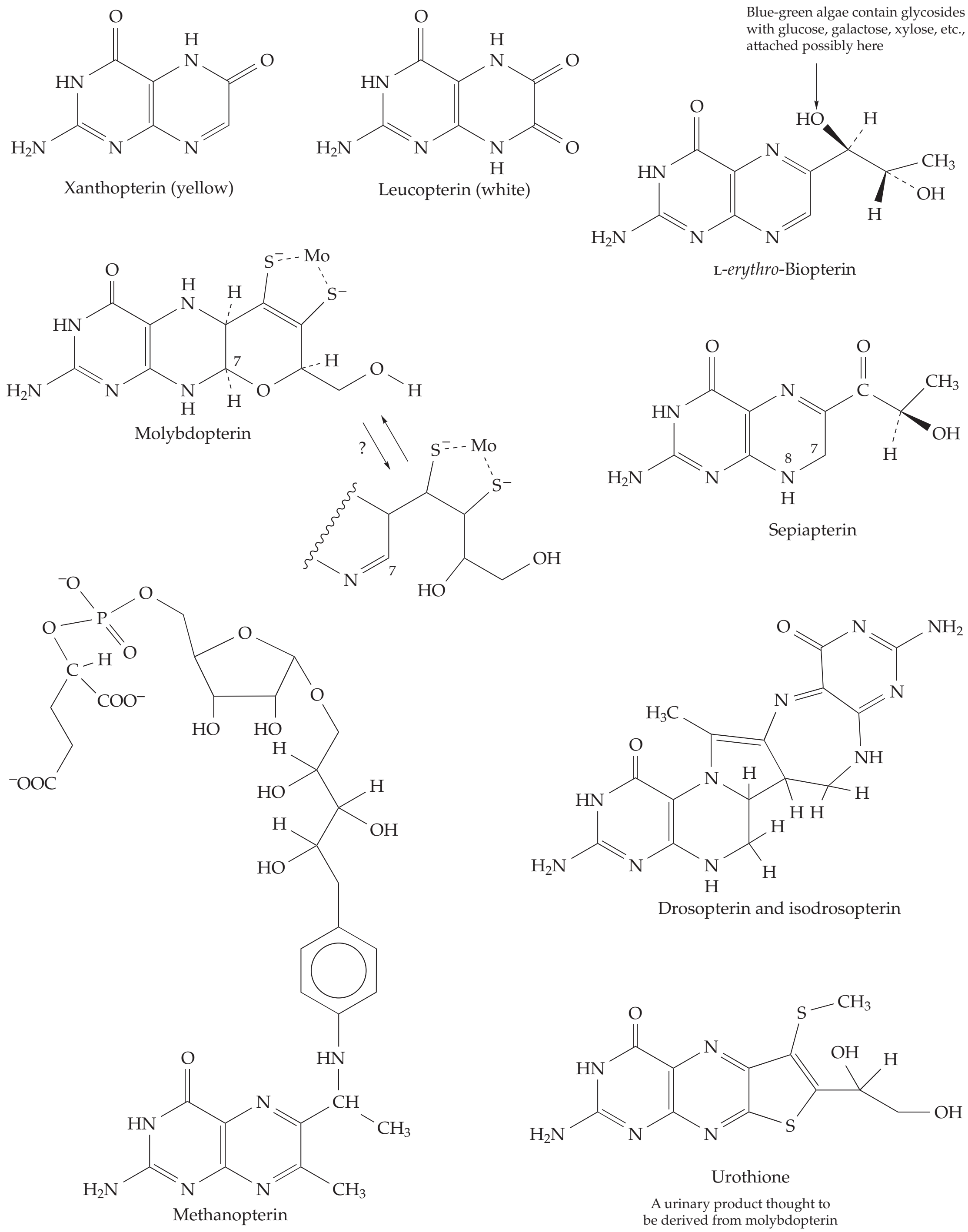


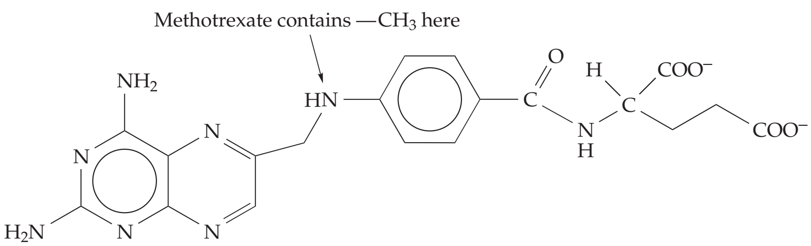
Figure 15-17 Structures of several biologically important pterins.

of many species contains only derivatives of folic acid itself but pteroylpentaglutamate is the major folate derivative present in rodent livers.[360,361] A large fraction of these pentaglutamates consists of the 5-methyl-THF derivative, while at the heptaglutamate level most consists of the free THF derivative. Folyl polyglutamate in its oxidized form is a component of some DNA photolyases (Chapter 23).[362]

3. Dihydrofolate Reductase

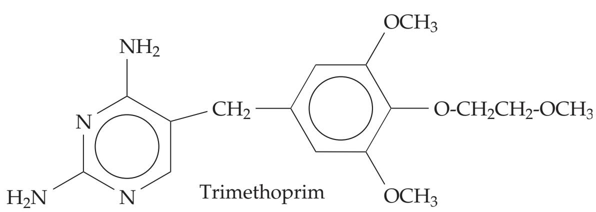
Folic acid and its polyglutamyl derivatives can be reduced to the THF coenzymes in two stages: the first step is a slow reduction with NADPH to 7,8-dihydrofolate (step *a*, Fig. 15-18). The same enzyme that catalyzes this reaction rapidly reduces the dihydrofolates to tetrahydrofolates (step *b*, Fig. 15-18). Again, NADPH is the reducing agent and the enzyme has been given the name **dihydrofolate reductase**. An unusual fact is that bacteriophage T4 not only carries a gene for its own dihydrofolate reductase but also the enzyme is a structural unit in the phage baseplate.[363]

Inhibitors. Aside from its role in providing reduced folate coenzymes for cells, this enzyme has attracted a great deal of attention because it appears to be a site of action of the important anticancer drugs **methotrexate** (amethopterin) and aminopterin.[293,364,365] These compounds inhibit dihydrofolate reductase in concentrations as low as 10^{-8} to 10^{-9} M. Methotrexate is also widely used as an **immunosuppresant** drug and in the treatment of parasitic infections.

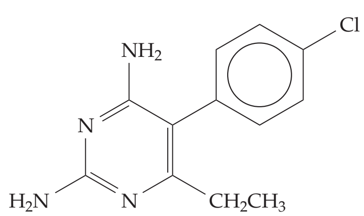


Aminopterin (4-amino-4-deoxyfolic acid)

Another dihydrofolate inhibitor **trimethoprim** is an important antibacterial drug, usually given together with a sulfonamide. Although it is not as close a structural analog of folic acid as is methotrexate, it is



Trimethoprim



Pyrimethamine

a potent inhibitor which binds far more tightly to the enzyme from bacteria than to that from humans.[366–368] The inhibitors **pyrimethamine** and **cycloguanil** are effective antimalarial drugs.[369,370]

An enormous number of synthetic compounds have been prepared in the hope of finding still more effective inhibitors. By 1984, over 1700 different inhibitors of dihydrofolate reductase had been studied.[367] However, methotrexate and trimethoprim remain outstanding. Methotrexate has been in clinical use for nearly 50 years and is very effective against leukemia and some other cancers.[371] Before 1960 persons with acute leukemia lived no more than 3–6 months. However, with antifolate treatment some have lived for 5 years or more and complete cures of the relatively rare choriocarcinoma have been achieved. New methods of chemotherapy use the antifolates in combination with other drugs.

Folate coenzymes are required in the biosynthesis of both purines and thymine. Consequently, rapidly growing cancer cells have a high requirement for activity of this enzyme. However, since all cells require the enzyme the antifolates are toxic and cannot be used for prolonged therapy. An even more serious problem is the development of resistance to the drug by cancer cells, often through "amplification" of the dihydrofolate reductase gene[365,372–374] as discussed in Chapter 27. Cells may also become resistant to methotrexate and other antifolates as a result of mutations that prevent efficient uptake of the drug,[375] cause increased action of efflux pumps in the cell,[376] reduce the affinity of dihydrofolate reductase for the drug,[377] or interfere with conversion of the drug to polyglutamate derivatives which are better inhibitors than free methotrexate.[378,379] Cell surface **folate receptors**, present in large numbers in some tumor cells, can be utilized to bring suitably designed antifolate drugs or even unrelated cytotoxic compounds into tumor cells.[379a] Some tumor cells are more active than normal cells in generating folyl polyglutamates, contributing to the effectiveness of methotrexate.[379] Resistance of *E. coli* cells to trimethoprin may result from acquisition by the bacteria of a new form of dihydrofolate reductase carried by a plasmid.[380]

Structure and mechanism. Dihydrofolate reductase from *E. coli* is a small 159-residue protein with a central parallel stranded sheet,[381–383] while that from higher animals is 20% larger. The three-dimensional structures of the enzymes from *Lactobacillus*,[384] chicken, mouse, and human are closely similar. The NADP binds at the C-terminal ends of β strands as in other dehydrogenases. In Fig. 15-19 the reduced nicotinamide end of NADP$^+$ is seen next to a molecule of bound dihydrofolate.[381] The side chain carboxylate of Asp 27 makes a pair of hydrogen bonds to the pterin ring as shown on the right-hand side of Eq. 15-42. As can be seen from Fig. 15-19, the nicotinamide ring of NADP$^+$ is correctly positioned to have donated a hydride ion to C-6 to form THF with the 6*S* configuration. Notice that in Eq. 15-42 the NH hydrogen atom at the 3' position

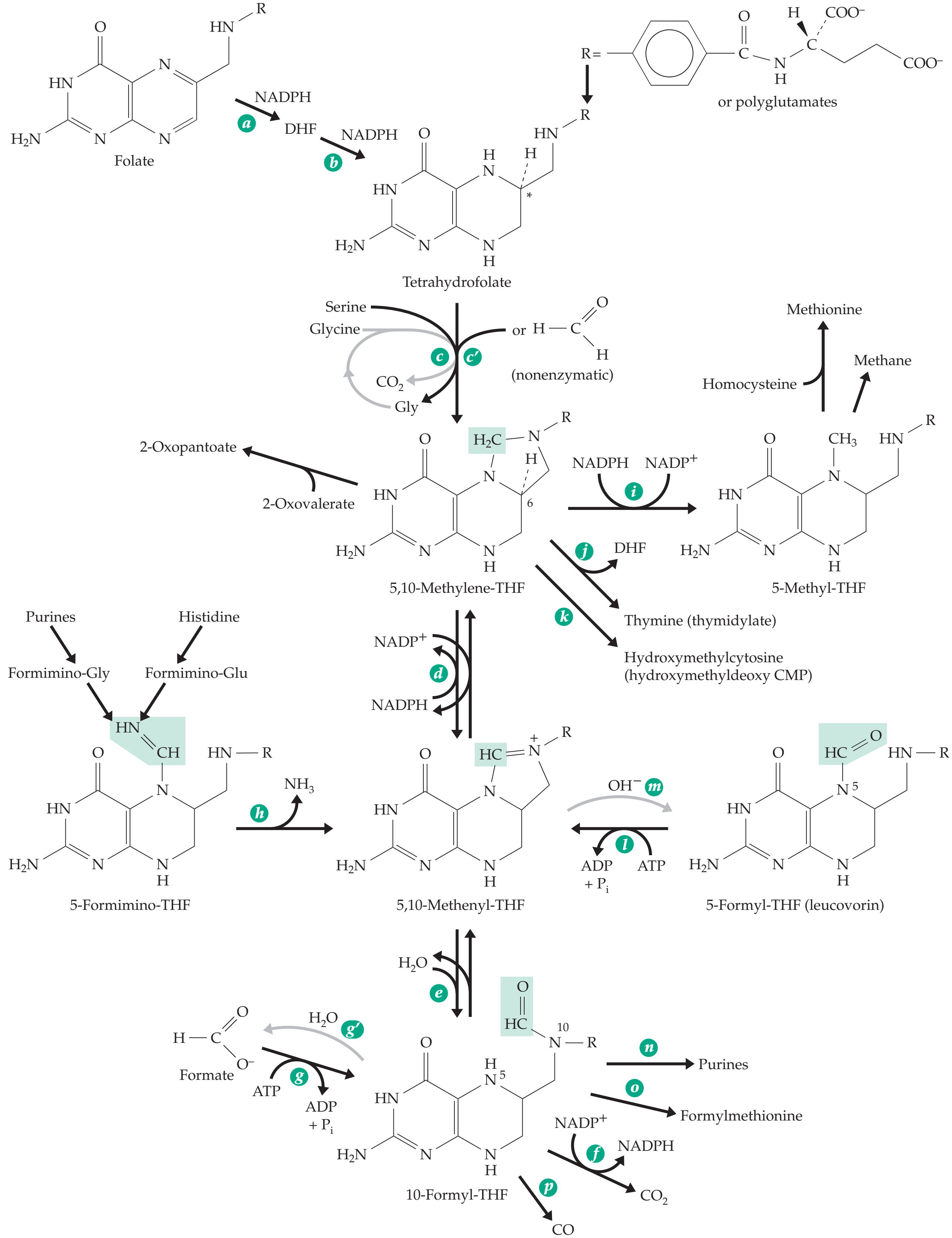
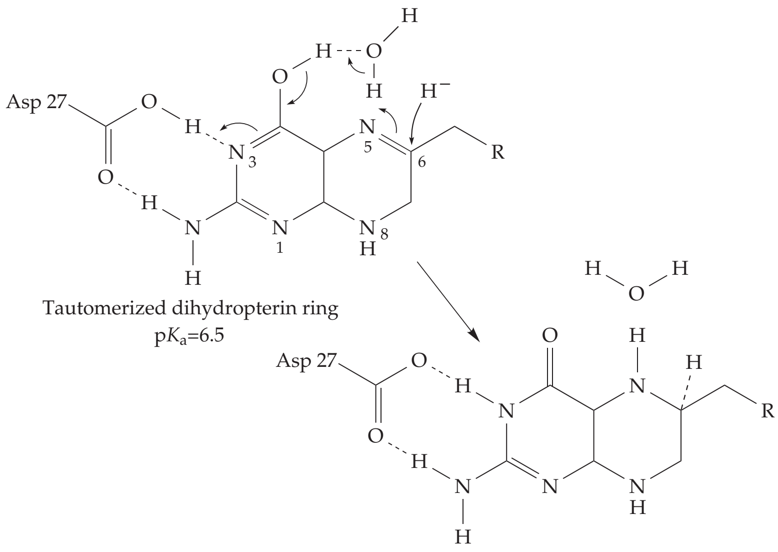


Figure 15-18 Tetrahydrofolic acid and its one-carbon derivatives.



(15-42)

of the pterin makes a hydrogen bond to Asp 27. However, X-ray studies have shown that the pteridine ring of methotrexate is turned 180° so that its 4-NH_2 group forms hydrogen bonds to backbone carbonyls of Ala 97 and Leu 4 at the edge of the central pleated sheet while a protonated N-1 interacts with the side chain carboxylate of Asp 27.

Because of its significance in cancer therapy and its small size, dihydrofolate reductase is one of the most studied of all enzymes. Numerous NMR studies[385–387] and investigations of catalytic mechanism and of other properties[388] have been conducted. Many mutant forms have been created.[389–391] For example, substitution of Asp 27 (Asp 26 in *L. casei*) of the *E. coli*

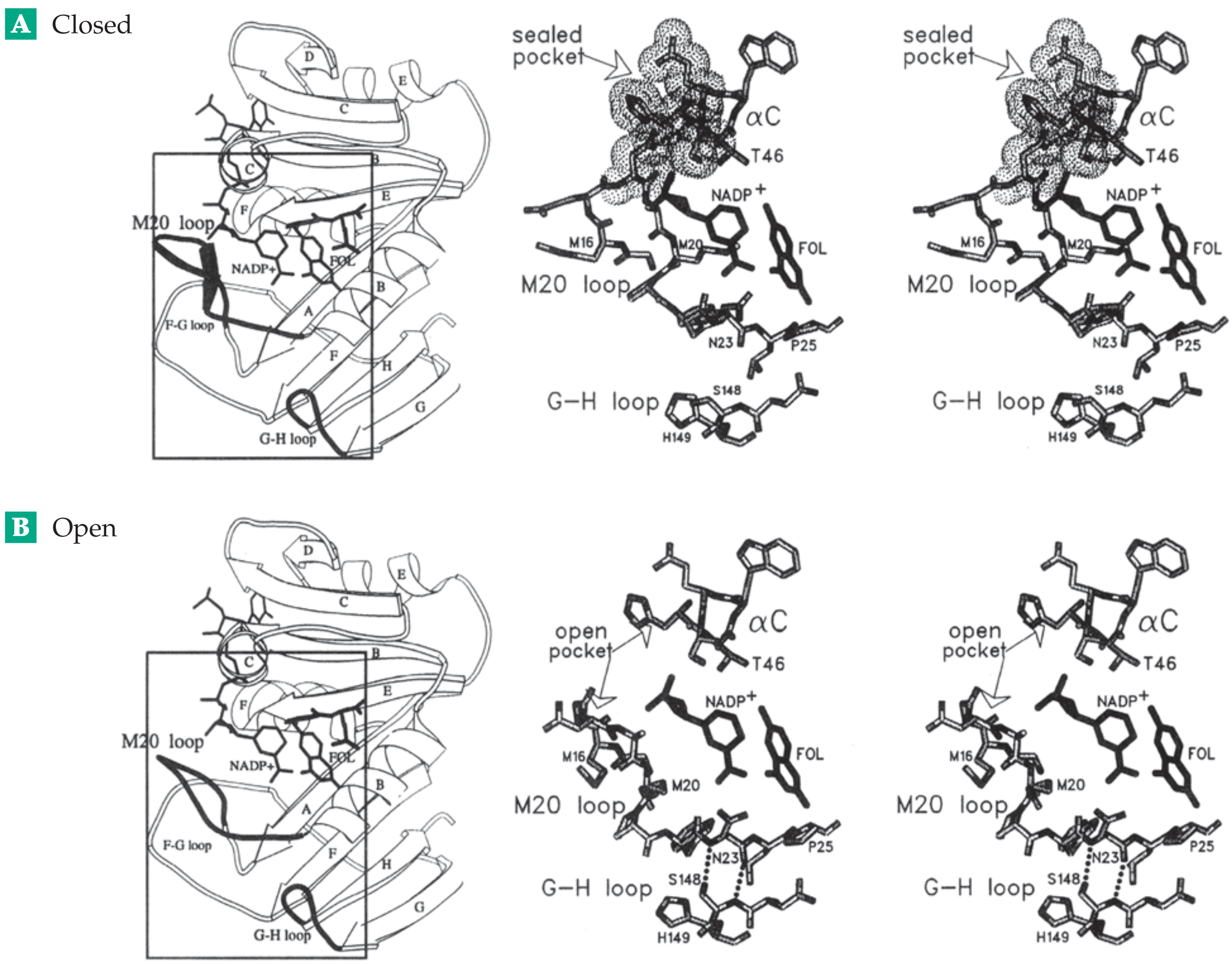


Figure 15-19 Drawings of the active site of *E. coli* dihydrofolate reductase showing the bound ligands NADP+ and tetrahydrofolate. Several key amino acid side chains are shown in the stereoscopic views on the right. The complete ribbon structures are on the left. (A) Closed form. (B) Open form into which substrates can enter and products can escape. From Sawaya and Kraut.[381] Courtesy of Joseph Kraut. Molscript drawings (Kraulis, 1991).

enzyme by Asn reduced the specific catalytic activity to 1/300 that of native enzyme indicating that this residue is important for catalysis.[390] The value of k_{cat} for reduction of dihydrofolate is highest at low pH and varies around a pK_a of ~6.5.[392–394] One interpretation is that this pK_a belongs to the Asp 27 carboxyl group and another is that it belongs to an N-5 protonated species of the coenzyme. As we have seen (Chapter 6) it is often impossible to assign a pK_a to a single group because there may be a mixture of interacting tautomeric species. Despite the enormous amount of study, we still don't know quite how the proton gets to N-5 in this reaction. Only one possibility is illustrated in Eq. 15-42. Asp 27 is protonated, the pterin ring is enolized, and a buried water molecule serves to relay a proton to N-5. The X-ray crystallographic studies on the *E. coli* enzyme show that after the binding of substrates a conformational change closes a lid over the active site (Fig. 15-19). This excludes water from N-5 but it may permit intermittent access, allowing transfer of a proton from a water molecule bound to O-4 as shown in Eq. 15-42.[381] During the reduction of folate to dihydrofolate a different mechanism of proton donation must be followed to allow protonation at C-7.

4. Single-Carbon Compounds and Groups in Metabolism

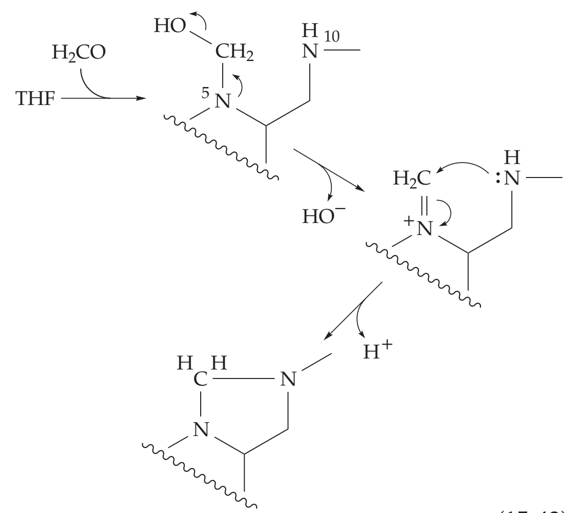
Some single-carbon compounds and groups important in metabolism are shown in Table 15-3 in order (from left to right) of increasing oxidation state of the carbon atom. Groups at three different oxidation levels, corresponding to **formic acid**, **formaldehyde**, and the **methyl group**, are carried by tetrahydrofolic acid coenzymes. While the most completely reduced compound, methane, cannot exist in a combined form, its biosynthesis depends upon reduced methanopterin, as does that of carbon monoxide. Figure 15-18 summarizes the metabolic interrelationships of the compounds and groups in this table.

Serine as a C1 donor. For many organisms, from *E. coli* to higher animals, **serine** is a major precursor of C-1 units.[395–397] The β-carbon of serine is removed as formaldehyde through direct transfer to tetrahydrofolate with formation of methylene-THF and glycine (Eq. 14-30, Fig. 15-18, step *c*). This is a stereospecific transfer in which the *pro-S* hydrogen on C-3 of serine enters the *pro-S* position also in methylene-THF.[398] The glycine formed in this reaction can, in turn, yield

TABLE 15-3
Single-Carbon Compounds in Order of Oxidation State of Carbon

CH_4 Methane	H_3C—Y Methyl groups bound to O, N, S	H—C(=O)—H Formaldehyde	H—C(=O)—H Formic acid	CO_2 H_2CO_3
		—CH_2OH Hydroxymethyl group	—C(=O)—H Formyl group	
		—CH_2— Methylene group	—C(=)—H Methenyl group	
			—C(=NH)—H Formimino group	
			C=O Carbon monoxide	

another single-carbon unit by loss of CO_2 under the influence of the THF and the PLP-requiring glycine cleavage system which is discussed in the next paragraph. Free formaldehyde in a low concentration can also combine with THF to form methylene-THF (Fig. 15-18, step *c′* and Eq. 15-43).[399]



(15-43)

The glycine decarboxylase–synthetase system. Glycine is cleaved reversibly within mitochondria of plants and animals and also by bacteria to CO_2, NH_3, and a methylene group which is carried by tetrahydrofolic acid[296,400–403] (Fig. 15-20). Four proteins are required. The P-protein consists of two identical 100-kDa subunits, each containing a molecule of PLP. This protein is a **glycine decarboxylase** which, however, replaces the lost CO_2 by an electrophilic sulfur of lipoate rather than by a proton. Serine hydroxymethyltransferase can also catalyze this step of the sequence.[404] The lipoate is bound to a second protein, the H-protein. A third protein, the T-protein, carries bound tetrahydrofolate which displaces the aminomethyl group from the dihydrolipoate and converts it to N^5,N^{10}-methylene tetrahydrofolate with release of ammonia. The dihydrolipoate is then reoxidized by NAD^+ and dihydrolipamide dehydrogenase (Fig. 15-20). Glycine can be oxidized completely in liver mitochondria with the methylene group of methylene-THF being converted to CO_2 through reaction steps *d*, *e*, and *f* of Fig. 15-18. The glycine cleavage system is reversible and is used by some organisms to synthesize glycine.

Whether it arises from the hydroxymethyl group of serine or from glycine, the single-carbon unit of methylene-THF (which is at the formaldehyde level of oxidation) can either be oxidized further to 5,10-methenyl-THF and 10-formyl-THF (steps *d* and *e*, Fig. 15-18)

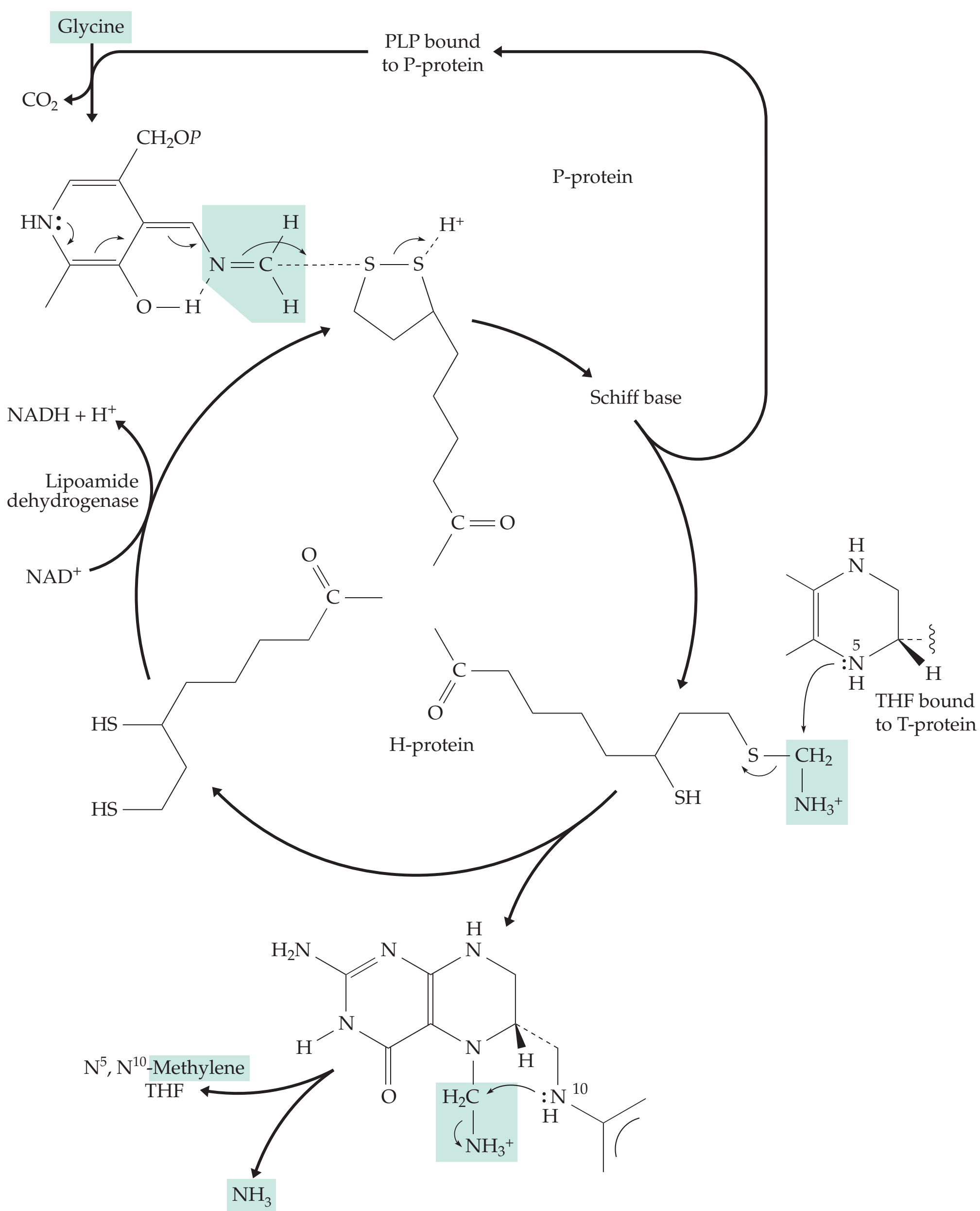


Figure 15-20 The reversible glycine cleavage system of mitochondria. Compare with Fig. 15-15.

or it can be reduced to methyl-THF (step *i*). The reactions of folate metabolism occurs in both cytoplasm and mitochondria at rates that are affected by the length of the polyglutamate chain. The many complexities of these pathways are not fully understood.[395,397]

Starting with formate or carbon dioxide. Most organisms can, to some extent, also utilize formate as a source of single-carbon units. Human beings have a very limited ability to metabolize formate and the accumulation of formic acid in the body following ingestion of methanol is often fatal. However, many bacteria are able to subsist on formate as a sole carbon source. Archaea may generate the formate by reduction of CO_2. Utilization of formate begins with formation of 10-formyl-THF (Fig. 15-18, step *g*, lower left corner). 10-Formyl-THF can be reduced to methylene-THF and the single-carbon unit can be transferred to glycine to form serine. In some bacteria three separate enzymes are required to convert formate to methylene-THF: 10-formyl-THF synthetase (Fig. 15-18, step *g*, presumably via formyl phosphate),[405] methenyl-THF cyclohydrolase (step *e*, reverse), and methylene-THF dehydrogenase (step *d*, reverse). NADH is used by the acetogens but in *E. coli* and in most higher organisms NADPH is the reductant. In some bacteria and in some tumor cells, two of the three enzymes are present as a bifunctional enzyme.[406–407a] In most eukaryotes all three of the enzymatic activities needed for converting formate to methylene-THF are present in a single large (200-kDa) dimeric trifunctional protein called **formyl-THF synthetase**.[407,408]

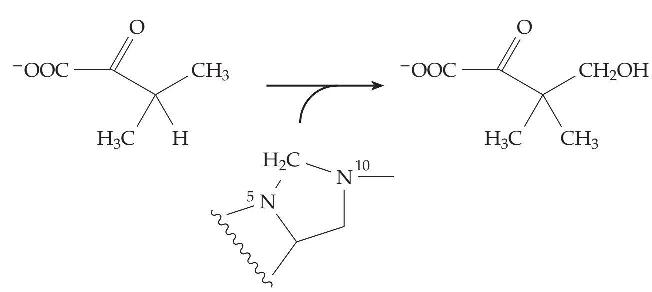
N^{10}-Formyl-THF serves as *a biological formylating agent* needed for two steps in the synthesis of purines[409–410b] (Chapter 25) and, in bacteria and in mitochondria and chloroplasts, for synthesis of formylmethionyl-tRNA[411] which initiates synthesis of all polypeptide chains in bacteria and in these organelles (Chapter 29).

N^5-Formyl-THF (leucovorin). The N^5-formyl derivative of THF (5-formyl-THF) is a growth factor for *Leuconostoc citrovorum.* Following this discovery in 1949 it was called the "citrovorum factor" or **leucovorin**. Its significance in metabolism is not clear. Perhaps it serves as a storage form of folate in cells that have a dormant stage, e.g., of seeds or spores.[412] It may also have a regulatory function.[413] In some ants and in certain beetles, it is stored and hydrolyzed to formic acid. The carabid beetle *Galerita lecontei* ejects a defensive spray that contains 80% formic acid.[414]

5-Formyl-THF can arise by transfer of a formyl group from formylglutamate and there is an enzyme that converts 5-formyl-THF to 5,10-methenyl-THF with concurrent cleavage of ATP.[412] 5-Formyl-THF is sometimes used in a remarkable way to treat certain highly malignant cancers. Following surgical removal of the tumor the patient is periodically given what would normally be a lethal dose of methotrexate, then, about 36 h later, the patient is rescued by injection of 5-formyl-THF. The mechanism of rescue is thought to depend upon the compound's rapid conversion to 10-formyl-THF within cells. Tumor cells are not rescued, perhaps because they have a higher capacity than normal cells for synthesis of polyglutamates of methotrexate.[379] This observation also suggests that the major anti-cancer effect of methotrexate may not be on dihydrofolate reductase but on enzymes of formyl group transfer that are inhibited by polyglutamate derivatives of methotrexate.

Catabolism of histidine and purines. Another source of single-carbon units in metabolism is the degradation of histidine which occurs both in bacteria and in animals via **formiminoglutamate**. The latter transfers the –CH=NH group to THF forming 5-formimino-THF, which is in turn converted (step *h*, Fig. 15-18) to 5,10-methenyl-THF and ammonia. In bacteria that ferment purines, **formiminoglycine** is an intermediate. Again, the formimino group is transferred to THF and deaminated, the eventual product being 10-formyl-THF. In these organisms, the enzyme 10-formyl-THF synthase also has a very high activity.[415] It probably operates in reverse as a mechanism for synthesis of ATP in this type of fermentation. In other organisms excess 10-formyl-THF may be oxidized to CO_2 via an $NADP^+$-dependent dehydrogenase (step *f*, Fig. 15-16), providing a mechanism for detoxifying formic acid. In some organisms excess 10-formyl-THF may simply be hydrolyzed with release of formate (step *q*).[415a]

Thymidylate synthase. Methylene-THF serves as the direct precursor of the 5-methyl group of thymine as well as of the hydroxymethyl groups of **hydroxymethylcytosine**[416] and of **2-oxopantoate**, an intermediate in the formation of pantothenate and coenzyme A.[417] The latter is a simple hydroxymethyl transfer reaction (Eq. 15-44) that is related to an aldol condensation and which may proceed through an imine of the kind shown in Eq. 15-43.



$N^{5,10}$-Methylene-THF (15-44)

During thymine formation the coenzyme is oxidized to dihydrofolate, which must be reduced by dihydrofolate reductase to complete the catalytic cycle. A possible mechanistic sequence for **thymidylate synthase**, an enzyme of known three-dimensional structure,[354,418–421a] is given in Fig. 15-21. In the first step (*a*) a thiolate anion, from the side chain of Cys 198 of the 316-residue *Lactobacillus* enzyme, adds to the 5 position of the substrate 2'-deoxyuridine monophosphate (dUMP). As a consequence the 6 position becomes a nucleophilic center which can combine with methylene-THF as shown in step *b* of Fig. 15-21. After tautomerization (step *c*) this adduct eliminates THF (step *d*) to give a 5-methylene derivative of the dUMP. The latter immediately oxidizes the THF by a hydride ion transfer to form thymidylate and dihydrofolate.[422,422a]

In protozoa thymidylate synthase and dihydrofolate reductase exist as a single bifunctional protein.

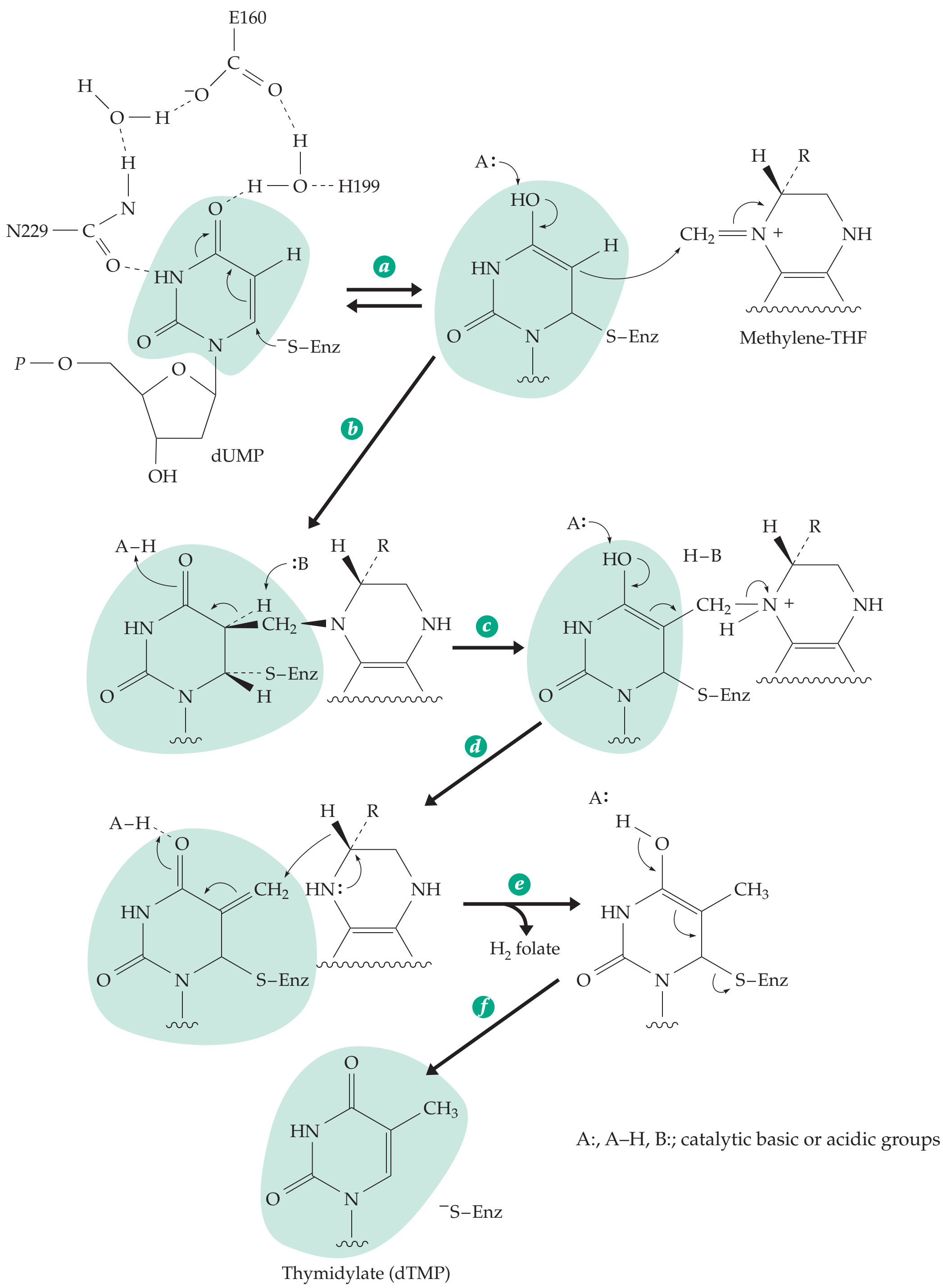
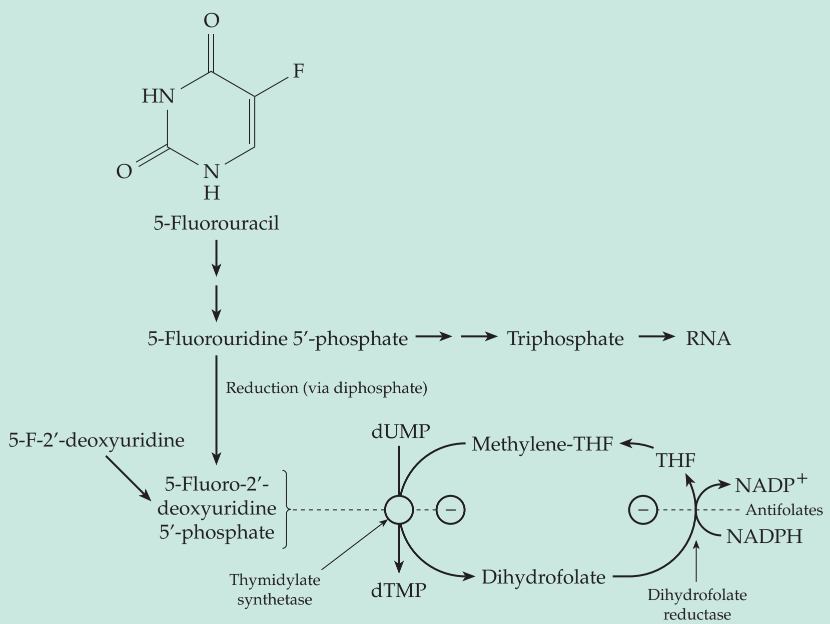


Figure 15-21 Probable mechanism of action of thymidylate synthase. After Huang and Santi[422] and Hyatt *et al.*[354]

Consequently, in methotrexate-resistant strains of *Leishmania* (and also in cancer cells) both enzyme activities are increased equally by gene amplification.[423] This presents a serious problem in the treatment of protozoal parasitic diseases for which few suitable drugs are known.

Cells of *E. coli*, when infected with T-even bacteriophage, convert dCMP to 5-hydroxymethyl-dCMP[423a]

BOX 15-E THYMIDYLATE SYNTHASE, A TARGET ENZYME IN CANCER CHEMOTHERAPY[a]



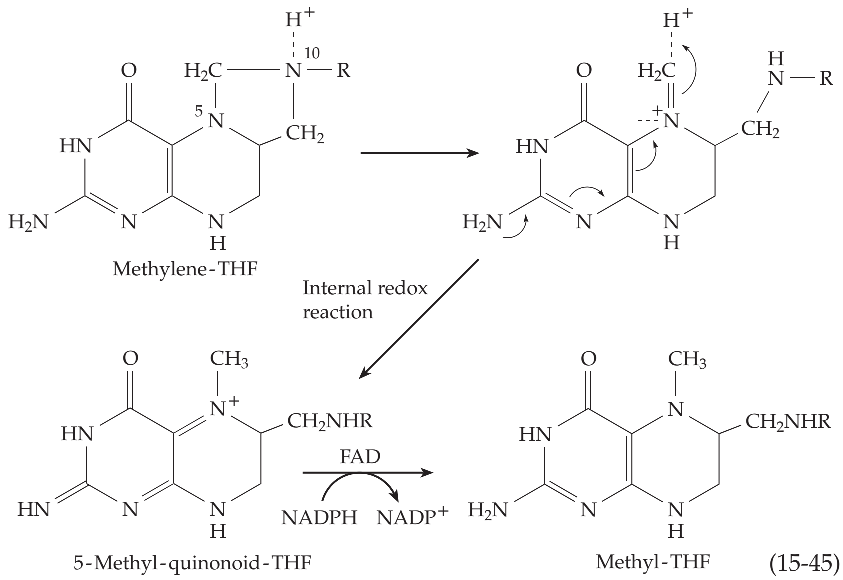
If an animal or bacterial cells are deprived of thymine they can no longer make DNA. However, synthesis of proteins and of RNA continues for some time. This can be demonstrated experimentally with thymine-requiring mutants. However, sooner or later such cells lose their vitality and die. The cause of this **thymineless death**[b] is not entirely clear. Perhaps thymine is needed to repair damage to DNA and if it is not available transcription eventually becomes faulty. Chromosome breakage is also observed.[c] Whatever the cause of death, the phenomenon provides the basis for some of the most effective chemotherapeutic attacks on cancer. Rapidly metabolizing cancer cells are especially vulnerable to thymineless death. Consequently, thymidylate synthase is an important target enzyme for inhibition. One powerful inhibitor is the monophosphate of **5-fluoro-2'-deoxyuridine**. The inhibition was originally discovered when 5-fluorouracil was recognized as a useful cancer chemotherapeutic agent.

Fluorouracil has many effects in cells, including incorporation into RNA,[b] but the inhibition of thymidylate synthase by the reduction product may be the most useful effect in chemotherapy. In fact, 5-fluoro-2'-deoxyuridine is a much less toxic and more potent drug than 5-fluorouracil. It binds into the active site of thymidylate synthase and reacts in the initial steps of catalysis. However, the 5-H of 2'-deoxyuridine, the normal substrate, must be removed as H^+ (step *c* of Fig. 15-21) in order for the reaction to continue. The 5-F atom cannot be removed in this way and a stable adduct is formed.[d,e] A crystal structure containing both the bound 5-fluorodeoxyuridine and methylene-THF supports this conclusion.[e]

Thymidylate synthase requires methylene tetrahydrofolate as a reductant and the reduction of dihydrofolate is also an important part of the process. In protozoa dihydrofolate reductase and thymidylate synthase occur as a single-chain bifunctional enzyme.[f] As has been pointed out in the main text, such folic acid analogs as methotrexate are among the most useful anticancer drugs. By inhibiting dihydrofolate reductase they deprive thymidylate synthase of an essential substrate.

Because 5-fluorouracil acts on normal cells as well as cancer cells, its usefulness is limited. Knowledge of the chemistry and three-dimensional structure of thymidylate synthase complexes is being used in an attempt to discover more specific and effective drugs that attack this enzyme.[g,h]

a Friedkin, M. (1973) *Adv. Enzymol.* **38**, 235–292
b Sahasrabudhe, P. V., and Gmeiner, W. H. (1997) *Biochemistry* **36**, 5981–5991
c Ayusawa, D., Shimizu, K., Koyama, H., Takeishi, K., and Seno, T. (1983) *J. Biol. Chem.* **258**, 12448–12454
d Huang, X. F., and Arvan, P. (1995) *J. Biol. Chem.* **270**, 20417–20423
e Hyatt, D. C., Maley, F., and Montfort, W. R. (1997) *Biochemistry* **36**, 4585–4594
f Ivanetich, K. M., and Santi, D. V. (1990) *FASEB J.* **4**, 1591–1597
g Shoichet, B. K., Stroud, R. M., Santi, D. V., Kuntz, I. D., and Perry, K. M. (1993) *Science* **259**, 1445–1449
h Weichsel, A., Montfort, W. R., Ciesla, J., and Maley, F. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 3493–3497



(15-45)

and suitably infected cells of *Bacillus subtilis* form 5-hydroxymethyl-dUMP.[424,425] These products can be formed by attack of OH^- on quinonoid intermediates of the type postulated for thymidylate synthetase (formed in step *d* of Fig. 15-21).[416] A related reaction is the posttranscriptional conversion of a single uracil residue in tRNA molecules of some bacteria to a thymine ring (a **ribothymidylic acid** residue). In this case, $FADH_2$ is used as a reducing agent to convert the initial adduct of Fig. 15-21 to the ribothymidylic acid and THF.[426]

Synthesis of methyl groups. The reduction of methylene-THF to 5-methyl-THF within all living organisms from bacteria to higher plants and animals provides the methyl groups needed in biosynthesis.[426a] These are required for formation of methionine and, from it, *S*-adenosylmethionine. The latter is used to modify proteins, nucleic acids, and other biochemicals through methylation of specific groups. In the methanogens, reduction of a corresponding methylene derivative of methanopterin gives rise to methane (Section F).

Mammalian methylene-THF reductase is a FAD-containing flavoprotein that utilizes NADPH for the reduction to 5-methyl-THF.[427,428] Matthews[429] suggested that the mechanism of this reaction involves an internal oxidation–reduction reaction that generates a 5-methyl-quinonoid dihydro-THF (Eq. 15-45). Methylene-THF reductase of acetogenic bacteria is also a flavoprotein but it contains Fe–S centers as well. The 237-kDa $\alpha_4\beta_4$ oligomer contains two molecules of FAD and four to six of both Fe and S^{2-} ions.[430,431]

The methyl group of methyl-THF is incorporated into methionine by the vitamin B_{12}-dependent methionine synthase which is discussed in Chapter 16. Matthews suggested that methionine synthase may also make use of the 5-methyl-quinonoid-THF of Eq. 15-45. An initial reduction step would precede transfer of the methyl group. Methyl-THF is a precursor to acetate in the acetogenic bacteria.[432] This corrinoid coenzyme-dependent process is also considered in Chapter 16.

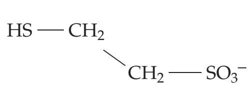
E. Specialized Coenzymes of Methanogenic Bacteria

Methane-producing bacteria[433,434] obtain energy by reducing CO_2 with molecular hydrogen:

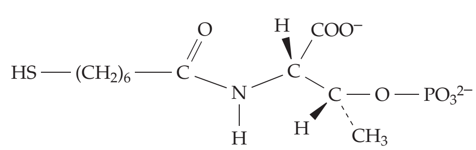
$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$$
$$\Delta G^\circ = -131 \text{ kJ / mol} \qquad (15\text{-}46)$$

Some species are also able to utilize formate or formaldehyde as reducing agents.[435] These compounds are oxidized to CO_2, the reducing equivalents formed being used to reduce CO_2 to methane. Carbon monoxide can also be converted to CO_2.

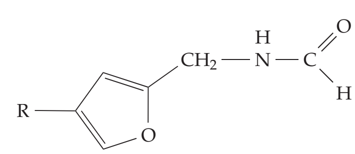
$$CO + H_2O \rightarrow CO_2 + H_2$$
$$\Delta G^\circ = -20 \text{ kJ / mol} \qquad (15\text{-}47)$$



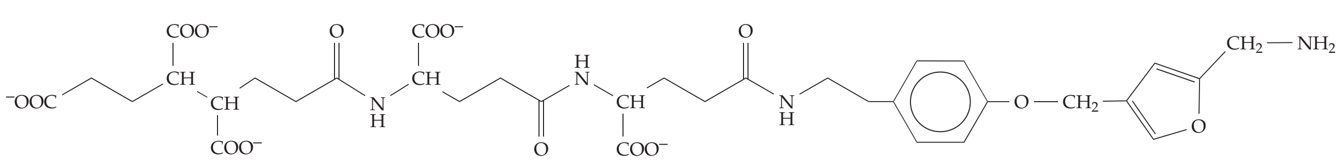
Coenzyme M



7-Mercaptoheptanoylthreonine phosphate



Formylmethanofuran



Methanofuran

By using the combined reactions of Eqs. 15-46 and 15-47 the bacteria can subsist on CO alone (Eq. 15-48):

$$4CO + 2\,H_2O \rightarrow 3CO_2 + CH_4 \quad \Delta G^{\circ} = -211 \text{ kJ / mol of methane formed} \qquad (15\text{-}48)$$

Some species reduce methanol to methane via Eqs. 15-49 or 15-50.

$$CH_3OH + H_2 \rightarrow CH_4 + H_2O \quad \Delta G^{\circ} = -113 \text{ kJ / mol} \qquad (15\text{-}49)$$

$$4CH_3OH \rightarrow 3CH_4 + CO_2 + 2\,H_2O \quad \Delta G^{\circ} = -107 \text{ kJ / mol of methane formed} \qquad (15\text{-}50)$$

To accomplish these reactions a surprising variety of specialized cofactors are needed.[351,352,434] The first of these, **coenzyme M**, 2-mercaptoethane sulfonate, was discovered in 1974.[436] It is the simplest known coenzyme. Later, the previously described **5-deazaflavin F_{420}** (Section B), a **nickel tetrapyrrole F_{430}** (Chapter 16), **methanopterin** (Fig. 15-17),[437] the "carbon dioxide reduction factor" called **methanofuran**,[352,438] and **7-mercaptoheptanoylthreonine phosphate**[439,440] were also identified.

A sketch of the metabolic pathways followed in methane formation is given in Fig. 15-22.[352,435] In the first step (*a*) the amino group of methanofuran is thought to add to CO_2 to form a carbamate which is reduced to formylmethanofuran by H_2 and an intermediate carrier H_2X in step *b*. The formyl group is then transferred to tetrahydromethanopterin (H_4MPT) (step *c*)[440a] and is cyclized and reduced in two stages in steps *d*, *e*, and *f*. The reductant is the deazaflavin F_{420} and the reactions parallel those for conversion of formyl-THF to methyl-THF (Fig. 15-18).[431,440b,441] The methyl group of methyl-H_4MPT is then transferred to the sulfur atom of the thiolate anion of coenzyme M, from which it is reduced off as CH_4. This is a complex process requiring the nickel-containing F_{430}, FAD, and 7-mercaptoheptanoylthreonine phosphate (HS-HTP).

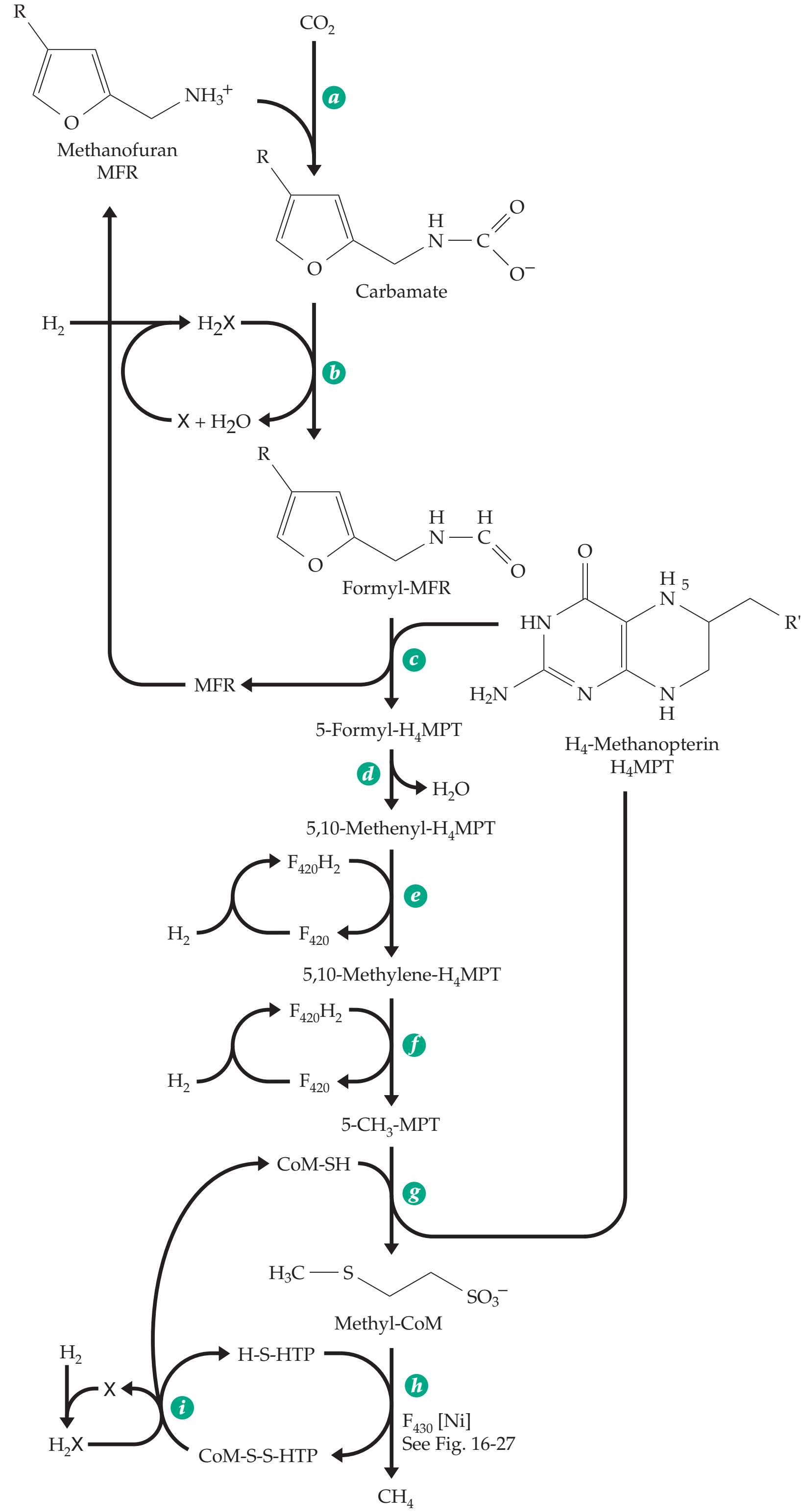


Figure 15-22 Tentative scheme for reduction of carbon dioxide to methane by methanogens. After Rouvière *et al*.[352] and Thauer *et al*.[435]

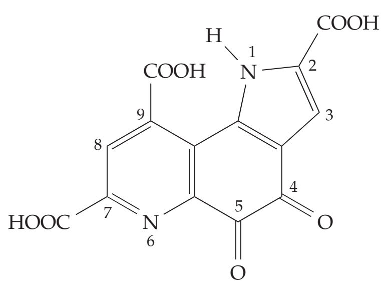
The HS-HTP may be the 2-electron donor for the reduction. The mixed-disulfide CoM-S-S-HTP appears to be an intermediate and also an allosteric effector for the first step, the reduction of CO_2 to formylmethanofuran. A complex of proteins that is unstable in oxygen is also needed. The principal component of the methyl reductase binds two molecules of the nickel-containing F_{430}. It also binds two moles of CoM-SH noncovalently. The binding process requires ATP as well as the presence of other proteins, notably those of the hydrogenase system.

Coenzyme M, previously found only in methanogenic archaeobacteria, has recently been discovered in both gram-negative and gram-positive alkene-oxidizing eubacteria. It seems to function in cleavage of epoxy rings and as a carrier of hydroxyalkyl groups.[441a] See also Chapter 17.

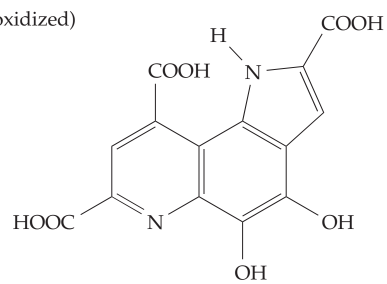
F. Quinones, Hydroquinones, and Tocopherols

1. Pyrroloquinoline Quinone (PQQ)

Bacteria that oxidize methane or methanol (**methylotrophs**) employ a periplasmic methanol dehydrogenase that contains as a bound coenzyme, the pyrroloquinoline quinone designated **PQQ** or methoxatin (Eq. 15-51).[442–444] This fluorescent *ortho*-quinone



PQQ (oxidized)

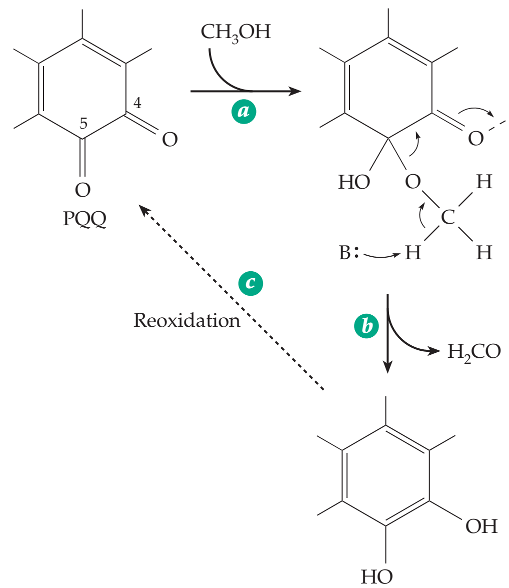


$PQQH_2$ (reduced)

is released when the protein is denatured. Some other bacterial alcohol dehydrogenases[445,445a] and glucose dehydrogenases[446-446d] contain the same cofactor.

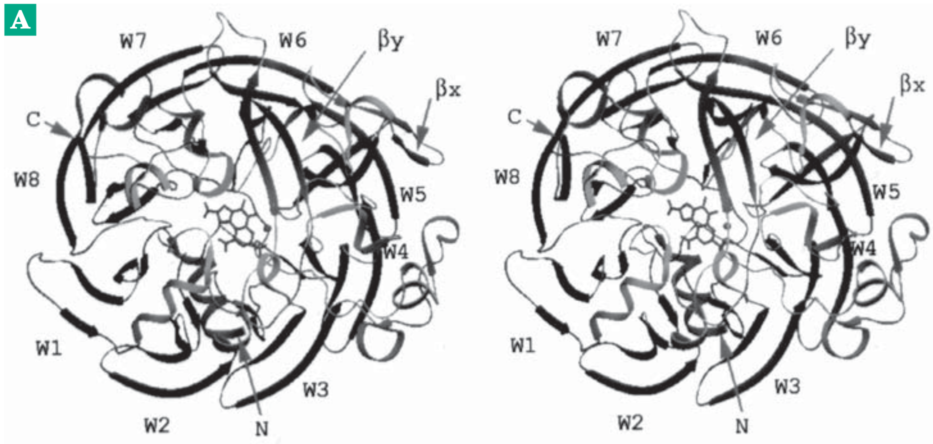
PQQ-containing methanol dehydrogenase from *Methylophilus* is a dimer of a 640-residue protein consisting of two disulfide-linked peptide chains with one noncovalently bound PQQ.[447] The large subunit contains a β propellor (Fig. 15-23), which is similar to that in the protein G_i shown in Fig. 11-7. The PQQ binding site lies above the propellor and is formed by a series of loops. The coenzyme interacts with several polar protein side chains and with a bound calcium ion.[448] Bacterial PQQ-dependent glucose dehydrogenases have also been studied intensively. The 450-residue soluble enzyme from *Acinetobacter calcoaceticus* is partially homologous to the methanol dehydrogenase.[446a,446b]

PQQ and the other quinone prosthetic groups described here all function in reactions that would be possible for pyridine nucleotide or flavin coenzymes. All of them, like the flavins, can exist in oxidized, half-reduced semiquinone and fully reduced dihydro forms. The questions to be asked are the same as we asked for flavins. How do the substrates react? How is the reduced cofactor reoxidized? In nonenzymatic reactions alcohols, amines, and enolate anions all add at C-5 of PQQ to give adducts such as that shown for methanol in Eq. 15-51, step *a*.[444,449,449a] Although many additional reactions are possible, this addition is a reasonable first step in the mechanism shown in Eq. 15-51. An enzymatic base could remove a proton as is indicated in step *b* to give $PQQH_2$. The pathway for reoxidation (step *c*) might involve a cytochrome *b*, cytochrome *c*, or bound ubiquinone.[445,446]



(15-51)

Although the soluble PQQ-dependent glucose dehydrogenase forms, with methylhydrazine, an adduct similar to that depicted in Eq. 15-51,[449b] the structure of a glucose complex of the $PPQH_2$- and Ca^{2+}-containing enzyme at a resolution of 0.2 nm suggests a direct hydride ion transfer. The only base close to glucose C1 is His 144. The C1-H proton lies directly above the $PQQH_2$ C5 atom. Oubrie *et al.*[446a] propose a deprotonation of the C1-OH by His 144 and H^- transfer to C5 of PQQ followed by tautomerization (Eq. 15-52). Theoretical calculations by Zheng and Bruice[449c] favor a simpler mechanism with hydride transfer to the oxygen atom of the C4 carbonyl (green arrows). In either case, the Ca^{2+} would assist by polarizing the C5 carbonyl group.



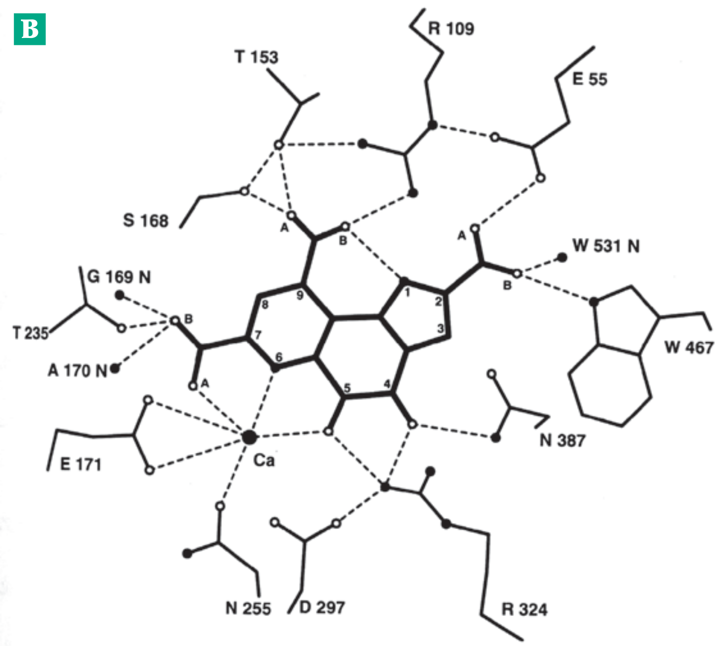
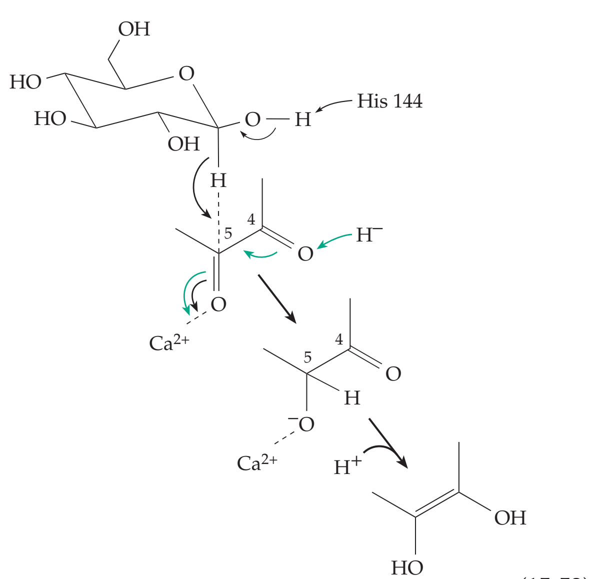


Figure 15-23 (A) Stereoscopic view of the H subunit of methanol dehydrogenase. Eight four-stranded antiparallel β sheets, labeled W1–W8, form the base of the subunit. Several helices and two additional β-sheet structures (labeled βx and βy) form a cap over the base. The PQQ is located in a funnel within the cap approximately on an eight-fold axis of pseudosymmetry. Courtesy of Xia *et al.*[447] (B) Schematic view of the active site. W467 is parallel to the plane of PQQ. All hydrogen-bond interactions between PQQ and its surrounding atoms, except for the three water molecules, are indicated. Courtesy of White *et al.*[448]

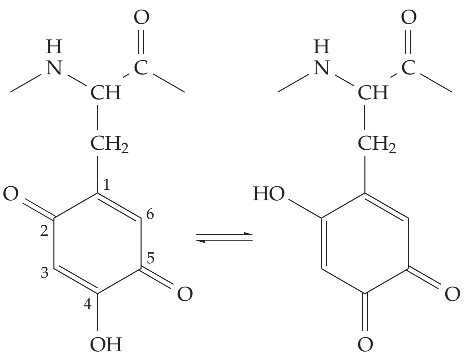


(15-52)

2. Copper Amine Oxidases

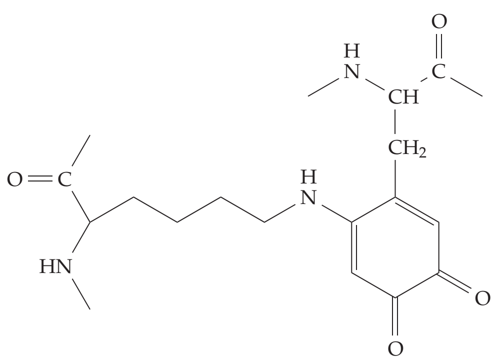
At first it appeared that PQQ had a broad distribution in enzymes, including eukaryotic amine oxidases. However, it was discovered, after considerable effort, that there are additional quinone cofactors that function in oxidation of amines. These are derivatives of tyrosyl groups of specific enzyme proteins. Together with enzymes containing bound PQQ they are often called **quinoproteins**.[450–454]

Topaquinone (TPQ). Both bacteria and eukaryotes contain amine oxidases that utilize bound copper ions and O_2 as electron acceptors and form an aldehyde, NH_3, and H_2O_2. The presence of an organic cofactor was suggested by the absorption spectra which was variously attributed to pyridoxal phosphate or PQQ. However, isolation from the active site of bovine serum amine oxidase established the structure of the reduced form of the cofactor as a trihydroxyphenylalanyl group covalently linked in the peptide chain.[455,456] Its oxidized form is called topaquinone:



Topaquinone (TPQ)

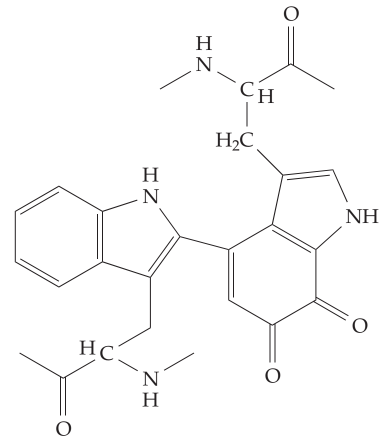
The same cofactor is present in an amine oxidase from *E. coli*[457–460] and in other bacterial[461–463] fungal,[463a] plant,[464] and mammalian[464a] amine oxidases.



Lysine tyrosylquinone (LTQ)

Lysine tyrosylquinone (LTQ). Another copper amine oxidase, **lysyl oxidase**, which oxidizes side chains of lysine in collagen and elastin (Eq. 8-8) contains a cofactor that has been identified as having a lysyl group of a different segment of the protein in place of the –OH in the 2 position of topaquinone.[465] Lysyl oxidase plays an essential role in the crosslinking of collagen and elastin.

Tryptophan tryptophanylquinone (TTQ). This recently discovered quinone cofactor is similar to the lysyl tyrosylquinone but is formed from two tryptophanyl side chains.[466] It has been found in **methylamine dehydrogenase** from methylotrophic gram-negative bacteria[467–469] and also in a bacterial aromatic amine dehydrogenase.[470]

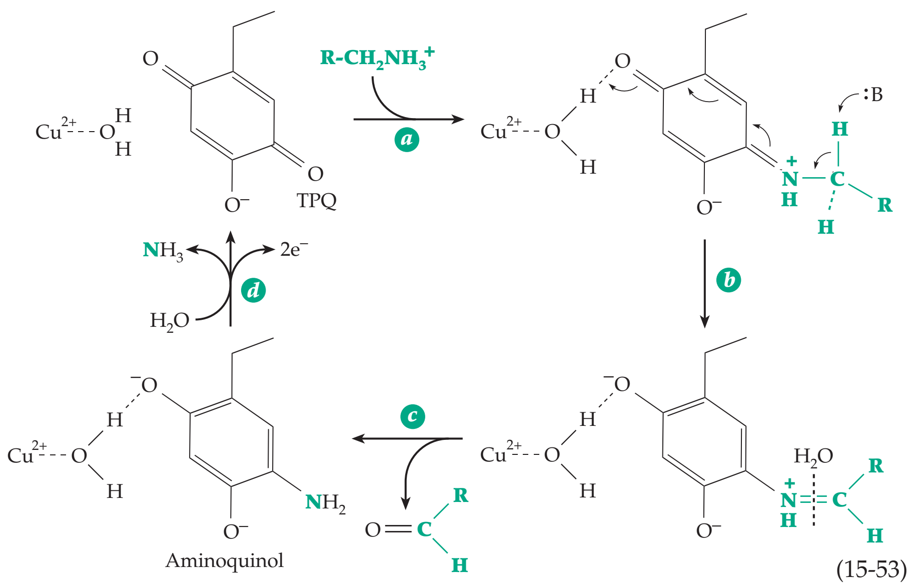


Tryptophan tryptophylquinone (TTQ)

Three-dimensional structures. The TPQ-containing amine oxidase from *E. coli* is a dimer of 727-residue subunits with one molecule of TPQ at position 402 in each subunit.[457,458] Methylamine dehydrogenase is also a large dimeric protein of two large 46.7-kDa subunits and two small 15.5-kDa subunits. Each large subunit contains a TTQ cofactor group.[468,471,471a] Reduced TTQ is reoxidized by the 12.5-kDa blue copper protein **amicyanin**. Crystal structures have been determined for complexes of methylamine dehydrogenase with amicyanin[471] and of these two proteins with a third protein, a small bacterial cytochrome *c*.[472,472a]

Mechanisms. Studies of model reactions[473–476] and of electronic, Raman,[466,477,478] ESR,[479,480] and NMR spectra and kinetics[481] have contributed to an understanding of these enzymes.[459,461,464,482,483] For these copper amine oxidases the experimental evidence suggests an aminotransferase mechanism.[450,453,474,474a-d] The structure of the *E.coli* oxidase shows that a single copper ion is bound by three histidine imidazoles and is located adjacent to the TPQ (Eq. 15-53). Asp 383 is a conserved residue that may be the catalytic base in Eq. 15-53.[474b] A similar mechanism can be invoked for LTQ and TTQ.

How is the reduced cofactor reoxidized? Presumably the copper ion adjacent to the TPQ functions in this process, passing electrons one at a time to the next carrier in a chain. There is no copper in the TTQ-containing subunits. Electrons apparently must jump about 1.6 nm to the copper ion of amicyanin, then another 2.5 nm to the iron ion of the cytochrome *c*.[472] Reoxidation of the aminoquinol formed in Eq. 15-53, step *d*, yields a Schiff base whose hydrolysis will release ammonia and regenerate the TTQ. Intermediate states with Cu^+ and a TTQ semiquinone radical have been observed.[483a]



(15-53)

A closely related series of **plastoquinones** occur in chloroplasts (Chapter 23). In these compounds the two methoxyl groups of ubiquinone are replaced by methyl groups (Fig. 15-24). The most abundant of these compounds, plastoquinone A, contains nine prenyl units.[493,494]

Addition of the hydroxyl group of the reduced ubiquinone or plastoquinone to the adjacent double bond leads to a chroman-6-ol structure. The compounds derived from ubiquinone in this way are called **ubichromanols** (Fig. 15-24). The corresponding ubichromenol (Fig. 15-24) has been isolated from human kidney. **Plastochromanols** are derived from plastoquinone. That from plastoquinone A, first isolated from tobacco, is also known as solanochromene.

A closely related and important family of chromanols are the **tocopherols** or vitamins E (Fig. 15-24, Box 15-G). Tocopherols are plant products found primarily in plant oils and are essential to proper nutrition of humans and other animals. α-Tocopherol is the most abundant form of the vitamin E family; smaller amounts of the β, δ, and γ forms occur, as do a series of **tocotrienols** which contain unsaturated isoprenoid units.[495] The configuration of α-tocopherol is 2*R*,4'*R*,8'*R* as indicated in Fig. 15-24. When α-tocopherol is oxidized, e.g., with ferric chloride, the ring can be opened by hydrolysis to give **tocopherolquinones** (Fig. 15-24), which can in turn be reduced to tocopherolhydroquinones. Large amounts of the tocopherolquinones have been found in chloroplasts.

Another important family of quinones, related in structure to those already discussed, are the **vitamins K** (Fig. 15-24, Box 15-F). These occur naturally as two families. The vitamins K_1 (**phylloquinones**) have only one double bond in the side chain and that is in the prenyl unit closest to the ring. This suggests again the possibility of chromanol formation. In the vitamin K_2 (**menaquinone**) series, a double bond is present in each of the prenyl units. A synthetic compound **menadione** completely lacks the polyprenyl side chain and bears a hydrogen in the corresponding position on the ring. Nevertheless, menadione serves as a synthetic vitamin K, apparently because it can be converted in the body to forms containing polyprenyl side chains.

3. Ubiquinones, Plastoquinones, Tocopherols, and Vitamin K

In 1955, R. A. Morton and associates in Liverpool announced the isolation of a quinone which they named ubiquinone for its ubiquitous occurrence.[484,485] It was characterized as a derivative of benzoquinone attached to an unsaturated polyprenyl (isoprenoid) side chain (Fig. 15-24). In fact, there is a family of ubiquinones: that from bacteria typically contains six prenyl units in its side chain, while most ubiquinones from mammalian mitochondria contain ten. Ubiquinone was also isolated by F. L. Crane and associates using isooctane extraction of mitochondria. These workers proposed that the new quinone, which they called **coenzyme Q**, might participate in electron transport. As is described in Chapter 18, this function has been fully established. Both the name ubiquinone and the abbreviation Q are in general use. A subscript indicates the number of prenyl units, e.g., Q_{10}. Ubiquinones can be reversibly reduced to the hydroquinone forms (Fig. 15-24), providing a basis for their function in electron transport within mitochondria and chloroplasts.[486–490]

While vitamin E (Box 15-G) and vitamin K (Box 15-F) are dietary essentials for humans, ubiquinone is apparently not. Animals are able to make ubiquinones in quantities adequate to meet the need for this essential component of mitochondria. However, an extra dietary supplement may sometimes be of value.[491] A reduced level of ubiquinone has been reported in gum tissues of patients with periodontal disease.[492] Ubiquinones are being tested as possible protectants against heart damage caused by lack of adequate oxygen or by drugs.[491]

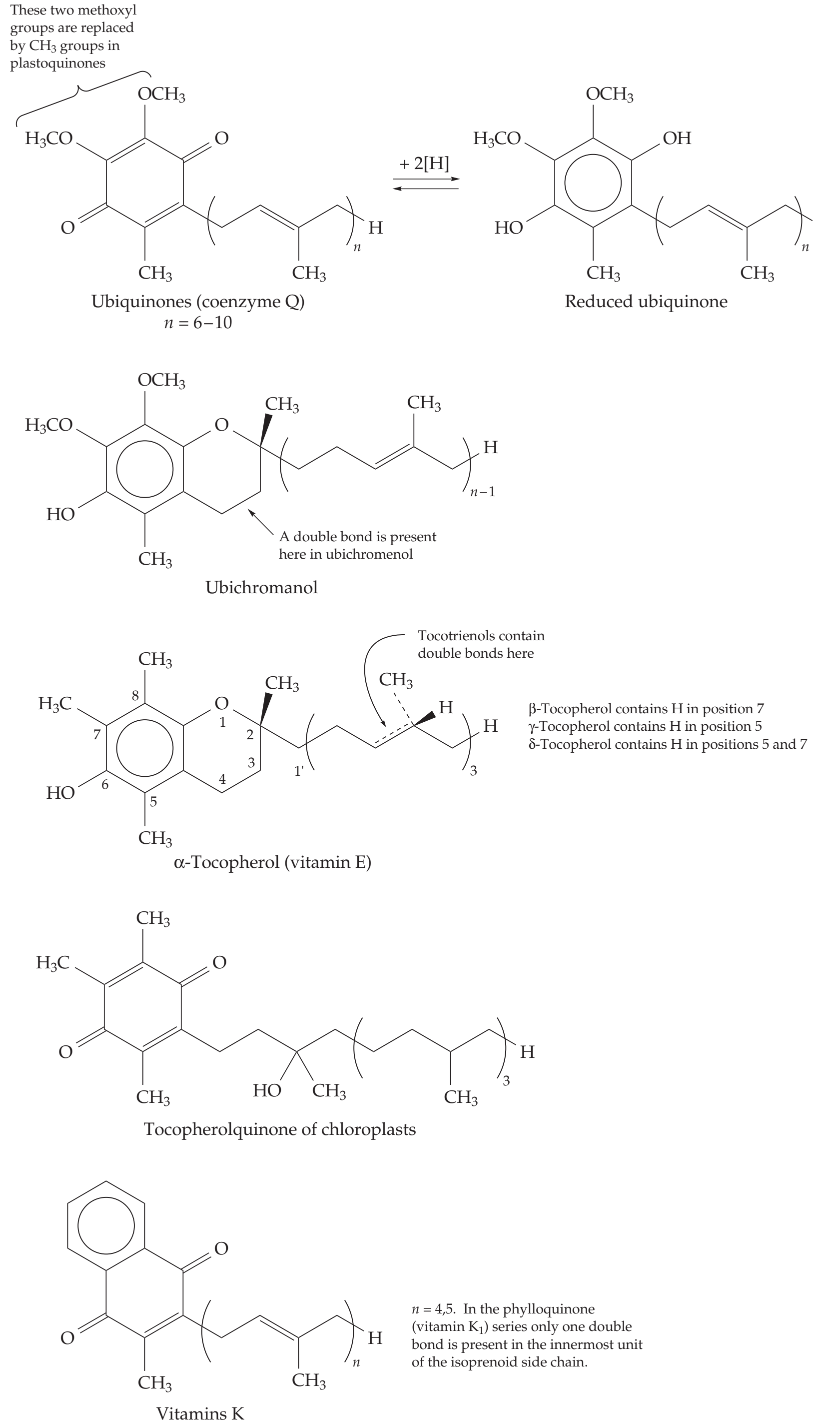


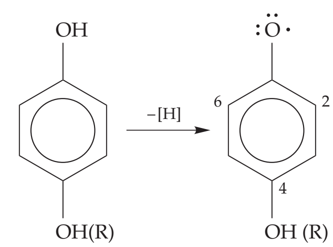
Figure 15-24 Structures of the isoprenoid quinones and vitamin E.

4. Quinones as Electron Carriers

Ubiquinones function as electron transport agents within the inner mitochondrial membranes[496] and also within the reaction centers of the photosynthetic membranes of bacteria (Eq. 23-32).[484,488,494] The plastoquinones also function in electron transport within these membranes. Within some mycobacteria vitamin K apparently participates in electron transport chains in the same way (see Chapter 18). Some bacteria contain both menaquinones and ubiquinones.

The vitamin E derivative **α-tocopherolquinone** (Fig. 15-24) can also serve as an electron carrier, being reversibly reduced to the hydroquinone form α-tocopherolquinol. Such a function has been proposed for the anaerobic rumen bacterium *Butyrovibrio fibrisolvens*.[497]

When a single hydrogen atom is removed from a hydroquinone or from a chromanol such as a tocopherol, a free radical is formed (Eq. 15-54). Phenols substituted in the 2, 4, and 6 positions give especially stable radicals.



(15-54)

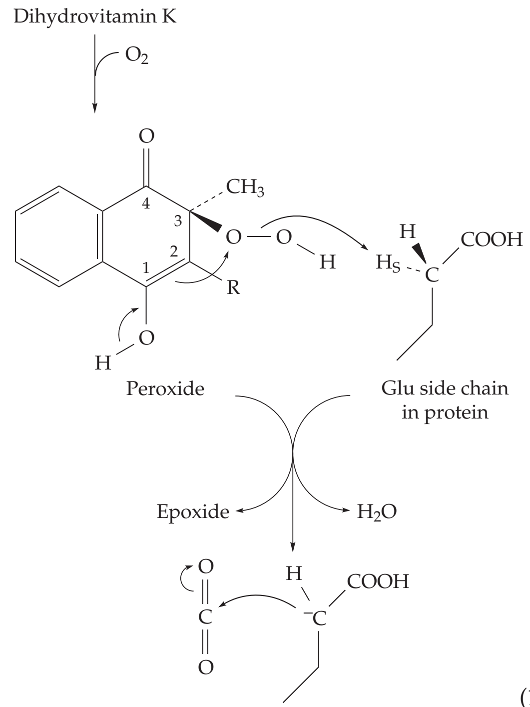
Both the presence of methyl substituents in the tocopherols and their chromanol structures increase the ability of these compounds to form relatively stable radicals.[498,499] This ability is doubtless probably important also in the function of ubiquinones and plastoquinones. Ubiquinone radicals (semiquinones) are probably intermediates in mitochondrial electron transport (Chapter 18) and radicals amounting to as much as 40% of the total ubiquinone in the NADH-ubiquinone reductase of heart mitochondria have been detected by EPR measurements.[500,501]

The equilibria governing semiquinone formation from quinones are similar to those for the flavin semiquinones which were discussed in Section B,6. Two consecutive one-electron redox steps can be defined. Their redox potentials will vary with pH because of a pK_a for the semiquinone in the pH 4.5 –6.5 region. For ubiquinone this pK_a is about 4.9 in water and 6.45 in methanol. A pK_a of over 13 in the hydroquinone form[502] will have little effect on redox potentials near pH 7. The potential for the one-electron reaction $Q + e^- \rightarrow Q^-$ is evaluated most readily. For this reaction $E°'$ (pH 7) is –0.074, –0.13, –0.17, and –0.23 for 2,3-dimethylbenzoquinone, plastoquinone, ubiquinone, and phylloquinone, respectively.

Why does this entire family of compounds have the long polyprenyl side chains? A simple answer is that they serve to anchor the compounds in the lipid portion of the cell membranes where they function. In the case of ubiquinones both the oxidized and the reduced forms may move freely through the lipid phase shuttling electrons between carriers.

5. Vitamin K and γ-Carboxyglutamate Formation

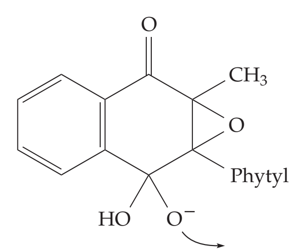
In higher animals the only known function of vitamin K is in the synthesis of γ-carboxyglutamate (Gla)-containing proteins, several of which are needed in blood clotting (Box 15-F, Chapter 12). Following the discovery of γ-carboxyglutamate, it was shown that liver microsomes were able to incorporate ^{14}C-containing bicarbonate or CO_2 into the Gla of prothrombin and could also generate Gla in certain simple peptides such as Phe-Leu-Glu-Val. Three enzymes are required. All are probably bound to the microsomal membranes.[503–507a] An NADPH-dependent reductase reduces vitamin K quinone to its hydroquinone form. Conversion of Glu residues to Gla residues requires this reduced vitamin K as well as O_2 and CO_2. During the carboxylation reaction the reduced vitamin K is converted into vitamin K 2,3-epoxide (Eq. 15-55).[508] The mechanism is uncertain but a peroxide intermediate such as that shown in Eq. 15-56 is probably involved. This could be used to generate a hydroxide ion adjacent to the *pro-S* -H of the glutamate side chain of the substrate. This hydrogen could be abstracted by the OH^- to form



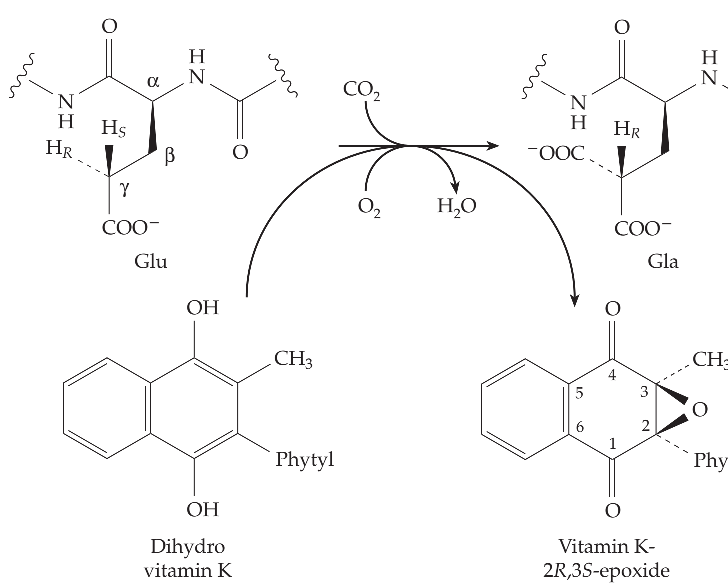
(15-56)

H_2O and a carbanion which would be stabilized by the adjacent carboxyl group[508–511] (Eq. 15-56).

Dowd and coworkers raised doubts that a hydroxide ion released in the active site in this manner is a strong enough base to generate the anion shown in Eq. 15-52.[512,513] They hypothesized a *"base strength amplification" mechanism* that begins with a peroxide formed at C-4 followed by ring closure to form a dioxetane and rearrangement to the following hypothetical strong base.[507,512,514]



Hypothetical strong base



(15-55)

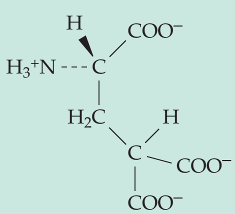
The proposal was supported by model experiments and also by observation of some incorporation of both atoms of $^{18}O_2$ into vitamin K epoxide. However, theoretical calculations support the simpler mechanism of Eq. 15-56.

The fact that L-*threo*-γ-fluoroglutamate residues, in which the fluorine atom is in the position corresponding to the *pro-S* hydrogen, are not carboxylated but that an *erythro*-γ-fluoroglutamate is carboxylated

BOX 15-F THE VITAMIN K FAMILY

The existence of an "antihemorrhagic factor" required in the diet of chicks to ensure rapid clotting of blood was reported in 1929 by Henrik Dam at the University of Copenhagen.[a,b] The fat-soluble material, later designated vitamin K, causes a prompt (2–6 h) decrease in the clotting time when administered to deficient animals and birds. The clotting time for a vitamin K-deficient chick may be greater than 240 s, but 6 h after injection of 2 µg of vitamin K_3 it falls to 50–100 s.[c] Pure vitamin K (Fig. 15-24), a 1,4-naphthoquinone, was isolated from alfalfa in 1939. Within a short time two series, the phylloquinones (vitamin K_1) and the menaquinones (vitamin K_2), were recognized. The most prominant phylloquinone contains the phytyl group, which is also present in the chlorophylls. For a human being a dietary intake of about 30 µg per day is recommended.[d] Additional vitamin K is normally supplied by intestinal bacteria.

The most obvious effect of a deficiency in vitamin K in animals is delayed blood clotting, which has been traced to a decrease in the activity of **prothrombin** and of clotting factors VII, IX, and X (Chapter 12, Fig. 12-17). Prothrombin formed by the liver in the absence of vitamin K lacks the ability to chelate calcium ions essential for the binding of prothrombin to phospholipids and to its activation to thrombin. The structural differences between this abnormal protein and the normal prothrombin have been pinpointed at the N terminus of the ~560 residue glycoprotein.[e,f] Tryptic peptides from the N termini differed in electrophoretic mobility. As detailed in Chapter 12, ten residues within the first 33, which were identified as glutamate residues by the sequence analysis on normal prothrombin, are actually **γ-carboxyglutamate** (Gla). The same amino acid is present near the N termini of clotting factors VII, IX, and X.

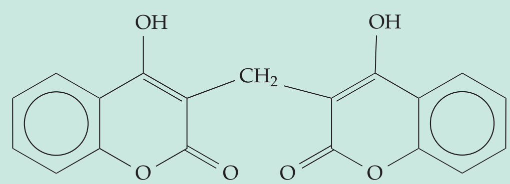


γ-Carboxyglutamate (Gla)

The fact that γ-carboxyglutamate had not been identified previously as a protein substituent is explained by its easy decarboxylation to glutamic acid during treatment with strong acid. The function of vitamin K is to assist in the incorporation of the additional carboxyl group into the glutamate residues of preformed prothrombin and other blood-clotting factors[g,h] with a resulting increase in calcium ion affinity.

Four other plasma proteins designated C, S, M, and Z contain γ-carboxyglutamate. The functions of proteins M and Z are unknown but protein C is a serine protease involved in regulation of blood coagulation and protein S is a cofactor that assists the action of protein C. Other proteins that require vitamin K for synthesis include the 49-residue **bone Gla protein** (or **osteocalcin**) and the 79-residue **matrix Gla protein** found in bone and cartilage.[i,j] These proteins contain three and five residues of Gla, respectively. Their possible functions in mineralization are considered in Chapter 8. At least two additional small human proline-rich Gla proteins of unknown function are synthesized in many tissues.[k] Gamma-carboxyglutamate also occurs in an invertebrate peptide from fish-hunting cone snails. This "sleeper peptide," which induces sleep in mice after intracerebral injection, has the sequence GEE*E*LQE*NQE*LIRE*KSN. Here E* designates the 5 residues of Gla.[l]

An interesting facet of vitamin K nutrition and metabolism was revealed by the observation that cattle fed on spoiled sweet clover develop a fatal hemorrhagic disease. The causative agent is **dicoumarol**, a compound arising from coumarin, a natural constituent of clover. Dicourmarol and the closely related synthetic **warfarin** are both potent vitamin K antagonists. Warfarin is used both as a rat poison and in the treatment of thromboembolic disease. As rodenticides hydroxycourmarin derivatives are usually safe because a single accidental ingestion by a child or pet does little harm, whereas regular ingestion by rodents is fatal.



Dicoumarol

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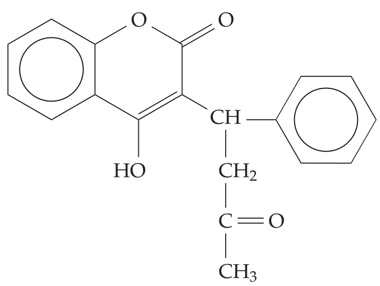
suggested the indicated stereospecificity.[515,516] This was confirmed by observation of a kinetic isotope effect when ^2H is present in the *pro-S* position.[517] Addition of the carbanion to CO_2 would generate the Gla residue. The two glutamates in the following sequence, in which X may be various amino acids, are carboxylated if the protein also carries a suitable N-terminal signal sequence: EXXXEXC. If a suitable glutamyl peptide is not available for carboxylation the postulated peroxide intermediate (Eq. 15-55) is still converted slowly to the 2,3-epoxide of vitamin K.

A third enzyme is required to reduce the epoxide to vitamin K (Eq. 15-57). The biological reductant is uncertain but dithiols such as dithiothreitol serve in the laboratory.[518] See also Eq. 18-47. Protonation of an intermediate enolate anion would give 3-hydroxy-2, 3-dihydrovitamin K, an observed side reaction product.

$$\text{Vitamin K epoxide} + 2[H] \rightarrow H_2O + \text{Vitamin K} \quad (15\text{-}57)$$

This reaction is of interest because of its specific inhibition by such coumarin derivatives as Warfarin:



This synthetic compound, as well as natural coumarin anticoagulants (Box 15-F), inhibits both the vitamin K reductase and the epoxide reductase.[518,519] The matter is of considerable practical importance because of the spread of warfarin-resistant rats in Europe and the United States. One resistance mutation has altered the vitamin K epoxide reductase so that it is much less susceptible to inhibition by warfarin.[519,520]

While glutamate residues in peptides of appropriate sequence are carboxylated by the vitamin K-dependent system, aspartate peptides scarcely react.[503] Beta-carboxyaspartate is present in protein C of the blood anticoagulant system (Fig. 12-17)[521] and in various other proteins containing EGF homology domains (Table 7-3),[522] but the mechanism of its formation is unknown.

6. Tocopherols (Vitamin E) as Antioxidants

The major function of the tocopherols is thought to be the protection of phospholipids of cell membranes against oxidative attack by free radicals and organic peroxides. Peroxidation of lipids, which is described in Chapter 21, can lead to rapid development of rancidity in fats and oils. However, the presence of a small amount of tocopherol inhibits this decomposition, presumably by trapping the intermediate radicals in the form of the more stable tocopherol radicals (Eq. 15-54) which may dimerize or react with other radicals to terminate the chain. Even though only one molecule of tocopherol is present for a thousand molecules of phospholipid, it is enough to protect membranes.[523] That vitamin E does function in this way is supported by the observation that much of the tocopherol requirement of some species can be replaced by *N,N'*-diphenyl-*p*-phenylenediamine, a synthetic antioxidant (see Table 18-5 for the structure of a related substance). Three generations of rats have been raised on a tocopherol-free diet containing this synthetic antioxidant. However, not all of the deficiency symptoms are prevented.

The antioxidant role of α-tocopherol in membranes is generally accepted.[524–526] It is thought to be critical to defense against oxidative injury and to help the body combat the development of tumors and to slow aging. Gamma-tocopherol may be more reactive than α-tocopherol in removing radicals created by NO and other nitrogen oxides.[526] Its actions are strongly linked to those of ascorbic acid (Box 18-D) and selenium. Ascorbate may reduce tocopherol semiquinone radicals, while selenium acts to enhance breakdown of peroxides as described in the next section.

G. Selenium-Containing Enzymes

In 1957, Schwartz and associates showed that the toxic element selenium was also a nutritional factor essential for prevention of the death of liver cells in rats.[527] Liver necrosis would be prevented by as little as 0.1 ppm of selenium in the diet. Similar amounts of selenium were shown to prevent a muscular dystrophy called "white muscle disease" in cattle and sheep grazing on selenium-deficient soil. Sodium selenite and other inorganic selenium compounds were more effective than organic compounds in which Se had replaced sulfur. **Keshan disease**, an often fatal heart condition that is prevalent among childen in Se-deficient regions of China, can be prevented by supplementation of the diet with $NaSeO_3$.[528] Even the little crustacean "water flea" *Daphnia* needs 0.1 part per billion of Se in its water.[529]

Selenium has long been known to enhance the antioxidant activity of vitamin E. Recent work suggests that vitamin E acts as a radical scavenger, preventing

BOX 15-G VITAMIN E: THE TOCOPHEROLS

Vitamin E was recognized in 1926 as a factor preventing sterility in rats that had been fed rancid lipids.[a–e] The curative factor, present in high concentration in wheat germ and lettuce seed oils, is a family of vitamin E compounds, the tocopherols (Fig. 15-24). The first of these was isolated by Evans and associates in 1936. Vitamin E deficiency in the rabbit or rat is accompanied by muscular degeneration (**nutritional muscular dystrophy**; see also Box 15-A) and a variety of other symptoms that vary from one species to another. Animals deficient in vitamin E display obvious physical deterioration followed by sudden death. Muscles of deficient rats show abnormally high rates of oxygen uptake, and abnormalities appear in the membranes of the endoplasmic reticulum as viewed with the electron microscope. It is thought that deterioration of lysosomal membranes may be the immediate cause of death.

The tocopherol requirement of humans is not known with certainty, but about 5 mg (7.5 IU) / day plus an additional 0.6 mg for each gram of polyunsaturated fatty acid consumed may be adequate. It is estimated that the average daily intake is about 14 mg, but the increasing use of highly refined foods may lead to dangerously low consumption. Recent interest[d–f] has been aroused by studies that show that much larger amounts of vitamin E (e.g., 100–400 mg/day) substantially reduce the risk of coronary disease and stroke in both women[g] and men[h] and also decrease oxidative modification of brain proteins.[i] The decrease in heart attacks and stroke may be in part an indirect effect of the anticlotting action of vitamin E quinone.[f] Plant oils are usually the richest sources of tocopherols, while animal products contain lower quantities.

To some extent the vitamin E requirement may be lessened by the presence in the diet of synthetic antioxidants and by selenium. Much evidence supports a relationship between the nutritional need for selenium and that for vitamin E. Lack of either causes muscular dystrophy in many animals as well as severe edema (exudative diathesis) in chicks. Since vitamin E-deficient rats have a low selenide (Se^{2-}) content, it has been suggested that vitamin E protects reduced selenium from oxidation.[j] Vitamin C (ascorbic acid), in turn, protects vitamin E.

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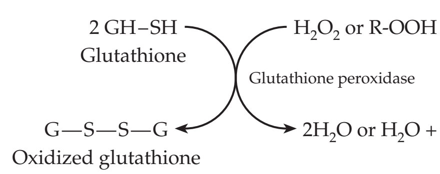
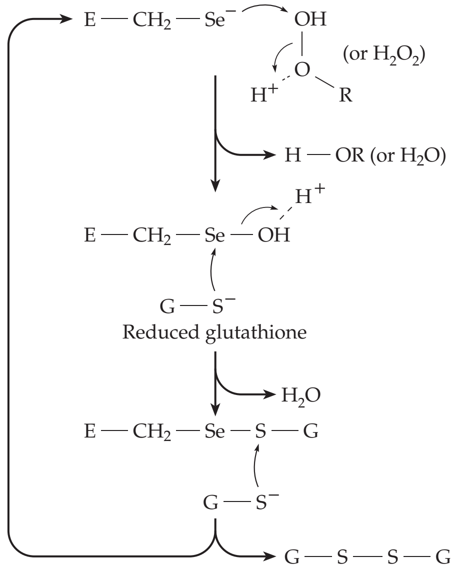
excessive peroxidation of membrane lipids, while selenium, in the enzyme **glutathione peroxidase**, acts to destroy the small amounts of peroxides that do form. This was the first established function of selenium in human beings, but there are others. If we include proteins from animals and bacteria, at least ten selenoproteins are known (Table 15-4).[530–534] Seven of these are enzymes and most catalyze redox processes. The active sites most often contain **selenocysteine**, whose selenol side chain is more acidic (pK_a ~5.2) than that of cysteine and exists as $-CH_2-Se^-$ at neutral pH.[530]

Glutathione peroxidases catalyze the reductive decomposition of H_2O_2 or of organic peroxides by glutathione (G–SH) according to Eq. 15-58. At least three isoenzyme forms have been identified in mammals: a cellular form,[531,535–537] a plasma form, and a



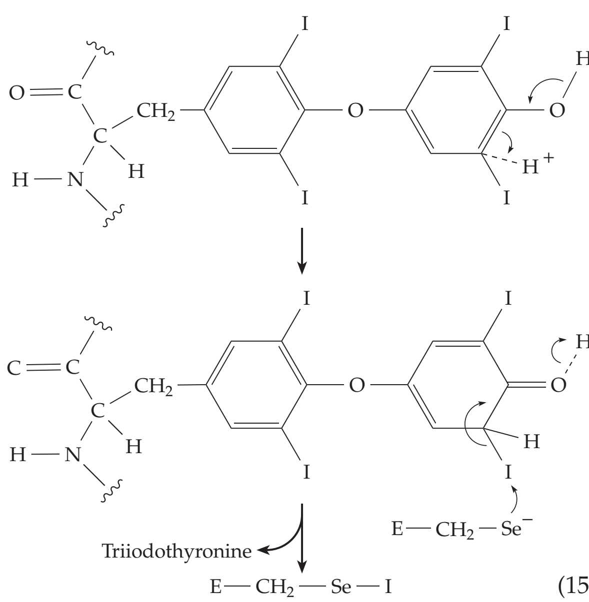
(15-58)

form with a preference for organic peroxides derived from phospholipids.[327,538–540] A related selenoprotein has been found in a human poxvirus.[540a] Selenocysteine is present at a position (residues 41–47) near the N terminus of an α helix, in the ~ 180-residue polypeptides. A possible reaction mechanism involves attack by the selenol on the peroxide to give a selenic acid intermediate which is reduced by glutathione in two nucleophilic displacement steps (Eq. 15-59).



(15-59)

Three types of **iodothyronine deiodinase** remove iodine atoms from thyroxine to form the active thyroid hormone triiodothyronine and also to inactivate the hormone by removing additional iodine[531,541–546] (see also Chapter 25). In this case the $-CH_2-Se^-$ may attach the iodine atom, removing it as I^+ to form $-CH_2-Se-I$. The process could be assisted by the phenolic –OH group if it were first tautomerized (Eq. 15-60).

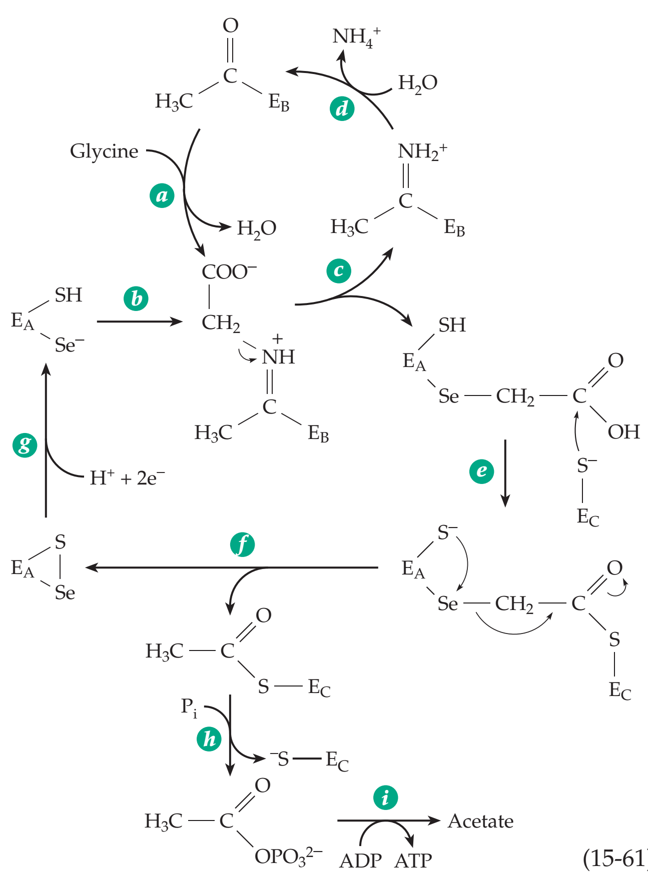


(15-60)

A recently discovered human selenoprotein is a **thioredoxin reductase** which is present in the T cells of the immune system as well as in placenta and other tissues.[189,547–549] The 55-kDa protein has one selenocysteine as the penultimate C-terminal residue. Another mammalian selenoprotein, of uncertain function, is the 57-kDa **selenoprotein P**. It contains over 60% of the selenium in rat plasma and is also present in the human body. Selenoprotein P contains ten selenocysteine residues.[550–552a] Some of these may be replaced by serine in a fraction of the molecules.[553] A smaller 9.6-kDa skeletal muscle protein, **selenoprotein W**, contains a single selenocysteine.[554–556] Another selenoprotein has been found in sperm cells, both in the tail and in a keratin-rich capsule that surrounds the mitochondria in the sperm midpiece.[557] Lack of this protein may be the cause of the abnormal immotile spermatozoa observed in Se-deficient rats and of reproductive difficulties among farm animals in Se-deficient regions.[558]

Several selenoproteins have been found in certain bacteria and archaea. A **hydrogenase** from *Methanococcus vannielii* contains selenocysteine.[559,560] This enzyme transfers electrons from H_2 to the C-5 *si* face of the 8-hydroxy-5-deazaflavin cofactor F_{420} (Section B,4). The same bacterium synthesizes two **formate dehydrogenases** (see Fig 15-23), one of which contains Se. Two Se-containing formate dehydrogenases are made by *E. coli*. One of them, which is coupled to a hydrogenase in the formate hydrogen-lyase system (see Eq. 15-37), is a 715-residue protein containing selenocysteine at position 140.[561–563] The second has selenocysteine at position 196 and functions with a nitrate reductase in anaerobic nitrate respiration.[561]

Glycine reductase is a complex enzyme[530,564–566] that catalyzes the reductive cleavage of glycine to acetyl phosphate and ammonia (Eq. 15-61) with the



(15-61)

subsequent synthesis of ATP (Eq. 14-43). Electrons for reduction of the disulfide that is formed are provided by NADH. A single selenocysteine residue is present in the small 12-kDa subunits. The enzyme contains a dehydroalanine residue (Chapter 14) in subunit B and a thiol group in subunit C.[566] An acetyl-enzyme derivative of subunit C, perhaps of its –SH group, has been identified.[546] The mechanism of action is uncertain but the steps in Eq. 15-61 have been suggested.[567] The subunits are designated E_A, E_B, and E_C. Step *e* is particularly hard to understand because formation of a thioester in this manner is not expected to occur spontaneously and must be linked in some way to other steps.

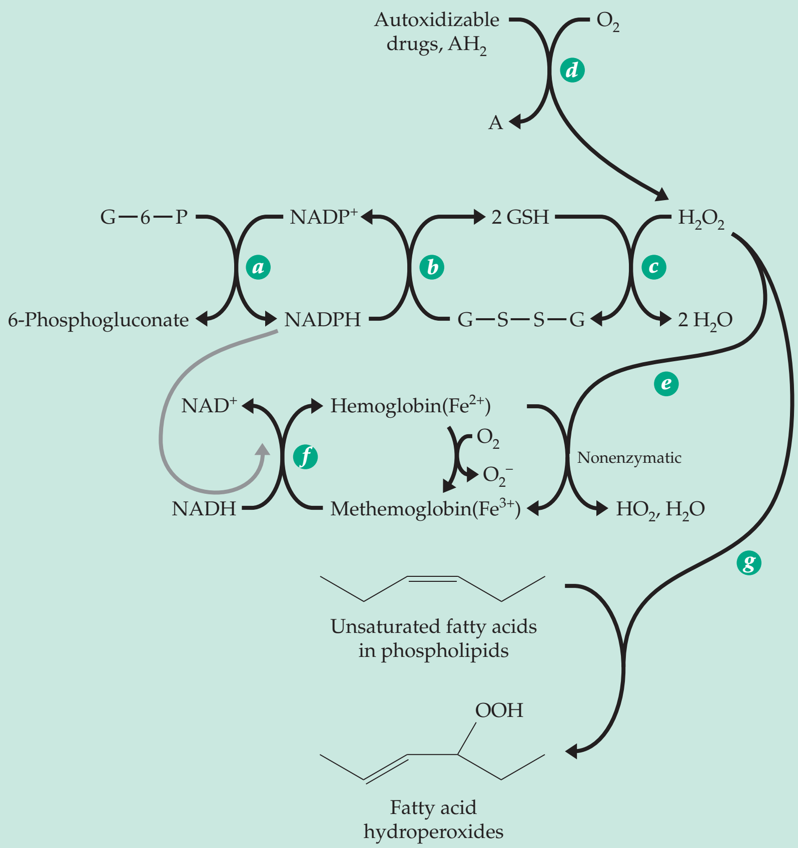
A selenium-containing **xanthine dehydrogenase** is present in purine-fermenting clostridia. Like other xanthine dehydrogenases (Chapter 16), it converts xanthine to uric acid and contains nonheme iron, molybdenum, FAD, and an Fe–S center. The selenium is probably present as Se^{2-} bound to Mo as is S^{2-} in xanthine oxidase (Fig. 16-32).[567a] A related reaction is catalyzed by the Fe–S protein **nicotinic acid hydroxylase** (Eq. 15-62) found in some clostridia.[568] Splitting of the Mo(V) EPR signal when ^{77}Se is present in the enzyme shows that the selenium is present as a ligand of molybdenum. Another member of the family is a purine hydroxylase that converts purine 2-hydroxypurine, or hypoxanthine to xanthine.[567a]

TABLE 15-4
Selenium-Containing Proteins

Enzyme	Source	Mass (kDa)	Subunit composition	Other cofactors
Glutathione peroxidases				None
Cellular	Mammals	21 x 4		
Plasma	Humans			
Phospholipid hydroperoxide	Pig, rat	18		
Iodothyronine deiodinases	Vertebrates			
Thioredoxin reductase	Humans	55 × 2		
Selenoprotein *P*	Mammals	57		None
Selenoprotein *W*	Rat	19.6		
Formate dehydrogenase	Bacteria - *E. coli*	600	$\alpha_2\beta_4\gamma_{2\text{-}4}$	Heme *b*
	Archaea			Mo; molybdopterin
Hydrogenase	Bacteria		$\alpha_2\beta_4\gamma_2$	FAD; NiFeSe
	Archaea			
Glycine reductase	Some clostridia		ABC	
Selenoprotein A		12		
Selenoprotein B		200		
Carbonyl protein C		250		Fe?
Nicotinic acid hydroxylase	Clostridia	300		FAD, FeS, Mo
Purine hydroxylase	Clostridia			FAD, FeS, Mo
Thiolase (contains selenomethionine)	*Clostridium kluyverii*	39 × 4		None
Carbon monoxide dehydrogenase	*Oligotropha carboxidovorans*	137 × 2	$\alpha_2\beta_2\gamma_2$	FAD; Mo; molybdopterin

BOX 15-H GLUTATHIONE PEROXIDASE AND ABNORMALITIES OF RED BLOOD CELLS

The processes by which hemoglobin is kept in the Fe(II) state and functioning normally within intact erythrocytes is vital to our health. Numerous hereditary defects leading to a tendency toward anemia have helped to unravel the biochemistry indicated in the accompanying scheme.[a]



About 90% of the glucose utilized by erythrocytes is converted by glycolysis to lactate, but about 10% is oxidized (via glucose 6-phosphate) to 6-phosphogluconate. The oxidation (reaction *a*) is catalyzed by glucose-6-phosphate dehydrogenase (Eq.15-10) using $NADP^+$. This is the principal reaction providing the red cell with NADPH for reduction of glutathione (Box 11-B) according to reaction *b*. Despite the important function of glucose-6-*P* dehydrogenase, ~400 million persons, principally in tropical and Mediterranean areas, have a hereditary deficiency of this enzyme. The genetic variations are numerous, with about 400 different ones having been identified. Although most individuals with this dificiency have no symptoms, the lack of the enzyme is truly detrimental and sometimes leads to excessive destruction of red cells and anemia during some sicknesses and in response to administration of various drugs.[b] The survival of the defective genes, like that for sickle cell hemoglobin (Box 7-B) is thought to result from increased resistance to malaria parasites.

Other erythrocyte defects that lead to drug sensitivity include a deficiency of glutathione (resulting from a decrease in its synthesis) and a deficiency of glutathione reductase (reaction *b*). The effects of drugs have been traced to the production of H_2O_2 (reaction *c*) in red blood cells; catalase, which converts H_2O_2 into H_2O and O_2, is thought to function in a similar way. Both enzymes are probably necessary for optimal health.

An excess of H_2O_2 can damage erythrocytes in two ways. It can cause excessive oxidation of functioning hemoglobin to the Fe(III)-containing methemoglobin. (Methemoglobin is also formed spontaneously during the course of the oxygen-carrying function of hemoglobin. It is estimated that normally as much as 3% of the hemoglobin may be oxidized to methemoglobin daily.) The methemoglobin formed is reduced back to hemoglobin through the action of **NADH-methemoglobin reductase** (reaction *f*). A smaller fraction of the methemoglobin is reduced by a similar enzyme requiring NADPH (as indicated by the colored arrow). A hereditary lack of the NADH-methemoglobin reductase is also known.

A second destructive function of H_2O_2 is attack on double bonds of unsaturated fatty acids of the phospholipids in cell membranes. The resulting fatty acid hydroperoxides can react further with C–C chain cleavage and disruption of the membrane. This is thought to be the principal cause of the hemolytic anemia induced by drugs in susceptible individuals. The selenium-containing glutathione peroxidase is thought to decompose these fatty acid hydroperoxides. Vitamin E (Box 15-G), acting as an antioxidant within membranes, is also needed for good health of erythrocytes.[c,d]

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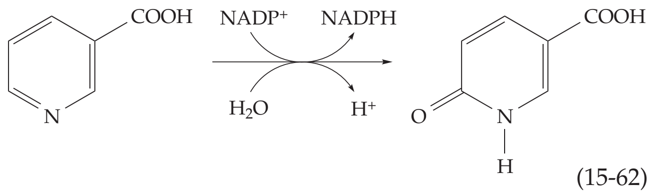
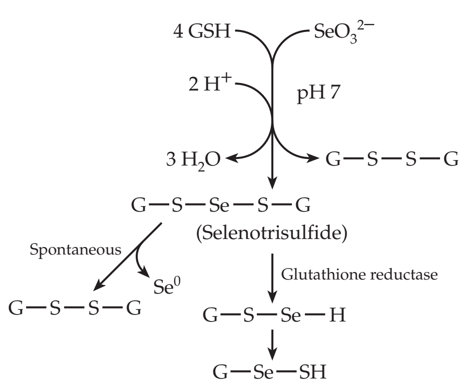


(15-62)

A **thiolase** (Eq. 13-35) from *Clostridium kluyveri* is one of only two known selenoproteins that contain selenomethionine.[569] However, the selenomethionine is incorporated randomly in place of methionine. This occurs in all proteins of all organisms to some extent and the toxicity of selenium may result in part from excessive incorporation of selenomethionine into various proteins.

Selenium is found to a minor extent wherever sulfur exists in nature. This includes the sulfur-containing modified bases of tRNA molecules. In addition to a small amount of nonspecific incorporation of Se into all S-containing bases there are, at least in bacteria, specific Se-containing tRNAs. In *E. coli* one of these is specific for lysine and one for glutamate. One of the modified bases has been identified as 5-methyl-amino-methyl-2-selenouridine.[570] It is present at the first position of the anticodon, the "wobble" position.[571]

Selenium has its own metabolism. Through the use of ^{75}Se as a tracer, normal rat liver has been shown to contain Se^{2-}, SeO_3^{2-}, and selenium in a higher oxidation state.[572] Glutathione may be involved in reduction of selenite to selenide.[573] The nonenzymatic reduction of selenite by glutathione yields a selenotrisulfide derivative (Eq. 15-63). The latter is spontaneously decomposed to oxidized glutathione and elemental selenium or by the action of glutathione reductase to glutathione and selenium. Selenocysteine can be converted to alanine + elemental selenium (Eq. 14-34). Some bacteria are able to oxidize elemental Se back to selenite.[574] Selenium undergoes biological methylation readily in bacteria, fungi, plants, and animals (Chapter 16).[575,576] This may in some way be related to the reported effect of selenium in protecting animals against the toxicity of mercury. Excess selenium may appear in the urine as trimethylselenonium ions.[577]



(15-63)

How is selenium incorporated into selenocysteine-containing proteins? This element does enter amino acids to a limited extent via the standard synthetic pathways for cysteine and methionine. However, the placement of selenocysteine into specific positions in selenoproteins occurs by the use of a minor serine-specific tRNA that acts as a suppressor of chain termination during protein synthesis.[532,533,578] (This topic is dealt with further in Chapter 29.) The genes for these and presumably for other selenocysteine-containing proteins have the "stop" codon TGA at the selenocysteine positions. However, when present in a suitable "context" the minor tRNA, carrying selenocysteine in place of serine, is utilized to place selenocysteine into the growing peptide chain. In bacteria, and presumably also in eukaryotes, the selenocysteinyl-tRNA is formed from the corresponding seryl-tRNA by a PLP-catalyzed β-replacement reaction. The selenium donor is not Se^{2-} but selenophosphate $Se-PO_3^{2-}$ in which the Se–P bond is quite weak.[579–581] After addition to the aminoacrylate intermediate in the PLP enzyme the Se–P bond may be hydrolytically cleaved to HPO_4^{2-} and selenocysteyl-tRNA.

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Study Questions

1. *S*-adenosylmethionine is also a biological methyl group donor. The product of its methyl transferase reactions is *S*-adenosylhomocysteine. This product is further degraded by *S*-adenosylhomocysteine hydrolase, an enzyme that contains tightly bound NAD^+, to form homocysteine and adenosine.

   Write a step-by-step mechanism for the action of this hydrolase.

2. Compare the chemical mechanisms of enzyme-catalyzed decarboxylation of the following:
   a) a β-oxo-acid such as acetoacetate or oxaloacetate
   b) an α-oxo-acid such as pyruvate
   c) an amino acid such as L-glutamate

3. Describe the subunit structure of the enzyme pyruvate dehydrogenase. Discuss the functioning of each of the coenzymes that are associated with these subunits and write detailed mechanisms for each step in the pyruvate dehydrogenase reaction.

4. Free **formate** can be assimilated by cells via the intermediate **10-formyl-tetrahydrofolate** (10-formyl-THF).
   a) Describe the mechanism of synthesis of this compound from formate and tetrahydrofolate.
   b) Diagram a hypothetical transition state for the first step of this reaction sequence.
   c) Describe two or more uses that the human body makes of 10-formyl-THF.

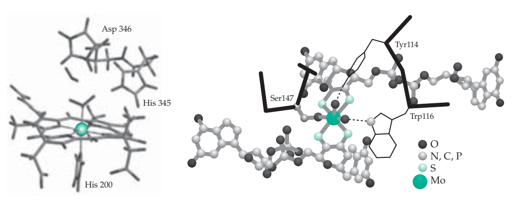
5. Using partial structural formulas, describe the reactions by which serine and methionine react to form *N*-formylmethionine needed for protein synthesis.

6. Write the equations for each of the reactions shown below. Using the $E^{0'}$ values below, calculate approximate Gibbs energies for each reaction, and show by the relative length of the arrows on which side of the reaction the equilibrium lies.
   a) The oxidation of malate by NAD^+
   b) The oxidation of succinate by NAD^+
   c) The oxidation of succinate by enzyme-bound FAD
   d) What can you say about the cofactor required for oxidation of succinate from your calculations?

   The values of $E^{0'}$ for several half reactions are given below. Everything has been rounded to one significant figure so that a calculator is unnecessary.

Reaction	$E^{0'}$ (volts)
$NAD^+ + H^+ + 2\ e^- \rightarrow NADH$	−0.3
enzyme bound FAD + $2\ H^+ + 2\ e^- \rightarrow$ enzyme bound $FADH_2$	0.0
fumarate + $2\ H^+ + 2\ e^- \rightarrow$ succinate	0.0
oxalacetate + $2\ H^+ + 2\ e^- \rightarrow$ malate	−0.2

7. Some acetogenic bacteria, which convert CO_2 to acetic acid, form pyruvate for synthesis of carbohydrates, etc., by formation of formaldehyde and conversion of the latter to glycine by reversal of the PLP and lipoic acid-dependent glycine decarboxylase, a 4-protein system. The glycine is then converted to serine, pyruvate, oxaloacetate, etc. Propose a detailed pathway for this sequence.



Transition metal ions function in many biological oxidation-reduction processes. (Left) A heme ring in **nitrite reductase** chelates an iron ion in its center, holding the Fe^{2+} or Fe^{3+} with bonds from four nitrogen atoms and a fifth bond from an imidazole ring below. Above the iron is a nitrite ion, NO_2^-, awaiting reduction to nitric oxide NO in a denitrifying bacterium. From Ranghino *et al.* (2000) *Biochemistry* **39**, 10958–10966. (Right) The active site of a bacterial **dimethylsulfoxide reductase** has an atom of molybdenum or tungsten at its center. The metal is held by four sulfur atoms from two molybdopterin molecules and an oxygen atom of a serine side chain. Two other oxygen atoms are bound as oxo groups and may participate in catalysis. From Stewart *et al.* (2000) *J. Mol. Biol.* **299**, 593–600.

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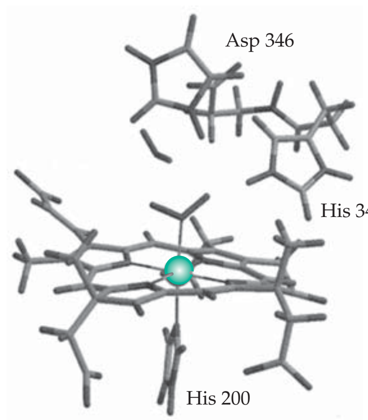
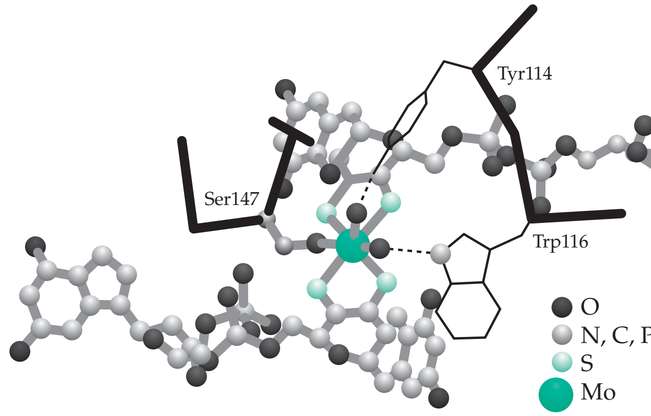




Although the amounts present within living cells are very small, the ions of the transition metals Fe, Co, Ni, Cu, and Mn are extremely active centers for catalysis, especially of reactions that take advantage of the ability of these metals to exist in more than one oxidation state.[1–4] Iron, copper, and nickel are also components of the electron carrier proteins that function as oxidants or reductants in many biochemical processes. These metals are all nutritionally essential, as are chromium and vanadium. Among the heavier transition elements molybdenum is a constituent of an important group of enzymes that includes the sulfite oxidase of human liver and nitrogenase of nitrogen-fixing bacteria. Tungsten occasionally substitutes for molybdenum.

A. Iron

Iron is one of the most abundant elements in the earth's crust, being present to the extent of ~4% in a typical soil. Its functions in living cells are numerous and diverse.[2,5–8] The average overall iron content of both bacteria and fungi is ~1 mmol / kg, but that of animal tissues is usually less. Seventy percent of the 3–5 g of iron present in the human body is located in the blood's erythrocytes, whose overall iron content is ~20 mM. In other tissues the total iron averages closer to 0.3 mM and consists principally of storage forms. The total concentration of iron in all of the iron-containing *enzymes* of tissues amounts to only about 0.01 mM. Although these concentrations are low, the iron is concentrated in oxidative enzymes of membranes and may attain much higher concentrations locally. Only a few parasitic or anaerobic bacteria, e.g. the lactic acid bacteria, possess no oxygen-requiring enzymes and are almost devoid of both iron and copper. All other organisms appear to require iron for life.

1. Uptake by Living Cells

A major problem for cells is posed by the relative insolubility of ferric hydroxide and other compounds from which iron must be extracted by the organism. A consequence is that iron is often taken up in a chelated form and is transferred from one organic ligand, often a protein, to another with little or no existence as free Fe^{3+} or Fe^{2+}. As can be calculated from the estimated solubility product of $Fe(OH)_3$ (Eq. 16-1),[7] the equilibrium concentration of Fe^{3+} at pH 7 is only 10^{-17} M.

$$K_{sp} = [Fe^{3+}][OH^-]^3 < 10^{-38}\ M^4 \quad (16\text{-}1a)$$

$$\text{or } [Fe^{3+}] / [H^+]^3 < 10^4\ M^4 \text{ at } 25°\ C \quad (16\text{-}1b)$$

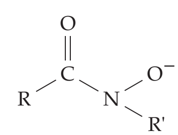
For a 2 μm^3 bacterial cell this amounts to just one free Fe^{3+} ion in almost 100 million cells at any single moment. The importance of chelated forms of iron becomes obvious. It is also evident from Eq. 16-1 that, in addition to chelation, a low external pH can also facilitate uptake of Fe^{3+} by organisms.

The values of the formation constants for chelates of Fe^{2+} typically lie between those of Mn^{2+} and Co^{2+} (Fig. 6-6, Table 6-9). For example, $K_1 = 10^{14.3}\ M^{-1}$ for formation of the Fe^{2+} chelate of EDTA. The smaller and more highly charged Fe^{3+} is bonded more strongly ($K_1 = 10^{25}\ M^{-1}$). These binding constants are independent of pH. However, the binding of any metal ion is affected by pH, as discussed in Chapter 6. A fact of considerable biochemical significance is the stronger binding of Fe^{3+} to oxygen-containing ligands than to nitrogen atoms, while Fe^{2+} tends to bind preferentially to nitrogen. It is also significant that Fe^{3+} bound to oxygen ligands tends to exchange readily with other ferric ions in the medium, whereas Fe^{3+} bound to nitrogen-containing ligands such as heme exchanges slowly. This fact is important for both iron-transport compounds and enzymes.

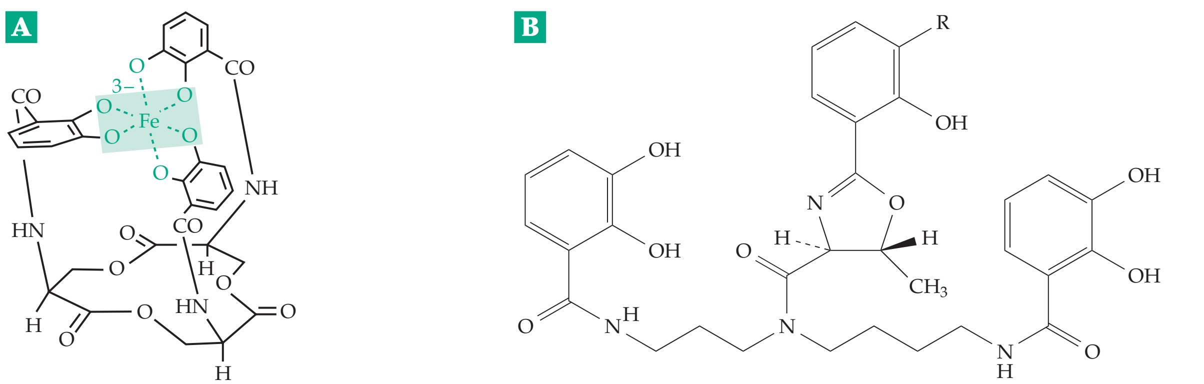
Siderophores. If a suitably high content of iron (e.g., 50 μM or more for *E. coli*) is maintained in the external medium, bacteria and other microorganisms have little problem with uptake of iron. However, when the external iron concentration is low, special compounds called siderophores are utilized to render the iron more soluble.[7–11c] For example, at iron concentrations below 2 μM, *E. coli* and other enterobacteria secrete large amounts of **enterobactin** (Fig. 16-1). The stable Fe^{3+}–enterobactin complex is taken up by a transport system that involves receptors on the outer bacterial membrane.[9,12,13] Siderophores from many bacteria have in common with enterobactin the presence of **catechol** (*ortho*-dihydroxybenzene) groups that chelate the iron.

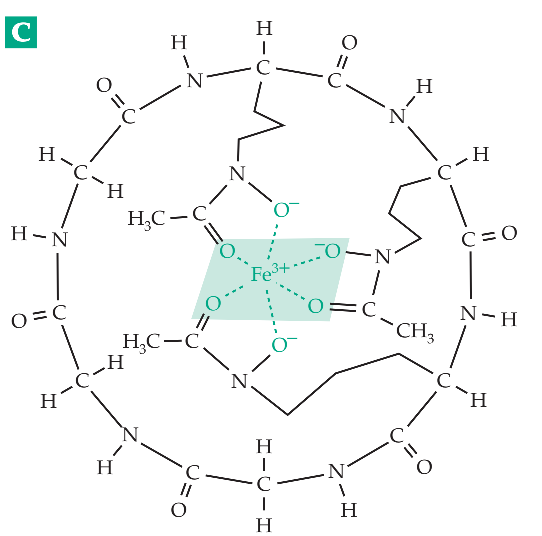
The three catechol groups of enterobactin are carried on a cyclic serine triester structure. A variety of both cyclic and linear structures are found among other catechol siderophores.[14–19] For example, **parabactin** and **agrobactin** (Fig. 16-1) contain a backbone of spermidine[20] (Chapter 24). After the Fe^{3+}–enterobactin complex enters a bacterial cell the ester linkages of a siderophore are cleaved by an esterase. Because of the extremely high formation constant of ~10^{52} M^{-1} for the complex[11] the only way for a cell to release the Fe^{3+} is through this irreversible destruction of the iron carrier.[11d] Reduction to Fe^{2+} may be involved in release of iron from some siderophores.[11e]

The first known siderophore, isolated in 1952 by Neilands,[22] is **ferrichrome** (Fig. 16-1), a cyclic hexapeptide containing **hydroxamate** groups at the iron-binding centers. Oxygen atoms form the bonds to iron



Hydroxamate group





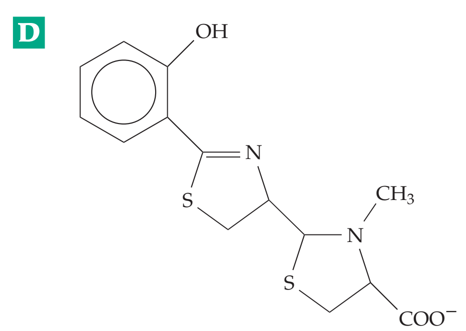


Figure 16-1 Structures of several siderophores and of their metal complexes. (A) Enterobactin of *E. coli* and other enteric bacteria;[12] (B) parabactin (R = H) from *Paracoccus denitrificans* and agrobactin (R = OH) from *Agrobacterium tumefaciens*;[20] (C) ferrichrome;[9] (D) pyochelin from *Pseudomonas aeruginosa*.[21]

in this compound also. Ferrichrome binds Fe^{3+} with a formation constant of ~ 10^{29} M^{-1}. The binding is not as tight as with enterobactin and the iron can be released by enzymatic reduction to Fe^{2+} which is much less tightly bound than is Fe^{3+}. The released ferrichrome can be secreted and used repeatedly to bring in more iron. Ferrichrome is produced by various fungi and bacilli and is only one of a series of known hydroxamate siderophores.[16] Since iron is essential to virtually all parasitic organisms the ability to obtain iron is often the limiting factor in establishing an infection.[23,24]

In *E. coli* there are seven different outer **membrane receptors** for siderophores.[25] One of these, the gated porin **FepA**, is specific for ferric enterochelin. With the assistance of another protein, **TonB**, it allows the ferric siderophore to penetrate the outer membrane.[26] A different receptor, **FhuA**, binds ferrichrome. Both FepA and FhaA are large 22-strand porins resembling the 16-strand porin shown in Fig. 8-20. However, they are nearly 7 nm long with internal diameters three times those of the 16-strand porins. In addition, loops of polypeptide chain on the outer edges can close while an N-terminal domain forms a "cork" that remains in place until the Fe^{3+}-siderophore complex enter the channel and binds. Like the hatches in an air lock on a spacecraft, the outer loops then close, after which the inner cork unwinds to allow the siderophore complex to enter the periplasmic space. The apparatus requires an energy supply, which apparently is provided by an additional complex consisting of proteins TonB, ExbA, and ExbD. They evidently couple the electrochemical gradient across the cell membrane (Chapter 8, B1 and C5 and Chapter 18) with the operation of the channel gates. Some bacteria use a different strategy for passage across the outer membrane.[11c] The channel of a receptor protein contains a molecule of an iron-free siderophore. When a molecule of Fe^{3+}-siderophore binds in the outside part of the channel the Fe^{3+} jumps to the inner siderophore, which then dissociates from the receptor, carrying the Fe^{3+}-siderophore complex into the periplasmic space. This mechanism also seems to be available in *E. coli*. The **ferric uptake regulation** (Fur) protein binds excess free Fe^{2+}, the resulting complex acting as a repressor of all of the iron uptake genes in *E. coli*.[11e]

Additional proteins are required for passage through the membrane.[11b,23,25,27] These are ABC transporters, which utilize hydrolysis of ATP as an energy source (Chapter 8, Section C,4). For uptake of the Fe^{3+}-ferrichrome complex protein **FhuD** is the periplasmic binding protein, **FhuC** is an integral membrane component, and the cytosolic **FhuC** contains the ATPase center.[11b] Another ABC transporter, found in many bacteria, carries unchelated Fe^{3+} across the inner membrane. The binding protein component for *Hemophilus influenzae*, designated Hit, resembles one lobe of mammalian transferrin (Fig. 16-2).[11b] The siderophore receptors of bacteria have been "parasitized" by various bacteriophages and toxic proteins. For example, FepA is also a receptor for the toxic colicins B and D (Box 8-D) and tonB is a receptor for bacteriophage T1.[9,13]

Some bacteria do not form siderophores but take up Fe^{2+}. Even *E. coli*, when grown anaerobically, synthesizes an uptake system for Fe^{2+}. It utilizes a 75-residue peptide encoded by gene *feoA* and a 773-residue protein encoded by *feoB*.[28,11b]

Uptake of iron by eukaryotic cells. The yeast *Saccharomyces cerevisiae* utilizes two systems for uptake of iron.[29,30] A low-affinity system transports Fe^{2+} with an apparent K_m of ~ 30 µM, while a high-affinity system has a K_m of ~ 0.15 µM. Study of these systems has been greatly assisted by the use of genetic methods developed for both bacteria and yeast (Chapter 26). The low-affinity iron uptake depends upon a protein transporter encoded by gene *FET*4 and on a reductase encoded by genes *FRE*1 and *FRE*2, proteins that are embedded in the cytoplasmic membrane.[29,31-33] It might seem reasonable that the *FET*3 copper oxidoreductase should keep Fe^{2+} *reduced* while it is transported. However, it appears to *oxidize* Fe^{2+} to Fe^{3+}. The high-affinity uptake system is more puzzling. It requires a permease encoded by *FTR I* and an additional protein encoded by *FET*3.[30,33–35] The Fet3 protein is a copper oxidoreductase related to **ceruloplasmin** (Section D). The protein **Fre1p** (encoded by *FRE*1) is a metalloreductase that reduces Cu^{2+} to Cu^{+}, as well as Fe^{3+} to Fe^{2+}. It is essential for copper uptake (Section D).[33] It has long been known that ceruloplasmin is required for mobilization of iron from mammalian tissues.[30] Hereditary ceruloplasmin deficiency causes accumulation of iron in tissues.[36] Yeast also contains both *mitochondrial* and *vacuolar* iron transporters.[37,37a,b]

The uptake of iron by animals is not as well understood[38–40] but it resembles that of yeast.[41] A general divalent cation transporter that is coupled to the membrane proton gradient is involved in intestinal iron uptake.[42,60] Ascorbic acid promotes the uptake of iron, presumably by reducing it to Fe(II), which is more readily absorbed than Fe(III), and also by promoting ferritin synthesis.[43] Uptake is also promoted by meat in the diet.[44] Within the body iron is probably transferred from one protein to another with only a transient existence as free Fe^{2+}. An average daily human diet contains ~ 15 mg of iron, of which ~ 1 mg is absorbed. This is usually enough to compensate for the small losses of the metal from the body, principally through the bile. Once it enters the body, iron is carefully conserved. The 9 billion red blood cells destroyed daily yield 20 – 25 mg of iron which is almost all reused or stored. The body apparently has no mechanism for excretion of large amounts of iron; a person's iron content is regulated almost entirely by the rate of uptake. This rate is increased during pregnancy and, in young women, to compensate for iron lost in menstrual bleeding. Nevertheless, control of iron uptake is imperfect and perhaps 500 million people around the world suffer from iron deficiency.[44,45] For others, an excessive intake of iron or a genetic defect lead to accumulation of iron to toxic levels, a condition called **hemochromatosis**.[44,46,46a,b] This condition may also arise in any disease that leads to excessive destruction of hemoglobin or accumulation of damaged erythrocytes. Examples are β-thalassemia (Chapter 28) and cerebral malaria.[47] Treatment with chelating agents designed to remove iron is often employed.[47,48]

Transferrins. Within the body iron is moved from one location to another while bound as Fe^{3+} to transferrins, a family of related 680- to 700-residue 80-kDa proteins.[38,49–53] Each transferrin molecule contains two Fe^{+3} binding sites, one located in each of two similar domains of the folded peptide chain. A dianion, usually CO_3^{2-}, is bound together with each Fe^{3+}. Milk transferrin (**lactoferrin**[51a,b,c] also found in leukocytes), hen egg transferrin (**ovotransferrin**),[52,52a] and rabbit and human serum transferrin[54,54a] all have similar structures. Each Fe^{3+} is bonded to oxygen anions from two tyrosine side chains, an aspartate carboxylate, an imidazole group, and the bound carbonate ion (Fig. 16-2B). Transferrin of blood plasma is encoded by a separate gene but has a similar structure. Transferrin of chickens appears to be identical to conalbumin of egg whites. The iron-binding proteins of body fluids are sometimes given the group name **siderophilins**. Transferrins may function not only in transport of iron throughout the body but also as iron buffers that provide a relatively constant iron concentration within tissues.

The entrance of iron into the body through the intestinal mucosal cells may involve the transferrin present in those cells[44] and the influx of iron may also be regulated by blood plasma transferrin. There is also a nontransferrin pathway.[42,55]

Transferrins bind Fe^{2+} weakly and it is likely that a transferrin– Fe^{2+}–HCO_3^- complex formed initially undergoes oxidation to the Fe^{3+}–CO_3^{2-} complex within cells and within the bloodstream. A conformational change closes the protein around the iron ions.[56] In yeast the previously mentioned copper oxidoreductase encoded by the *FET*3 gene appears to not only oxidize Fe^{2+} but also transfer the resulting Fe^{3+} to transferrin. Ceruloplasmin may play a similar role in mammals.[33]

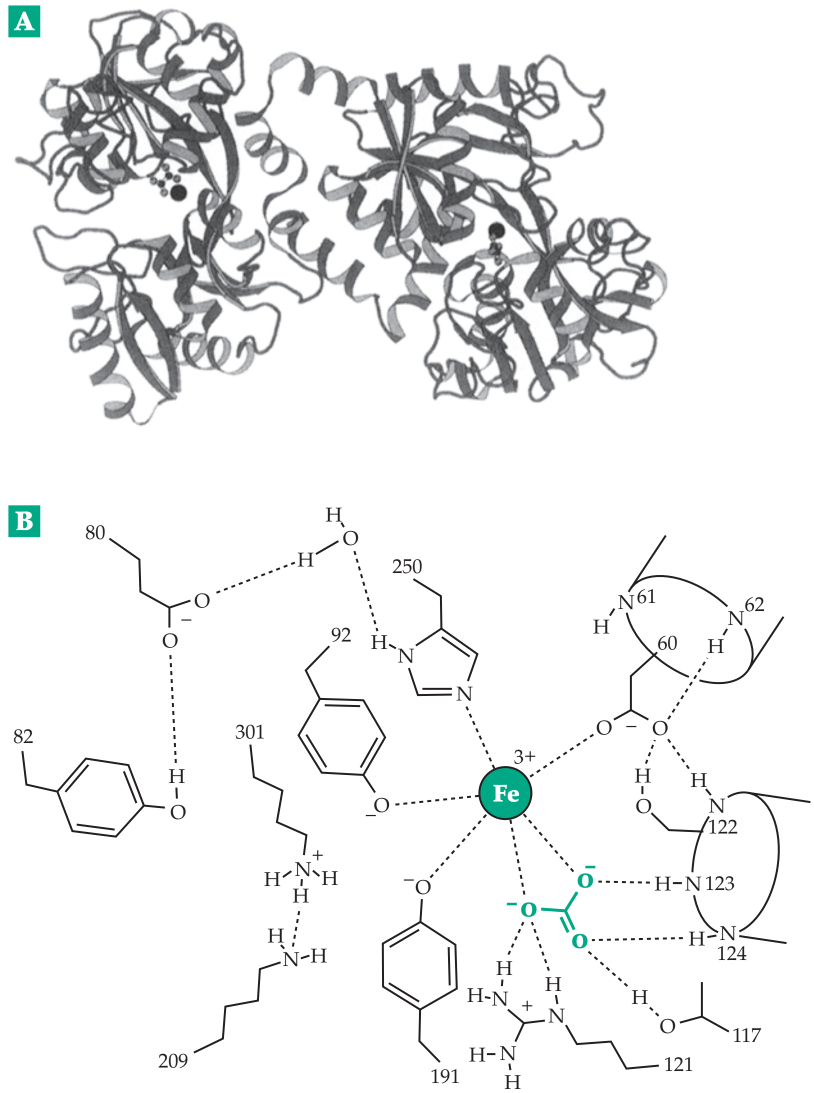


Figure 16-2 (A) Ribbon drawing of the polypeptide chain of a transferrin, human lactoferrin. The N lobe is to the left and the C lobe to the right. Each active site contains bound Fe^{3+} and a molecule of oxalate dianion which replaces the physiological CO_3^{2-}. From Baker *et al.*[51] Courtesy of Edward Baker. (B) Schematic diagram showing part of the hydrogen-bond network involved in binding the Fe^{3+} in the N lobe of hen ovotransferrin. Some side chain groups and water molecules have been omitted. The positions of hydrogen atoms and the charge state of acid–base groups are uncertain. Most of the hydrogen-bond distances (O - - O, O - - N, N - - N) indicated by dashed lines are between 0.27 and 0.3 nm. Release of the bound Fe^{3+} may be accomplished in part by protonation of the bound CO_3^{2-} to form HCO_3^-. See Kurokawa *et al.*[52]

Iron is transferred from the plasma transferrin into cells of the body following binding of the Fe^{3+}–transferrin complex to specific receptors. The surface of an immature red blood cell (reticulocyte) may contain 300,000 transferrin receptors, each capable of catalyzing the entry of ~36 iron ions per hour.[38] The receptor is a 180-kDa dimeric glycoprotein. When the Fe^{3+}–transferrin complex is bound, the receptors aggregate in coated pits and are internalized. The mechanism of release of the Fe^{3+} may occur by different mechanisms in the two lobes.[57] The pH of the endocytic vesicles containing the receptor complex is probably lowered to ~5.6 as in lysosomes. This may protonate the bound CO_3^{2-} in the complex[51c,54a,58] and assists in the release of the Fe^{3+}, possibly after reduction to Fe^{2+}. Both the apotransferrin and its receptor are returned to the cell surface for reuse, the apotransferrin being released into the blood. Chelating agents such as pyrophosphate, ATP, and citrate as well as simple anions[59] may also assist in removal of iron from transferrin. The same transmembrane transporter that is involved in intestinal iron uptake[42] is needed to remove iron from the endosome after release.[60]

2. Storage of Iron

Within tissues of animals, plants, and fungi much of the iron is packaged into the red-brown water-soluble protein **ferritin**, which stores Fe(III) in a soluble, nontoxic, and readily available form.[61–64] Although bacteria store very little iron,[65] some of them also contain a type of ferritin.[66–67a] On the other hand, the yeast *S. cerevisieae* stores iron in polyphosphate-rich granules, even though a ferritin is also present.[65] Ferritin contains 17–23% iron as a dense core of hydrated ferric oxide ~7 nm in diameter surrounded by a protein coat made up of twenty-four subunits of mo-

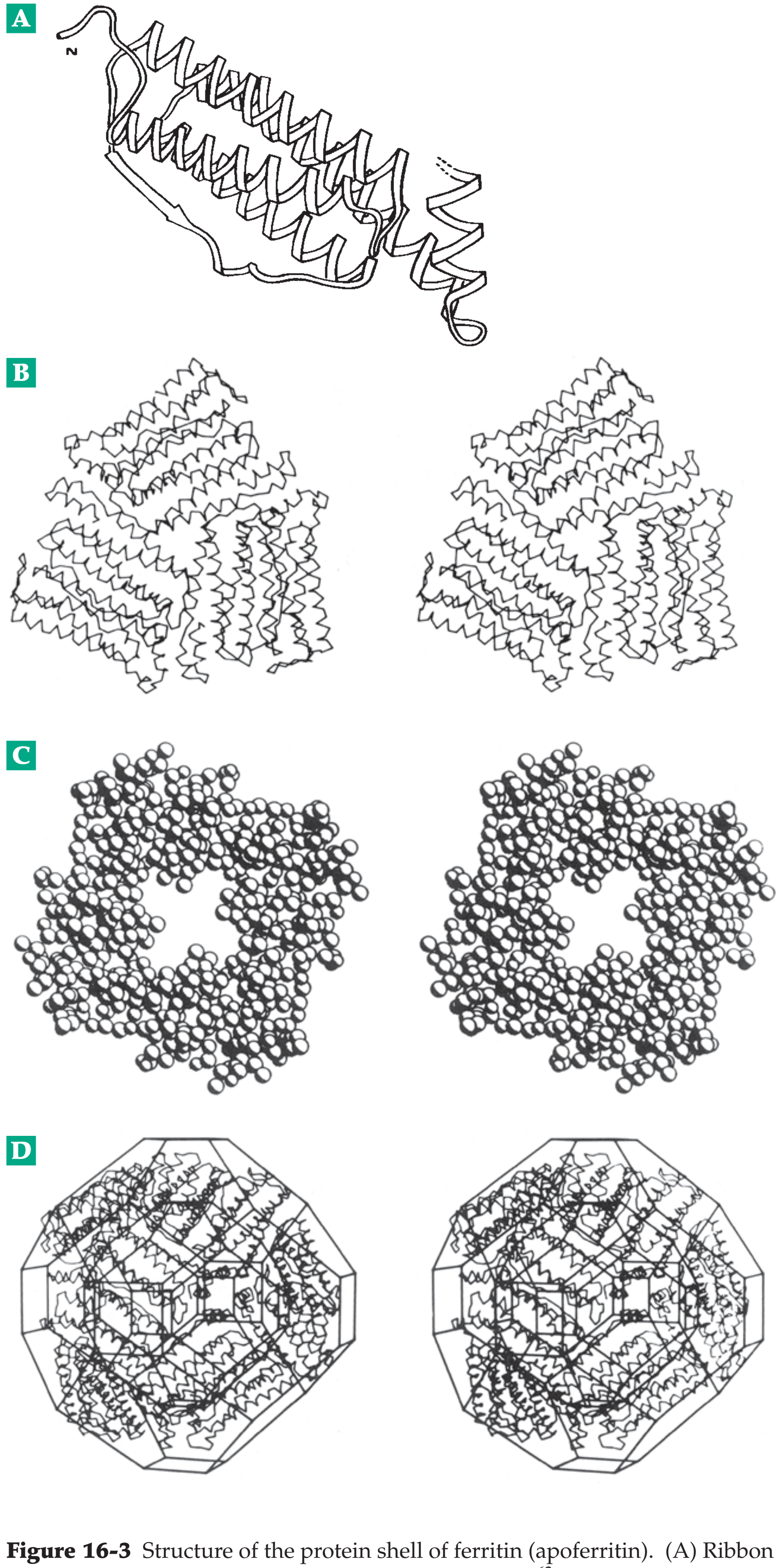


Figure 16-3 Structure of the protein shell of ferritin (apoferritin). (A) Ribbon drawing of the 163-residue monomer. From Crichton.[62] (B) Stereo drawing of a hexamer composed of three dimers. (C) A tetrad of four subunits drawn as a space-filling diagram and viewed down the four-fold axis from the exterior of the molecule. (D) A half molecule composed of 12 subunits inscribed within a truncated rhombic dodecahedron. B–D from Bourne *et al*.[74]

lecular mass 17- to 21-kDa. Each subunit is folded as a four-helix bundle (Fig. 16-3). Mammalian ferritins consist of combinations of subunits of two or more types. For example, human ferritins contain similarly folded 19-kDa L (light) and 21-kDa H (heavy) subunits.[68] The twenty-four subunits are arranged in a cubic array (Fig. 7-13, Fig. 16-3). The outer diameter of the 444-kDa apoferritin is ~12 nm. The completely filled ferritin molecule contains 23% Fe and over 2000 atoms of iron in a crystalline lattice. Larger ferritins may contain as many as 4500 atoms of iron with the approximate composition $[Fe(O)OH]_8FeOPO_3H_2$.[69] Phosphate ions are sometimes bound into surface layers of the ferritin cores.[69a] Ferritin cores are readily visible in the electron microscope, and ferritin is often used as a labeling reagent in microscopy. Another storage form of iron, **hemosiderin**, seems to consist of ferritin partially degraded by lysosomes and containing a higher iron content than does ferritin. Depositions of hemosiderin in the liver can rise to toxic levels if excessive amounts of iron are absorbed.

Iron can be deposited in ferritin by allowing apoferritin to stand with an Fe(II) salt and a suitable oxidant, which may be O_2. Physiological transfer of Fe(III) from transferrin to ferritin is thought to require prior reduction to Fe(II). The reoxidation by O_2 to Fe(III) for deposition in the ferritin core (Eq. 16-2) is catalyzed by **ferrooxidase sites** located in the centers of the helical bundles of the H-chains.[70–73]

$$2\,Fe^{2+} + O_2 + 4\,H_2O \rightarrow 2\,Fe(O)OH_{core} + H_2O_2 + 4\,H^+ \qquad (16\text{-}2)$$

BOX 16-A MAGNETIC IRON OXIDE IN ORGANISMS

An unusual form of stored iron is the magnetic iron oxide **magnetite** (Fe_3O_4). Honeybees,[a,b] monarch butterflies,[b–c] homing pigeons,[c–e] migrating birds, and even magnetotactic bacteria[f] contain deposits of Fe_3O_4 that are suspected of being used in nagivation.[c,g] Some bacteria have magnetic iron sulfide particles.[h-j] Human beings have magnetic bones in their sinuses[k] and in their brains[l] and may be able to sense direction magnetically. A set of possible magnetoreceptor cells, as well as associated nerve pathways, have been identified in trout.[j] In the magnetotactic bacteria found in the Northern Hemisphere the magnetic domains are oriented parallel with the axis of motility of the bacteria which tend to swim toward the geomagnetic North and downward into sediments. Similar bacteria from the Southern Hemisphere prefer to swim south and downward. The magnetic polarity of the bacterial magnetite crystals can be reversed by strong magnetic pulses, after which the bacteria swim in the direction opposite to their natural one.[m] Magnetic ferritin can be produced artificially in the laboratory.[n] The resulting particles may have practical uses, for example, in medical magnetic imaging. Magnetic materials in the human body are of interest not only in terms of a possible sensory function but also because of possible effects of electromagnetic fields on human health and behavior.[o]

Magnetotactic soil bacterium containing 36 magnetite-containing magnetosomes. Courtesy of Dennis Bazylinski.

a Hsu, C.-Y., and Li, C.-W. (1994) *Science* **265**, 95–97

b Nichol, H., and Locke, M. (1995) *Science* **269**, 1888–1889

bc Etheredge, J. A., Perez, S. M., Taylor, O. R., and Jander, R. (1999) *Proc. Natl. Acad. Sci. U.S.A.* **96**, 13845–13846

c Gould, J. L. (1982) *Nature (London)* **296**, 205–211

d Guilford, T. (1993) *Nature (London)* **363**, 112–113

e Moore, B. R. (1980) *Nature (London)* **285**, 69–70

f Blakemore, R. P., and Frankel, R. B. (1981) *Sci. Am.* **245** (Dec), 58–65

g Maugh, T. H., II,(1982) *Science* **215**, 1492 – 1493

h Dunin-Borkowski, R. E., McCartney, M. R., Frankel, R. B., Bazylinski, D. A., Pósfai, M., and Buseck, P. R. (1998) *Science* **282**, 1868–1870

i Pósfai, M., Buseck, P. R., Bazylinski, D. A., and Frankel, R. B. (1998) *Science* **280**, 880–883

j Walker, M. M., Diebel, C. E., Haugh, C. V., Pankhurst, P. M., Montgomery, J. C., and Green, C. R. (1997) *Nature (London)* **390**, 371–376

k Baker, R. R., Mather, J. G., and Kennaugh, J. H. (1983) *Nature (London)* **301**, 78–80

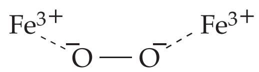
l Kirschvink, J. L., Kobayashi-Kirschvink, A., and Woodford, B. J. (1992) *Proc. Natl. Acad. Sci. U.S.A.* **89**, 7683–7687

m Blakemore, R. P., Frankel, R. B., and Kalmijn, A. J. (1980) *Nature (London)* **286**, 384–385

n Meldrum, F. C., Heywood, B. R., and Mann, S. (1992) *Science* **257**, 522–523

o Barinaga, M. (1992) *Science* **256**, 967

The ferrooxidase site is a dinuclear iron center (see Section 8) in which two iron ions (probably Fe^{2+}) are bound as in Fig. 16-4. They are then converted to Fe^{3+} ions by O_2, which may bind initially to the Fe^{2+}, forming a transient blue intermediate that is thought to have a peroxodiferric structure, perhaps of the following type.[71–73] Reaction of this intermediate with H_2O



may yield H_2O_2 plus a biomineral precursor Fe^{2+} – O – Fe^{2+}, which is incorporated into the core.[72a] Ferritin H subunits predominate in tissues with high oxygen levels, e.g., heart and blood cells, while the L subunits predominate in tissues with slower turnover of iron, e.g., liver.[72] The L subunits lack ferrooxidase activity but, in the centers of their helical bundles, contain polar side chains that may help to initiate growth of the mineral core.[64]

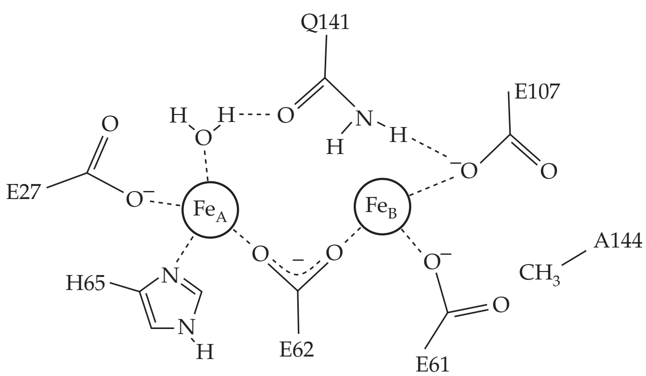


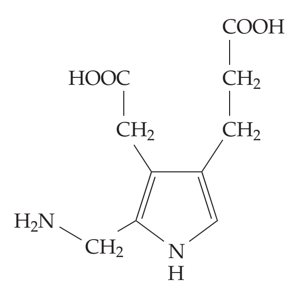
Figure 16-4 The dinuclear iron center or ferroxidase center of human ferritin based on the structure of a terbium(III) derivative.[73] Courtesy of Pauline Harrison.

Removal of Fe(III) from storage in ferritin cores may require reduction to Fe(II) again, possibly by ascorbic acid[75] or glutathione. Some bacterial ferritins contain a bound cytochrome *b* which may assist in reduction.[67,67a] Released iron in the Fe^{2+} state can be incorporated into iron-containing proteins or into heme. The enzyme **ferrochelatase**[76–76b] catalyzes the transfer of free Fe^{2+} into protoporphyrin IX (Section 3) to form protoheme (Fig. 16-5). Iron in the Fe(II) state may also be oxidized to Fe^{3+} through action of the copper-containing ceruloplasmin (Section D) and be incorporated into heme by direct transfer from ferritin.[53]

3. Heme Proteins

In 1879, German physiological chemist Hoppe-Seyler showed that two of the most striking pigments of nature are related. The red iron-containing **heme** from blood and the green magnesium complex **chlorophyll a** of leaves have similar ring structures. Later, H. Fischer proved their structures and provided them with the names and numbering systems that are used today. This information is summarized in the following section.

Some names to remember. **Porphins** are planar molecules which contain large rings made by joining four pyrrole rings with methine bridges. In the **chlorins**, found in the chlorophylls, one of the rings (ring D in chlorophyll, Fig. 23-20) is reduced. The specific class of porphins known as **porphyrins** have eight substituents around the periphery of the large ring. Like the chlorins and the **corrins** of vitamin B_{12} (Section B), the porphyrins are all formed biosynthetically from **porphobilinogen**. This compound is polymerized in two ways (see Fig. 24-21) to give porphyrins of types I and III (Fig. 16-5). In type I porphyrins, polymerization of porphobilinogen has taken place in a regular way so that the sequences of the carboxymethyl and carboxyethyl side chains (often referred to as acetic acid and propionic acid side chains, respectively) are the same all the way around the outside of the molecule. However, most biologically important porphyrins belong to type III, in which the first three rings A, B, and C have the same sequence of carboxymethyl and carboxyethyl side chains, but in which ring D has been incorporated in a reverse fashion. Thus, the carboxyethyl side chains of rings C and D are adjacent to each other (Fig. 16-5). Porphyrins containing all four carboxymethyl and four carboxyethyl side chains intact are known as **uroporphyrins**.

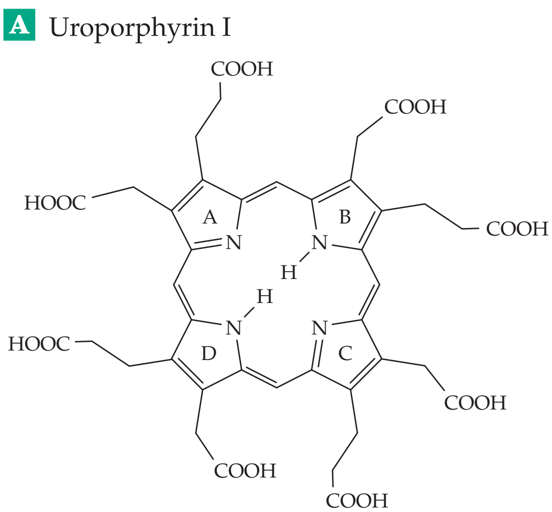


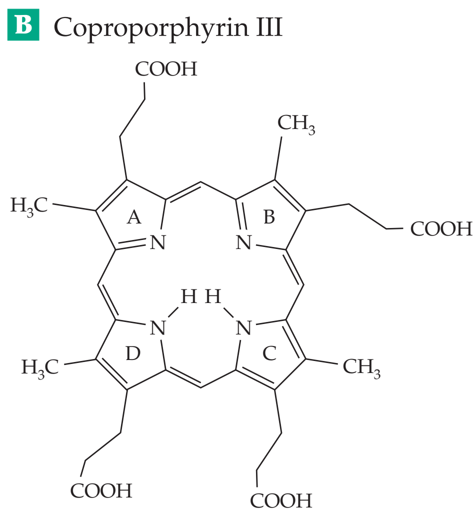
Porphobilinogen

Uroporphyrins I and III are both excreted in small amounts in the urine. Another excretion product is **coproporphyrin** III, in which all of the carboxymethyl side chains have been decarboxylated to methyl groups. The feathers of the tropical touraco are colored with copper(II) complex of coproporphyrin III and this porphyrin as well as others are commonly found in birds' eggs. The heme proteins are all derived from protoporphyrin IX, which is formed by decarboxylation and dehydrogenation of two of the carboxyethyl side chains of uroporphyrin III to vinyl groups (Fig. 16-5).

Hemes and heme proteins. Protoporphyrin IX contains a completely conjugated system of double bonds. In the center two hydrogen atoms are attached, one each to two of the nitrogens; they are free to move to other nitrogens in the center with rearrangement of the double bonds. Thus, there is tautomerism as well as resonance within the heme ring.[77–78a] The two central hydrogens can be replaced by many metal ions to form stable chelates. The complexes with Fe^{2+} are known as **hemes** and the Fe^{2+} complex with protoporphyrin IX as **protoheme**. Heme complexes of Fe^{2+} may be designated as **ferrohemes** and the Fe^{3+} compounds as **ferrihemes**. The Fe^{3+} protoporphyrin IX is also called **hemin**, and may be crystallized as a chloride salt.[78b] Iron tends to have a coordination number of six, and other ligands can attach to the iron from the two axial positions on opposite sides of the planar heme. If these are nitrogen ligands, such as pyridine or imidazole, the resulting compounds, called **hemochromes**, have characteristic absorption spectra. An example is cytochrome b_5, which contains two axial imidazole groups.

Several modifications of protoheme are indicated in Fig. 16-5. To determine which type of heme exists in a particular protein, it is customary to split off the heme by treatment with acetone and hydrochloric acid and to convert it by addition of pyridine to the pyridine hemochrome for spectral analysis. By this means, protoheme was shown to occur in hemoglobin, myoglobin, cytochromes of the *b* and P450 types, and catalases and many peroxidases. Cytochromes *a* and a_3 contain ***heme a***, while one of the terminal oxidase





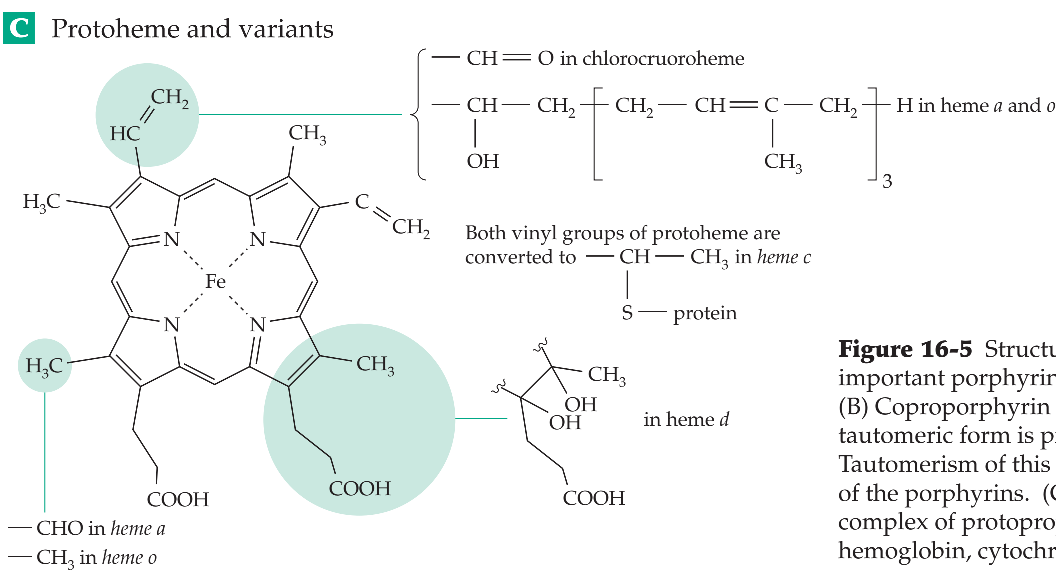


Figure 16-5 Structures of some biologically important porphyrins. (A) Uroporphyrin I. (B) Coproporphyrin III. Note that a different tautomeric form is pictured in B than in A. Tautomerism of this kind occurs within all of the porphyrins. (C) Protoheme, the Fe^{2+} complex of protoprophyrin IX, present in hemoglobin, cytochromes *b*, and other proteins.

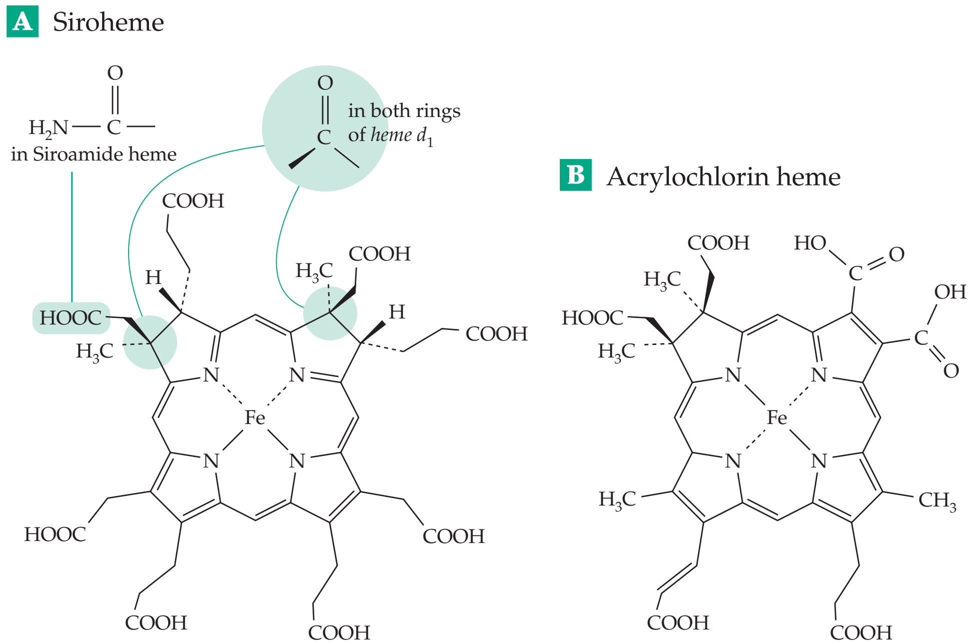


Figure 16-6 Structures of isobacteriochlorin prosthetic groups. (A) Siroheme from nitrite and sulfite reductases; (B) acrylochlorin heme from dissimilatory nitrite reductases of *Pseudomonas* and *Paracoccus*.

systems of enteric bacteria contains the closely related heme *o* (Fig. 16-5).[79,80] A second terminal oxidase of those same bacteria contains **heme *d*** (formerly a_2). **Heme *c*** (present in cytochromes *c* and *f*) is a variation in which two SH groups of the protein have added to the vinyl groups of protoheme to form two thioether linkages (Fig. 16-5). A few cytochromes *c* have only one such linkage. In myeloperoxidase (Section 6) three covalent linkages, different than those in cytochrome *c*, join the heme to the protein.[81,82] There is a possibility that heme *a* may sometimes form a Schiff base with a lysyl amino group through its formyl group.[83,84]

Heme *d* is a chlorin,[85] as is **acrylochlorin heme** from certain bacterial nitrite reductases (Fig. 16-6).[86,87] **Siroheme** (Fig. 16-6), which is found in both nitrite and sulfite reductases of bacteria (Chapter 24),[88,89] is an isobacteriochlorin in which both the A and B rings are reduced. It apparently occurs as an amide **siroamide** (Fig. 16-6) in *Desulfovibrio*.[90] Heme d_1 of nitrite reductases of denitrifying bacteria is a dioxo-bacteriochlorin derivative (Fig. 16-6).[91,92]

As in myoglobin, hemoglobin (Fig. 7-23), and cytochrome *c* (see Fig 16-8), one axial coordination position on the iron of most heme proteins (customarily called the *proximal* position) is occupied by an imidazole group of a histidine side chain. However, in cytochrome P450 and chloroperoxidase a thiolate ($-S^-$) group from a cysteinyl side chain, and in catalase a phenolate anion from a tyrosyl side chain, occupies the proximal position. The sixth or *distal* coordination position is occupied by the sulfur atom of methionine in cytochrome *c* and most other cytochromes with low-spin iron but cytochromes b_5 and c_3 have histidine. The high-spin heme proteins, such as cytochromes *c*′, globins, peroxidase, and catalase, usually have no ligand other than weakly bound H_2O in the distal position.[93]

Hemes are found in all organisms except the anaerobic clostridia and lactic acid bacteria. Heme proteins of blood carry oxygen reversibly, whereas those of **terminal oxidase systems**, **hydroxylases**, and **oxygenases** "activate" oxygen, catalyzing reactions with hydrogen ions and electrons or with carbon compounds. The heme-containing **peroxidases** and **catalases** catalyze reactions not with O_2 but with H_2O_2. Another group of heme proteins includes most of the cytochromes, which are purely electron-transferring compounds.

4. The Cytochromes

The iron in the small proteins known as cytochromes acts as an electron carrier, undergoing alternate reduction to the +2 state and oxidation to the +3 state. The cytochromes, discovered in 1884 by McMunn,[94] were first studied systematically in the 1920s by Keilin (Chapter 18) and have been isolated from many sources.[95–97] The classification into groups *a*, *b*, and *c* according to the position of the longest wavelength light absorption band (the α band; Fig. 16-7) follows a practice introduced by Keilin. However, it is now customary to designate a new cytochrome by giving the heme type (*a*, *b*, *c* or *d*) together with the wavelength of the α band, e.g., cytochrome c_{552} or cyt $b_{557.5}$.

Cytochromes of the *b* type including bacterial cytochrome *o* contain protoheme. Because the sixth position is ligated, most cytochromes *b* do not react with O_2. However, cytochromes *o* and *d* serve as terminal electron acceptors (cytochrome oxidases) and are oxidizable by O_2. Another protoheme-containing cytochrome, involved in hydroxylation (Chapter 18), is **cytochrome P450**. Here the 450 refers to the position of the intense "Soret band" (also called the γ band) of the spectrum (Fig. 16-7) in a difference spectrum run in the presence and absence of CO. Other properties are also used in arriving at designations for cytochromes. For example, cytochrome a_3 has a spectrum similar to that of cytochrome *a* but it reacts readily with both CO and O_2.

Another property that distinguishes various cytochromes is the redox potential $E°'$ (Table 6-8), which in this discussion is given for pH 7.0. Cytochromes carry electrons between other oxidoreductase proteins of widely varying values of $E^{0'}$. Because of the various heme environments cytochromes have greatly differing values of $E^{0'}$, allowing them to function in many different biochemical systems.[97a,97b] For mitochondrial cytochrome *c* the value of $E°'$ is ~ + 0.265 V but for the closely related cytochrome *f* of chloroplasts it is ~ +0.365 V and for cytochrome c_3 of *Desulfovibrio* about −0.330 V. There is more than an 0.6-volt difference between $E°'$ of cytochromes *f* and c_3. Cytochromes *b* tend to have lower $E°'$ values, close to zero, than most cytochromes *c*, while cytochrome a_3 has $E°'$ ~ +0.385 V.

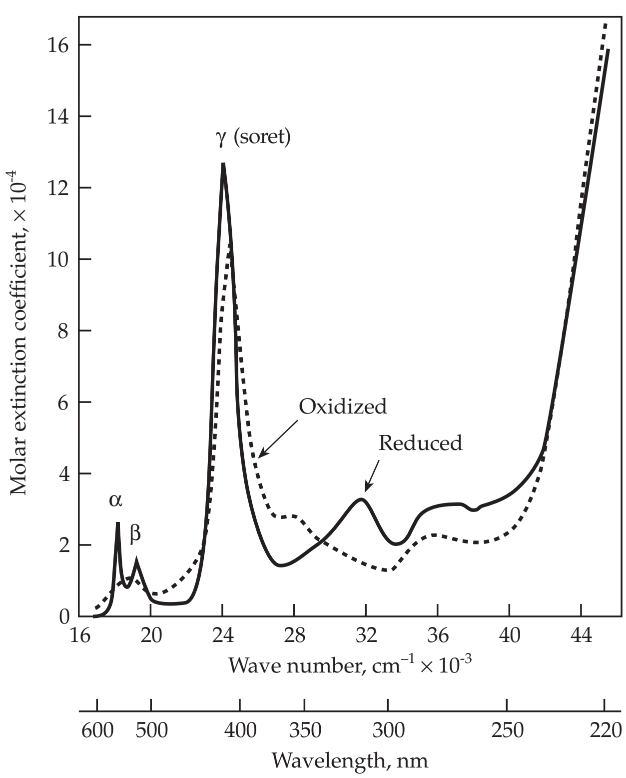


Figure 16-7 Absorption spectra of oxidized and reduced horse heart cytochrome *c* at pH 6.8. From data of Margoliash and Frohwirt.[98]

The c-type cytochromes. Mitochondrial cytochrome *c* is one of the few intracellular heme pigments that is soluble in water and that can be removed easily from membranes. A small 13-kDa protein typically containing about 104 amino acid residues, cytochrome *c* has been isolated from plants, animals, and eukaryotic microorganisms.[95–97,99,100] Complete amino acid sequences have been determined for over 100 species. Within the peptide chain 28 positions are invariant and a number of other positions contain only conservative substitutions. Cytochrome *c* was one of the first proteins to be used in attempting to trace evolutionary relationships between species by observing differences in sequence. Humans and chimpanzees have identical cytochrome *c*, but 12 differences in amino acid sequence occur between humans and the horses and 44 between human and *Neurospora*.[96] The related cytochrome c_2 of the photosynthetic bacterium *Rhodospillum rubrum* is thought to have diverged in evolution 2 x 10^9 years ago from the precursor of mammalian cytochrome *c*. Even so, 15 residues remain invariant.[101]

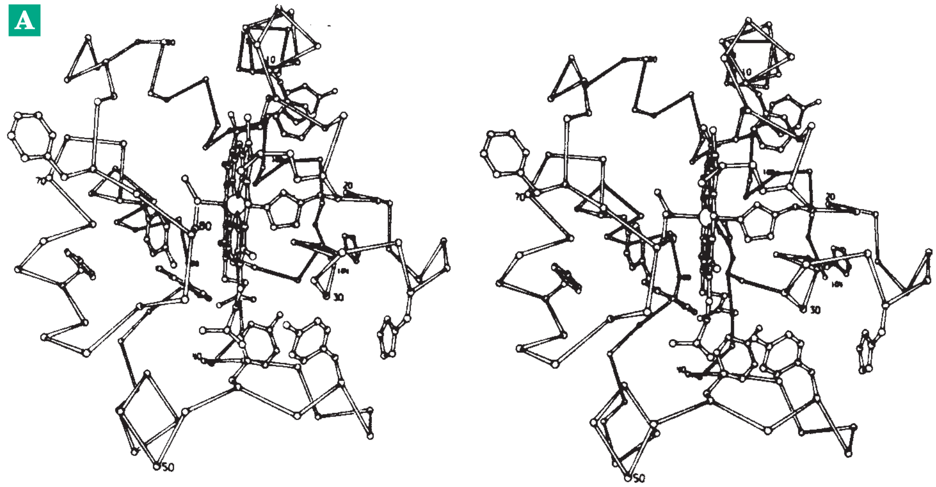
Structural studies[95–97,101–103] on cytochromes of the *c* and c_2 types show that the heme group provides a core around which the peptide chain is wound. The 104 residues of mitochondrial cytochrome *c* are enough to do little more than envelope the heme. In both the oxidized and reduced forms of the protein, methionine 80 (to the left in Fig. 16-8A) and histidine 18 (to the right) fill the axial coordination positions of the iron. The heme is nearly "buried" and inaccessible to the surrounding solvent.

The shorter chains of the 82- to 86-residue cytochromes c_{550} (from *Pseudomonas*[102]), c_{553},[102a] and c_{555} (from *Chlorobium*[105]) as well as the longer 112-residue polypeptide of cytochrome c_2 from *Rhodospirillum rubrum*[93] have nearly the same folding pattern as that in mitochondrial cytochrome *c*. However, the 128-residue chain of the dimeric cytochrome *c*′ from *Rhodospirillum molischianum* forms an antiparallel four-helix bundle (Fig. 16-8).[106–109] This is the same folding pattern present in the ferritin monomer (Fig. 16-3), hemerythrin (Fig. 2-22), and many other proteins including cytochrome b_{562} of *E. coli*.[110] Cytochrome *f*, which functions in photosynthetic electron transport, is also a *c*-type cytochrome but with a unique protein fold.[111,112]

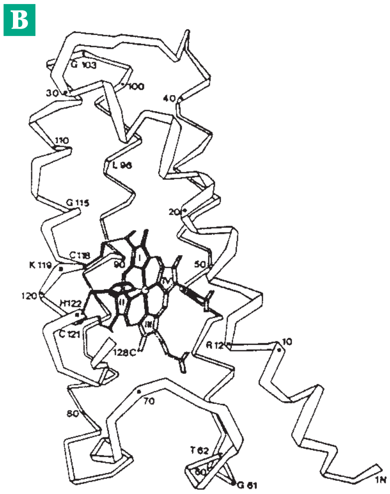
Most cytochromes have only one heme group per polypeptide chain,[112] but the 115-residue cytochrome c_3 from the sulfate-reducing bacterium *Desulfovibrio* binds four hemes (Fig. 16-8C).[104,113 – 115] Each one seems to have a different redox potential in the −0.20 to −0.38 V range.[114] Another *c*-type cytochrome, also from *Desulfovibrio*, contains six hemes in a much larger 66-kDa protein and functions as a nitrite reductase.[116]

Figure 16-8 Structures of three cytochromes of the *c* type. (A) Horse mitochondrial cytochrome *c*; (B) a subunit of the dimeric cytochrome *c'* from *Rhodospirillum molischianum*; (C) cytochrome c_3 from *Desulfovibrio desulfuricans*. (A) and (B) courtesy of Salemme;[101] (C) from Simões *et al.*[104] Courtesy of Maria A. Carrondo.

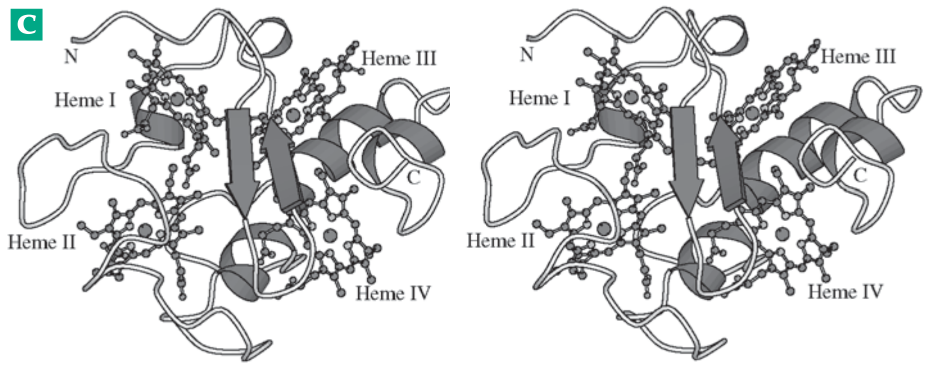
A



B



C



Triheme and octaheme proteins are also known.[117]

Many cytochromes *c* are soluble but others are bound to membranes or to other proteins. A well-studied tetraheme protein binds to the reaction centers of many purple and green bacteria and transfers electrons to those photosynthetic centers.[118–120] Cytochrome c_2 plays a similar role in *Rhodobacter*, forming a complex of known three-dimensional structure.[121] Additional cytochromes participate in both cyclic and noncyclic electron transport in photosynthetic bacteria and algae (see Chapter 23).[120,122–124] Some bacterial membranes as well as those of mitochondria contain a **cytochrome *bc*$_1$ complex** whose structure is shown in Fig. 18-8.[125,126]

Cytochromes b, a, and o. Protoheme-containing cytochromes *b* are widely distributed.[127,128] There are at least five of them in *E. coli*. Whether in bacteria, mitochondria, or chloroplasts, the cytochromes *b* function within electron transport chains, often gathering electrons from dehydrogenases and passing them on to *c*-type cytochromes or to iron–sulfur proteins. Most cytochromes *b* are bound to or embedded within membranes of bacteria, mitochondria, chloroplasts, or endoplasmic reticulum (ER). For example, cytochrome b_5[129,129a] delivers electrons to a fatty acid desaturating system located in the ER of liver cells and to many other reductive biosynthetic enzymes.[130–132] The protein contains 132 amino acid residues plus another 85 largely hydrophobic N-terminal residues that provide a nonpolar tail which is thought to be buried in the ER membranes.[130] Solubilization of the protein causes loss of this N-terminal sequence. The heme in cytochrome b_5 is not covalently bonded to the protein but is held tightly between two histidine side chains. The polypeptide chain is folded differently than in either cytochrome *c* or myoglobin.

The folding pattern of cytochrome b_5 is also found in the complex heme protein **flavocytochrome b$_2$** from yeast (Chapter 15)[133] and probably also in liver **sulfite oxidase**[134,135] Both are 58-kDa peptides which can be cleaved by trypsin to 11-kDa fragments that have spectroscopic similarities and sequence homology with cytochrome b_5. Sulfite oxidase also has a molybdenum center (Section H). The 100-residue N-terminal portion of flavocytochrome b_2 has the cytochrome b_5 folding pattern but the next 386 residues form an eight-stranded $(\alpha / \beta)_8$ barrel that binds a molecule of FMN.[133,136] All of these proteins pass electrons to cytochrome *c*. In contrast, the folding of **cytochrome**

b_{562} of *E. coli* resembles that of cytochrome *c*′ (Fig. 16-8).[110,137] However, it has methionine side chains as both the fifth and sixth iron ligands.

Cytochromes *b* of mitochondrial membranes are involved in passing electrons from succinate to ubiquinone in complex II[138] and also from reduced ubiquinone to cytochrome c_1 in the 248-kDa complex III (Fig. 18-8). A similar complex is present in photosynthetic purple bacteria.[123,139] Cytochrome b_{560} functions in the transport of electrons from succinate dehydrogenase to ubiquinone,[138] and cytochrome b_{561} of secretory vesicle membranes has a specific role in reducing ascorbic acid radicals.[140]

In bacteria some cytochromes *b* and d_1 serve as terminal electron carriers able to react with O_2, nitrite, or nitrate, while others act as carriers between redox systems.[141–143a] The aldehyde heme *a* is utilized by animals and by some bacteria in **cytochrome *c* oxidase**, a complex enzyme whose three-dimensional structure is known (see Fig. 18-10) and which is discussed further in Chapter 18.

5. Mechanisms of Biological Electron Transfer

The heme groups of the cytochromes as well as many other transition metal centers act as carriers of electrons. For example, cytochrome *c* may accept an electron from reduced cytochrome c_1 and pass it to cytochrome oxidase or cytochrome *c* peroxidase. The electron moves from one heme group to another over distances as great as 2 nm. Similar electron-transfer reactions between defined redox sites are met in photosynthetic reaction centers (Fig. 23-31), in metalloflavoproteins (Fig. 15-9), and in mitochondrial membranes.

What are the factors that determine the probability of an electron transfer reaction and the rate at which it may occur? They include: (1) The distance from the electron donor to the acceptor. (2) The thermodynamic driving force $\Delta G°$ for the reaction. This can be approximated using the difference in standard electrode potentials (as in Table 6-1) between donor and acceptor. $\Delta G° = -96.5\ \Delta E°$ kJ/mol at 25° C. (3) The chemical makeup of the material through which the electron transfer takes place. (4) Any changes in the geometry or charge state of the donor or acceptor that accompany the transfer. (5) The orientation of the acceptor and donor groups.[144–146] It is usually assumed that the Franck–Condon principle is obeyed, i.e., that the electron jump occurs so rapidly ($<10^{-12}$ s) that there is no change in the positions of atomic nuclei (see also Chapter 23). Subsequent rearrangement of nuclear positions may occur at rates that allow a rapid overall reaction.

The various factors that affect the rate of electron transfer were incorporated by Marcus into a quantitative theory. Electron transfer is often discussed in terms of this classic Marcus theory together with effects of quantum mechanical tunneling.[144,147–150] According to Marcus the electron transfer rate from a donor to an acceptor at a fixed separation depends upon $\Delta G°$, a nuclear reorganization parameter (λ), and the electronic coupling strength $|H_{AB}|$ between reactant and product in the transition state (Eq. 16-3):

$$k_{ET} = (4\pi^3 / h^2\lambda k_B T)|H_{AB}|^2 \exp[-(\Delta G° + \lambda)^2 / 4\lambda k_B T] \quad (16\text{-}3)$$

Here $|H_{AB}|$ is a quantum mechanical matrix whose strength decreases exponentially with the distance of separation *R* as $e^{-\beta R}$ where β is a coefficient of the order of 9–14 nm^{-1}. At the closest contact (R = 0) the rate k_{ET}, by extrapolation from experimental data on small synthetic compounds, is close to the molecular vibration frequency of $10^{13}\ s^{-1}$.[151,152] At distances greater than 2 nm the rate would be negligible were it not for other factors.

Using mutant proteins as well as a variety of redox pairs and electron-transfer distances the validity of the Marcus equation with respect to the thermodynamic driving force and distance dependence has been verified.[153] This is even true for cytochrome *c* mutants functioning in living yeast cells.[146]

A huge amount of experimental work with proteins has been done to test and refine the theories of electron transport. For example, electron donor groups with various reduction potentials have been attached to various sites on the surface of a protein containing a heme or other electron accepting group. Ruthenium complexes such as Ru(III) $(NH_3)_5^{3+}$ form tight covalent linkages to imidazole nitrogens[154,155] such as that of His 33 of horse heart cytochrome *c*. This metal can be reduced rapidly to Ru(II) by an external reagent, after which the transfer of an electron from the Ru(II) across a distance of 1.2 nm to the heme Fe(III) can be followed spectroscopically. The reduction potentials *E*° for the Ru(III)/Ru(II) and Fe(III)/Fe(II) couples at pH 7 in these compounds are 0.16 and 0.27 V, respectively. Thus, an electron will jump spontaneously from the Ru(II) to the Fe(III) with $\Delta G° = -15.4$ kJ/mol. A rate constant of ~5 s^{-1}, which was nearly independent of temperature, was observed. Since the structures of Fe(II) and Fe(III) forms of cytochrome *c* differ only slightly,[156] the electron transfer apparently occurs with only a small amount of geometric rearrangement. The distribution of charges and dipoles within the protein may be such that the Fe^{2+} and Fe^{3+} complexes have almost equal thermodynamic stability.

Electron-transfer pathways? In spite of the success of the Marcus theory, rates of electron-transfer from the iron of cytochrome *c* have been found to vary for different pathways.[150,153,155] For example, transfer of an electron from Fe(II) in reduced cytochrome *c* to an Ru(III) complex on His 33 was fast (~440 s^{-1})[157] but

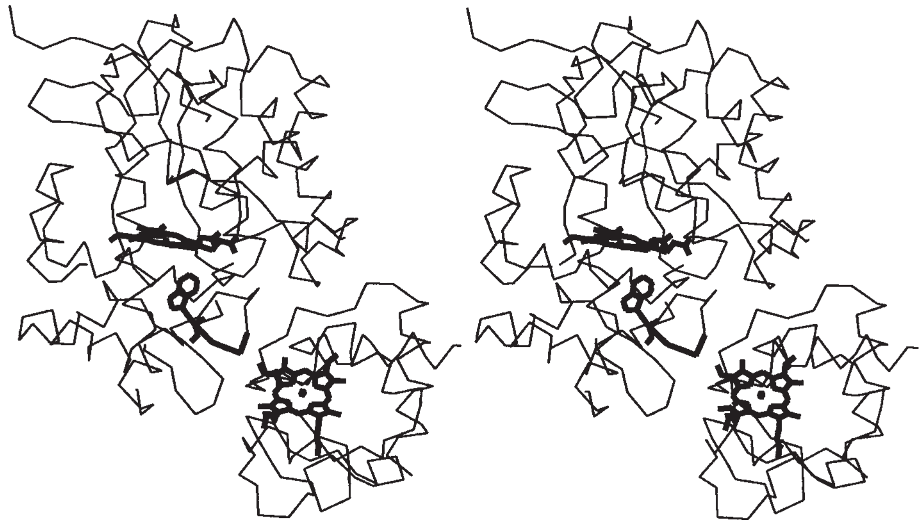


Figure 16-9 Stereoscopic α-carbon plot of yeast cytochrome *c* peroxidase (top) and yeast cytochrome *c* (below) as determined from a cocrystal by Pelletier and Kraut.[164] The heme rings of the two proteins appear in bold lines, as does the ring of tryptophan 191 and the backbone of residues 191–193 of the cytochrome *c* peroxidase. Drawing from Miller *et al.*[165]

the rate of transfer to an equidistant Fe(III) ion on Met 65 was at most 0.6 s^{-1}. These results suggested that distinct electron-transfer pathways exist. One suggestion was that the sulfur atom of Met 80 donates an electron to Fe^{3+} leaving an electron-deficient radical. The "hole" so created could be filled by an electron jumping in from the –OH group of the adjacent Tyr 67, which might then accept an electron from an external acceptor via Tyr 74 at the protein surface. Do electrons flow singly or as pairs from the surface through hydrogen-bonded paths? Use of both semisynthesis[155] and directed mutation[153] of cytochromes *c* is permitting a detailed study of these effects. A striking result is that substitution of the conserved residue phenylalanine 82 in a yeast cytochrome *c* with leucine or isoleucine retards electron transfer by a factor of ~10^4.

"Docking." It is now recognized that there are distinct "docking sites" on the surface of electron-transport proteins. For rapid electron transfer to occur the two electron carriers must be properly oriented and docked by formation of correct polar and nonpolar interactions. Early indications of the importance of docking came from study of modified cytochromes *c.* Each one of the 19 lysine side chains was individually altered by acylation or alkylation to remove the positive charge or to replace it with a negative charge. The rate of electron transfer into cytochrome *c* from hexacyanoferrate was decreased by a factor of 1.3–2.0 when any one lysine at positions 8, 13, 27, 72, or 79, which are clustered around the heme edge, was modified.[158] Modification of Lys 22, 55, 99, or 100, distant from this edge, had no effect. Electron transfer *into* cytochrome *c* from its natural electron donor ubiquinol:cytochrome *c* reductase was also strongly inhibited by modification of lysines that surround the heme edge.[159] Modification of these lysines also inhibited electron transfer *out of* cytochrome *c* into its natural acceptors, cytochrome *c* oxidase and cytochrome *c* peroxidase (Fig. 16-9).[160] A major factor in these effects is probably the large dipole moment in the cytochrome *c* that arises from the unequal distribution of surface charges. This charge distribution must assist in the proper docking of the cytochrome with its natural electron donors and acceptors. The positive surface charges presumably also facilitate the reaction with hexacyanoferrate (II) or ascorbate, both of which are negatively charged reductants that react rapidly with cytochrome *c*.

Measurements of many kinds have been made between natural donor–acceptor pairs such as cytochrome *c*–cytochrome b_5,[161,162] cytochrome *c*–cytochrome *c* peroxidase (Fig. 16-9),[153,163–166] trimethylamine dehydrogenase–FMN to Fe_4S_4 center (Fig. 15-9),[167] and methylamine dehydrogenase (TTQ radical)–amicyanin (Cu^{2+}).[168] Designed metalloproteins are being studied as well.[169] Femtosecond laser spectroscopy is providing a new approach.[169a]

Coupling and gating of electron transfer. Electrons are thought to be transferred into or out of cytochrome *c* through the exposed edge of the heme. The rate depends upon effective coupling, which in turn may depend upon orientation as well as the structure and dynamics of the protein.[170] Proteins with much β structure appear to provide stronger coupling than those that are largely composed of α helices.[144,171] The high nuclear reorganization energy λ of α helices may block electron transfer along some pathways.[153] A conformational change,[167,169] transfer of a proton, or binding of some other specific ion[172] before electron transfer occurs can be the "gating" process that determines the rate of electron transfer.[162,167,173] Electron transfer can also be "coupled" to an unfavorable, but fast, equilibrium.

Effects of ionic equilibria on electron transfer. The charge on an ion of Fe^{2+} in a heme is exactly balanced by two negative charges on the porphyrin ring. However, when the Fe^{2+} loses an electron to become Fe^{3+} an extra positive charge is suddenly present in the center of the protein. This change in charge will have a powerful electrostatic effect on charged groups in the immediate vicinity of the iron and even at the outer surface of the molecule. For example, an anion from the medium or from a neighboring protein molecule might become bound to the heme protein (Eq. 16-4).

$$X^- + [\mathrm{Fe(II)\ (heme)}] \rightarrow [\mathrm{Fe(III)\ (heme)}]^+ X^- + e^- \quad (16\text{-}4)$$

In this case the presence of a high concentration of X^- in the medium would favor the oxidation of Fe(II) to Fe(III). The reduced heme would be a better reducing agent and the oxidized form a weaker oxidant than in the absence of X^-. If –YH were a group in the protein the loss of an electron could cause –YH to dissociate so that Y^- and the Fe(III) would interact more tightly (Eq. 16-5).

$$\begin{gathered}\mathrm{HY} — [\mathrm{\ Fe\ (II)\ (heme)\ }] \rightarrow \\ Y^- — [\mathrm{\ Fe\ (III)\ (heme)\ }]^+ + H^+ + e^- \end{gathered} \quad (16\text{-}5)$$

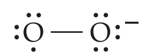
We see that electron transfer can be accompanied by loss of a proton and that $E°'$ may become pH dependent. (See also Eq. 16-18.) Even with cytochrome *c*, although there is little structural change upon electron transfer, there is an increased structural mobility in the oxidized form.[156] This may be important for coupling and could also facilitate associated proton-transfer reactions. For example, it is possible that in some cytochromes the imidazole ring in the fifth coordination position may become deprotonated upon oxidation. This possibility is of special interest because cytochromes are components of proton pumps in mitochondrial membranes (Chapter 18).

6. Reactions of Heme Proteins with Oxygen or Hydrogen Peroxide

As Ingraham remarked,[174] "Living in a bath of 20% oxygen, we tend to forget how reactive it is." From a thermodynamic viewpoint, all living matter is extremely unstable with respect to combustion by oxygen. Ordinarily, a high temperature is required and if we are careful with fire, we can expect to escape a catastrophe. However, one mole of properly chelated copper could catalyze consumption of all of the air in an average room within one second.[174] Biochemists are interested in both the fact that O_2 is kinetically stable and unreactive and also that oxidative enzymes such as cytochrome *c* oxidase are able to promote rapid reactions. Two oxygen atoms, each with six valence electrons, might reasonably be expected to form dioxygen, O_2, as a double-bonded structure with one σ and one π bond as follows (left):

Dioxygen

However, O_2 is paramagnetic and contains two unpaired electrons.[175] From this evidence O_2 might be assigned the structure on the right.[175] The oxygen molecule is very stable, and it is relatively difficult to add an electron to form the reactive **superoxide anion radical** O_2^-.

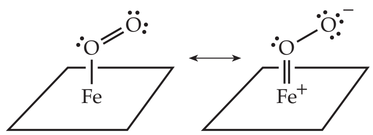


Superoxide anion radical

For this reason, oxidative attack by O_2 tends to be slow. However, once an electron has been acquired, it is easy for additional electrons to be added to the structure and further reduction occurs more easily. The biochemical question suggested is, "How can some heme proteins carry O_2 reversibly without any oxidation of the iron contained in them while others *activate* oxygen toward reaction with substrates?" Among this latter group, cytochrome *c* oxidase transfers electrons to both oxygen atoms so that only H_2O is a product, whereas the hydroxylases and oxygenases, which are discussed in Chapter 18, incorporate either one or two of the atoms of O_2, respectively, into an organic substrate. Before examining these reactions let us reconsider the heme oxygen carriers.

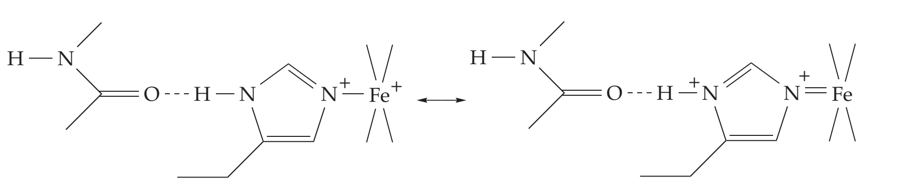
Oxygen-carrying proteins. In Chapter 7, we examined the behavior of hemoglobin in the cooperative binding of four molecules of O_2 and studied its structural relationship to the monomeric muscle protein myoglobin.[176] The iron in functional hemoglobin and myoglobin is always Fe(II) and is only very slowly converted by O_2 into the Fe(III) forms methemoglobin or metmyoglobin.[177,178] Erythrocytes contain an enzyme system for immediately reducing methemoglobin back to the Fe(II) state (see Box 15-H).

Binding of O_2 to the iron in the heme is usually considered not to cause a change in the oxidation state of the metal. However, oxygenated heme has some of the electronic characteristics of an $Fe^{3+}–OO^-$ peroxide anion. Bonding of the heme iron to oxygen is thought to occur by donation of a pair of electrons by the oxygen to the metal. In deoxyhemoglobin the Fe(II) ion is in the "high-spin" state; four of the five 3*d* orbitals in the valence shell of the iron contain one unpaired electron and the fifth orbital contains two paired electrons. The binding of oxygen causes the iron to revert to the "low-spin" state in which all of the electrons are paired and the paramagnetism of hemoglobin is lost. The stability of heme–oxygen complexes is thought to be enhanced by "back-bonding," i.e., the donation of an electron pair from one of the filled *d* orbitals of the iron atom to form a bond with the adjacent oxygen.[179] This can be indicated symbolically as follows:



These structures, which have been formulated by assuming that one of the unshared electron pairs on O_2 forms the initial bond to the metal, are expected to lead to an angular geometry which has been observed in X-ray structures of model compounds,[180] in oxymyoglobin (Fig. 16-10),[181,182] and in oxyhemoglobin.[183] Neutron diffraction studies have shown that the outermost oxygen atom of the bound O_2 is hydrogen bonded to the H atom on the N$^{\varepsilon}$ atom of the distal imidazole ring of His E7 (Fig. 16-10). Carbon monoxide binds with the $C{\equiv}O$ axis perpendicular to the heme plane and unable to form a corresponding hydrogen bond.[184] This decreases the affinity for CO and helps to protect us from carbon monoxide poisoning.

All oxygen-carrying heme proteins have another imidazole group that binds to iron on the side opposite the oxygen site. Without this proximal imidazole group, heme does not combine with oxygen. Coordination with heterocyclic nitrogen compounds favors formation of low-spin iron complexes and simple synthetic compounds that closely mimic the behavior of myoglobin have been prepared by attaching an imidazole group by a chain of appropriate length to the edge of a heme ring.[179,185] Similar compounds bearing a pyridine ring in the fifth coordination position have a low affinity for oxygen. Thus, the polarizable imidazole ring itself seems to play a role in promoting oxygen binding. The π electrons of the imidazole ring may also participate in bonding to the iron as is indicated in the following structures.[179] The π bonding to the iron would allow the iron to back-bond more strongly to an O_2 atom entering the sixth coordination position. These diagrams illustrate another feature found frequently in heme proteins: The N–H group of the imidazole is hydrogen bonded to a peptide backbone carbonyl group.



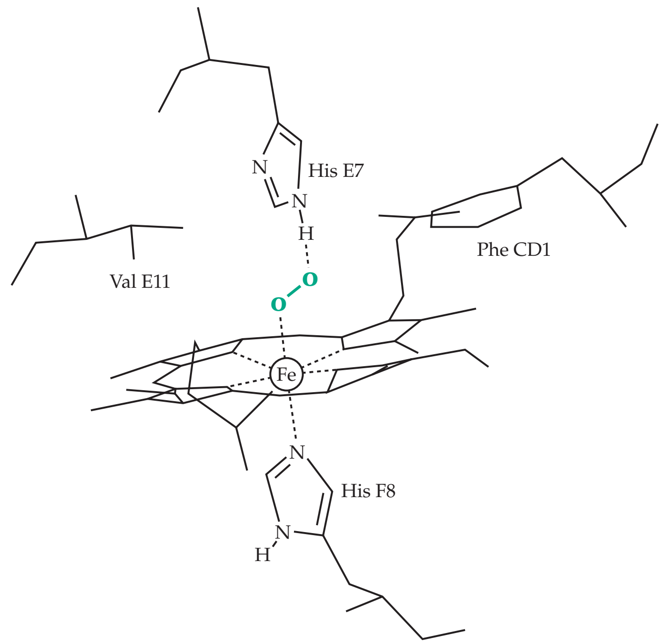


Figure 16-10 Geometry of bonding of O_2 to myoglobin and position of hydrogen bond to N$^{\varepsilon}$ of the distal histidine E7 side chain. After Perutz.[182]

The coordination of the heme iron to histidine also appears to provide the basis for the cooperativity in binding of oxygen by hemoglobin.[186] The radius of high-spin iron, whether Fe(II) or Fe(III), is so large that the iron cannot fit into the center of the porphyrin ring but is displaced toward the coordinated imidazole group by a distance of ~0.04 nm for Fe(II).[187] Thus, in deoxyhemoglobin both iron and the imidazole group lie further from the center of the ring than they do in oxyhemoglobin. In the latter, the iron lies in the center of the porphyrin ring because the change to the low-spin state is accompanied by a decrease in ionic radius.[186,188] The change in protein conformation induced by this small shift in the position of the iron ion was described in Chapter 7. However, the exact nature of the linkage between the Fe position and the conformational changes is not clear.

The mechanical response to the movement of the iron and proximal histidine, described in Chapter 7, may explain this linkage. However, oxygenation may also induce a change in the charge distribution within the hydrogen-bond network of the protein. The carbonyl group shown in the foregoing structure is attached to the F helix (see Fig. 7-23) and is also hydrogen bonded to other amide groups. Electron withdrawal into the heme–oxygen complex would tend to strengthen the hydrogen bond as indicated by the resonance forms shown and also to weaken competing hydrogen bonds.[189,190] This could affect the charge distribution in the upper end of the F helix and could conceivably induce a momentary conformational change that could facilitate the rearrangement of structure that was discussed in Chapter 7 (Fig. 7-25). The $\alpha_1\beta_2$ contact in which a change of hydrogen bonding takes place is located nearby behind the F and G helices. In any event, it is remarkable that nature has so effectively made use of the subtle differences in the properties of iron induced by changes in the electron distribution within the *d* orbitals of this transition metal.

A few groups of invertebrates, e.g., the sipunculid worms, use a nonheme iron-containing protein, **hemerythrin**, as an oxygen carrier.[191,192] Its 113-residue subunits are often associated as octamers of C_4 symmetry, each peptide chain having a four-helix bundle structure (Fig. 2-22). Instead of a heme group, each monomer contains two atoms of high-spin Fe(II) held by a cluster of histidine and carboxylate side chains (see Fig. 16-20).[193,194] Hemerythrin is a member of a group of such diiron oxoproteins which are considered further in Section 8. The copper oxygen carrier **hemocyanin** is discussed in Section D.

Catalases and peroxidases. Many iron and copper proteins do not bind O_2 reversibly but "activate" it for further reaction. We will look at such metalloprotein oxidases in Chapter 18. Here we will consider heme enzymes that react not with O_2 but with peroxides. The peroxidases,[194a] which occur in plants, animals, and fungi, catalyze the following reactions (Eq. 16-6, 16-7):

$$H_2O_2 + AH_2 \rightarrow 2\ H_2O + A \qquad (16\text{-}6)$$

$$ROOH + AH_2 \rightarrow R{-}OH + H_2O + A \qquad (16\text{-}7)$$

Here AH_2 is an oxidizable organic compound such as an alcohol or a pair of one-electron donor molecules. Catalases, which are found in almost all aerobic cells,[194b] may sometimes account for as much as 1% of the dry weight of bacteria. The enzyme catalyzes the breakdown of H_2O_2 to water and oxygen by a mechanism similar to that employed by peroxidases. If Eq. 16-7 is rewritten with H_2O_2 for AH_2 and O_2 for A, we have the following equation:

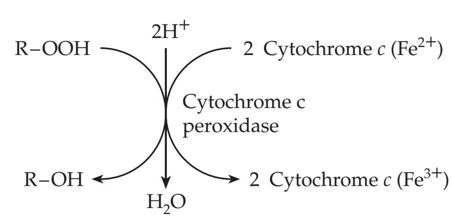
$$2\ H_2O_2 \xrightarrow{\text{Catalase}} 2\ H_2O + O_2 \qquad (16\text{-}8)$$

The action of catalase is very fast, almost 10^4 times faster than that of peroxidases. The molecular activity per catalytic center is about $2 \times 10^5\ s^{-1}$.

Catalase exerts a protective function by preventing the accumulation of H_2O_2 which might be harmful to cell constituents. The complete intolerance of obligate anaerobes to oxygen may result from their lack of this enzyme. Support for this protective function comes from the existence of the human hereditary condition **acatalasemia**.[195,196] Persons with extremely low catalase activity are found worldwide but are especially numerous in Korea. In Japan it is estimated that there are 1800 persons lacking catalase. Because about half of them have no symptoms, catalase might be judged unessential. However, many of the individuals affected develop ulcers around their teeth. Apparently, hydrogen peroxide produced by bacteria accumulates and oxidizes hemoglobin to methemoglobin (Box 15-H) depriving the tissues of oxygen.

Catalase from most eukaryotic species is tetrameric.[197] The protein from beef liver consists of 506-residue subunits.[198] Human catalase is similar.[198a] The proximal ligand to the heme Fe^{3+} is a tyrosinate anion (Tyr 358), while side chains of His 75 and Asn 148 lie close to the heme on the distal H_2O_2-binding side (Fig. 16-11). Larger ~650-residue fungal and bacterial catalases have a similar folding pattern but an extra C-terminal domain with a flavodoxin-like structure.[197,199] Catalase is gradually inactivated by its very reactive substrate. As isolated, beef liver catalase usually contains about two subunits in which the heme ring has been oxidatively cleaved to **biliverdin**[200] (Fig. 24-24) and various other alterations have been found.[197] Each subunit of mammalian catalases normally contains a bound molecule of NADPH which helps to protect against inactivation by H_2O_2.[201,202] Catalases from *Neurospora* and from *E. coli* contain heme *d* rather than protoporphyrin.[197,203] Some lactobacilli, lacking heme altogether, form a manganese-containing pseudocatalase.[204,205]

Of the plant peroxidases, which are found in abundance in the peroxisomes, the 40-kDa monomeric **horseradish peroxidase** has been studied the most.[206–208a] It occurs in over 30 isoforms and has an extracellular role in generating free radical intermediates for polymerization and crosslinking of plant cell wall components.[209] Secreted fungal peroxidases, e.g., such as those from *Coprinus*[210] and *Arthromyces*,[211] form a second class of peroxidases with related structues.[212] A third class is represented by **ascorbate peroxidase** from the cytosol of the pea[212–214] and by the small 34-kDa **cytochrome c peroxidase** from yeast mitochondria[215] (Fig. 16-11). The latter has a strong preference for reduced cytochrome *c* as a substrate (Eq. 16-9).[216,217]



(16-9)

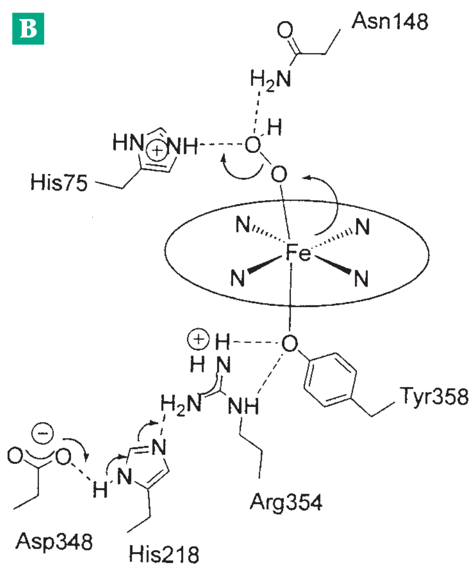


Figure 16-11 (A) Stereo drawing showing folding pattern for beef liver catalase and the positions of the NADPH (upper left) and heme (center). From Fita and Rossmann.[198] (B) Diagram of proposed structure of an Fe(III)–OOH ferric peroxide complex of human catalase (see also Fig. 16-14). A possible mechanism by which the peroxide is cleaved (step *b* in Fig. 16-14) is indicated by the arrows. His 75 and Asn 148 are directly involved, and a charge relay system below the ring may also participate. From Putnam *et al.*[198a]

Because the three-dimensional structures of the peroxidase, its reductant cytochrome *c*, and the complex of the two (Fig. 16-9) are known, cytochrome *c* peroxidase is the subject of much experimental study. Other fungal peroxidases, some of which contain manganese rather than iron, act to degrade lignin (Chapter 25).[218] A lignin peroxidase from the white wood-rot fungus *Phanerochaete chrysosporium* has a surface tryptophan with a specifically hydroxylated Cβ carbon atom which may have a functional role in catalysis.[218a,b]

The human body contains **lactoperoxidase**, a product of exocrine secretion into milk, saliva, tears, etc., and peroxidases with specialized functions in **saliva**, the **thyroid, eosinophils**,[219] and **neutrophils**.[220] The functions are largely protective but the enzymes also participate in biosynthesis. Mammalian peroxidases have heme covalently linked to the proteins, as indicated in Fig. 16-12.[220–222a]

The active site structure of peroxidases (Fig. 16-13) is quite highly conserved. As in myoglobin, an imidazole group is the proximal heme ligand, but it is usually hydrogen bonded to an aspartate carboxylate as a catalytic diad (Fig. 16-13).[223] In cytochrome *c* peroxidase there is also a buried tryptophan, which has already been highlighted in Fig. 16-9. A conserved and essential feature on the distal side is another histidine, which is hydrogen bonded to an asparagine[224] and which can also hydrogen bond to the substrate H_2O_2. Fungal peroxidases also have a conserved arginine on the distal side. However, even an octapeptide with a bound heme cut from cytochrome *c* acts as a "microperoxidase" with properties similar to those of natural peroxidases.[225]

Peroxidases and catalases contain high-spin Fe(III) and resemble metmyoglobin in properties. The enzymes are reducible to the Fe(II) state in which form they are able to combine (irreversibly) with O_2. We see that the same active center found in myoglobin and hemoglobin is present but its chemistry has been modified by the proteins. The affinity for O_2 has been altered drastically and a new group of catalytic activities for ferriheme-containing proteins has emerged.

Mechanisms of catalase and peroxidase catalysis. Attention has been focused on a series of strikingly colored intermediates formed in the presence of substrates. When a slight excess of H_2O_2 is added to a solution of horseradish peroxidase, the dark brown enzyme first turns olive green as **compound I** is formed, and then pale red as it turns into **compound II**. The latter reacts slowly with substrate AH_2 or with another H_2O_2 molecule to regenerate the original enzyme. This sequence of reactions is indicated by the colored arrows in Fig. 16-14, steps *a*–*d*.

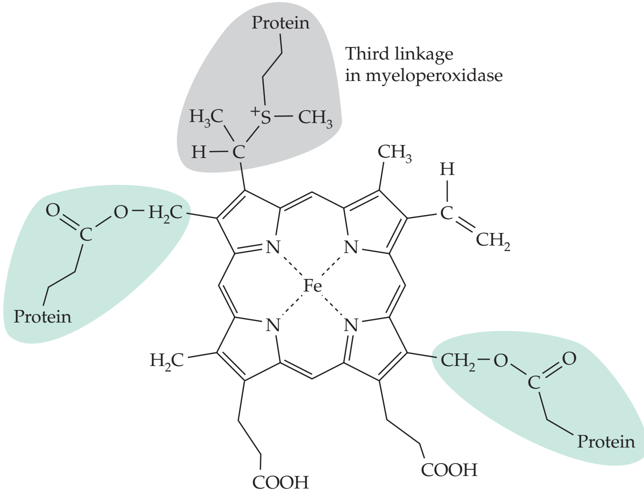


Figure 16-12 Linkage of heme to mammalian peroxidases. There are two ester linkages to carboxylate side chains from the protein.[220,221] Myeloperoxidase contains a third linkage.[222,222a]

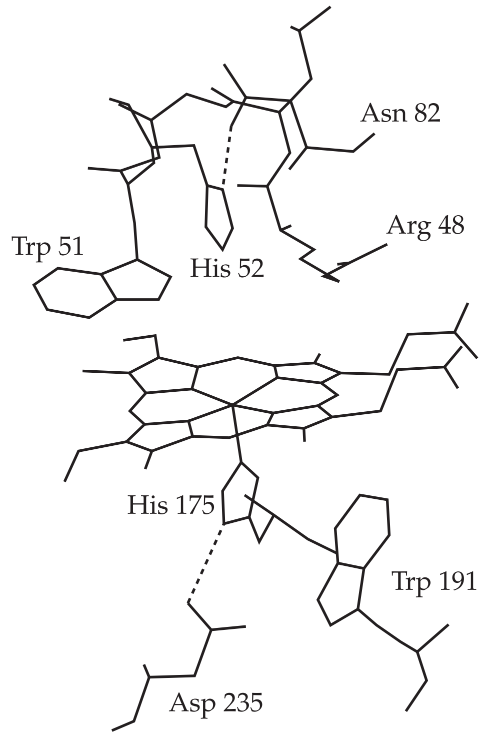


Figure 16-13 The active site of yeast cytochrome *c* peroxidase. Access for substrates is through a channel above the front edge of the heme ring as viewed by the reader. A pathway for entrance of electrons may be via Trp 191 and His 175. From Holzbaur *et al.*[206] Based on coordinates of Finzel *et al.*[215]

Titrations with such reducing agents as ferrocyanide or K_2IrBr_6 have established that compound I is converted into compound II by a one-electron reduction and compound II to free peroxidase by another one-electron reduction. Thus, the iron in compound I may formally be designated Fe(V) and that in II as Fe(IV). However, this does not tell us whether or not the oxygen atoms of H_2O_2 are present in compounds I and II. The enzyme in the Fe^{3+} form can be reduced to Fe^{2+} (Fig. 16-14, step *e*), as previously mentioned, and when the Fe^{2+} enzyme reacts with H_2O_2 it is apparently converted into compound II (Fig. 16-14, step *f*). This suggests that the latter is an Fe^{2+} complex of the peroxide anion. Here, P^{2-} represents the porphyrin ring:

$$P^{2-}\text{—}Fe^{+}(II)\text{—}OOH$$

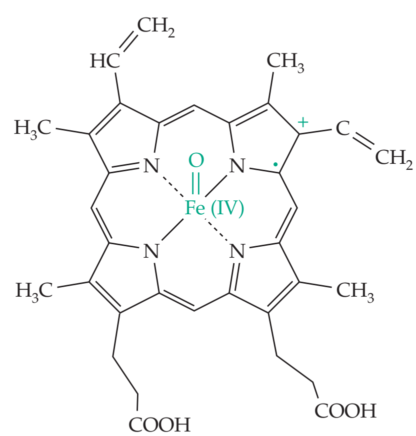
However, spectroscopic evidence suggests that compound II is a **ferryl iron** complex which could be derived from the preceding structure by addition of a proton and loss of water.[226,227]

$$P^{2-}\text{—}Fe^{2+}(IV){=}O$$

Compound II

High concentrations of H_2O_2 convert II into compound III, which is thought to be the same as the **oxyperoxidase** that is formed upon addition of O_2 to the Fe(II) form of the free enzyme (Fig. 16-14, step *g*) and corresponds in structure to oxyhemoglobin.[228]

Compound I was at one time thought to be a complex of H_2O_2 or its anion with Fe(III), but its magnetic and spectral properties are inconsistent with this structure. Rather, it too appears to contain ferryl iron bound to an electron-deficient porphyrin π-cation radical.[229] The reaction with peroxide probably involves initial formation of a peroxide anion complex (Fig. 16-14, step *a*) which is cleaved with release of water (step *b*).[215,230] The resulting $Fe(V)^{+}{=}O$ compound is converted to compound I by transfer of a single elec-



Compound I. The unpaired electron and the positive charge are delocalized over the porphyrin ring and perhaps into the proximal histidine ring tron from the porphyrin to the iron. In cytochrome *c* peroxidase compound I contains a free radical on the nearby Trp 191 ring instead of on the porphyrin radical.[216] Consistent with this is the fact that horseradish peroxidase contains phenylalanine in place of Trp 191.

If we consider the fate of substrate AH_2 during the action of a peroxidase, we see that donation of an electron to compound I to convert it into II (Fig. 16-14, step *c*) will generate a free radical •AH as well as a proton. The radical may then donate a second electron to II to form the free enzyme. Alternatively, a second molecule of AH_2 may react (Fig. 16-14, step *c*) to form a second radical •AH. The two •AH radicals may then disproportionate to form A and AH_2 or they may leave the enzyme and react with other molecules in their environment. Compound II of horse radish peroxidase is able to exchange the oxygen atom of its Fe(IV)=O center with water rapidly at pH 7, presumably by donation of a proton from the nearby histidine side chain (corresponding to His 52 of Fig. 16-13).[227,230a,b] This histidine presumably also functions in proton transfer during reactions with substrates (see Fig. 16-11B).[224]

The catalase compound I appears to be converted in a two-electron reduction by H_2O_2 directly to free ferricatalase without intervention of compound II (Fig. 16-14, step *c′*). The catalytic histidine probably donates a proton to help form water from one of the oxygen

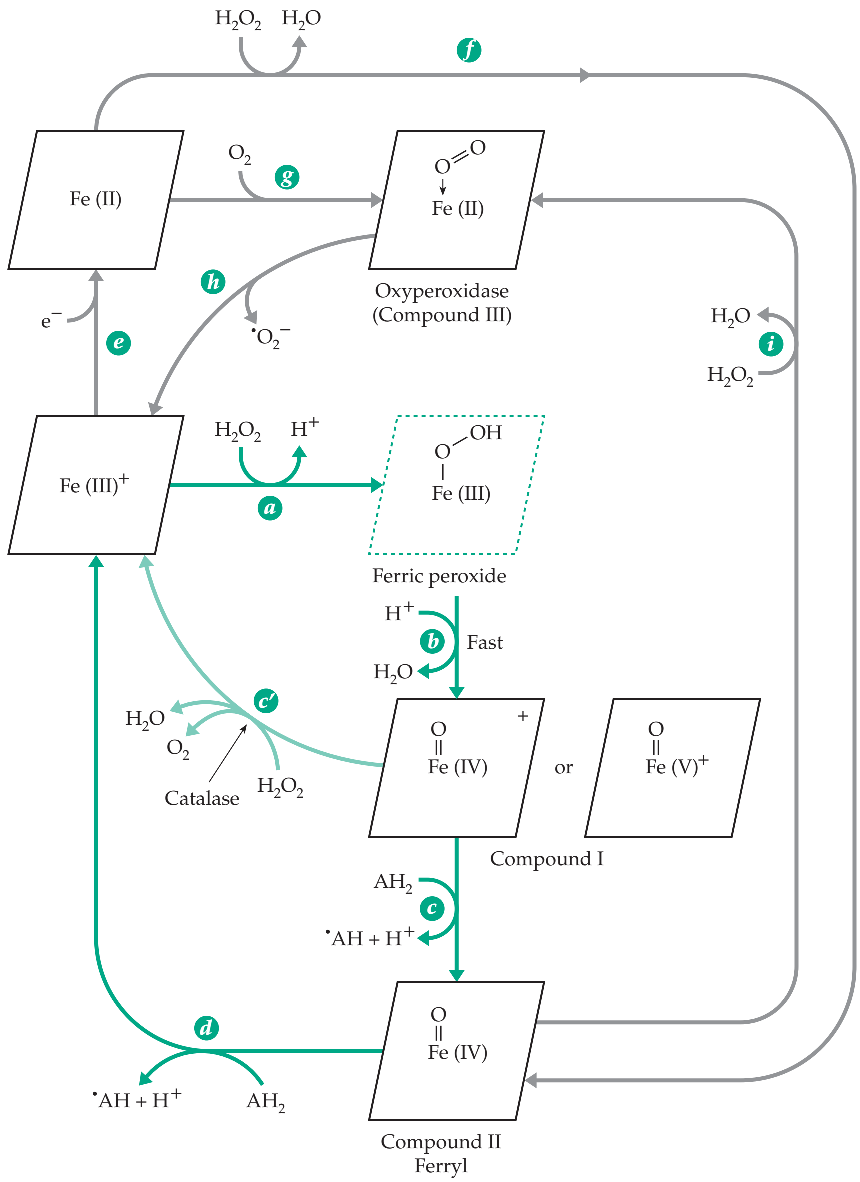


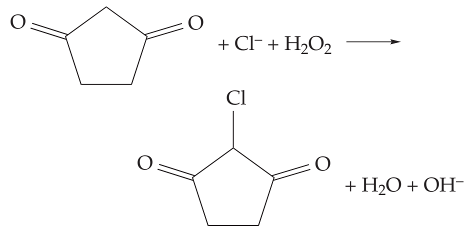
Figure 16-14 The catalytic cycles and other reactions of peroxidases and catalases. The principal cycle for peroxidases is given by the colored arrows. That of catalases is smaller, making use of step *a, b,* and *c′*, which is marked by a light green line.

$$•P^{+}\text{—Fe(IV)=O} + H_2O_2 \rightarrow H_2O + O_2 + \text{P—Fe(III)}^{+} \quad (16\text{-}10)$$
Compound I

atoms of the H_2O_2. Nevertheless, compound II does form slowly, especially if a slow substrate such as ethanol is present. The previously mentioned bound NADPH apparently reduces compound II formed in this way, converting the inactivated enzyme back to active catalase.[201,231] This may involve unusual one-electron oxidation steps for the NADPH.[231] Under some circumstances compound II of peroxidases reacts with substrates in a two-electron process[232–235] with transfer of an oxygen atom to the substrate, a characteristic also of reactions catalyzed by cytochromes P450 (see Eq. 18-57).

Haloperoxidases. Many specialized peroxidases are active in halogenation reactions. **Chloroperoxidases** from fungi[236,237] catalyze chlorination reactions like that of Eq. 16-11 using H_2O_2 and Cl^- as well as the usual peroxidase reaction.

Chloroperoxidase is isolated in a low-spin Fe(III) state. The reduced Fe(II) enzyme is a high-spin form

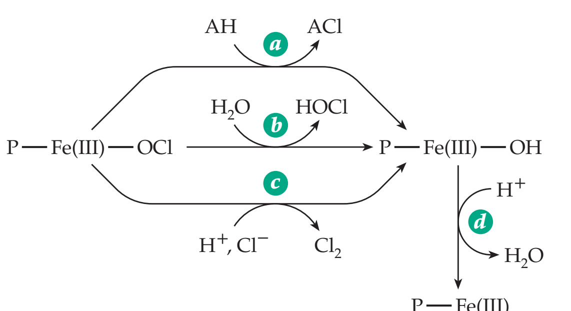


(16-11)

with spectroscopic properties similar to those of the oxygenases of the cytochrome P450 family, which are discussed in Chapter 18.[238,239] Like the cytochromes P450 chloroperoxidase contains a thiolate group of a cysteine side chain as the fifth iron ligand.[235,239,240] Chloroperoxidase forms compounds I and II, as do other peroxidases. A chloride ion may combine with compound I to form a complex of hypochlorite with the Fe(III) heme.

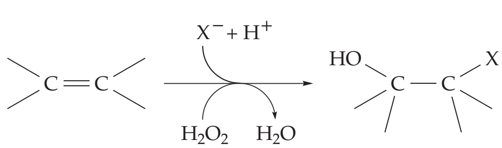
$$^{\bullet}P^{+}—Fe(IV)=O + Cl^{-} \rightarrow P—Fe(III)^{+}—OCl \quad (16\text{-}12)$$

This intermediate could then halogenate substrate AH (Eq. 16-13, step *a*), lose HOCl (Eq. 16-13, step *b*), or generate Cl_2 by reaction with Cl^- (Eq. 16-13, step *c*). These are all well-established reactions for the enzyme. In each case the chlorine in the peroxidase complex can be viewed as an electrophile which is transferred to an attacking nucleophile. A fourth reaction that can go



(16-13)

through the same intermediate is conversion of alkenes to α,β-halohydrins (Eq. 16-14).[241] **Lactoperoxidase** of milk, reacting with $I^- + H_2O_2$, promotes an analogous iodination of tyrosine and histidine residues of proteins. With radioactive $^{125}I^-$ or $^{131}I^-$ it provides a convenient and much used method for labeling of proteins in



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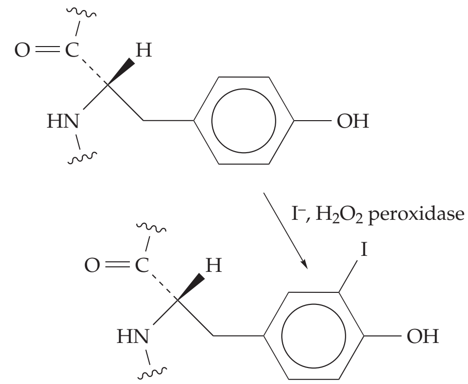


(16-15)

exposed surfaces of membranes (Eq. 16-15).[242] Iodinated tyrosine derivatives are formed in the thyroid gland by a similar reaction catalyzed by **thyroid peroxidase**.[232] Even horseradish peroxidase can oxidize iodide ions but neither it nor lactoperoxidase will carry out chlorination or bromination reactions.

Myeloperoxidase, present in specialized lysosomes of polymorphonuclear leukocytes (neutrophils),[243] utilizes H_2O_2 and a halide ion to kill ingested bacteria.[244,245] Phagocytosis induces increased respiration by the leukocyte and generation of H_2O_2, partly by the membrane-bound NADPH oxidase described in Chapter 18. Some of the H_2O_2 is used by myeloperoxidase to attack the bacteria, apparently through generation of HOCl by peroxidation of Cl^-. Human myeloperoxidase is a tetramer of two 466-residue chains and two 108-residue chains, which carry the covalently linked heme.[222a,246] Another oxygen-dependent killing mechanism that may also be used by neutrophils is the generation of the reactive **singlet oxygen**.[247] This can occur by reaction of hypochlorite with H_2O_2 (Eq. 16-16) or from an enzyme-bound hypochlorite intermediate such as that shown in Eq. 16-13. Hereditary deficiency of myeloperoxidase is relatively common.[245]

$$OCl^- + H_2O_2 \rightarrow \underset{\text{singlet } O_2}{O_2(^1\Delta g)} + H_2O + Cl^- \quad (16\text{-}16)$$

Lactoperoxidase[248] and chloroperoxidase[249] also generate singlet oxygen. The possible biological significance is discussed in Chapter 18. Eosinophil peroxidase appears to promote formation of **hydroxyl radicals**.[250] **Bromoperoxidases** are found in many red and green marine algae.[251] Many of them contain **vanadium** and function by a mechanism different than that used by heme peroxidases (see Section G).[252,253]

Another related nonheme enzyme is the selenoprotein glutathione peroxidase. It reacts by a mechanism very different (Eq. 15-59) from those discussed here, as does **NADPH peroxidase**, a flavoprotein with a cysteine sulfinate side chain in the active site.[254–255a] A lignin-degrading peroxidase from the white wood rot fungus *Phanerochaete chrysosporium* is a simple heme protein,[256] while other peroxidases secreted by this organism contain Mn.[257,258]

7. The Iron–Sulfur Proteins

Not all of the iron within cells is chelated by porphyrin groups. Hemerythrin (see Fig. 16-20) has been known for many years, but the general significance of nonheme iron proteins was not appreciated until large-scale preparation of mitochondria was developed by Crane in about 1945. The iron content of mitochondria was found to far exceed that of the heme proteins present. In 1960, Beinert, who was studying the mitochondrial dehydrogenase systems for succinate and for NADH, observed that when the electron transport chain was partially reduced by these substrates and the solutions were frozen at low temperature and examined, a strong EPR signal was observed at $g = 1.94$. The signal was obtained only upon reduction by substrate, and fractionation pointed to the nonheme iron proteins. Six or more proteins of this type are involved in the mitochondrial electron transport chain (Eq. 18-1), and numerous others have become recognized as members of the same large family of **iron–sulfur proteins** (Fe–S proteins).[259,260]

Ferredoxins, high-potential iron proteins, and rubredoxins. The presence of nonheme iron proteins is most evident in the anaerobic clostridia, which contain no heme. It was from these bacteria that the first Fe–S protein was isolated and named **ferredoxin**. This protein has a very low reduction potential of $E°'$(pH 7) = –0.41 V. It participates in the pyruvate–ferredoxin oxidoreductase reaction (Eq. 15-35), in nitrogen fixation in some species, and in formation of H_2. A small green-brown protein, the ferredoxin of *Peptococcus aerogenes* contains only 54 amino acids but complexes eight atoms of iron. If the pH is lowered to ~1, eight molecules of H_2S are released. Thus, the protein contains eight "labile sulfur" atoms in an iron sulfide linkage. There are also eight iron atoms.

Another group of related electron carriers, the **high-potential iron proteins** (HIPIP) contain four labile sulfur and four iron atoms per peptide chain.[261–266] X-ray studies showed that the 86-residue polypeptide chain of the HIPIP of *Chromatium* is wrapped around a single **iron–sulfur cluster** which contains the side chains of four cysteine residues plus the four iron and four sulfur atoms (Fig. 16-15D).[261] This kind of cluster is referred to as [4Fe–4S], or as Fe_4S_4. Each cysteine sulfur is attached to one atom of Fe, with the four iron atoms forming an irregular tetrahedron with an Fe–Fe

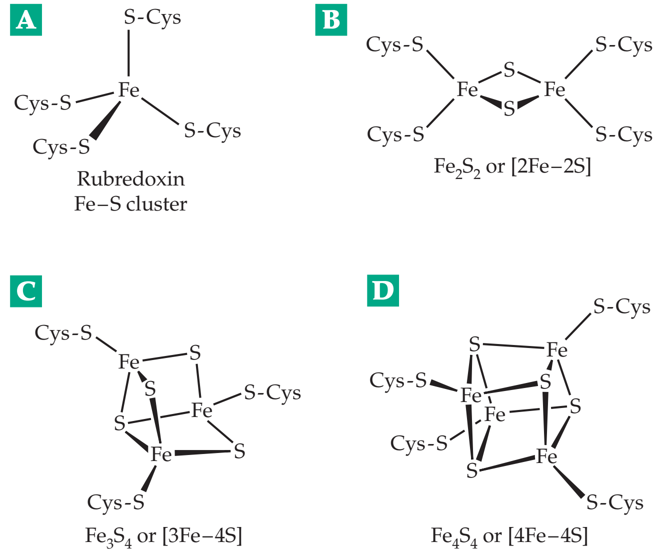


Figure 16-15 Four different iron–sulfur clusters of a type found in many proteins. From Beinert[259] with permission.

distance of ~0.28 nm. The four labile sulfur atoms (S^{2-}) form an interpenetrating tetrahedron 0.35 nm on a side with each of the sulfur atoms bonded to three iron atoms. The cluster is ordinarily able to accept only a single electron. The iron–sulfur cluster structure was a surprise, but after its discovery it was found that ions such as $[Fe_4S_4(S{-}CH_2CH_2COO^-)_4]^{6-}$ assemble spontaneously from their components and have a similar cluster structure.[259,266a] Thus, living things have simply improved upon a natural bonding arrangement.

The bacterial ferredoxins from *Peptococcus, Clostridium* (Fig. 16-16B),[267,268] *Desulfovibrio,* and other anaerobes each contain two Fe_4S_4 clusters with essentially the same structure as that of the *Chromatium* HIPIP.[267,269] Each cluster can accept one electron. Much of the amino acid sequence in the first half of the ferredoxin chain is repeated in the second half, suggesting that the chain may have originated as a result of gene duplication. Many invariant positions are present in the sequence, including those of the cysteine residues forming the Fe–S cluster. Ferredoxins with single Fe_4S_4 clusters are also known.[270]

The ferredoxins have reduction potentials $E°'$ (pH 7) from about – 0.4 V to as low as – 0.6 V. However, the corresponding values for HIPIP proteins range from +0.05 to +0.50 V at pH 7.[271] This wide range of potentials initially seemed strange because the structures of the active centers of both the clostridial ferredoxins and the *Chromatium* HIPIP appear virtually identical.[272] Part of the explanation lies in the fact that Fe_4S_4 clusters can exist in three oxidation states (Eq. 16-17) that differ, one from another, by a single electron.[273]

$$[Fe_4S_4]^{+} \xrightarrow{-e^-} [Fe_4S_4]^{2+} \xrightarrow{-e^-} [Fe_4S_4]^{3+} \quad (16\text{-}17)$$

Reduced Fd	Oxidized Fd; reduced high potential iron protein	Oxidized high potential iron protein; Super-oxidized Fd

Here the charges shown are those on the cluster. The cysteine ligands from the protein each add an additional negative charge. The *Chromatium* HIPIP and the clostridial ferredoxins have the middle oxidation state in common. The cluster is a little smaller in the more oxidized states; in the *Chromatium* HIPIP the Fe–Fe distance changes from 0.281 to 0.272 nm upon oxidation.

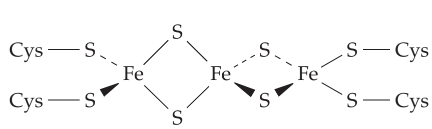
Rubredoxins. The simplest of the Fe–S proteins are the rubredoxins. These proteins contain iron but no labile sulfur. The rubredoxin of *Clostridium pasteurianum* is a 54-residue peptide containing four cysteines whose side chains form a distorted tetrahedron about a single iron atom (Fig. 16-16A).[274] Not shown for any

Figure 16-16 (A) Superimposed stereoscopic α-carbon traces of the peptide chain of rubredoxin from *Clostridium pasteurianum* with either Fe^{3+} (solid circles) or Zn^{2+} (open circles) bound by four cysteine side chains. From Dauter *et al.*[274] (B) Alpha-carbon trace for ferredoxin from *Clostridium acidurici*. The two Fe_4S_4 clusters attached to eight cysteine side chains are also shown. The open circles are water molecules. Based on a high-resolution X-ray structure by Duée *et al.*[267] Courtesy of E. D. Duée. (C) Polypeptide chain of a chloroplast-type ferredoxin from the cyanobacterium *Spirulina platensis*. The Fe_2S_2 cluster is visible at the top of the molecule. From Fukuyama *et al.*[276] Courtesy of K. Fukuyama.

of the structures in Fig. 16-16 are NH—S hydrogen bonds that connect backbone NH groups of the peptide chain to the sulfur atoms of the cysteine groups, forming the clusters.[275] These bonds may have important effects on properties of the cluster. Rubredoxins also participate in electron transport and can substitute for ferredoxins in some reactions. Larger 14-kDa and 18-kDa rubredoxins able to bind two iron ions participate in electron transport in a hydroxylase system of *Pseudomonas* (Chapter 18).[277] A smaller 7.9-kDa 2-Fe **desulforedoxin** functions in sulfate-reducing bacteria.[278]

Chloroplast-type ferredoxins. Members of a large class of [2Fe–2S] or Fe_2S_2 ferredoxins each contain two iron atoms and two labile sulfur atoms with the linear structure of Fig. 16-15B.[279,279a] Best known are the chloroplast ferredoxins, which transfer electrons from photosynthetic centers of chloroplasts to the flavoprotein reductase that reduces $NADP^+$ to NADPH.[280,280a] The structure of a cyanobacterial protein of this type is shown in Fig. 16-16C.[276] A second group of Fe_2S_2 ferredoxins are found in bacteria including *E. coli*[281] and in human mitochondria.[282] For example, in steroid hormone-forming tissues the ferredoxin **adrenodoxin**[282a] carries electrons to cytochromes P450. Its Fe_2S_2 center receives electrons from adrenodoxin reductase (Chapter 15 banner).[282b] The nitrogen-fixing *Clostridium pasteurianum* also contains a ferredoxin of this class.[283] In *Pseudomonas putida* (Chapter 18) the related 106-residue **putidaredoxin** transfers electrons to cytochromes P450.[284]

The 3Fe–4S clusters. A 106-residue ferredoxin from *Azotobacter vinelandii* contains seven iron atoms in two Fe–S clusters that operate at very different redox potentials of –0.42 and +0.32 V.[285] Other similar seven-iron proteins are known.[286] From EPR measurements it appeared that both clusters function between the 2^+ and 3^+ (oxidized and superoxidized) states of Eq. 16-17, despite the widely differing potentials. A super-reduced all-Fe(II) form with $E^{\circ\prime}$ (pH 7) = –0.70 V can also be formed.[286] X-ray crystallographic studies have revealed that the protein contains one Fe_4S_4 cluster and one Fe_3S_4 cluster (Fig. 16-15C).[286–288] The structures and environments of the Fe_3S_4 clusters are similar to those of Fe_4S_4 clusters but they lack one iron and one cysteine side chain. An Fe_4S_4 cluster may sometimes lose S^{2-} to form an Fe_3S_4 cluster such as the one in the *A. vinelandii* ferredoxin. A less likely possibility is isomerization to a linear Fe_3S_4 structure.[289]



Aconitase (Eq. 13-17) isolated under aerobic conditions contains an Fe_3S_4 cluster and is catalytically inactive. Incubation with Fe^{2+} activates the enzyme and reconverts the Fe_3S_4 to an Fe_4S_4 cluster (Fig. 13-4).[260,289,290]

Properties of iron–sulfur clusters. These clusters were viewed for many years as unstable and unable to exist outside of a protein. However, if protected from oxygen and manipulated in the presence of soluble organic thiols they are stable and "cofactor-like."[260] Intact clusters can be "extruded" from proteins by treatment with thiols in nonaqueous media. Both Fe_4S_4 and Fe_2S_2 clusters as well as more complex forms have been synthesized[291] and nonenzymatic cluster interconversions have been demonstrated. Binding to proteins stabilizes the clusters further, but some (Fe–S) proteins are labile and difficult to study. This is evidently because of partial exposure of the cluster to the surrounding solvent. Not only can O_2 cause oxidation of exposed clusters[291a] but also superoxide,[292] nitric oxide, and peroxynitrite can react with the iron. Aconitase has only three cysteine side chains available for coordination with Fe and the protein is unstable. Apparently, a superoxidized $[Fe_4S_4]^{3+}$ cluster is formed in the presence of O_2 but loses Fe^{2+} to give an $[Fe_3S_4]^+$ cluster.[293]

Another interesting cluster conversion is the joining of two Fe_2S_2 clusters in a protein to form a single Fe_4S_4 cluster at the interface between a dimeric protein. Such a cluster is present in the nitrogenase iron protein (Fig. 24-2) and probably also in biotin synthase.[294] The clusters in such proteins can also be split to release the monomers.

Synthetic iron–sulfur clusters have weakly basic properties[273] and accept protons with a pK_a of from 3.9 to 7.4. Similarly, one clostridial ferredoxin, in the oxidized form, has a pK_a of 7.4; it is shifted to 8.9 in the reduced form.[295] If we designate the low-pH oxidized form of such a protein as HOx^+ and the reduced form as HRed, we can depict the reduction of each Fe_4S_4 cluster as follows.

$$HOx^+ + e^- \rightarrow HRed \tag{16-18}$$

Comparing with Eq. 6-64 and using the Michaelis pH functions (first two terms of Eq. 7-13) for HOx^+ and HRed, it is easy to show that the value of $E^{\circ\prime}$ ($E_{1/2}$) at which equal amounts of oxidized protein (HOx^+ + Ox) and reduced protein (HRed + Red^-) are present is given by Eq. 16-19, in which K_{ox} and K_{red} are the K_a values for dissociation of the protonated oxidized and reduced forms, respectively.

$$E_{1/2} = E^0 \text{ (low pH)} + 0.0592 \log\left[\frac{(1 + K_{red}/[H^+])}{(1 + K_{ox}/[H^+])}\right] \text{V} \tag{16-19}$$

At the high pH limit this becomes $E_{1/2} = E°$ (low pH) + 0.0592 ($pK_{ox} - pK_{red}$) and V = −0.371 + 0.0592 (7.4−8.9) V = −0.431 V. Thus, the value of $E_{1/2}$ changes from −0.371 to −0.460 V as the pH is increased. In the pH range between the pK_a values of 7.4 and 8.9 reduction of the protein will lead to binding of a proton from the medium and oxidation to loss of a proton. Human and other vertebrate ferredoxins also show pH-dependent redox potentials.[282] This suggests, as with the cytochromes, a possible role of Fe−S centers in the operation of proton pumps in membranes. Nevertheless, many ferredoxins, such as that of *C. pasteurianum*, show a constant value of $E°'$ from pH 6.3 to 10[296] and appear to be purely electron carriers.

Both the iron and labile sulfur can be removed from Fe−S proteins and the active proteins can often be reconstituted by adding sulfide and Fe^{2+} ions. Using this approach, the natural isotope ^{56}Fe (nuclear spin zero) has been exchanged with ^{57}Fe, which has a magnetic nucleus[297] and ^{32}S has been replaced by ^{77}Se. The resulting proteins appear to function naturally and give EPR spectra containing hyperfine lines that result from interaction of these nuclei with unpaired electrons in the Fe_4S_4 clusters (Fig. 16-17). These observations suggest that electrons accepted by Fe_4S_4 clusters are not localized on a single type of atom but interact with nuclei of both Fe and S. The native proteins as well as many mutant forms are being studied by NMR[299,300] and other spectroscopic techniques (Fig. 16-18),[260] by theoretical computations,[301–304] and by protein engineering and "rational design."[305,306]

Functions of iron−sulfur enzymes. Numerous iron−sulfur clusters are present within the membrane-bound electron transport chains discussed in Chapter 18. Of special interest is the Fe_2S_2 cluster present in a protein isolated from the cytochrome *bc* complex (complex III) of mitochondria. First purified by Rieske *et al.*,[307] this protein is often called the **Rieske iron−sulfur protein**.[308] Similar proteins are found in cytochrome *bc* complexes of chloroplasts.[125,300,309,310] In

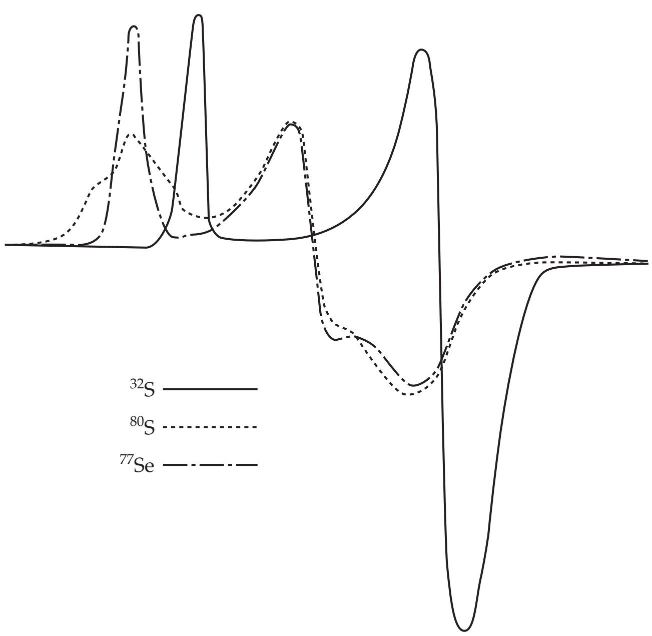


Figure 16-17 Electron paramagnetic resonance spectrum of the Fe−S protein putidaredoxin in the natural form (^{32}S) and with labile sulfur replaced by selenium isotopes. Well-developed shoulders are seen in the low-field end of the spectrum of the ^{77}Se (spin = 1/2)-containing protein. From Orme-Johnson *et al.*[298] Courtesy of W. H. Orme-Johnson.

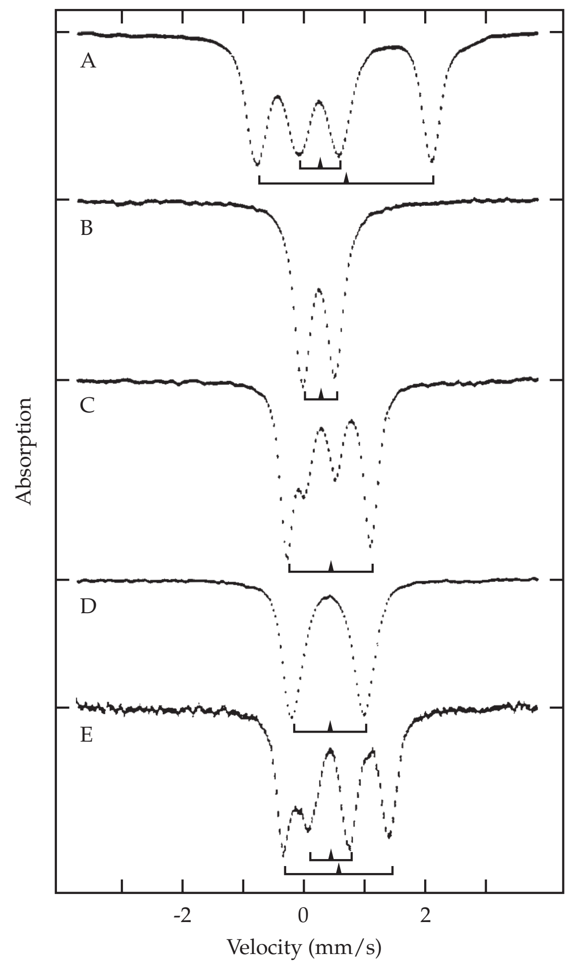
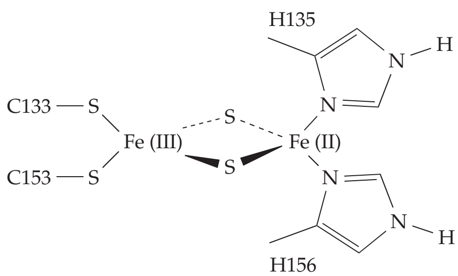


Figure 16-18 Mössbauer X-ray absorption spectra of iron−sulfur clusters. (See Chapter 23 for a brief description of the method.) Quadrupole doublets are indicated by brackets and isomer shifts are marked by triangles. (A) $[Fe_2S_2]^{1+}$ cluster of the Rieske protein from *Pseudomonas mendocina*, at temperature T = 200 K. (B) $[Fe_3S_4]^{1+}$ state of *D. gigas* ferredoxin II, T = 90 K. (C) $[Fe_3S_4]^0$ state of *D. gigas* ferredoxin II, T = 15 K. (D) $[Fe_4S_4]^{2+}$ cluster of *E. coli* FNR protein, T = 4.2 K. (E) $[Fe_4S_4]^{1+}$ cluster of *E. coli* sulfite reductase, T = 110 K. From Beinert *et al.*[260]

some bacteria Rieske-type proteins deliver electrons to oxygenases.[300] The 196-residue mitochondrial protein has an unusually high midpoint potential, E_m of ~ 0.30 V. The Fe_2S_2 cluster, which is visible in the atomic structure of complex III shown in Fig. 18-8, is coordinated by two cysteine thiolates and two histidine side chains. In a *Rhodobacter* protein they occur in the following conserved sequences:[311] C133-T-H-L-G-C138 and C153-P-C-H-G-S158. One iron is bound by C133 and C153 and the other by H135 and H156 as follows:

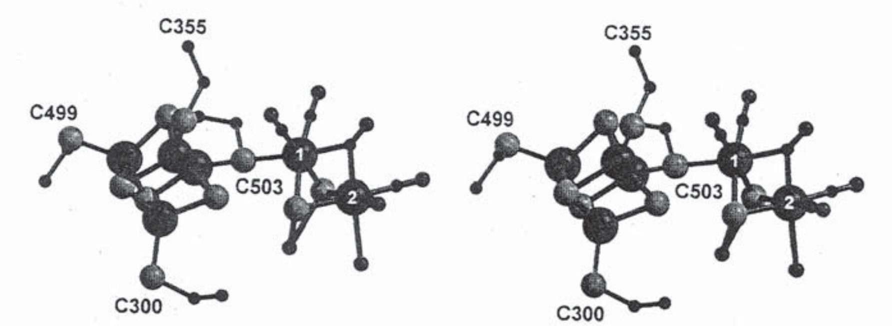


These proteins may also have an ionizable group with a pK_a of ~8.0, perhaps from one of the histidines that is linked to the oxidation–reduction reaction.

Iron–sulfur clusters are found in flavoproteins such as NADH dehydrogenase (Chapter 18) and trimethylamine dehydrogenase (Fig. 15-9) and in the siroheme-containing **sulfite reductases** and **nitrite reductases**.[312] These two reductases are found both in bacteria and in green plants. Spinach nitrite reductase,[313] which is considered further in Chapter 24, utilizes reduced ferredoxin to carry out a six-electron reduction of NO_2^- to NH_3 or of SO_3^{2-} to S^{2-}. The 61-kDa monomeric enzyme contains one siroheme and one Fe_4S_4 cluster. A sulfite reductase from *E. coli* utilizes NADPH as the reductant. It is a large $\beta_8\alpha_4$ oligomer.[312] The 66-kDa α chains contain bound flavin (4 FAD + 4 FMN),[312,314,315] while the 64-kDa β subunits contain both siroheme and a neighboring Fe_4S_4 cluster (Fig. 16-19).[89,304,312,316] The iron of the siroheme and the closest iron atom of the cluster are bridged by a single sulfur atom of a cysteine side chain.

A somewhat similar double cluster is present in **all-Fe hydrogenases** from *Clostridium pasteurianum*.[316a,b,c] These enzymes also contain two or three Fe_4S_4 clusters and, in one case, an Fe_2S_2 cluster.[316a] At the presumed active site a special **H cluster** consists of an Fe_3S_4 cluster with one cysteine sulfur atom shared by an adjoining Fe_2S_2 cluster: Because many hydrogenases contain nickel, their chemistry and functions are discussed in Section C,2.

The role of the iron–sulfur clusters in many of the proteins that we have just considered is primarily one of single-electron transfer. The Fe–S cluster is a place for an electron to rest while waiting for a chance to react. There may sometimes be an associated proton pumping action. In a second group of enzymes, exemplified by aconitase (Fig. 13-4), an iron atom of a cluster functions as a **Lewis acid** in facilitating removal of an –OH group in an α,β dehydration of a carboxylic acid (Chapter 13). A substantial number of other bacterial dehydratases as well as an important plant dihydroxyacid dehydratase also apparently use Fe–S clusters in a catalytic fashion.[317] Fumarases A and B from *E. coli*,[317] L-serine dehydratase of a *Peptostreptococcus* species,[317–319] and the dihydroxyacid



The hydrogenase Fe–S H cluster

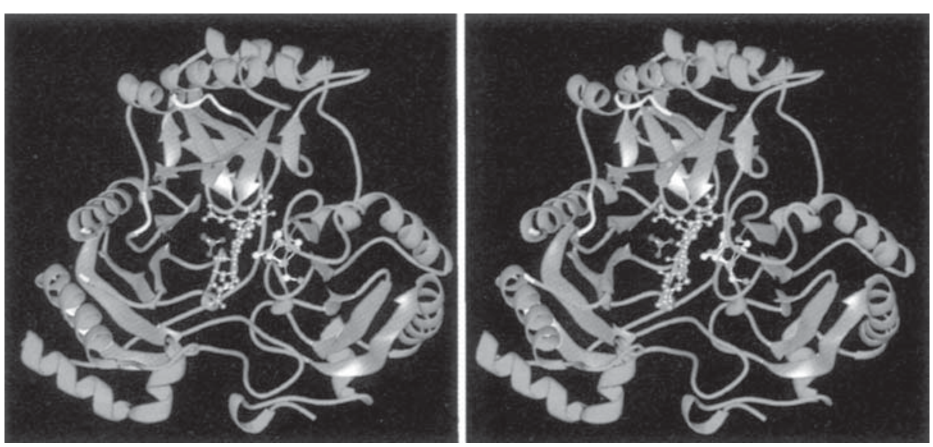


Figure 16-19 Stereoscopic view of *E. coli* assimilatory sulfite reductase. The siroheme (Fig. 16-6) is in the center with one edge toward the viewer and the Fe_4S_4 cluster is visible on its right side. A single S atom from a cysteine side chain bridges between the Fe of the siroheme and the Fe_4S_4 cluster. A phosphate ion is visible in the sulfite-binding pocket at the left center of the siroheme. From Crane *et al.*[312] Courtesy of E. D. Getzoff.

dehydratase[320,321] may all use their Fe_4S_4 clusters in a manner similar to that of aconitase (Eq. 13-17). However, the Fe–S enzymes that dehydrate *R*-lactyl-CoA to crotonyl-CoA,[322] 4-hydroxybutyryl-CoA to crotonyl-CoA,[323,324] and *R*-2-hydroxyglutaryl-CoA to *E*-glutaconyl-CoA[325] must act by quite different mechanisms, perhaps similar to those utilized in vitamin B_{12}-dependent reactions. In these enzymes, as in pyruvate formate lyase (Eq. 15-38), the Fe–S center may act as a radical generator.

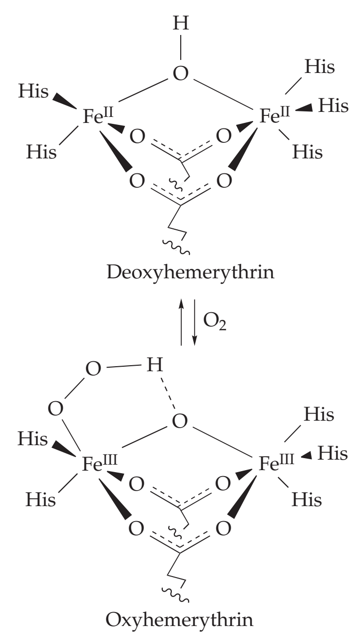
The molybdenum-containing enzymes considered in Section F also contain Fe–S clusters. Nitrogenases (Chapter 24) contain a more complex Fe–S–Mo cluster. Carbon monoxide dehydrogenase (Section C) contains 2 Ni, ~11 Fe, and 14 S^{2-} as well as Zn in a dimeric structure. In these enzymes the Fe–S clusters appear to participate in catalysis by undergoing alternate reduction and oxidation.

For a few enzymes such as aconitase and amido-transferases,[326] an Fe–S cluster plays a **regulatory role** in addition to or instead of a catalytic function.[327] Cytosolic aconitase is identical to **iron regulatory proteins 1** (IRP1), which binds to iron responsive elements in RNA to inhibit translation of genes asso-ciated with iron uptake. A high iron concentration promotes assembly of the Fe_4S_4 cluster (see Chapter 28, Section C,6). Another example is provided by the *E. coli* transcription factor SOXR. This protein, which controls a cellular defense system against oxygen-derived superoxide radicals, contains two Fe_2S_2 centers. Oxidation of these centers by superoxide radicals appears to induce the transcription of genes encoding superoxide dismutase and other proteins involved in protecting cells against oxidative injury (Chapter 18).[328–330]

8. The (μ-oxo) Diiron Proteins

Both the ferroxidase center of ferritin (Fig. 16-4) and the oxygen-carrying hemerythrin are members of a family of diiron proteins with similar active site structures.[331,332] The pair of iron atoms, with the bridging (μ) ligands such as O_2, HO^-, HOO^-, O_2^{2-}, is often held between four α helices, as can be seen in Fig. 2-22 and in Fig. 16-20C. In most cases each iron is ligated by at least one histidine, one glutamate side chain, and frequently a tyrosinate side chain. As many as three side chains may bind to each iron. In addition, one or two carboxylate groups from glutamate or aspartate side chains bridge to both irons, as does the μ-oxo group, which is typically H_2O or ^-OH. Examples of this structure are illustrated in Figs. 16-4 and 16-20. Although the active sites of all of the diiron proteins appear similar, the chemistry of the catalyzed reactions is varied.

Hemerythrin. When both iron atoms are in the Fe(II) state, hemerythrin, like hemoglobin, functions as a carrier of O_2. In the oxidized Fe(III) **methemerythrin** form the iron atoms are only 0.32 nm apart. Three bridging (μ) groups lie between them: two carboxylate groups and a single oxygen atom which may be either O^{2-} or OH^-. One coordination position on one of the hexacoordinate iron atoms is open and appears to be the site of binding of oxygen. The O_2 is thought to accept two electrons, oxidizing the two iron atoms to Fe(III) and itself becoming a peroxide dianion O_2^{2-}. The process is completely reversible. The conversion of the oxygen to a bound peroxide ion is supported by studies of resonance Raman spectra (see Chapter 23) which also suggest that the peroxide group is protonated. In the diferrous protein the μ-oxo bridge is thought to be an ^-OH group. Upon oxygenation the proton could be shared with or donated to the peroxo group (Eq. 16-20).[334] A similar binding of O_2 as a peroxide dianion appears to occur in the copper-containing hemocyanins (Section C,3).



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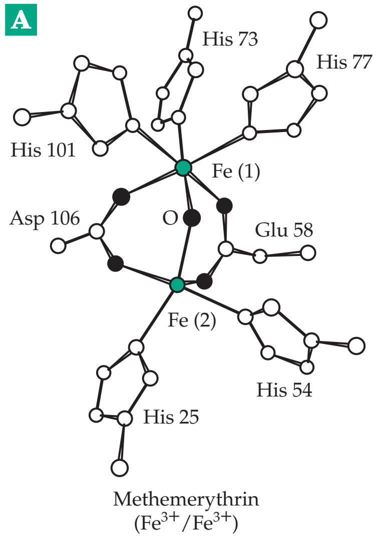
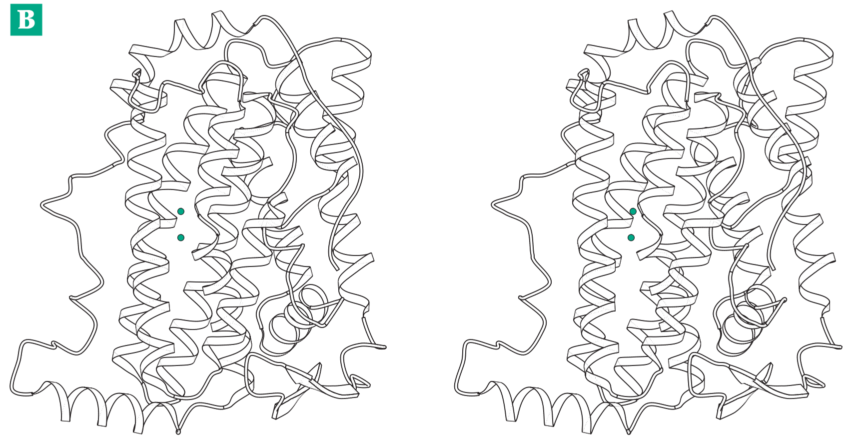
Purple acid phosphatases. Diiron-tyrosinate proteins with acid phosphatase activity occur in mammals, plants, and bacteria. Most are basic glycoproteins with an intense 510- to 550-nm light absorption band. Well-studied members come from beef spleen, from the uterine fluid of pregnant sows (**uteroferrin**),[335] and from human macrophages and osteoclasts.[336–336b] One of the two iron atoms is usually in the Fe(III) oxidation state, but the second can be reduced to Fe(II) by mild reductants such as ascorbate. This half-reduced form is enzymatically active and has a pink color and a characteristic EPR signal. Treatment with oxidants such as H_2O_2 or hexacyannoferrate (III)





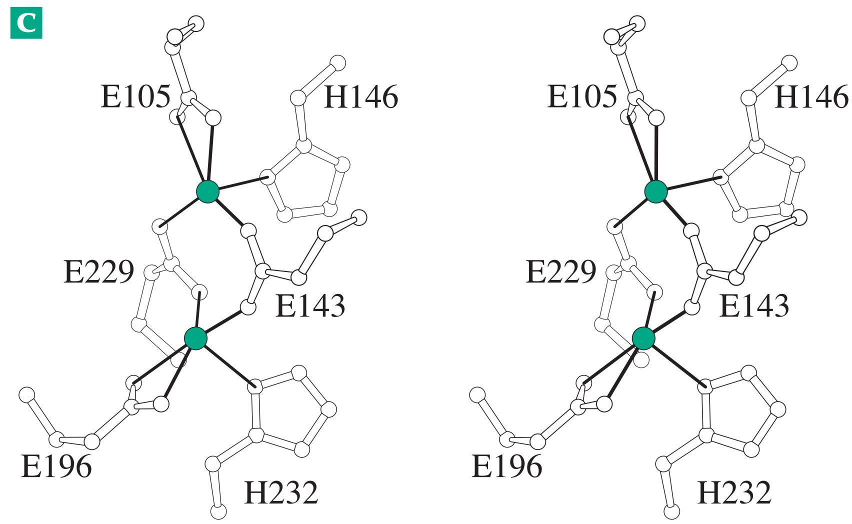


Figure 16-20 (A) The active site of hemerythrin showing the two iron atoms (green) and their ligands which include the μ oxo bridge and two bridging carboxylate groups. From Lukat *et al.*[193] The active site is between four parallel helices as shown in Fig. 2-22. (B) Stereoscopic view of the backbone structure of a Δ^9 stearoyl-acyl carrier protein desaturase which also contains a diiron center. Notice the nine-antiparallel-helix bundle. The diiron center is between four helices as in hemerythrin. (C) Stereoscopic view of the diiron center of the desaturase.[333] (B) and (C) courtesy of Ylva Lindqvist.

generates purple inactive forms which lack a detectable EPR signal.[337,338] The Fe^{3+} of the active enzyme can be replaced[339] with Ga^{3+} and the Fe^{2+} with Zn^{2+} with retention of activity; also, some plants contain phosphatases with Fe–Zn centers.[340,341] The catalytic mechanism resembles those of other metallophosphatases (Chapter 12) and the change of oxidation state of the Fe may play a regulatory role. On the other hand, a principal function of uteroferrin may be in transplacental transport of iron to the fetus.[342]

Diiron oxygenases and desaturases. In the (μ-oxo) diiron oxygenases O_2 is initially bound in a manner similar to that in hemerythrin but one atom of the bound O_2 is reduced to H_2O using electrons supplied by a cosubstrate such as NADPH. The other oxygen atom enters the substrate. This is illustrated by Eq. 16-21 for methane monooxygenase.[332,343–345] A toluene monooxygenase has similar properties.[346]

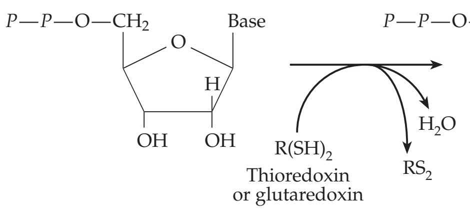
$$CH_4 + NADH + H^+ + O_2 \rightarrow CH_3OH + NAD^+ + H_2O \quad (16\text{-}21)$$

In green plants a soluble Δ^9 stearoyl-acyl carrier protein desaturase uses O_2 and NADH or NADPH to introduce a double bond into fatty acids. The structure of this protein (Fig. 16-20B,C) is related to those of methane oxygenase and ribonucleotide reductase.[333,347] The desaturase mechanism is discussed in Chapter 21.

9. Ribonucleotide Reductases

Ribonucleotides are reduced to the 2'-deoxyribonucleotides (Eq. 16-21) that are needed for DNA synthesis by enzymes that act on either the di- or triphosphates of the purine and pyrimidine nucleosides[348–351] (Chapter 25). These ribonucleotide reductases utilize either thioredoxin or glutaredoxin (Box 15-C) as the immediate hydrogen donors (Eq. 16-22). The pair of closely spaced –SH groups in the reduced thioredoxin or glutaredoxin are converted into a disulfide bridge at the same time that the 2'-OH of the ribonucleotide di- (or tri-) phosphate is converted to H_2O. While some organisms employ a vitamin B_{12}-

dependent enzyme for this purpose, most utilize iron–tyrosinate enzymes (Class I ribonucleotide reductases). These are two-protein complexes of composition $\alpha_2\beta_2$. The enzyme from *E. coli* contains 761-residue α chains and 375-residue β chains. That from *Salmonella typhimurium*[351a] is similar, as are corresponding mammalian enzymes and a virus-encoded ribonucleotide reductase formed in *E. coli* following infection by T4 bacteriophage.[352] In every case the larger α_2 dimer, which is usually called the **R1 protein**, contains the substrate binding sites, allosteric effector sites, and redox-active SH groups. Each α chain is folded into an unusual $(\alpha / \beta)_{10}$ barrel.[353,354] As in the more familiar $(\alpha / \beta)_8$ barrels, the active site is at the N termini of the β strands.

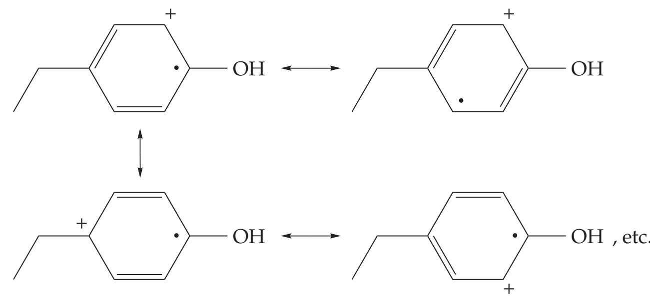


(16-22)

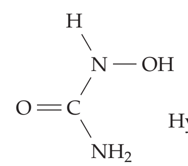
Each polypeptide chain of the β_2 dimer or **R2 protein** contains a diiron center which serves as a free radical generator.[354a,b,c] A few bacteria utilize a dimanganese center.[355] Oxygenation of this center is linked to the uptake of both a proton and an electron and to the removal of a hydrogen atom from the ring of tyrosine 166 to form H_2O and an organic radical (Eq. 16-23):[356–360]

$$2\,Fe(II)^{2+} + O_2 + H^+ + e^- + Y122\text{-}OH \rightarrow Fe(III)^{3+} - O^{2-} - Fe(III)^{3+} - Y122 - O\bullet + H_2O \tag{16-23}$$

The tyrosyl radical is used to initiate the ribonucleotide reduction at the active site in the R1 protein ~3.5 nm away. The tyrosyl radical is very stable and was discovered by a characteristic EPR spectrum of isolated enzyme. Alteration of this spectrum when bacteria were grown in deuterated tyrosine indicated that the radical is located on a tyrosyl side chain and that the spin density is delocalized over the tyrosyl ring.[361,362] Using protein engineering techniques the ring was located as Tyr 122 of the *E. coli* enzyme. A few of the resonance structures that can be used to depict the radical are the following:



A chain of hydrogen-bonded side chains apparently provides a pathway for transfer of an unpaired electron from the active site to the Tyr 122 radical and from there to the radical generating center.[363] The tyrosyl radical can be destroyed by removal of the iron by exposure to O_2 or by treatment of ribonucleotide reductases with hydroxyurea, which reduces the radical and also destroys catalytic activity:



Hydroxyurea

A second group of ribonucleotide reductases (Class II), found in many bacteria, depend upon the cobalt-containing **vitamin B_{12} coenzyme** which is discussed in Section B. These enzymes are monomeric or homodimeric proteins of about the size of the larger α subunits of the Class I enzymes. The radical generating center is the 5'-deoxyadenosyl coenzyme.[350,364,365]

Class III or **anaerobic nucleotide reductases** are used by various anaerobic bacteria including *E. coli* when grown anaerobically[350,366–369a] and also by some bacteriophages.[370] Like the Class I reductases, they have an $\alpha_2\beta_2$ structure but each β subunit contains an Fe_4S_4 cluster which serves as the free radical generator,[369] that forms a stable glycyl radical at G580.[369a] In this respect the enzyme resembles pyruvate formate-lyase (Eq. 15-40). As with other enzymes using Fe_4S_4 clusters as radical generators, *S*-adenosylmethionine is also required. All ribonucleotide reductases may operate by similar radical mechanisms.[350,351]

When a 2'-Cl or -F analog of UDP was used in place of the substrate an irreversible side reaction occurred by which Cl^- or F^-, inorganic pyrophosphate, and uracil were released.[349] When one of these enzyme-activated inhibitors containing ^3H in the 3' position was tested, the tritium was shifted to the 2' position with loss of Cl^- and formation of a reactive 3'-carbonyl compound (Eq. 16-24) that can undergo β elimination at each end to give an unsaturated ketone which inactivates the enzyme. This suggested that the Fe-tyrosyl radical abstracts an electron (through a

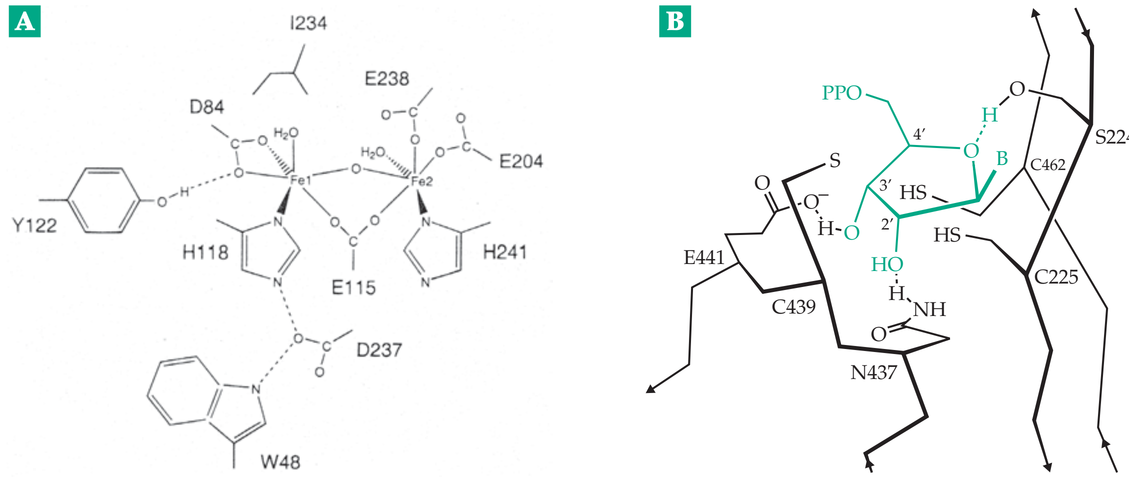
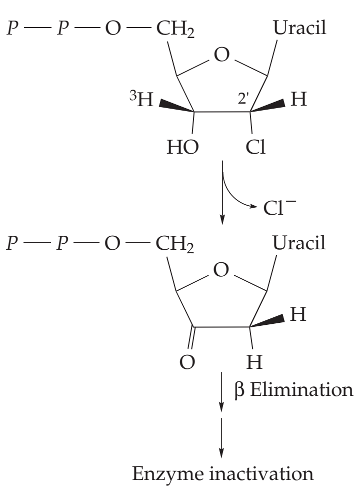


Figure 16-21 (A) Scheme showing the diiron center of the R2 subunit of *E. coli* ribonucleotide reductase. Included are the side chains of tyrosine 122, which loses an electron to form a radical, and of histidine 118, aspartate 237, and tryptophan 48. These side chains provide a pathway for radical transfer to the R1 subunit where the chain continues to tyrosines 738 and 737 and cysteine 429.[354a–c] From Andersson *et al.*[354c] (B) Schematic drawing of the active site region of the *E. coli* class III ribonucleotide reductase with a plausible position for a model-built substrate molecule. Redrawn from Lenz and Giese[373] with permission.

chain of intermediate groups) from an –SH group, now identified as C439 in the *E. coli* enzyme[371–372] (Fig. 16-21). The resulting thiyl radical is thought to abstract a hydrogen atom from C′-3 of a true substrate to form a substrate radical (Eq. 16-25, step *a*) which, with the help of the C462 –SH group, facilitates the loss of $^-$OH from C-2 in step *b*. The resulting C2 radical would be reduced by the nearby redox-active thiol pair C462 and C225 (Fig. 16-21 and Eq. 16-25, step *c*). In Eq. 16-25 the reaction is shown as a hydride transfer with an associated one-electron shift but the mechanism is uncertain. In step *d* of Eq. 16-25 the thiyl radical is regenerated and continues to function in subsequent rounds of catalysis. In the final step (step *e*) the redox active pair is reduced by reduced thioredoxin or glutaredoxin. The active site must open to release the product and to permit this reduction, which may involve participation of still other –SH groups in the protein.



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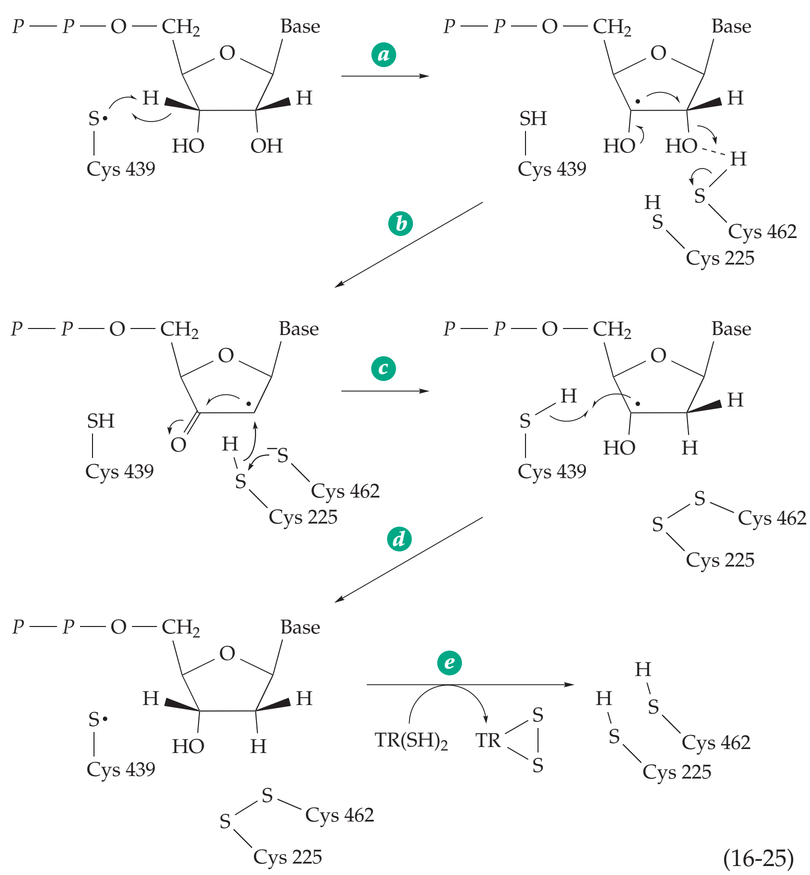


(16-25)

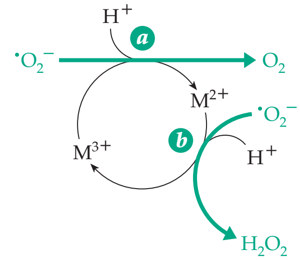
10. Superoxide Dismutases

Metalloenzymes of at least three different types catalyze the destruction of superoxide radicals that arise from reactions of oxygen with heme proteins, reduced flavoproteins, and other metalloenzymes. These superoxide dismutases (SODs) convert superoxide anion radicals $^{\bullet}O_2^-$ into H_2O_2 and O_2 (Eq. 16-26). The H_2O_2 can then be destroyed by catalase (Eq. 16-8).

$$2\,O_2^- + 2\,H^+ \longrightarrow H_2O_2 + O_2 \qquad (16\text{-}26)$$

The much studied Cu / Zn superoxide dismutase of eukaryotic cytoplasm is described in Section D. However, eukaryotic mitochondria contain manganese SOD and some eukaryotes also synthesize an iron-containing SOD. For example, the protozoan *Leishmania tropica*, which takes up residence in the phagolysosomes of a victim's macrophages, synthesizes an iron-containing SOD[374] to protect itself against superoxide generated by the macrophages. *Mycobacterium tuberculosis* secretes an iron SOD which assists its survival in living tissues and is also a target for the immune response of human hosts.[375]

Iron and manganese SODs have ~20-kDa subunits in each of which a single ion of Fe or Mn is bound by three imidazole groups and a carboxylate group.[376,377] The metal ion undergoes a cyclic change in oxidation state as illustrated by Eq. 16-27. Notice that two protons must be taken up for formation of H_2O_2. In Cu / Zn SOD the copper cycles between Cu^{2+} and Cu^+. The structure of the active site of an Fe SOD is shown in Fig. 16-22. That of Mn SOD is almost identical.[376] In addition to the histidine and carboxylate ligands, the metal binds a hydroxyl ion ^-OH or H_2O and has a site



(16-27)

open for binding of $^{\bullet}O_2^-$. As indicated in Eq. 16-27, uptake of one proton is associated with each reaction step. As illustrated in Fig 16-22 for step *a* of Eq. 16-27, the first proton may be taken up to convert the bound ^-OH to H_2O. The enzyme in the Fe^{2+} form has a pK_a of 8.5 that has been associated with tyrosine 34. Perhaps this residue is involved in the proton uptake process.[378] A similarly located Tyr 41 from *Sulfolobus* is covalently modified, perhaps by phosphorylation.[378a]

B. Cobalt and Vitamin B_{12}

The human body contains only about 1.5 mg of cobalt, almost all of it is in the form of **cobalamin**, vitamin B_{12}. Ruminant animals, such as cattle and sheep, have a relatively high nutritional need for cobalt and in regions with a low soil cobalt content, such as Australia, cobalt deficiency in these animals is a serious problem. This need for cobalt largely reflects the high requirement of the microorganisms of the rumen (paunch) for vitamin B_{12}. All bacteria require vitamin B_{12} but not all are able to synthesize it. For example, *E. coli* lacks one enzyme in the biosynthetic

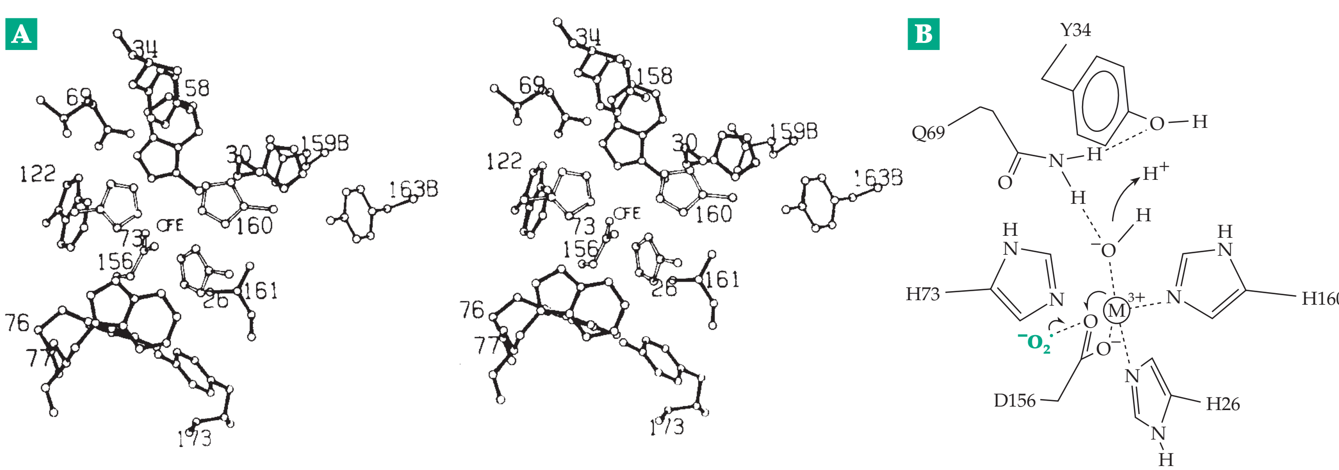
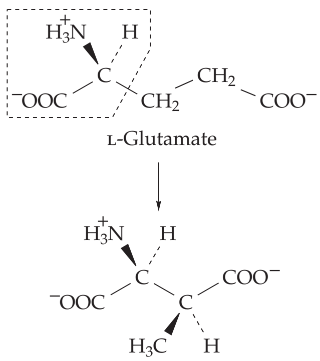


Figure 16-22 (A) Structure of the active site of iron superoxide dismutase from *E. coli*. From Carlioz *et al.*[379] Courtesy of M. Ludwig. (B) Interpretive drawing illustrating the single-electron transfer from a superoxide molecule to the Fe^{3+} of superoxide dismutase and associated proton uptake. Based on Lah *et al.*[376]

pathway and must depend upon other bacteria to complete the synthesis.

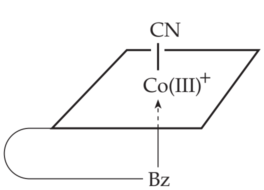
1. Coenzyme Forms

For several years after the discovery of cobalamin its biochemical function remained a mystery, a major reason being the extreme sensitivity of the coenzymes to decomposition by light. Progress came after Barker and associates discovered that the initial step in the anaerobic fermentation of glutamate by *Clostridium tetanomorphum* is rearrangement to β-methylaspartate[380,381] (Eq. 16-28).

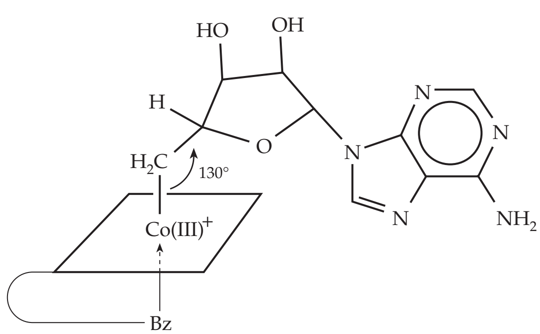


threo-β-Methylaspartate (16-28)

The latter compound can be catabolized by reactions that cannot be used on glutamate itself. Thus, the initial rearrangement is an indispensable step in the energy metabolism of the bacterium. A new coenzyme required for this reaction was isolated in 1958 after it was found that protection from light during the preparation was necessary. The coenzyme was characterized in 1961 by X-ray diffraction[382] as **5'-deoxyadenosylcobalamin**. It is related to cyanocobalamin (Box 16-B) by replacement of the CN group by a 5'-deoxyadenosyl group as indicated in the following abbreviated formulas.[383–385] Here the planes represent the corrin ring system and Bz the dimethylbenzimidazole that is coordinated with the cobalt from below the ring.



Cyanocobalamin

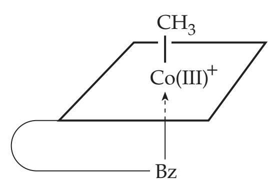


5'-Deoxyadenosylcobalamin

The most surprising structural feature is the Co–C single bond of length 0.205 nm. Thus, the coenzyme is an alkyl cobalt, the first such compound found in nature. In fact, alkyl cobalts were previously thought to be unstable. Vitamin B_{12} contains Co(III), and cyanocobalamin can be imagined as arising by replacement of the single hydrogen on the inside of the corrin ring by Co^{3+} plus CN^-. However, bear in mind that three other nitrogens of the corrin ring and a nitrogen of dimethylbenzimidazole also bind to the cobalt. Each nitrogen atom donates an electron pair to form coordinate covalent linkages. Because of resonance in the conjugated double-bond system of the corrin, all four of the Co–N bonds in the ring are nearly equivalent and the positive charge is distributed over the nitrogen atoms surrounding the cobalt.

The strength of the axial Co–C bond is directly influenced by the strength of bonding of the dimethylimidazole whose conjugate base has a microscopic pK_a of 5.5.[386] Protonation of this base breaks its bond to cobalt and may thereby strengthen the Co–C bond.[387] Steric factors are also important in determining the strength of this bond. NMR techniques are now playing an important role in investigation of these factors.[386–388]

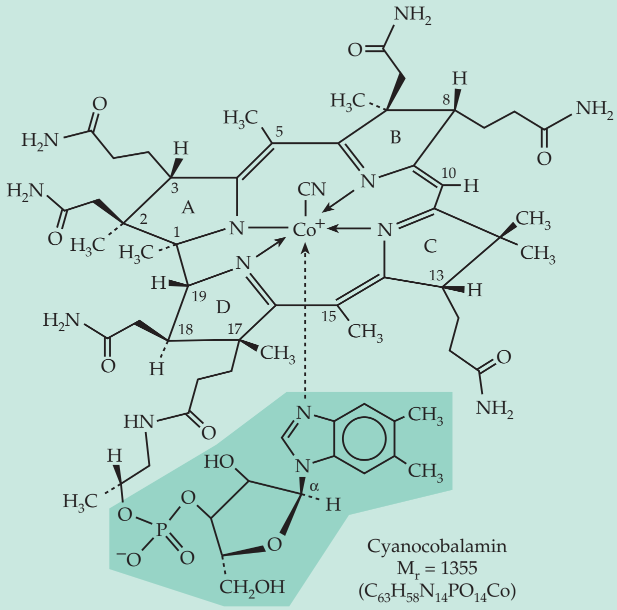
In both bacteria and liver, the 5'-deoxyadenosyl coenzyme is the most abundant form of vitamin B_{12}, while lesser amounts of **methylcobalamin** are present. Other naturally occurring analogs of the coenzymes include **pseudo vitamin B_{12}** which contains adenine in place of the dimethylbenzimidazole.



Methylcobalamin

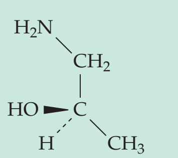
Like dimethylbenzimidazole, it is combined with ribose in the unusual α linkage. A compound called factor A is the vitamin B_{12} analog with 2-methyladenine. Related compounds have been isolated from such sources as sewage sludge which abounds in anaerobic bacteria. It has been suggested that plants may contain vitamin B_{12}-like materials which do not support growth of bacteria. Thus, we may not have discovered all of the alkyl cobalt coenzymes.

BOX 16-B COBALAMIN (VITAMIN B_{12})

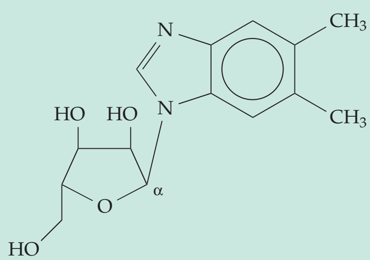


Cyanocobalamin
$M_r = 1355$
($C_{63}H_{58}N_{14}PO_{14}Co$)

The story of vitamin B_{12} began with pernicious anemia, a disease that usually affects only persons of age 60 or more but which occasionally strikes children.[a] Before 1926 the disease was incurable and usually fatal. Abnormally large, immature, and fragile red blood cells are produced but the total number of erythrocytes is much reduced from 4–6 x 10^6 mm^{-3} to 1– 3 x 10^6 mm^{-3}. Within the bone marrow mitosis appears to be blocked and DNA synthesis is suppressed. The disease also affects other rapidly growing tissues such as the gastric mucous membranes (which stop secreting HCl) and nervous tissues. Demyelination of the central nervous system with loss of muscular coordination (ataxia) and psychotic symptoms is often observed.



R-1-Amino-2-propanol



α-Ribofuranoside of dimethylimidazole

In 1926, Minot and Murphy discovered that pernicious anemia could be controlled by eating one-half pound of raw or lightly cooked liver per day, a treatment which not all patients accepted with enthusiasm. Twenty-two years later vitamin B_{12} was isolated (as the crystalline derivative cyanocobalamin) and was shown to be the curative agent. It is present in liver to the extent of 1 mg kg^{-1} or ~10^{-6} M. Although much effort was expended in preparation of concentrated liver extracts for the treatment of pernicious anemia, the lack of an assay other than treatment of human patients made progress slow.

In the early 1940s nutritional studies of young animals raised on diets lacking animal proteins and maintained out of contact with their own excreta (which contained vitamin B_{12}) demonstrated the need for "animal protein factor" which was soon shown to be the same as vitamin B_{12}. The animal feeding experiments also demonstrated that waste liquors from streptomyces fermentations used in production of antibiotics were extremely rich in vitamin B_{12}. Later this vitamin was recognized as a growth factor for a strain of *Lactobacillus lactis* which responded with half-maximum growth to as little as 0.013 μg / l (10^{-11} M).

In 1948, red cobalt-containing crystals of vitamin B_{12} were obtained almost simultaneously by two pharmaceutical firms. Charcoal adsorption from liver extracts was followed by elution with alcohol and numerous other separation steps. Later fermentation broths provided a richer source. Chemical studies revealed that the new vitamin had an enormous molecular weight, that it contained one atom of phosphorus which could be released as P_i, a molecule of aminopropanol, and a ribofuranoside of dimethyl benzimidazole with the unusual α configuration.

Note the relationship of the dimethylbenzimidazole to the ring system of riboflavin (Box 15-B). Several molecules of ammonia could be released from amide linkages by hydrolysis, but all attempts to remove the cobalt reversibly from the ring system were unsuccessful. The structure was determined in 1956 by Dorothy C. Hodgkin and coworkers using X-ray diffraction.[b] At that time, it was the largest organic structure determined by X-ray diffraction. The complete laboratory synthesis was accomplished in 1972.[c]

The ring system of vitamin B_{12}, like that of porphyrins (Fig. 16-5), is made up of four pyrrole rings whose biosynthetic relationship to the corresponding rings in porphyrins is obvious from the structures. In addition, a number of "extra" methyl groups are present. A less extensive conjugated system of double bonds is present in the **corrin** ring of vitamin B_{12} than in porphyrins, and as a result, many chiral centers are found around the periphery

BOX 16-B (continued)

of the somewhat nonplanar rings.

Cyanocobalamin, the form of vitamin B_{12} isolated initially, contains cyanide attached to cobalt. It occurs only in minor amounts, if at all, in nature but is generated through the addition of cyanide during the isolation. **Hydroxocobalamin** (vitamin B_{12a}) containing OH^- in place of CN^- does occur in nature. However, the predominant forms of the vitamin are the coenzymes in which an alkyl group replaces the CN^- of cyanocobalamin.

Intramuscular injection of as little as 3–6 µg of crystalline vitamin B_{12} is sufficient to bring about a remission of pernicious anemia and 1 µg daily provides a suitable maintenance dose (often administered as hydroxocobalamin injected once every 2 weeks). For a normal person a dietary intake of 2–5 µg / day is adequate. There is rarely any difficulty in meeting this requirement from ordinary diets. Vitamin B_{12} has the distinction of being synthesized only by bacteria, and plants apparently contain none. Consequently, strict vegetarians sometimes have symptoms of vitamin B_{12} deficiency.

Pernicious anemia is usually caused by poor absorption of the vitamin. Absorption depends upon the **intrinsic factor**, a mucoprotein (or mucoproteins) synthesized by the stomach lining.[a,d–f] Pernicious anemia patients often have a genetic predilection toward decreased synthesis of the intrinsic factor. Gastrectomy, which decreases synthesis of the intrinsic factor, or infection with fish tapeworms, which compete for available vitamin B_{12} and interfere with absorption, can also induce the disease. Also essential are a plasma membrane receptor[g,h] and two blood transport proteins **transcobalamin**[d,i] and **cobalophilin**. The latter is a glycoprotein found in virtually every human biological fluid and which may protect the vitamin from photodegradation by light that penetrates tissues.[j] A variety of genetic defects involving uptake, transport, and conversion to vitamin B_{12} coenzyme forms are known.[f,k]

Normal blood levels of vitamin B_{12} are $\sim 2 \times 10^{-10}$ M or a little more, but in vegetarians the level may drop to less than one-half this value. A deficiency of folic acid can also cause megaloblastic anemia, and a large excess of folic acid can, to some extent, reverse the anemia of pernicious anemia and mask the disease.

a Karlson, P. (1979) *Trends Biochem. Sci.* **4**, 286

b Hodgkin, D. C. (1965) *Science* **150**, 979–988

c Maugh, T. H., II (1973) *Science* **179**, 266–267

d Gräsbeck, R., and Kouvonen, I. (1983) *Trends Biochem. Sci.* **8**, 203–205

e Allen, R. H., Stabler, S. P., Savage, D. G., and Lindenbaum, J. (1993) *FASEB J.* **7**, 1344–1353

f Fenton, W. A., and Rosenberg, L. E. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 2 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 3129–3149, McGraw-Hill, New York

g Seetharam, S., Ramanujam, K. S., and Seetharam, B. (1992) *J. Biol. Chem.* **267**, 7421–7427

h Birn, H., Verroust, P. J., Nexo, E., Hager, H., Jacobsen, C., Christensen, E. I., and Moestrup, S. K. (1997) *J. Biol. Chem.* **272**, 26497–26504

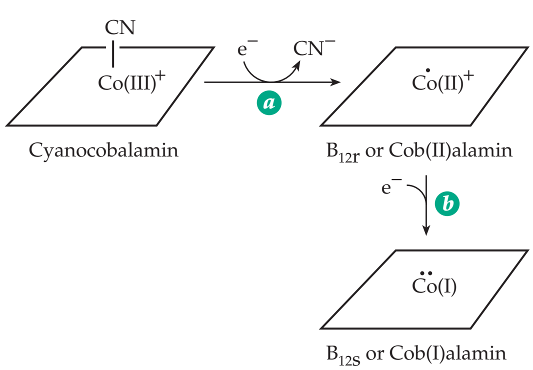
i Fedosov, S. N., Berglund, L., Nexo, E., and Petersen, T. E. (1999) *J. Biol. Chem.* **274**, 26015–26020

j Frisbie, S. M., and Chance, M. R. (1993) *Biochemistry* **32**, 13886–13892

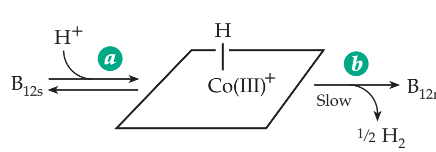
k Rosenblatt, D. S., Hosack, A., Matiaszuk, N. V., Cooper, B. A., and Laframboise, R. (1985) *Science* **228**, 1319–1321

2. Reduction of Cyanocobalamin and Synthesis of Alkyl Cobalamins

Cyanocobalamin can be reduced in two one-electron steps (Eq. 16-29).[385,388] The cyanide ion is lost in the first step (Eq. 16-29, step *a*), which may be accomplished with chromous acetate at pH 5 or by catalytic hydrogenation. The product is the brown paramagnetic compound B_{12r}, a tetragonal low-spin cobalt(II) complex. In the second step (Eq. 16-29, step *b*), an additional electron is added, e.g., from sodium borohydride or from chromous acetate at pH 9.5, to give the gray-green exceedingly reactive B_{12s}. The latter is thought to be in equilibrium with cobalt(III) hydride, as shown in Eq. 16-30, step *a*. The hydride is unstable and breaks down slowly to H_2 and B_{12r} (Eq. 16-30, step *b*).[389]

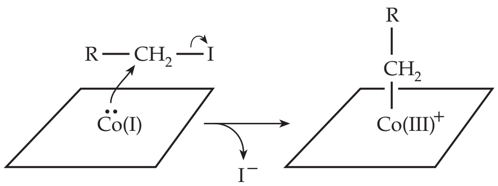


(16-29)



(16-30)

Vitamin B_{12s} reacts rapidly with alkyl iodides (e.g., methyl iodide or a 5'-chloro derivative of adenosine) via nucleophilic displacement to form the alkyl cobalt forms of vitamin B_{12} (Eq. 16-31). These reactions provide a convenient way of preparing isotopically labeled alkyl cobalamins, including those selectively

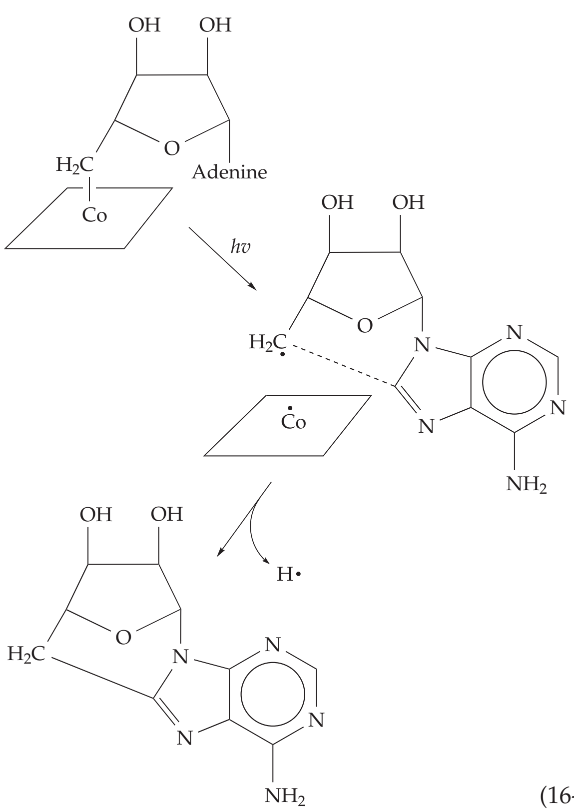


(16-31)

enriched in ^{13}C for use in NMR studies.[390] The biosynthesis of 5'-deoxyadenosylcobalamin utilizes the same type of reaction with ATP as a substrate.[391] A **B_{12s} adenosyltransferase** catalyzes nucleophilic displacement on the 5' carbon of ATP with formation of the coenzyme and displacement of inorganic tripolyphosphate PPP_i.

3. Three Nonenzymatic Cleavage Reactions of Vitamin B_{12} Coenzymes

The 5'-deoxyadenosyl coenzyme is easily decomposed by a variety of agents. Anaerobic irradiation with visible light yields principally vitamin B_{12r} and a cyclic 5',8-deoxyadenosine which is probably formed through an intermediate radical[391a] (Eq. 16-32):

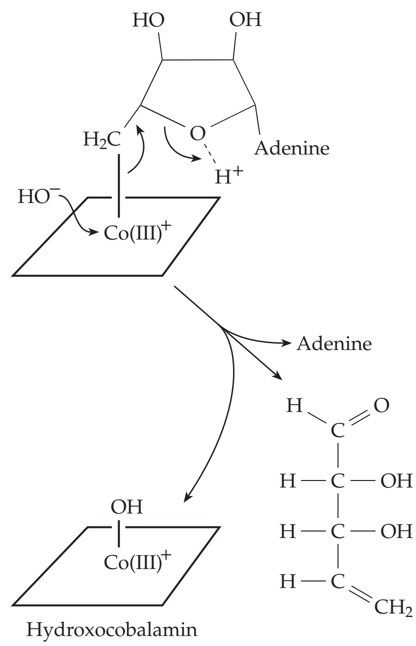


(16-32)

Irradiation in the presence of air gives a variety of products.

Hydrolysis of deoxyadenosylcobalamin by acid (1 M HCl, 100°, 90 min) yields hydroxycobalamin, adenine, and an unsaturated sugar (Eq. 16-33). The initial reaction step is thought to be protonation of the oxygen of the ribose ring.

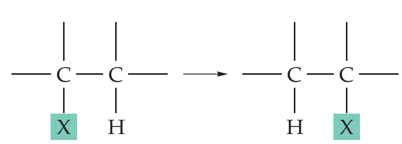
A related cleavage by alkaline cyanide can be viewed as a nucleophilic displacement of the deoxyadenosyl anion by cyanide. The end product is **dicyanocobalamin**, in which the loosely bound nucleotide containing dimethyl benzimidazole is replaced by a second cyanide ion. Methyl and other simple alkyl cobalamins are stable to alkaline cyanide. A number of other cleavage reactions of alkyl cobalamins are known.[392,393]



(16-33)

4. Enzymatic Functions of B_{12} Coenzymes

Three types of enzymatic reactions depend upon alkyl corrin coenzymes. The first is the reduction of ribonucleotide triphosphates by cobalamin-dependent ribonucleotide reductase, a process involving *intermolecular* hydrogen transfer (Eq. 16-21). The second type of reaction encompasses the series of isomerizations shown in Table 16-1. These can all be depicted as in Eq. 16-34. Some group X, which may be attached by a C–C, C–O, or C–N bond, is transferred to an adjacent carbon atom bearing a hydrogen. At the same time,



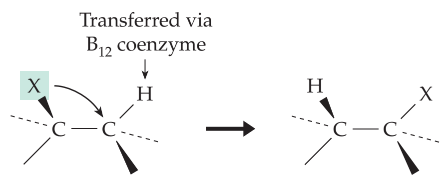
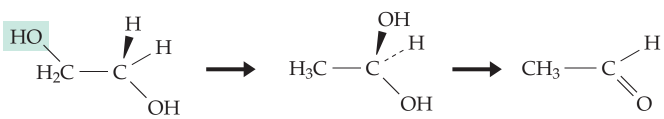
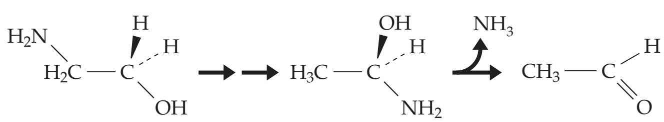
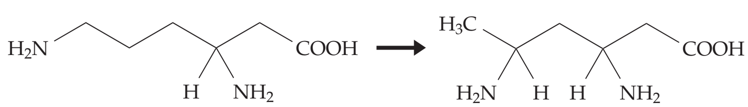
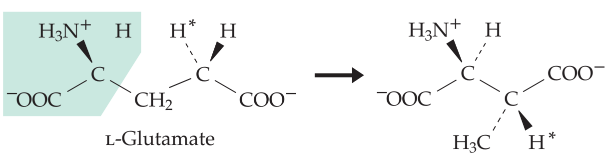
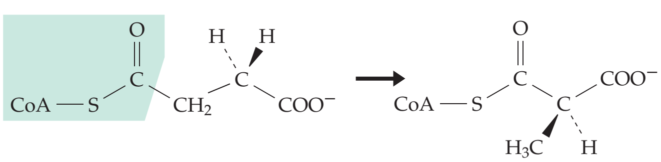
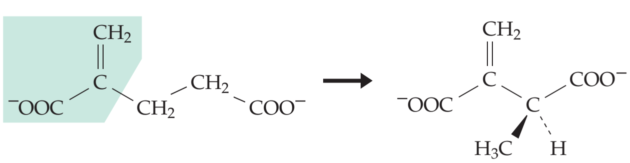
(16-34)

the hydrogen is transferred to the carbon to which X was originally attached. The third type of reaction is the transfer of methyl groups via methylcobalamin and some related bacterial metabolic reactions.

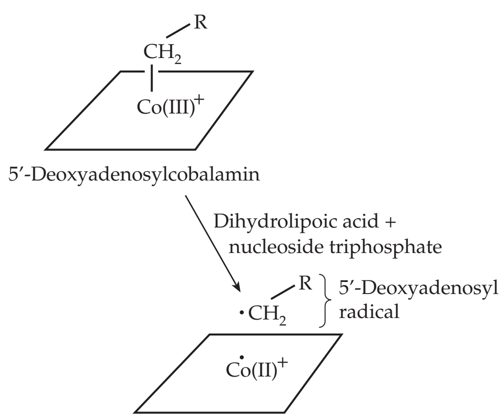
Cobalamin-dependent ribonucleotide reductase. Lactic acid bacteria such as *Lactobacillus leichmanni* and many other bacteria utilize a 5'-deoxyadenosylcobalamin-containing enzyme to reduce nucleoside triphosphates according to Eq. 16-21. Thioredoxin or dihydrolipoic acid can serve as the hydrogen donor. Early experiments showed that protons from water are reversibly incorporated at C-2' of the reduced nucleotide with retention of configuration. A more important finding was a large kinetic isotope effect of 1.8 when 3'-^{3}H-containing UTP was reduced by the enzyme.[394]

Reaction of the reductase with dihydrolipoic acid in the presence of deoxy-GTP, which apparently serves as an allosteric activator, leads to formation, within a few milliseconds, of a radical with a characteristic EPR spectrum that can be studied when the reaction mixture is rapidly cooled to 130°K. When GTP (a true substrate) is used instead of dGTP, the radical signal reaches a maximum in about 20 ms and then decays. Of the various oxidation states of cobalt (3+, 2+, and 1+)

TABLE 16-1
Isomerization Reactions Involving Hydrogen Transfer and Dependent upon a Vitamin B_{12} Coenzyme

	General reaction Migrating group is enclosed in a box
	Dioldehydratase Glycerol dehydratase catalyzes the same type of reaction
	Ethanolamine ammonia-lyase
	L-β-Lysine mutase D-α-Lysine mutase and ornithine mutase catalyze the same type of reaction
 L-Glutamate → *threo*-L-β-Methylaspartate	Glutamate mutase
 (R) Methylmalonyl-CoA	Methylmalonyl-CoA mutase Isobutyryl-CoA mutase catalyzes the same type of reaction
	α-Methyleneglutarate mutase

only the 2+ state of vitamin B_{12r} is paramagnetic and gives rise to an EPR signal. The electronic absorption spectrum of the coenzyme of ribonucleotide reductase is also changed rapidly by substrate in a way that suggests formation of B_{12r}. Thus, it was proposed that a *homolytic* cleavage occurs to form B_{12r} and a stabilized 5'-deoxyadenosyl radical (Eq. 16-35).[395–396a] However, H^3 is not transferred from the 3' position of the substrate into the deoxyadenosyl part of the coenzyme.[394] The enzyme has many properties in common with the previously discussed iron–tyrosinate ribonucleotide reductases (Fig. 16-21) including limited peptide sequence homology.[350] Stubbe and coworkers suggested that the deoxyadenosyl radical is formed as a radical chain initiator[394,396b] and that the mechanism of ribonucleotide reduction is as shown in Eq. 16-25. Studies of enzyme-activated inhibitors support this mechanism.[364]



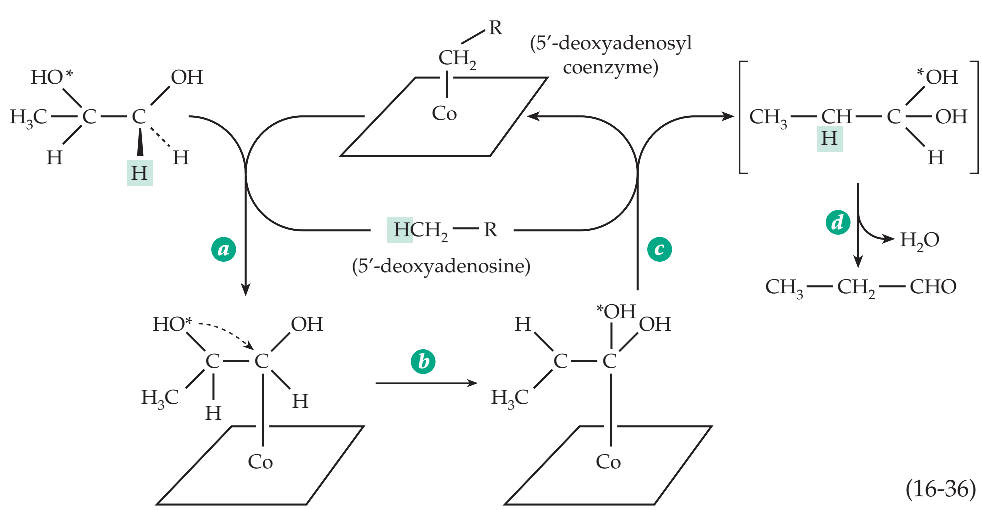
(16-35)

The isomerization reactions. At least 10 reactions of the type described by Eq. 16-34 are known[397] (Table 16-1). They can be subdivided into three groups. First, X = OH or NH_2 in Eq. 16-34; isomerization gives a *geminal*-diol or aminoalcohol that can eliminate H_2O or NH_3 to give an aldehyde. All of these enzymes, which are called **hydro-lyases** or **ammonia-lyases**, specifically require K^+ as well as the vitamin B_{12} coenzyme. Second, X = NH_2 in Eq. 16-34; For this group of **aminomutases** PLP is required as a second coenzyme. Third, X is attached via a carbon atom; the enzymes are called **mutases**. **Methylmalonyl-Co mutase** is required for catabolism of propionate in the human body, and is one of only two known vitamin B_{12}-dependent enzymes. The related isobutyryl-CoA mutase participates in the microbial synthesis of such polyether antibiotics as monensin A.[397a,b] Other mutases are involved in anaerobic bacterial fermentations.

In these reactions the hydrogen is always transferred via the B_{12} coenzyme. No exchange with the medium takes place. Since X may be an electronegative group such as OH, the reactions could all be treated formally as hydride ion transfers but it is more likely that they occur via homolytic cleavages. Such cleavage is indicated by the observation of EPR signals for several of the enzymes in the presence of their substrates.[398]

Abeles and associates showed that when **dioldehydratase** (Table 16-1) catalyzes the conversion of 1,2-[1-^{3}H]propanediol to propionaldehyde, tritium appears in the coenzyme as well as in the final product. When ^{3}H-containing coenzyme is incubated with unlabeled propanediol, the product also contains ^{3}H, which was shown by chemical degradation to be exclusively on C-5'. Synthetic 5'-deoxyadenosyl coenzyme containing ^{3}H in the 5' position transferred ^{3}H to product. Most important, using a mixture of propanediol and ethylene glycol, a small amount of *intermolecular* transfer was demonstrated; that is, ^{3}H was transferred into acetaldehyde, the product of dehydration of ethylene glycol. Similar results were also obtained with ethanolamine ammonia-lyase.[399]

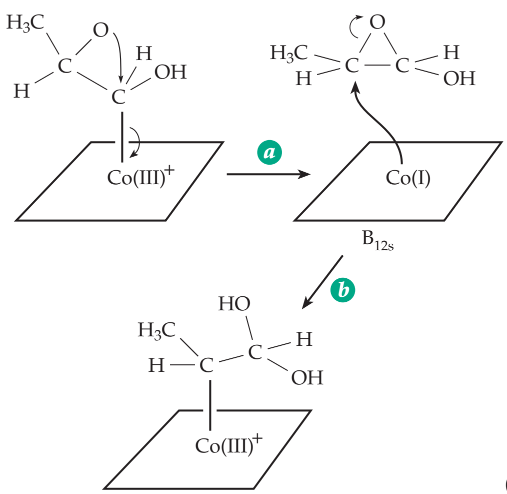
Another important experiment[398] showed that ^{18}O from [2-^{18}O]propanediol was transferred into the 1 position without exchange with solvent. Furthermore, ^{18}O from (*S*)-[1-^{18}O]propanediol was retained in the product while that from the (*R*) isomer was not. Thus, it appears that the enzyme stereospecifically dehydrates the final intermediate. From these and other experiments, it was concluded that initially a 5'-deoxyadenosyl radical is formed via Eq. 16-35. This radical then abstracts the hydrogen atom marked by a shaded box in Eq. 16-36 to form a substrate radical and 5'-deoxyadenosine. One proposal, illustrated in Eq. 16-36, is that the substrate radical immediately recombines with



(16-36)

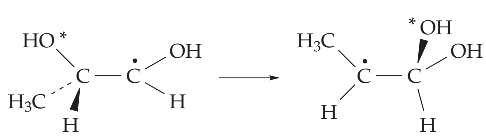
the Co(II) of the coenzyme to form an organo-cobalt substrate compound (step *a* of Eq. 16-36). The 5'-deoxyadenosine now contains hydrogen from the substrate; because of rotation of the methyl group this hydrogen becomes equivalent to the two already present in the coenzyme. The substrate–cobalamin compound formed in this step then undergoes isomerization, which, in the case of dioldehydratase, leads to intramolecular transfer of the OH group (step *b*). In step *c* the hydrogen atom is transferred back from the 5'-deoxyadenosine to its new location in the product and in step *d* the resulting *gem*-diol is dehydrated to form the aldehyde product.[401a]

Carbocation, carboanion, and free radical intermediates have all been proposed for the isomerization step in the reaction. A carbocation would presumably be cyclized to an epoxide which could react with the B_{12s} (Eq. 16-37, step *b*) to complete the isomerization.



(16-37)

At present most evidence favors, for the isomerization reactions, an enzyme-catalyzed rearrangement of the substrate radical produced initially during formation of the 5'-deoxyadenosine (Eq. 16-38).[400–401c]

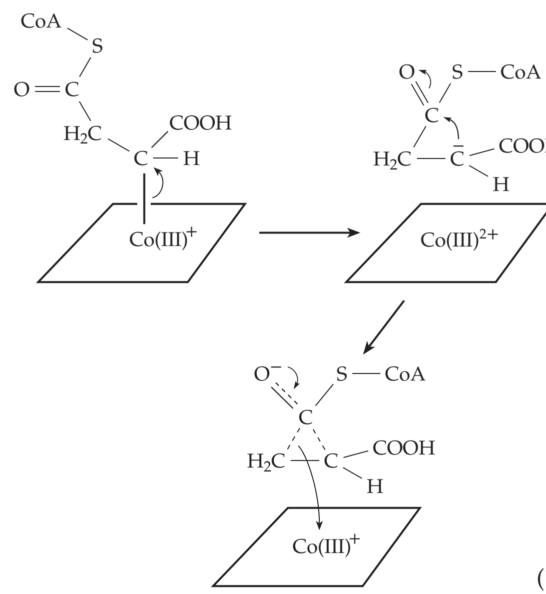


(16-38)

According to this mechanism the Co(II) of the B_{12r} formed in Eq. 16-35 has no active role in the isomerization and does not form an organocobalt intermediate as in Eq. 16-36. Its only role is to be available to recombine with the 5'-deoxyadenosyl radical at the end of the reaction sequence. Support for this interpretation has been obtained from study of model reactions and of organic radicals generated in other ways.[400]

Recently, EPR spectroscopy with ^{2}H- and ^{13}C-labeled glutamates as substrates for glutamate mutase permitted identification of a 4-glutamyl radical as a probable intermediate for that enzyme.[402,402a,b]

On the other hand, for methylmalonyl-CoA mutase Ingraham suggested cleavage of the Co–C bond of an organocobalt intermediate to form a carbanion, the substrate–cobalamin compound serving as a sort of "biological Grignard reagent." The carbanion would be stabilized by the carbonyl group of the thioester forming a "homoenolate anion" (Eq. 16-39). The latter could break up in either of two ways reforming a C–Co bond and causing the isomerization.[403] Some experimental results also favor an ionic or organocobalt pathway.[404]



(16-39)

The three-dimensional structure of methylmalonyl-CoA mutase from *Propionibacterium shermanii* shows that the vitamin B_{12} is bound in a base-off conformation with the dimethylbenzimidazole group bound to the protein far from the corrin ring (Fig. 16-23; see also Fig. 16-24).[405,405a] A histidine of the protein coordinates the cobalt, as also in methionine synthase. The entrance to the deeply buried active site is blocked by the coenzyme part of the substrate. The buried active site may be favorable for free radical rearrangement reactions. The structure of substrate complexes shows that the coenzyme is in the cob(II)alamin (B_{12r}) form with the 5'-deoxyadenosyl group detached from the cobalt, rotated, shifted, and weakly bound to the protein.[405a] Side chains from neighboring Y89, R207, and H244 all hydrogen bond to the substrate. The R207 guanidinium group makes an ion pair with the substrate carboxylate, and the phenolic and imidazole groups may have catalytic functions.[405b,c,d,406] Studies are also in progress on crystalline glutamate mutase.[406,406a]

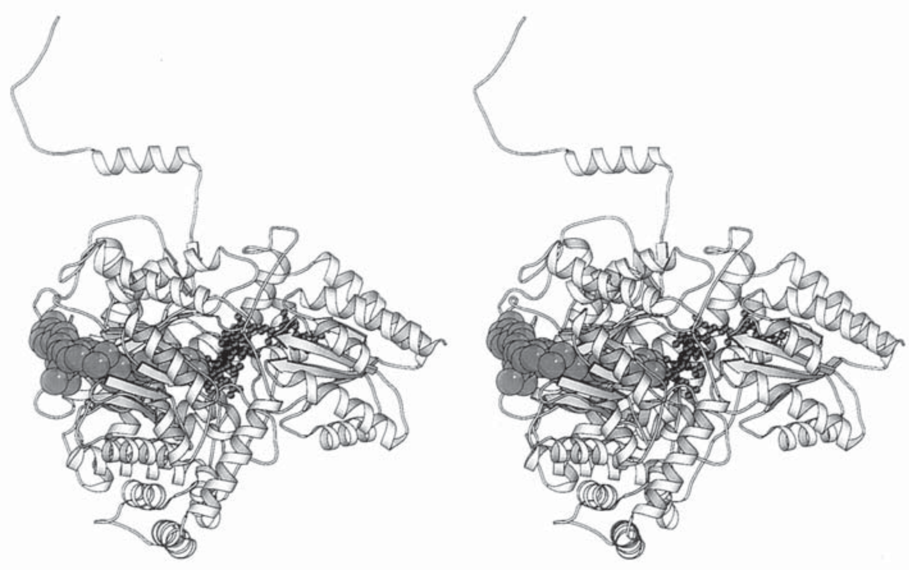
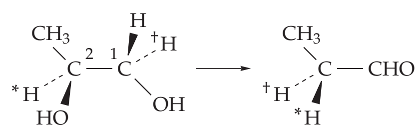


Figure 16-23 Three-dimensional structure of methylmalonyl-CoA mutase from *Propionobacterium shermanii*. The B_{12} coenzyme is deeply buried, as is the active site. A molecule of bound desulfo-coenzyme A, a substrate analog, blocks the active site entrance on the left side. From Mancia *et al*.[405] Courtesy of Philip R. Evans.

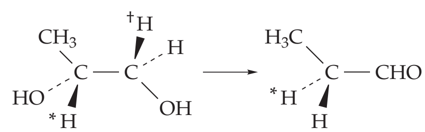
According to all of these mechanisms, 5'-deoxyadenosine is freed from its bond to cobalt during the action of the enzymes. Why then does the deoxyadenosine not escape from the coenzyme entirely, leading to its inactivation? Substrate-induced inactivation is not ordinarily observed with coenzyme B_{12}-dependent reactions, but some quasi-substrates do inactivate their enzymes. Thus, glycolaldehyde converts the coenzyme of dioldehydratase to 5'-deoxyadenosine and ethylene glycol does the same with ethanolamine deaminase. When 5'-deoxyinosine replaces the 5'-deoxyadenosine of the normal coenzyme in dioldehydratase, 5'-deoxyinosine is released quantitatively by the substrate. This suggests that the dehydratase may normally hold the adenine group of 5'-deoxyadenosine through hydrogen bonding to the amino group. Because the OH group of inosine tautomerizes to C=O, inosine may not be held as tightly. The deeply buried active site (Fig. 16-23) may also prevent escape of the deoxyadenosine.

Despite the evidence in its favor, there was initially some reluctance to accept 5'-deoxyadenosine as an intermediate in vitamin B_{12}-dependent isomerization reactions. It was hard to believe that a methyl group could exchange hydrogen atoms so rapidly. It was suggested that protonation of the oxygen of the ribose ring as in Eq. 16-33 might facilitate release of a hydrogen atom. However, substitution of the ring oxygen by CH_2 in a synthetic analog did not destroy the coenzymatic activity.[407] Another possibility is that the methyl group has an unusual reactivity if the cobalt is reduced to Co(II).

Stereochemistry of the isomerization reactions. Dioldehydratase acts on either the (*R*) or (*S*) isomers of 1,2-propanediol (Eqs. 16-40 and 16-41; asterisks and daggers mark positions of labeled atoms in specific experiments).



(*S*)-1,2-Propanediol (16-40)

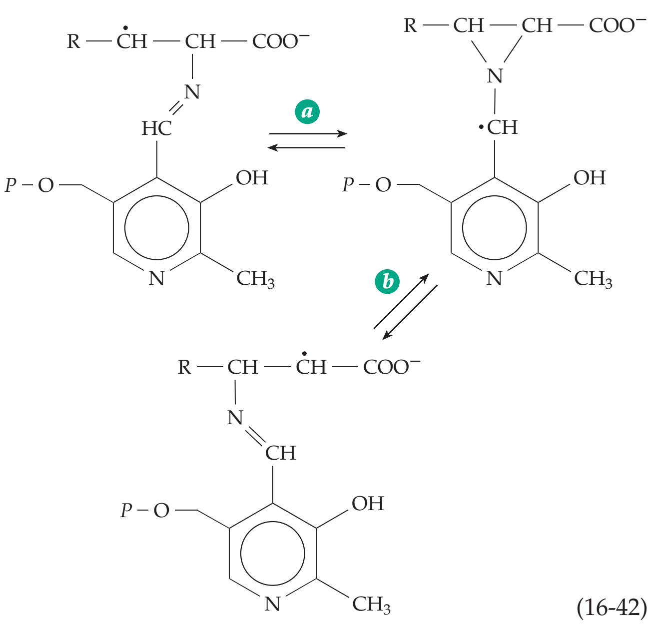


(*R*)-1,2-Propanediol (16-41)

In both cases the reaction proceeds with retention of configuration at C-2 and with stereochemical specificity[408] for one of the two hydrogens at C-1. The reaction catalyzed by methylmalonyl-CoA mutase likewise proceeds with retention of configuration at C-2 (Table 16-1)[409] but the glutamate mutase reaction is accompanied by inversion (Eq. 16-28).

Aminomutases. The enzymes **L-β-lysine mutase** (which is also **D-α-lysine mutase**) and **D-ornithine mutase** catalyze the transfer of an ω-amino group to an adjacent carbon atom[410] (Table 16-1). Two proteins are needed for the reaction; pyridoxal phosphate is required and is apparently directly involved in the amino group migration. In the β-lysine mutase the 6-amino group of L-β-lysine replaces the pro-*S* hydrogen at C-5 but with inversion at C-5 to yield (3*S*, 5*S*)-3,5-diaminohexanoic acid.[411] A bacterial D-lysine 5,6-aminomutase interconverts D-lysine with 2,5-diaminohexanoic acid.[411a] Another related enzyme is **L-leucine 2,3-aminomutase**, which catalyzes the reversible interconversion of L-leucine and β-leucine.[410,412] It was reported to be present in plants and also in the human body, but the latter could not be confirmed.[413]

The interconversion of L-α-lysine and L-β-lysine is catalyzed by a lysine 2,3-aminomutase found in certain clostridia.[414-415a] This enzyme also requires pyridoxal phosphate and catalyzes a reaction with the same stereochemistry as that of β-lysine mutase. However, it does not contain vitamin B_{12} but depends upon *S*-adenosylmethionine (AdoMet) and an iron–sulfur cluster. The adenosyl group of AdoMet may function in the same manner as does the deoxyadenosyl group of the adenosylcobalamin coenzyme. In these mutases a 5'-deoxyadenosyl radical may abstract a hydrogen atom from the β position of a Schiff base of PLP with the amino acid substrate. The radical isomerizes (Eq. 16-42) and then accepts a hydrogen atom back from the 5'-deoxyadenosine to complete the reaction. Its



(16-42)

properties suggest that lysine 2,3-aminomutase is related to the pyruvate formate-lyase of *E. coli* (Eq. 15-37),[410] class I ribonucleotide reductases, and other enzymes that act by homolytic mechanisms.

Transfer reactions of methyl groups. The generation and utilization of methyl groups is a quantitatively important aspect of the metabolism of all cells. As we have seen (Fig. 15-18), methyl groups can be created by the reduction of one-carbon compounds attached to tetrahydrofolic acid. Methyl groups of methyltetrahydrofolic acid (N^5-CH_3THF) can then be transferred to the sulfur atom of homocysteine to form methionine (Eq. 16-43). The latter is converted to *S*-adenosylmethionine, the nearly universal methyl group donor for transmethylation reactions (Eq. 12-3). In some bacteria, fungi, and higher plants, the methyl-THF-homocysteine transmethylase does not depend upon vitamin B_{12} but is a metalloenzyme with zinc at the active center.[416] However, human beings share with certain strains of *E. coli* and other bacteria the need for methylcobalamin.

The structure of the *E. coli* enzyme (Fig. 16-24) shows methylcobalamin bound in a base-off conformation, with histidine 759 of the protein replacing dimethylbenzimidazole in the distal coordination position on the cobalt. This histidine is part of a sequence Asp-X-His-X-X-Gly that is found not only in methionine synthase but also in methylmalonyl-CoA mutase, glutamate mutase, and 2-methyleneglutarate mutase. However, diol dehydratase lacks this sequence and binds adenosylcobalamin with the dimethylbenzimidazole–cobalt bond intact.[417]

The coenzyme evidently functions in a cyclic process. The cobalt alternates between the +1 and +3 oxidation states as shown in Eq. 16-43. The first indication of such a cyclic process was the report by Weissbach that ^{14}C-labeled methylcobalamin could be isolated following treatment of the enzyme with such methyl donors as AdoMet and methyl iodide

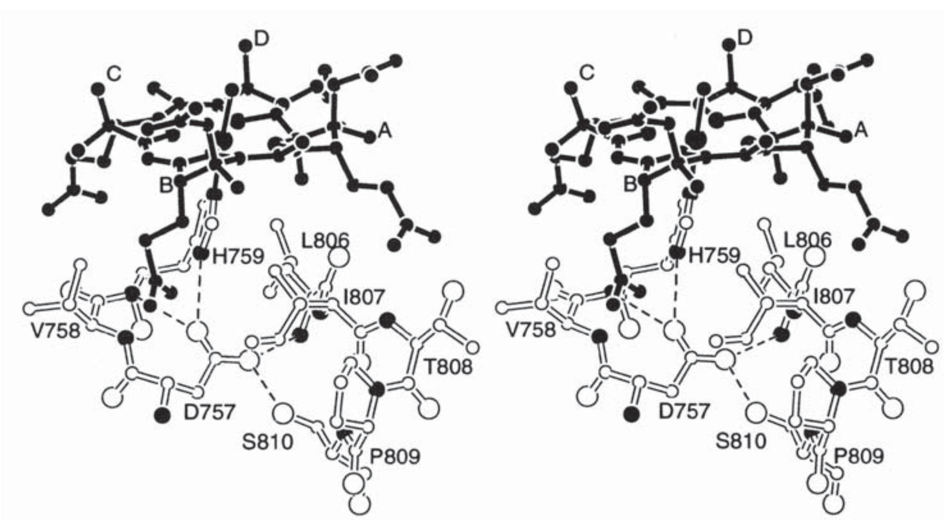
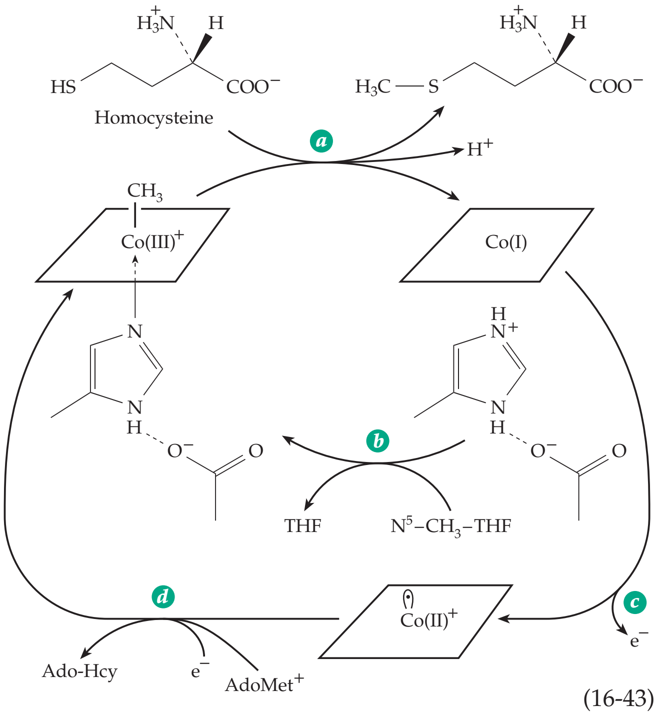


Figure 16-24 Stereoscopic views of the active site of methionine synthase from *E. coli*. Methylcobalamin (black) is in the active site with His 759 of the protein in the distal position of the coenzyme in a base-off conformation. The dimethylbenzimidole nucleotide has been omitted for clarity. Notice the hydrogen-bonded His 759 – D757 – S810 triad. From Jarret *et al*.[418a] Courtesy of R. G. Matthews.

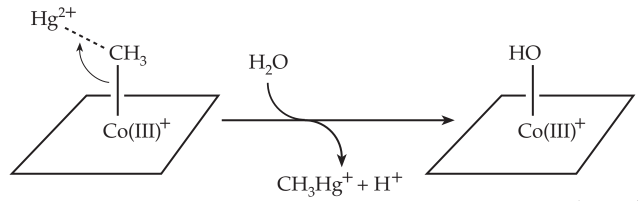


(16-43)

after reduction (e.g., with reduced riboflavin phosphate). The sequence parallels that of Eq. 16-31 for the laboratory synthesis of methylcobalamin. Nevertheless, the transmethylase demonstrates some complexities. Initially, it must be "activated" by AdoMet or methyl iodide, after which it cycles according to Eq. 16-43, steps *a* and *b,* but is gradually inactivated. This apparently happens by oxidation to a Co(II) form of the enzyme (Eq. 16-43, step *c*) that must be reductively methylated with AdoMet and reduced flavodoxin (step *d*) to regenerate the active form.[418] It is also possible that the methyl group is not transferred as a formal CH_3^+ as pictured in Eq. 16-43 but as a $^\bullet CH_3$ radical generated by homolytic cleavage of methylcobalamin to cob(II)alamin. Whatever the mechanism, a chiral methyl group is transferred from 5-methyl-THF to homocysteine with overall retention of configuration.[419]

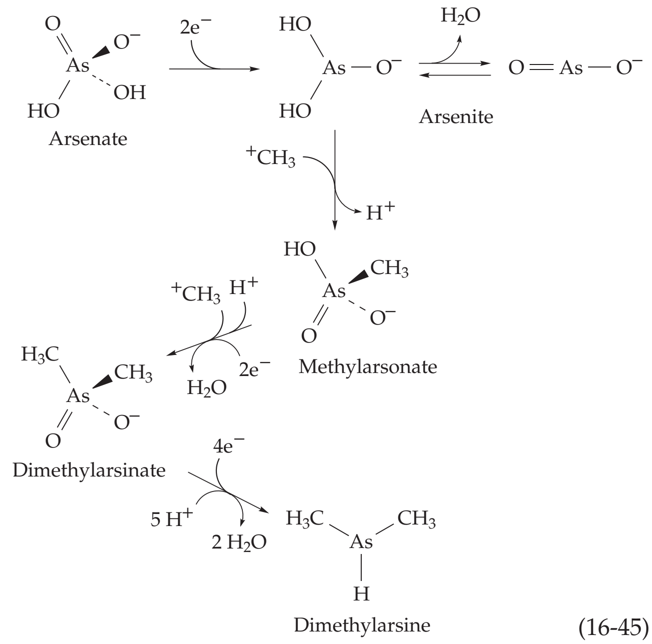
Another important group of methyl transfer reactions are those from methyl corrinoids to mercury, tin, arsenic, selenium, and tellurium. For example, Eq. 16-44 describes the methylation of Hg^{2+}. These reactions are of special interest because of the generation of toxic methyl and dimethyl mercury and dimethylarsine.

Notice that whereas in Eq. 16-43 the methyl group is transferred as CH_3^+ by nucleophilic displacement on a carbon atom, the transfer to Hg^{2+} in Eq. 16-44 is that of a carbanion, CH_3^-, with no valence change occurring in the cobalt. However, it is also possible that transfer occurs as a methyl radical.[420] Methyl corrinoids are able to undergo this type of reaction nonenzymatically, and the ability to transfer a methyl anion is a property of methyl corrinoids not shared by other transmethylating agents such as AdoMet. At the conclusion of the reaction in Eq. 16-44 the cobalt is in the +3 state. To be remethylated, it must presumably be reduced to Co(II). A second methyl group can be transferred by the same type of reaction to form $(CH_3)_2Hg$.



(16-44)

Methylation of arsenic is an important pollution problem because of the widespread use of arsenic compounds in insecticides and because of the presence of arsenate in the phosphate used in household detergents.[421,422] After reduction to arsenite, methylation occurs in two steps (Eq. 16-45). Additional reduction steps result in the formation of **dimethylarsine**, one of the principal products of action of methanogenic bacteria on arsenate. The methyl transfer is shown as occurring through CH_3^+, with an accompanying loss of a proton from the substrate. However, a CH_3 radical may be transferred with formation of a cobalt(II) corrinoid.[423]



(16-45)

Corrinoid-dependent synthesis of acetyl-CoA. The anaerobic bacterium *Clostridium thermoaceticum* obtains its energy for growth by reduction of CO_2 with hydrogen (Eq. 16-46). One of the CO_2 molecules is reduced to formate which is converted via 5-methyl-THF to the methyl corrinoid 5-methoxybenzimidazolylcobamide. The methyl group of the latter

$$4\ H_2 + 2\ CO_2 \longrightarrow CH_3COO^- + H^+ + 2\ H_2O$$
$$\Delta G'\ (\text{pH } 7) = -94.9\ \text{kJ mol}^{-1} \quad (16\text{-}46)$$

combines with another CO_2 to form acetyl-CoA. The process, which requires the nickel-containing carbon monoxide dehydrogenase, is discussed in Section C.

C. Nickel

It was not until the 1970s that nickel was first recognized as a dietary essential for animals.[424–426] Nickel-deficient chicks grew poorly, had thickened legs, and developed dermatitis. Tissues of deficient animals contained swollen mitochondria and swollen perinuclear space suggesting a function for Ni in membranes. However, doubts have been raised about the conclusions based on these experiments.[427] Within tissues the nickel content ranges from 1 to 5 µg / l. Some of the metal in serum is present as complexes of low molecular mass and some is bound to serum albumin[428] and to a specific nickel-containing protein of the macroglobulin class, known as **nickeloplasmin**.[429] Nickel is also present in plants; in some, e.g., *Allysum*, it accumulates to high concentrations.[430] It is essential for legumes and possibly for all plants.[431] Nickel uptake proteins have been identified in bacteria and fungi.[432,432a] Because of its ubiquitous occurrence it is difficult to prepare a totally Ni-free diet. The acute toxicity of orally ingested Ni(II) is low, and homeostatic mechanisms exist in the animal body for regulating its concentration. However, the volatile nickel carbonyl $Ni(CO)_4$ is very toxic[426] and Ni from jewelry is a common cause of dermatitis.[428,433]

In its compounds nickel usually has the +2 oxidation state but the +3 and +4 states occur rarely in complexes. The Ni^{2+} ion contains eight 3d electrons, a configuration that favors square-planar coordination of four ligands. However, the ion is also able to form a complex with six ligands and an octahedral geometry. It has been suggested that this "ambivalence" may be of biochemical significance.

Nickel is found in at least four enzymes: **urease**, certain **hydrogenases, methyl-CoM reductase** (in its **cofactor F_{430}**) of methanogenic bacteria, and **carbon monoxide dehydrogenase** of acetogenic and methanogenic bacteria.[434]

1. Urease

Urease, which was first isolated from the jack bean has a special place in biochemical history as the first enzyme to be crystallized. This was accomplished by J. B. Sumner in 1926, and although Sumner eventually obtained the Nobel Prize, his first reports were greeted with skepticism and outright disbelief. The presence of two atoms of nickel in each molecule of urease[435] was not discovered until 1975. The metal ions had been overlooked previously, despite the fact that the absorption spectrum of the purified enzyme contains an absorption "tail" extending into the visible region with a shoulder at 425 nm and weak maxima at 725 and 1060 nm. Urease catalyzes the hydrolytic cleavage of urea to two molecules of ammonia and one of bicarbonate and is useful in the analytical determination of urea.

Jack bean urease is a trimer or hexamer of identical 91-kDa subunits while that of the bacterium *Klebsiella* has an $(\alpha\beta_2\gamma_2)_2$ stoichiometry. Nevertheless, the enzymes are homologous and both contain the same binickel catalytic center (Fig. 16-25).[435–437a] The three-dimensional structure of the *Klebsiella* enzyme revealed that the two nickel ions are bridged by a carbamyl group of a carbamylated lysine. Like ribulose bisphosphate carboxylase (Fig. 13-10), urease also requires CO_2 for formation of the active enzyme.[438] Formation of the metallocenter also requires four additional proteins, including a chaperonin and a nickel-binding protein.[438,439]

The mechanism of urease action is probably related to those of metalloproteases such as carboxypeptidase A (Fig. 12-16) and of the zinc-dependent carbonic

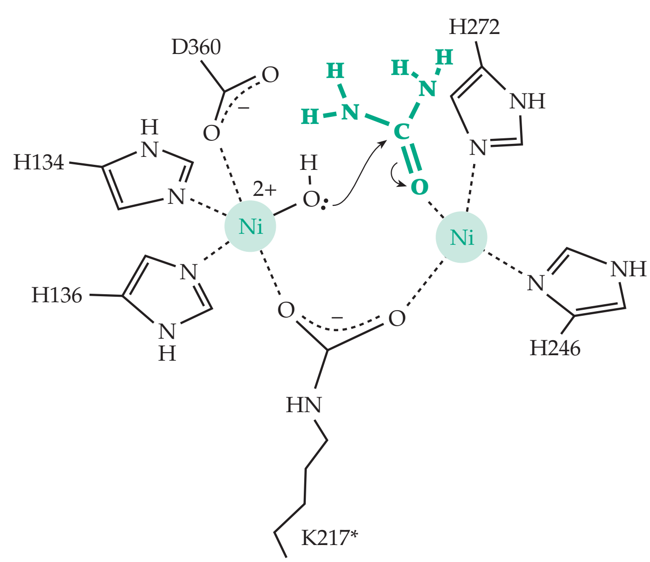
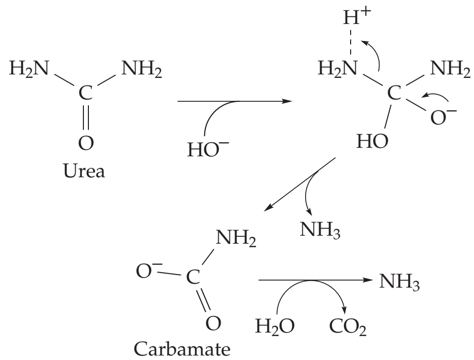


Figure 16-25 The active site of urease showing the two Ni^+ ions held by histidine side chains and bridged by a carbamylated lysine (K217*). A bound urea molecule is shown in green. It has been placed in an open coordination position on one nickel and is shown being attacked for hydrolytic cleavage by a hydroxyl group bound to the other nickel. Based on a structure by Jabri *et al.*[436] and drawing by Lippard.[437]



(16-47)

anhydrase (Fig. 13-1). In Fig. 16-25 one nickel ion is shown polarizing the carbonyl group, while the second provides a bound hydroxyl ion that serves as the attacking nucleophile. A probable intermediate product is a carbamate ion (Eq. 16-47).

Urease is an essential enzyme for bacteria and other organisms that use urea as a primary source of nitrogen. The peptic ulcer bacterium *Helicobacter pylori* uses urease to hydrolyze urea in order to defend against the high acidity of the stomach.[439a] The enzyme is also present in plant leaves and may play a necessary role in nitrogen metabolism.[431] In nitrogen-fixing legumes urea derivatives, the **ureides**, have an important function (Chapter 25) but urease may not be involved in the catabolism of these compounds.[440]

2. Hydrogenases

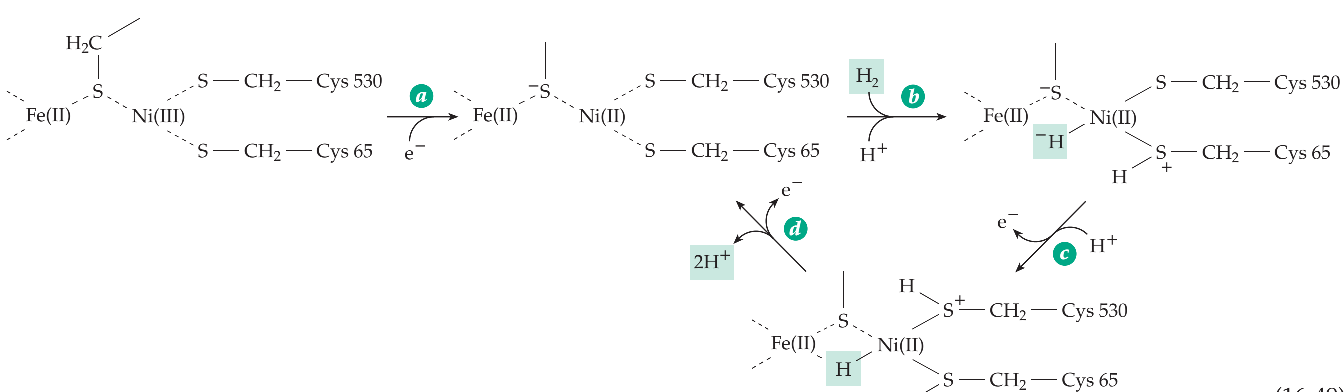
Many plants, animals, and microorganisms are able to evolve H_2 by reduction of hydrogen ions (Eq. 16-48) or to oxidize H_2 by the reverse of this reaction.[441,442]

$$2H^+ + 2e^- \rightarrow H_2 \qquad (16\text{-}48)$$

Hydrogenases have been classified into two main types: **Fe-hydrogenases**, which contain iron as the only metal,[443] and **Ni-hydrogenases**, which contain both iron and nickel.[444] In a few Ni-hydrogenases a selenocysteine residue replaces a conserved cysteine side chain.[445,446] Fe-hydrogenases are often extremely active and are utilized to rid organisms of an excess of electrons by evolution of H_2. Since they may also be used to acquire electrons by oxidation of H_2, they are often described as *bidirectional*. Ni-hydrogenases, as well as some Fe-hydrogenases, are involved primarily in *uptake* of H_2.[447] All hydrogenases contain one or more Fe–S centers in addition to the H_2-forming catalytic center.[441] Some hydrogenases are membrane bound and are often coupled through unidentified carriers to formate dehydrogenase (Chapter 17). In the strict anaerobes such as clostridia, hydrogenases are linked to ferredoxins. Hydrogenases are inactivated readily by O_2, which oxidizes the catalytic centers but can sometimes be reactivated by treatment with reducing agents.[448]

The 60-kDa all-iron monomeric hydrogenase I of *Clostridium pasteurianum* (mentioned on p. 861) contains ferredoxin-like Fe_4S_4 clusters plus additional Fe and sulfur atoms organized as a special H cluster. The EPR spectrum of the catalytic center, recognized because the spectrum is altered by the binding of carbon monoxide, is unusual. Its *g* values of 2.00, 2.04, and 2.10 are similar to those of oxidized high-potential iron proteins.[449] The *C. pasteurianum* hydrogenase II is a 53-kDa monomer containing eight Fe and eight S^{2-} ions. These are organized into one ferredoxin-like Fe_4S_4 cluster plus a three-Fe cluster and one iron ion in a unique environment.[449]

In contrast to these iron-only hydrogenases, the large periplasmic hydrogenase of *Desulfovibrio gigas* consists of one 28-kDa subunit and one 60-kDa subunit and contains two Fe_4S_4 clusters, one Fe_3S_4 cluster, and another dimetal center containing a single atom of Ni.[450,451] These are seen clearly in the three-dimensional structure depicted in Fig. 16-26. The three Fe–S clusters, at 5- to 6-nm intervals, form a chain from the external surface to the deeply buried nickel–iron center.[450–453a] A plausible pathway for transport of protons to the active center can also be seen.[450,451] The nickel center also contains an atom of Fe. Four cysteine side chains participate in forming the Ni–Fe cluster, two of them provide sulfur atoms that bridge between the metals while two others are ligands to Ni (Fig. 16-26).



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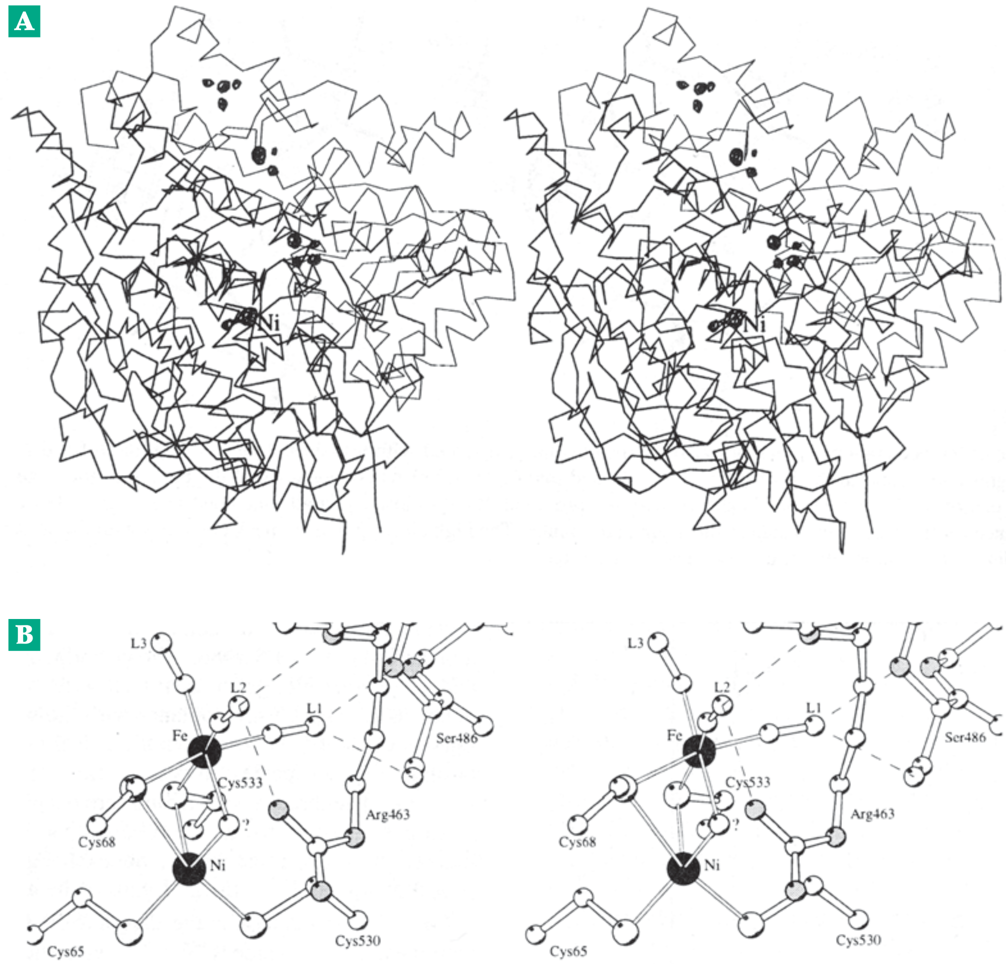


Figure 16-26 (A) Stereoscopic view of the structure of the *Desulfovibrio gigas* hydrogenase as an α-carbon plot. The electron density map at the high level of 8σ is superimposed and consists of dark spheres representing the Fe and Ni atoms. The iron atoms of the two Fe_4S_4 and one Fe_3S_4 clusters are seen clearly forming a chain from the surface of the protein to the Ni–Fe center. (B) The structure of the active site Ni–Fe pair. The two metals are bridged by two cysteine sulfur atoms and an unidentified atom, perhaps O, and the nickel is also coordinated by two additional cysteine sulfurs. Unidentified small molecules L1, L2, and L3 are also present. From Volbeda *et al.*[453] Courtesy of M. Frey.

A hydrogenase from *chromatium vinosum* has a similar structure.[453b] There are still uncertainties about other nonprotein ligands such as H_2O.[452–453a]

All of the Ni-hydrogenases display an EPR signal that can be assigned to Ni(III).[452] However, the active enzyme from *D. gigas* contains Ni(II). A proposed mechanism[452] is indicated in Eq. 16-49. Step *a* of this equation is a reductive activation. In step *b* a molecule of H_2 is bound as a hydride ion on Ni and a proton on a nearby sulfur. Protonation of a second sulfur ligand to Ni is needed to promote the cleavage of H_2 prior to the two-step oxidation of the bound H^-. One of two Ni-containing hydrogenases of *Methanobacterium thermoautotrophicum* contains FAD as well as Fe–S clusters.[454] It specifically reduces the 5-deazaflavin cofactor F_{420} (Chapter 15). A major function of this deazaflavin is reduction of the nickel-containing cofactor F_{430}.

3. Cofactor F_{430} and Methyl-Coenzyme M Reductase

Cofactor F_{430} is a nickel tetrapyrrole with a structure (Fig. 16-27)[455,456] similar to those of vitamin B_{12} and of siroheme. The tetrapyrrole ring is the most highly reduced in cofactor F_{430}, which functions in reduction of methyl-CoM to methane in methanogens (Fig. 16-28). The methyl CoM reductase of *Methanobacterium thermoautotrophicum* is a large 300-kDa protein with subunit composition $\alpha_2\beta_2\gamma_2$ and containing two molecules of bound F_{430}. The nickel in F_{430} is first thought to be reduced, in an activation step, to Ni(I),[456a] which may attack the methyl group of methyl-CoM homolytically to yield a methyl nickel complex and a sulfur radical.[457,458] Alkyl nickel compounds react with protons, and in this case they would yield methane and would regenerate the Ni(II) form of the cofactor. The CoM radical could be reduced back to free CoM.

High-resolution crystal structures of the enzyme in two inactive Ni(II) forms[458] show the two F_{430} molecules. Each is bound in an identical channel about 3 nm in length and extending from the surface deep into the interior of the protein. The F_{430} lies at the bottom of this channel with its nickel atom coordinated with the oxygen atom of a glutamine side chain. In one form CoM lies directly above the nickel, with its

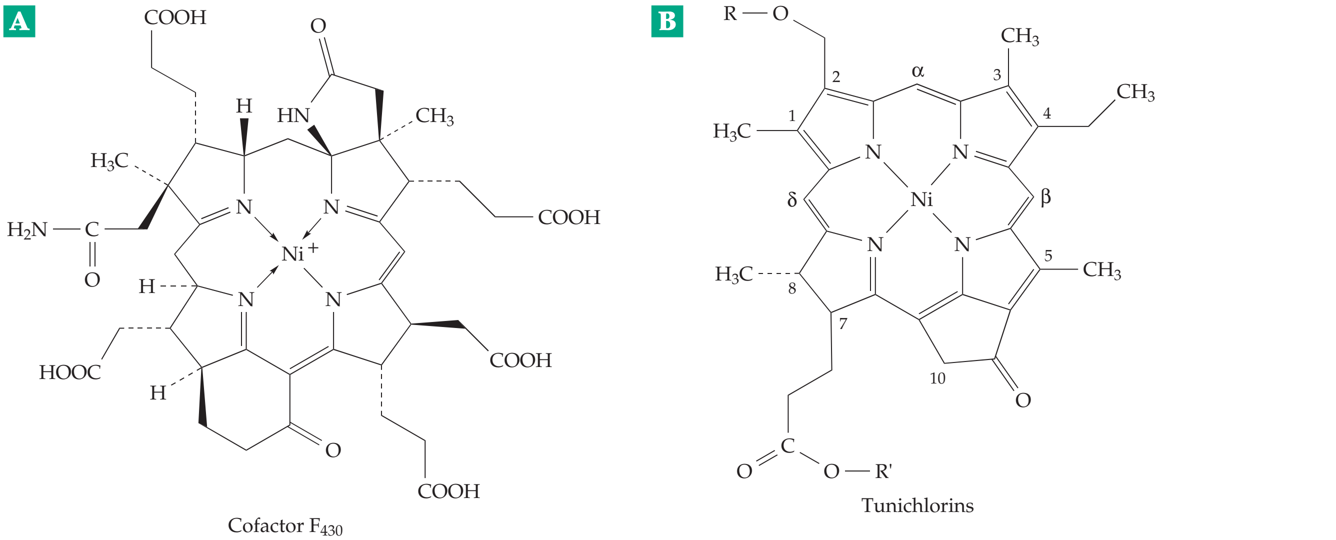


Figure 16-27 (A) Structure of the nickel-containing prosthetic group F_{430} as isolated in the esterified (methylated) form. From Pfaltz *et al.*[459] The "front" face, which reacts with methyl-coenzyme M, is toward the reader.[458] (B) Structure of a representative member of a family of tunichlorins isolated from marine tunicates.[460] For tunichlorin R = R′ = H. Related compounds have R′ = CH_3 and/or R = an alkyl group with 13–21 carbon atoms and up to six double bonds.

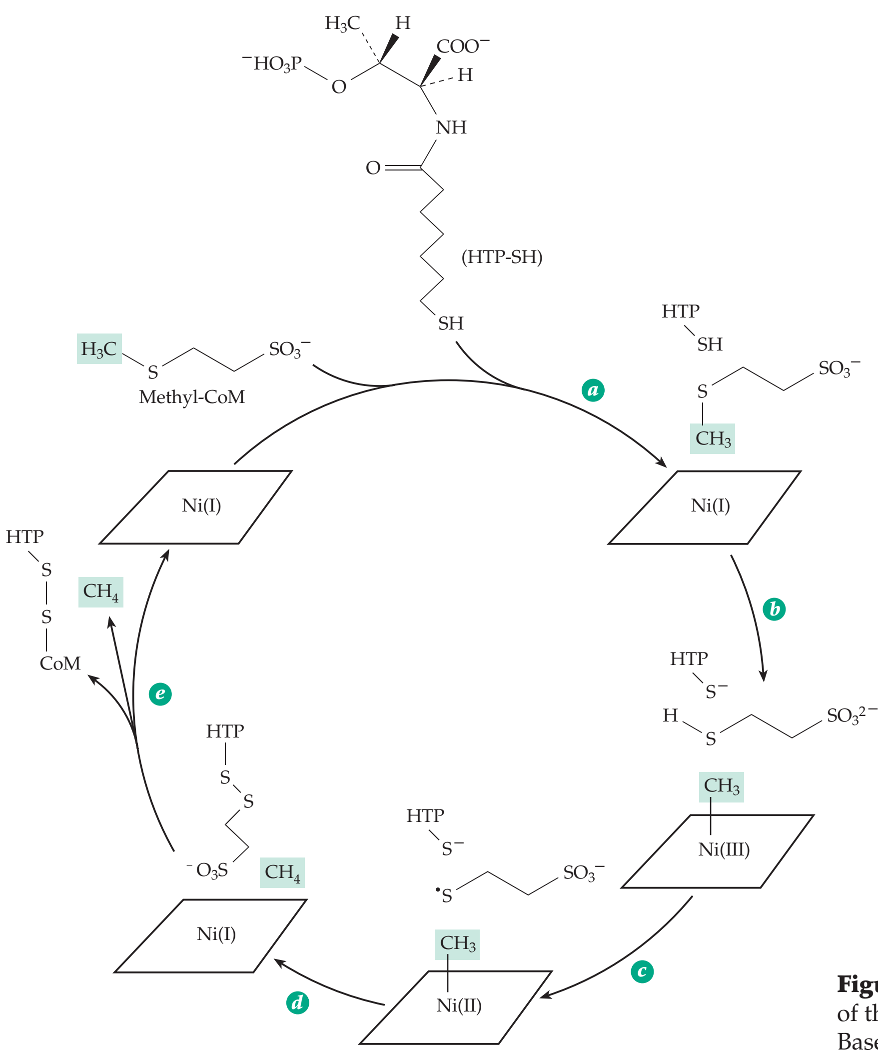


Figure 16-28 Proposed mechanism of action of the methane-forming coenzyme M reductase. Based on the crystal structure.

thiolate sulfur providing the sixth ligand for the nickel. The long –SH-containing side chain of **heptanoyl threonine phosphate** (HTP; also called coenzyme B; Chapter 15, Section E) also lies within the channel with its amino acid head group blocking the entrance. In a second crystal form the mixed disulfide HTP–S–S–CoM, an expected product of the reaction (Fig. 16-28), is present in the channel.[458] Because of the distance from the –SH group of HTP and the nickel atom it is clear that there must be some motion of the methyl-CoM and that the methane formed may stay trapped in the active site until the HTP–S–S–CoA product leaves.

A proposed mechanism for the catalytic cycle based on the X-ray results as well as previous chemical studies and EPR spectroscopy is shown in Fig. 16-28. The substrates enter in step *a*. The position of the HTP, with its extended side chain, is probably the same as that seen in the X-ray structures of the Ni(II) complexes but the conformation of the methyl-CoM is different. The methyl transfer in step *b* is reminiscent of that of methionine synthase (Eq. 16-43). Although the distance from the CoM sulfur and the HTP–SH is too great for direct proton transfer between the two, as indicated for step *b*, there are two tyrosine hydroxyls that could provide a pathway for proton transfer. The region around the surface of the nickel coenzyme is largely hydrophobic and could facilitate formation of the thiyl radical in step *c*. In the structure with the bound HTP–S–S–CoM heterodisulfide an oxygen atom of the sulfonate group of CoM is bonded tightly to the Ni(II). However, in the active Ni(I) form the nickel is nucleophilic and would probably repel the sulfonate, perhaps assisting the product release.[458]

The enzyme contains five posttranslationally modified amino acids near the active site: *N*-methylhistidine, 5-methylarginine, 2-methylglutamine, 2-methylcysteine, and thioglycine in a thiopeptide bond. The latter may be the site of radical formation.[458a,b]

4. Tunichlorins

Nickel is found in various marine invertebrates. In the tunicates (sea squirts and their relatives) it occurs in a fixed ratio with cobalt, suggesting a metabolic role.[460] A new class of nickel chelates called tunichlorins have been isolated. An example is shown in Fig. 16-27B. The function of tunichlorins is unknown but their existence suggests the possibility of unidentified biochemical roles for nickel.

5. Carbon Monoxide Dehydrogenases and Carbon Monoxide Dehydrogenase/Acetyl-CoA Synthase

There are several bacterial carbon monoxide dehydrogenases that catalyze the reversible oxidation of CO to CO_2:

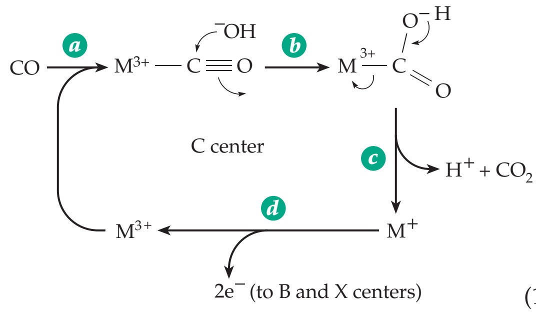
$$CO + H_2O \rightarrow CO_2 + H^+ + 2\,e^- \qquad (16\text{-}50a)$$

$$\begin{gathered} CO + H_2O \rightarrow CO_2 + H_2 \\ \Delta G^\circ = -38.1 \text{ kJ / mol} \end{gathered} \qquad (16\text{-}50b)$$

Some bacteria use CO as both a source of energy and for synthesis of carbon compounds. The purple photosynthetic bacterium *Rhodospirillum rubrum* employs a relatively simple monomeric Ni-containing CO dehydrogenase containing one atom of Ni and seven or eight iron atoms, apparently arranged in Fe_4S_4 clusters. These bacteria can grow anaerobically with CO as the sole source of both energy and carbon.[461] Some aerobic bacteria oxidize CO using a molybdenum-containing enzyme (Section H). However, the most studied CO dehydrogenase is a complex enzyme that also synthesizes, reversibly, acetyl-CoA from CO and a methyl corrin. Employed by methanogens, acetogens, and sulfate-reducing bacteria, it is at the heart of the **Wood–Ljundahl pathway** of autotrophic metabolism, which is discussed further in Chapter 17.

Both oxidation of CO to CO_2 and reduction of CO_2 to CO are important activities of CO dehydrogenase / acetyl-CoA synthase. During growth on CO, some CO must be oxidized to CO_2 and then reduced by the pathways of Fig. 15-22 to form a methyl-tetrahydropterin which can be used to form the methyl group of acetyl-CoA. During growth on any other carbon compound CO_2 must be reduced to CO to form the carbonyl group of acetyl-CoA which can serve as a precursor to all other carbon compounds. Native CO dehydrogenase/acetyl CoA synthase was isolated from cells of *Clostridium thermoaceticum* grown in the presence of radioactive ^{63}Ni. The protein is a 310-kDa $\alpha_2\beta_2$ oligomer. Each $\alpha\beta$ dimer contains 2 atoms of Ni, 1 of Zn, ~12 of Fe and ~12 sulfide ions,[462–464] which are organized into three metal clusters referred to as A, B, and C. Each cluster contains 4 Fe atoms and clusters A and C also contain 1 Ni each. Oxidation of CO occurs in the β subunits, each of which contains both cluster B, an Fe_4S_4 ferredoxin-type cluster, and cluster C, where the oxidation of CO is thought to occur. Cluster C contains 1 nickel ion as well as an Fe_4S_4 cluster that resembles that of aconitase (Fig. 13-4). Cluster A, which is in the α subunit, also contains 1 atom of Ni and 4 Fe ions and is probably the site of synthesis of acetyl-CoA.

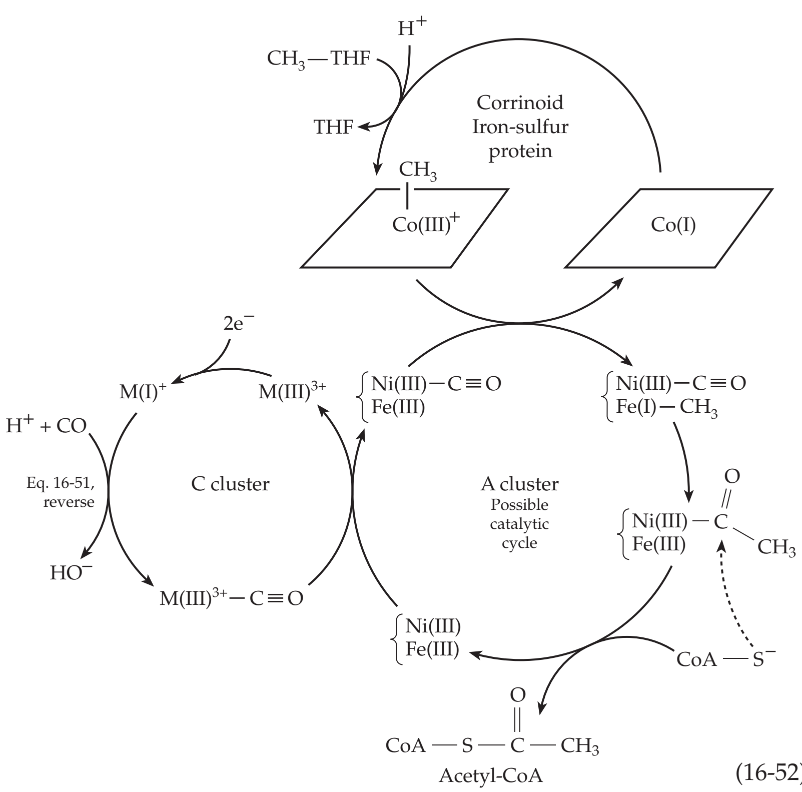
Oxidation of CO may require cooperation of the nickel ion and the Fe_4S_4 group within the C cluster.



(16-51)

CO probably binds to one of these metals and is attacked by a hydroxyl ion (Eq. 16-51, step *b*) which may be donated by the other metal of the pair.[465,465a] CO_2 and a proton are released rapidly (step *c*), after which the reduced metal center is reoxidized (step *d*). One electron is thought to be transferred directly to the Fe_4S_4 cluster B and the second by an alternative route.[465] A multienzyme complex isolated from the methanogen *Methanosarcina* has an $(\alpha\beta\gamma\delta\varepsilon)_6$ structure, with the subunits having masses of 89, 60, 50, 48, and 20 kDa, respectively.[466] In this complex the CO dehydrogenase/acetyl-CoA synthase activity appears to reside in the $\alpha_2\varepsilon_2$ complex, the γδ complex has a tetrahydropteridine: cob(I)amide–protein transferase, and the β subunit has an acetyltransferase that binds acetyl-CoA and transfers the acetyl group to a group on the β subunit.[466]

Although the *Clostridium* and *Methanosarcina* systems are not identical, similar mechanisms are presumably involved.[467,467a] To generate acetyl-CoA a methyl group is first transferred from a tetrahydropterin such as tetrahydrofolate or, in methanogens, tetrahydromethanopterin or tetrahydrosarcinapterin (Fig. 15-17) to form a methylcorrinoid. At the A center of the CO dehydrogenase a molecule of CO, which may be bound to the Ni, equilibrates with the methyl group, and with acetyl-CoA. As depicted in Eq. 16-52, acetyl-Ni may be an intermediate. Other details shown here are hypothetical. It is possible that the methyl group is transferred to the Ni atom in the M cluster before reaction with the CO, which might be bound to either Ni of cluster C or to Fe. This reaction of two transition-metal-bound ligands parallels a proposed industrial process for synthesis of acetic acid from methanol and CO and involving catalysis by rhodium metal and methyl iodide. It is thought that rhodium-bound CO is inserted into bound $Rh-CH_3$ to form an intermediate $Rh-\overset{\displaystyle O}{\overset{\|}{C}}-CH_3$. The acetyl group is released as acetyl iodide which is hydrolyzed to acetic acid. An acetyl-nickel intermediate may be involved in the corresponding biological reaction of Eq. 16-52. The stereochemistry of the sequence has been investigated using methyl-THF containing a chiral methyl group. Overall retention of the configuration of the methyl group in acetyl-CoA was observed.[468,469]



(16-52)

D. Copper

Copper was recognized as nutritionally essential by 1924 and has since been found to function in many cellular proteins.[470–474] Copper is so broadly distributed in foods that a deficiency has only rarely been observed in humans.[474a] However, animals may sometimes receive inadequate amounts because absorption of Cu^{2+} is antagonized by Zn^{2+} and because copper may be tied up by molybdate as an inert complex. There are copper-deficient desert areas of Australia where neither plants nor animals survive. Copper-deficient animals have bone defects, hair color is lacking, and hemoglobin synthesis is impaired. Cytochrome oxidase activity is low. The protein elastin of arterial walls is poorly crosslinked and the arteries are weak. Genetic defects in copper metabolism can have similar effects.

An adult human ingests ~2–5 mg of copper per day, about 30% of which is absorbed. The total body content of copper is ~100 mg (~2 x 10^{-4} mol / kg), and both uptake and excretion (via the bile) are regulated. Since an excess of copper is toxic, regulation is important.

Because Cu^{2+} is the most tightly bound metal ion in most chelating centers (Table 6-9), almost all of the copper present in living cells is complexed with proteins. Copper is transported in the blood by a 132-kDa, 1046-residue sky-blue glycoprotein called **ceruloplasmin**.[471,475–477] This one protein contains 3% of the total body copper.

Regulation of copper uptake has been studied in most detail in the yeast *Saccharomyces cerevisiae*. Uptake of Cu^{2+} is similar to that of Fe^{3+}. The same plasma membrane reductase system, consisting of proteins Fre1p and Fre2p (encoded by genes *FRE*1 and *FRE*2), acts to reduce both Fe^{3+} and Cu^{2+}.[478–481] These two genes are controlled in part by a transcriptional activator that responds to the internal copper concentration.[410,482,483] Similar regulation is thought to occur in both plants[484] and animals.

The human hereditary disorders **Wilson's disease** and **Menkes' disease** have provided further insight into copper metabolism.[485,486] In Wilson's disease the ceruloplasmin content is low and copper gradually accumulates to high levels in the liver and brain. In Menkes syndrome, there is also a low ceruloplasmin level and an accumulation of copper in the form of copper metallothionein.[487,488] Persons with this disease have abnormalities of hair, arteries, and bones and die in childhood of cerebral degeneration.[489,490] Similar symptoms are seen in some patients with Ehlers–Danlos syndrome (Box 8-E).[491] Genes for the proteins that are defective in both Wilson's and Menkes' diseases have been cloned and both proteins have been identified as P-type ATPase cation transporters (Chapter 8).[492–495c] The two proteins must be similar in structure as indicated by a 55% sequence identity.[496] Homologous genes involved in copper homeostasis have been located in both yeast[497] and the cyanobacterium *Synechococcus.*[498] The transporter encoded by this yeast gene, designated *Ccc*2, apparently functions to export copper from the cytosol into an extracytosolic compartment. In a similar way the Wilson and Menkes disease proteins, which reside in the *trans*-Golgi network, are thought to export copper or to provide copper for incorporation into essential proteins.[493,499] The Wilson disease protein is also found in a shortened form in mitochondrial membranes.[492] Other proteins associated with intracellular copper metabolism seem to be chaperones for Cu(I).[500–501a]

The ability of copper ions to undergo reversible changes in oxidation state permits them to function in a variety of oxidation–reduction processes. Like iron, copper also provides sites for reaction with O_2, with superoxide radicals, and with nitrite ions.

1. Electron-Transferring Copper Proteins

A large group of small, intensely blue copper proteins function as single-electron carriers within bacteria and plants. Best known is **plastocyanin**, which is ubiquitous in green plants and functions in the electron transport chain between the light-absorbing photosynthetic centers I and II of chloroplasts (Chapter 23). The bacterial **azurins**[502] are thought to carry electrons between cytochrome c_{441} and cytochrome oxidase. **Amicyanin** accepts electrons from the coenzyme TTQ of methylamine dehydrogenase of methylotrophic bacteria and passes them to a cytochrome *c* (Chapter 15).[168,503,504] A basic blue copper protein **phytocyanin** of uncertain function occurs in cucumber seeds.[505]

The 10.5-kDa peptide chain of plastocyanin is folded into an eight-stranded β barrel (Fig. 2-16), which contains a single copper atom. In poplar plastocyanin, the Cu is coordinated by the side chains of His 37, His 87, Met 92, and Cys 84 in a tetrahedral but distorted toward a trigonal bipyramidal geometry. Since copper-free apoplastocyanin has essentially the same structure, this geometry may be imposed by the protein onto the Cu^{2+}, which usually prefers square-planar or tetrahedral coordination (Chapter 7).[506] Calculations suggest that there is little or no strain and that the "Franck–Condon barrier" to electron transfer is low.[507,508] The three-dimensional structure and copper environment of azurin are similar to those of plastocyanin.[509–511] Messerschmidt *et al.*[512] suggested that the copper site in these proteins is perfectly adapted to its function because its geometry is a compromise between the optimal geometries of the Cu(I) and Cu(II) states between which it alternates. **Stellacyanin**, present in the Japanese lac tree and some other plants, is a mucoprotein; the 108-residue protein is over 40% carbohydrate.[513] While its spectrum resembles that of plastocyanins and azurins, stellacyanin contains no methionine and this amino acid cannot be a ligand to copper.[514] **Rusticyanin** functions in the periplasmic space of some chemolithotropic sulfur bacteria to transfer electrons from Fe^{2+} to cytochrome *c* as part of the energy-providing reaction for these organisms (Eq. 18-23).[515–517] **Halocyanin** functions in membranes of the archaeobacterium *Natronobacterium*,[518] and **aurocyanin** functions in green photosynthetic bacteria.[519]

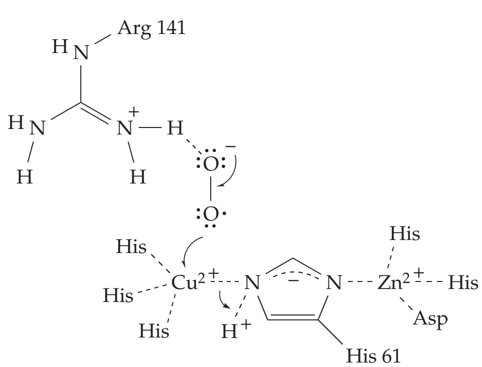
The blue color of these "type 1" copper proteins is much more intense than are the well known colors of the hydrated ion $Cu(H_2O)_4{}^{2+}$ or of the more strongly absorbing $Cu(NH_3)_4{}^{2+}$. The blue color of these simple complexes arises from a transition of an electron from one *d* orbital to another within the copper atom. The absorption is somewhat more intense in copper peptide chelates of the type shown in Eq. 6-85. However, the ~600 nm absorption bands of the blue proteins are an order of magnitude more intense, as is illustrated by the absorption spectrum of azurin (Fig. 23-8). The intense blue is thought to arise as a result of transfer of electronic charge from the cysteine thiolate to the Cu^{2+} ion.[520,521]

A third type of copper center, first recognized in cytochrome *c* oxidase (see Fig. 18-10) is called Cu_A or **purple CuA**. Each copper ion is bonded to an imidazole and two cysteines serve as bridging ligands. The two copper ions are about 0.24 nm apart, and the two Cu^{2+} ions together can accept a single electron from an external donor such as cytochrome *c* or azurin to give a half-reduced form.[521a,b]

2. Copper, Zinc-Superoxide Dismutase

Although of similar topology to the blue electron-transferring proteins, Cu, Zn-superoxide dismutase, has a different function. This dimeric 153-residue protein has been demonstrated in the cytoplasm of virtually all eukaryotic cells[522] and in the periplasmic space of some bacteria[523] where it converts superoxide ions $^{\bullet}O_2^-$ to O_2 and H_2O_2. The enzyme, which has a major protective role against oxidative damage to cells, presumably functions in a manner similar to that indicated in Eq. 16-27 for iron or manganese. However, copper cycles between Cu^{2+} and Cu^+, alternately accepting and donating electrons.

The active site of cytosolic superoxide dismutase (SOD) contains both Cu^{2+} and Zn^{2+}. The copper ion is of "type 2": nonblue and paramagnetic. It is surrounded by four imidazole groups with an irregular square planar geometry.[524–527] One of these imidazole groups (that of His 61) is shared with the Zn^{2+}, which is also bonded to two additional imidazole groups and a side chain carboxylate. The metal ions have evidently replaced the hydrogen atom that would otherwise be present on the imidazole of His 61 (see the following diagram). It has been suggested, as is also indicated in the diagram, that when the bound superoxide



donates an electron to the Cu(II) to become O_2 (first step of Eq. 16-27), a proton becomes attached to the bridging imidazole with breakage of its linkage to the Cu(I). The structure of a new crystalline form of reduced yeast SOD shows that the Cu(I) has moved 0.1 nm away from the bridging imidazole in agreement with this possibility.[528] In the second half reaction the imidazole proton, together with a second proton from the medium and an electron from the Cu(I), would react to convert the second O_2^- into H_2O_2. The role of the Zn^{2+} may be in part structural but it may also serve to ensure that His 61 is protonated on the correct nitrogen atom. Arg 141 may assist in binding the O_2^- as is shown in the diagram. However, the fact that a mutant containing leucine in place of the active site arginine has over 10% of the activity of the native enzyme shows that the arginine is not absolutely essential.[529] Additional nearby positively charged arginine and lysine side chains may provide "electrostatic guidance" that increases the velocity of reaction of superoxide ions.[530,531] Cu, Zn-SOD is one of the fastest enzymes known.

In addition to the cytosolic SOD there is a longer ~222-residue extracellular form that binds to the proteoglycans found on cell surfaces.[522,527] Manganese SODs are found in mitochondria and in bacteria and iron SODs in plants and bacteria. They all appear to be important in protecting cells from superoxide radicals.[522,532,533] This importance was dramatically emphasized when it was found that a defective SOD is present in persons (about 1 in 100,000) with a hereditary form of **amylotrophic lateral sclerosis** (ALS), which is also called Lou Gehrig's disease after the baseball hero who was stricken with this terrible disease of motor neurons in 1939 at the age of 36.[534–536]

3. Nitrite and Nitrous Oxide Reductases

Copper enzymes participate in two important reactions catalyzed by denitrifying bacteria. Nitrite reductases from species of *Achromobacter*[537,538] and *Alcaligenes*[539–542] are trimeric proteins[543] made up of 37-kDa subunits, each of which contains one type 1 (blue) copper and one type 2 (nonblue) copper. The first copper serves as an electron acceptor from a small blue **pseudoazurin**.[544,544a] The second copper, which is in the active site, is thought to bind to nitrite throught its nitrogen atom and to reduce it to NO.

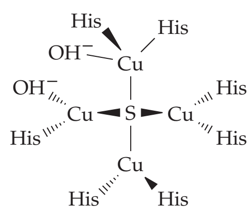
$$NO_2^- + 2H^+ + e^- \rightarrow NO + H_2O \qquad (16\text{-}53)$$

Crystallographic studies on the 343-residue *Alcaligenes* enzyme reveal two β barrel domains with the type 1 copper embedded in one of them and the type 2 copper in an interface between the domains. Studies of EPR[541] and ENDOR[545] spectra and of various mutant forms have shown that, as for other copper enzymes, the type 1 copper is an electron-transferring center, accepting electrons from the pseudoazurin and passing them to the type 2 copper which binds and reduces nitrite.[540,545a]

The immediate product of nitrite reductase is NO, which is reduced in two one-electron steps to N_2O, and then to N_2. The second of these steps is catalyzed by another copper enzyme, nitrous oxide reductase.[546]

$$N_2O + 2H^+ + 2e^- \rightarrow N_2 + H_2O \qquad (16\text{-}54)$$

The biological significance of these reactions is considered further in Chapters 18 and 24. The 132-kDa dimeric N_2O reductase from *Pseudomonas stutzeri* contains four copper atoms per subunit.[546] One of its copper centers resembles the Cu_A centers of cytochrome *c* oxidase. A second copper center consists of four copper ions, held by seven histidine side chains in a roughly tetrahedral array around one sulfide (S^{2-}) ion. Rasmussen *et al.* speculate that this copper-sulfide cluster may be an acceptor of the oxygen atoms of N_2O in the formation of N_2.[546a] There is also a cytochrome cd_1 type of nitrite reductase.[143a]



4. Hemocyanins

While many copper proteins are catalysts for oxidative reactions of O_2, hemocyanin reacts with O_2 reversibly. This water-soluble O_2 carrier is found in the blue blood of many molluscs and arthropods, including snails, crabs, spiders, and scorpions. Hemocyanins are large oligomers ranging in molecular mass from 450 to 13,000 kDa. Molluscan hemocyanins are cylindrical oligomers which have a striking appearance under the electron microscope. Simpler hemocyanins, found in arthropods, are hexamers of 660-residue 75-kDa subunits. Each subunit of the hemocyanin from the spiny lobster is folded into three distinct domains, one of which contains a pair of Cu(I) atoms which bind the O_2. Each copper ion is held by three imidazole groups without any bridging groups between them, the Cu–Cu distance being 0.36–0.46 nm.[547] *Octopus* hemocyanin has a different fold and forms oligomers of ten subunits. However, the active sites are very similar.[548] The O_2 is thought to bind between the two copper atoms. An allosteric mechanism may involve changes in the distance between the copper atoms.[549] The oxygenated compound is distinctly blue with a molar extinction coefficient 5–10 times greater than that of cupric complexes. This fact suggests that the Cu(I) has been oxidized to Cu(II) and that the O_2 has been reduced to the peroxide dianion O_2^{2-} in the complex.[550–552] Further support for this idea comes from the observation that treatment of oxygenated hemocyanin with glacial acetic acid leads to the formation of equal amounts of Cu^{2+} and Cu^+ and protonated superoxide.

$$(CuO_2Cu)^{2+} + H^+ \longrightarrow Cu^{2+} + Cu^+ + HO_2 \qquad (16\text{-}55)$$

5. Copper Oxidases

A large group of copper-containing proteins activate oxygen toward chemical reactions of dehydrogenation, hydroxylation, or oxygenation. **Galactose oxidase** (Fig. 16-29), from the mushroom *Polyporus,* is a dehydrogenase which converts the 6-hydroxymethyl group of galactose to an aldehyde while O_2 is reduced to H_2O_2.

$$R-CH_2OH \xrightarrow[O_2 \;\; H_2O_2]{} R-CHO \qquad (16\text{-}56)$$

Galactose oxidase has been used frequently to label glycoproteins of external cell membrane surfaces. Exposed terminal galactosyl or *N*-acetylgalactosaminyl residues are oxidized to the corresponding C-6 aldehydes and the latter are reduced under mild conditions with tritiated sodium borohydride.[553]

The single 639-residue polypeptide chain contains one type 2 copper ion.[554] Neither oxygen nor galactose affects the absorption spectrum of the light murky green enzyme, but the combination of the two does, suggesting that both substrates bind to the enzyme before a reaction takes place. A side reaction releases

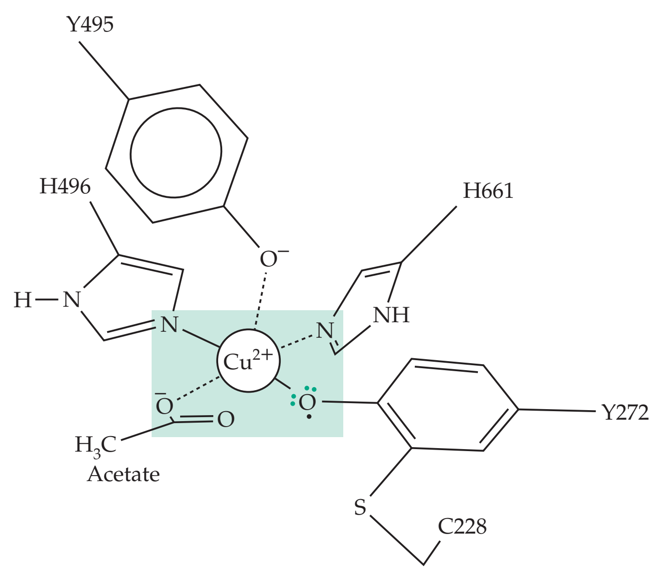


Figure 16-29 Drawing of the active site of galactose oxidase showing both the Cu(II) atom and the neighboring free radical on tyrosine 272, which has been modified by addition of the thiol of cysteine 228 and oxidation. See Halfen *et al.*[557] Based on a crystal structure of Ito *et al.*[558]

superoxide ion and leaves the enzyme in the inactive Cu(II) state. EPR spectroscopic observations on the enzyme were puzzling. The active enzyme shows no EPR signal but a one-electron reduction gives an inactive form with an EPR signal that arises from Cu(II). Experimental studies eventually pointed to the presence of a second reducible center which contains an organic free radical. In the active form this radical is **antiferromagnetically coupled** (spin-coupled) giving an "EPR-silent" enzyme able to accept two electrons.[555–557]

Another surprise was the discovery, from the X-ray structure,[558–559] that a tyrosine side chain at the active site has been modified by addition of a thiolate group from a cysteine residue. This structure, which is also the site of the organic free radical, is shown in Fig. 16-29. A possible mechanism of action is portrayed in Fig. 16-30. The substrate binds in step *a* and its –OH group is deprotonated, perhaps by transfer of H^+ to the phenolate oxygen of tyrosine 495 (Fig. 16-29) in step *b* of Fig. 16-30. In step *c* a free radical hydrogen transfer occurs to form a ketyl radical which is immediately oxidized by Cu(II) in step *d*. In steps *c* and *f* the oxidant O_2 is converted to H_2O and the aldehyde product is also released. A fungal glyoxal oxidase has similar characteristics.[560] The galactose oxidase proenzyme is self-processing. The Tyr-Cys cofactor arises as a result of copper-catalyzed oxidative modification via a tyrosine free radical.[560a]

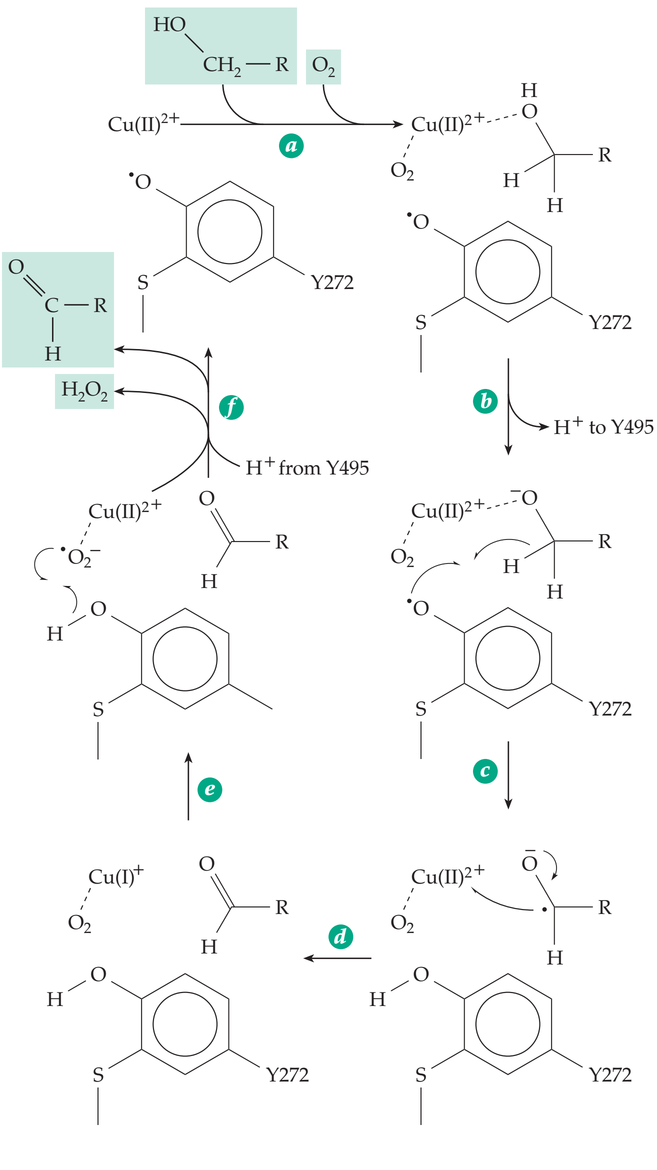
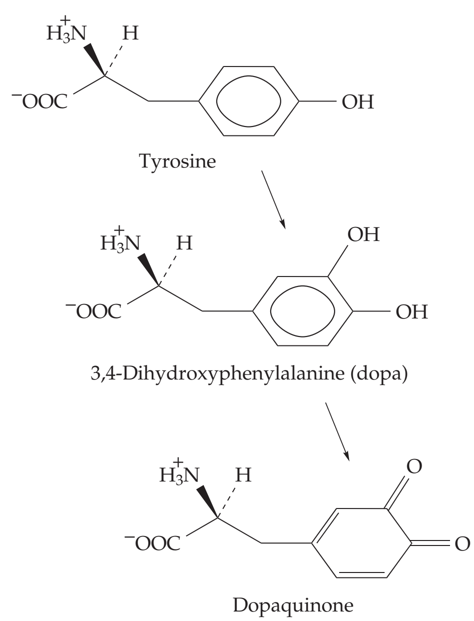


Figure 16-30 Possible reaction cycle for catalysis of the 6–OH of D-galactose or other suitable alcohol substrate by galactose oxidase. See Wachter *et al.*[566] and Whittaker *et al.*[567]

Several copper-containing amine oxidases[561,561a] convert amines to aldehydes and H_2O_2. They also contain one of the organic quinone cofactors discussed in Chapter 15. The dimeric **plasma amine oxidase** contains a molecule of the coenzyme TPQ and one Cu^{2+} per 90-kDa subunit.[562–563a] Whether the O_2 binds only to the single atom Cu(I) or also interacts directly with a cofactor radical in step *d* of Eq. 15-53 is uncertain.[562] **Lysyl oxidase**, which is responsible for conversion of ε-amino groups of side chains of lysine into aldehyde groups in collagen and elastin (Chapter 8) contains coenzyme LTQ as well as Cu.[564] The enzyme is specifically inhibited by β-aminopropionitrile (Box 8-G) and its activity is decreased in genetic diseases of copper metabolism. The **glycerol oxidase** of *Aspergillus* is a large 400 kDa protein containing one heme and two atoms of Cu.[565] It converts glycerol + O_2 into glyceraldehyde and H_2O_2. Also containing copper is **urate oxidase**, whose action is indicated in Fig. 25-18.

Tyrosinase catalyzes hydroxylation followed by dehydrogenation (Eq. 16-57). First identified in mushrooms, the enzyme has a widespread distribution in



(16-57)

nature. It is present in large amounts in plant tissues and is responsible for the darkening of cut fruits. In animals tyrosinase participates in the synthesis of **dihydroxyphenylalanine** (dopa) and in the formation of the black **melanin** pigment of skin and hair. Either a lack of or inhibition of this enzyme in the melanin-producing melanocytes causes **albinism** (Chapter 25).

The 46-kDa monomeric tyrosinase of *Neurospora* contains a pair of spin-coupled Cu(II) ions.[568,569] The structure of this copper pair (**type 3 copper**) has many properties in common with the copper pair in hemocyanin.[569a] For example, in the absence of other substrates, tyrosinase binds O_2 to form "oxytyrosinase," a compound with properties resembling those of oxyhemocyanin and containing a bound peroxide dianion.[569]

Tyrosinase is both an oxidase and a hydroxylase. Some other copper enzymes have only a hydroxylase function. One of the best understood of these is the **peptidylglycine α-hydroxylating monoxygenase**, which catalyzes the first step of the reaction of Eq. 10-11. The enzyme is a colorless two-copper protein but the copper atoms are 1.1 nm apart and do not form a binuclear center.[570] Ascorbate is an essential cosubstrate, with two molecules being oxidized to the semidehydroascorbate radical as both coppers are reduced to Cu(I). A ternary complex of reduced enzyme, peptide, and O_2 is formed and reacts to give the hydroxylated product.[570] A related two-copper enzyme is **dopamine β-monooxygenase**, which utilizes O_2 and ascorbate to hydroxylate dopamine to noradrenaline (Chapter 25).[571,572] These and other types of hydroxylases are compared in Chapter 18.

The **blue multicopper oxidases** couple the oxidation of substrates to the four-electron reduction of molecular oxygen to H_2O.[573] In this respect they resemble cytochrome *c* oxidase, which also contains copper. However, they do not contain iron. The best known member of this group is the plant enzyme **ascorbate oxidase**, which dehydrogenates ascorbic acid to dehydroascorbic acid (See Box 18-D). It is a dimeric blue copper with identical 70-kDa subunits. The three-dimensional structure revealed one type 1 copper ion held in a typical blue copper environment as in plastocyanin or azurin and also a **three-copper center**. In this center a pair of copper ions, each held by three imidazole groups, and bridged by a μ-oxo group as in hemerythrin (Fig. 16-20), lie 0.51 nm apart and 0.41–0.44 nm away from the third copper, which is held by two other imidazoles.[574] The type 1 copper shows typical intense 600-nm absorption and characteristic EPR signal, while the pair with the oxo bridge are antiferromagnetically coupled and EPR silent but with strong near ultraviolet light absorption (type 3 copper). The additional metal ion in the trinuclear center is a type 2 copper which lacks characteristic spectroscopic features.[575,576] Reduction of O_2 to 2 H_2O is thought to procede via superoxide radical intermediates. When substrate is added to the enzyme, the blue color fades and it can be shown that the copper is reduced to the +1 state. The reduced enzyme then reacts with O_2, converting it into two molecules of H_2O. Similar to ascorbate oxidase in structure and properties are **laccase**, found in the latex of the Japanese lac tree and in the mushroom *Polyporus*, and the previously discussed **ceruloplasmin.**[477] Laccase is a catalyst for oxidation of phenolic compounds by a free radical mechanism involving the trinuclear copper center.[577] Studied by Gabriel Bertrand in the 1890s, it was one of the first oxidative enzymes investigated.[578]

In addition to its previously mentioned role in copper transport, ceruloplasmin is an amine oxidase, a superoxide dismutase, and a ferrooxidase able to catalyze the oxidation of Fe^{2+} to Fe^{3+}. Ceruloplasmin contains three consecutive homologous 350-residue sequences which may have originated from an ancestral copper oxidase gene. Like ascorbate oxidase, this blue protein contains copper of the three different types. Blood clotting factors V and VIII (Fig. 12-17), and the iron uptake protein Fet3 (Section A,1) are also closely related.

6. Cytochrome *c* Oxidase

The most studied of all copper-containing oxidases is cytochrome *c* oxidase of mitochondria. This multisubunit membrane-embedded enzyme accepts four electrons from cytochrome *c* and uses them to reduce O_2 to 2 H_2O. It is also a proton pump. Its structure and functions are considered in Chapter 18. However, it is appropriate to mention here that the essential catalytic centers consist of two molecules of heme *a* (*a* and a_3) and three Cu^+ ions. In the fully oxidized enzyme two metal centers, one Cu^{2+} (of the two-copper center Cu_A) and one Fe^{3+} (heme *a*), can be detected by EPR spectroscopy. The other Cu^{2+} (Cu_B) and heme a_3 exist as an EPR-silent exchange-coupled pair just as do the two copper ions of hemocyanin and of other type 3 binuclear copper centers.

E. Manganese

Tissues usually contain less than one part per million of manganese on a dry weight basis, less than 0.01 mM in fresh tissues. This compares with a total content in animal tissues of the more abundant Mg^{2+} of 10 mM. A somewhat higher Mn content (3.5 ppm) is found in bone. Nevertheless, manganese is nutritionally essential[579,580] and its deficiency leads to well-defined symptoms. These include ovarian and testicular degeneration, shortening and bowing of legs, and other skeletal abnormalities such as the "slipped tendon disease" of chicks. In Mn deficiency the organic matrix of bones and cartilage develops poorly. The galactosamine, hexuronic acids, and chondroitin sulfates content of cartilage is decreased. Manganese is also essential for plant growth and plays a unique and essential role in the photosynthetic reaction centers of chloroplasts. Two magnetically coupled pairs of manganese ions bound in a protein act as the O_2 evolving center in photosynthetic system II. This function is considered in Chapter 23. An ABC transporter for managanese uptake has been identified in the cyanobacterium *Synechocystis*.[581]

Manganese lies in the center of the first transition series of elements. The stable Mn^{2+} (manganous ion) contains five 3*d* electrons in a high-spin configuration. The less stable Mn^{3+} (manganic ion) appears to be of importance in some enzymes and is essential to the photosynthetic evolution of oxygen. Many enzymes specifically require or prefer Mn^{2+}. These include galactosyl and *N*-acetylgalactosaminyltransferases[582] needed for synthesis of mucopolysaccharides (Chapter 20), lactose synthetase (Eq. 20-15), and a muconate-lactonizing enzyme (Eq. 13-23).[583]

Arginase, essential to the production of urea in the human body (Fig. 24-11), specifically requires Mn^{2+} which exists as a spin-coupled dimetal center with a bridging water or ^-OH ion. The Mn^{2+} may act much as does the Ni^{2+} ions of urease (Fig. 16-25).[584,585] Pyruvate carboxylase (Eq. 14-3) contains four atoms of tightly bound Mn^{2+}, one for each biotin molecule present. This manganese is essential for the transcarboxylation step in the action of this enzyme. Either Mn^{2+} or Mg^{2+} is also needed in the initial step of carboxylation of biotin (Eq. 14-5). Another Mn^{2+}-containing protein is the lectin concanavalin A (Chapter 4). The joining of *O*-linked oligosaccharides to secreted glycoproteins also seems to require manganese.[586]

Manganese is a component of a "pseudocatalase" of *Lactobacillus*,[204] of lignin-degrading peroxidases,[257,258] and of the wine-red superoxide dismutases found in bacteria and in the mitochondria of eukaryotes.[376,587] The dimeric dismutase from *E. coli* has a structure nearly identical to that of bacterial iron SOD (Fig. 16-22). The manganese ions are presumed to alternate between the +3 and +2 states during catalysis (Eq. 16-27). "Knockout" mice with inactivated Mn SOD genes live no more than three weeks, indicating that this enzyme is essential to life. However, mice lacking CuZn SOD appear normal in most circumstances.[588] Some dioxygenases contain manganese.[589] Many enzymes that require Mg^{2+} can utilize Mn^{2+} in its place, a fact that has been exploited in study of the active sites of enzymes.[590] The highly paramagnetic Mn^{2+} is the most useful ion for EPR studies (Box 8-C) and for investigations of paramagnetic relaxation of NMR signals. Manganese can also replace Zn^{2+} in some enzymes and may alter catalytic properties.

Manganese may function in the regulation of some enzymes. For example, glutamine synthetase (Fig. 24-7) in one form requires Mg^{2+} for activity but upon adenylylation binds Mn^{2+} tightly.[591] Nucleases and DNA polymerases often show altered specificity when Mn^{2+} substitutes for Mg^{2+}. However, the significance of these differences *in vivo* is uncertain. Manganese is mutagenic in living organisms, apparently because it diminishes the fidelity of DNA replication.[592]

A striking accumulation of Mn^{2+} often occurs within bacterial spores (Chapter 32). *Bacillus subtilus* absolutely requires Mn^{2+} for initiation of sporulation. During logarithmic growth the bacteria can concentrate Mn^{2+} from 1 μM in the external medium to 0.2 mM internally; during sporulation the concentrations become much higher.[593]

F. Chromium

Animals deficient in chromium grow poorly and have a reduced life span.[594–596] They also have decreased "glucose tolerance," i.e., glucose injected into the blood stream is removed only half as fast as it is normally.[597,598] This is similar to the effect of a deficiency of insulin. Fractionation of yeast led to the isolation of a chromium-containing **glucose tolerance factor** which appeared to be a complex of Cr^{3+}, nicotinic acid, and amino acids.[597] The chromium in this material is apparently well absorbed by the body but is probably not an essential cofactor.[596] Nevertheless, dietary supplementation with chromium appears to improve glucose utilization, apparently by enhancing the action of insulin.[596] Ingestion of glucose not only increases insulin levels in blood but also causes increased urinary loss of chromium,[599] perhaps as a result of insulin-induced mobilization of stored chromium.[596] It has been suggested that a specific **chromium-binding oligopeptide** isolated from mammalian liver[596,600] may be released in response to insulin and may activate a membrane phosphotyrosine phosphatase.[596]

Chromium concentrations in animal tissues are usually less than 2 μM but tend to be much higher in the caudate nucleus of the brain. High concentrations of Cr^{3+} have also been found in RNA–protein complexes.[601] While several oxidation states, including +2, +3, and +6, are known for chromium, only Cr(III) is found to a significant extent in tissues. The Cr(VI) complex ions, chromate and dichromate, are toxic and chronic exposure to chromate-containing dust can lead to lung cancer. Ascorbate is a principal biological reductant of chromate and can create mutagenic Cr(V) compounds that include a Cr(V)–ascorbate–peroxo complex.[602] However, Cr(III) compounds administered orally are not significantly toxic. Evidently, the Cr(VI) compounds can cross cell membranes and be reduced to Cr(III), which forms stable complexes with many constituents of cells including DNA.[603] The use of such "exchange-inert" Cr(III) complexes of ATP in enzymology was considered in Chapter 12.

Most forms of Cr(III) are not absorbed and utilized by the body. For this reason, and because of the increased use of sucrose and other refined foods, a marginal human chromium deficiency may be widespread.[604,605] This may result not only in poor utilization of glucose but also in other effects on lipid and protein metabolism.[597] However, questions have been raised about the use of chromium picolinate as a dietary supplement. High concentrations have been reported to cause chromosome damage[606] and there may be danger of excessive accumulation of chromium in the body.[607]

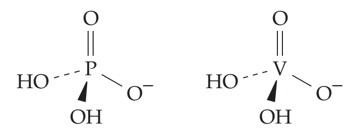
G. Vanadium

Vanadium is a dietary essential for goats and presumably also for human beings,[427] who typically consume ~2 mg / day. However, because vanadium compounds have powerful pharmacological effects it has been difficult to establish the nutritional requirement for animals.[427,604,607a] The adult body contains only about 0.1 mg of vanadium. Typical tissue concentrations are 0.1–0.7 µM[608,609] and serum concentration may be 10 nM or less. Vanadium can assume oxidation states ranging from +2 to +5, the vanadate ion VO_4^{3-} being the predominant form of V(V) in basic solution and in dilute solutions at pH 7. However, at millimolar concentrations, $V_3O_9^{3-}$, $V_4O_{12}^{4-}$, and other polynuclear forms predominate.[610–612] In plasma most exists as metavanadate, VO_2^-, but within cells it is reducted to the vanadyl cation VO^{2+} which is an especially stable double-bonded unit in compounds of V(IV).[609] Only at very low pH is V(III) stable.

The first suggestion of a possible biochemical function for vanadium came from the discovery that **vanadocytes**, the green blood cells of tunicates (sea squirts), contain ~1.0 M V(III) and 1.5–2 M H_2SO_4.[613] It was proposed that a V-containing protein is an oxygen carrier. However, the V^{3+} appears not to be associated with proteins[612] and it does not carry O_2. It may be there to poison predators.[614] The vanadium-accumulating species also synthesize several complex, yellow catechol-type chelating agents (somewhat similar to enterobactin; Fig. 16-1) which presumably complex V(V) and perhaps also reduce it to V(III).[615] Vanadium is also accumulated by other marine organisms and by the mushroom *Amanita muscaria*.

Vanadoproteins are found in most marine algae and seaweed and in some lichens.[616] Among these are **haloperoxidases**,[252,253,617–618b] enzymes that are quite different from the corresponding heme peroxidases discussed in Section A,6. The vanadium is bound as hydrogen vanadate, HVO_4^{2-}, in trigonal bipyrimidal coordination with the three oxygens in equatorial positions and a histidine in one axial position. In the crystal structure an azide (N_3^-) ion occupies the other axial position, but it is presumably the site of interaction with peroxide.[619] The structure is similar to that of acid phosphatases inhibited by vanadate.[620,621] Many nitrogen-fixing bacteria contain genes for a vanadium-dependent nitrogenase that is formed only if molybdenum is not available.[622] The nitrogenases are discussed in Chapter 24.

Much of current interest in vanadium stems from the discovery that vanadate (HVO_4^{2-} at pH 7) is a powerful inhibitor of ATPases such as the sodium pump protein ($Na^+ + K^+$)ATPase (Chapter 8), of phosphatases,[623] and of kinases.[624] This can be readily understood from comparison of the structure of phosphate and vanadate ions.



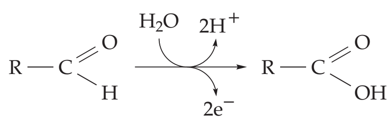
Phosphate Vanadate

Other enzymes such as the cyclic AMP-dependent protein kinase are *stimulated* by vanadium.[624] Vanadate seems to inhibit most strongly those enzymes that form a phosphoenzyme intermediate. This inhibition may be diminished within cells because vanadate is readily reduced by glutathione and other intracellular reductants. The resulting vanadyl ion is a much weaker inhibitor and also stimulates several metabolic processes.[608]

Also of great interest is an insulin-like action of vanadium[607a,625] and evidence that vanadium may be essential to proper cardiac function.[626] A role in lipid metabolism was suggested by the observation that in high doses vanadium inhibits cholesterol synthesis and lowers the phospholipid and cholesterol content of blood. Vanadium is reported to inhibit development of caries by stimulating mineralization of teeth. Unlike tungsten, vanadium does not compete with molybdenum in the animal body.[627] The sometimes dramatic effects of vanadate as an inhibitor, activator, and metabolic regulator are shared also by molybdate and tungstate.[628,629] Even greater effects are observed with vanadate, molybdate, or tungstate plus H_2O_2.[630] The resulting **pervanadate**, **permolybdate**, and **pertungstate** are often assumed to be monoperoxo compounds, e.g., vanodyl hydroperoxide. However, there is some uncertainty.[631]

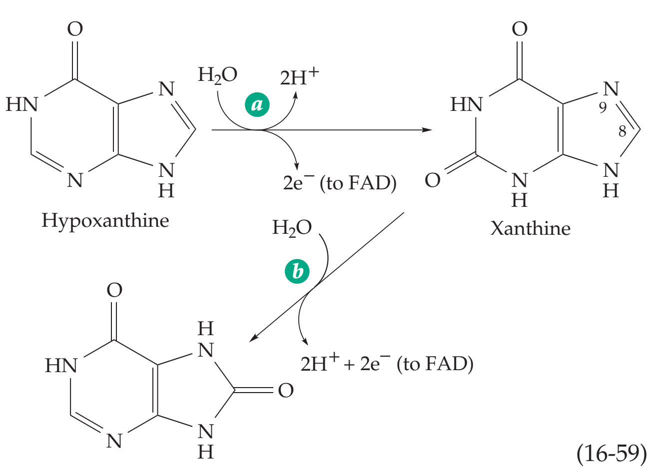
H. Molybdenum

Long recognized as an essential element for the growth of plants, molybdenum has never been directly demonstrated as a necessary animal nutrient. Nevertheless, it is found in several enzymes of the human body, as well as in 30 or more additional enzymes of bacteria and plants.[632] **Aldehyde oxidases**,[633] **xanthine oxidase** of liver and the related **xanthine dehydrogenase**, catalyze the reactions of Eqs. 16-58 and 16-59 and contain molybdenum that is essential for catalytic activity. Xanthine oxidase also contains two Fe_2S_2 clusters and bound FAD. The enzymes can also



(16-58)

oxidize xanthine further (Eq. 16-59, step *b*) by a repetition of the same type of oxidation process at positions 8 and 9 to form **uric acid**. The much studied xanthine dehydrogenase has been isolated from milk,[634,635] liver, fungi,[636] and some bacteria.[637] In the dehydrogenase NAD^+ is the electron acceptor that oxidizes the bound $FADH_2$ formed in Eq. 16-59. Xanthine dehydrogenase, in the absence of thiol compounds, is converted spontaneously into xanthine oxidase, probably as a result of a conformational change and formation of a disulfide bridge within the protein. Treatment with thiol compounds such as dithiothreitol reconverts the enzyme to the dehydrogenase. Evidently in the oxidase form the NAD^+ binding site has moved away from the FAD, permitting oxidation of $FADH_2$ by O_2 with formation of hydrogen peroxide.[635,638]



(16-59)

A purine hydroxylase from fungi,[639] bacterial quinoline and isoquinoline oxidoreductases,[640,641] and a selenium-containing nicotinic acid hydroxylase from *Clostridium barberei*[642] are members of the **xanthine oxidase family** (or molybdenum hydroxylase family).[632,641,643] Also included in the family are aldehyde oxidoreductases from the sulfate-reducing *Desulfovibrio gigas*[633] and from the tomato.[644]

Two other families of molybdoenzymes are the **sulfite oxidase family**[646a,b] and the **dimethylsulfoxide reductase family**.[632,641] **Nitrogenase** (Chapter 24) constitutes a fourth family. Sulfite oxidase (Eq. 16-60) is an essential human liver enzyme (see also Chapter 24).[645,646]

$$SO_3^{2-} + H_2O \rightarrow SO_4^{2-} + 2\,e^- + 2\,H^+ \qquad (16\text{-}60)$$

The **assimilatory nitrate reductase** (Eq. 16-61) of fungi and green plants (Chapter 24) also belongs to the sulfite oxidase family.

$$NO_3^{2} + 2\,e^- + 2\,H^+ \rightarrow NO_2^- + H_2O \qquad (16\text{-}61)$$

DMSO reductase reduces dimethylsulfoxide to dimethylsulfide (Eq. 16-62) as part of the biological sulfur cycle.[647–648d]

$$(H_3C)_2S{=}O + 2\,H^+ + 2\,e^- \rightarrow (CH_3)_2S \qquad (16\text{-}62)$$

A number of other reductases and dehydrogenases, including **dissimilatory nitrate reductases** of *E. coli* and of denitrifying bacteria (Chapter 18), belong to the DMSO reductase family. Other members are reductases for biotin *S*-oxide,[649] trimethylamine *N*-oxide, and polysulfides as well as **formate dehydrogenases** (Eq. 16-63), formylmethanofuran dehydrogenase (Fig. 15-22,

$$HCOO^- + 2\,e^- + 3\,H^+ \rightarrow CO_2 + 2\,H_2O \qquad (16\text{-}63)$$

step *b*), and arsenite oxidase.[632] Several other molybdoenzymes, such as pyridoxal oxidase, had not been classified by 1996.[632]

1. Molybdenum Ions and Coenzyme Forms

Molybdenum is a metal of the second transition series, one of the few heavy elements known to be essential to life. Its most stable oxidation state, Mo(VI), has 4*d* orbitals available for coordination with anionic ligands. Coordination numbers of 4 and 6 are preferred, but molybdenum can accommodate up to eight ligands. Most of the complexes are formed from the oxycation $Mo(VI)O_2^{2+}$. If two molecules of water are coordinated with this ion, the protons are so acidic that they dissociate completely to give $Mo(VI)O_4^{2-}$, the molybdate ion. Other oxidation states vary from Mo(III) to Mo(V). In these lower oxidation states, the tendency for protons to dissociate from coordinated ligands is less, e.g., $Mo(III)(H_2O)_6^{3+}$ does not lose protons even in a very basic medium. Molybdenum tends to form dimeric

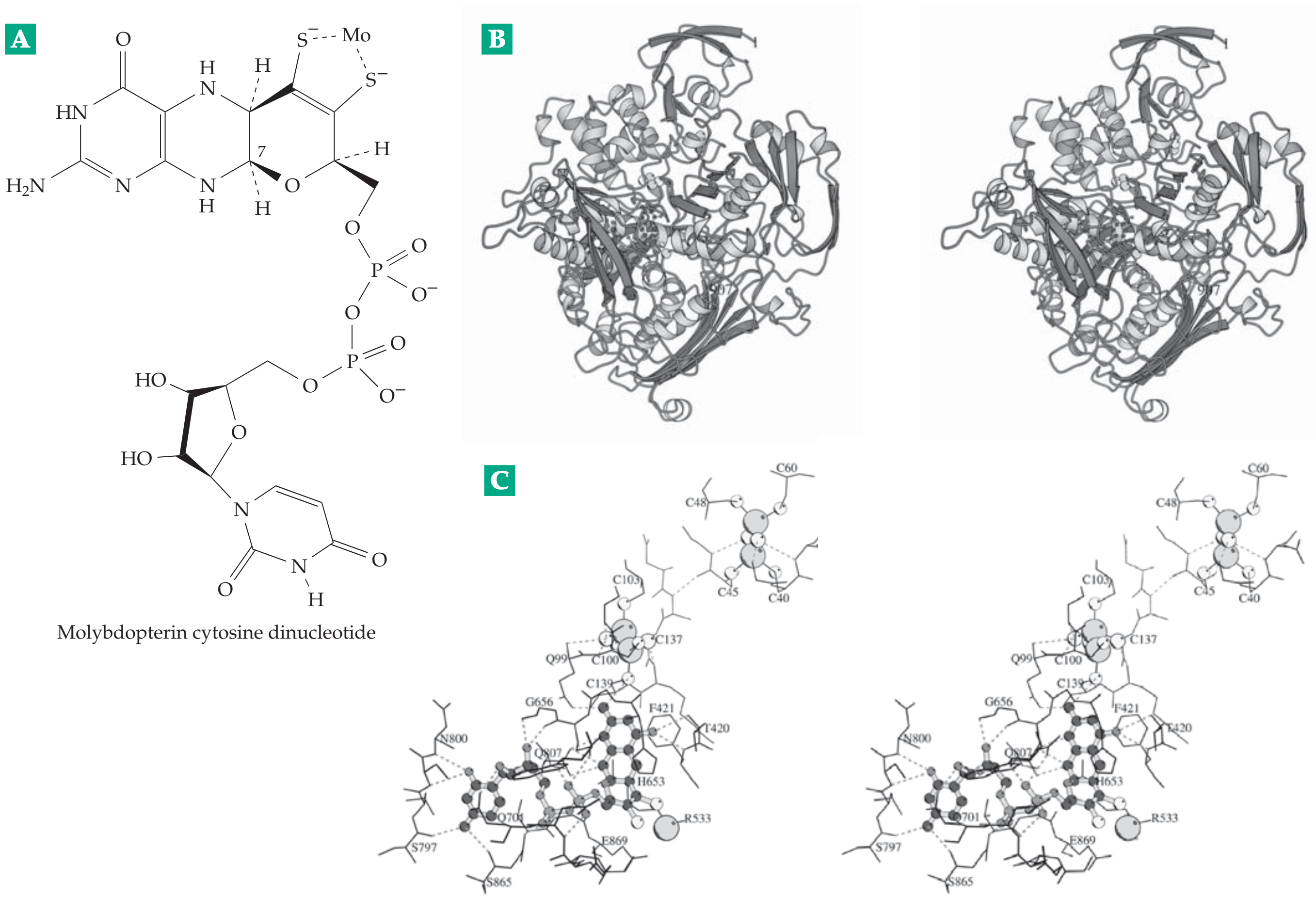
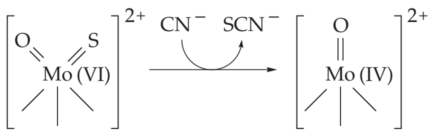


Figure 16-31 (A) Structure of molybdopterin cytosine dinucleotide complexed with an atom of molybdenum. (B) Stereoscopic ribbon drawing of the structure of one subunit of the xanthine oxidase-related aldehyde oxidoreductase from *Desulfovibrio gigas*. Each 907-residue subunit of the homodimeric protein contains two Fe_2S_2 clusters visible at the top and the molybdenum–molybdopterin coenzyme buried in the center. (C) Alpha-carbon plot of portions of the protein surrounding the molybdenum–molybdopterin cytosine dinucleotide and (at the top) the two plant-ferredoxin-like Fe_2S_2 clusters. Each of these is held by a separate structural domain of the protein. Two additional domains bind the molybdopterin coenzyme and there is also an intermediate connecting domain. In xanthine oxidase the latter presumably has the FAD binding site which is lacking in the *D. gigas* enzyme. From Romão *et al.*[633] Courtesy of R. Huber.

or polymeric oxygen-bridged ions. However, within the enzymes it exists as the unique **molybdenum coenzymes.** The Mo-containing enzymes usually also contain additional bound cofactors, including Fe–S clusters and flavin coenzymes or heme.

The recognition that the Mo in the molybdoproteins exists in organic cofactor forms came from studies of mutants of *Aspergillus* and *Neurospora*.[650] In 1964, Pateman and associates discovered mutants that lacked both nitrate reductase and xanthine dehydrogenase. Later, it was shown that acid-treated molybdoenzymes released a material that would restore activity to the inactived nitrate reductase from the mutant organisms. This new coenzyme, a phosphate ester of molybdopterin (Fig. 15-17), was characterized by Rajagopalan and coworkers.[650,651] A more complex form of the coenzyme, **molybdopterin cytosine dinucleotide** (Fig. 16-31), is found in the *D. gigas* aldehyde oxidoreductase. Related coenzyme forms include nucleotides of adenine, guanine (see chapter banner, p. 837), and hypoxanthine.[651a,651b] The structure of molybdopterin is related to that of **urothione** (Fig. 15-17), a normal urinary constituent. The relationship to urothione was strengthened by the fact that several children with severe neurological and other symptoms were found to lack both sulfite oxidase and xanthine dehydrogenase as well as the molybdenum cofactor and urinary urothione.[646,646a,646b]

Study by X-ray absorption spectroscopy of the extended **X-ray absorption fine structure** (EXAFS) has provided estimates of both the nature and the number of the nearest neighboring atoms around the Mo. The EXAFS spectra of xanthine dehydrogenase and of nitrate reductase from **Chlorella** confirmed the presence of both the Mo(VI)O_2 unit with Mo–O distances of 0.17 nm and two or three sulfur atoms at distances of 0.24 nm.[652,653] The two sulfur atoms were presumed to come from the molybdopterin. A peculiarity of the xanthine oxidase family is the presence on the molybdenum of a "cyanolyzable" sulfur.[654] This is a sulfide attached to the molybdenum, which is present as Mo(VI)OS rather than Mo(VI)O_2. Reaction with cyanide produces thiocyanate (Eq. 16-64).



(16-64)

The active site structures of the three classes of molybdenum-containing enzymes are compared in Fig. 16-32. In the DMSO reductase family there are two identical molybdopterin dinucleotide coenzymes complexed with one molybdenum. However, only one of these appears to be functionally linked to the Fe_2S_2 center.

Nitrogenase, which catalyzes the reduction of N_2 to two molecules of NH_3, has a different **molybdenum–iron cofactor (FeMo-co)**. It can be obtained by acid denaturation of the very oxygen-labile iron–molybdenum protein of nitrogenase followed by extraction with dimethylformamide.[655,656] The coenzyme is a complex Fe–S–Mo cluster also containing **homocitrate** with a composition $MoFe_7S_9$–homocitrate (see Fig. 24-3). Nitrogenase and this coenzyme are considered further in Chapter 24.

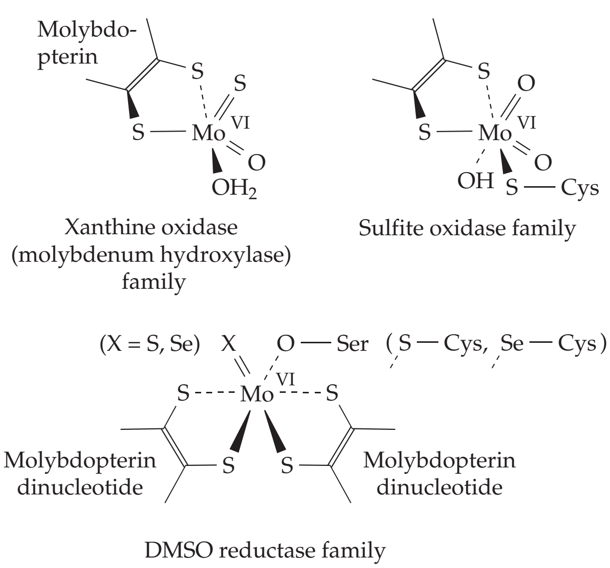


Figure 16-32 Structures surrounding molybdenum in three families of molybdoenzymes. See Hille.[632]

2. Enzymatic Mechanisms

Although several of the reactions catalyzed by molybdoenzymes are classified as dehydrogenases, all of them except nitrogenase involve H_2O as either a reactant or a product. The EXAFS spectra suggest that the Mo(VI)O_2 unit is converted to Mo(IV)O during reaction with a substrate Sub (Eq. 16-65, step *a*). Reaction of the Mo(IV)O with water (step *b*) completes the catalysis.

$$[\mathrm{Mo(VI)O_2}]^{2+} + \mathrm{Sub} \xrightarrow{a} [\mathrm{Mo(IV)O}]^{2+} + \mathrm{Sub\text{–}O}$$

$$[\mathrm{Mo(IV)O}]^{2+} + \mathrm{H_2O} \xrightarrow{b} 2\mathrm{H}^+ + 2e^- + [\mathrm{Mo(VI)}]^{2+}\mathrm{O_2}$$

$$\mathrm{SumSub} + \mathrm{H_2O} \rightarrow \mathrm{Sub\text{–}O} + 2\mathrm{H}^+ + 2e^- \quad (16\text{-}65)$$

Step *a* of all of these reactions can be regarded as an **oxo-transfer**.[653]

To complete the reaction, two electrons must be passed from Mo(IV) to a suitable acceptor, usually an Fe–S cluster or a bound heme group. FAD is also often present. Xanthine oxidase[634,635,643,657,657a] contains two Fe_2S_2 clusters and a FAD for each of the two atoms of Mo in the dimer. Since this enzyme acts like a typical flavin oxidase that generates H_2O_2 from O_2, it may be that electrons pass from Mo to the Fe–S center and then to the flavin. Since the EPR signal of the paramagnetic Mo(V), with its characteristic six-line hyperfine structure, is seen during the action of xanthine oxidase and other molybdenum-containing enzymes, single-electron transfers are probably involved.

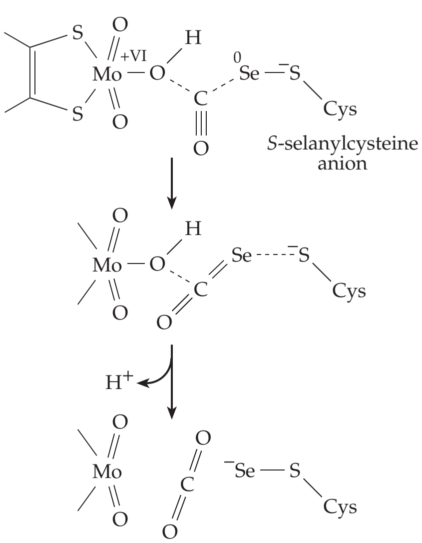
In bacteria such as *E. coli* a dissimilatory nitrate reductase allows nitrate to serve as an oxidant in place of O_2. An oxygen atom is removed from the nitrate to form nitrite as two electrons are accepted from a membrane-bound cytochrome *b*. The nitrate reductase consists of a 139-kDa Mo-containing catalytic subunit, a 58-kDa electron-transferring subunit that contains both Fe_3S_4 and Fe_4S_4 centers, and a 26-kDa heme-containing membrane anchor subunit.[658–660] The assimilatory nitrate reductase of fungi, green algae, and higher plants contains both a *b*-type cytochrome and FAD and a molybdenum coenzyme in a large oligomeric complex.[661–663a]

Formate dehydrogenases from many bacteria contain molybdopterin and also often selenium (Table 15-4).[664,665] A membrane-bound Mo-containing formate dehydrogenase is produced by *E. coli* grown anaerobically in the presence of nitrate. Under these circumstances it is coupled to nitrate reductase via an electron-transport chain in the membranes which permits oxidation of formate by nitrate (Eq. 18-26). This enzyme is also a multisubunit protein.[665,666] Two other Mo- and Se- containing formate dehydrogenases are produced by *E. coli*.[667,668] The three-dimensional structure is known for one of them, **formate dehydrogenase H**, a component of the anaerobic formate hydrogen lyase complex (Eq. 17-25).[669,670] The structure shows Mo held by the sulfur atoms of two molybdopterin molecules, as in DMSO reductase. The Se atom of SeCys 140 is also coordinated with the Mo atom, and the imidazole of His 141 is in close proximity. When ^{13}C-labeled formate was oxidized in ^{18}O-enriched water no ^{18}O was found in the released product, CO_2.[671] This suggested that formate may be bound to Mo and dehydrogenated, with Mo(VI) being reduced to Mo(IV). The formate hydrogen might be transferred as H^+ to the His 140 side chain. Mo(IV) could then be reoxidized by electron transfer in two one-electron steps.[670] However, recent X-ray absorption spectra suggest the presence in the enzyme of a selenosulfide ligand to Mo.[672] Mechanistic uncertainties remain!

A flavin-dependent formate dehydrogenase system found in *Methanobacterium* passes electrons from dehydrogenation of formate to FAD and then to the deazaflavin coenzyme F_{420}.[673] In contrast to these Mo-containing enzymes, the formate dehydrogenase from *Pseudomonas oxalaticus*, which oxidizes formate with NAD^+ (Eq. 16-66), contains neither Mo or Se.[674]

$$HCOO^- + NAD^+ \longrightarrow CO_2 + NADH \qquad (16\text{-}66)$$

It is a large 315-kDa oligomer containing 2 FMN and ~20 Fe / S. Formate dehydrogenases of green plants and yeasts are smaller 70- to 80-kDa proteins lacking bound prosthetic groups.[674] A key enzyme in the metabolism of carbon monoxide-oxidizing bacteria is CO oxidase, another membrane-bound molybdoenzyme.[675–676c] It also contains selenium, which is attached to a cysteine side chain as **S-selanylcysteine.** A proposed reaction requence[676a] is shown in Eq. 16-67.



(16-67)

3. Nutritional Need for Mo

The first hint of an essential role of molybdenum in metabolism came from the discovery that animals raised on a diet deficient in molybdenum had decreased liver xanthine oxidase activity. There is no evidence that xanthine oxidase is essential for all life, but a human genetic deficiency of sulfite oxidase or of its molybdopterin coenzyme can be lethal.[646,646a,b] The conversion of molybdate into the molybdopterin cofactor in *E. coli* depends upon at least five genes.[677] In *Drosophila* the addition of the cyanolyzable sulfur (Eq. 16-64) is the final step in formation of xanthine dehydrogenase.[678] It is of interest that sulfur (S^0) can be transferred from rhodanese (see Eq. 24-45), or from a related mercaptopyruvate sulfurtransferase[679] into the desulfo form of xanthine oxidase to generate an active enzyme.[680]

Uptake of molybdate by cells of *E. coli* is accomplished by an ABC-type transport system.[681] In some bacteria, e.g., the nitrogen-fixing *Azotobacter*, molybdenum can be stored in protein-bound forms.[682]

I. Tungsten

For many years tungsten was considered only as a potential antagonist for molybdenum. However, in 1970 growth stimulation by tungsten compounds was observed for some acetogens, some methanogens, and a few hyperthermophilic bacteria. Since then over a dozen tungstoenzymes have been isolated.[683,683a] These can be classified into three categories: aldehyde oxidoreductases, formaldehyde oxidoreductases, and the single enzyme acetylene hydratase. In most cases the tungstoenzymes resemble the corresponding molybdoenzymes and in most instances organisms containing a tungsten-requiring enzyme also contain the corresponding molybdenum enzyme. However, a few hyperthermophilic archaea appear to require W and are unable to use Mo.

The aldehyde ferredoxin oxidoreductase from the hyperthermophile *Pyrococcus furiosus* was the first molybdopterin-dependent enzyme for which a three-dimensional structure became available.[683,684] The tungstoenzyme resembles that of the related molybdoenzyme (Fig. 16-31). A similar ferredoxin-dependent enzyme reduces glyceraldehyde-3-phosphate.[685] Another member of the tungstoenzyme aldehyde oxidoreductase family is **carboxylic acid reductase**, an enzyme found in certain acetogenic clostridia. It is able to use reduced ferredoxin to convert unactivated carboxylic acids into aldehydes, even though $E^{o'}$ for the acetaldehyde / acetate couple is –0.58 V.[686]

Tungsten- and sometimes Se-containing formate dehydrogenases together with *N*-formylmethanofuran dehydrogenases (Fig. 15-22, step *b*) form a second family.

Again, these appear to resemble the corresponding Mo-dependent enzymes. The unique **acetylene hydratase** from the acetylene-utilizing *Pelobacter acetylenicus* catalyzes the hydration of acetylene to acetaldehyde.[687]

$$H{-}C{\equiv}C{-}H + H_2O \rightarrow H_3C{-}CHO \qquad (16\text{-}68)$$

In *Thermotoga maritima*, the most thermophilic organism known, tungsten promotes synthesis of an Fe-containing hydrogenase as well as some other enzymes but seems to have a regulatory rather than a structural role.[688]

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Study Questions

1. Describe one or more metabolic functions of ions or chelate complexes of ions derived from each of the following metallic elements: Ca, Mg, Fe, Cu, Ni, Co.

2. If the concentration of Cu, Zn-superoxide dismutase (SOD) in a yeast cell is 10 µM, the total copper (bound and free) is 70 µM, and the dissociation constant for loss of Cu^+ from SOD is 6 fM, what will be the concentration of free Cu^+ within the cell? If the cell volume is 10^{-14} liter, how many copper ions will be present in a single cell? See Roe *et al.* (1999) *Science* **284**, 805–808.

3. Outline the metabolic pathways that are utilized by acetic acid-producing bacteria (acetogens) in the stoichiometric conversion of one molecule of glucose into three molecules of acetic acid. Indicate briefly the nature of any unusual coenzymes or metalloproteins that are required.

4. Suppose that you could add a solution containing micromolar concentrations of Cu^{2+}, Mn^{2+}, Fe^{3+}, Co^{2+}, Zn^{2+}, and MoO_4^{2-} and millimolar amounts of Mg^{2+}, Ca^{2+}, and K^+ to a solution that contains a large excess of a mixture of many cellular proteins. What would be the characteristics of the sites that would become occupied by each of these metal ions? How tightly do you think they would be bound?

5. What factors affect the rate of electron transfer from an electron **donor** (atom or molecule) to an **acceptor**?

6. List some mechanism that cells can use to combat the toxicity of metal ions?

7. NADH peroxidase (p. 857, Eq. 15-59) does not contain a transition metal ion. Propose a reasonable detailed mechanism for its action and compare it with mechanisms of action of heme peroxidases.

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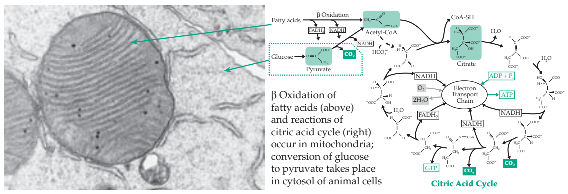
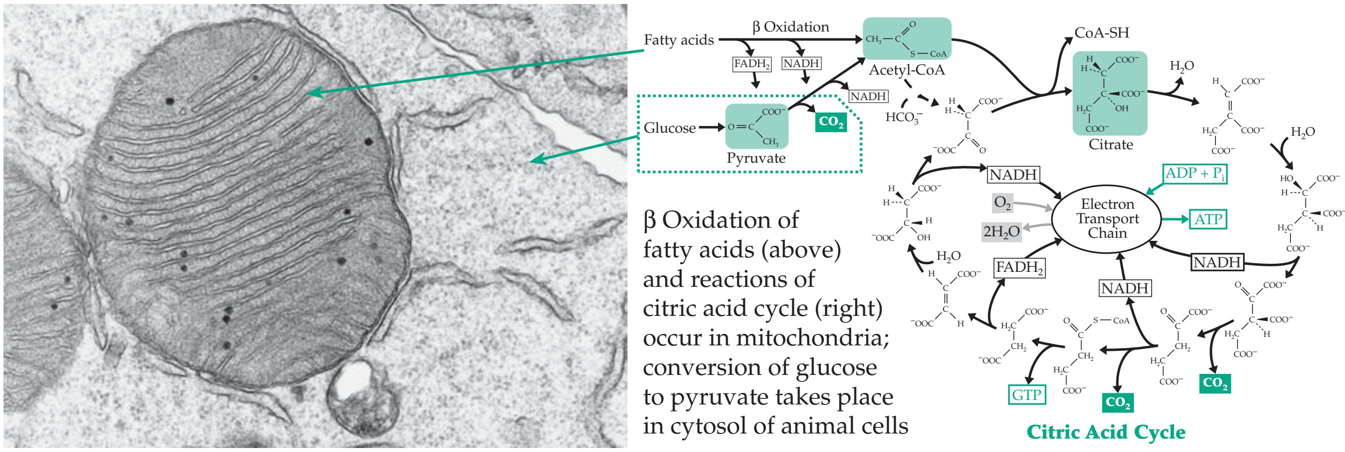


Metabolism, a complex network of chemical reactions, occurs in several different compartments in eukaryotic cells. Fatty acids, a major source of energy for many human cells, are oxidized in the mitochondria via β oxidation and the citric acid cycle. Glucose, a primary source of energy, is converted to pyruvate in the cytosol. Biosynthetic reactions occurring in both compartments form proteins, nucleic acids, storage polymers such as glycogen, and sparingly soluble lipid materials which aggregate to form membranes. Hydrophobic groups in proteins and other polymers also promote self-assembly of the cell. At the same time, oxidative processes, initiated by O_2, increase the water solubility of molecules, leading to metabolic turnover. Micrograph courtesy of Kenneth Moore.

Contents

17 The Organization of Metabolism



β Oxidation of fatty acids (above) and reactions of citric acid cycle (right) occur in mitochondria; conversion of glucose to pyruvate takes place in cytosol of animal cells

Metabolism involves a bewildering array of chemical reactions, many of them organized as complex cycles which may appear difficult to understand. Yet, there is logic and orderliness. With few exceptions, metabolic pathways can be regarded as sequences of the reactions considered in Chapters 12–16 (and summarized in the table inside the back cover) which are organized to accomplish specific chemical goals. In this chapter we will examine the chemical logic of the major pathways of catabolism of foods and of cell constituents as well as some reactions of biosynthesis (anabolism). A few of the sequences have already been discussed briefly in Chapter 10.

A. The Oxidation of Fatty Acids

Hydrocarbons yield more energy upon combustion than do most other organic compounds, and it is, therefore, not surprising that one important type of food reserve, the fats, is essentially hydrocarbon in nature. In terms of energy content the component fatty acids are the most important. Most aerobic cells can oxidize fatty acids completely to CO_2 and water, a process that takes place within many bacteria, in the matrix space of animal mitochondria, in the peroxisomes of most eukaryotic cells, and to a lesser extent in the endoplasmic reticulum.

The carboxyl group of a fatty acid provides a point for chemical attack. The first step is a priming reaction in which the fatty acid is converted to a water-soluble acyl-CoA derivative in which the α hydrogens of the fatty acyl radicals are "activated" (step *a*, Fig. 17-1). This synthetic reaction is catalyzed by **acyl-CoA synthetases** (fatty acid:CoA ligases). It is driven by the hydrolysis of ATP to AMP and two inorganic phosphate ions using the sequence shown in Eq. 10-1 (p. 508). There are isoenzymes that act on short-, medium-, and long-chain fatty acids. Yeast contains at least five of these.[1] In every case the acyl group is activated through formation of an intermediate acyl adenylate; hydrolysis of the released pyrophosphate helps to carry the reaction to completion (see discussion in Section H).

1. Beta Oxidation

The reaction steps in the oxidation of long-chain acyl-CoA molecules to acetyl-CoA were outlined in Fig. 10-4. Because of the great importance of this β oxidation sequence in metabolism the steps are shown again in Fig. 17-1 (steps *b–e*). The chemical logic becomes clear if we examine the structure of the acyl-CoA molecule and consider the types of biochemical reactions available. If the direct use of O_2 is to be avoided, the only reasonable mode of attack on an acyl-CoA molecule is dehydrogenation. Removal of the α hydrogen as a proton is made possible by the activating effect of the carbonyl group of the thioester. The β hydrogen can be transferred from the intermediate enolate, as a hydride ion, to the bound FAD present in the **acyl-CoA dehydrogenases** that catalyze this reaction[2–5] (step *b*, Fig. 17-1; see also Eq. 15-23). These enzymes contain FAD, and the reduced coenzyme $FADH_2$ that is formed is reoxidized by an **electron transferring flavoprotein** (Chapter 15), which also contains FAD. This protein carries the electrons abstracted in the oxidation process to the inner membrane of the mitochondrion where they enter the mitochondrial electron transport system,[5a] as depicted in Fig. 10-5 and as discussed in detail in

Chapter 18.

The product of step *b* is always a **trans-Δ²-enoyl-CoA**. One of the few possible reactions of this unsaturated compound is nucleophilic addition at the β position. The reacting nucleophile is an HO^- ion from water. This reaction step (step *c*, Fig. 17-1) is completed by addition of H^+ at the α position. The resulting **β-hydroxyacyl-CoA** (3-hydroxyacyl-CoA) is dehydrogenated to a ketone by NAD^+ (step *d*).[5b] This series of three reactions is the β oxidation sequence.

At the end of this sequence, the β-oxoacyl-CoA derivative is cleaved (Fig. 17-1, step *e*) by a **thiolase** (see also Eq. 13-35). One of the products is acetyl-CoA, which can be catabolized to CO_2 through the citric acid cycle. The other product of the thiolytic cleavage is an acyl-CoA derivative that is *two carbon atoms shorter than the original acyl-CoA*. This molecule is recycled through the β oxidation process, a two-carbon acetyl unit being removed as acetyl-CoA during each turn of the cycle (Fig. 17-1). The process continues until the fatty acid chain is completely degraded. If the original fatty acid contained an *even* number of carbon atoms in a straight chain, acetyl-CoA is the only product. However, if the original fatty acid contained an *odd* number of carbon atoms, **propionyl-CoA** is formed at the end.

For every step of the β oxidation sequence there is a small family of enzymes with differing chain length preferences.[6,7] For example, in liver mitochondria one acyl-CoA dehydrogenase acts most rapidly on *n*-butyryl and other short-chain acyl-CoA; a second prefers a substrate of medium chain length such as *n*-octanoyl-CoA; a third prefers long-chain substrates such as palmitoyl-CoA; and a fourth, substrates with 2-methyl branches. A fifth enzyme acts specifically on isovaleryl-CoA. Similar preferences exist for the other enzymes of the β oxidation pathway. In *Escherichia coli*

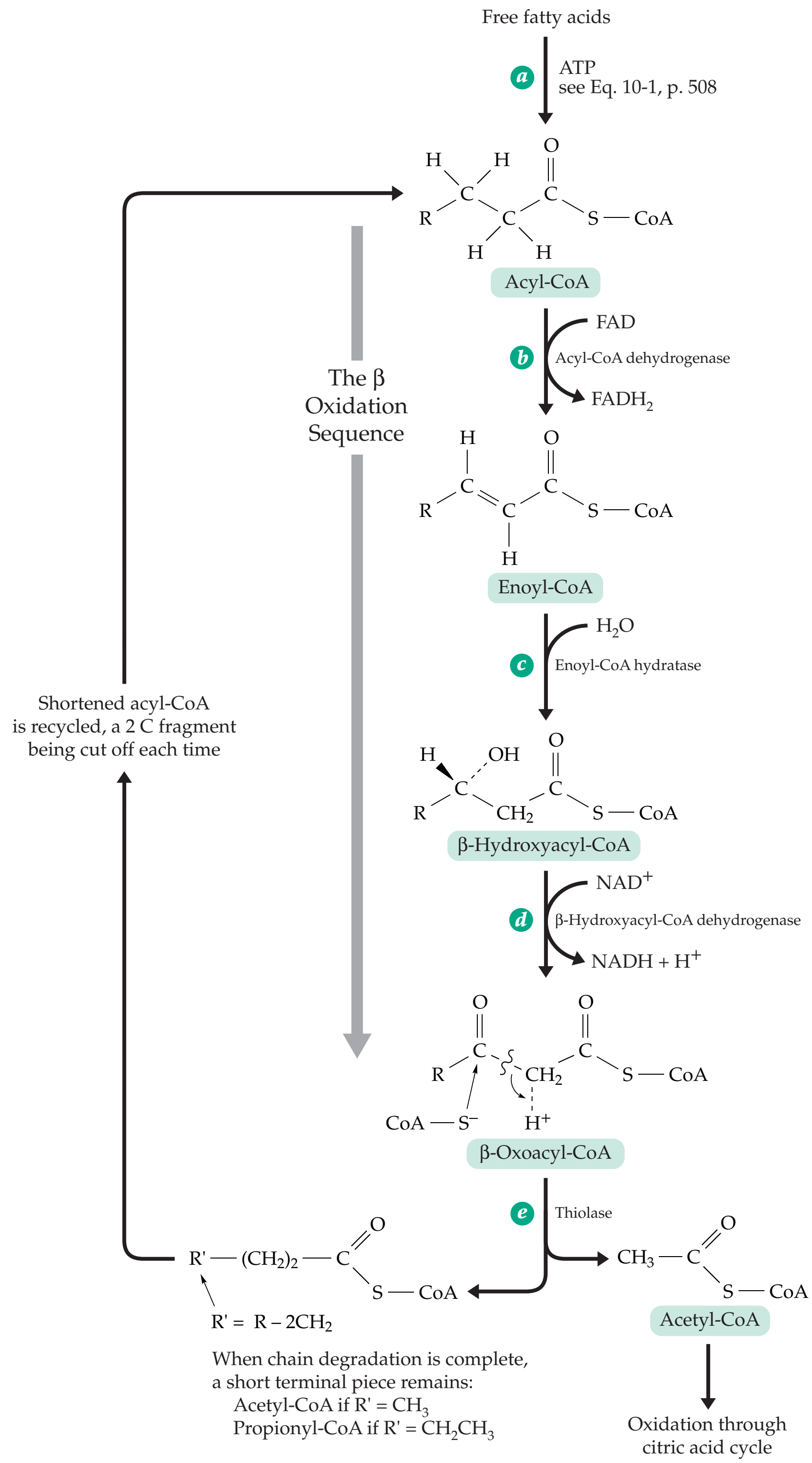


Figure 17-1 The β oxidation cycle for fatty acids. Fatty acids are converted to acyl-CoA derivatives from which 2-carbon atoms are cut off as acetyl-CoA to give a shortened chain which is repeatedly sent back through the cycle until only a 2- or 3-carbon acyl-CoA remains. The sequence of steps *b*, *c*, and *d* also occurs in many other places in metabolism.

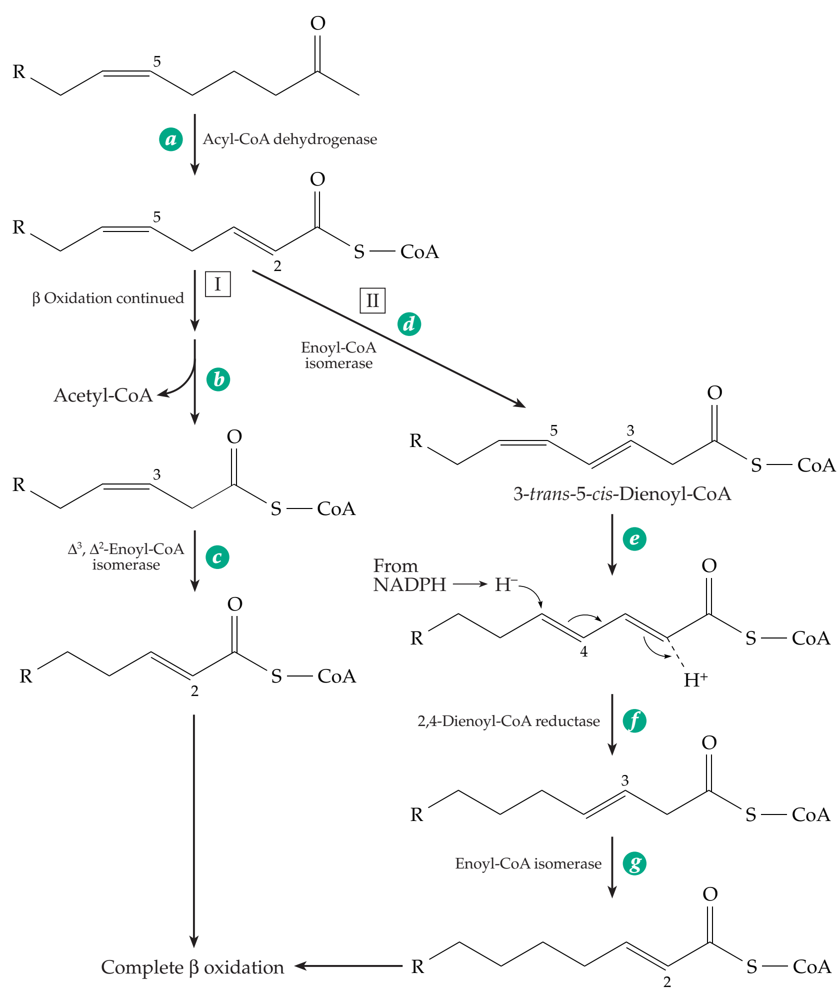
most of these enzymes are present as a complex of multifunctional proteins[8] while the mitochondrial enzymes may be organized as a multiprotein complex.[9,10]

Peroxisomal beta oxidation. In animal cells β oxidation is primarily a mitochondrial process,[5] but it also takes place to a limited extent within peroxisomes and within the endoplasmic reticulum.[11–14] This "division of labor" is still not understood well. Straight-chain fatty acids up to 18 carbons in length appear to be metabolized primarily in mitochondria, but in the liver fatty acids with very long chains are processed largely in peroxisomes.[13] There, a very long-chain acyl-CoA synthetase acts on fatty acids that contain 22 or more carbon atoms.[15] In yeast all β oxidation takes place in peroxisomes,[15,16] and in most organisms, including green plants,[17–18a] the peroxisomes are the most active sites of fatty acid oxidation. However, animal peroxisomes cannot oxidize short-chain acyl-CoA molecules; they must be returned to the mitochondria.[16] The activity of peroxisomes in β oxidation is greatly increased by the presence of a variety of compounds known as **peroxisome proliferators**. Among them are drugs such as aspirin and clofibrate and environmental xenobiotics such as the plasticizer bis-(2-ethyl-hexyl)-phthalate. They may induce as much as a tenfold increase in peroxisomal β oxidation.[11,12,19,19a]

Several other features also distinguish β oxidation in peroxisomes. The peroxisomal flavoproteins that catalyze the dehydrogenation of acyl-CoA molecules to unsaturated enoyl-CoAs (step *b* of Fig. 17-1) are **oxidases** in which the $FADH_2$ that is formed is reoxidized by O_2 to form H_2O_2.[13,20] In peroxisomes the enoyl-hydratase and the NAD^+-dependent dehydrogenase catalyzing steps *c* and *d* of Fig. 17-1 are present together with an enoyl-CoA isomerase (next section) as a trifunctional enzyme consisting of a single polypeptide chain.[21] As in mitochondrial β oxidation the 3-hydroxyacyl-CoA intermediates formed in both animal peroxisomes and plant peroxisomes (glyoxysomes) have the L configuration. However, in fungal peroxisomes as well as in *E. coli* they have the D configuration.[22,23] Further metabolism in these organisms requires an epimerase that converts the D-hydroxyacyl-CoA molecules to L.[24] In the past it has often been assumed that peroxisomal membranes are freely permeable to NAD^+, NADH, and acyl-CoA molecules. However, genetic experiments with yeast and other recent evidence indicate that they are impermeable *in vivo* and that carrier and shuttle mechanisms similar to those in mitochondria may be required.[14,25]

Unsaturated fatty acids. Mitochondrial β oxidation of such unsaturated acids as the Δ^9-oleic acid begins with removal of two molecules of acetyl-CoA to form a Δ^5-acyl-CoA. However, further metabolism is slow. Two pathways have been identified (Eq. 17-1).[26–29b] The first step for both is a normal dehydrogenation to a 2-*trans*-5-*cis*-dienoyl-CoA. In pathway I this intermediate reacts slowly by the normal β oxidation sequence to form a 3-*cis*-enoyl-CoA intermediate which must then be acted upon by an auxiliary enzyme, a *cis*-Δ^3-*trans*-Δ^2-enoyl-CoA isomerase (Eq. 17-1, step *c*), before β oxidation can continue.

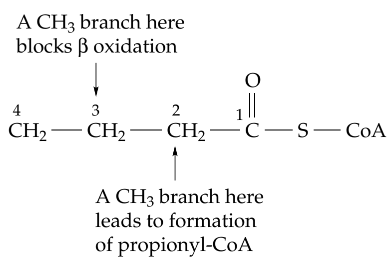
The alternative reductase pathway (II in Eq. 17-1) is often faster. It makes use of an additional isomerase which converts 3-*trans*, 5-*cis*-dienoyl-CoA into the 2-*trans*, 4-*trans* isomer in which the double bonds are conjugated with the carbonyl group.[29] This permits removal of one double bond by reduction with NADPH as shown (Eq. 17-1, step *f*).[29a,29b] The peroxisomal



(17-1)

pathway is similar.[21] However, the intermediate formed in step *e* of Eq. 17-1 may sometimes have the 2-*trans*, 4-*cis* configuration.[17] The NADH for the reductive step *f* may be supplied by an NADP-dependent isocitrate dehydrogenase.[29c] Repetition of steps *a*, *d*, *e*, and *f* of Eq. 17-1 will lead to β oxidation of the entire chain of polyunsaturated fatty acids such as linoleoyl-CoA or arachidonoyl-CoA. Important additional metabolic routes for polyunsaturated fatty acid derivatives are described in Chapter 21.

Branched-chain fatty acids. Most of the fatty acids in animal and plant fats have straight unbranched chains. However, branches, usually consisting of methyl groups, are present in lipids of some microorganisms, in waxes of plant surfaces, and also in polyprenyl chains. As long as there are not too many branches and if they occur only in the even-numbered positions (i.e., on carbons 2, 4, etc.) β oxidation proceeds normally. Propionyl-CoA is formed in addition to acetyl-CoA as a product of the chain degradation. On the other hand, if methyl groups occur in positions 3, 5, etc., β oxidation is blocked at step *d* of Fig. 17-1. A striking example of the effect of such blockage was

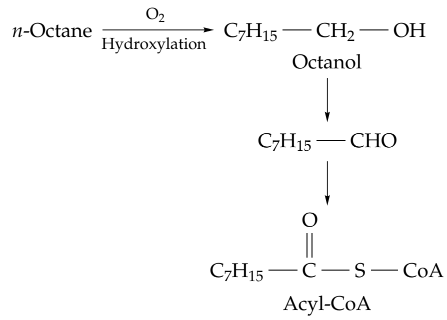


provided by the synthetic detergents in common use until about 1966. These detergents contained a hydrocarbon chain with methyl groups distributed more or less randomly along the chain. Beta oxidation was blocked at many points and the result was a foamy pollution crisis in sewage plants in the United States and in some other countries. Since 1966, only biodegradable detergents having straight hydrocarbon chains have been sold.

In fact, cells *are* able to deal with small amounts of these hard-to-oxidize substrates. The O_2-dependent reactions called α oxidation and ω oxidation are used. These are related also to the oxidation of hydrocarbons which we will consider next.

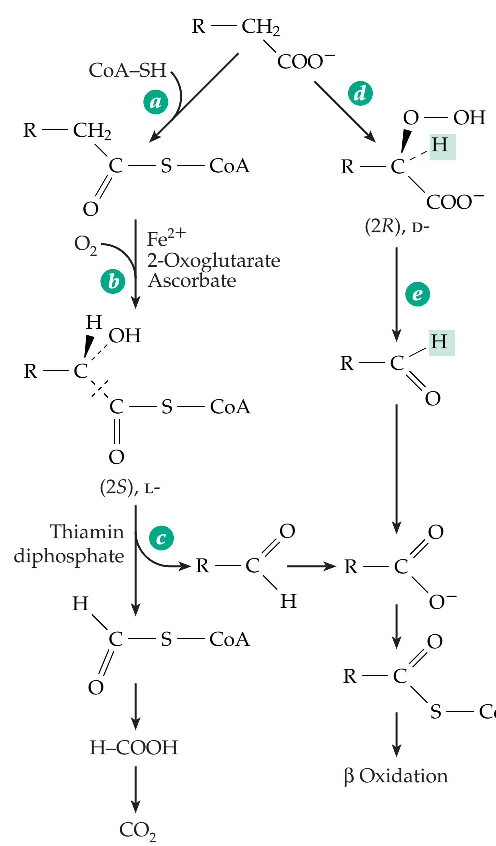
Oxidation of saturated hydrocarbons. Although the initial oxidation step is chemically difficult, the tissues of our bodies are able to metabolize saturated hydrocarbons such as *n*-heptane slowly, and some microorganisms oxidize straight-chain hydrocarbons rapidly.[30,31] Strains of *Pseudomonas* and of the yeast *Candida* have been used to convert petroleum into edible proteins.[9]

The first step in oxidation of alkanes is usually an O_2-requiring **hydroxylation** (Chapter 18) to a primary alcohol. Further oxidation of the alcohol to an acyl-CoA derivative, presumably via the aldehyde (Eq. 17-2), is a frequently encountered biochemical oxidation sequence.



(17-2)

Alpha oxidation and omega oxidation. Animal tissues degrade such straight-chain fatty acids as palmitic acid, stearic acid, and oleic acid almost entirely by β oxidation, but plant cells often oxidize fatty acids one carbon at a time. The initial attack may involve hydroxylation on the α-carbon atom (Eq. 17-3) to form either the D- or the L-2-hydroxy acid.[17,18,32,32a] The L-hydroxy acids are oxidized rapidly, perhaps by dehydrogenation to the oxo acids (Eq. 17-3, step *b*) and oxidative decarboxylation, possibly utilizing H_2O_2 (see Eq. 15-36). The D-hydroxy acids tend to accumulate



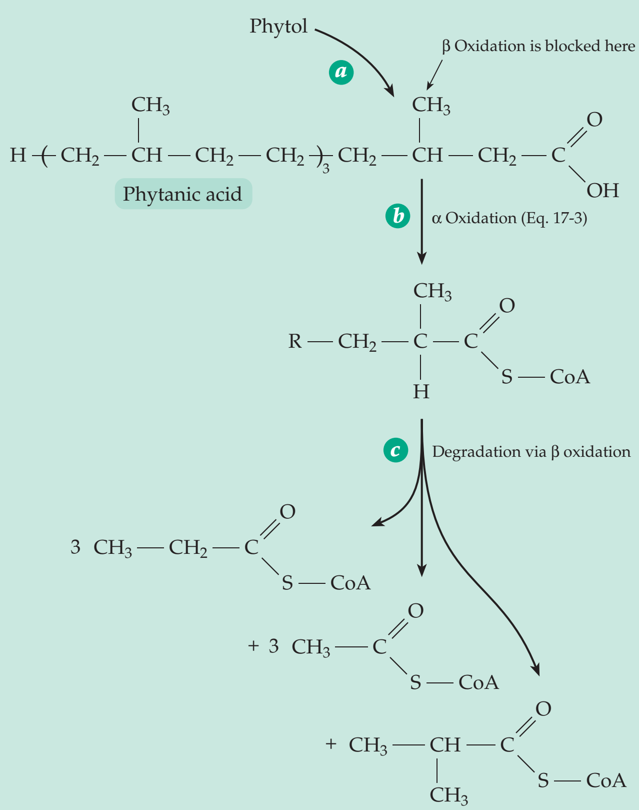
(17-3)

and are normally present in green leaves. However, they too are oxidized further, with retention of the α hydrogen as indicated by the shaded squares in Eq. 17-3, step *e*. This suggests a new type of dehydrogenation with concurrent decarboxylation. Alpha oxidation also occurs to some extent in animal tissues. For example, when β oxidation is blocked by the presence of a methyl side chain, the body may use α oxidation to get past the block (see **Refsum disease**, Box 17-A). As in plants, this occurs principally in the peroxisomes[33–35] and is important for degradation not only of polyprenyl chains but also bile acids. In the brain some of the fatty acyl groups of sphingolipids are hydroxylated to α-hydroxyacyl groups.[36] Alpha oxidation in animal cells occurs after conversion of free fatty acids to their acyl-CoA derivatives (Eq. 7-3, step *a*). This is followed by a 2-oxoglutarate-dependent hydroxylation (step *b*, see also Eq. 18-51) to form the 2-hydroxyacyl-CoA, which is cleaved in a standard thiamin diphosphate-requiring α cleavage (step *c*). The products are formyl-CoA, which is hydrolyzed and oxidized to CO_2, and a fatty aldehyde which is metabolized further by β oxidation.[34a]

In plants α-dioxygenases (Chapter 18) convert free fatty acids into 2(*R*)-hydroperoxy derivatives (Eq. 7-3, step *d*).[32a] These may be decarboxylated to fatty aldehydes (step *e*, see also Eq. 15-36) but may also give rise to a variety of other products. Compounds arising from linoleic and linolenic acids are numerous and include epoxides, epoxy alcohols, dihydroxy acids, short-chain aldehydes, divinyl ethers, and jasmonic acid (Eq. 21-18).[32a]

On other occasions, **omega (ω) oxidation** occurs at the opposite end of the chain to yield a dicarboxylic acid. Within the human body 3,6-dimethyloctanoic acid and other branched-chain acids are degraded largely via ω oxidation. The initial oxidative attack is by a hydroxylase of the cytochrome P450 group (Chapter 18). These enzymes act not only on fatty acids but also on prostaglandins, sterols, and many other lipids. In the animal body fatty acids are sometimes hydroxylated both at the terminal (ω) position and at the next (ω-2 or ω2) carbon. In plants hydroxylation may occur at the ω2, ω3, and ω4 positions as well.[17,37] Dicarboxylates resulting from ω oxidation of straight-chain fatty acids

BOX 17-A REFSUM DISEASE



In this autosomally inherited disorder of lipid metabolism the 20-carbon branched-chain fatty acid **phytanic acid** accumulates in tissues. Phytanic acid is normally formed in the body (step *a* in the accompanying scheme) from the polyprenyl plant alcohol **phytol**, which is found as an ester in the chlorophyll present in the diet (Fig. 23-20). Although only a small fraction of the ingested phytol is oxidized to phytanic acid, this acid accumulates to a certain extent in animal fats and is present in dairy products. Because β oxidation is blocked, the first step (step *b*) in degradation of phytanic acid is α oxidation in peroxisomes.[a] The remainder of the molecule undergoes β oxidation (step *c*) to three molecules of propionyl-CoA, three of acetyl-CoA, and one of isobutyryl-CoA. The disease, which was described by Refsum in 1946, causes severe damage to nerves and brain as well as lipid accumulation and early death.[b–d] This rare disorder apparently results from a defect in the initial hydroxylation. The causes of the neurological symptoms of Refsum disease are not clear, but it is possible that the isoprenoid phytanic acid interferes with prenylation of membrane proteins.[b]

[a] Singh, I., Pahan, K., Dhaunsi, G. S., Lazo, O., and Ozand, P. (1993) *J. Biol. Chem.* **268**, 9972–9979

[b] Steinberg, D. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 2 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 2351–2369, McGraw-Hill, New York

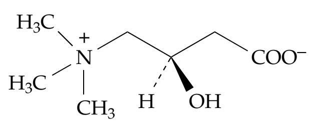
[c] Steinberg, D., Herndon, J. H., Jr., Uhlendorf, B. W., Mize, C. E., Avigan, J., and Milne, G. W. A. (1967) *Science* **156**, 1740–1742

[d] Muralidharan, V. B., and Kishimoto, Y. (1984) *J. Biol. Chem.* **259**, 13021–13026

can undergo β oxidation from both ends. The resulting short-chain dicarboxylates, which appear to be formed primarily in the peroxisomes,[38] may be converted by further β oxidation into succinyl-CoA and free succinate.[39] Incomplete β oxidation in mitochondria (Fig. 17-1) releases small amounts of 3(β)-hydroxy fatty acids, which also undergo ω oxidation and give rise to free 3-hydroxydicarboxylic acids which may be excreted in the urine.[40]

2. Carnitine and Mitochondrial Permeability

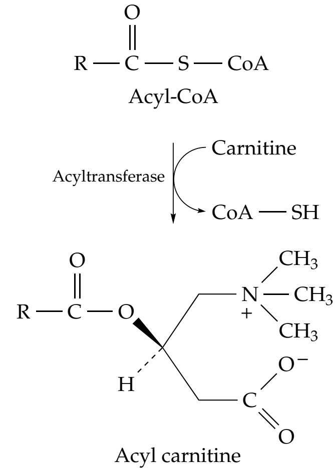
A major factor controlling the oxidation of fatty acids is the rate of entry into the mitochondria. While some long-chain fatty acids (perhaps 30% of the total) enter mitochondria as such and are converted to CoA derivatives in the matrix, the majority are "activated" to acyl-CoA derivatives on the inner surface of the outer membranes of the mitochondria. Penetration of these acyl-CoA derivatives through the mitochondrial inner membrane is facilitated by **L-carnitine**.[41–44]



L-Carnitine

Carnitine is present in nearly all organisms and in all animal tissues. The highest concentration is found in muscle where it accounts for almost 0.1% of the dry matter. Carnitine was first isolated from meat extracts in 1905 but the first clue to its biological action was obtained in 1948 when Fraenkel and associates described a new dietary factor required by the mealworm, *Tenebrio molitor*. At first designated **vitamin B_t**, it was identified in 1952 as carnitine. Most organisms synthesize their own carnitine from lysine side chains (Eq. 24-30). The inner membrane of mitochondria contains a long-chain acyltransferase (carnitine palmitoyltransferase I) that catalyzes transfer of the fatty acyl group from CoA to the hydroxyl group of carnitine (Eq. 17-4).[45–47a] Perhaps acyl carnitine derivatives pass through the membrane more easily than do acyl-CoA derivatives because the positive and negative charges can swing together and neutralize each other as shown in Eq. 17-4. Inside the mitochondrion the acyl group is transferred back from carnitine onto CoA (Eq. 17-4, reverse) by carnitine palmitoyltransferase II prior to initiation of β oxidation.

Tissues contain not only long-chain acylcarnitines but also **acetylcarnitine** and other short-chain acylcarnitines, some with branched chains.[41] By accepting acetyl groups from acetyl-CoA, carnitine causes the release of free coenzyme A which can then be reused.



(17-4)

Thus, carnitine may have a regulatory function. In flight muscles of insects acetylcarnitine serves as a reservoir for acetyl groups. Carnitine acyltransferases that act on short-chain acyl-CoA molecules are also present in peroxisomes and microsomes, suggesting that carnitine may assist in transferring acetyl groups and other short acyl groups between cell compartments. For example, acetyl groups from peroxisomal β oxidation can be transferred into mitochondria where they can be oxidized in the citric acid cycle.[41]

3. Human Disorders of Fatty Acid Oxidation

Mitochondrial β oxidation of fatty acids is the principal source of energy for the heart. Consequently, inherited defects of fatty acid oxidation or of carnitine-assisted transport often appear as serious heart disease (inherited cardiomyopathy). These may involve heart failure, pulmonary edema, or sudden infant death. As many as 1 in 10,000 persons may inherit such problems.[48–50a] The proteins that may be defective include a plasma membrane carnitine transporter; carnitine palmitoyltransferases; carnitine/acylcarnitine translocase; long-chain, medium-chain, and short-chain acyl-CoA dehydrogenases; 2,4-dienoyl-CoA reductase (Eq. 17-1); and long-chain 3-hydroxyacyl-CoA dehydrogenase. Some of these are indicated in Fig. 17-2.

Several cases of genetically transmitted carnitine deficiency in children have been recorded. These children have weak muscles and their mitochondria oxidize long-chain fatty acids slowly. If the inner mitochondrial membrane carnitine palmitoyltransferase II is lacking, long-chain acylcarnitines accumulate in the mitochondria and appear to have damaging effects on membranes. In the unrelated condition of **acute myocardial ischemia** (lack of oxygen, e.g., during a heart attack) there is also a large accumulation of long-chain acylcarnitines.[51,52] These compounds may induce cardiac arrhythmia and may also account for sudden death from deficiency of carnitine palmitoyltransferase II. Treatment of disorders of carnitine metabolism with daily oral ingestion of several grams of carnitine is helpful, especially for deficiency of the plasma membrane transporter.[50a,53] Metabolic abnormalities may be corrected completely.[50a]

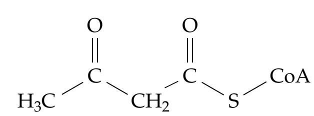
One of the most frequent defects of fatty acid oxidation is deficiency of a mitochondrial acyl-CoA dehydrogenase.[50] If the long-chain-specific enzyme is lacking, the rate of β oxidation of such substrates as octanoate is much less than normal and afflicted individuals excrete in their urine hexanedioic (adipic), octanedioic, and decanedioic acids, all products of ω oxidation.[54] Much more common is the lack of the mitochondrial *medium-chain* acyl-CoA dehydrogenase. Again, dicarboxylic acids, which are presumably generated by ω oxidation in the peroxisomes, are present in blood and urine. Patients must avoid fasting and may benefit from extra carnitine.

A deficiency of very long-chain fatty acid oxidation in peroxisomes is apparently caused by a defective transporter of the ABC type (Chapter 8).[55] The disease, **X-linked adrenoleukodystrophy (ALD)**, has received considerable publicity because of attempts to treat it with "Lorenzo's oil," a mixture of triglycerides of oleic and the C_{22} monoenoic **erucic acid**. The hope has been that these acids would flush out the very long-chain fatty acids that accumulate in the myelin sheath of neurons in the central nervous system and may be responsible for the worst consequences of the disease. However, there has been only limited success.[56,57]

Several genetic diseases involve the development of peroxisomes.[14,35,58,59] Most serious is the **Zellweger syndrome** in which there are no functional peroxisomes. Only "ghosts" of peroxisomes are present and they fail to take up proteins containing the C-terminal peroxisome-targeting sequence SKL.[60,60a] There are many symptoms and death occurs within the first year. Less serious disorders include the presence of catalaseless peroxisomes.[60a]

4. Ketone Bodies

When a fatty acid with an even number of carbon atoms is broken down through β oxidation the last intermediate before complete conversion to acetyl-CoA is the four-carbon **acetoacetyl-CoA**:



Acetoacetyl-CoA

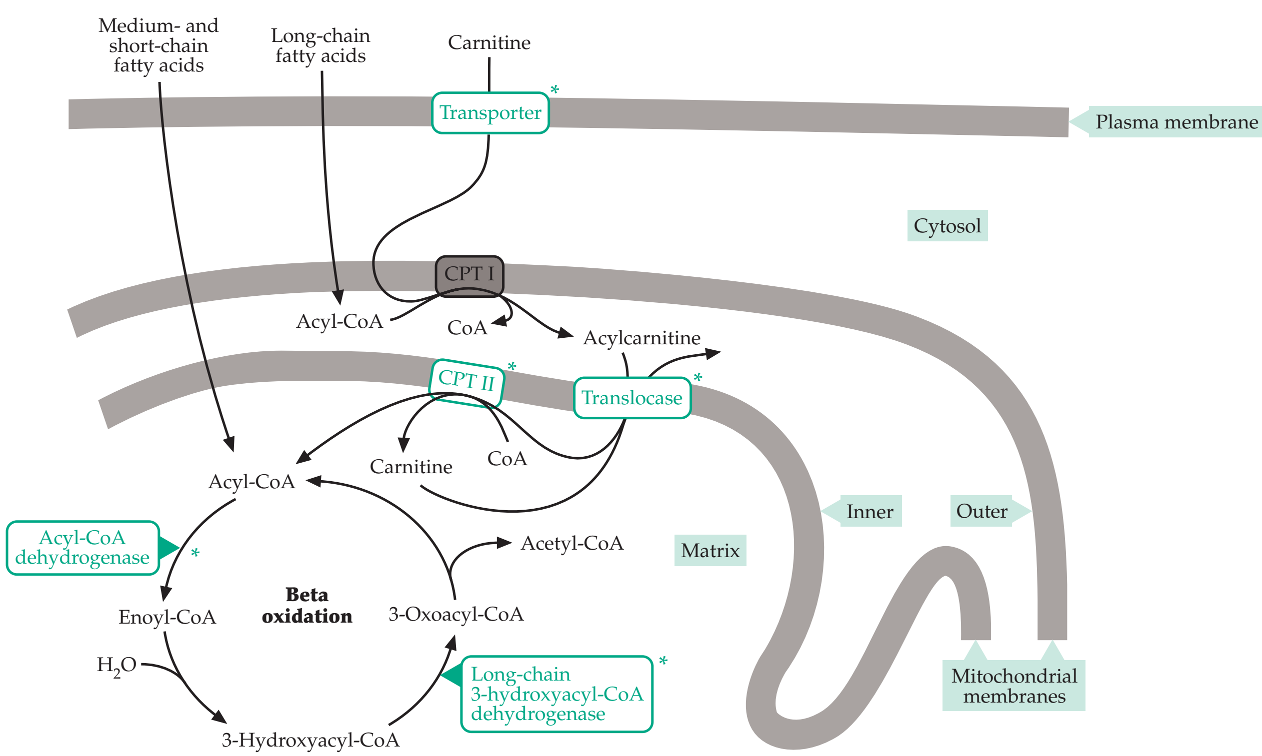
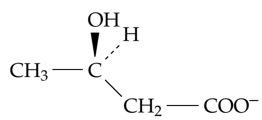


Figure 17-2 Some specific defects in proteins of β oxidation and acyl-carnitine transport causing cardiomyopathy are indicated by the green asterisks. CPT I and CPT II are carnitine palmitoyltransferases I and II. After Kelly and Strauss.[48]

Acetoacetyl-CoA appears to be in equilibrium with acetyl-CoA within the body and is an important metabolic intermediate. It can be cleaved to two molecules of acetyl-CoA which can enter the citric acid cycle. It is also a precursor for synthesis of polyprenyl (isoprenoid) compounds, and it can give rise to free **acetoacetate**, an important constituent of blood. Acetoacetate is a β-oxoacid that can undergo decarboxylation to acetone or can be reduced by an NADH-dependent dehydrogenase to D-3-hydroxybutyrate. Notice that the configuration of this compound is opposite to that of L-3-hydroxybutyryl-CoA which is



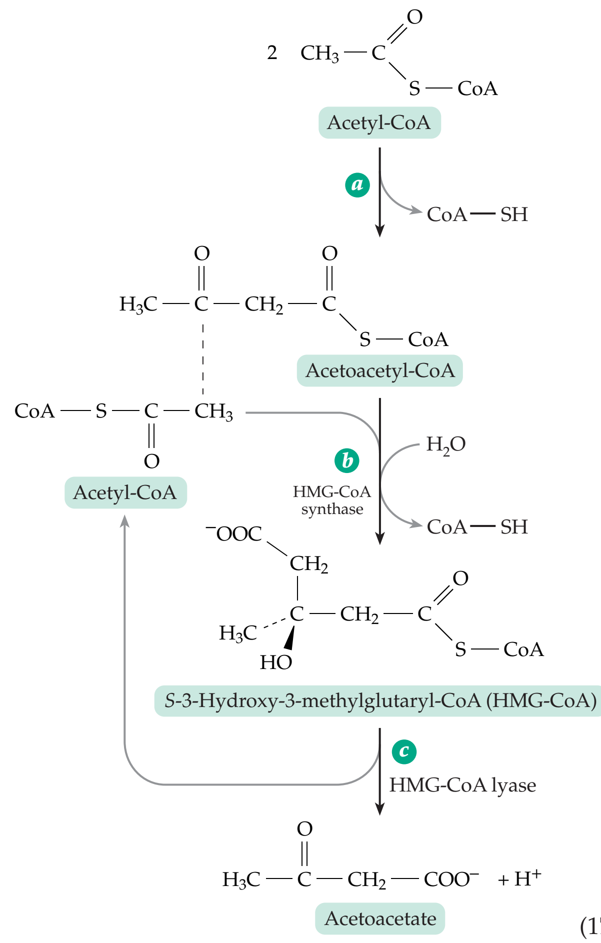
D-3-Hydroxybutyrate

formed during β oxidation of fatty acids (Fig. 17-1). D-3-Hydroxybutyrate is sometimes stored as a polymer in bacteria (Box 21-D).

The three compounds, acetoacetate, acetone, and 3-hydroxybutyrate, are known as **ketone bodies**.[60b] The inability of the animal body to form the glucose precursors, pyruvate or oxaloacetate, from acetyl units sometimes causes severe metabolic problems. The condition known as **ketosis**, in which excessive amounts of ketone bodies are present in the blood, develops when too much acetyl-CoA is produced and its combustion in the critic acid cycle is slow. Ketosis often develops in patients with Type I **diabetes mellitus** (Box 17-G), in anyone with high fevers, and during starvation. Ketosis is dangerous, if severe, because formation of ketone bodies produces hydrogen ions (Eq. 17-5) and acidifies the blood. Thousands of young persons with insulin-dependent diabetes die annually from ketoacidosis.

Rat blood normally contains about 0.07 mM acetoacetate, 0.18 mM hydroxybutyrate, and a variable amount of acetone. These amounts increase to 0.5 mM acetoacetate and 1.6 mM hydroxybutyrate after 48 h of starvation. On the other hand, the blood glucose concentration falls from 6 to 4 mM after 48 h starvation.[61] Under these conditions acetoacetate and hydroxybutyrate are an important alternative energy source for muscle and other tissues.[62,63] Acetoacetate can be thought of as a transport form of acetyl units, which can be reconverted to acetyl-CoA and oxidized in the citric acid cycle.

Some free acetoacetate is formed by direct hydrolysis of acetoacetyl-CoA. In rats, ~11% of the hydroxybutyrate that is excreted in the urine comes from acetoacetate generated in this way.[64] However, most acetoacetate arises in the liver indirectly in a two-step process (Eq. 17-5) that is closely related to the synthesis

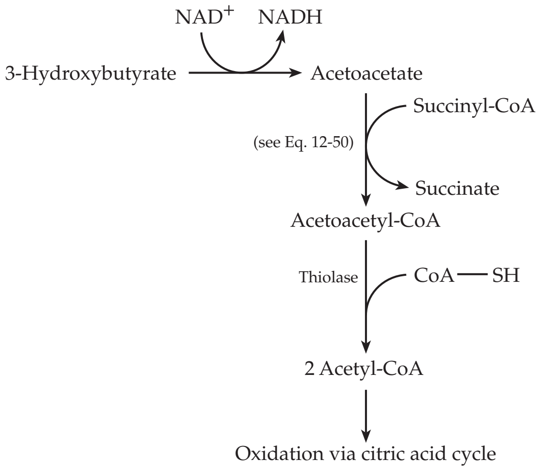


(17-5)

of cholesterol and other polyprenyl compounds. Step *a* of this sequence is a Claisen condensation, catalyzed by 3-hydroxy-3-methyl-glutaryl CoA synthase (HMG-CoA synthase)[64a–c] and followed by hydrolysis of one thioester linkage. It is therefore similar to the citrate synthase reaction (Eq. 13-38). Step *c* is a simple aldol cleavage. The overall reaction has the stoichiometry of a direct hydrolysis of acetoacetyl-CoA. Liver mitochondria contain most of the body's HMG-CoA synthase and are the major site of ketone body formation (**ketogenesis**). Cholesterol is synthesized from HMG-CoA that is formed in the cytoplasm (Chapter 22).

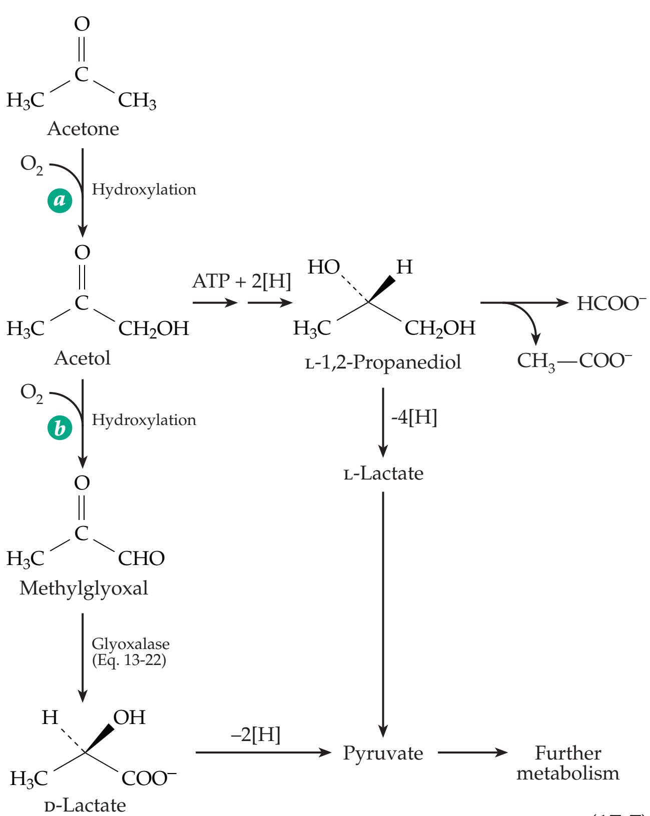
Utilization of 3-hydroxybutyrate or acetoacetate for energy requires their reconversion to acetyl-CoA as indicated in Eq. 17-6. All of the reactions of this sequence may be nearly at equilibrium in tissues that use ketone bodies for energy.[61]

Acetone, in the small amounts normally present in the body, is metabolized by hydroxylation to acetol (Eq. 17-7, step *a*), hydroxylation and dehydration to methylglyoxal (step *b*), and conversion to D-lactate and pyruvate. A second pathway via 1,2-propanediol and L-lactate is also shown in Eq. 17-7. During fasting the acetone content of human blood may rise to as much as 1.6 mM. As much as two-thirds of this may be converted to glucose.[65–69] Accumulation of acetone



(17-6)

appears to induce the synthesis of the hydroxylases needed for methylglyoxal formation,[68] and the pyruvate formed by Eq. 17-7 may give rise to glucose by the gluconeogenic pathway. However, at high acetone concentrations most metabolism may take place through a poorly understood conversion of 1,2-propanediol to acetate and formate or CO_2.[69] No net conversion of acetate into glucose can occur in animals, but isotopic labels from acetate can enter glucose via acetyl-CoA and the citric acid cycle.



(17-7)

B. Catabolism of Propionyl Coenzyme A and Propionate

Beta oxidation of fatty acids with an odd number of carbon atoms leads to the formation of propionyl-CoA as well as acetyl-CoA. The three-carbon propionyl unit is also produced by degradation of cholesterol and other isoprenoid compounds and of isoleucine, valine, threonine, and methionine. Human beings ingest small amounts of free propionic acid, e.g., from Swiss cheese (which is cultivated with propionic acid-producing bacteria) and from propionate added to bread as a fungicide. In **ruminant** animals, such as cattle and sheep, the ingested food undergoes extensive fermentation in the **rumen**, a large digestive organ containing cellulose-digesting bacteria and protozoa. Major products of the rumen fermentations include acetate, propionate, and butyrate. Propionate is an important source of energy for these animals.

1. The Malonic Semialdehyde Pathways

The most obvious route of metabolism of propionyl-CoA is further β oxidation which leads to 3-hydroxypropionyl-CoA (Fig. 17-3, step *a*). This appears to be the major pathway in green plants.[17] Continuation of the β oxidation via steps *a*–*c* of Fig. 17-3 produces the CoA derivative of malonic semialdehyde. The latter can, in turn, be oxidized to malonyl-CoA, a β-oxoacid which can be decarboxylated to acetyl-CoA. The necessary enzymes have been found in *Clostridium kluyveri*,[70] but the pathway appears to be little used.

Nevertheless, malonyl-CoA is a major metabolite. It is an intermediate in fatty acid synthesis (see Fig. 17-12) and is formed in the peroxisomal β oxidation of odd chain-length dicarboxylic acids.[70a] Excess malonyl-CoA is decarboxylated in peroxisomes, and lack of the decarboxylase enzyme in mammals causes the lethal **malonic aciduria**.[70a] Some propionyl-CoA may also be metabolized by this pathway. The modified β oxidation sequence indicated on the left side of Fig. 17-3 is used in green plants and in many microorganisms. 3-Hydroxypropionyl-CoA is hydrolyzed to *free* β-hydroxypropionate, which is then oxidized to malonic semialdehyde and converted to acetyl-CoA by reactions that have not been completely described. Another possible pathway of propionate metabolism is the direct conversion to pyruvate via α oxidation into lactate, a mechanism that may be employed by some bacteria. Another route to lactate is through addition of water to acrylyl-CoA, the product of step *a* of Fig. 17-3. The water molecule adds in the "wrong way," the OH^- ion going to the α carbon instead of the β (Eq. 17-8). An enzyme with an active site similar to that of histidine ammonia-lyase (Eq. 14-48) could

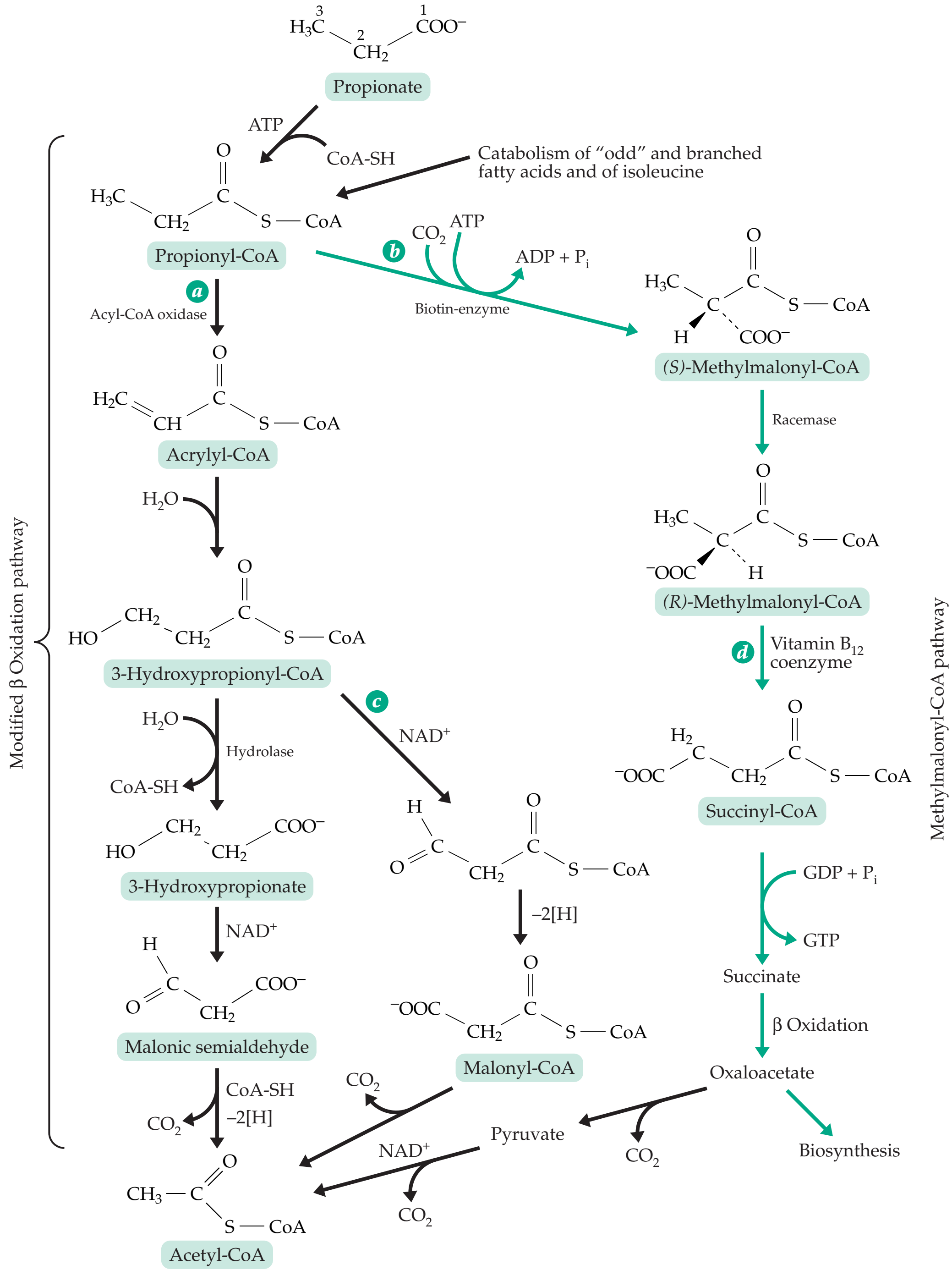
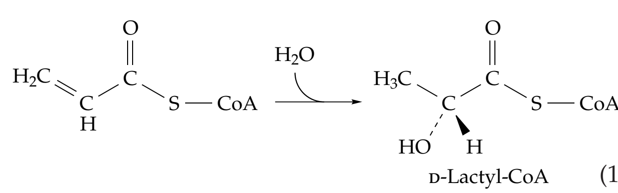


Figure 17-3 Catabolism of propionate and propionyl-CoA. In the names for methylmalonyl-CoA the *R* and *S* refer to the methylmalonyl part of the structure. Coenzyme A is also chiral.

presumably catalyze such a reaction. Lactyl-CoA could be converted to pyruvate readily. *Clostridium propionicum* does interconvert propionate, lactate, and pyruvate via acrylyl-CoA and lactyl-CoA as part of a fermentation of alanine (Fig. 24-19).[71–74] The enzyme that catalyzes hydration of acrylyl-CoA in this case is a complex flavoprotein that may function via a free radical mechanism.[71,72,74]



(17-8)

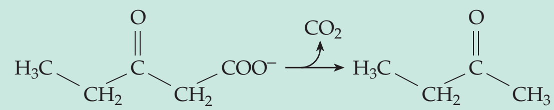
BOX 17-B METHYLMALONIC ACIDURIA

In this hereditary disease up to 1–2 g of methylmalonic acid per day (compared to a normal of <5 mg/day) is excreted in the urine, and a high level of the compound is present in blood. Two causes of the rare disease are known.[a–d] One is the lack of functional vitamin B_{12}-containing coenzyme. This can be a result of a mutation in any one of several different genes involved in the synthesis and transport of the cobalamin coenzyme.[e] Cultured fibroblasts from patients with this form of the disease contain a very low level of the vitamin B_{12} coenzyme (Chapter 16), and addition of excess vitamin B_{12} to the diet may restore coenzyme synthesis to normal. Among elderly patients a smaller increase in methylmalonic acid excretion is a good indicator of vitamin B_{12} deficiency. A second form of the disease, which does not respond to vitamin B_{12}, arises from a defect in the methylmalonyl mutase protein. Methylmalonic aciduria is often a very severe disease, frequently resulting in death in infancy. Surprisingly, some children with the condition are healthy and develop normally.[a,f]

A closely related disease is caused by a deficiency of propionyl-CoA carboxylase.[a] This may be a result of a defective structural gene for one of the two subunits of the enzyme, of a defect in the enzyme that attaches biotin to carboxylases, or of biotinitase, the enzyme that hydrolytically releases biotin from linkage with lysine (Chapter 14). The latter two defects lead to a multiple carboxylase deficiency and to methylmalonyl aciduria as well as ketoacidosis and propionic acidemia.[g]

Both methylmalonic aciduria and propionyl-CoA decarboxylase deficiency are usually accompanied by severe ketosis, hypoglycemia, and hyperglycinemia. The cause of these conditions is not entirely clear. However, methylmalonyl-CoA, which accumulates in methylmalonic aciduria, is a known inhibitor of pyruvate carboxylase. Therefore, ketosis may develop because of impaired conversion of pyruvate to oxaloacetate.

Patients with propionic or methylmalonic acidemia also secrete 2,3-butanediols (D-,L- or meso) and usually also 1,2-propandiol in their urine. Secretion of 1,2-propanediol is also observed during starvation and in diabetic ketoacidosis. Propanediol may be formed from acetone (Eq. 17-7), and butanediols may originate from acetoin, which is a side reaction product of pyruvate dehydrogenase. However, in the metabolic defects under consideration here, acetoin may be formed by hydroxylation of methylethyl ketone which can arise, as does acetone, by decarboxylation of an oxoacid precursor formed by β oxidation.[h]



Methylmalonic aciduria is rare and can be diagnosed incorrectly. In 1989 a woman in St. Louis, Missouri, was convicted and sentenced to life in prison for murdering her 5-month-old son by poisoning with ethylene glycol. While in prison she gave birth to another son who soon fell ill of methylmalonyl aciduria and was successfully treated. Reexamination of the evidence revealed that the first boy had died of the same disease and the mother was released.[i]

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[a] Fenton, W. A., and Rosenberg, L. E. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 1 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 1423–1449, McGraw-Hill, New York

[b] Matsui, S. M., Mahoney, M. J., and Rosenberg, L. E. (1983) *N. Engl. J. Med.* **308**, 857–861

[c] Hubbard, S. R., Wei, L., Ellis, L., and Hendrickson, W. A. (1994) *Nature (London)* **372**, 746–754

[d] Luschinsky Drennan, C., Matthews, R. G., Rosenblatt, D. S., Ledley, F. D., Fenton, W. A., and Ludwig, M. L. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 5550–5555

[e] Fenton, W. A., and Rosenberg, L. E. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 2 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 3129–3149, McGraw-Hill, New York

[f] Ledley, F. D., Levy, H. L., Shih, V. E., Benjamin, R., and Mahoney, M. J. (1984) *N. Engl. J. Med.* **311**, 1015–1018

[g] Wolf, B. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 2 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 3151–3177, McGraw-Hill, New York

[h] Casazza, J. P., Song, B. J., and Veech, R. L. (1990) *Trends Biochem. Sci.* **15**, 26–30

[i] Zurer, P. (1991) *Chem. Eng. News* **69** Sep 30, 7–8

2. The Methylmalonyl-CoA Pathway of Propionate Utilization

Despite the simplicity and logic of the β oxidation pathway of propionate metabolism, higher animals use primarily the more complex methylmalonyl-CoA pathway (Fig. 17-3, step *b*). This is one of the two processes in higher animals presently known to depend upon vitamin B_{12}. This vitamin has never been found in higher plants, nor does the methylmalonyl pathway occur in plants. The pathway (Fig. 17-3) begins with the biotin- and ATP-dependent carboxylation of propionyl-CoA. The *S*-methylmalonyl-CoA so formed is isomerized to *R*-methylmalonyl-CoA, after which the methylmalonyl-CoA is converted to succinyl-CoA in a vitamin B_{12} coenzyme-requiring reaction step *d* (Table 16-1). The succinyl-CoA is converted to free succinate (with the formation of GTP compensating for the ATP used initially). The succinate, by β oxidation, is converted to oxaloacetate which is decarboxylated to pyruvate. This, in effect, removes the carboxyl group that was put on at the beginning of the sequence in the ATP-dependent step. Pyruvate is converted by oxidative decarboxylation to acetyl-CoA.

A natural question is "Why has this complex pathway evolved to do something that could have been done much more directly?" One possibility is that the presence of too much malonyl-CoA, the product of the β oxidation pathway of propionate metabolism (Fig. 17-3, pathways *a* and *c*), would interfere with lipid metabolism. Malonyl-CoA is formed in the cytosol during fatty acid biosynthesis and retards mitochondrial β oxidation by inhibiting carnitine palmitoyltransferase I.[46,70a,75] However, a relationship to mitochondrial propionate catabolism is not clear. On the other hand, the tacking on of an extra CO_2 and the use of ATP at the beginning suggests that the *methylmalonyl-CoA pathway* (Fig. 17-3) *is a biosynthetic rather than a catabolic route* (see Section H,4). The methylmalonyl pathway provides a means for converting propionate to oxaloacetate, a transformation that is chemically difficult.

In this context it is of interest that cows, whose metabolism is based much more on acetate than is ours, often develop a severe ketosis spontaneously. A standard treatment is the administration of a large dose of propionate which is presumably effective because of the ease of its conversion to oxaloacetate via the methylmalonyl-CoA pathway. It is possible that this pathway was developed by animals as a means of capturing propionyl units, scanty though they may be, for conversion to oxaloacetate and use in biosynthesis. In ruminant animals, the pathway is especially important. Whereas we have 5.5 mM glucose in our blood, the cow has only half as much, and a substantial fraction of this glucose is derived, in the liver, from the propionate provided by rumen microorganisms.[76] The need for vitamin B_{12} in the formation of propionate by these organisms also accounts for the high requirement for cobalt in the ruminant diet (Chapter 16).

C. The Citric Acid Cycle

To complete the oxidation of fatty acids the acetyl units of acetyl-CoA generated in the β oxidation sequence must be oxidized to carbon dioxide and water.[77] The citric acid (or tricarboxylic acid) cycle by which this oxidation is accomplished is a vital part of the metabolism of almost all aerobic creatures. It occupies a central position in metabolism because of the fact that acetyl-CoA is also an intermediate in the catabolism of carbohydrates and of many amino acids and other compounds. The cycle is depicted in detail in Fig. 10-6 and in an abbreviated form, but with more context, in Fig. 17-4.

1. A Clever Way to Cleave a Reluctant Bond

Oxidation of the chemically resistant two-carbon acetyl group to CO_2 presents a chemical problem. As we have seen (Chapter 13), cleavage of a C–C bond occurs most frequently between atoms that are α and β to a carbonyl group. Such β cleavage is clearly impossible within the acetyl group. The only other common type of cleavage is that of a C–C bond adjacent to a carbonyl group (α cleavage), a thiamin-dependent process (Chapter 14). However, α cleavage would require the prior oxidation (hydroxylation) of the methyl group of acetate. Although many biological hydroxylation reactions occur, they are rarely used in the major pathways of rapid catabolism. Perhaps this is because the overall yield of energy obtainable via hydroxylation is less than that gained from dehydrogenation and use of an electron transport chain.[78]

The solution to the chemical problem of oxidizing acetyl groups efficiently is one very commonly found in nature; a catalytic cycle. Although direct cleavage is impossible, the two-carbon acetyl group of acetyl-CoA *can* undergo a Claisen condensation with a second compound that contains a carbonyl group. The condensation product has more than two carbon atoms, and a β cleavage to yield CO_2 is now possible. Since the cycle is designed to oxidize acetyl units we can regard acetyl-CoA as the **primary substrate** for the cycle. The carbonyl compound with which it condenses can be described as the **regenerating substrate**. To complete the catalytic cycle it is necessary that two carbon atoms be removed as CO_2 from the compound formed by condensation of the two substrates and that the remaining molecule be reconvertible to the original regenerating substrate. The reader may wish to play a

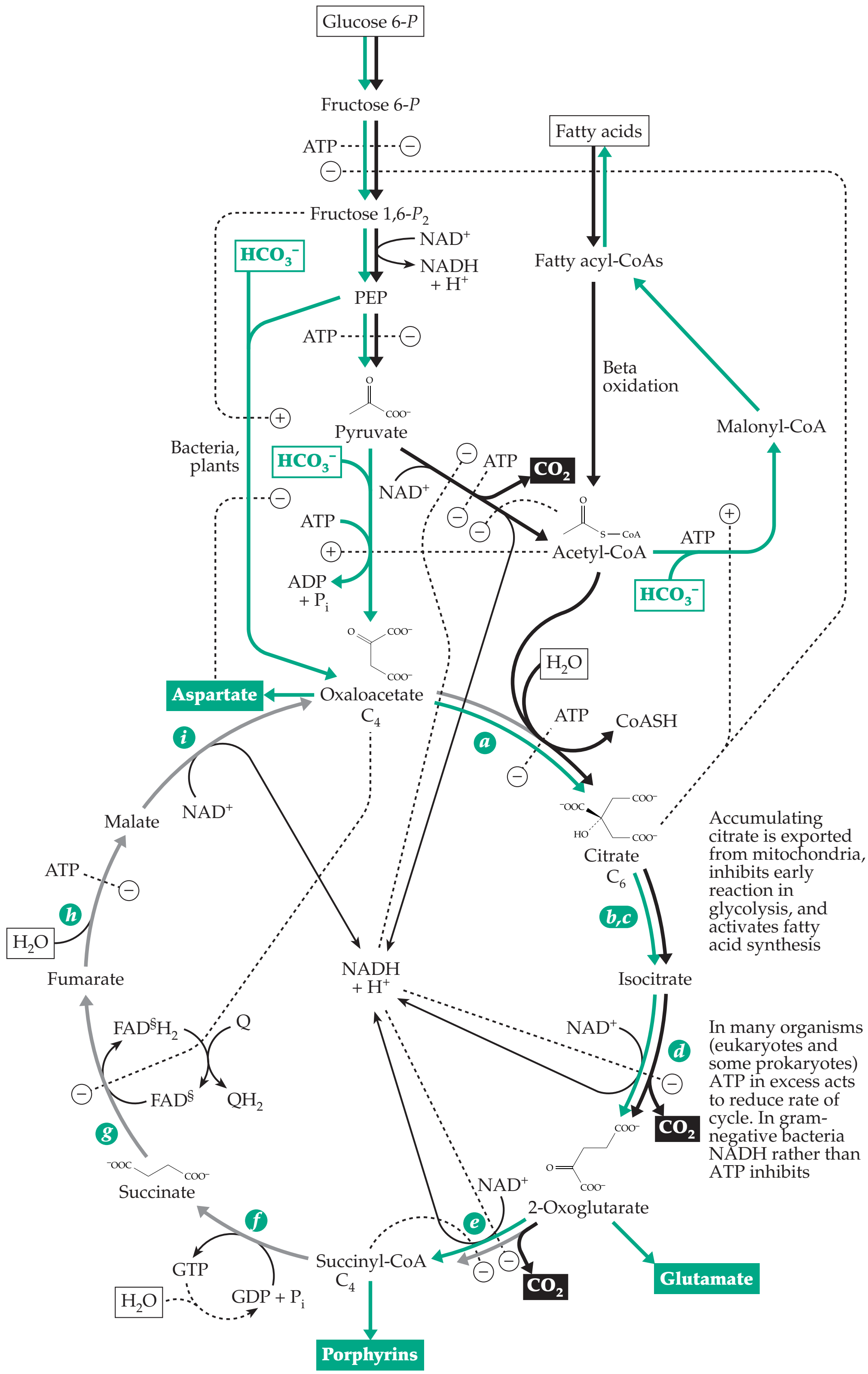


Figure 17-4 The Krebs citric acid cycle. Some of its controlling interactions and its relationship to glycolysis. See also Figure 10-6. Positive and negative regulatory influences, whether arising by allosteric effects or via covalent modification, are indicated by ⊕ or ⊖. Some biosynthetic reaction pathways related to the cycle are shown in green. Steps are lettered to correspond to the numbering in Fig. 10-6, which shows more complete structural formulas. Three molecules of H_2O (boxed) enter the cycle at each turn, providing hydrogen atoms for generation of NADH + H$^+$ and reduced ubiquinone (QH$_2$). The covalently attached FAD is designated FAD§.

game by devising suitable sequences of reactions for an acetyl-oxidizing cycle and finding the simplest possible regenerating substrate. Ask yourself whether nature could have used anything simpler than **oxaloacetate**, the molecule actually employed in the citric acid cycle.

The first step in the citric acid cycle (step *a*, Fig. 17-4) is the condensation of acetyl-CoA with oxaloacetate to form citrate. The synthase that catalyzes this condensation also removes the CoA by hydrolysis after it has served its function of activating a methyl hydrogen. This hydrolysis also helps to drive the cycle by virtue of the high group transfer potential of the thioester linkage that is cleaved. Before the citrate can be degraded through β cleavage, the hydroxyl group must be moved from its tertiary position to an adjacent carbon where, as a secondary alcohol, it can be oxidized to a carbonyl group. This is accomplished through steps *b* and *c*, both catalyzed by the enzyme aconitase (Eq. 13-17). Isocitrate can then be oxidized to the β-oxoacid **oxalosuccinate**, which does not leave the enzyme surface but undergoes decarboxylation while still bound (step *d*; also see Eq. 13-45).

The second carbon to be removed from citrate is released as CO_2 through catalysis by the thiamin diphosphate dependent **oxidative decarboxylation** of **2-oxoglutarate** (α-ketoglutarate; Chapter 15). To complete the cycle the four-carbon succinyl unit of succinyl-CoA must be converted back to oxaloacetate through a pathway requiring two more oxidation steps: Succinyl-CoA is converted to free succinate (step *f*) followed by a β oxidation sequence (steps *g*–*i*; Figs. 10-6 and 17-4). Steps *e* and *f* accomplish a substrate-level phosphorylation (Fig. 15-16). Succinyl-CoA is an unstable thioester with a high group transfer potential. Therefore, step *f* could be accomplished by simple hydrolysis. However, this would be energetically wasteful. The cleavage of succinyl-CoA is coupled to synthesis of ATP in *E. coli* and higher plants and to GTP in mammals. Some of the succinyl-CoA formed in mitochondria is used in other ways, e.g., as in Eq. 17-6 and for biosynthesis of porphyrins.

2. Synthesis of the Regenerating Substrate Oxaloacetate

The primary substrate of the citric acid cycle is acetyl-CoA. Despite many references in the biochemical literature to substrates "entering" the cycle as oxaloacetate (or as one of the immediate precursors succinate, fumarate, or malate), *these compounds are not consumed* by the cycle but are completely regenerated; hence the term *regenerating substrate*, which can be applied to any of these four substances. A prerequisite for the operation of a catalytic cycle is that a regenerating substrate be readily available and that its concentration be increased if necessary to accommodate a more rapid rate of reaction of the cycle. Oxaloacetate can normally be formed in any amount needed for operation of the citric acid cycle from **PEP** or from **pyruvate**, both of these compounds being available from metabolism of sugars.

In bacteria and green plants **PEP carboxylase** (Eq. 13-53), a highly regulated enzyme, is responsible for synthesizing oxaloacetate. In animal tissues **pyruvate carboxylase** (Eq. 14-3) plays the same role. The latter enzyme is almost inactive in the absence of the allosteric effector acetyl-CoA. For this reason, it went undetected for many years. In the presence of high concentrations of acetyl-CoA the enzyme is fully activated and provides for synthesis of a high enough concentration of oxaloacetate to permit the cycle to function. Even so, the oxaloacetate concentration in mitochondria is low, only 0.1 to 0.4 x 10^{-6} M (10–40 molecules per mitochondrion), and is relatively constant.[65,79]

3. Common Features of Catalytic Cycles

The citric acid cycle is not only one of the most important metabolic cycles in aerobic organisms, including bacteria, protozoa, fungi, higher plants, and animals, but also *a typical catalytic cycle*. Other cycles also have one or more primary substrates and at least one regenerating substrate. Associated with every catalytic cycle there must be a metabolic pathway that provides for synthesis of the regenerating substrate. Although it usually needs to operate only slowly to replenish regenerating substrate lost in side reactions, the pathway also provides *a mechanism for the net biosynthesis of any desired quantity of any intermediate in the cycle*. Cells draw off from the citric acid cycle considerable amounts of oxaloacetate, 2-oxoglutarate, and succinyl-CoA for synthesis of other compounds. For example, aspartate and glutamate are formed directly from oxaloacetate and 2-oxoglutarate, respectively, by transamination (Eq. 14-25).[79a,b] Citrate itself is exported from mitochondria and used for synthesis of fatty acids. It is often stated that the citric acid cycle functions in biosynthesis, but when intermediates in the cycle are drawn off for synthesis the complete cycle does not operate. Rather, *the pathway for synthesis of the regenerating substrate, together with some of the enzymes of the cycle, is used to construct a biosynthetic pathway.*

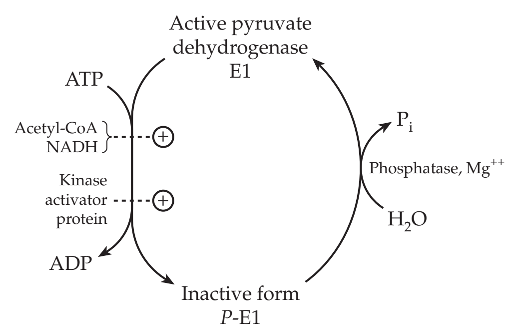
The word **amphibolic** is often applied to those metabolic sequences that are part of a catabolic cycle and at the same time are involved in a biosynthetic (anabolic) pathway. Another term, **anaplerotic**, is sometimes used to describe pathways for the synthesis of regenerating substrates. This word, which was suggested by H. L. Kornberg, comes from a Greek root meaning "filling up."[80]

4. Control of the Cycle

What factors determine the rate of oxidation by the citric acid cycle? As with most other important pathways of metabolism, several control mechanisms operate and different steps may become rate limiting under different conditions.[81] Factors influencing the flux through the cycle include (1) the rate of generation of acetyl groups, (2) the availability of oxaloacetate, and (3) the rate of reoxidation of NADH to NAD^+ in the electron transport chain. As indicated in Fig. 17-4, acetyl-CoA is a positive effector for conversion of pyruvate to oxaloacetate. Thus, acetyl-CoA "turns on" the formation of a substance required for its own further metabolism. However, when no pyruvate is available operation of the cycle may be impaired by lack of oxaloacetate. This may be the case when liver metabolizes high concentrations of ethanol. The latter is oxidized to acetate but it cannot provide oxaloacetate. Accumulating NADH reduces pyruvate to lactate, further interfering with formation of oxaloacetate.[82] In some individuals the accumulating acetyl units cannot all be oxidized in the cycle and instead are converted to the ketone bodies (Section A,4). A similar problem arises during metabolism of fatty acids by diabetic individuals with inadequate insulin. The accelerated breakdown of fatty acids in the liver overwhelms the system and results in ketosis, even though the oxaloacetate concentration remains normal.[83]

The rates of the oxidative steps in the citric acid cycle are limited by the rate of reoxidation of NADH and reduced ubiquinone in the electron transport chain which may sometimes be restricted by the availability of O_2. However, in aerobic organisms this rate is usually determined by the concentration of ADP and/or P_i available for conversion to ATP in the oxidative phosphorylation process (Chapter 18). If catabolism supplies an excess of ATP over that needed to meet the cell's energy needs, the concentration of ADP falls to a low level, cutting off phosphorylation. At the same time, ATP is present in high concentration and acts as a feedback inhibitor for the catabolism of carbohydrates and fats. This inhibition is exerted at many points, a few of which are indicated in Fig. 17-4. Important sites of inhibition are the **pyruvate dehydrogenase complex**,[84–85a] which converts pyruvate into acetyl-CoA; **isocitrate dehydrogenase**,[86,86a] which converts isocitrate into 2-oxoglutarate; and **2-oxoglutarate dehydrogenase**.[87] The enzyme **citrate synthase**, which catalyzes the first reaction of the cycle, is also inhibited by ATP.[88,89]

Mitochondrial pyruvate dehydrogenase, which contains a 60-subunit icosohedral core of dihydrolipoyltransacylase (Fig. 15-14), is associated with three molecules of a two-subunit kinase as well as six molecules of a structural **binding protein** which contains a lipoyl group that can be reduced and acetylated by other subunits of the core protein. The binding protein is apparently essential to the functioning of the dehydrogenase complex but not through its lipoyl group.[90,91] The specific pyruvate dehydrogenase kinase is thought to be one of the most important regulatory proteins involved in controlling energy metabolism in most organisms.[92–92b] Phosphorylation of up to three specific serine hydroxyl groups in the thiamin-containing decarboxylase subunit (designated E1) converts the enzyme into an inactive form (Eq. 17-9). A specific phosphatase reverses the inhibition. The kinase is most active on enzymes whose core lipoyl (E2) subunits are reduced and acetylated, a condition favored by high ratios of [acetyl-CoA] to free [CoASH] and of [NADH] to [NAD^+]. Since the kinase inactivates the enzyme the effect is to decrease the pyruvate dehydrogenase action when the system becomes saturated and NADH and acetyl-CoA accumulate. Conversely, a high [pyruvate] inhibits the kinase and increases the action of the dehydrogenase complex. This system also permits various external signals to be felt. For example, insulin has a pronounced stimulatory effect on mitochondrial energy.[65,92,93] One way in which this may be accomplished is through stimulation of the pyruvate dehydrogenase phosphatase, as indicated in Eq. 17-9. A **kinase activator protein** (Eq. 17-9) may also respond to various external stimuli and may be inhibited by insulin.[92]



(17-9)

The activities of 2-oxoglutarate dehydrogenase,[94] and to a lesser extent of pyruvate and isocitrate dehydrogenases, are increased by increases in the free Ca^{2+} concentration.[87] Calcium ions stimulate the phosphatase that dephosphorylates the deactivated phosphorylated pyruvate dehydrogenase and activate the other two dehydrogenases allosterically, increasing the affinities for the substrates.[87] Phosphorylation of the NAD^+-dependent isocitrate dehydrogenase also decreases its activity. In *E. coli* the isocitrate dehydrogenase kinase and a protein phosphatase exist as a bifunctional protein able to both deactivate the dehydrogenase and restore its activity.[86] For this organism, the decrease in activity forces substrate into the glyoxylate pathway (Section J,4) instead of the citric acid cycle.

BOX 17-C USE OF ISOTOPIC TRACERS IN STUDY OF THE TRICARBOXYLIC ACID CYCLE

The first use of isotopic labeling in the study of the citric acid cycle and one of the first in the history of biochemistry was carried out by Harland G. Wood and C. H. Werkman in 1941.[a,b] The aim was to study the fermentation of glycerol by propionic acid bacteria, a process that was not obviously related to the citric acid cycle. Some succinate was also formed in

$$\text{Glycerol} \rightarrow \text{propionate}^- + H^+ + H_2O$$
$$\Delta G' = -69 \text{ kJ/mol}$$

the fermentation, and on the basis of simple measurements of the fermentation balance reported in 1938 it was suggested that CO_2 was incorporated into oxaloacetate, which was then reduced to succinate. As we now know, this is indeed an essential step in the propionic acid fermentation (Section F,3). At the time ^{14}C was not available but the mass spectrometer, newly developed by A. O. Nier, permitted the use of the stable ^{13}C as a tracer. Wood and Werkman constructed a thermal diffusion column and used it to prepare bicarbonate enriched in ^{13}C and also built a mass spectrometer. By 1941 it was established unequivocally that carbon dioxide was incorporated into succinate by the bacteria.[c]

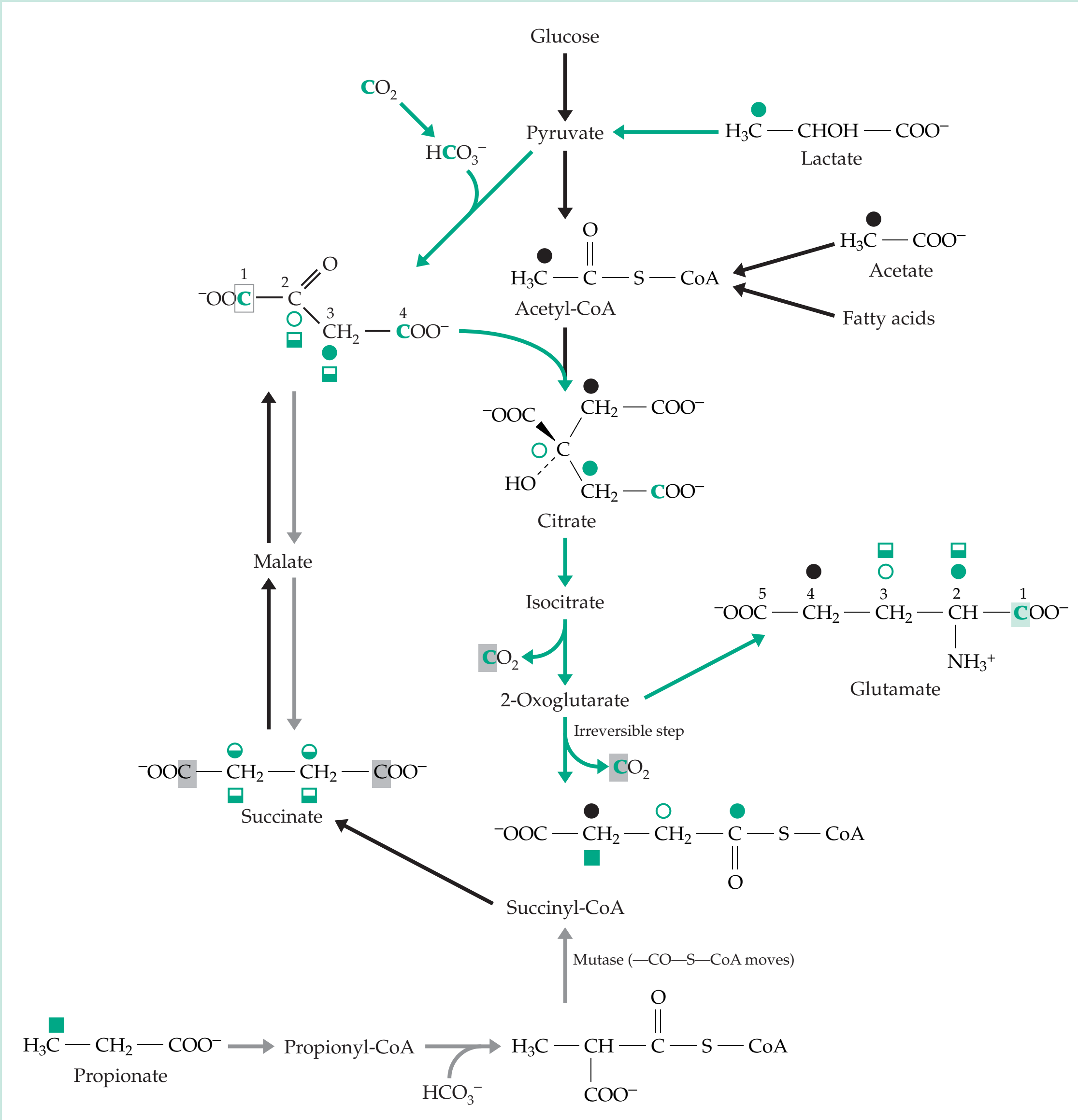
To test the idea that animal tissues could also incorporate CO_2 into succinate Wood examined the metabolism of a pigeon liver preparation to which malonate had been added to block succinate dehydrogenase (Box 10-B). Surprisingly, the accumulating succinate, which arose from oxalocetate via citrate, isocitrate, and 2-oxoglutarate (traced by green arrows in accompanying scheme), contained no ^{13}C. Soon, however, it was shown that CO_2 was incorporated into the carboxyl group of 2-oxoglutarate that is adjacent to the carbonyl group. That carboxyl is lost in conversion to succinate (Fig. 10-6) explains the lack of ^{13}C in succinate. It is of historical interest that these observations were incorrectly interpreted by many of the biochemists of the time. They agreed that *citrate could not be a member of the tricarboxylic acid cycle*. Since citrate is a symmetric compound it was thought that any ^{13}C incorporated into citrate would be present in equal amounts in both terminal carboxyl groups. This would necessarily result in incorporation of ^{13}C into succinate. It was not until 1948 that Ogston popularized the concept that by binding with substrates at three points, enzymes were capable of asymmetric attack upon symmetric substrates.[d] In other words, an enzyme could synthesize citrate with the carbon atoms from acetyl-CoA occupying one of the two $-CH_2COOH$ groups surrounding the prochiral center. Later, the complete stereochemistry of the citric acid cycle was elucidated through the use of a variety of isotopic labels (see p. 704). Some of the results are indicated by the asterisks and daggers in the structures in Fig. 10-6.

The operation of the citric acid cycle in living cells, organs, and whole animals has also been observed using NMR and mass spectroscopy with ^{13}C-containing compounds. For example, a heart can be perfused with a suitable oxygenated perfusion fluid[e] containing various ^{13}C-enriched substrates such as [U-^{13}C]fatty acids, [2-^{13}C]acetate, [3-^{13}C]L-lactate, or [2,3-^{13}C]propionate.[e–k] NMR spectroscopy allows direct and repeated observation of the ^{13}C nuclei from a given substrate and its entry into a variety of metabolic pathways. Because of the high dispersion of chemical shift values for ^{13}C the NMR resonance for the isotope can be seen at each position within a single compound.

A compound that is especially easy to observe is glutamate. This amino acid, most of which is found in the cytoplasm, is nevertheless in relatively rapid equilibrium with 2-oxoglutarate of the citric acid cycle in the mitochondria. The accompanying scheme shows where isotopic carbon from certain compounds will be located *when it first enters* the citric acid cycle and traces some of the labels into glutamate. For example, uniformly enriched fatty acids will introduce label into the two atoms of the *pro-S* arm of citrate and into 4- and 5-positions of glutamate whereas [2-^{13}C]acetate will introduce label only into the C4 position as marked by ● in the scheme. In the NMR spectrum a singlet resonance at 32.4 ppm will be observed. However, as successive turns of the citric acid cycle occur the isotope will appear in increasing amounts in the adjacent 3-position of glutamate. They will be recognized readily by the appearance of a multiplet. The initial singlet will be flanked by a pair of peaks that arise from spin–spin coupling with the adjacent 3-^{13}C of the [2,3-^{13}C]isotopomer (see accompanying figure). After longer periods of time the central resonance will weaken and the outer pair strengthen as the recycling occurs.

Metabolism with [U-^{13}C]fatty acids gives a labeling pattern similar to that with [2,3-^{13}C]acetate and it has been deduced that heart muscle normally metabolizes principally fatty acids for energy. What will happen to the glutamate C4 resonance if [3-^{13}C]lactate is added to the perfusion solution? It will enter both acetyl-CoA and oxaloacetate as indicated by ● in the following scheme. That will also intro-duce ^{13}C at C3 of glutamate. By looking at spectra at short times the relative amounts of lactate being oxidized via the cycle and that being converted

BOX 17-C (continued)

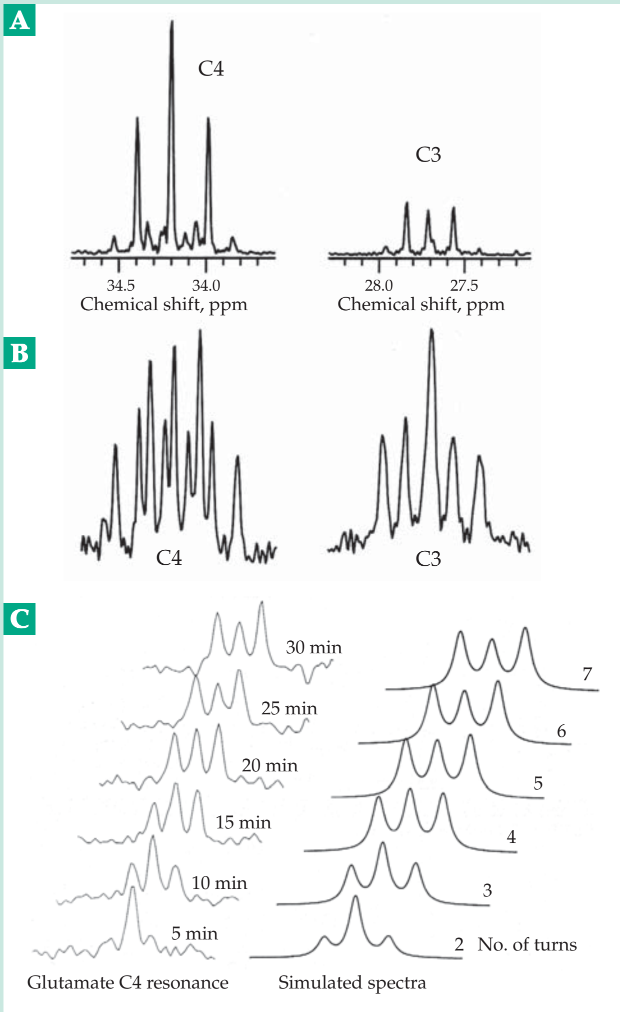


biosynthetically (anaplerotically) to glutamate can be estimated. There is a complication that has long been recognized. Oxaloacetate can be converted by exchange processes to succinate. Since succinate is symmetric the effect is to put 50% of the label into each of the central atoms of succinate (◒ in scheme). The exchange will then transfer label back into the C2 position of oxaloacetate (○) and through citric acid cycle reactions into C3 of glutamate. Now the C4 NMR resonance will contain an additional pair of peaks arising from spin–spin coupling with C2 but which will have a different coupling constant than that for coupling to C3.

If uniformly labeled [U-^{13}C]acetate is introduced the additional isotopomers, [3,4 - ^{13}C]glutamate and [3,4,5 - ^{13}C]glutamate, will be formed as will others with ^{13}C in the C1 and C5 positions but which will not affect the C4 resonance. A total of nine lines will be seen as illustrated in curve *a* of the accompanying figure. We see that the multiplet patterns arising from mass isotopomers are complex, but they can be predicted accurately with a computer program.[f] Isotopomers of succinate have also been analyzed.[g]

It is also of interest to introduce ^{13}C from propionate labeled in various positions. One of these is illustrated in the scheme. In this case the appearance of multiplets arising from [3,4-^{13}C] glutamate verifies the existence of end-to-end scrambling of the isotope in succinate. However, is the scrambling complete or are some molecules

BOX 17-C USE OF ISOTOPIC TRACERS IN STUDY OF THE TRICARBOXYLIC ACID CYCLE (cont.)



A. ^{13}C-NMR spectrum of extracts of Langendorff-perfused rat hearts perfused for 5 min with [1,2^{13}C] acetate, [3^{13}C]-lactate and glucose. Only the glutamate C4 (left) and C3 (right) resonances are shown. B. Spectrum after perfusion for 30 min. From Malloy *et al*.[r] C. The glutamate C4 resonance of an intact Langendorff-perfused rat heart supplied with 2 mM [2-^{13}C]acetate showing evolution of the multiplet as a function of time after introducing the label. The right panel shows glutamate C4 resonances generated by a computer simulation after turnover of citric acid cycle pools the indicated number of time. From Jeffrey *et al*.[f]

efficiently "channeled" through enzyme–enzyme complexes in such a way as to avoid scrambling? As shown in the scheme, full scrambling would give equal labeling of C2 and C3 of oxaloacetate and of glutamate. Experimentally greater labeling was seen at C3 than at C2 during the first few turns of the cycle suggesting that some channeling does occur.[e]

Isotopomer analysis can also be conducted by mass spectroscopy, which is more sensitive than NMR, using ^{13}C[h,k,l] or ^{2}H labeling.[j] Making use of a technique like that employed by Knoop (Box 10-A), a "chemical biopsy" can be performed on animals or on human beings, who may ingest gram quantities of sodium phenylacetate without harm. The phenylacetate is converted to an amide with glutamine (phenylacetylglutamine) which is excreted in the urine, from which it can easily be recovered for analysis.[l–n] This provides a non-invasive way of studying the operation of the citric acid cycle in the human body. Direct measurement on animal brains[o,p] and on human limbs or brain has also been accomplished by NMR spectroscopy[q] and may become more routine as instrumentation is improved.

[a] Wood, H. G. (1972) in *The Molecular Basis of Biological Transport* (Woessner, J. F., and Huijing, F., eds), pp. 1–54, Academic Press, New York

[b] Krampitz, L. O. (1988) *Trends Biochem. Sci.* **13**, 152–155

[c] Wood, H. G., Werkman, C. H., Hemingway, A., and Nier, A. O. (1941) *J. Biol. Chem.* **139**, 377–381

[d] Ogston, A. G. (1948) *Nature (London)* **162**, 936

[e] Sherry, A. D., Sumegi, B., Miller, B., Cottam, G. L., Gavva, S., Jones, J. G., and Malloy, C. R. (1994) *Biochemistry* **33**, 6268–6275

[f] Jeffrey, F. M. H., Rajagopal, A., Malloy, C. R., and Sherry, A. D. (1991) *Trends Biochem. Sci.* **16**, 5–10

[g] Jones, J. G., Sherry, A. D., Jeffrey, F. M. H., Storey, C. J., and Malloy, C. R. (1993) *Biochemistry* **32**, 12240–12244

[h] Des Rosiers, C., Di Donato, L., Comte, B., Laplante, A., Marcoux, C., David, F., Fernandez, C. A., and Brunengraber, H. (1995) *J. Biol. Chem.* **270**, 10027–10036

[i] Sherry, A. D., and Malloy, C. R. (1996) *Cell Biochem. Funct.* **14**, 259–268

[j] Yudkoff, M., Nelson, D., Daikhin, Y., and Erecinska, M. (1994) *J. Biol. Chem.* **269**, 27414–27420

[k] Beylot, M., Soloviev, M. V., David, F., Landau, B. R., and Brunengraber, H. (1995) *J. Biol. Chem.* **270**, 1509–1514

[l] Di Donato, L., Des Rosiers, C., Montgomery, J. A., David, F., Garneau, M., and Brunengraber, H. (1993) *J. Biol. Chem.* **268**, 4170–4180

[m] Magnusson, I., Schumann, W. C., Bartsch, G. E., Chandramouli, V., Kumaran, K., Wahren, J., and Landau, B. R. (1991) *J. Biol. Chem.* **266**, 6975–6984

[n] Chervitz, S. A., and Falke, J. J. (1995) *J. Biol. Chem.* **270**, 24043–24053

[o] Hyder, F., Chase, J. R., Behar, K. L., Mason, G. F., Siddeek, M., Rothman, D. L., and Shulman, R. G. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 7612–7617

[p] Cerdan, S., Künnecke, B., and Seelig, J. (1990) *J. Biol. Chem.* **265**, 12916–12926

[q] Rothman, D. L., Novotny, E. J., Shulman, G. I., Howseman, A. M., Petroff, O. A. C., Mason, G., Nixon, T., Hanstock, C. C., Prichard, J. W., and Shulman, R. G. (1992) *Proc. Natl. Acad. Sci. U.S.A.* **89**, 9603–9606

[r] Malloy, C. R., Thompson, J. R., Jeffery, F. M. H., and Sherry, A. D. (1990) *Biochemistry* **29**, 6756–6761

Acting to counteract any drop in ATP level, accumulating ADP acts as a positive effector for isocitrate dehydrogenases.

Another way in which the phosphorylation state of the adenylate system can regulate the cycle depends upon the need for GDP in step *f* of the cycle (Fig. 17-4). Within mitochondria, GTP is used largely to reconvert AMP to ADP. Consequently, formation of GDP is promoted by AMP, a compound that arises in mitochondria from the utilization of ATP for activation of fatty acids (Eq. 13-44) and activation of amino acids for protein synthesis (Eq. 17-36).

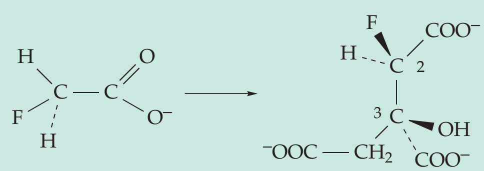
In *E. coli* and some other bacteria ATP does not inhibit citrate synthase but NADH does; the control is via the redox potential of the NAD^+ system rather than by the level of phosphorylation of the adenine nucleotide system.[95] Succinic dehydrogenase may be regulated by the redox state of ubiquinone (Chapter 15). Another mechanism of regulation may be the formation of specific protein–protein complexes between enzymes catalyzing reactions of the cycle.[96–97a] This may permit one enzyme to efficiently have a product of its action transferred to the enzyme catalyzing the next step in the cycle.

5. Catabolism of Intermediates of the Citric Acid Cycle

Acetyl-CoA is the only substrate that can be completely oxidized to CO_2 by the reaction of the citric acid cycle alone. Nevertheless, cells must sometimes

BOX 17-D FLUOROACETATE AND "LETHAL SYNTHESIS"

Among the most deadly of simple compounds is sodium fluoroacetate. The LD_{50} (the dose lethal for 50% of animals receiving it) is only 0.2 mg/kg for rats, over tenfold less than that of the nerve poison diisopropylphosphofluoridate (Chapter 12).[a,b] Popular, but controversial, as the rodent poison "1080," fluoroacetate is also found in the leaves of several poisonous plants in Africa, Australia, and South America. Surprisingly, difluoroacetate $HCF_2–COO^-$ is nontoxic and biochemical studies reveal that monofluoroacetate has no toxic effect on cells until it is converted metabolically in a "lethal synthesis" to 2*R*,3*R*-2-fluorocitrate, which is a competitive inhibitor of aconitase (aconitate hydratase, Eq. 13-17).[b–g] This fact was difficult to understand since citrate formed by the reaction of fluorooxaloacetate and acetyl-CoA has only weak inhibitory activity toward the same enzyme. Yet, it is the fluorocitrate formed from fluorooxaloacetate that contains a fluorine atom at a site that is attacked by aconitase in the citric acid cycle.



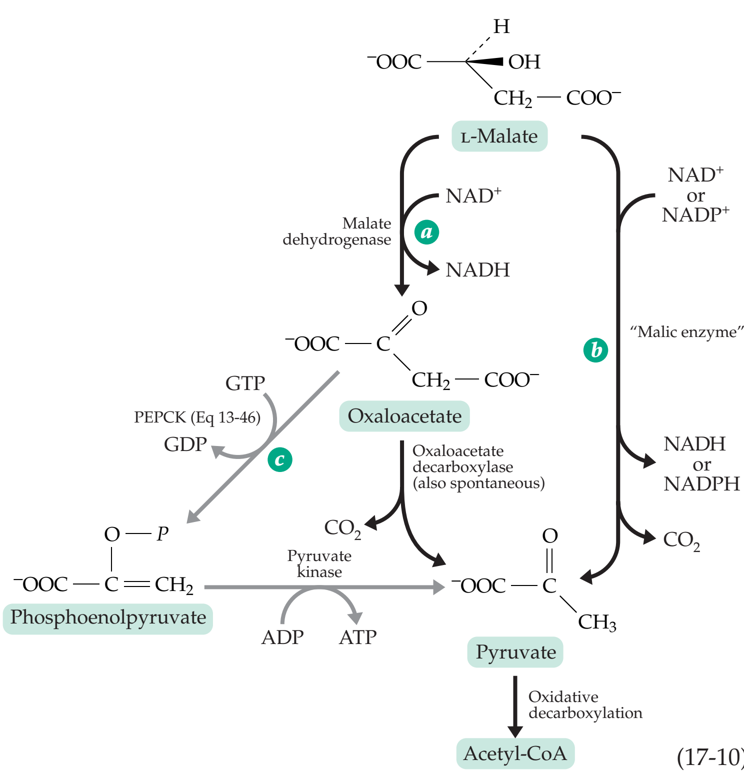
2*R*,3*R* 2-Fluoroacetate

The small van der Waals radius of fluorine (0.135 nm), comparable to that of hydrogen (0.12 nm), is often cited as the basis for the ability of fluoro compounds to "deceive" enzymes. However, the high electronegativity and ability to enter into hydrogen bonds may make F more comparable to –OH in metabolic effects. In the case of fluorocitrate it was proposed that the inhibitory isomer binds in the "wrong way" to aconitase in such a manner that the fluorine atom is coordinated with the ferric ion at the catalytic center.[c] However, 2*R*,3*R*-2-fluorocitrate is a simple competitive inhibitor of aconitase but an irreversible poison. It is especially toxic to nerves and also appears to affect mitochondrial membranes. Therefore, this poison may affect some other target, such as a citrate transporter.[d] Fluoroacetate is only one of many known naturally occurring fluorine compounds.[c]

Another example of lethal synthesis is seen in the use of 5-fluorouracil in cancer therapy (Box 15-E). In this compound and in many other fluorine-containing inhibitors the F atom replaces the H atom that is normally removed as H^+ in the enzymatic reaction. The corresponding F^+ cannot be formed.[h] Because of the high electronegativity of fluorine a C–F bond is polarized: $C^{\delta+}–F^{\delta-}$. This may have very large effects on reactivity at adjacent positions. For example, the reactivity of 2-fluoroglycosyl groups toward glycosyl transfer is decreased by several orders of magnitude (p. 597).

a Gibble, G. W. (1973) *J. Chem. Educ.* **50**, 460–462
b Elliott, K., and Birch, J., eds. (1972) *Carbon–Fluorine Compounds*, Elsevier, Amsterdam
c Glusker, J. P. (1971) in *The Enzymes*, 3rd ed., Vol. 5 (Boyer, P. D., ed), pp. 413–439, Academic Press, New York
d Kun, E. (1976) in *Biochemistry Involving Carbon–Fluorine Bonds* (Filler, R., ed), pp. 1–22, American Chemical Society, Washington, DC
e Marletta, M. A., Srere, P. A., and Walsh, C. (1981) *Biochemistry* **20**, 3719–3723
f Rokita, S. E., and Walsh, C. T. (1983) *Biochemistry* **22**, 2821–2828
g Peters, R. A. (1957) *Adv. Enzymol.* **18**, 113–159
h Abeles, R. H., and Alston, T. A. (1990) *J. Biol. Chem.* **265**, 16705–16708

oxidize large amounts of one of the compounds found in the citric acid cycle to CO_2.[98,99] For example, bacteria subsisting on succinate as a carbon source must oxidize it for energy as well as convert some of it to carbohydrates, lipids, and other materials. Complete combustion of *any citric acid cycle intermediate* can be accomplished by conversion to malate followed by oxidation of malate to oxaloacetate (Eq. 17-10, step *a*) and decarboxylation (β cleavage) to pyruvate, or (Eq. 17-10, step *b*) oxidation and decarboxylation of malate by the **malic enzyme** (Eq. 13-45) without free oxaloacetate as an intermediate. Pathway *b* is probably the most important. It is catalyzed by two different malic enzymes present in animal mitochondria. One is specific for $NADP^+$ while the other reacts with NAD^+ as well.[100,101] They both have complex regulatory properties. For example, the less specific NAD^+-utilizing enzyme is allosterically inhibited by ATP but is activated by fumarate, succinate, or isocitrate.[100] Thus, accumulation of citric acid cycle intermediates "turns on" the malic enzyme, allowing the excess to leave the cycle and reenter as acetyl groups. Since the Michaelis constant for malate is high, this will not happen unless malate accumulates, signaling a need for acetyl-CoA. The $NADP^+$-dependent enzyme is activated by a high concentration of free CoA and is inhibited by NADH. Perhaps when glycolysis becomes slow the free CoA level rises and turns on malate oxidation.[101] On the other hand, rapid glycolysis increases the NADH concentration which inhibits the malic enzyme. The result is a buildup of the oxaloacetate concentration and an increase in activity of the citric acid cycle. The malic enzymes are also present in the cytoplasm, where one of them functions as part of an NADPH-generating cycle (Eq. 17-46).



(17-10)

D. Oxidative Pathways Related to the Citric Acid Cycle

In this section we will consider some other catalytic cycles as well as some noncyclic pathways of oxidation of one- and two-carbon substrates that are utilized by microorganisms.

1. The γ-Aminobutyrate Cycle

A modification of the citric acid cycle which involves glutamate and gamma (γ) aminobutyrate (GABA) has an important function in the brain (Fig. 17-5). Both glutamate and γ-aminobutyrate occur in high concentrations in brain (10 and 0.8 mM, respectively). Both are important neurotransmitters, γ-aminobutyrate being a principal neuronal inhibitory substance[102,103] (Chapter 30). In the γ-aminobutyrate cycle acetyl-CoA and oxaloacetate are converted into citrate (step *a*) in the usual way and the citrate is then converted into 2-oxoglutarate. The latter is transformed to L-glutamate either by direct amination (*b*) or by transamination (*c*), the amino donor being γ-aminobutyrate.

γ-Aminobutyrate is formed by decarboxylation of glutamate (Fig. 17-5, step *d*)[104] and is catabolized via transamination (step *e*)[105] to succinic semialdehyde, which is oxidized to succinate[106] and oxaloacetate. The two transamination steps in the pathways may be linked, as indicated in Fig. 17-5, to form a complete cycle that parallels the citric acid cycle but in which 2-oxoglutarate is oxidized to succinate via glutamate and γ-aminobutyrate. No thiamin diphosphate is required, but 2-oxoglutarate is reductively aminated to glutamate. The cycle is sometimes called the **γ-aminobutyrate shunt**, and it plays a significant role in the overall oxidative processes of brain tissue. This pathway is also prominent in green plants.[107–109] For example, under anaerobic conditions the radish *Raphanus sativus* accumulates large amounts of γ-aminobutyrate.[110] Most animal tissues contain very little γ-aminobutyrate, although it has been found in the oviducts of rats at concentrations that exceed those in the brain.[111]

2. The Dicarboxylic Acid Cycle

Some bacteria can subsist solely on glycolate, glycine, or oxalate, all of which

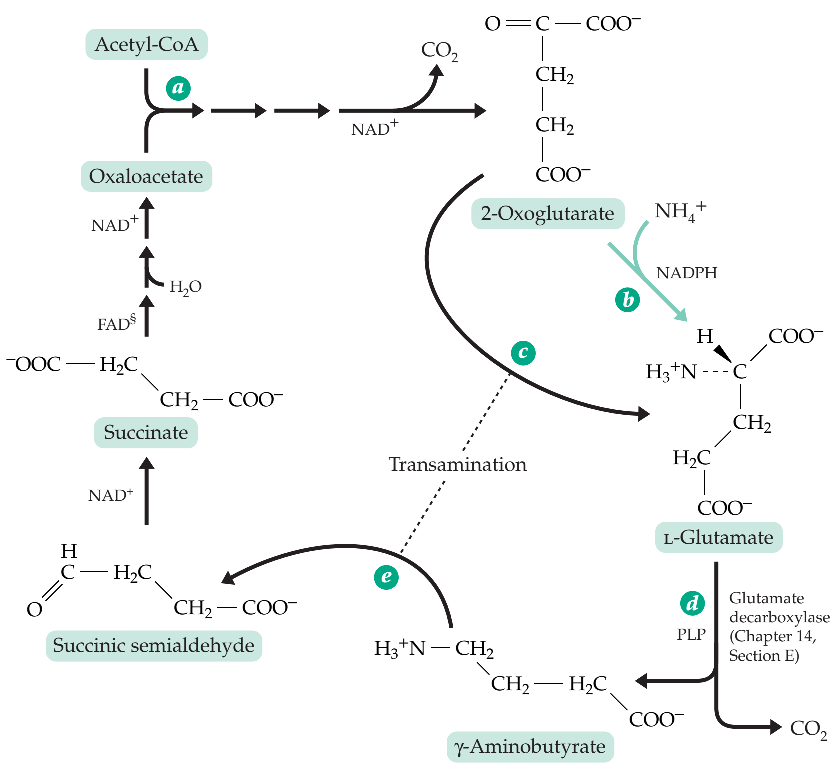


Figure 17-5 Reactions of the γ-aminobutyrate (GABA) cycle.

are converted to glyoxylate (Eq. 17-11). Glyoxylate is oxidized to CO_2 and water to provide energy to the bacteria and is also utilized for biosynthetic purposes. The energy-yielding process is found in the **dicarboxylic acid cycle** (Fig. 17-6), which catalyzes the complete oxidation of glyoxylate. Four hydrogen atoms are removed with generation of two molecules of NADH which can be oxidized by the respiratory chain to provide energy.[112,113] In the dicarboxylic acid cycle glyoxylate is the principal substrate and acetyl-CoA is the regenerating substrate rather than the principal substrate as it is for the citric acid cycle.

The logic of the dicarboxylic acid cycle is simple. Acetyl-CoA contains a potentially free carboxyl group. After the acetyl group of acetyl-CoA has been condensed with glyoxylate and the resulting hydroxyl group has been oxidized, the free carboxyl group appears in oxaloacetate in a position β to the carbonyl group. The carboxyl

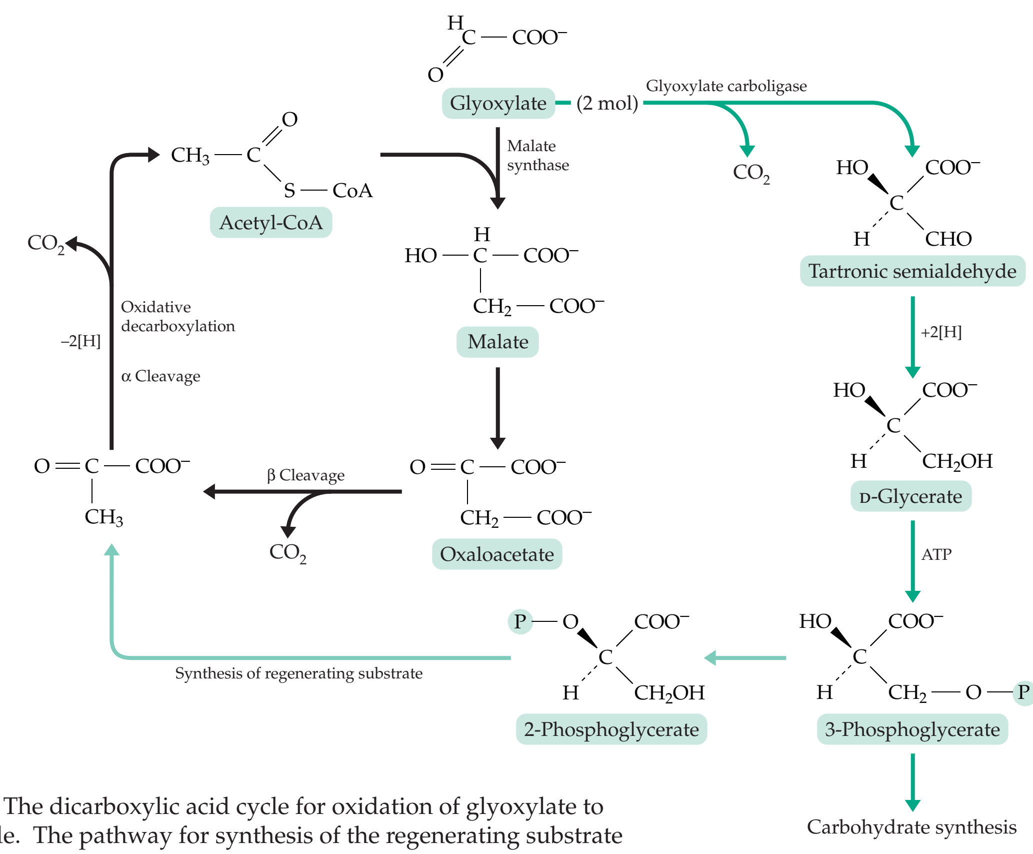
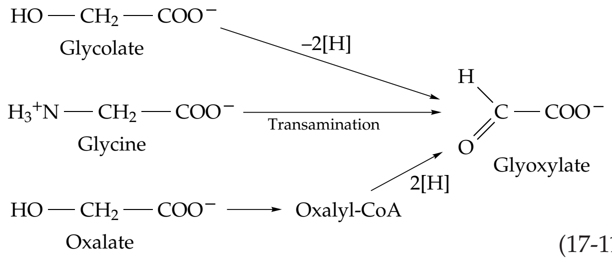


Figure 17-6 The dicarboxylic acid cycle for oxidation of glyoxylate to carbon dioxide. The pathway for synthesis of the regenerating substrate is indicated by green lines. This pathway is also needed for synthesis of carbohydrates and all other cell constituents.



(17-11)

donated by the glyoxylate is still in the α position. A consecutive β cleavage and an oxidative α cleavage release the two carboxyl groups as carbon dioxide to reform the regenerating substrate. The cycle is simple and efficient. Like the citric acid cycle, it depends upon thiamin diphosphate, without which the α cleavage would be impossible. Comparing the citric acid cycle (Fig. 17-2) with the simpler dicarboxylic acid cycle, we see that in the former the initial condensation product citrate contains a hydroxyl group attached to a tertiary carbon atom. With no adjacent hydrogen it is impossible to oxidize it directly to the carbonyl group which is essential for subsequent chain cleavage; hence the dependence on aconitase to shift the OH to an adjacent carbon. Both cycles involve oxidation of a hydroxy acid to a ketone followed by β cleavage and oxidative α cleavage. In the citric acid cycle additional oxidation steps are needed to convert succinate back to oxaloacetate, corresponding to the fact that the citric acid cycle deals with a more reduced substrate than does the dicarboxylic acid cycle.

The synthetic pathway for the regenerating substrate of the dicarboxylic acid cycle is quite complex. Two molecules of glyoxylate undergo α condensation with decarboxylation by glyoxylate carboligase[114] (see also Chapter 14, Section D) to form **tartronic semialdehyde**. The latter is reduced to D-glycerate, which is phosphorylated to 3-phosphoglycerate and 2-phosphoglycerate. Since the phosphoglycerates are carbohydrate precursors, this **glycerate pathway** provides the organisms with a means for synthesis of carbohydrates and other complex materials from glyoxylate alone. At the same time, 2-phosphoglycerate can be converted to pyruvate and the pyruvate, by oxidative decarboxylation, to the regenerating substrate acetyl-CoA.

E. Catabolism of Sugars

In most sugars each carbon atom bears an oxygen atom which facilitates chemical attack by oxidation at any point in the carbon chain. Every sugar contains a potentially free aldehyde or ketone group, and the carbonyl function can be moved readily to adjacent positions by isomerases. Consequently, aldol cleavage is also possible at many points. For these reasons, the metabolism of carbohydrates is complex and varied.

A sugar chain can be cut in several places giving rise to a variety of metabolic pathways. However, in the energy economy of most organisms, including human beings, the **Embden–Meyerhof–Parnas** or **glycolysis pathway** by which hexoses are converted to pyruvate (Fig. 17-7) stands out above all others. We have already considered this pathway, which is also outlined in Figs. 10-2 and 10-3. Some history and additional important details follow.

1. The Glycolysis Pathway

The discovery of glycolysis followed directly the early observations of Buchner and of Harden and Young on fermentation of sugar by yeast juice (p. 767). Another line of research, the study of muscle, soon converged with the investigations of alcoholic fermentation. Physiologists were interested in the process by which an isolated muscle could obtain energy for contraction in the absence of oxygen. It was shown by A. V. Hill that glycogen was converted to lactate to supply the energy, and Otto Meyerhof later demonstrated that the chemical reactions were related to those of alcoholic fermentation. The establishment of the structures and functions of the pyridine nucleotides in 1934 (Chapter 15) coincided with important studies by G. Embden in Frankfurt and of J. K. Parnas in Poland. The sequence of reactions in glycolysis soon became clear. All of the 15 enzymes catalyzing the individual steps in the sequence have been isolated and crystallized and are being studied in detail.[115]

Formation of pyruvate. The conversion of glucose to pyruvate requires ten enzymes (Fig. 17-7), and the sequence can be divided into four stages: preparation for chain cleavage (reactions *1*–*3*), cleavage and equilibration of triose phosphates (reactions *4* and *5*), oxidative generation of ATP (reactions *6* and *7*), and conversion of 3-phosphoglycerate to pyruvate (reactions *8*–*10*).

In preparation for chain cleavage, free glucose is phosphorylated to glucose 6-phosphate by ATP under the action of hexokinase (reaction *1*). Glucose 6-phosphate can also arise by cleavage of a glucosyl unit from glycogen by the consecutive action of glycogen phosphorylase (reaction 1*a*) and phosphoglucomutase, which transfers a phospho group from the oxygen at C-1 to that at C-6 (reaction 1*b*) (see also Eq. 12-39 and associated discussion of the mechanism of this enzyme). Why do cells attach phospho groups to sugars to initiate metabolism of the sugars? Four reasons can be given:

(a) The phospho group constitutes an electrically charged handle for binding the sugar phosphate to enzymes.

(b) There is a kinetic advantage in initiating a reaction sequence with a highly irreversible reaction

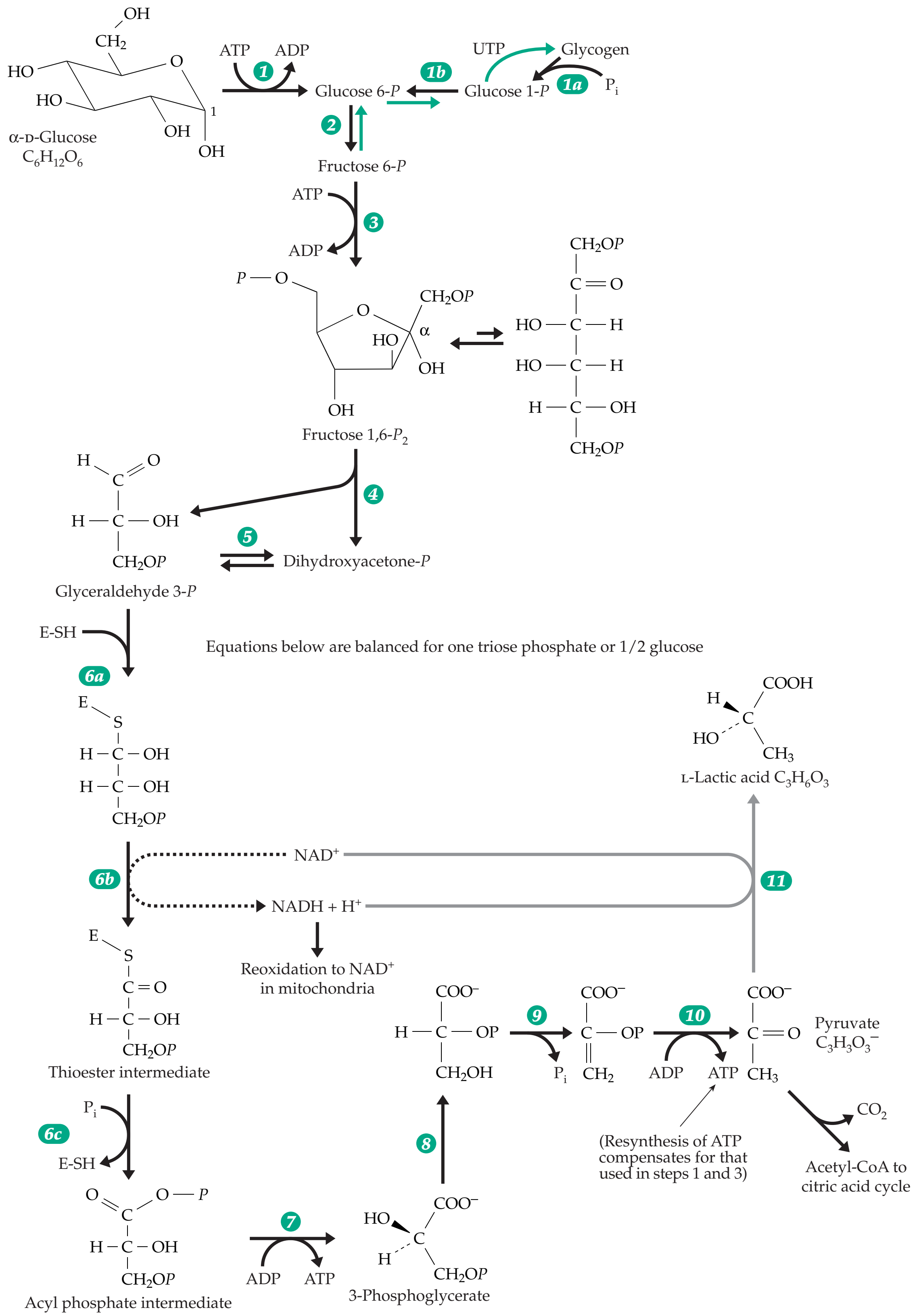


Figure 17-7 Outline of the glycolysis pathway by which hexoses are broken down to pyruvate. The ten enzymes needed to convert D-glucose to pyruvate are numbered. The pathway from glycogen using glycogen phosphorylase is also included, as is the reduction of pyruvate to lactate (step *11*). Steps *6a–7*, which are involved in ATP synthesis via thioester and acyl phosphate intermediates, are emphasized. See also Figures 10-2 and 10-3, which contain some additional information.

such as the phosphorylation of glucose.

(c) Phosphate esters are unable to diffuse out of cells easily and be lost.

(d) There is at least a possibility that the phospho groups may function in catalysis.

Reaction *2* of Fig. 17-7 is a simple isomerization that moves the carbonyl group to C-2 so that β cleavage to two three-carbon fragments can occur. Before cleavage a second phosphorylation (reaction *3*) takes place to form fructose 1,6-bisphosphate. This ensures that when fructose bisphosphate is cleaved by aldolase each of the two halves will have a phosphate handle. This second priming reaction (reaction *3*) is the first step in the series that is unique to glycolysis. The catalyst for the reaction, **phosphofructokinase**, is carefully controlled, as discussed in Chapter 11 (see Fig. 11-2).

Fructose bisphosphate is cleaved by action of an aldolase (reaction *4*) to give glyceraldehyde 3-phosphate and dihydroxyacetone phosphate. These two triose phosphates are then equilibrated by triose phosphate isomerase (reaction *5*; see also Chapter 13). As a result, both halves of the hexose can be metabolized further via glyceraldehyde 3-*P* to pyruvate. The oxidation of glyceraldehyde 3-*P* to the corresponding carboxylic acid, 3-phosphoglyceric acid (Fig. 17-7, reactions *6* and *7*), is coupled to synthesis of a molecule of ATP from ADP and P_i. This means that two molecules of ATP are formed per hexose cleaved, and that two molecules of NAD^+ are converted to NADH in the process.

The conversion of 3-phosphoglycerate to pyruvate begins with transfer of a phospho group from the C-3 to the C-2 oxygen (reaction *8*) and is followed by dehydration through an α, β elimination catalyzed by **enolase** (reaction *9*). The product, phosphoenolpyruvate (PEP), has a high group transfer potential. Its phospho group can be transferred easily to ADP via the action of the enzyme **pyruvate kinase**, to leave the enol of pyruvic acid which is spontaneously converted to the much more stable pyruvate ion (see Eq. 7-59). Because two molecules of PEP are formed from each glucose molecule, the process provides for the recovery of the two molecules of ATP that were expended in the initial formation of fructose 1,6-bisphosphate from glucose. Several isoenzyme forms exist in mammals. Most of these are allosterically activated by fructose 1,6-bisphosphate.[115a,b] However, the enzyme from trypanosomes is activated by fructose 2,6-P_2.[115c]

The further metabolism of pyruvate. In the aerobic metabolism that is characteristic of most tissues of our bodies, pyruvate is oxidatively decarboxylated to acetyl-CoA, which can then be completely oxidized in the citric acid cycle (Fig. 17-4). The NADH produced in reaction *6* of Fig. 17-7, as well as in the oxidative decarboxylation of pyruvate and in subsequent reactions of the citric acid cycle, is reoxidized in the electron transport chain of the mitochondria as described in detail in Chapter 18 (see Fig. 18-5). An important alternative fate of pyruvate is to enter into fermentation reactions. For example, the enzyme lactate dehydrogenase (Fig. 17-7, reaction *11*) catalyzes reduction by NADH of pyruvate to L-lactate, or, for some bacteria, to D-lactate. This reaction can be coupled to the NADH-producing reaction *6* to give a balanced process by which glucose is fermented to lactic acid in the absence of oxygen (see also Eq. 10-3). In a similar process, yeast cells decarboxylate pyruvate (α cleavage) to acetaldehyde which is reduced to ethanol using the NADH produced in reaction *6*. These fermentation reactions are summarized in Fig. 10-3 and, along with many others, are discussed further in Section F of this chapter.

2. Generation of ATP by Substrate Oxidation

The formation of ATP from ADP and P_i is a vital process for all cells. It is usually referred to as "phosphorylation" and includes **oxidative phosphorylation** associated with the passage of electrons through an electron transport chain—usually in mitochondria; **photosynthetic phosphorylation**, a similar process occurring in chloroplasts under the influence of light; and **substrate-level phosphorylation**. Only the last is fully understood chemically. The dehydrogenation of glyceraldehyde 3-*P* and the accompanying ATP formation (reactions *6* and *7*, Fig. 17-7; Fig. 15-6) is the best known example of substrate-level phosphorylation and is tremendously important for yeasts and other microorganisms that live anaerobically. They depend upon this one reaction for their entire supply of energy. The conversion of glucose either to lactate or to ethanol and CO_2 is accompanied by a net synthesis of only two molecules of ATP and it is most logical to view these as arising from oxidation of glyceraldehyde 3-*P*. The formation of ATP from PEP and ADP in reaction *10* of Fig. 17-7 can be regarded as the recapturing of ATP "spent" in the priming reactions of steps *1* and *3*. With a gain of only two molecules of ATP for each molecule of hexose fermented, it is not surprising that yeast must ferment enormous amounts of sugar to meet its energy needs.

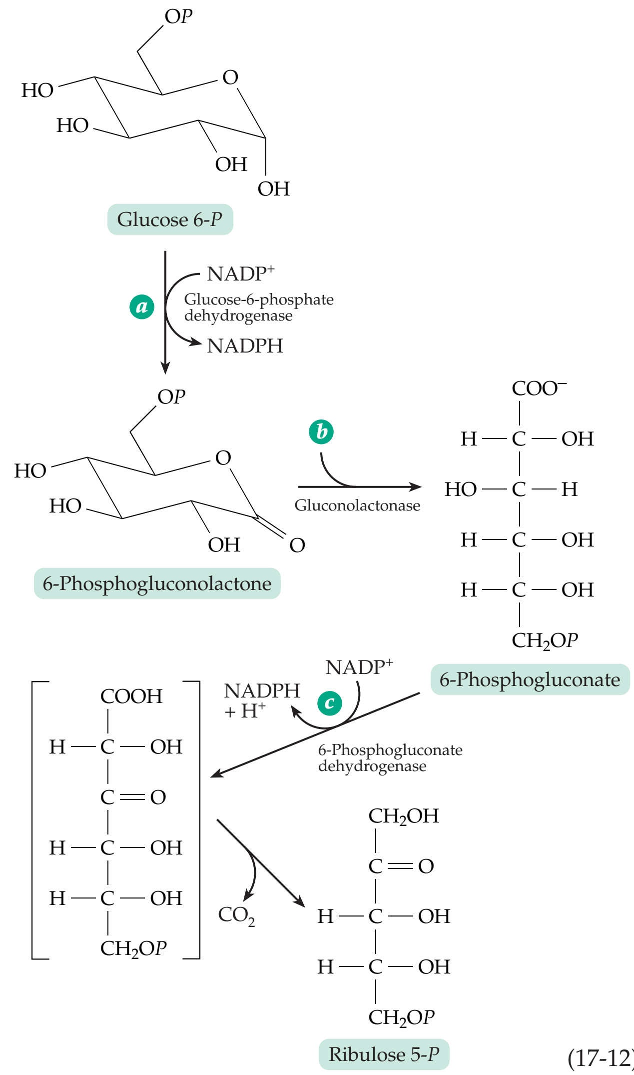
Each glucose unit of glycogen stored in our bodies can be converted to pyruvate with an apparent net gain of *three* molecules of ATP. However, two molecules of ATP were needed for the initial synthesis of each hexose unit of glycogen (Fig. 12-2). Therefore, the overall net yield for fermentation of stored polysaccharide is still only two ATP per hexose. The fermentation of glycogen accounts for the very rapid generation of lactic acid during intense muscular activity. However, in most circumstances within aerobic tissues reoxidation of NADH occurs via the electron transport chain of mitochondria with a much higher yield of ATP. Substrate-level phosphorylation can also follow oxidative decarboxylation of an α-oxoacid. For example, in the citric acid cycle GTP is formed following oxidative decarboxylation of 2-oxoglutarate (Fig. 17-4, steps *e* and *f*).

3. The Pentose Phosphate Pathways

A second way of cleaving glucose 6-phosphate utilizes sequences involving the five-carbon pentose sugars. They are referred to as **pentose phosphate pathways**, the phosphogluconate pathway, or the hexose monophosphate shunt. Historically, the evidence for such routes dates from the experiments of Warburg on the oxidation of glucose 6-*P* to 6-phosphogluconate (Chapter 15). For many years the oxidation remained an enzymatic reaction without a defined pathway. However, it was assumed to be part of an alternative method of degradation of glucose. Supporting evidence was found in the observation that tissues continue to respire in the presence of a high concentration of fluoride ion, a known inhibitor of the enolase reaction and capable of almost completely blocking glycolysis. Some tissues, e.g., liver, are especially active in respiration through this alternative pathway, whose details were elucidated by Horecker and associates.[116,117] We now know that the pentose phosphate pathways are multiple as well as multipurpose. They function in catabolism and also, when operating in the reverse direction, as a **reductive pentose phosphate pathway** that lies at the heart of the sugar-forming reactions of photosynthesis.

The oxidative pentose pathway provides *a means for cutting the chain of a sugar molecule one carbon at a time*, with the carbon removed appearing as CO_2. The enzymes required can be grouped into three distinct systems, all of which are found in the cytosol of animal cells: (i) a dehydrogenation–decarboxylation system, (ii) an isomerizing system, and (iii) a sugar rearrangement system. The dehydrogenation–decarboxylation system cleaves glucose 6-*P* to CO_2 and the pentose phosphate, ribulose 5-*P* (Eq. 17-12). Three enzymes are required, the first being glucose 6-*P* dehydrogenase[117a] (Eq. 17-12, step *a*; see also Eq. 15-10). The immediate product, a lactone, undergoes spontaneous hydrolysis. However, the action of **gluconolactonase** (Eq. 17-12, step *b*) causes a more rapid ring opening. A second dehydrogenation is catalyzed by **6-phosphogluconate dehydrogenase** (Eq. 17-12, step *c*),[117b] and this reaction is immediately followed by a β decarboxylation catalyzed by the same enzyme (as in Eq. 13-45). The value of $\Delta G°$ for oxidation of glucose 6-*P* to ribulose 5-*P* by $NADP^+$ according to Eq. 17-12 is $-30.8 \text{ kJ mol}^{-1}$, a negative enough value to drive the $[NADPH]/[NADP^+]$ ratio to an equilibrium value of over 2000 at a CO_2 tension of 0.05 atm.

The isomerizing system, consisting of two enzymes,



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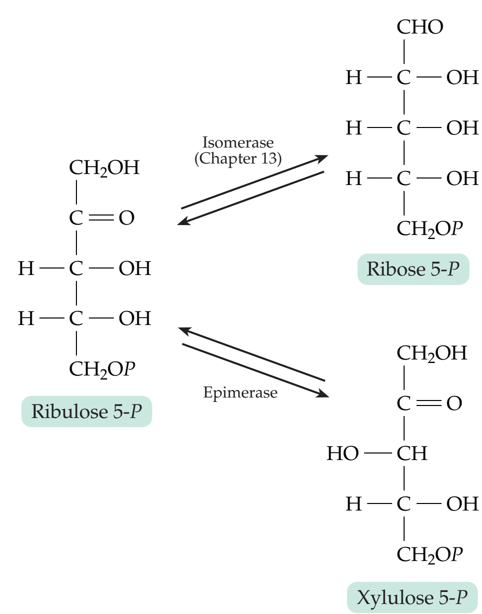

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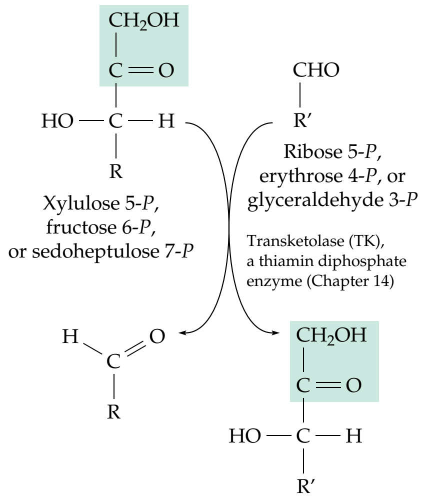

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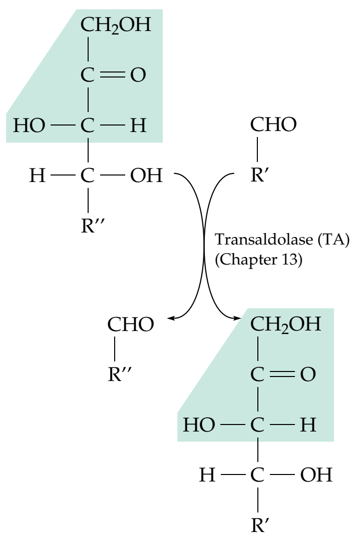


(17-15)

interconverts three pentose phosphates (Eq. 17-13). As a consequence the three compounds exist as an equilibrium mixture. Both xylulose 5-*P* and ribose 5-*P* are needed for further reactions in the pathways.

The ingenious sugar rearrangement system uses two enzymes, **transketolase** and **transaldolase**. Both catalyze chain cleavage and transfer reactions (Eqs. 17-14 and 17-15) that involve the same group of substrates. These enzymes use the two basic types of C–C bond cleavage, adjacent to a carbonyl group (α) and one carbon removed from a carbonyl group (β). Both types are needed in the pentose phosphate pathways just as they are in the citric acid cycle. The enzymes of the pentose phosphate pathway are found in the cytoplasm of both animal and plant cells.[117c] Mammalian cells appear to have an additional set that is active in the endoplasmic reticulum and plants have another set in the chloroplasts.[117c]

An oxidative pentose phosphate cycle. Putting the three enzyme systems together, we can form a cycle that oxidizes hexose phosphates. Three carbon atoms are chopped off one at a time (Fig. 17-8A) leaving a three-carbon triose phosphate as the product. Since the dehydrogenation system works only on glucose 6-*P*, a part of the sugar rearrangement system must be utilized between each of the three oxidation steps. Notice that a C_5 unit (ribose 5-*P*) is used in the first reaction with transketolase but is regenerated at the end of the sequence. This C_5 unit is the regenerating substrate for the cycle. As indicated by the dashed arrows, it is formed readily in any quantity needed by oxidation of glucose 6-*P*. Before the C_5 unit that is formed in each oxidation step can be processed by the sugar rearrangement reactions, it must be isomerized[117c,118,118a,b] from ribulose 5-*P* to xylulose 5-*P*; before the C_5 unit, produced at the end of the sequence in Fig. 17-8, can be reutilized as a regenerating substrate, it must be isomerized to ribose 5-*P*. Thus, the cycle is quite complex. The same C_5 substrates appear at several points in Fig. 17-8A and substrates from different parts of the cycle become scrambled and the pathway does not degrade all the hexose molecules in a uniform manner. For this reason, Zubay described the pentose phosphate pathways as a "swamp."[119]

The oxidative pentose phosphate cycle is often presented as a means for complete oxidation of hexoses to CO_2. For this to happen the C_3 unit indicated as the product in Fig. 17-8A must be converted (through the action of aldolase, a phosphatase, and hexose phosphate isomerase) back to one-half of a molecule of glucose-6-*P* which can enter the cycle at the beginning. On the other hand, alternative ways of degrading the C_3 product glyceraldehyde-*P* are available. For example, using glycolytic enzymes, it can be oxidized to pyruvate and to CO_2 via the citric acid cycle.

As a general rule, NAD^+ is associated with catabolic reactions and it is somewhat unusual to find $NADP^+$ acting as an oxidant. However, in mammals the enzymes of the pentose phosphate pathway are specific for $NADP^+$. The reason is thought to lie in the need of NADPH for biosynthesis (Section I). On this basis, the occurrence of the pentose phosphate pathway in tissues having an unusually active biosynthetic function (liver and mammary gland) is understandable. In these tissues the cycle may operate as indicated in Fig. 17-8A with the C_3 product also being used in biosynthesis. Furthermore, any of the products from C_4 to C_7 may be withdrawn in any desired amounts without disrupting the smooth operation of the cycle. For example, the C_4 intermediate **erythrose 4-*P*** is required in synthesis of aromatic amino acids by bacteria and plants (but not in animals). **Ribose 5-*P*** is needed for formation of several amino acids and of nucleic acids by all organisms. In some circumstances the formation of ribose 5-*P* may be the only essential function for the pentose phosphate pathway.[120]

Several studies of the metabolism of isotopically labeled glucose[121–122a] have been in accord with

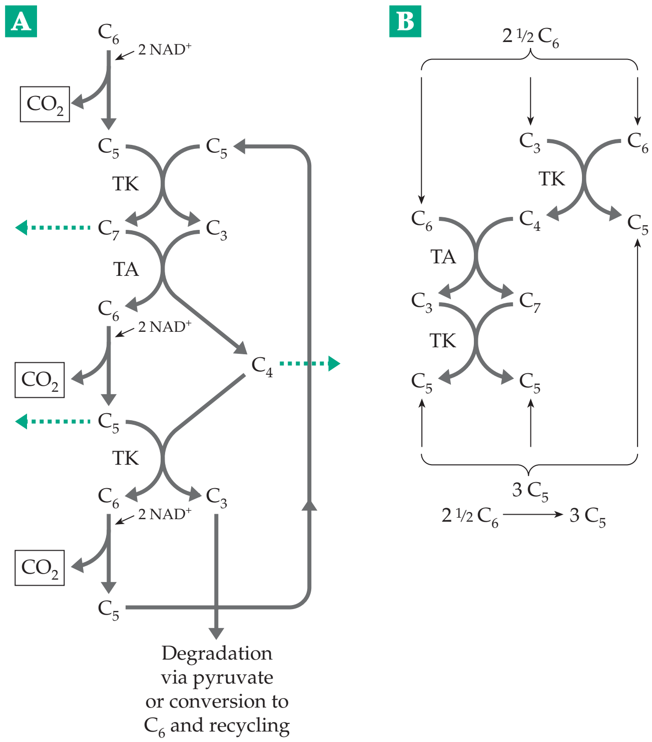


Figure 17-8 The pentose phosphate pathways. (A) Oxidation of a hexose (C_6) to three molecules of CO_2 and a three-carbon fragment with the option of removing C_3, C_4, C_5, and C_7 units for biosynthesis (dashed arrows). (B) Nonoxidative pentose pathways: $2\,1/2\,C_6 \rightarrow 3\,C_5$ or $2\,C_6 \rightarrow 3\,C_4$ or $3\,1/2\,C_6 \rightarrow 3\,C_7$.

operation of the pentose phosphate pathway as is depicted in Fig. 17-8. However, Williams and associates proposed a modification in the sugar rearrangement sequence in liver[123–126] to include the formation of arabinose 5-*P* (from ribose 5-*P*), an octulose bisphosphate, and an octulose 8-monophosphate. Many investigators argue that these additional reactions are of minor significance.[121,122,127] The measured concentrations of pentose phosphate pathway intermediates in rat livers are close to those predicated for a near-equilibrium state from equilibrium constants measured for the individual steps of Fig. 17-8.[128] Most of the concentrations are in the 4- to 10-µM range but the level of erythrose 4-*P*, which is predicted to be ~ 0.2 µM, is too low to measure.

In contrast to animals, the resurrection plant *Craterostigma plantaginenm* accumulates large amounts of a 2-oxo-octulose. This plant is one of a small group of angiosperms that can withstand severe dehydration and are able to rehydrate and resume normal metabolism within a few hours. During desiccation much of the octulose is converted into sucrose. The plant has extra transketolase genes which may be essential for this rapid interconversion of sugars.[129]

Nonoxidative pentose phosphate pathways. The sugar rearrangement system together with the glycolytic enzymes that convert glucose 6-*P* to glyceraldehyde 3-*P* can function to transform hexose phosphates into pentose phosphates (Fig. 17-8B; Eq. 17-16) which may be utilized for nucleic acid synthesis in erythrocytes and other cells.[130,131]

$$2\tfrac{1}{2}\ C_6 \rightarrow 3\,C_5 \qquad (17\text{-}16)$$

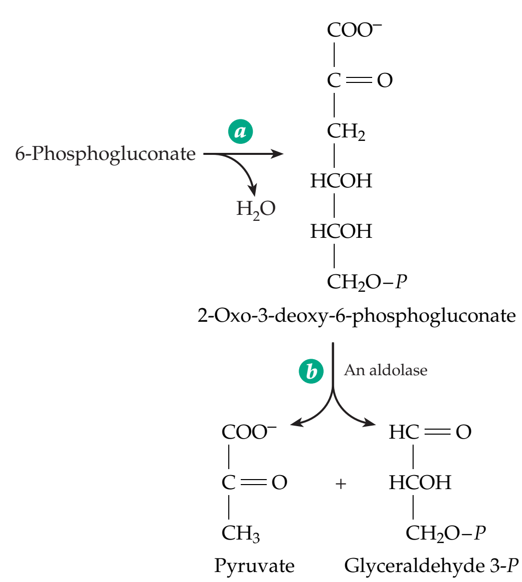
The reader can easily show that the same enzymes will catalyze the net conversion of hexose phosphate to erythrose 4-phosphate or to sedoheptulose 7-phosphate (Eq. 17-17):

$$2\,C_6 \rightarrow 3\,C_4;\ \ 3\tfrac{1}{2}\,C_6 \rightarrow 3\,C_7 \qquad (17\text{-}17)$$

An investigation of metabolism of the red lipid-forming yeast *Rhodotorula gracilis* (which lacks phosphofructokinase and is thus unable to break down sugars through the glycolytic pathway) indicated that 20% of the glucose is *oxidized* through the pentose phosphate pathways while 80% is *altered* by the nonoxidative pentose phosphate pathway.[100] However, it is not clear how the C_3 unit used in the nonoxidative pathway (Fig. 17-8B) is formed if glycolysis is blocked. A number of fermentations are also based on the pentose phosphate pathways (Section F,5).

4. The Entner–Doudoroff Pathway

An additional way of cleaving a six-carbon sugar chain provides the basis for the **Entner–Doudoroff pathway** which is used by *Zymomonas lindneri* and many other species of bacteria. Glucose 6-*P* is oxidized first to 6-phosphogluconate, which is converted by dehydration to a 2-oxo-3-deoxy derivative (Eq. 17-18,



(17-18)

step *a*). The resulting 2-oxo-3-deoxy sugar is cleaved by an aldolase (Eq. 17-18, step *b*) to pyruvate and glyceraldehyde 3-*P*, which are then metabolized in standard ways.

F. Fermentation: "Life without Oxygen"

Pasteur recognized in 1860 that fermentation was not a spontaneous process but a result of life in the near absence of air.[132] He realized that yeasts decompose much more sugar under anaerobic conditions than they do aerobically, and that the anaerobic fermentation was essential to the life of these organisms. In addition to the alcoholic fermentation of yeast, there are many other fermentations which have been attractive subjects for biochemical study. If life evolved at a time when no oxygen was available, the most primitive organisms must have used fermentations. They may be the oldest as well as the simplest ways in which cells obtain energy. The enzymes of the glycolysis pathway are found in the small genomes of *Mycoplasma, Haemophila*, and *Methanococcus*.[133,134]

Fermentation is also a vital process in the human body. Our muscles usually receive enough oxygen to oxidize pyruvate and to obtain ATP through aerobic metabolism, but there are circumstances in which the oxygen supply is inadequate. During extreme exertion, after most oxygen is consumed, muscle cells produce lactate by fermentation. White muscle of fish and fowl has little aerobic metabolism and normally yields L-lactic acid as a principal end product. Likewise, a variety of tissues within the human body, including the transparent lens and cornea of our eyes, are poorly supplied with blood and depend upon fermentation of glucose to lactic acid. Red blood cells and skin and sometimes adipose tissue are also major producers of lactic acid.[135] Of the ~115 g of lactic acid present in a 70-kg human body, about 29% comes from erythrocytes, 29% from skin, 17% from the brain, and 16% from skeletal muscle.[136] Because lactic acid lowers the pH of cells it must be removed efficiently.

Some of the lactic acid formed in muscle and most of the lactate formed in less aerobic tissues (e.g., adipose tissue)[136a] enters the bloodstream, which normally contains 1–2 mM lactate,[136] and is carried to the liver where it is reoxidized to pyruvate. Part of the pyruvate is then oxidized via the citric acid cycle while a larger part is reconverted to glucose (Section J,5). This glucose may be released into the bloodstream and returned to the muscles. The overall process is known as the **Cori cycle**. Lactic acid accumulates in muscle after vigorous exercise. It is exported to the liver slowly, but if mild exercise continues the lactate may be largely oxidized within muscle via the tricarboxylic acid cycle. Recent NMR studies have shown that lactic acid is formed rapidly during muscular contraction, even when exercise is mild.[136b] During the initial 15 ms of contraction the ATP utilized is regenerated from creatine phosphate (Eq. 6-67). During the remainder of the contraction (up to ~100 ms) glycogen is converted to lactic acid to provide ATP and to replenish the creatine phosphate. In the resting period following contraction most of the lactate is either dehydrogenated to pyruvate and oxidized in mitochondria or exported to other tissues. The glycogen stores in muscle are renewed by synthesis from blood glucose. Lactic acid is a convenient energy carrier and a precursor for gluconeogenesis which can be transferred between tissues easily.[136c] Cancer cells often take advantage of this opportunity to grow rapidly using fermentation of glucose to lactic acid as a source of energy.[136d]

Alcoholic fermentation allows roots of some plants to survive short periods of flooding. Ethanol does not acidify the tissues as does lactic acid, avoiding possible damage from low pH.[137,138] Goldfish can also use the ethanolic fermentation for short times, excreting the ethanol.[139]

1. Fermentations Based on the Embden–Meyerhof Pathway

Homolactic and alcoholic fermentations. The reactions by which glucose can be converted to lactate and, by yeast cells, to ethanol and CO_2 (Figs. 10-3 and 17-7) illustrate several features common to all fermentations. The NADH produced in the oxidation step is reoxidized in a reaction by which substrate is reduced to the final end product. The NAD alternates between oxidized and reduced forms. This coupling of oxidation steps with reduction steps in exact equivalence is characteristic of all true (anaerobic) fermentations. The formation of ATP from ADP and P_i by substrate-level phosphorylation is also common to all fermentations. The stoichiometry is often nearly exact and simple. For example, according to the reaction of Eq. 17-19, which is outlined step-by-step in Fig. 17-7, a net total of two moles of ATP is formed per mole of glucose fermented.

Energy relationships. If we disregard the synthesis of ATP, the equations for the lactic acid and ethanol fermentations are given by Eqs. 17-19 and 17-20.

$$\text{Glucose} \rightarrow 2\ \text{lactate}^- + 2\ \text{H}^+$$
$$\Delta G'\ (\text{pH 7}) = -196\ \text{kJ mol}^{-1}\ (-46.8\ \text{kcal mol}^{-1}) \quad (17\text{-}19)$$

$$\text{Glucose} \rightarrow 2\ CO_2 + 2\ \text{ethanol}$$
$$\Delta G° = -235\ \text{kJ mol}^{-1} \quad (17\text{-}20)$$

The Gibbs energy changes are negative and of sufficient magnitude that the reactions will unquestionably go to completion. However, the synthesis of two molecules of ATP from inorganic phosphate and ADP, a reaction

(Eq. 17-21) for which $\Delta G'$ is substantially positive, is coupled to the fermentation.

$$ADP^{3-} + HPO_4^{2-} + H^+ \rightarrow ATP^{4-} + H_2O$$
$$\Delta G' \text{ (pH 7)} = +34.5 \text{ kJ mol}^{-1} \text{ (Table 6-5)} \quad (17\text{-}21)$$

To obtain the net Gibbs energy change for the complete reaction we must add 2 x 34.5 = +69.0 kJ to the values of $\Delta G'$ for Eqs. 17-19 and 17-20. When this is done we see that the net Gibbs energy changes are still highly negative, that the reactions will proceed to completion, and that these fermentations can serve as an usable energy source for organisms.

Biochemists sometimes divide ΔG for the ATP synthesis in a coupled reaction sequence (in this case +69 kJ) by the overall Gibbs energy decrease for the coupled process (196 or 235 kJ mol^{-1}) to obtain an "efficiency." In the present case the efficiency would be 35% and 29% for coupling of Eq. 17-21 (for 2 mol of ATP) to Eqs. 17-19 and 17-20, respectively. According to this calculation, nature is approximately one-third efficient in the utilization of available metabolic Gibbs energy for ATP synthesis. However, it is important to realize that this calculation of efficiency has no exact thermodynamic meaning. Furthermore, the utilization of ATP formed by a cell for various purposes is far from 100% efficient.

Why are the Gibbs energy decreases for Eqs. 17-19 and 17-20 so large? No overall oxidation takes place; there is only a rearrangement of the existing bonds between atoms of the substrate. Why does this rearrangement of bonds lead to a substantial negative ΔG? An answer is suggested by an examination of the numbers of each type of bond in the substrate and in the products. During the conversion from glucose to two molecules of lactate one C–C bond, one C–O bond, and one O–H bond are lost and one C–H bond and one C=O are gained. If we add up the bond energies for these bonds (Table 6-6) we find that the difference (ΔH) between substrate and products amounts to only about 20 kJ/mol. However, lactic acid contains a carboxyl group, and carboxyl groups have a special stability as a result of resonance. The extra resonance energy of a carboxyl group (Table 6-6) is ~117 kJ (28 kcal) per mole or 234 kJ/mol for two carboxyl groups. This is approximately the same as the Gibbs energy change (Eq. 17-19) for fermentation of glucose to lactate. Thus, the energy available results largely from the rearrangement of bonds by which the carboxyl groups of lactate are formed. Likewise, the resonance stabilization of CO_2 is given by Pauling as 151 kJ/mol, again of just the right magnitude to explain ΔG in alcoholic fermentations (Eq. 17-20).

On this basis we can state as a general rule that fermentations can occur when substrates consisting of largely singly bonded atoms and groups, such as the carbonyl groups that are not highly stabilized by resonance, are converted to products containing carboxyl groups or to CO_2. If we assume an efficiency of ~30%, the energy available will be about sufficient for synthesis of one ATP molecule for each carboxyl group or CO_2 created. Bear in mind that generation of ATP also depends upon availability of a mechanism. It is of interest that most synthesis of ATP is linked directly to the chemical processes by which carboxyl groups or CO_2 molecules are created in a fermentation process. The most important single reaction is the oxidation of the aldehyde group of glyceraldehyde 3-*P* to the carboxyl group of 3-phosphoglycerate (steps *6a*– *6c* and *7* in Fig. 17-7; see also Fig. 15-6).

Compare the fermentation of glucose with the complete oxidation of the sugar to carbon dioxide and water (Eq. 17-22), a process which yeast cells (as well as our own cells) carry out in the presence of air. The overall Gibbs energy change is over 10 times greater than that for fermentation, a fact that permits the cell

$$\text{Glucose} + 6\, O_2 \rightarrow 6\, CO_2 + 6\, H_2O$$
$$\Delta G' = -2872 \text{ kJ } (-686.5 \text{ kcal}) \text{ mol}^{-1} \quad (17\text{-}22)$$

to form an enormously greater quantity of ATP. The net gain in ATP synthesis, accompanying Eq. 17-22, is usually about 38 mol of ATP—19 times more than is available from fermentation of glucose. Thus, the explanation of Pasteur's observation that yeast decomposes much less sugar in the presence of air than in its absence is clear. Also, we can understand why a cell, living anaerobically, must metabolize a very large amount of substrate to grow. (Recall from Chapter 6 that ~1 mol of ATP energy is needed to produce 10 g of cells.)

Variations of the alcoholic and homolactic fermentations. The course of a fermentation is often affected drastically by changes in conditions. Many variations can be visualized by reference to Fig. 17-9, which shows a number of available metabolic sequences. We have already discussed the conversion of glucose to triose phosphate and via reaction pathway *a* to pyruvate, via reaction *c* to lactate, and via reaction *d* to ethanol.

If bisulfite is added to a fermenting culture of yeast, the acetaldehyde formed through reaction *d* is trapped as the bisulfite adduct blocking the reduction of acetaldehyde to ethanol, an essential part of the fermentation. Yeast cells accommodate this change by using the accumulating NADH to reduce half of the triose-*P* to glycerol through pathway *b*. Two enzymes are needed, a dehydrogenase and a phosphatase, to hydrolytically cleave off the phosphate. The balanced reaction is given by Eq. 17-23:

$$\text{Glucose} \rightarrow \text{glycerol} + \text{acetaldehyde (trapped)} + CO_2$$
$$\Delta G' \text{ (pH 7)} = -105 \text{ kJ mol}^{-1} \quad (17\text{-}23)$$

In this reaction only one molecule of CO_2 is produced but the overall Gibbs energy change is still adequate to make the reaction highly spontaneous. However (referring to Fig. 17-9), we see that the net synthesis of ATP is now apparently zero. The fermentation apparently does not permit cell growth. Nevertheless, it has been used industrially for production of glycerol.

Reduction of dihydroxyacetone phosphate to glycerol phosphate also occurs in insect flight muscle and apparently operates as an alternative to lactic acid formation in that tissue. There is no net gain of ATP in the conversion of free glucose to glycerol phosphate and pyruvate, but using stored glycogen in muscle as the starting material, the dismutation of triose-*P* to glycerol-*P* and pyruvate provides one ATP per glucose unit rapidly during the vigorous contraction of the powerful insect flight muscle. During the slower recovery phase, glycerol-*P* is thought to be reoxidized after entering the mitochondria of these highly aerobic cells. Thus, the transport of glycerol-*P* into mitochondria serves as a means for transporting reducing equivalents derived from reoxidation of NADH into the mitochondria. Indeed, the significance of glycerol-*P* to muscle metabolism may be more related to this function than to the rapid formation of ATP (see Chapter 18).

Why does the glycolysis sequence begin with phosphorylation of glucose by ATP? The phospho groups probably provide convenient handles and doubtless assist in substrate recognition. There may be a kinetic advantage but also a danger. Unless there is adequate regulation the "turbo design," in which ATP is used at the outset to drive glycolysis, may lead to accumulation of phosphorylated intermediates and to inadequate concentrations of ATP and inorganic phosphate.[139a,b] Yeast cells guard against this problem by synthesizing trehalose 6-phosphate, which acts as a feedback inhibitor of hexokinase.[139a] Trypanosomes utilize a different type of control. The enzymes that convert glucose into 3-phosphoglycerate are present in membrane-bounded organelles called **glycosomes**. Phosphoglycerate is exported from them into the cytosol where glycolysis is completed.[139b] Since inorganic phosphate is essential for ATP formation, if the P_i concentration falls too low the rate of fermentation by yeast juice is greatly decreased, an observation made by Harden and Young[139c] in 1906.

2. The Mixed Acid Fermentation

Enterobacteria, including *E. coli*, convert glucose to ethanol and acetic acid and either formic acid or CO_2 and H_2 derived from it. The stoichiometry is variable but the fermentation can be described in an idealized form as follows:

$$\text{Glucose} + H_2O \rightarrow \text{ethanol} + \text{acetate}^- + H^+ + 2\,H_2 + 2\,CO_2$$

$$\Delta G' \text{ (pH 7)} = -225 \text{ kJ mol}^{-1} \quad (17\text{-}24)$$

The details of the process and the oxidation–reduction balance can be pictured as in Eq. 17-25. Pyruvate is cleaved by the pyruvate formate–lyase reaction (Eq. 15-37) to acetyl-CoA and formic acid. Half of the acetyl-CoA is cleaved to acetate via acetyl-*P* with generation of ATP, while the other half is reduced in two steps to ethanol using the two molecules of NADH produced in the initial oxidation of triose phosphate (Eq. 17-25). The overall energy yield is three molecules of ATP per glucose. The "efficiency" is thus (3 x 34.5) ÷ 225 = 46%. Some of the glucose is converted to D-lactic and to succinic acids (pathway *f*, Fig. 17-9); hence the name **mixed acid fermentation**. Table 17-1 gives typical yields of the mixed acid fermentation of *E. coli*. Among the four major products are acetate, ethanol, H_2, and CO_2, as shown in Eq. 17-25. However, at high pH formate accumulated instead of CO_2.

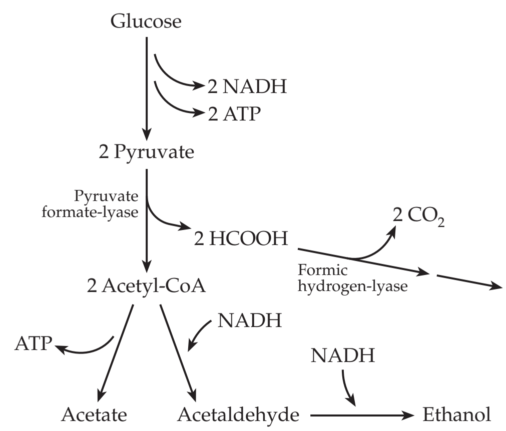


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TABLE 17-1
Products of the Mixed Acid Fermentation by *E. coli* at Low and High Values of pH[a]

Product (Millimole formed from 100 mmol of glucose)	pH 6.2	pH 7.8
Acetate	36	39
Ethanol	50	51
H_2	70	0.3
CO_2	88	1.7
Formate	2.4	86
Lactate	79	70
Succinate	11	15
Glycerol	1.4	0.3
Acetoin	0.1	0.2
Butanediol	0.3	0.2

[a] From Tempest, D. W. and Neijssel, O. M.[140] Based on data of Blackwood.[141]

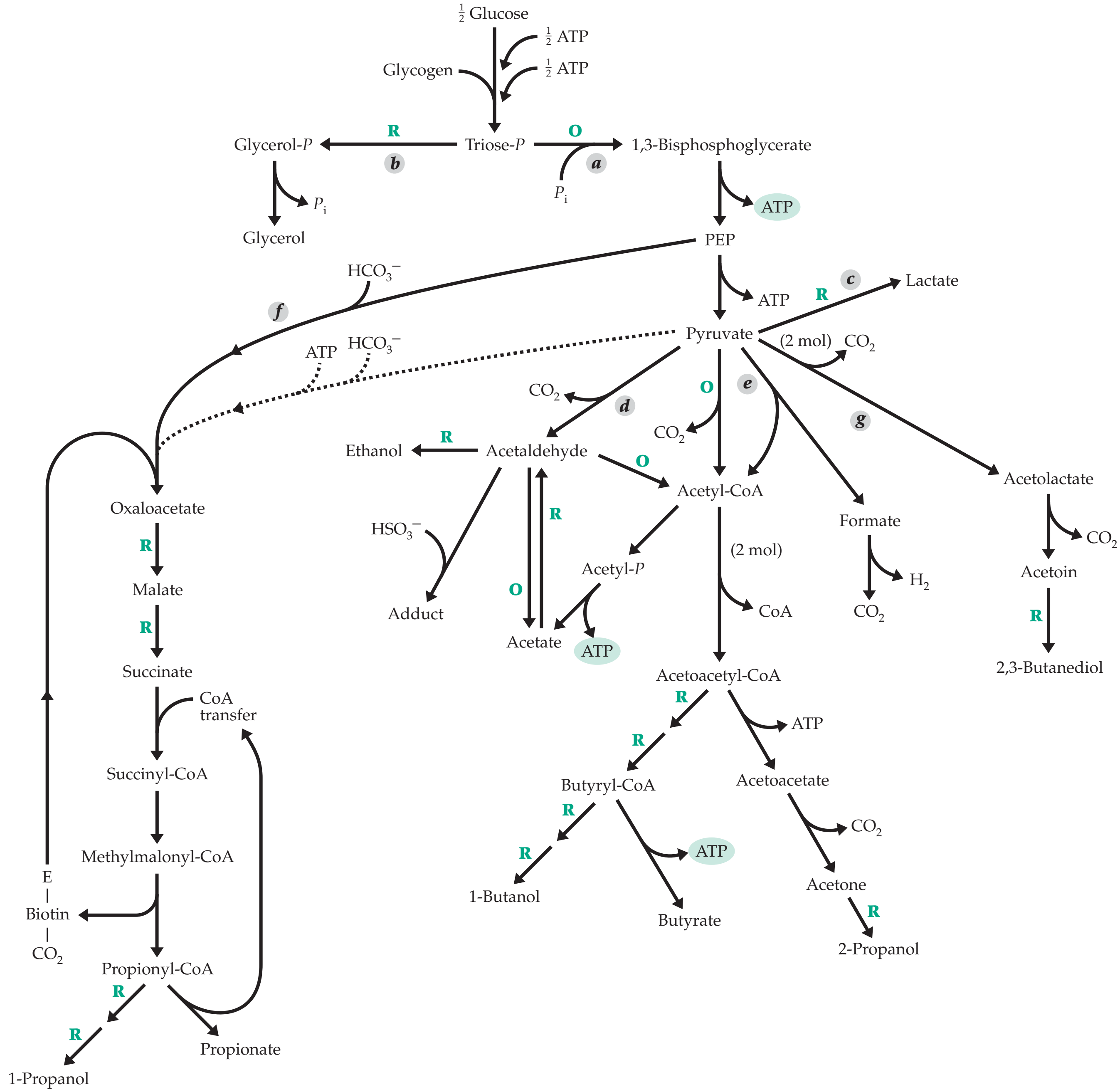
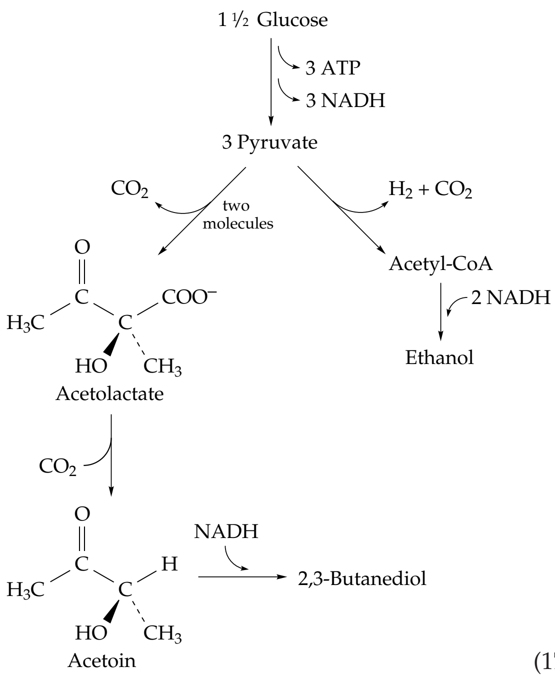


Figure 17-9 Reaction sequences in fermentation based on the Embden–Meyerhof–Parnas pathway. Oxidation steps (producing NADH + H^+) are marked "O"; reduction steps (using NADH + H^+) are marked "R."

Over one-third of the glucose was fermented to lactate in both cases.

In some mixed acid fermentations (e.g., that of *Shigella*) formic acid accumulates, but in other cases (e.g., with *E. coli* at pH 6) it is converted to CO_2 and H_2 (Eq. 17-25). The equilibration of formic acid with CO_2 and hydrogen is catalyzed by the **formic hydrogen–lyase** system which consists of two iron–sulfur enzymes. The selenium-containing **formate dehydrogenase** (Eq. 16-63) catalyzes oxidation of formate to CO_2 by NAD^+, while a membrane-bound **hydrogenase** (Eq. 16-48) equilibrates NADH + H^+ with NAD^+ + H_2. Hydrogenase also serves to release H_2 from excess NADH. Krebs pointed out that an excess of NADH may arise because growth of cells requires biosynthesis of many components such as amino acids. When glucose is the sole source of carbon, biosynthetic reactions involve an excess of oxidation steps over reduction steps.[142] The excess of reducing equivalents may be released as H_2 or may be used to form highly reduced products such as succinate.

Among such genera as *Aerobacter* and *Serratia* part of the pyruvate formed is condensed with decarboxylation to form ***S* acetolactate**,[143] which is decarboxylated to acetoin (Eq. 17-26; pathway *g* of Fig. 17-9). The acetoin is reduced with NADH to **2,3-butanediol**, while a third molecule of pyruvate is converted to ethanol, hydrogen, and CO_2 (Eq. 17-26). The reaction provides the basis for industrial production of butanediol, which can be dehydrated nonenzymatically to butadiene.



(17-26)

Mixed acid fermentations are not limited to bacteria. For example, trichomonads, parasitic flagellated protozoa, have no mitochondria. They export pyruvate into the bloodstreams of their hosts and also contain particles called **hydrogenosomes** which can convert pyruvate to acetate, succinate, CO_2, and H_2.[144] Hydrogenosomes are bounded by double membranes and have a common evolutionary relationship with both mitochondria and bacteria. The enzyme that catalyzes pyruvate cleavage in hydrogenosomes apparently does not contain lipoate and may be related to the pyruvate–ferredoxin oxidoreductase of clostridia (Eq. 15-35). The hydrogenosomes also contain an active hydrogenase.

Many invertebrate animals are true facultative anaerobes, able to survive for long periods, sometimes indefinitely, without oxygen.[145–147] Among these are *Ascaris* (Fig. 1-14), oysters, and other molluscs. Succinate and alanine are among the main end products of anaerobic metabolism. The former may arise by a mixed acid fermentation that also produces pyruvate. The pyruvate is converted to acetate to balance the fermentation in *Ascaris lumbricoides*, which is in effect an obligate anaerobe. However, in molluscs the pyruvate may undergo transamination with glutamate to form alanine and 2-oxoglutarate; the oxoglutarate may be oxidatively decarboxylated to succinate. The reactions depend upon the availability of a store of glutamate or of other amino acids, such as arginine, that can give rise to glutamate.

3. The Propionic Acid Fermentation

Propionic (propanoic) acid-producing bacteria are numerous in the digestive tract of ruminants. Within the rumen some bacteria digest cellulose to form glucose, which is then converted to lactate and other products. The propionic acid bacteria can convert either glucose or lactate into propionic and acetic acids which are absorbed into the bloodstream of the host. Usually some succinic acid is also formed.

The basis of the propionic acid fermentation is conversion of pyruvate to oxaloacetate by carboxylation and the further conversion through succinate and succinyl-CoA to methylmalonyl-CoA and propionyl-CoA, reactions which are almost the exact reverse of those for the oxidation of propionate in the animal body (Fig. 17-3, pathway *d*). However, whereas the carboxylation of pyruvate to oxaloacetate in the animal body requires ATP, the propionic acid bacteria save one equivalent of ATP by using a carboxyltransferase (p. 725). This enzyme donates a carboxyl group from a preformed carboxybiotin compound generated in the decarboxylation of methylmalonyl-CoA in the next to final step of the reaction sequence (Fig. 17-10). A second molecule of ATP is saved by linking directly the conversion of succinate to succinyl-CoA to the cleavage of propionyl-CoA to propionate through the use of a CoA transferase (Eq. 12-50). To provide for oxidation–reduction balance, two-thirds of the glucose goes to propionate and one-third to acetate (Eq. 17-27):

$$1\tfrac{1}{2}\,\text{Glucose} \rightarrow$$
$$2\ \text{propionate}^- + \text{acetate}^- + 3\,H^+ + CO_2 + H_2O$$
$$\Delta G'\ (\text{pH } 7) = -465\ \text{kJ per } 1\tfrac{1}{2}\ \text{mol of glucose} \quad (17\text{-}27)$$

More carboxyl groups and CO_2 molecules are formed in this fermentation ($2\tfrac{2}{3}$ per glucose molecule) than in the regular lactic acid fermentation. The yield of ATP (also $2\,^2/_3$ mol/mol of glucose fermented) is correspondingly greater and $\Delta G'$ is more negative.

Using the same mechanism (Fig. 17-10), propionic acid bacteria are also able to ferment lactate, the product of fermentation by other bacteria, to propionate and acetate (Eq. 17-28). The net gain is one molecule of ATP. This reaction probably accounts for the niche

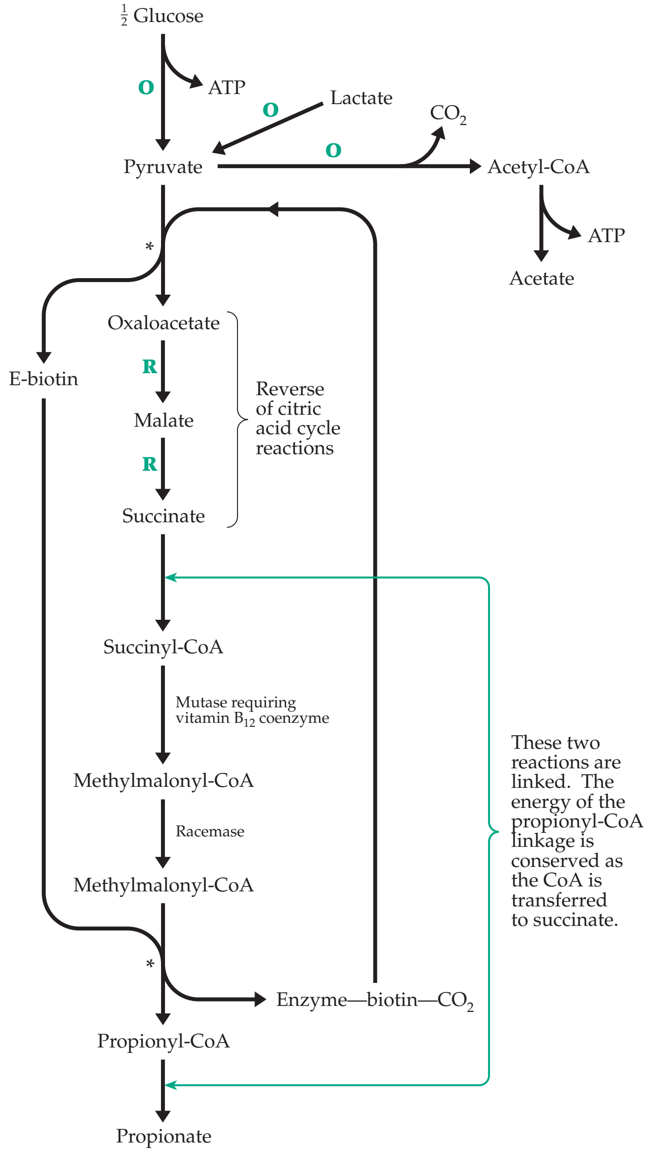


Figure 17-10 Propionic acid fermentation of *Propionobacteria* and *Veillonella*. Oxidation steps are designated by the symbol "O" and reduction steps by "R." The two coupled reactions marked by asterisks are catalyzed by carboxyltransferase.

$$3\ \text{Lactate}^- \rightarrow 2\ \text{propionate}^- + \text{acetate}^- + H_2O + CO_2$$
$$\Delta G'\ (\text{pH 7}) = -171\ \text{kJ per 3 mol of lactate} \quad (17\text{-}28)$$

in the ecology of the animal rumen that is occupied by propionic acid bacteria.

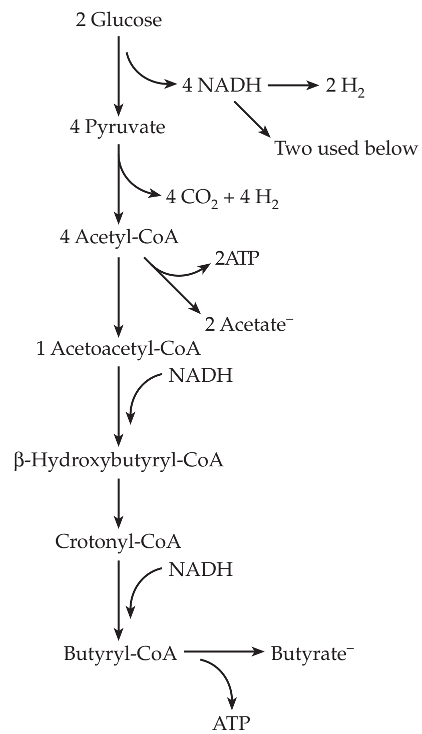
4. Butyric Acid and Butanol-Forming Fermentations

A variety of fermentations are carried out by bacteria of the genus *Clostridium* and by the rumen organisms *Eubacterium (Butyribacterium)* and *Butyrivibrio*.

For example, glucose may be converted to butyric and acetic acids together with CO_2 and H_2 (Eqs. 17-29 and 17-30).

$$2\ \text{Glucose} + 2\ H_2O \rightarrow \text{butyrate}^- + 2\ \text{acetate}^- + 4\ CO_2 + 6\ H_2 + 3\ H^+$$
$$\Delta G'\ (\text{pH 7}) = -479\ \text{kJ per 2 mol of glucose} \quad (17\text{-}29)$$

The yield of ATP (3½ mol/mol of glucose) is the highest we have discussed giving an efficiency of 50%. Another fermentation yields butanol, isopropanol, ethanol, and acetone.

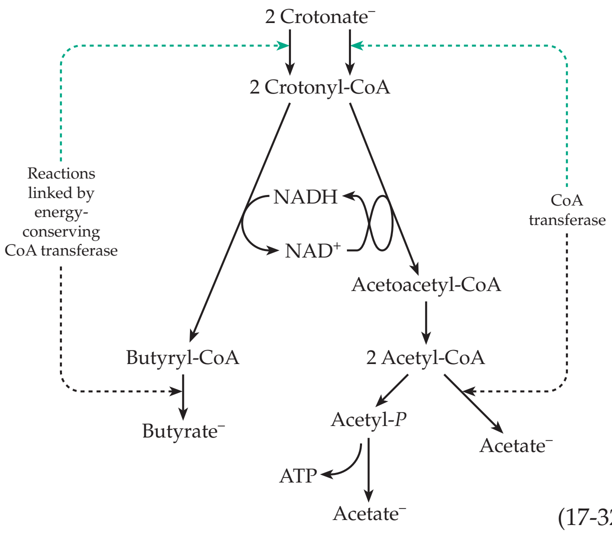


(17-30)

The fermentation of Eq. 17-31 is catalyzed by *Clostridium kluyveri*. The value of $-\Delta G'$ is one of the lowest that we have considered but is still enough to provide easily for the synthesis of one molecule of ATP.

$$\underset{\text{Crotonate}}{2\ CH_3CH{=}CH{-}COO^-} + 2\ H_2O \rightarrow \text{butyrate}^- + 2\ \text{acetate}^- + H^+$$
$$\Delta G'\ (\text{pH 7}) = -105\ \text{kJ mol}^{-1} \quad (17\text{-}31)$$

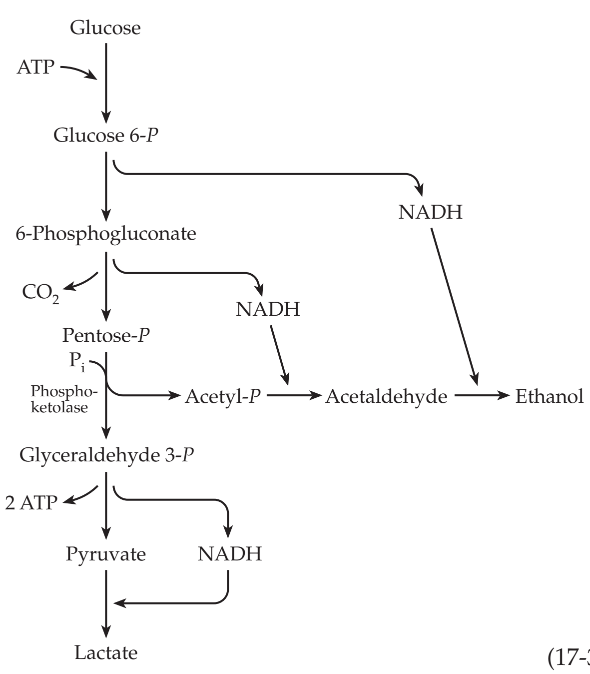
The energy of the butyryl-CoA linkage and of one of the acetyl-CoA linkages is conserved and utilized in the initial formation of crotonyl-CoA (Eq. 17-32). That leaves one acetyl-CoA which can be converted via acetyl-*P* to acetate with formation of ATP.



(17-32)

5. Fermentations Based on the Phosphogluconate and Pentose Phosphate Pathways

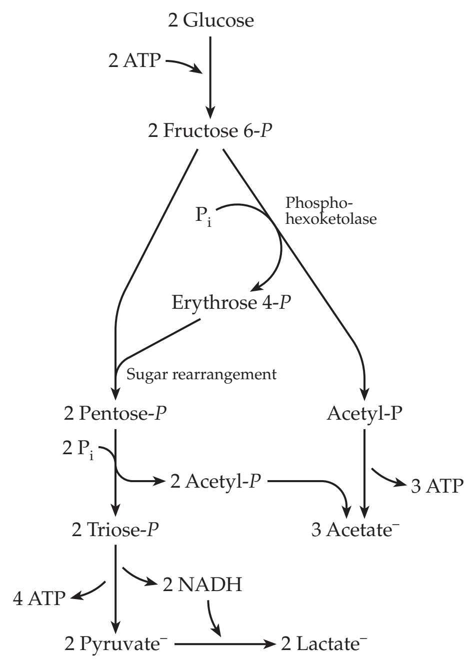
Some lactic acid bacteria of the genus *Lactobacillus*, as well as *Leuconostoc mesenteroides* and *Zymomonas mobilis*, carry out the **heterolactic** fermentation (Eq. 17-33) which is based on the reactions of the pentose phosphate pathway. These organisms lack aldolase, the key enzyme necessary for cleavage of fructose 1,6-bisphosphate to the triose phosphates. Glucose is converted to ribulose 5-*P* using the oxidative reactions of the pentose phosphate pathway. The ribulose-phosphate is cleaved by phosphoketolase (Eq. 14-23) to acetyl-phosphate and glyceraldehyde 3-phosphate, which are converted to ethanol and lactate, respectively. The overall yield is only one ATP per glucose fermented.



(17-33)

This is generated in the substrate level oxidative phosphorylation catalyzed by phosphoketolase. Metabolic engineering of *Zymomonas* was accomplished by transferring from other bacteria two operons that provide for assimilation of xylose and a complete set of enzymes for the pentose phosphate pathway. The engineered bacteria are able to convert pentose phosphates nonoxidatively (see Fig. 17-8) into glyceraldehyde 3-phosphate, which is converted to ethanol in high yield and with much greater synthesis of ATP than according to Eq. 17-33.[148]

A variation of the heterolactic fermentation is used by *Bifidobacterium* (Eq. 17-34).[149] Phosphoketolase and a **phosphohexoketolase**, which cleaves fructose 6-*P* to erythrose 4-*P* and acetyl-*P*, are required, as are the enzymes of the sugar rearrangement system (Section E,3). The net yield of ATP is 2 ½ molecules per molecule of glucose.



(17-34)

G. Biosynthesis

In this section and sections H – K the general principles and strategy of synthesis of the many carbon compounds found in living things will be considered. Since green plants and autotrophic bacteria are able to assemble all of their needed carbon compounds from CO_2, let us first examine the mechanisms by which this is accomplished. We will also need to ask how some organisms are able to subsist on such simple compounds as methane, formate, or acetate.

1. Metabolic Loops and Biosynthetic Families

As was pointed out in Chapter 10, routes of biosynthesis (anabolism) often closely parallel pathways of biodegradation (catabolism). Thus, catabolism begins with hydrolytic breakdown of polymeric molecules; the resulting monomers are then cleaved into small two- and three-carbon fragments. Biosynthesis begins with formation of monomeric units from small pieces followed by assembly of the monomers into polymers. The mechanisms of the individual reactions of biosynthesis and biodegradation are also often closely parallel. However, in most instances, there are clear-cut differences. A first principle of biosynthesis is that *biosynthetic pathways, although related to catabolic pathways, differ from them in distinct ways and are often catalyzed by completely different sets of enzymes*.

The sum of the pathways of biosynthesis and biodegradation form a continuous loop – a series of reactions that take place concurrently and often within the same part of a cell. Metabolic loops often begin in the central pathways of carbohydrate metabolism with three- or four-carbon compounds such as phosphoglycerate, pyruvate, and oxaloacetate. After loss of some atoms as CO_2 the remainder of the compound rejoins the "mainstream" of metabolism by entering a catabolic pathway leading to acetyl-CoA and oxidation in the citric acid cycle. Not all of the loops are closed within a given species. For example, *human beings are unable to synthesize the vitamins and the "essential amino acids."* We depend upon other organisms to make these compounds, but we do degrade them. Some metabolites, such as uric acid, are excreted by humans and are further catabolized by bacteria. From a chemical viewpoint the whole of nature can be regarded as an enormously complex set of branching and interconnecting metabolic cycles. Thus, the synthetic pathways used by autotrophs are all parts of metabolic loops terminating in oxidation back to CO_2.

It is often not possible to state at what point in a metabolic loop biosynthesis has been completed and biodegradation begins. An end product X that serves one need of a cell may be a precursor to another cell component Y which is then degraded to complete the loop. The reactions that convert X to Y can be regarded as either biosynthetic (for Y) or catabolic (for X).

2. Key Intermediates and Biosynthetic Families

In examining routes of biosynthesis it is helpful to identify some key intermediates. One of these is **3-phosphoglycerate**. This compound is a primary product of photosynthesis and may reasonably be regarded as the starting material from which all other carbon compounds in nature are formed. Phosphoglycerate, in most organisms, is readily interconvertible with both **glucose** and **phosphoenolpyruvate (PEP)**. Any of these three compounds can serve as the precursor for synthesis of other organic materials. A first stage in biosynthesis consists of those reactions by which 3-phosphoglycerate or PEP arise, whether it be from CO_2, formate, acetate, lipids, or polysaccharides.

The further biosynthetic pathways from 3-phosphoglycerate to the myriad amino acids, nucleotides, lipids, and miscellaneous compounds found in cells are complex and numerous. However, the basic features are relatively simple. Figure 17-11 indicates the origins of many substances including the 20 amino acids present in proteins, nucleotides, and lipids. Among the additional key biosynthetic precursors that can be identified from this chart are **glucose 6-phosphate**, **pyruvate**, **oxaloacetate**, **acetyl-CoA**, **2-oxoglutarate**, and **succinyl-CoA**.

The amino acid **serine** originates almost directly from 3-phosphoglycerate. **Aspartate** arises from oxaloacetate and **glutamate** from 2-oxoglutarate. These three amino acids each are converted to "families" of other compounds.[150] A little attention paid to establishing correct family relationships will make the study of biochemistry easier. Besides the serine, aspartate, and glutamate–oxoglutarate families, a fourth large family originates directly from pyruvate and a fifth (mostly lipids) from acetyl-CoA. The aromatic amino acids are formed from erythrose 4-*P* and PEP via the key intermediate **chorismic acid** (Box 9-E; Fig. 25-1). Other families of compounds arise from glucose 6-*P* and from the **pentose phosphates**. These groups have been set off roughly by the boxes outlined in green in Fig. 17-11.

H. Harnessing the Energy of ATP for Biosynthesis

In the past it seemed reasonable to think that some biosynthetic pathways involved exact reversal of catabolic pathways. For example, it was observed that glycogen phosphorylase catalyzed elongation of glycogen branches by transfer of glycosyl groups from glucose 1-phosphate. Likewise, the enzymes needed for the β oxidation of fatty acid derivatives, when isolated from mitochondria, catalyze formation of fatty acyl-CoA derivatives from acetyl-CoA and a reducing agent such as NADH. However, reactant concentrations within cells are rarely appropriate for reversal of a catabolic sequence. For catabolic sequences the Gibbs energy change is usually distinctly negative and reversal requires high concentrations of end products. However, the latter are often removed promptly from the cells. For example, NADH produced in degradation of fatty acids is oxidized to NAD^+ and is therefore never available in sufficient concentrations to reverse the β oxidation sequence.

Nature's answer to the problem of reversing a catabolic pathway lies in the coupling of cleavage of ATP to the biosynthetic reaction. The concept was introduced in Chapter 10, in which one sequence for linking hydrolysis of ATP to biosynthesis was discussed. However, living cells employ several different methods of harnessing the Gibbs energy of hydrolysis of ATP to drive biosynthetic processes. Many otherwise strange aspects of metabolism become clear if it is recognized that they provide a means for coupling ATP cleavage to biosynthesis. A few of the most important of these coupling mechanisms are summarized in this section.

1. Group Activation

Consider the formation of an ester (or of an amide) from a free carboxylic acid and an alcohol (or amine) by elimination of a molecule of water (Eq. 17-35). The reaction is thermodynamically unfavorable with values of $\Delta G'$ (pH 7) ranging from ~ +10 to 30 kJ mol^{-1} depending on conditions and structures of the specific compounds. Long ago, organic chemists learned that such reactions can be made to proceed by careful removal of the water that is generated (Eq. 17-35).

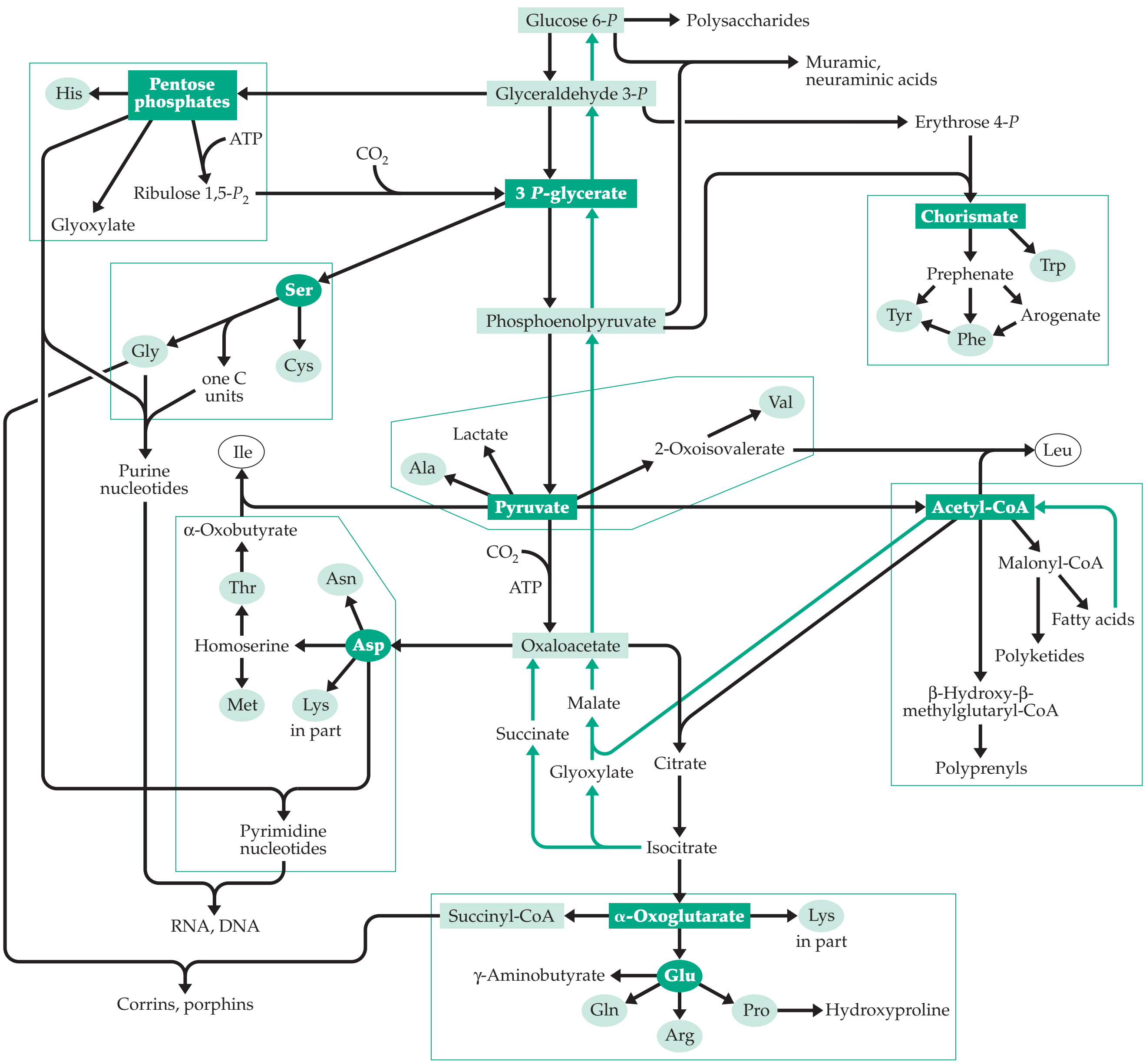
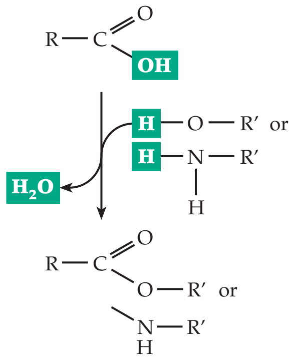
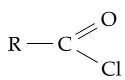


Figure 17-11 Some major biosynthetic pathways. Some key intermediates are enclosed in boxes and the 20 common amino acids of proteins are encircled. Key intermediates for each family are in shaded boxes or elipses. Green lines trace the reactions of the glyoxylate pathway and of glucogenesis.

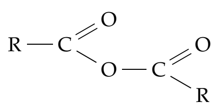


(17-35)

However, it is often better to "activate" the carboxylic acid by conversion to an acyl chloride or an anhydride:



Acyl chloride



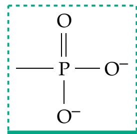
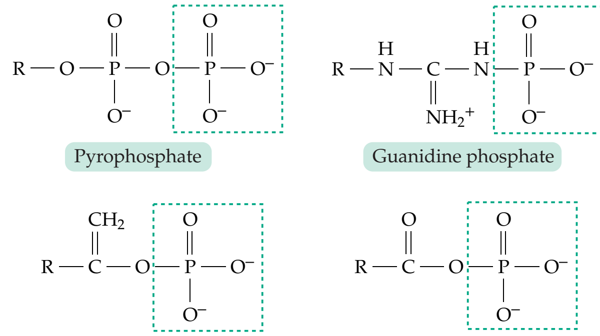
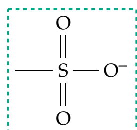
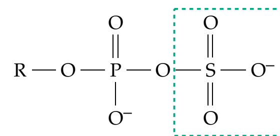
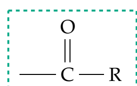
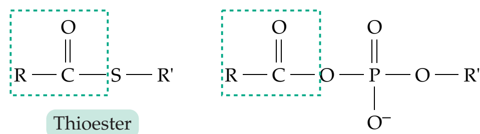
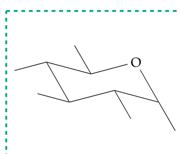
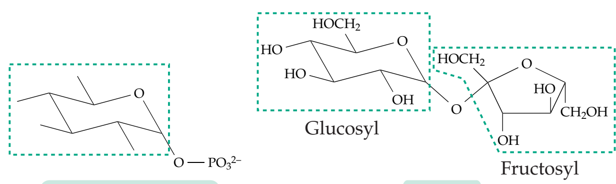
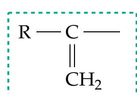
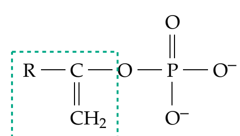
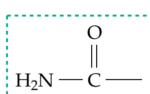
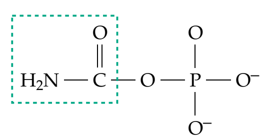
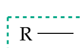
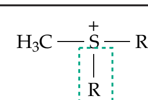
Acid anhydride

Nucleophilic attack on the carbonyl group of such a compound results in displacement of a good leaving group, Cl^- or $R{-}COO^-$. Nature has followed the same approach in forming from carboxylic acids **acyl phosphates** or **acyl-CoA** derivatives.

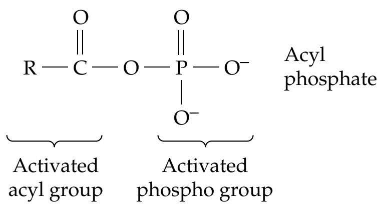
The virtue of these "activated" acyl compounds in biosynthetic reactions was considered in Chapter 12 and Table 15-1. Just as a carboxylic acid can be converted to an active acyl derivative, so other groups can be activated. ATP and other compounds with phospho groups of high group transfer potential are **active phospho** compounds. Sulfate is converted to a phosphosulfate anhydride, an **active sulfo** derivative. Sugars are converted to compounds such as glucose 1-*P* or sucrose, which contain **active glycosyl** groups. The group transfer potentials of the latter, though not as great as that of the phospho groups of ATP, are still high enough to make glucose 1-phosphate and sucrose effective glycosylating reagents. Table 17-2 lists several of the more important activated groups.

Group activation usually takes place at the expense of ATP cleavage.

TABLE 17-2
"Activated" Groups Used in Biosynthesis

Group	Typical activated forms
 Phospho	 Pyrophosphate; Guanidine phosphate; Enol phosphate; Acyl phosphate
 Sulfo	 in PAPS (3′-Phosphoadenosine-5′-phosphosulfate)
 Acyl	 Thioester; Acyl phosphate
 Glycosyl	 Glycosyl phosphate; Sucrose (Glucosyl, Fructosyl)
 Enoyl	 Enol phosphate
 Carbamoyl	 Carbamoyl phosphate
 Alkyl	 *S*-Adenosylmethionine

As pointed out in Chapter 12, acyl phosphates play a central role in metabolism by virtue of the fact that they contain both an activated acyl group and an activated phospho group. The high group transfer potential can be conserved in subsequent reactions *in either one group or the other* (but not in both). Thus, displacement on P by an oxygen of ADP will regenerate ATP and attack on C by an – SH will give a thioester.

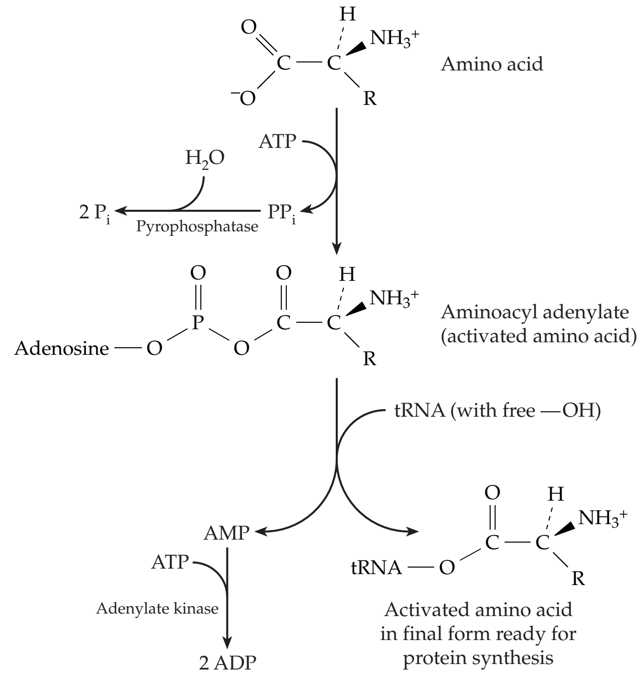


Several of the other compounds in Table 17-2 can also be split in two ways to yield different activated groups, e.g., the phosphosulfate anhydride, enoyl phosphate, and carbamoyl phosphate. It is probably only through intermediates of this type that cleavage of ATP can be coupled to synthesis of activated groups. Such **common intermediates** are essential to the synthesis of ATP by substrate-level phosphorylation (Fig. 15-6).

2. Hydrolysis of Pyrophosphate

The splitting of inorganic pyrophosphate (PP_i) into two inorganic phosphate ions is catalyzed by **pyrophosphatases** (p. 636)[150a,b] that apparently occur universally. Their function appears to be simply to remove the product PP_i from reactions that produce it, shifting the equilibrium toward formation of a desired compound. An example is the formation of **aminoacyl-tRNA** molecules needed for protein synthesis. As shown in Eq. 17-36, the process requires the use of two ATP molecules to activate one amino acid. While the "spending" of two ATPs for the addition of one monomer unit to a polymer does not appear necessary from a thermodynamic viewpoint, it is frequently observed, and there is no doubt that hydrolysis of PP_i ensures that the reaction will go virtually to completion. Transfer RNAs tend to become saturated with amino acids according to Eq. 17-36 even if the concentration of free amino acid in the cytoplasm is low. On the other hand, kinetic considerations may be involved. Perhaps the biosynthetic sequence would move too slowly if it were not for the extra boost given by the removal of PP_i. Part of the explanation for the complexity may depend on control mechanisms which are only incompletely understood.

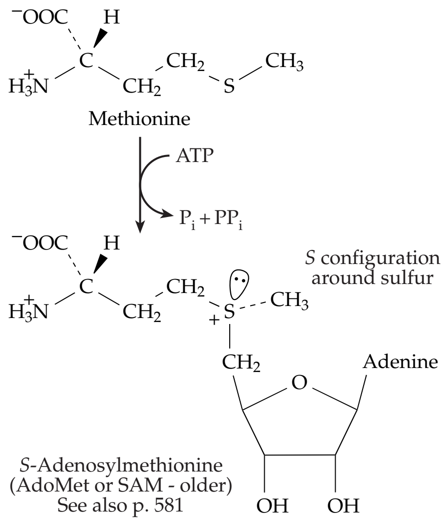
In some metabolic reactions pyrophosphate esters are formed by consecutive transfer of the terminal phospho groups of two ATP molecules onto a hydroxyl



$$\text{Net: } 2\ \text{ATP} + \text{amino acid} + \text{tRNA} \longrightarrow \text{aminoacyl tRNA} + 2\ \text{ADP} + 2\ P_i \qquad (17\text{-}36)$$

group. Such esters often react with elimination of PP_i, e.g., in polymerization of prenyl units (reaction type 6B, Table 10-1; Fig. 22-1). Again, hydrolysis to P_i follows. Thus, *cleavage of pyrophosphate is a second very general method for coupling ATP cleavage to synthetic reactions*.

Although pyrophosphatases are ubiquitous, there are organisms in which PP_i is conserved by the cell and replaces ATP in several glycolytic reactions. These include *Propionibacterium*,[151,152] sulfate-reducing bacteria,[153] the photosynthetic *Rhodospirillum*, and the parasitic *Entamoeba histolytica*.[152,154] In the latter the internal concentration of PP_i is about 0.2 mM. Green plants also accumulate PP_i at concentrations of up to 0.2 mM.[155] Apparently, pyrophosphate is not always hydrolyzed immediately. Another mystery of metabolism is the accumulation of inorganic **polyphosphate** in chains of tens to many hundreds of phospho groups linked, as in pyrophosphate, by phosphoanhydride bonds. These polyphosphates are present in many bacteria, including *E. coli*, and also in fungi, plants, and animals.[156–156b] They constitute a store of energy as well as of phosphate. Various other functions have also been proposed. A polyphosphate kinase transfers a terminal phospho group from polyphosphate chains onto ADP to form ATP. This source of metabolic energy is evidently essential to the ability of *Pseudomonas aeruginosa* to form biofilms.[156a] Both endophosphatases and exophosphatases, of uncertain function, can degrade the chains hydrolytically. An exophosphatase from *E. coli* can completely hydrolyze polyphosphate chains of 1000 units processively without release of intermediates.[156b]

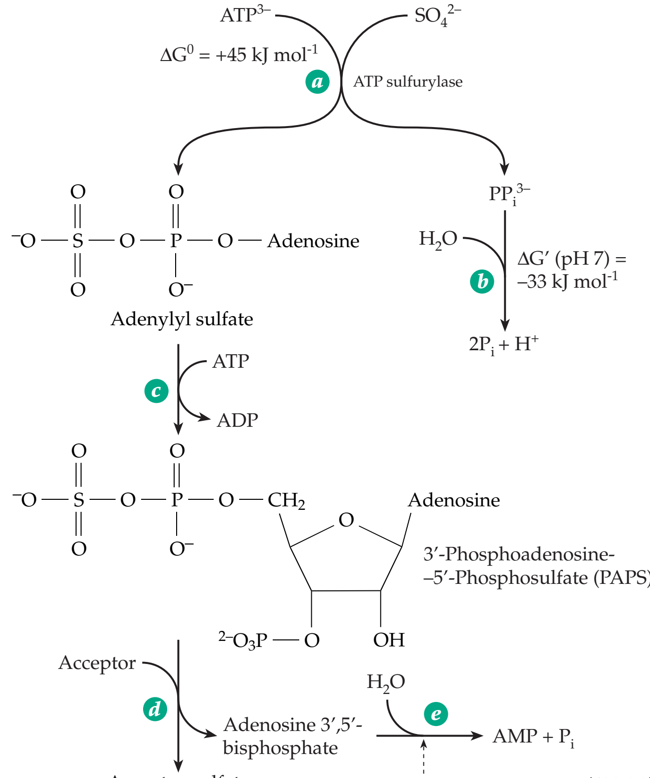


(17-37)

In a few instances group activation is coupled to cleavage of ATP at C-5' presumably with formation of bound tripolyphosphate (PPP_i). The latter is hydrolyzed to P_i and PP_i and ultimately to *three* molecules of P_i. An example is the formation of *S*-adenosylmethionine[157] shown in Eq. 17-37. The reaction is a displacement on the 5'-methylene group of ATP by the sulfur atom of methionine. While the initial product may be enzyme-bound PPP_i, it is P_i and PP_i that are released from the enzyme, the P_i arising from the terminal phosphorus (P_γ) of ATP.[157] The *S*-adenosylmethionine formed has the *S* configuration around the sulfur.[158]

3. Coupling by Phosphorylation and Subsequent Cleavage by a Phosphatase

A third general method for coupling the hydrolysis of ATP to drive a synthetic sequence is to transfer the terminal phospho group from ATP to a hydroxyl group *somewhere* on a substrate. Then, after the substrate has undergone a synthetic reaction, the phosphate is removed by action of a phosphatase. For example, in the activation of sulfate (Eq. 17-38),[159] the overall standard Gibbs energy change for steps *a* (catalyzed by **ATP sulfurylase**[160,161]) and *b* is distinctly positive (+12 kJ mol^{-1}). The equilibrium concentration of adenylyl sulfate formed in this group activation process is extremely low. Nature's solution to this problem is to spend another molecule of ATP to phosphorylate the 3' – OH of adenosine phosphosulfate. As the latter is formed, it is converted to 3'-phosphoadenosine-5'-phosphosulfate (Eq. 17-38, step *c*) by a kinase, which is often part of a bifunctional enzyme that also contains the active site of ATP sulfurylase.[162–163a] Since the equilibrium in this step lies far toward the right, the product accumulates in a substantial concentration (up to 1 mM in cell-free systems)



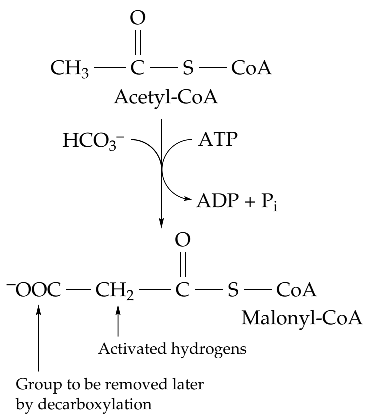
(17-38)

and serves as the active sulfo group donor in formation of sulfate esters. The reaction cycle is completed by two more reactions. In Eq. 17-38, step *d*, the sulfo group is transferred to an acceptor, and in step *e* the extra phosphate group is removed from adenosine 3',5'-bisphosphate by a specific phosphatase. Since the reconversion of AMP to ADP requires expenditure of still a third high-energy linkage of ATP, the overall process makes use of three high-energy phosphate linkages for formation of one sulfate ester.

An analogous use of ATP is found in photosynthetic reduction of carbon dioxide in which ATP phosphorylates ribulose 5-*P* to ribulose bisphosphate and the phosphate groups are removed later by phosphatase action on fructose bisphosphate and sedoheptulose bisphosphate (Section J,2). Phosphatases involved in synthetic pathways usually have a high substrate specificity and are to be distinguished from nonspecific phosphatases which are essentially digestive enzymes (Chapter 12).

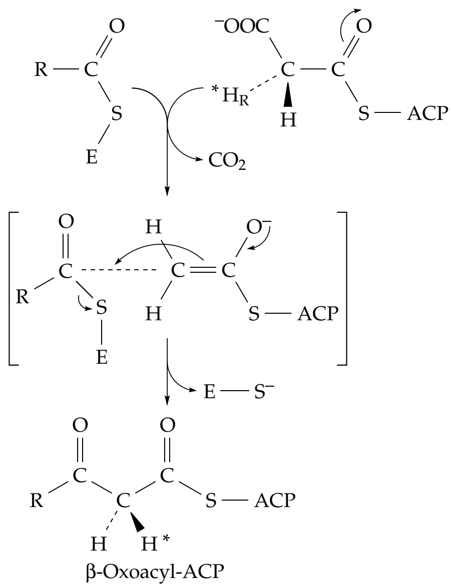
4. Carboxylation and Decarboxylation: Synthesis of Fatty Acids

A fourth way in which cleavage of ATP can be coupled to biosynthesis was recognized in about 1958 when Wakil and coworkers discovered that synthesis of fatty acids in animal cytoplasm is stimulated by carbon dioxide. However, when $^{14}CO_2$ was used in the experiment no radioactivity appeared in the fatty acids formed. Rather, it was found that acetyl-CoA was carboxylated to **malonyl-CoA** in an ATP- and biotin-requiring process (Eq. 17-39; see also Chapter 13). The carboxyl group formed in this reaction is later converted back to CO_2 in a decarboxylation (Fig. 17-12).



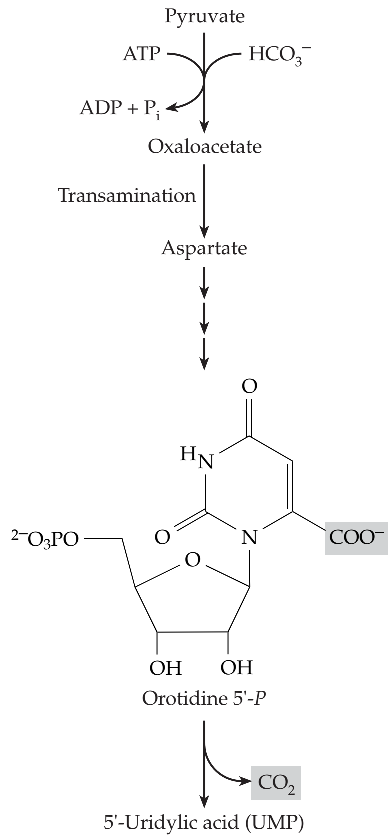
(17-39)

We know now that in most bacteria and green plants both an acetyl group of acetyl-CoA and a malonyl group of malonyl-CoA are transferred (steps *a* and *d* of Fig. 17-12) to the sulfur atoms of the phosphopantetheine groups of a low-molecular-weight **acyl carrier protein** (ACP; Chapter 14). The malonyl group of the malonyl-ACP is then condensed (step *f* of Fig. 17-12) with an acetyl group, which has been transferred from acetyl-ACP onto a thiol group of the enzyme (E in Eq. 17-40). The enolate anion indicated in this equation is generated by decarboxylation of the malonyl-ACP. It is this decarboxylation that drives the reaction to completion and, in effect, links C–C bond formation to the cleavage of the ATP required for the carboxylation step. A related sequence involving multifunctional proteins is used by animals and fungi[164] (Section J,6).



(17-40)

Carboxylation followed by a later decarboxylation is an important pattern in other biosynthetic pathways, too. Sometimes the decarboxylation follows the carboxylation by many steps. For example, pyruvate (or PEP) is converted to uridylic acid (Eq. 17-41; details are shown in Fig. 25-14):



(17-41)

I. Reducing Agents for Biosynthesis

Still another difference between biosynthesis of fatty acids and oxidation (in mammals) is that the former has an absolute requirement for NADPH (Fig. 17-12) while the latter requires NAD^+ and flavoproteins (Fig. 17-1). This fact, together with many other observations, has led to the generalization that *biosynthetic reduction reactions usually require NADPH rather than NADH*. Many measurements have shown that in the cytosol of eukaryotic cells the ratio $[NADPH]/[NADP^+]$ is high, whereas the ratio $[NADH]/[NAD^+]$ is low. Thus, the $NAD^+/NADH$ system is kept highly oxidized, in line with the role of NAD^+ as a principal biochemical oxidant, while the $NADP^+/NADPH$ system is kept reduced.

The use of NADPH in step *g* of Fig. 17-12 ensures that significant amounts of the β-oxoacyl-ACP derivative are reduced to the alcohol. Another difference between β oxidation and biosynthesis is that the alcohol formed in this reduction step in the biosynthetic process

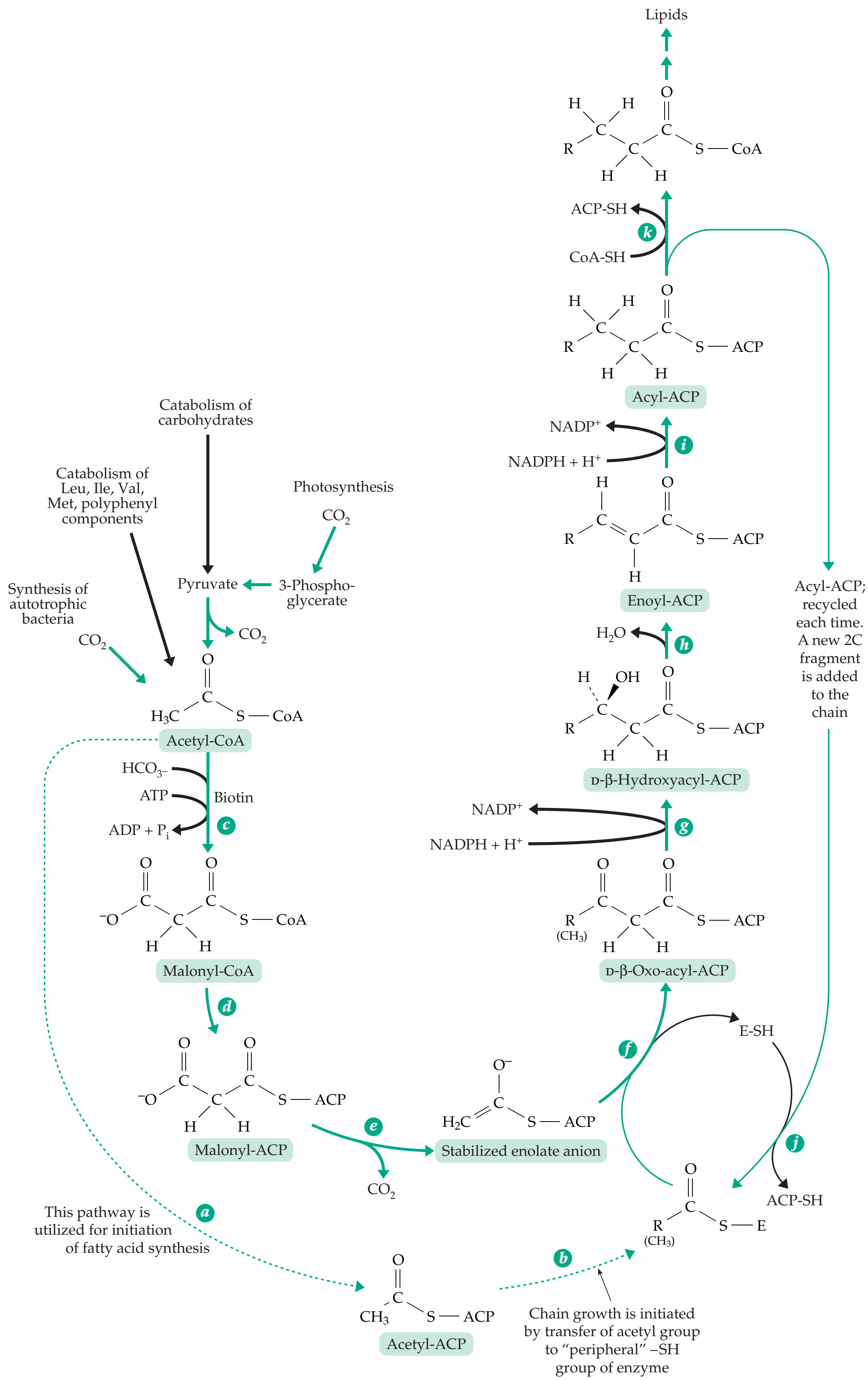


Figure 17-12 The reactions of cytoplasmic biosynthesis of saturated fatty acids. Compare with pathway of β oxidation (Fig. 17-1).

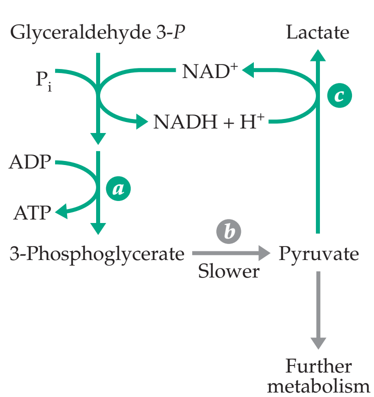
has the D configuration while the corresponding alcohol in β oxidation has the L configuration.

1. Reversing an Oxidative Step with a Strong Reducing Agent

The second reduction step in biosynthesis of fatty acids in the rat liver (step *i*) also required NADPH. The corresponding step in β oxidation utilizes FAD, but NADPH is a stronger reducing agent than $FADH_2$. Therefore, use of a reduced pyridine nucleotide again provides a thermodynamic advantage in pushing the reaction in the biosynthetic direction. Interesting variations have been observed among different species. For example, fatty acid synthesis in the rat requires only NADPH, but the multienzyme complexes from *Mycobacterium phlei*, *Euglena gracilis*, and the yeast *Saccharomyces cerevisiae* all give much better synthesis with a mixture of NADPH and NADH than with NADPH alone.[165] Apparently, NADPH is required in step *g* and NADH in step *i*. This seems reasonable because the equilibrium in step *i* lies far toward the product formation, and NADH at a very low concentration could carry out the reduction.

2. Regulation of the State of Reduction of the NAD and NADP Systems

The ratio $[NAD^+]/[NADH]$ appears to be maintained at a relatively constant value and in equilibrium with a series of different reduced and oxidized substrate pairs. Thus, it was observed that in the cytoplasm of rat liver cells, the dehydrogenations catalyzed by lactate dehydrogenase, *sn*-glycerol 3-phosphate dehydrogenase, and malate dehydrogenase are all at equilibrium with the same ratio of $[NAD^+]/[NADH]$.[166] In one experiment rat livers were removed and frozen in less than 8 s by "freeze-clamping" (Section L,2) and the concentrations of different components of the cytoplasm determined[167]; the ratio $[NAD^+]/[NADH]$ was found to be 634, while the ratio of [lactate]/[pyruvate] was 14.2. From these values an



(17-42)

apparent equilibrium constant for reaction *c* of Eq. 17-42 was calculated as $K_c' = 9.0 \times 10^3$. The known equilibrium constant for the reaction (from *in vitro* experiments) is 8.8×10^3 (Eq. 17-43). In a similar way it was shown that several other dehydrogenation reactions are nearly at equilibrium. This conclusion has been confirmed more recently by NMR observations.[168]

$$K_c'\ (\text{pH } 7,\ 38°\text{C}) = \frac{[\text{lactate}]}{[\text{pyruvate}]} \times \frac{[\text{NAD}^+]}{[\text{NADH}]} = 8.8 \times 10^3 \quad (17\text{-}43)$$

Now consider Eq. 17-42, step *a*, the ADP- and P_i-requiring oxidation of glyceraldehyde 3-phosphate (Fig. 15-6). Experimental measurements indicated that this reaction is also at equilibrium in the cytoplasm. In one series of experiments the measured phosphorylation state ratio $[ATP]/[ADP][P_i]$ was 709, while the ratio [3-phosphoglycerate]/[glyceraldehyde 3-phosphate] was 55.5. The overall equilibrium constant for Eq. 17-42*a* is given by Eq. 17-44. That calculated from known equilibrium constants is 60.

$$K_a'\ (\text{pH } 7,\ 38°\text{C}) = \frac{[\text{ATP}]}{[\text{ADP}][\text{P}_i]} \times \frac{[\text{3-phosphoglycerate}]}{[\text{glyceraldehyde phosphate}]} \times \frac{[\text{NADH}]}{[\text{NAD}^+]} = 709 \times 55.5 \times 1/634 = 62 \quad (17\text{-}44)$$

From these data Krebs and Veech concluded that the oxidation state of the NAD system is determined largely by the phosphorylation state ratio of the adenylate system.[169] If the ATP level is high the equilibrium in Eq. 17-42*a* will be reached at a higher $[NAD^+]/[NADH]$ ratio and lactate may be oxidized to pyruvate to adjust the [lactate]/[pyruvate] ratio.

It is important not to confuse the reactions of Eq. 17-42 as they occur in an aerobic cell with the tightly coupled pair of redox reactions in the homolactate fermentation (Fig. 10-3; Eq. 17-19). The reactions of steps *a* and *c* of Eq. 17-42 are essentially at equilibrium, but the reaction of step *b* may be relatively slow. Furthermore, pyruvate is utilized in many other metabolic pathways and ATP is hydrolyzed and converted to ADP through innumerable processes taking place within the cell. Reduced NAD does not cycle between the two enzymes in a stoichiometric way and the "reducing equivalents" of NADH formed are, in large measure, transferred to the mitochondria. The proper view of the reactions of Eq. 17-42 is that the redox pairs represent a kind of **redox buffer system** that poises the $NAD^+/NADH$ couple at a ratio appropriate for its metabolic function.

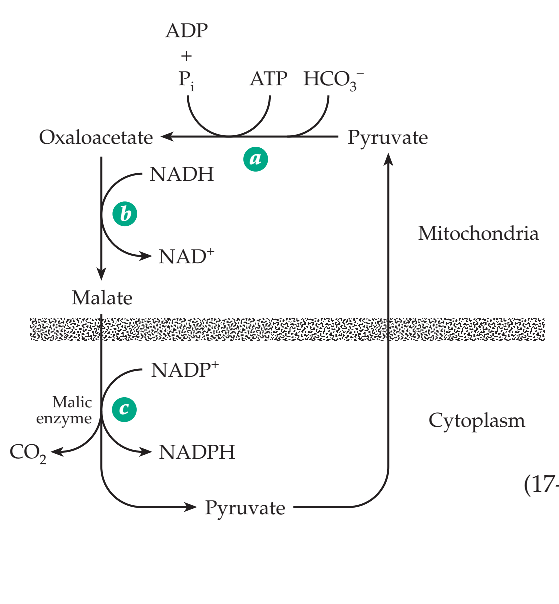
Somewhat surprisingly, within the mitochondria the ratio $[NAD^+]/[NADH]$ is 100 times lower than in the cytoplasm. Even though mitochondria are the site of oxidation of NADH to NAD^+, the intense catabolic activity occurring in the β oxidation pathway and the citric acid cycle ensure extremely rapid production of NADH. Furthermore, the reduction state of NAD is apparently buffered by the low potential of the β-hydroxybutyrate–acetoacetate couple (Chapter 18, Section C,2). Mitochondrial pyridine nucleotides also appear to be at equilibrium with glutamate dehydrogenase.[169]

How is the cytoplasmic $[NADPH]/[NADP^+]$ ratio maintained at a value higher than that of $[NADH]/[NAD^+]$? Part of the answer is from operation of the pentose phosphate pathway (Section E,3). The reactions of Eq. 17-12, if they attained equilibrium, would give a ratio of cytosolic $[NADPH]/[NADP^+] > 2000$ at 0.05 atm CO_2. Compare this with the ratio 1/634 for $[NADH/[NAD^+]$ deduced from the observation on the reactions of Eq. 17-42.

Consider also the following **transhydrogenation** reaction (Eq. 17-45):

$$NADH + NADP^+ \rightarrow NAD^+ + NADPH \quad (17\text{-}45)$$

There are soluble enzymes that catalyze this reaction for which K equals ~ 1. Within mitochondria an energy-linked system (Chapter 18) involving the membrane shifts the equilibrium to favor NADPH. However, within the cytoplasm, the reaction of Eq. 17-45 is driven by coupling ATP cleavage to the transhydrogenation via carboxylation followed by eventual decarboxylation. One cycle that accomplishes this is given in Eq. 17-46. The first step (step *a*) is ATP-dependent carboxylation of pyruvate to oxaloacetate, a reaction that occurs only within mitochondria (Eq. 14-3). Oxaloacetate can be reduced by malate dehydrogenase using NADH (Eq. 17-46, step *b*), and the resulting malate can be exported from the mitochondria. In the cytoplasm the malate is oxidized to pyruvate, with decarboxylation, by the



(17-46)

malic enzyme (Eq. 13-45). The malic enzyme (Eq. 17-46, step *c*) is specific for $NADP^+$, is very active, and also appears to operate at or near equilibrium within the cytoplasm. On this basis, using known equilibrium constants, it is easy to show that the ratio $[NADPH]/[NADP^+]$ will be ~ 10^5 times higher at equilibrium than the ratio $[NADH]/[NAD^+]$.[169,170]

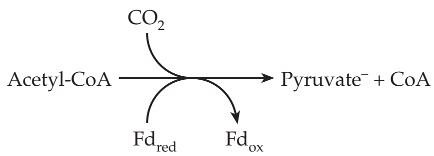
Since NADPH is continuously used in biosynthetic reactions, and is thereby reconverted to $NADP^+$, the cycle of Eq. 17-46 must operate continuously. As in Eq. 17-42, a true equilibrium does not exist but steps *b* and *c* are both essentially at equilibrium. These equilibria, together with those of Eq. 17-42 for the NAD system, ensure the correct redox potential of both pyridine nucleotide coenzymes in the cytoplasm.

Malate is not the only form in which C_4 compounds are exported from mitochondria. Much oxaloacetate is combined with acetyl-CoA to form citrate; the latter leaves the mitochondria and is cleaved by the ATP-dependent citrate-cleaving enzymes (Eq. 13-39). This, in effect, exports both acetyl-CoA (needed for lipid synthesis) and oxaloacetate which is reduced to malate within the cytoplasm. Alternatively, oxaloacetate may be transaminated to aspartate. The aspartate, after leaving the mitochondria, may be converted in another transamination reaction back to oxaloacetate. All of these are part of the nonequilibrium process by which C_4 compounds diffuse out of the mitochondria before completing the reaction sequence of Eq. 17-46 and entering into other metabolic processes. Note that the reaction of Eq. 17-46 leads to the *export* of reducing equivalents from mitochondria, the opposite of the process catalyzed by the malate–aspartate shuttle which is discussed in Chapter 18 (Fig. 18-18). The two processes are presumably active under different conditions.

While the difference in the redox potential of the two pyridine nucleotide systems is clear-cut in mammalian tissues, in *E. coli* the apparent potentials of the two systems are more nearly the same.[171]

3. Reduced Ferredoxin in Reductive Biosynthesis

Both the NAD^+ and $NADP^+$ systems have standard electrode potentials $E°'$ (pH 7) of −0.32 V. However, because of the differences in concentration ratios, the NAD^+ system operates at a less negative potential (−0.24 V) and the $NADP^+$ system at a more negative potential (−0.38 V) within the cytoplasm of eukaryotes. In green plants and in many bacteria a still more powerful reducing agent is available in the form of reduced ferredoxin. The value of $E°'$ (pH 7) for clostridial ferredoxin is −0.41 V, corresponding to a Gibbs energy change for the two-electron reduction of a substrate ~18 kJ mol^{-1} more negative than the corresponding value of $\Delta G'$ for reduction by NADPH. Using reduced ferredoxin (Fd) some photosynthetic bacteria and anaerobic bacteria are able to carry out reductions that are virtually impossible with the pyridine nucleotide system. For example, pyruvate and 2-oxoglutarate can be formed from acetyl-CoA (Eq. 15-35) and succinyl-CoA, respectively (Eq. 17-47).[172–173a] In our bodies the reaction of Eq. 17-47, with NAD^+ as the oxidant, goes only in the opposite direction and is essentially irreversible.



$\Delta G'$ (pH 7) = +17 kJ mol^{-1} (17-47)

J. Constructing the Monomer Units

Now let us consider the synthesis of the monomeric units from which biopolymers are made. How can simple one-carbon compounds such as CO_2 and formic acid be incorporated into complex carbon compounds? How can carbon chains grow in length or be shortened? How are branched chains and rings formed?

1. Carbonyl Groups in Chain Formation and Cleavage

Except for some vitamin B_{12}-dependent reactions, the cleavage or formation of carbon–carbon bonds usually depends upon the participation of carbonyl groups. For this reason, carbonyl groups have a central mechanistic role in biosynthesis. The activation of hydrogen atoms β to carbonyl groups permits β condensations to occur during biosynthesis. Aldol or Claisen condensations require the participation of two carbonyl compounds. Carbonyl compounds are also essential to thiamin diphosphate-dependent condensations and the aldehyde pyridoxal phosphate is needed for most C–C bond cleavage or formation within amino acids.

Because of the importance of carbonyl groups to the mechanism of condensation reactions, much of the assembly of either straight-chain or branched-carbon skeletons takes place between compounds in which the average oxidation state of the carbon atoms is similar to that in carbohydrates (or in formaldehyde, H_2CO). The diversity of chemical reactions possible with compounds at this state of oxidation is a maximum, a fact that may explain why carbohydrates and closely related substances are major biosynthetic precursors and why the average state of oxidation of the carbon in most living things is similar to that in carbohydrates.[174] This fact may also be related to the presumed occurrence of formaldehyde as a principal component of the earth's atmosphere in the past and to the ability of formaldehyde to condense to form carbohydrates.

In Fig. 17-13 several biochemicals have been arranged according to the oxidation state of carbon. Most of the important biosynthetic intermediates lie within ± 2 electrons per carbon atom of the oxidation state of carbohydrates. As the chain length grows, they tend to fall even closer. It is extremely difficult to move through enzymatic processes between 2C, 3C, and 4C compounds (i.e., vertically in Fig. 17-13) except at the oxidation level of carbohydrates or somewhat to its right, at a slightly higher oxidation level. On the other hand, it is often possible to move horizontally with ease using oxidation–reduction reactions. Thus, fatty acids are assembled from acetate units, which lie at the same oxidation state as carbohydrates and, after assembly, are reduced.

Among compounds of the same overall oxidation state, e.g., acetic acid and sugars, the oxidation states of individual carbon atoms can be quite different. Thus, in a sugar every carbon atom can be regarded as immediately derived from formaldehyde, but in acetic acid one end has been oxidized to a carboxyl group and the other has been reduced to a methyl group. Such internal oxidation–reduction reactions play an important role in the chemical manipulations necessary to assemble the carbon skeletons needed by a cell. Decarboxylation is a feature of many biosynthetic routes. Referring again to Fig. 17-13, notice that many of the biosynthetic intermediates such as pyruvate, oxoglutarate, and oxaloacetate are more oxidized than the carbohydrate level. However, their decarboxylation products, which become incorporated into the compounds being synthesized, are closer to the oxidation level of carbohydrates.

2. Starting with CO_2

There are three known pathways by which autotrophic organisms can use CO_2 to synthesize triose phosphates or 3-phosphoglycerate, three-carbon compounds from which all other biochemical substances can be formed.[175–177] The first of these is the **reductive tricarboxylic cycle.** This is a reversal of the oxidative citric acid cycle in which reduced ferredoxin is used as a reductant in the reaction of Eq. 17-47 to incorporate CO_2 into pyruvate. Succinyl-CoA can react with CO_2 in the same type of reaction to form 2-oxoglutarate, accomplishing the reversal of the only irreversible step in the citric acid cycle. Using these reactions photosynthetic bacteria and some anaerobes that can generate a high ratio of reduced to oxidized ferredoxin carry out the reductive tricarboxylic acid cycle. Together with

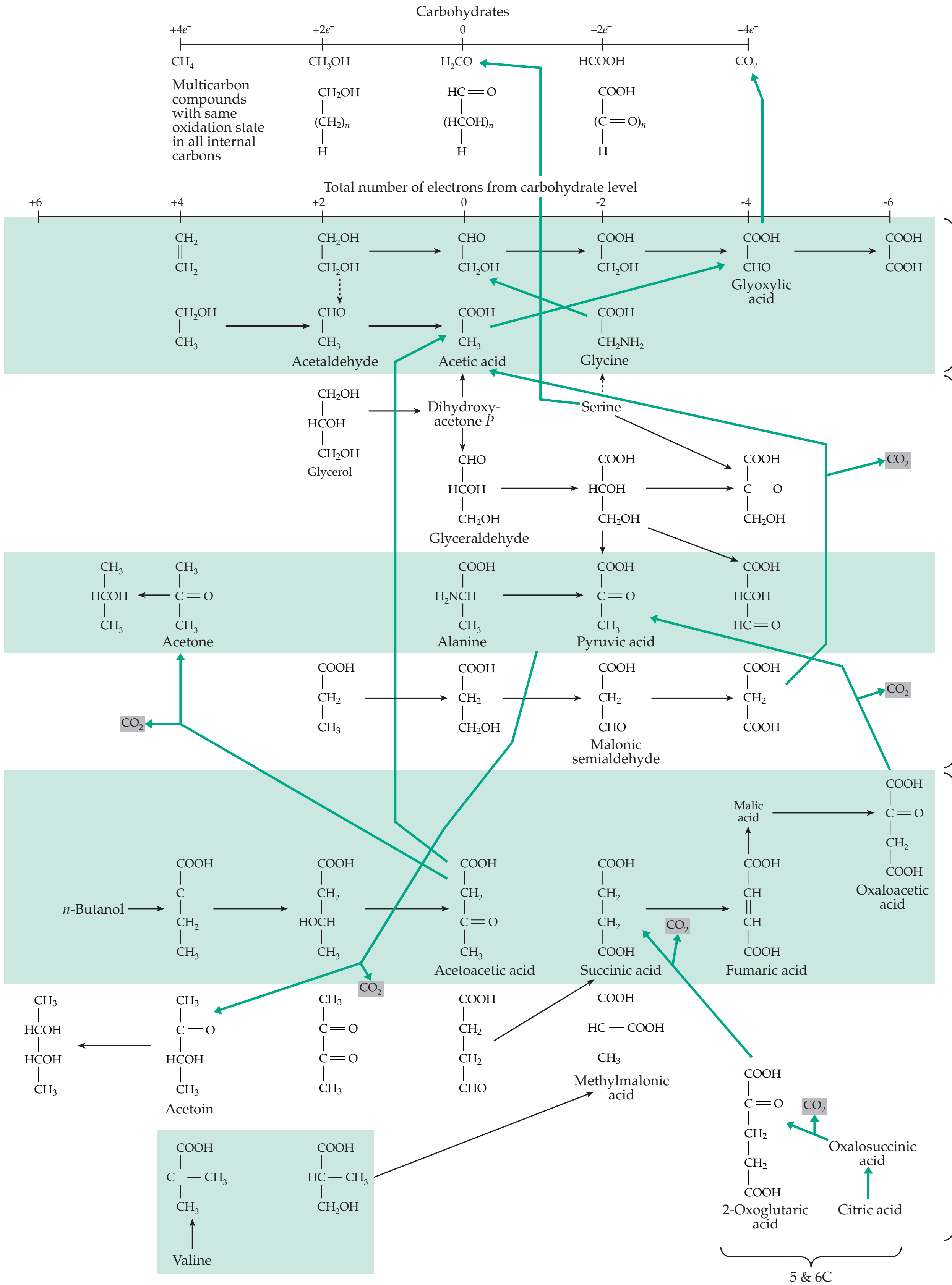


Figure 17-13 Some biochemical compounds arranged in order of average oxidation state of the carbon atoms and by carbon-chain lengths. Black horizontal arrows mark some biological interconversions among compounds with the same chain length, while green lines show changes in chain length and are often accompanied by decarboxylation.

Eq. 17-47, the cycle provides for the complete synthesis of pyruvate from CO_2.[178,179]

A quantitatively much more important pathway of CO_2 fixation is the **reductive pentose phosphate pathway** (ribulose bisphosphate cycle or **Calvin–Benson cycle**; Fig. 17-14). This sequence of reactions, which takes place in the chloroplasts of green plants and also in many chemiautotrophic bacteria, is essentially a way of reversing the oxidative pentose phosphate pathway (Fig. 17-8). The latter accomplishes the complete oxidation of glucose or of glucose 1-phosphate by $NADP^+$ (Eq. 17-48):

$$\begin{aligned}&\text{Glucose 1-}P^{2-} + ATP^{4-} + 8\,H_2O + 12\,NADP^+ \rightarrow \\ &\quad 6\,CO_2 + 12\,NADPH + ADP^{3-} + 2HPO_4^{2-} + 13\,H^+ \\ &\quad \Delta G' \text{ (pH 7)} = -299 \text{ kJ mol}^{-1}\end{aligned} \tag{17-48}$$

It would be almost impossible for a green plant to fix CO_2 using photochemically generated NADPH by an exact reversal of Eq. 17-48 because of the high positive Gibbs energy change. To solve this thermodynamic problem the reductive pentose phosphate pathway has been modified in a way that couples ATP cleavage to the synthesis.

The **reductive carboxylation** system is shown within the green shaded box of Fig. 17-14. Ribulose 5-phosphate is the starting compound and in the first step one molecule of ATP is expended to form **ribulose 1,5-bisphosphate**. The latter is carboxylated and cleaved to two molecules of 3-phosphoglycerate. This reaction was discussed in Chapter 13. The reductive step (step *c*) of the system employs NADPH together with ATP. Except for the use of the NADP system instead of the NAD system, it is exactly the reverse of the glyceraldehyde phosphate dehydrogenase reaction of glycolysis. Looking at the first three steps of Fig. 17-14 it is clear that in the reductive pentose phosphate pathway three molecules of ATP are utilized for each CO_2 incorporated. On the other hand, in the oxidative direction *no* ATP is generated by the operation of the pentose phosphate pathway.

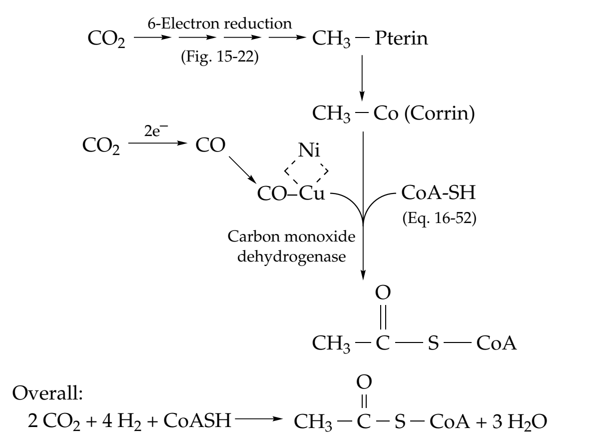
The reactions enclosed within the shaded box of Fig. 17-14 do not give the whole story about the coupling mechanism. A phospho group was transferred from ATP in step *a* and to complete the hydrolysis it must be removed in some future step. This is indicated in a general way in Fig. 17-14 by the reaction steps *d*, *e*, and *f*. Step *f* represents the action of specific phosphatases that remove phospho groups from the seven-carbon sedoheptulose bisphosphate and from fructose bisphosphate. In either case the resulting ketose monophosphate reacts with an aldose (via transketolase, step *g*) to regenerate ribulose 5-phosphate, the CO_2 acceptor. The overall reductive pentose phosphate cycle (Fig. 17-14B) is easy to understand as a reversal of the oxidative pentose phosphate pathway in which the oxidative decarboxylation system of Eq. 17-12 is replaced by the reductive carboxylation system of Fig. 17-14A. The scheme as written in Fig. 17-14B shows the incorporation of three molecules of CO_2. The reductive carboxylation system operates three times with a net production of one molecule of triose phosphate. As with other biosynthetic cycles, any amount of any of the intermediate metabolites may be withdrawn into various biosynthetic pathways without disruption of the flow through the cycle.

The overall reaction of carbon dioxide reduction in the Calvin–Benson cycle (Fig. 17-14) becomes

$$\begin{aligned}&6\,CO_2 + 12\,NADPH + 18\,ATP^{4-} + 11\,H_2O \rightarrow \\ &\quad \text{glucose-1-}P^{2-} + 12\,NADP^+ + 18\,ADP^{3-} \\ &\qquad 17\,HPO_4^{2-} + 6\,H^+\end{aligned} \tag{17-49}$$

The Gibbs energy change $\Delta G'$ (pH 7) is now -357 kJ mol^{-1} instead of the $+299$ kJ mol^{-1} required to reverse the reaction of Eq. 17-48.

The third pathway for reduction of CO_2 to acetyl-CoA is utilized by acetogenic bacteria, by methanogens, and probably by sulfate-reducing bacteria.[179–181] This **acetyl-CoA pathway** (or **Wood–Ljungdahl pathway**) involves reduction by H_2 of one of the two molecules of CO_2 to the methyl group of methyltetrahydromethanopterin in methanogens and of methyltetra-hydrofolate in acetogens. The pathway utilized by methanogens is illustrated in Fig. 15-22.[182–184] A similar process utilizing H_2 as the reductant is employed by acetogens.[179,185–188a] In both cases a methyl corrinoid is formed and its methyl group is condensed with a molecule of carbon monoxide bound to a copper ion in a Ni–Cu cluster.[189a,b] The resulting acetyl group is transferred to a molecule of coenzyme A as illustrated in Eq. 16-52.[189] The bound CO is formed by reduction of CO_2, again using H_2 as the reductant.[190] The overall reaction for acetyl-CoA synthesis is given by Eq. 17-50. Conversion of acetyl-CoA to pyruvate via Eq. 17-47 leads into the glucogenic pathway.



(17-50)

An alternative pathway by which some acetogenic bacteria form acetate is via reversal of the glycine decarboxylase reaction of Fig. 15-20. Methylene-THF is formed by reduction of CO_2, and together with NH_3 and CO_2 a lipoamide group of the enzyme and PLP forms glycine. The latter reacts with a second methylene-THF to form serine, which can be deaminated to pyruvate and assimilated. Methanogens may use similar pathways but ones that involve methanopterin (Fig. 15-17).[191]

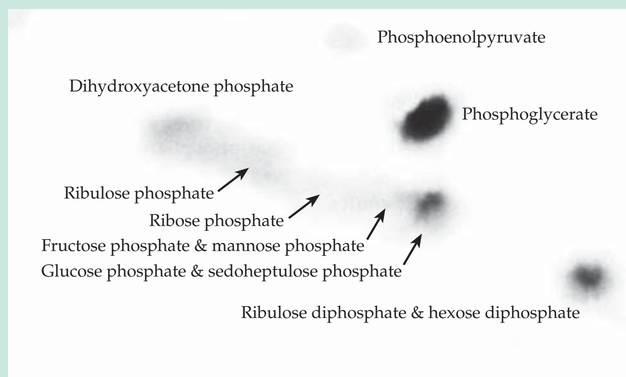
3. Biosynthesis from Other Single-Carbon Compounds

Various bacteria and fungi are able to subsist on such one-carbon compounds as methane, methanol, methylamine, formaldehyde, and formate.[192–197] Energy is obtained by oxidation to CO_2. **Methylotrophic bacteria** initiate oxidation of methane by hydroxylation (Chapter 18) and dehydrogenate the resulting methanol or exogenous methanol using the PPQ cofactor (Eq. 15-51).[198] Further dehydrogenation to formate and of formate to CO_2 via formate dehydrogenase (Eq. 16-63) completes the process. Some methylotrophic bacteria incorporate CO_2 for biosynthetic purposes via the ribulose bisphosphate (Calvin–Benson) cycle but many use pathways that begin with formaldehyde (or methylene-THF). Others employ variations of the reductive pentose phosphate pathway to convert formaldehyde to triose phosphate. In one of these, the **ribulose monophosphate cycle** or Quayle cycle,[192,193] ribulose 5-*P* undergoes an aldol condensation with formaldehyde to give a 3-oxo-hexulose 6-phosphate (Eq. 17-51, step *a*). The latter is isomerized to fructose 6-*P* (Eq. 17-51, step *b*). If this equation is applied to the

BOX 17-E ^{14}C AND THE CALVIN–BENSON CYCLE

The chemical nature of photosynthesis had intrigued chemists for decades but little was learned about the details until radioactive ^{14}C became available. Discovered in 1940 by Ruben and Kamen, the isotope was available in quantity by 1946 as a product of nuclear reactors. Initial studies of photosynthesis had been conducted by Ruben and Kamen using ^{11}C but ^{14}C made rapid progress possible. In 1946 Melvin Calvin and Andrew A. Benson began their studies that elucidated the mechanism of incorporation of CO_2 into organic materials.

A key development was two-dimensional paper chromatography with radioautography (Box 3-C). A suspension of the alga *Chlorella* (Fig. 1-11) was allowed to photosynthesize in air. At a certain time, a portion of $H^{14}CO_3$ was injected into the system, and after a few seconds of photosynthesis with ^{14}C present the suspension of algae was run into hot methanol to denature proteins and to stop the reaction. The soluble materials extracted from the algal cells were concentrated and chromatographed; radioautographs were then prepared. It was found that after 10 s of photosynthesis in the presence of $^{14}CO_2$, the algae contained a dozen or more ^{14}C labeled compounds. These included malic acid, aspartic acid, phosphoenolpyruvate, alanine, triose phosphates, and other sugar phosphates and diphosphates. However, *during the first five seconds a single compound, 3-phosphoglycerate, contained most of the radioactivity*.[a,b] This finding suggested that a two-carbon regenerating substrate might be carboxylated by $^{14}CO_2$ to phosphoglycerate. Search for this two-carbon compound was unsuccessful, but Benson, in Calvin's laboratory, soon identified ribulose bisphosphate,[c] which kinetic studies proved to be the true regenerating substrate.[c,d] Its carboxylation and cleavage[e] represent the first step in what has come to be known as the Calvin–Benson cycle (Fig. 17-14).[f]



Chromatogram of extract of the alga *Scenedesmus* after photosynthesis in the presence of $^{14}CO_2$ for 10 s. Courtesy of J. A. Bassham. The origin of the chromatogram is at the lower right corner.

[a] Benson, A. A., Bassham, J. A., Calvin, M., Goodale, T. C., Haas, V. A., and Stepka, W. (1950) *J. Am. Chem. Soc.* **72**, 1710–1718
[b] Benson, A. A. (1951) *J. Am. Chem. Soc.* **73**, 2971–2972
[c] Benson, A. A., Kawaguchi, S., Hayes, P., and Calvin, M. (1952) *J. Am. Chem. Soc.* **74**, 4477–4482
[d] Bassham, J. A., Benson, A. A., Kay, L. D., Harris, A. Z., Wilson, A. T., and Calvin, M. (1954) *J. Am. Chem. Soc.* **76**, 1760–1770
[e] Calvin, M., and Bassham, J. A. (1962) *The Photosynthesis of Carbon Compounds*, Benjamin, New York
[f] Fuller, R. C. (1999) *Photosynth Res.* **62**, 1–29

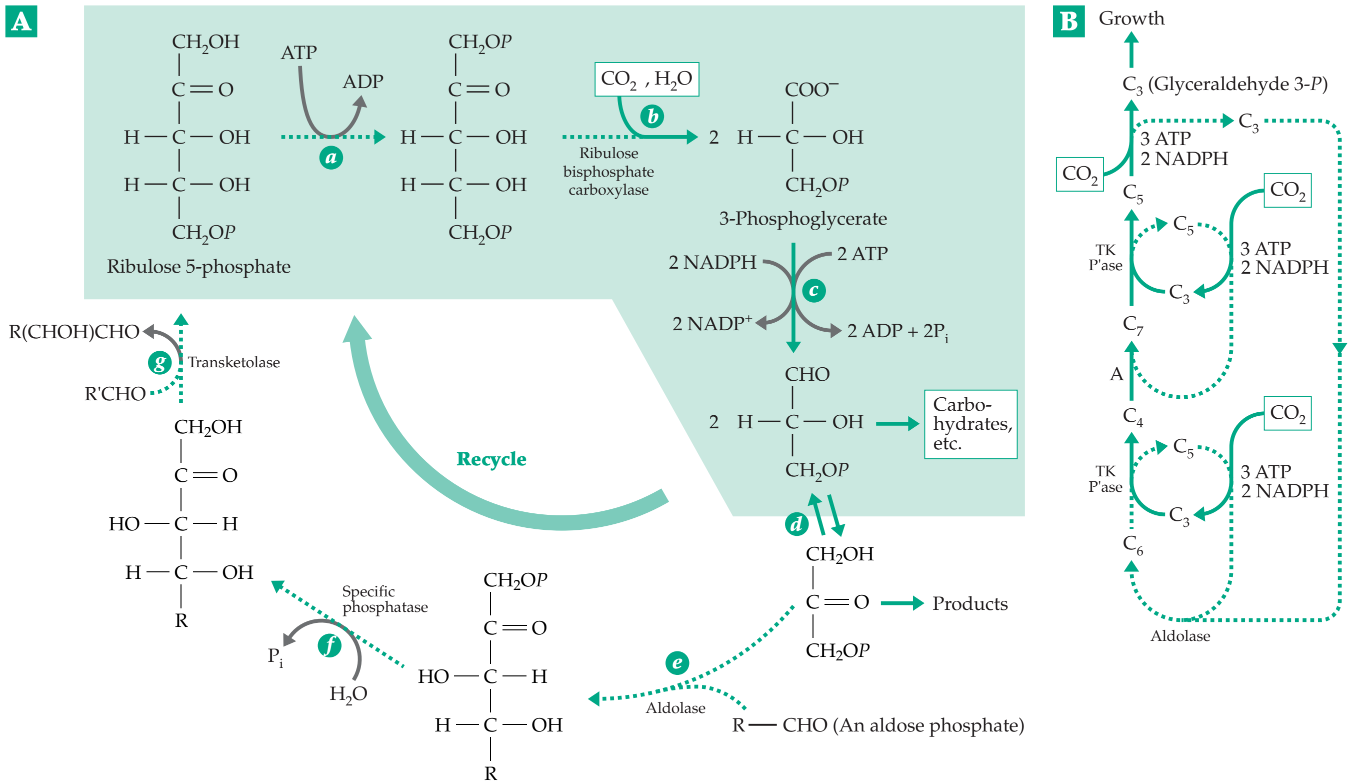


Figure 17-14 (A) The reductive carboxylation system used in reductive pentose phosphate pathway (Calvin–Benson cycle). The essential reactions of this system are enclosed within the dashed box. Typical subsequent reactions follow. The phosphatase action completes the phosphorylation–dephosphorylation cycle. (B) The reductive pentose phosphate cycle arranged to show the combining of three CO_2 molecules to form one molecule of triose phosphate. Abbreviations are RCS, reductive carboxylation system (from above); A, aldolase, Pase, specific phosphatase; and TK, transketolase.

three C_5 sugars three molecules of fructose 6-phosphate will be formed. One of these can be phosphorylated by ATP to fructose 1,6-bisphosphate, which can be cleaved by aldolase. One of the resulting triose phosphates can then be removed for biosynthesis and the second, together with the other two molecules of fructose 6-*P*, can be recycled through the sugar rearrangement sequence of Fig. 17-8B to regenerate the three ribulose 5-*P* molecules that serve as the regenerating substrate.

In bacteria, which lack formate dehydrogenase, formaldehyde can be oxidized to CO_2 to provide energy beginning with the reactions of Eq. 17-51. The resulting fructose 6-*P* is isomerized to glucose 6-*P*, which is then dehydrogenated via Eq. 17-12 to form CO_2 and the regenerating substrate ribulose 5-phosphate.

A number of pseudomonads and other bacteria convert C_1 compounds to acetate via tetrahydrofolic acid-bound intermediates and CO_2 using the **serine pathway**[179,192,193] shown in Fig. 17-15. This is a cyclic process for converting one molecule of formaldehyde (bound to tetrahydrofolate) plus one of CO_2 into acetate. The regenerating substrate is **glyoxylate**. Before condensation with the "active formaldehyde" of methylene THF, the glyoxylate undergoes transamination to glycine (Fig. 17-15, step *a*). The glycine plus formaldehyde forms serine (step *b*), which is then transaminated to hydroxypyruvate, again using step *a*. Glyoxylate plus formaldehyde could have been joined in a thiamin-dependent condensation. However, as in the γ-aminobutyrate shunt (Fig. 17-5), the coupled transamination step of Fig 17-15 permits use of PLP-dependent C–C bond formation.

Conversion of hydroxypyruvate to PEP (Fig. 17-15) involves reduction by NADH and phosphorylation by ATP to form 3-phosphoglycerate, which is converted to PEP as in glycolysis. The conversion of malate to acetate and glyoxylate via malyl-CoA and isocitrate lyase (Eq. 13-40) forms the product acetate and regenerates glyoxylate. As with other metabolic cycles, various intermediates, such as PEP, can be withdrawn for biosynthesis. However, there must be an independent route of synthesis of the regenerating substrate glyoxylate. One way in which this can be accomplished is to form glycine via the reversal of the glycine decarboxylase pathway as is indicated by the shaded green lines in Fig. 17-15.

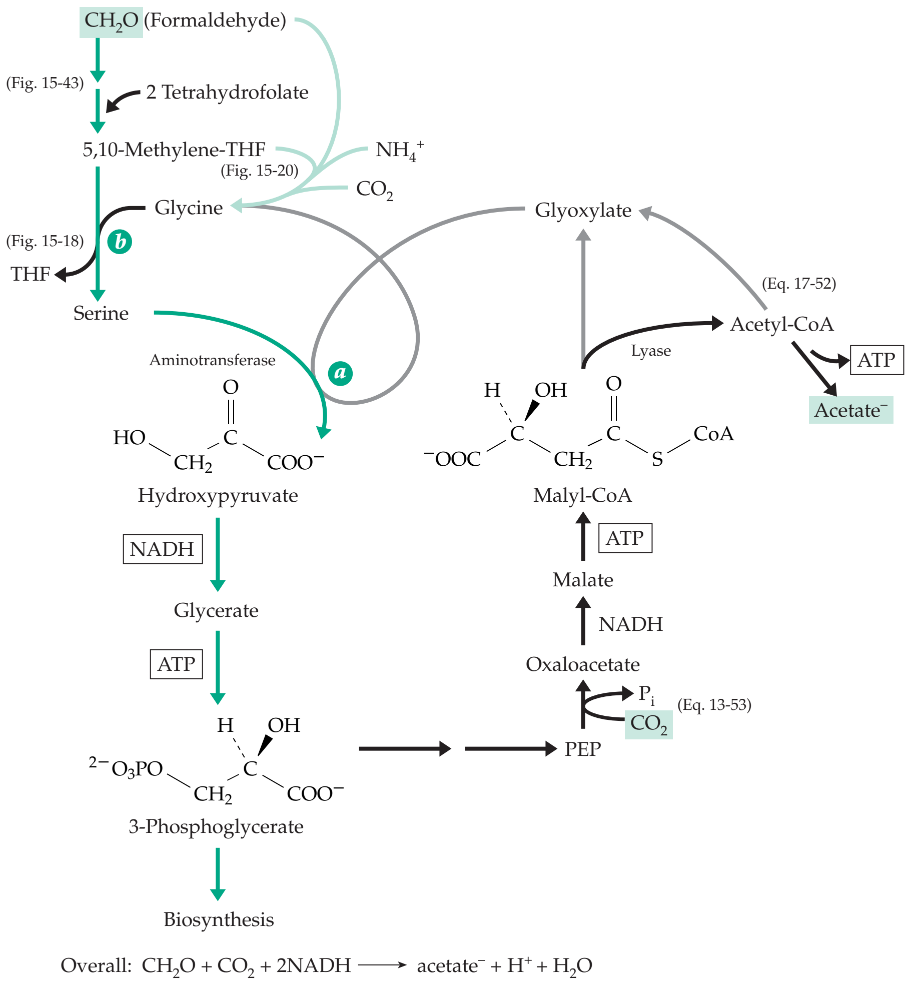
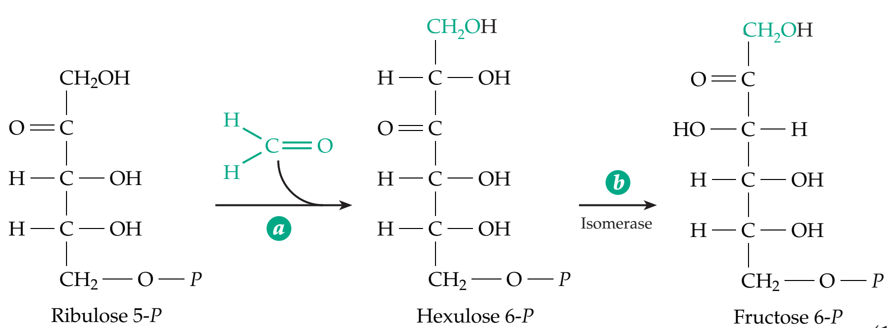


Figure 17-15 One of the serine pathways for assimilation of one-carbon compounds.



(17-51)

4. The Glyoxylate Pathways

The reductive carboxylation of acetyl-CoA to pyruvate (Eq. 17-47) occurs only in a few types of bacteria. For most species, from microorganisms to animals, the oxidative decarboxylation of pyruvate to acetyl-CoA is irreversible. This fact has many important consequences. For example, carbohydrate is readily converted to fat; because of the irreversibility of this process, excess calories lead to the deposition of fats. However, in animals fat cannot be used to generate most of the biosynthetic intermediates needed for formation of carbohydrates and proteins because those intermediates originate largely from C_3 units.

This limitation on the conversion of C_2 acetyl units to C_3 metabolites is overcome in many organisms by the **glyoxylate cycle** (Fig. 17-16), which converts *two* acetyl units into one C_4 unit. The cycle provides a way for organisms, such as *E. coli*,[111,199] *Saccharomyces*,[200] *Tetrahymena*, and the nematode *Caenorhabditis*,[201] to subsist on acetate as a sole or major carbon source. It is especially prominent in plants that store large amounts of fat in their seeds (**oil seeds**). In the germinating oil seed the glyoxylate cycle allows fat to be converted rapidly to sucrose, cellulose, and other carbohydrates needed for growth.

A key enzyme in the glyoxylate cycle is **isocitrate lyase**, which cleaves isocitrate (Eq. 13-40) to succinate and glyoxylate. The latter is condensed with a second acetyl group by the action of **malate synthase** (Eq. 13-38). The L-malate formed in this reaction is dehydrogenated to the regenerating substrate oxaloacetate. Some of the reaction steps are those of the citric acid cycle and it appears that in bacteria there is no spatial separation of the citric acid cycle and glyoxylate pathway. However, in plants the enzymes of the glyoxylate cycle are present in specialized peroxisomes known as **glyoxysomes**.[60] The glyoxysomes also contain the enzymes for β oxidation of fatty acids, allowing for efficient conversion of fatty acids to **succinate**. This compound is exported from the glyoxysomes and enters the mitochondria where it undergoes β oxidation to oxaloacetate. The latter can be converted by PEP carboxylase (Eq. 13-53) or by PEP carboxykinase (Eq. 13-46) to PEP.

An **acetyl-CoA-glyoxylate** cycle, which catalyzes oxidation of acetyl groups to glyoxylate, can also be constructed from isocitrate lyase and citric acid cycle

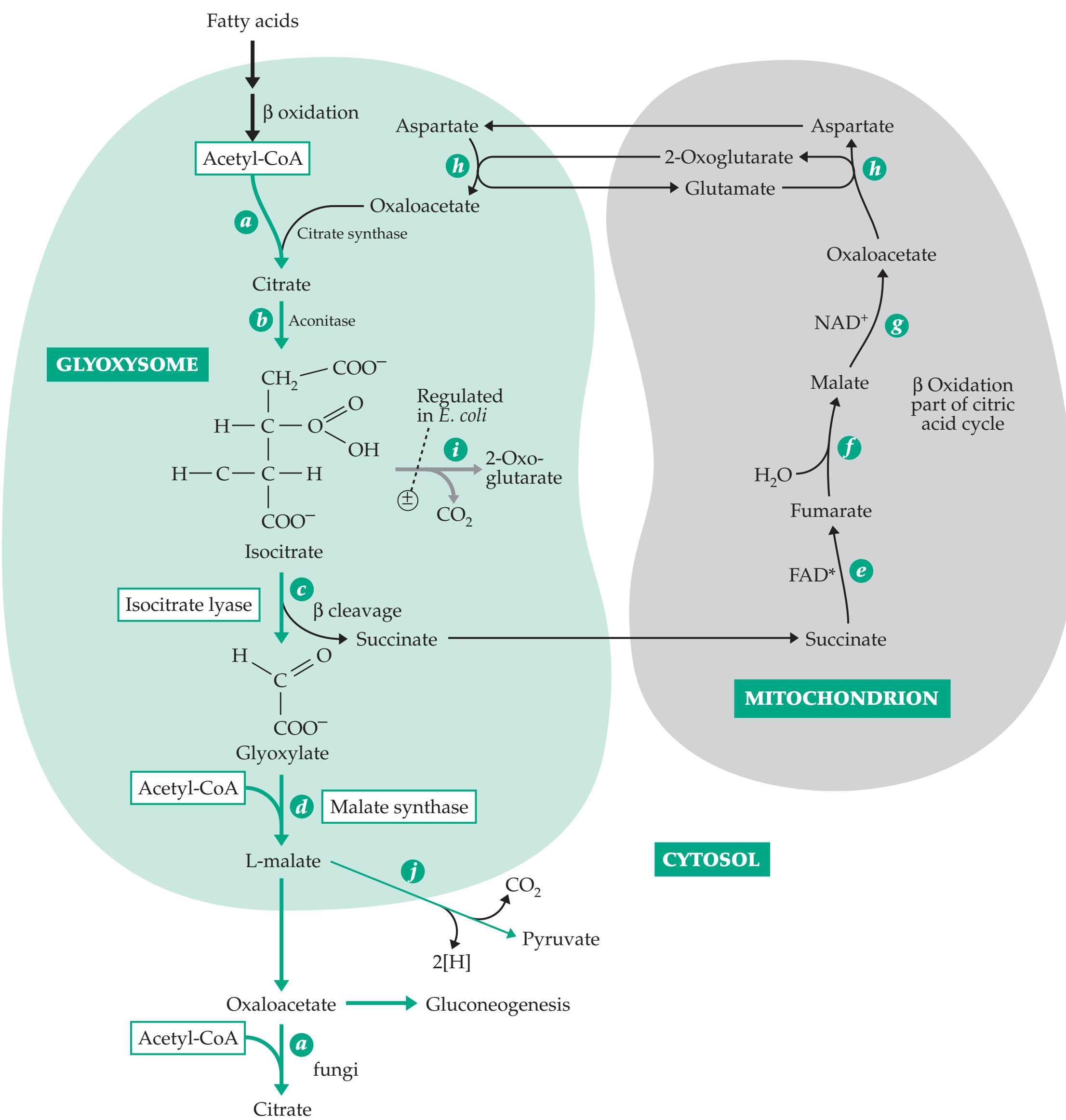
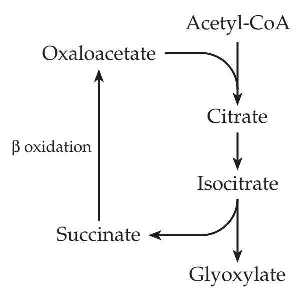


Figure 17-16 The glyoxylate pathway. The green line traces the pathway of labeled carbon from fatty acids or acetyl-CoA into malate and other products.

enzymes. Glyoxylate is taken out of the cycle as the product and succinate is recycled (Eq. 17-52). The independent pathway for synthesis of the regenerating substrate oxaloacetate is condensation of glyoxylate with acetyl-CoA (malate synthetase) to form malate and oxidation of the latter to oxaloacetate as in the main cycle of Fig. 17-16.



(17-52)

5. Biosynthesis of Glucose from Three-Carbon Compounds

Now let us consider the further conversion of PEP and of the triose phosphates to **glucose 1-phosphate**, the key intermediate in biosynthesis of other sugars and polysaccharides. The conversion of PEP to glucose 1-*P* represents a reversal of part of the glycolysis sequence. It is convenient to discuss this along with **gluconeogenesis**, the reversal of the complete glycolysis sequence from lactic acid. This is an essential part of the Cori cycle (Section F) in our own bodies, and the same process may be used to convert pyruvate derived from deamination of alanine or serine (Chapter 24) into carbohydrates.

Just as with the pentose phosphate cycle, an exact reversal of the glycolysis sequence (Eq. 17-53) is precluded on thermodynamic grounds. Even at very high values of the phosphorylation state ratio R_p, the reaction:

$$2\ \text{Lactate}^- + 3\ \text{ATP}^{4-} + 2\ H_2O \rightarrow C_6H_{10}O_5\ (\text{glycogen}) + 3\ \text{ADP}^{3-} + 3\ HPO_4^{2-} + H^+$$
$$\Delta G'\ (\text{pH 7}) = +107\ \text{kJ per glycosyl unit} \quad (17\text{-}53)$$

would be unlikely to go to completion. The actual pathways used for gluconeogenesis (Fig. 17-17, green lines) differ from those of glycolysis (black lines) in three significant ways. First, while glycogen breakdown is initiated by the reaction with inorganic phosphate catalyzed by phosphorylase (Fig. 17-17, step *a*), the biosynthetic sequence from glucose 1-*P*, via uridine diphosphate glucose (Fig. 17-17, step *b*; see also Eq. 17-56), is coupled to cleavage of ATP. Second, in the catabolic process (glycolysis) fructose 6-*P* is converted to fructose 1,6-P_2 through the action of a kinase (Fig. 17-17, step *c*), which is then cleaved by aldolase. The resulting triose phosphate is degraded to PEP. In glucogenesis a phosphatase is used to form fructose *P* from fructose P_2 (Fig. 17-17, step *d*). Third, during gly-colysis PEP is converted to pyruvate by a kinase with generation of ATP (Fig. 17-17, step *e*). During glucogenesis pyruvate is converted to PEP indirectly via oxaloacetate (Fig. 17-17, steps *f* and *g*) using pyruvate carboxylase (Eq. 14-3) and PEP carboxykinase (Eq. 13-46). This is another example of the coupling of ATP cleavage through a carboxylation–decarboxylation sequence. The net effect is to use two molecules of ATP (actually one ATP and one GTP) rather than *one* to convert pyruvate to PEP.

The overall reaction for reversal of glycolysis to form glycogen (Eq. 17-54) now has a comfortably negative standard Gibbs energy change as a result of coupling the cleavage of 7 ATP to the reaction.

$$2\ \text{Lactate}^- + 7\ \text{ATP}^{4-} + 6\ H_2O \rightarrow \text{glycogen} + 7\ \text{ADP}^{3-} + 7\ HPO_4^{2-} + 5\ H^+$$

$$\Delta G'\ (\text{pH 7}) = -31\ \text{kJ mol}^{-1}\ \text{per glycosyl unit} \quad (17\text{-}54)$$

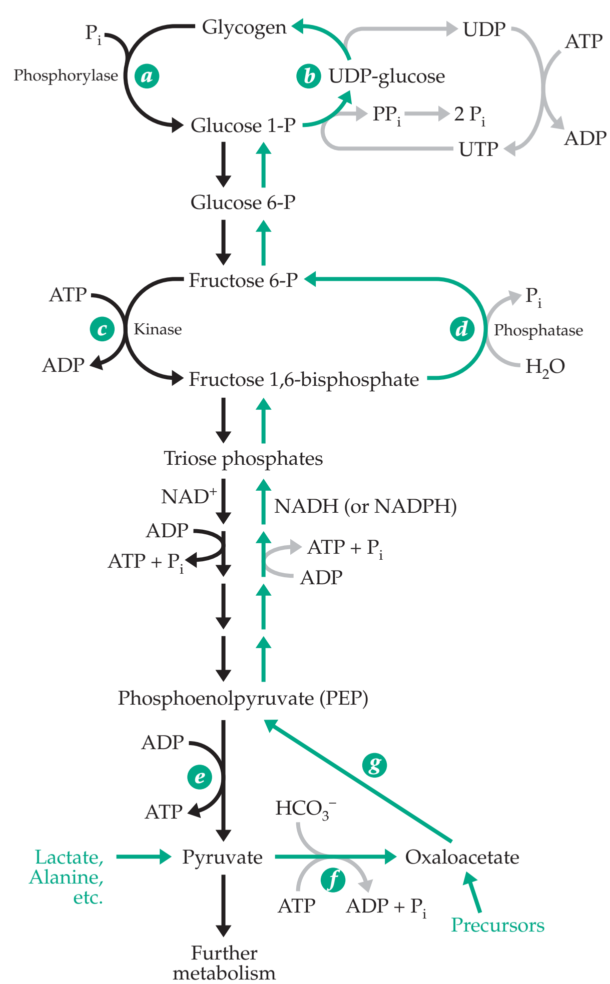
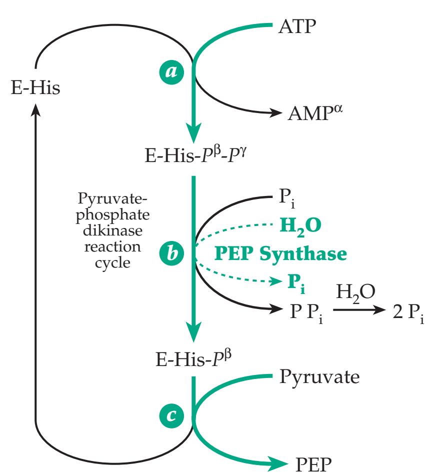


Figure 17-17 Comparison of glycolytic pathway (left) with pathway of gluconeogenesis (right, green arrows).

Two enzymes that are able to convert pyruvate directly to PEP are found in some bacteria and plants. In each case, as in the animal enzyme system discussed in the preceding paragraph, the conversion involves expenditure of two high-energy linkages of ATP. The **PEP synthase** of *E. coli* first transfers a pyrophospho group from ATP onto an imidazole group of histidine in the enzyme (Eq. 17-55). A phospho group is hydrolyzed from this intermediate (dashed green line in Eq. 17-55, step *b*), ensuring that sufficient intermediate E–His–*P* is present. The latter reacts with pyruvate to form PEP.[202,203] **Pyruvate–phosphate dikinase** is a similar enzyme first identified in tropical grasses and known to play an important role in the CO_2 concentrating system of the so-called "C_4 plants" (Chapter 23).[204] The same enzyme participates in gluconeogenesis in *Acetobacter*. The reaction cycle for this enzyme is also portrayed in Eq. 17-55. In this case P_i, rather than water, is the attacking nucleophile in Eq. 17-55 and PP_i is a product. The latter is probably hydrolyzed by pyrophosphatase action, the end result being an overall reaction that is the same as with PEP synthase. Kinetic and positional isotope exchange studies suggest that the P_i must be bound to pyruvate–phosphate dikinase before the bound ATP can react with the imidazole group.[202] Likewise, AMP doesn't dissociate until P_i has reacted to form PP_i.



(17-55)

6. Building Hydrocarbon Chains with Two-Carbon Units

Fatty acid chains are taken apart two carbon atoms at a time by β oxidation. Biosynthesis of fatty acids reverses this process by using the two-carbon acetyl unit of acetyl-CoA as a starting material. The coupling of ATP cleavage to this process by a carboxylation–decarboxylation sequence, the role of acyl carrier protein (Section H,4), and the use of NADPH as a reductant (Section I) have been discussed and are summarized in Fig. 17-12, which gives the complete sequence of reactions for fatty acid biosynthesis. Why does β oxidation require CoA derivatives while biosynthesis requires the more complex acyl carrier protein (ACP)? The reason may involve control. ACP is a complex handle able to hold the growing fatty acid chain and to guide it from one enzyme to the next. In *E. coli* the various enzymes catalyzing the reactions of Fig. 17-12 are found in the cytosol and behave as independent proteins. The same is true for fatty acid synthases of higher plants which resemble those of bacteria.[205,205a]

It is thought that the ACP molecule lies at the center of the complex and that the growing fatty acid chain on the end of the phosphopantetheine prosthetic group moves from one subunit to the other.[164,206] The process is started by a **primer** which is usually acetyl-CoA in *E. coli*. Its acyl group is transferred first to the central molecule of ACP (step *a*, Fig. 17-12) and then to a "peripheral" thiol group, probably that of a cysteine side chain on a separate protein subunit (step *b*, Fig. 17-12). Next, a malonyl group is transferred (step *d*) from malonyl-CoA to the free thiol group on the ACP. The condensation (steps *e* and *f*) occurs with the freeing of the peripheral thiol group. The latter does not come into use again until the β-oxoacyl group formed has undergone the complete sequence of reduction reactions (steps *g–i*). Then the growing chain is again transferred to the peripheral –SH (step *j*) and a new malonyl unit is introduced on the central ACP.

After the chain reaches a length of 12 carbon atoms, the acyl group tends to be transferred off to a CoA molecule (step *k*) rather than to pass around the cycle again. Thus, chain growth is terminated. This tendency systematically increases as the chain grows longer.

In higher animals as well as in *Mycobacterium*,[207] yeast,[208] and *Euglena*, the **fatty acid synthase** consists of only one or two multifunctional proteins. The synthase from animal tissues has seven catalytic activities in a single 263-kDa 2500-residue protein.[209] The protein consists of a series of domains that contain the various catalytic activities needed for the entire synthetic sequence. One domain contains an ACP-like site with a bound 4'-phosphopantetheine as well as a cysteine side chain in the second acylation site. This synthase produces free fatty acids, principally the C_{16} palmitate. The final step is cleavage of the acyl-CoA by a thioesterase, one of the seven enzymatic activities of the synthase. See Chapter 21 for further discussion.

7. The Oxoacid Chain Elongation Process

As mentioned in Section 4, glyoxylate can be converted to oxaloacetate by condensation with acetyl-CoA (Fig. 17-16) and the oxaloacetate can be decarboxylated to pyruvate. This sequence of reactions resembles that of the conversion of oxaloacetate to 2-oxoglutarate in the citric acid cycle (Fig. 17-4). *Both* *are examples of a frequently used general chain elongation process for α-oxo acids*. This sequence, which is illustrated in Fig. 17-18, has four steps: (1) condensation of the α-oxo acid with an acetyl group, (2) isomerization by dehydration and rehydration (catalyzed by aconitase in the case of the citric acid cycle), (3) dehydrogenation, and (4) β decarboxylation. In many cases steps 3 and 4 are combined as a single enzymatic reaction. The isomerization of the intermediate hydroxy acid in step 2 is required because the hydroxyl group, which is attached to a tertiary carbon bearing no hydrogen, must be moved to the adjacent carbon atom before oxidation to a ketone can take place. However, in the case of glyoxylate, isomerization is not necessary because R = H.

It may be protested that the reaction of the citric acid cycle by which oxaloacetate is converted to oxoglutarate does not follow exactly the pattern of Fig. 17-18. The carbon dioxide removed in the decarboxylation step does not come from the part of the molecule donated by the acetyl group but from that formed from oxaloacetate. However, the end result is the same. Furthermore, there are two known citrate-forming enzymes with different stereospecificities (Chapter 13), one of which leads to a biosynthetic pathway strictly according to the sequence of Fig. 17-18.

At the bottom of Fig. 17-18 several stages of the α-oxo acid elongation process are arranged in tandem. We see that glyoxylate (a product of the acetyl-CoA–glyoxylate cycle) can be built up systematically to

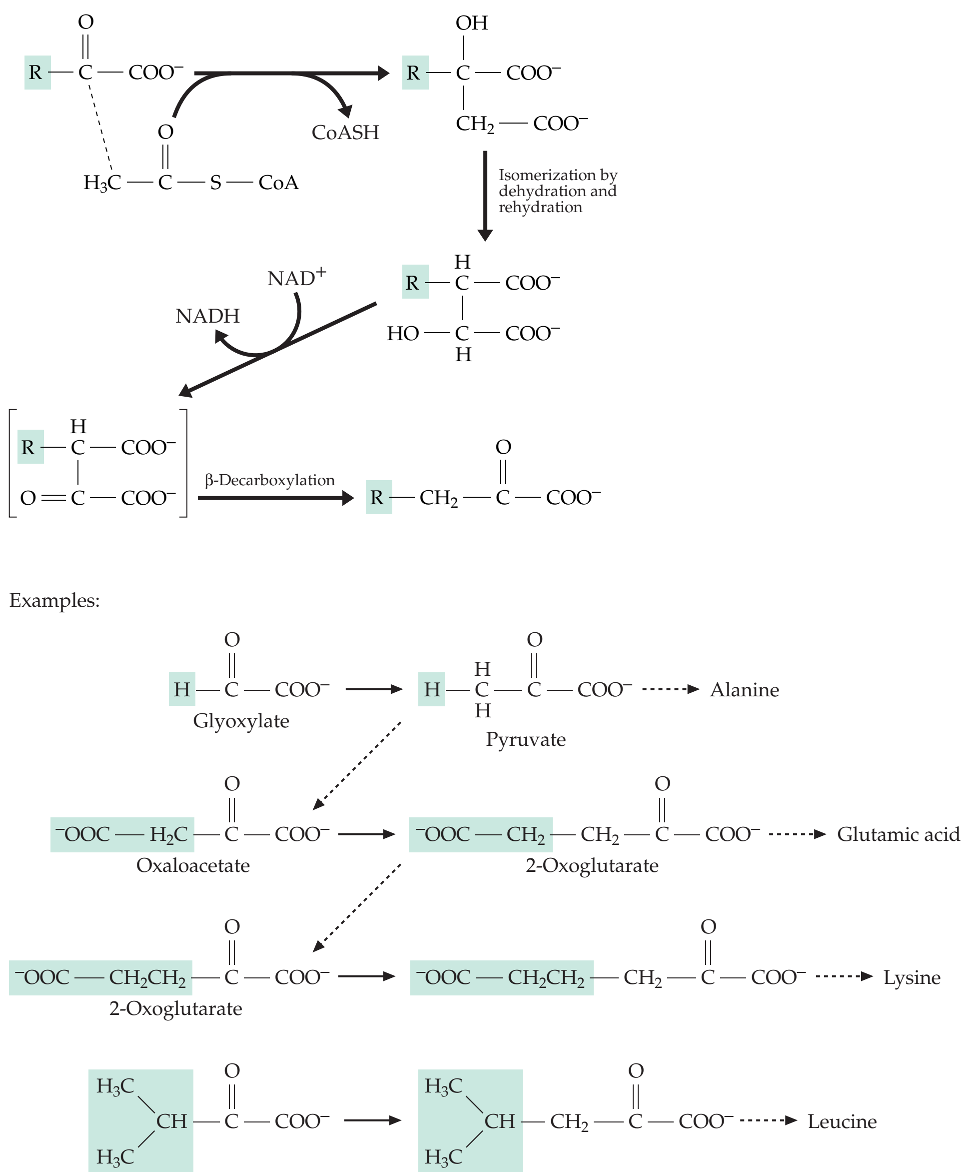
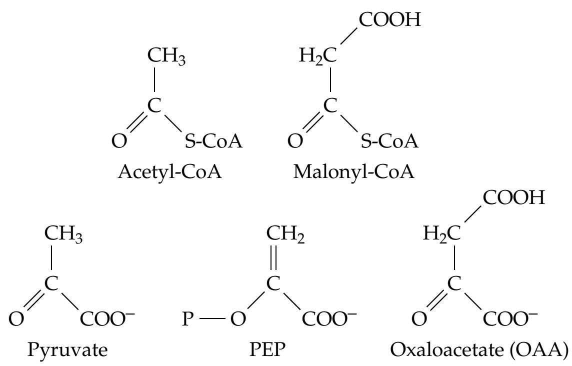


Figure 17-18 The oxoacid chain elongation process.

pyruvate, oxaloacetate, 2-oxoglutarate, and 2-oxoadipate (a precursor of lysine) using this one reaction sequence. Methanogens elongate 2-oxoadipate by one and two carbon atoms using the same sequence to give 7- and 8-carbon dicarboxylates.[210]

8. Decarboxylation as a Driving Force in Biosynthesis

Consider the relationship of the following prominent biosynthetic intermediates one to another:



Utilization of acetyl-CoA for the synthesis of long-chain fatty acids occurs via carboxylation to malonyl-CoA. *We can think of the malonyl group as a β-carboxylated acetyl group*. During synthesis of a fatty acid the carboxyl group is lost, and only the acetyl group is ultimately incorporated into the fatty acid. In a similar way *pyruvate can be thought of as an α-carboxylated acetaldehyde and oxaloacetate as an α- and β-dicarboxylated acetaldehyde*. During biosynthetic reactions these three- and four-carbon compounds also often undergo decarboxylation. Thus, they both can be regarded as "activated acetaldehyde units." Phosphoenolpyruvate is an α-carboxylated phosphoenol form of acetaldehyde and undergoes both decarboxylation and dephosphorylation before contributing a two-carbon unit to the final product.

It is of interest to compare two chain elongation processes by which two-carbon units are combined. In the synthesis of fatty acids the acetyl units are condensed and then are reduced to form straight hydrocarbon chains. In the oxo-acid chain elongation mechanism, the acetyl unit is introduced but is later decarboxylated. Thus, the chain is increased in length by one carbon atom at a time. These two mechanisms account for a great deal of the biosynthesis by chain extension. However, there are other variations. For example, glycine (a carboxylated methylamine), under the influence of pyridoxal phosphate and with accompanying decarboxylation, condenses with succinyl-CoA (Eq. 14-32) to extend the carbon chain and at the same time to introduce an amino group. Likewise, serine (a carboxylated ethanolamine) condenses with palmitoyl-CoA in biosynthesis of sphingosine (as in Eq. 14-32). Phosphatidylserine is decarboxylated to phosphatidylethanolamine in the final synthetic step for that phospholipid (Fig. 21-5).

9. Stabilization and Termination of Chain Growth by Ring Formation

Biochemical substances frequently undergo cyclization to form stable five- and six-atom ring structures. The three-carbon glyceraldehyde phosphate exists in solution primarily as the free aldehyde (and its covalent hydrate) but glucose 6-phosphate exists largely as the cyclic hemiacetal. In this ring form no carbonyl group is present and further chain elongation is inhibited. When the hemiacetal of glucose 6-*P* is enzymatically isomerized to glucose 1-*P* the ring is firmly locked. Glucose 1-*P*, in turn, serves as the biosynthetic precursor of polysaccharides and related compounds, in all of which the sugar rings are stable. Ring formation can occur in lipid biosynthesis, too. Among the **polyketides** (Chapter 21), polyprenyl compounds (Chapter 22), and aromatic amino acids (Chapter 25) are many substances in which ring formation has occurred by ester or aldol condensations followed by reduction and elimination processes. This is a typical sequence for biosynthesis of highly stable aromatic rings.

10. Branched Carbon Chains

Branched carbon skeletons are formed by standard reaction types but sometimes with addition of rearrangement steps. Compare the biosynthetic routes to three different branched five-carbon units (Fig. 17-19) The first is the use of a **propionyl group** to initiate formation of a branched-chain fatty acid. Propionyl-CoA is carboxylated to methylmalonyl-CoA, whose acyl group is transferred to the acyl carrier protein before condensation. Decarboxylation and reduction yields an acyl-CoA derivative with a methyl group in the 3-position.

The second five-carbon branched unit, in which the branch is one carbon further down the chain, is an intermediate in the biosynthesis of **polyprenyl** (isoprenoid) compounds and steroids. Three two-carbon units are used as the starting material with decarboxylation of one unit. Two acetyl units are first condensed to form acetoacetyl-CoA. Then a third acetyl unit, which has been transferred from acetyl-CoA onto an SH group of the enzyme, is combined with the acetoacetyl-CoA through an ester condensation. The thioester linkage to the enzyme is hydrolyzed to free the product **3-hydroxy-3-methylglutaryl-CoA** (HMG-CoA). This sequence is illustrated in Eq. 17-5. The thioester group of HMG-CoA is reduced to the alcohol **mevalonic acid**, a direct precursor to isopentenyl pyrophosphate, from which the polyprenyl compounds are formed (Fig. 22-1).

The third type of carbon-branched unit is 2-oxoisovalerate, from which valine is formed by transamination. The starting units are two molecules of pyruvate which combine in a thiamin diphosphate-dependent α condensation with decarboxylation. The resulting α-acetolactate contains a branched chain but is quite unsuitable for formation of an α amino acid. A rearrangement moves the methyl group to the β position (Fig. 24-17), and elimination of water from the diol forms the enol of the desired α-oxo acid (Fig. 17-19). The precursor of isoleucine is formed in an analogous way by condensation, with decarboxylation of one molecule of pyruvate with one of 2-oxobutyrate.

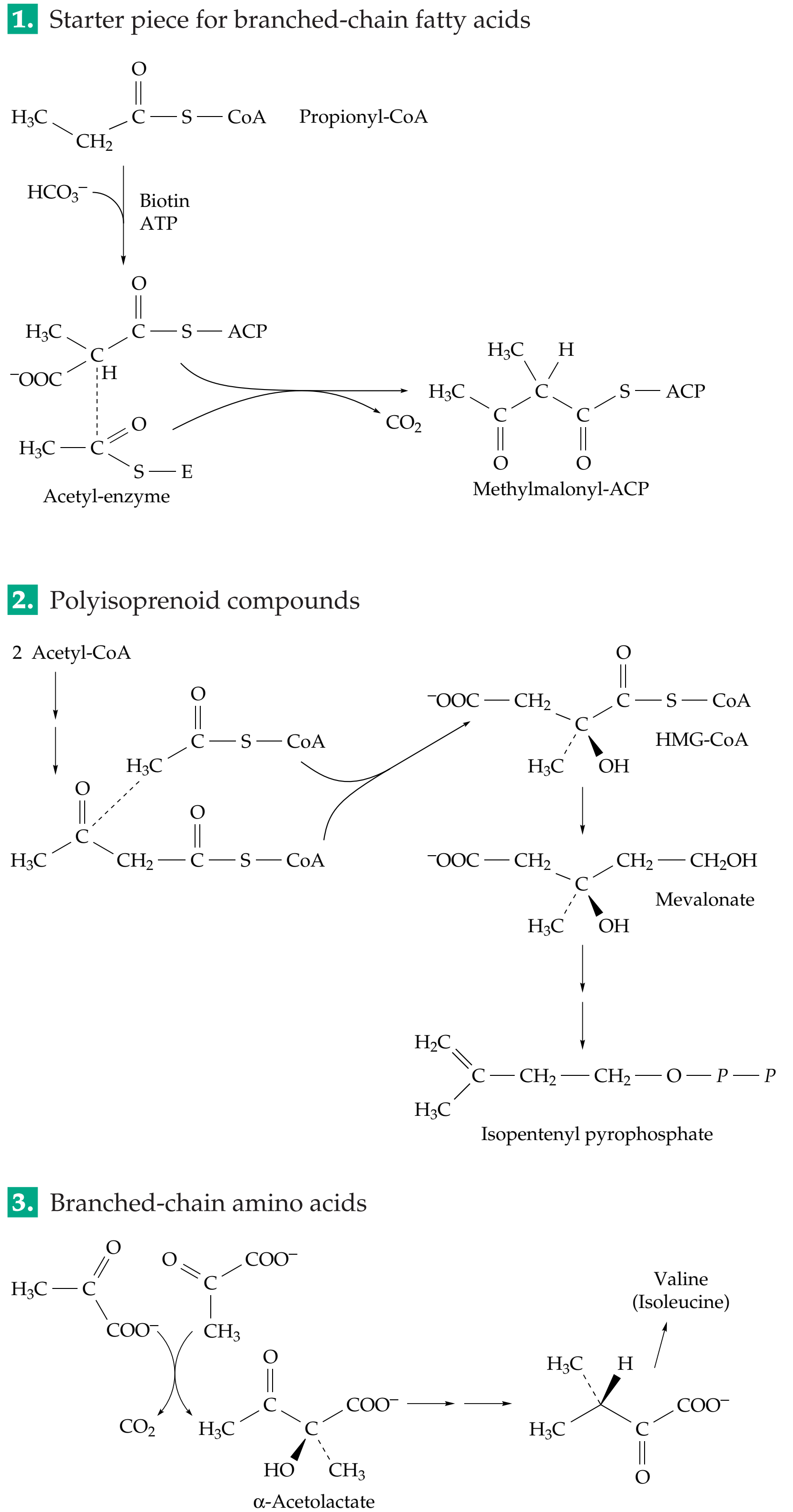


Figure 17-19 Biosynthetic origins of three five-carbon branched structural units. Notice that decarboxylation is involved in driving each sequence.

K. Biosynthesis and Modification of Polymers

There are three chemical problems associated with the assembly of a protein, nucleic acid, or other biopolymer. The first is *to overcome thermodynamic barriers*. The second is *to control the rate of synthesis*, and the third is *to establish the pattern or sequence in which the monomer units are linked together*. Let us look briefly at how these three problems are dealt with by living cells.

1. Peptides and Proteins

Activation of amino acids for incorporation into oligopeptides and proteins can occur via two routes of acyl activation. In the first of these an **acyl phosphate** (or acyl adenylate) is formed and reacts with an amino group to form a peptide linkage (Eq. 13-4). The tripeptide **glutathione** is formed in two steps of this type (Box 11-B). In the second method of activation **aminoacyl**

adenylates are formed. They transfer their activated aminoacyl groups onto specific tRNA molecules during synthesis of proteins (Eq. 17-36). In other cases activated aminoacyl groups are transferred onto –SH groups to form intermediate **thioesters**. An example is the synthesis of the antibiotic **gramicidin S** formed by *Bacillus brevis*. The antibiotic is a cyclic decapeptide with the following five-amino-acid sequence repeated twice in the ringlike molecule[211]:

$$(–\text{D-Phe}–\text{L-Pro}–\text{L-Val}–\text{L-Orn}–\text{L-Leu}–)_2$$

The soluble enzyme system responsible for its synthesis contains a large 280-kDa protein that not only activates the amino acids as aminoacyl adenylates and transfers them to thiol groups of 4'-phosphopantetheine groups covalently attached to the enzyme but also serves as a template for joining the amino acids in proper sequence.[211–214] Four amino acids—proline, valine, ornithine (Orn), and leucine—are all bound. A second enzyme (of mass 100 kDa) is needed for activation of phenylalanine. It is apparently the activated phenylalanine (which at some point in the process is isomerized from L- to D-phenylalanine) that initiates polymer formation in a manner analogous to that of fatty acid elongation (Fig. 17-12). Initiation occurs when the amino group of the activated phenylalanine (on the second enzyme) attacks the acyl group of the aminoacyl thioester by which the activated proline is held. Next, the freed imino group of proline attacks the activated valine, etc., to form the pentapeptide. Then two pentapeptides are joined and cyclized to give the antibiotic. The sequence is absolutely specific, and it is remarkable that this relatively small enzyme system is able to carry out each step in the proper sequence. Many other peptide antibiotics, such as the bacitracins, tyrocidines,[215] and enniatins, are synthesized in a similar way,[213,216,217] as are depsipeptides and the immunosuppresant cyclosporin. A virtually identical pattern is observed for formation of **polyketides**,[218,219] whose chemistry is considered in Chapter 21.

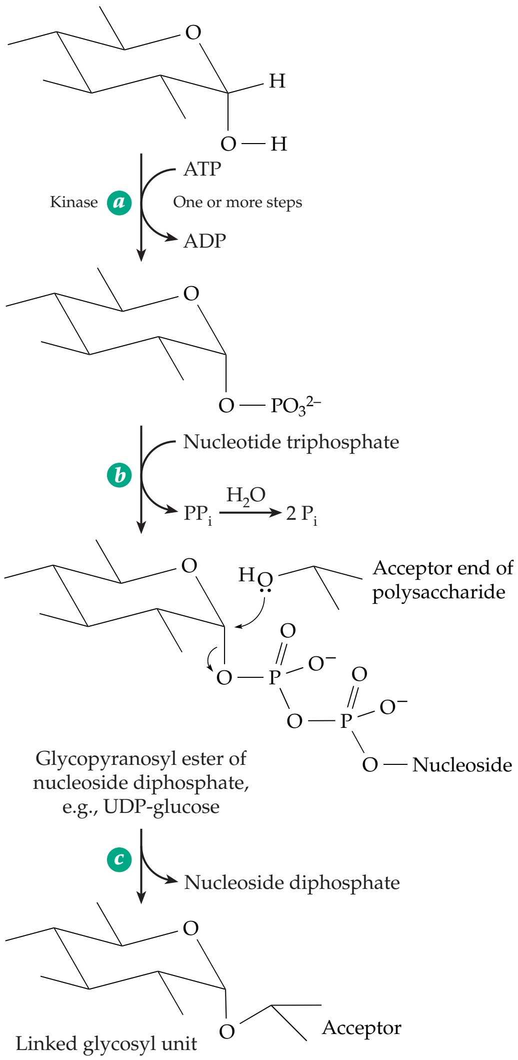
While peptide antibiotics are synthesized according to enzyme-controlled polymerization patterns, both proteins and nucleic acids are made by **template mechanisms**. The sequence of their monomer units is determined by genetically encoded information. A key reaction in the formation of proteins is the transfer of activated aminoacyl groups to molecules of tRNA (Eq. 17-36). The tRNAs act as carriers or adapters as explained in detail in Chapter 29. Each **aminoacyl-tRNA synthetase** must recognize the correct tRNA and attach the correct amino acid to it. The tRNA then carries the activated amino acid to a ribosome, where it is placed, at the correct moment, in the active site. **Peptidyltransferase**, using a transacylation reaction, in an *insertion mechanism* transfers the C terminus of the growing peptide chain onto the amino group of the new amino acid to give a tRNA-bound peptide one unit longer than before.

2. Polysaccharides

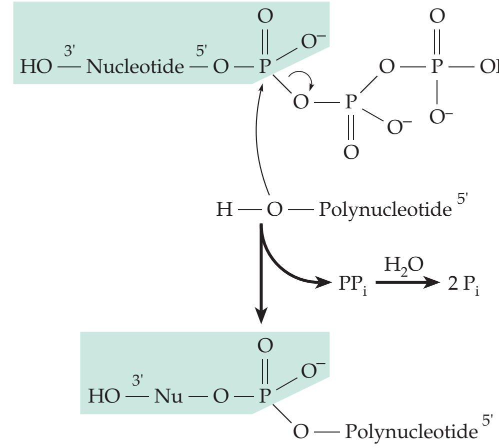
Incorporation of a sugar monomer into a polysaccharide also involves cleavage of two high-energy phosphate linkages of ATP. However, the activation process has its own distinctive pattern (Eq. 17-56). Usually a sugar is first phosphorylated by a kinase or a kinase plus a phosphomutase (Eq. 17-56, step *a*). Then a nucleoside triphosphate (NuTP) reacts under the influence of a second enzyme with elimination of pyrophosphate and formation of a **glycopyranosyl ester** of the nucleoside diphosphate, more often known as a **sugar nucleotide** (Eq. 17-56, step *b*). The inorganic pyrophosphate is hydrolyzed by pyrophosphatase while the sugar nucleotide donates the activated glycosyl group for polymerization (Eq. 17-56, step *c*). In this step the glycosyl group is transferred with displacement of the nucleoside diphosphate. Thus, the overall process involves first the cleavage of ATP to ADP and P_i, and then the cleavage of a nucleoside triphosphate to a nucleoside diphosphate plus P_i. The nucleoside triphosphate in Eq. 17-56, step *b* is sometimes ATP, in which case the overall result is the splitting of two molecules of ATP to ADP. However, as detailed in Chapter 20, the whole series of nucleotide "handles" serve to carry various activated glycosyl units.

What determines the pattern of incorporation of sugar units into polysaccharides? Homopolysaccharides, like cellulose and the linear amylose form of starch, contain only one monosaccharide component in only one type of linkage. A single synthetase enzyme can add unit after unit of an activated sugar (UDP glucose or other sugar nucleotide) to the growing end. However, at least two enzymes are needed to assemble a branched molecule such as that of the glycogen molecule. One is the synthetase; the second is a **branching enzyme**, a transglycosylase. After the chain ends attain a length of about ten monosaccharide units the branching enzyme attacks a glycosidic linkage somewhere in the chain. Acting much like a hydrolase, it forms a glycosyl enzyme (or a stabilized carbocation) intermediate. The enzyme does not release the severed chain fragment but transfers it to another nearby site on the branched polymer. In the synthesis of glycogen, the chain fragment is joined to a free 6-hydroxyl group of the glycogen, creating a new branch attached by an α-1,6-linkage.

Other carbohydrate polymers consist of **repeating oligosaccharide units**. Thus, in hyaluronan units of glucuronic acid and N-acetyl-D-glucosamine alternate (Fig. 4-11). The "O antigens" of bacterial cell coats (p. 180) contain repeating subunits made up of a "block" of four or five different sugars. In these and



(17-56)



(17-57)

many other cases the pattern of polymerization is established by the specificities of individual enzymes. An enzyme capable of joining an activated glucosyl unit to a growing polysaccharide will do so only if the proper structure has been built up to that point. In cases where a block of sugar units is transferred it is usually *inserted* at the nonreducing end of the polymer, which may be covalently attached to a protein. Notice that the insertion mode of chain growth exists for lipids, polysaccharides, and proteins.

3. Nucleic Acids

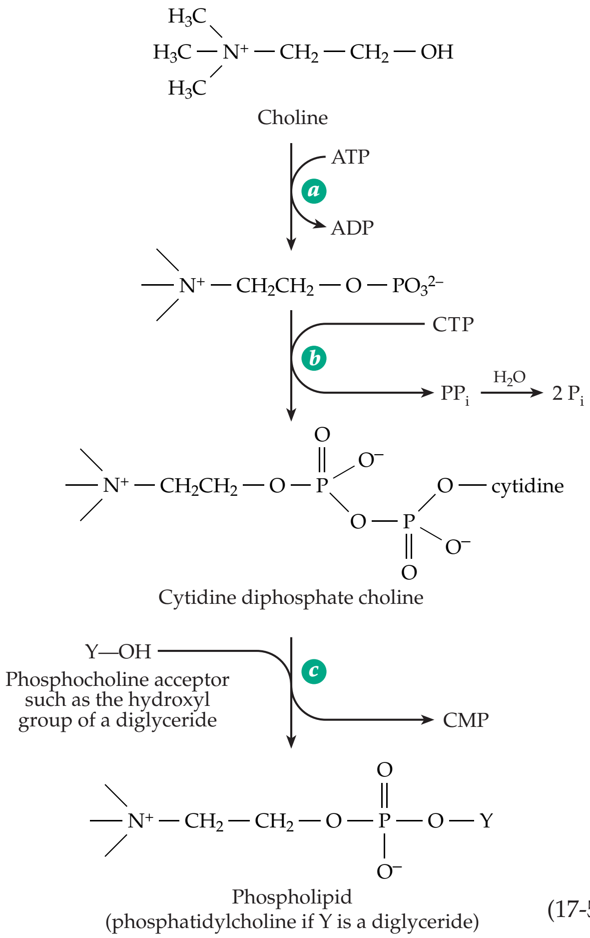
The activated nucleotides are the nucleoside 5'-triphosphates. The ribonucleotides ATP, GTP, UTP, and CTP are needed for RNA synthesis and the 2'-deoxyribonucleotide triphosphates, dATP, dTTP, dGTP, and dCTP for DNA synthesis. In every case, the addition of activated monomer units to a growing polynucleotide chain is catalyzed by an enzyme that binds to the template nucleic acid. The choice of the proper nucleotide unit to place next in the growing strand is determined by the nucleotide already in place in the complementary strand, a matter that is dealt with in Chapters 27 and 28. The chemistry is a simple displacement of pyrophosphate (Eq. 17-57). The 3'-hydroxyl of the polynucleotide attacks the phosphorus atom of the activated nucleoside triphosphate. Thus, *nucleotide chains always grow from the 5′ end, with new units being added at the 3′ end*.

4. Phospholipids and Phosphate–Sugar Alcohol Polymers

Choline and ethanolamine are activated in much the same way as are sugars. For example, choline can be phosphorylated using ATP (Eq. 17-58, step *a*) and the phosphocholine formed can be further converted (Eq. 17-58, step *b*) to **cytidine diphosphate choline**. Phosphocholine is transferred from the latter onto a suitable acceptor to form the final product (Eq. 17-58, step *c*). The polymerization pattern differs from that for polysaccharide synthesis. When the sugar nucleotides react, the entire nucleoside diphosphate is eliminated (Eq. 17-56), but CDP-choline and CDP-ethanolamine react with elimination of CMP (Eq. 17-58, step *c*), leaving one phospho group in the final product. The same thing is true in the synthesis of the bacterial teichoic acids (Chapter 8). Either CDP-glycerol or CDP-ribitol is formed first and polymerization takes place with elimination of CMP to form the alternating phosphate–sugar alcohol polymer.[220]

5. Irreversible Modification and Catabolism of Polymers

While polymers are being synthesized continuously by cells, they are also being modified and torn down. Nothing within a cell is static. As discussed in Chapters



(17-58)

10 and 29, everything turns over at a slower or faster rate. Hydrolases attack all of the polymers of which cells are composed, and active catabolic reactions degrade the monomers formed. Membrane surfaces are also altered, for example, by hydroxylation and glycosylation of both glycoproteins and lipid head groups. It is impossible to list all of the known modification reactions of biopolymers. They include hydrolysis, methylation, acylation, isopentenylation, phosphorylation, sulfation, and hydroxylation. Precursor molecules are cut and trimmed and often modified further to form functional proteins or nucleic acids. Phosphotransferase reactions splice RNA transcripts to form mRNA and a host of alterations convert precursors into mature tRNA molecules (Chapter 28). Even DNA, which remains relatively unaltered, undergoes a barrage of chemical attacks. Only because of the presence of an array of repair enzymes (Chapter 27) does our DNA remain nearly unchanged so that faithful copies can be provided to each cell in our bodies and can be passed on to new generations.

L. Regulation of Biosynthesis

A simplified view of metabolism is to consider a cell as a "bag of enzymes." Indeed, much of metabolism can be explained by the action of several thousand enzymes promoting specific reactions of their substrates. These reactions are based upon the natural chemical reactivities of the substrates. However, the enzymes, through the specificity of their actions and through association with each other,[96,221–223] channel the reactions into a selected series of metabolic pathways. The reactions are often organized as cycles which are inherently stable. We have seen that biosynthesis often involves ATP-dependent reductive reactions. *It is these reductive processes that produce the less reactive nonpolar lipid groupings and amino acid side chains so essential to the assembly of insoluble intracellular structures*. Oligomeric proteins, membranes, microtubules, and filaments are all the natural result of aggregation caused largely by hydrophobic interactions with electrostatic forces and hydrogen bonding providing specificity. A major part of metabolism is the creation of complex molecules that aggregate spontaneously to generate structure. This structure includes the lipid-rich cytoplasmic membranes which, together with embedded carrier proteins, control the entry of substances into cells. Clearly, the cell is now much more than a bag of enzymes, containing several compartments, each of which contains its own array of enzymes and other components. Metabolite concentrations may vary greatly from one compartment to another.

The reactions that modify lipids and glycoproteins provide a driving force that assists in moving membrane materials generated internally into the outer surface of cells. Other processes, including the breakdown by lysosomal enzymes, help to recycle membrane materials. Oxidative attack on hydrophobic materials such as the sterols and the fatty acids of membrane lipids results in their conversion into more soluble substances which can be degraded and completely oxidized. The flow of matter within cells tends to occur in metabolic loops and some of these loops lead to formation of membranes and organelles and to their turnover. This flow of matter, which is responsible for growth and development of cells, is driven both by hydrolysis of ATP coupled to biosynthesis and by irreversible degradative alterations of polymers and lipid materials. It also provides for transient formation and breakup of complexes of macromolecules, which may be very large, in response to varying metabolic needs.

Anything that affects the rate of a reaction involved in either biosynthesis or degradation of any component of the cell will affect the overall picture in some way. Thus, every chemical reaction that contributes to a quantitatively significant extent to metabolism has some controlling influence. Since molecules interact with each other in so many ways, reactions of metabolic control are innumerable. Small molecules act on macromolecules as effectors that influence conformation and reactivity. Enzymes act on each other to break covalent bonds, to oxidize, and to crosslink. Transferases add phospho, glycosyl, methyl, and other groups to various sites. The resulting alterations often affect catalytic activities. The number of such interactions significant to metabolic control within an organism may be in the millions. Small wonder that biochemical journals are filled with a confusing number of postulated control mechanisms.

Despite this complexity, some regulatory mechanisms stand out clearly. The control of enzyme synthesis through feedback repression and the rapid control of activity by feedback inhibition (Chapter 11) have been considered previously. Under some circumstances, in which there is a constant growth rate, these controls may be sufficient to ensure the harmonious and proportional increase of all constitutents of a cell. Such may be the case for bacteria during logarithmic growth (Box 9-B) or for a mammalian embryo growing rapidly and drawing all its nutrients from the relatively constant supply in the maternal blood.

Contrast the situation in an adult. Little growth takes place, but the metabolism must vary with time and physiological state. The body must make drastic readjustments from normal feeding to a starvation situation and from resting to heavy exercise. The metabolism needed for rapid exertion is different from that needed for sustained work. A fatty diet requires different metabolism than a high-carbohydrate diet. The necessary control mechanisms must be rapid and sensitive.

1. Glycogen and Blood Glucose

Two special features of glucose metabolism in animals are dominant.[224] The first is the storage of glycogen for use in providing muscular energy rapidly. This is a relatively short-term matter but the rate of glycolysis can be intense: The entire glycogen content of muscle could be exhausted in only 20 s of anaerobic fermentation or in 3.5 min of oxidative metabolism.[225] There must be a way to turn on glycolysis quickly and to turn it off when it is no longer needed. At the same time, it must be possible to reconvert lactate to glucose or glycogen (gluconeogenesis). The glycogen stores of the muscle must be repleted from glucose of the blood. If insufficient glucose is available from the diet or from the glycogen stores of the liver, it must be synthesized from amino acids.

The second special feature of glucose metabolism is that certain tissues, including brain, blood cells, kidney medulla, and testis, ordinarily obtain most of their energy through oxidation of glucose.[226,227] For this reason, the glucose level of blood cannot be allowed to drop much below the normal 5 mM. The mechanism of regulation of the blood glucose level is complex and incompletely understood. A series of hormones are involved.

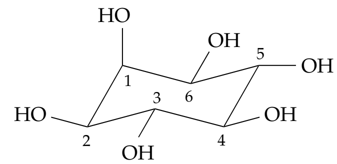
Insulin. This 51-residue cross-linked polypeptide (Fig. 7-17) is synthesized in the pancreatic islets of Langerhans, a tissue specialized for synthesis and secretion into the bloodstream of a series of small peptide hormones. One type of islet cells, the β cells, forms primarily insulin which is secreted in response to high (> 5mm) blood [glucose].[228] Insulin has a wide range of effects on metabolism,[228a] which are discussed in Chapter 11, Section G. Most of these effects are thought to arise from binding to insulin receptors (Figs. 11-11 and 11-12) and are mediated by cascades such as that pictured in Fig. 11-13.[229–232c] The end result is to increase or decrease activities of a large number of enzymes as is indicated in Table 17-3. Some of those are also shown in Fig. 17-20, which indicates interactions with the tricarboxylic acid cycle and lipid metabolism. Binding of insulin to the extracellular domain of its dimeric receptor induces a conformational change that activates the intracellular tyrosine kinase domains of the two subunits. Recent studies suggest that in the activated receptor the two transmembrane helices and the internal tyrosine kinase domains move closer together, inducing the essential autophosphosphorylation.[232b] The kinase domain of the phosphorylated receptor, in turn, phosphorylates several additional proteins, the most important of which seem to be the insulin receptor substrates IRS-1 and IRS-2. Both appear to be essential in different tissues.[232d,e] Phosphorylated forms of these proteins initiate a confusing variety of signaling cascades.[232f–i]

One of the most immediate effects of insulin is to stimulate an increased rate of uptake of glucose by muscle and adipocytes (fat cells) and other insulin-sensitive tissues. This uptake is accomplished largely by movement of the glucose transporter GLUT4 (Chapter 8) from internal "sequestered" storage vesicles located near the cell membrane into functioning positions in the membranes.[232f,j–m] Activation of this translocation process apparently involves IRS-1 and phosphatidylinositol (PI) 3-kinase, which generates PI-3,4,5-P_3 (Fig. 11-9).[232c,n,o] The latter induces the translocation. However, the mechanism remains obscure. The process may also require a second signaling pathway which involves action of the insulin receptor kinase on an adapter protein known as **CAP**, a transmembrane caveolar protein **flotillin**, and a third protein **Cbl**, a known cellular protooncogene. Phosphorylated Cbl forms a complex with CAP and flotillin in a "lipid raft" which induces the exocytosis of the sequestered GLUT4 molecules.[232o,p]

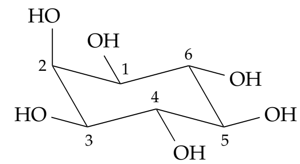
TABLE 17-3
Some Effects of Insulin on Enzymes

Name of Enzyme	Type of Regulation
A. Activity increased	
Enzymes of glycolysis	
Glucokinase	Transcription induced
Phosphofructokinase	via 2,6-fructose P_2
Pyruvate kinase	Dephosphorylation
6-Phosphofructo-2-kinase	Dephosphorylation
Enzymes of glycogen synthesis	
Glucokinase	Transcription
Glycogen synthase (muscle)	Dephosphorylation
Enzymes of lipid synthesis	
Pyruvate dehydrogenase (adipose)	Dephosphorylation (Eq. 17-9)
Acetyl-CoA carboxylase	Dephosphorylation
ATP-citrate lyase	Phosphorylation
Fatty acid synthase	
Lipoprotein lipase	
Hydroxymethylglutaryl-CoA reductase	
B. Activity decreased	
Enzymes of gluconeogenesis	
Pyruvate carboxylase	
PEP carboxykinase	Transcription inhibited
Fructose 1,6-bisphosphate	
Glucose 6-phosphatase	
Enzymes of lipolysis	
Triglyceride lipase	
(hormone-sensitive lipase)	Dephosphorylation
Enzymes of glycogenolysis	
Glycogen phosphorylase	
C. Other proteins affected by insulin	
Glucose transporter GLUT4	Redistribution
Ribosomal protein S6	Phosphorylation by p90^{rsk}
IGF-II receptor	Redistribution
Transferrin receptor	Redistribution
Calmodulin	Phosphorylation

A clue to another possible unrecognized mechanism of action for insulin comes from the observation that urine of patients with non-insulin-dependent diabetes contains an unusual isomer of inositol, D-*chiro*-inositol.[233,234]



D-*chiro*-Inositol



myo-Inositol
(numbered as D)

Plasma of such individuals contains an antagonist of insulin action, an inositol phosphoglycan containing *myo*-inositol as a cyclic 1,2-phosphate ester and galactosamine and mannose in a 1:1:3 ratio.[235] This appears to be related to the glycosyl phosphatidylinositol (GPI) membrane anchors (Fig. 8-13). It has been suggested that such a glycan, perhaps containing *chiro*-inositol, is released in response to insulin and serves as a second messenger for insulin.[235–236a] This hypothesis remains unproved.[237] However, insulin does greatly stimulate a GPI-specific phospholipase C, at least in yeast.[237a] Another uncertainty surrounds the possible cooperation of chromium (Chapter 16) in the action of insulin.

How do the insulin-secreting pancreatic β cells sense a high blood glucose concentration? Two specialized proteins appear to be involved. The sugar transporter **GLUT2** allows the glucose in blood to equilibrate with the free glucose in the β cells,[237b] while **glucokinase** (hexokinase IV or D) apparently serves as the glucose sensor.[228,238] Despite the fact that glucokinase is a monomer, it displays a cooperative behavior toward glucose binding, having a low affinity at low [glucose] and a high affinity at high [glucose]. Mutant mice lacking the glucokinase gene develop early onset diabetes which is mild in heterozygotes but severe and fatal within a week of birth for homozygotes.[239,240] These facts alone do not explain how the sensor works and there are doubtless other components to the signaling system.

A current theory is that the increased rate of glucose catabolism in the β cells when blood [glucose] is high leads to a high ratio of [ATP]/[ADP] which induces closure of ATP-sensitive K^+ channels and opening of voltage-gated Ca^{2+} channels.[241] This could explain the increase in [Ca^{2+}] within β cells which has been associated with secretion of insulin[242,243] and which is thought to induce the exocytosis in insulin storage granules. The internal [Ca^{2+}] in pancreatic islet cells is observed to oscillate in a characteristic way that is synchronized with insulin secretion.[243]

Glucagon. This 29-residue peptide is the principal hormone that counteracts the action of insulin. Glucagon acts primarily on liver cells (hepatocytes) and adipose tissue and is secreted by the α cells of the islets of Langerhans in the pancreas, the same tissue whose β cells produce insulin, if the blood glucose concentration falls much below 2 mM.[244–250] Like the insulin-secreting β cells, the pancreatic α cells contain glucokinase, which may be involved in sensing the drop in glucose concentration. However, the carrier GLUT2 is not present and there is scant information on the sensing mechanism.[248]

Glucagon promotes an increase in the blood glucose level by stimulating breakdown of liver glycogen, by inhibiting its synthesis, and by stimulating gluconeogenesis. All of these effects are mediated by cyclic AMP through cAMP-activated protein kinase (Fig. 11-4) and through fructose 2,6-P_2 (Fig. 11-4 and next section). Glucagon also has a strong effect in promoting the release of glucose into the bloodstream. **Adrenaline** has similar effects, again mediated by cAMP. However, glucagon affects the liver, while adrenaline affects many tissues. **Glucocorticoids** such as cortisol (Chapter 22) also promote gluconeogenesis and the accumulation of glycogen in the liver through their action on gene transcription.

The release of glucose from the glycogen stores in the liver is mediated by **glucose 6-phosphatase**, which is apparently embedded within the membranes of the endoplasmic reticulum. A labile enzyme, it consists of a 357-residue catalytic subunit,[251,252] which may be associated with other subunits that participate in transport.[252,253] A deficiency of this enzyme causes the very severe type 1a **glycogen storage disease** (see Box 20-D).[251,253] Only hepatocytes have significant glucose 6-phosphatase activity.

2. Phosphofructo-1-Kinase in the Regulation of Glycolysis

The metabolic interconversions of glucose 1-*P*, glucose 6-*P*, and fructose 6-*P* are thought to be at or near equilibrium within most cells. However, the phosphorylation by ATP of fructose 6-*P* to fructose 1,6-P_2 catalyzed by phosphofructose-1-kinase (Fig. 11-2, step *b*; Fig. 17-17, top center) is usually far from equilibrium. This fact was established by comparing the mass action ratio [fructose 1,6-P_2] [ADP]/[fructose 6-P] [ATP] measured within tissues with the known equilibrium constant for the reaction. At equilibrium this mass action ratio should be equal to the equilibrium constant (Section I,2). The experimental techniques for determining the four metabolite concentrations that are needed for evaluation of the mass action ratio in tissues are of interest. The tissues must be frozen very rapidly. This can be done by compressing them between large liquid nitrogen-cooled aluminum clamps. For details see Newsholme and Start,[225] pp. 30 – 32. Tissues can be cooled to – 80°C in less than 0.1 s in this manner. The frozen tissue is then powdered, treated with a frozen protein denaturant such as perchloric acid, and analyzed. From data obtained in this way, a mass action ratio of 0.03 was found for the phosphofructo-1-kinase reaction in heart muscle.[225] This is much lower than the equilibrium constant of over 3000 calculated from the value of $\Delta G'$ (pH 7) = –20.1 kJ mol^{-1}. Thus, like other biochemical reactions that are nearly irreversible thermodynamically, this reaction is far from equilibrium in tissues.

The effects of ATP, AMP, and fructose 2,6-bisphosphate on phosphofructokinase have been discussed in Chapter 11, Section C. Fructose 2,6-P_2 is a potent allosteric activator of phosphofructokinase and a strong competitive inhibitor of fructose 1,6-bisphosphatase (Fig. 11-2). It is formed from fructose 6-*P* and ATP by the 90-kDa bifunctional phosphofructo-2-kinase/fructose 2,6-bisphosphatase. Thus, the same protein forms and destroys this allosteric effector. Since the bifunctional enzyme is present in very small amounts, the rate of ATP destruction from the substrate cycling is small.

Glucagon causes the concentration of *liver* fructose 2,6-P_2 to drop precipitously from its normal value. This, in turn, causes a rapid drop in glycolysis rate and shifts metabolism toward gluconeogenesis. At the same time, liver glycogen breakdown is inceased and glucose is released into the bloodstream more rapidly. The effect on fructose 2,6-P_2 is mediated by a cAMP-dependent protein kinase which phosphorylates the bifunctional kinase/phosphatase in the liver.[254] This modification greatly reduces the kinase activity and strongly activates the phosphatase, thereby destroying the fructose 2,6-P_2. The changes in activity appear to be largely a result of changes in the appropriate K_m values which are increased for fructose 6-*P* and decreased for fructose 2,6-P_2.[255]

3. Gluconeogenesis

If a large amount of lactate enters the liver, it is oxidized to pyruvate which enters the mitochondria. There, part of it is oxidized through the tricarboxylic acid cycle. However, if [ATP] is high, pyruvate dehydrogenase is inactivated by phosphorylation (Eq. 17-9) and the amount of pyruvate converted to oxaloacetate and malate (Eq. 17-46) may increase. Malate may leave the mitochondrion to be reoxidized to oxaloacetate, which is then converted to PEP and on to glycogen (heavy green arrows in Fig. 17-20). When [ATP] is high, phosphofructokinase is also blocked, but the fructose 1,6-bisphosphatase, which hydrolyzes one phosphate group from fructose 1,6-P_2 (Fig. 11-2, step *d*), is active. If the glucose content of blood is low, the glucose 6-*P* in the liver is hydrolyzed and free glucose is secreted. Otherwise, most of the glucose 6-*P* is converted to glycogen. Muscle is almost devoid of glucose 6-phosphatase, the export of glucose not being a normal activity of that tissue.

Gluconeogenesis in liver is strongly promoted by glucagon and adrenaline. The effects, mediated by cAMP, include stimulation of fructose 1,6-bisphosphatase and inhibition of phosphofructo-1-kinase, both caused by the drop in the level of fructose 2,6-P_2.[254,256] The conversion of pyruvate to PEP via oxaloacetate is also promoted by glucagon. This occurs primarily by stimulation of pyruvate carboxylase (Eq. 14-3).[257,258] However, it has been suggested that the most important mechanism by which glucagon enhances gluconeogenesis is through stimulation of mitochondrial respiration, which in turn may promote gluconeogenesis.[257]

The conversion of oxaloacetate to PEP by PEP-carboxykinase (PEPCK, Eq. 14-43; Fig. 17-20) is another control point in gluconeogenesis. Insulin inhibits gluconeogenesis by decreasing transcription of the mRNA for this enzyme.[259–261a] Glucagon and cAMP stimulate its transcription. The activity of PEP carboxykinase[262] is also enhanced by Mn^{2+} and by very low concentrations of Fe^{2+}. However, the enzyme is readily inactivated by Fe^{2+} and oxygen.[263] Any regulatory significance is uncertain.

Although the regulation of gluconeogenesis in the liver may appear to be well understood, some data indicate that the process can occur efficiently in the presence of high average concentrations of fructose 2,6-P_2. A possible explanation is that liver consists of several types of cells, which may contain differing concentrations of this inhibitor of gluconeogenesis.[264] However, mass spectroscopic studies suggest that glucose metabolism is similar throughout the liver.[265]

4. Substrate Cycles

The joint actions of phosphofructokinase and fructose 1,6-bisphosphatase (Fig. 11-2, steps *b* and *c*; see also Fig. 17-20) create a substrate cycle of the type discussed in Chapter 11, Section F. Such cycles apparently accomplish nothing but the cleavage of ATP to ADP and P_i (ATPase activity). There are many cycles of this type in metabolism and the fact that they do not ordinarily cause a disastrously rapid loss of ATP is a consequence of the tight control of the metabolic pathways involved. In general, only one of the two enzymes of Fig. 11-2, steps *b* and *c*, is fully activated at any time. Depending upon the metabolic state of the cell, degradation may occur with little biosynthesis or biosynthesis with little degradation. Other obvious substrate cycles involve the conversion of glucose to glucose 6-*P* and hydrolysis of the latter back to glucose (Fig. 17-20, upper left-hand corner), the synthesis and breakdown of glycogen (upper right), and the conversion of PEP to pyruvate and the reconversion of the latter to PEP via oxaloacetate and malate (partially within the mitochondria).

While one might suppose that cells always keep substrate cycling to a bare minimum, experimental measurements on tissues *in vivo* have indicated surprisingly high rates for the fructose 1,6-bisphosphatase–phosphofructokinase cycle in mammalian tissues when glycolytic flux rates are low and also for the pyruvate → oxaloacetate → PEP → pyruvate cycle.[266] As pointed out in Chapter 11, by maintaining a low rate of substrate cycling under conditions in which the carbon flux is low (in either the glycolytic or glucogenic direction) the system is more sensitive to allosteric effectors than it would be otherwise. However, when the flux through the glycolysis pathway is high the relative amount of cycling is much less and the amount of ATP formed approaches the theoretical 2.0 per glucose.[267]

Substrate cycles generate heat, a property that is apparently put to good use by cold bumblebees whose thoracic temperature must reach at least 30°C before they can fly. The insects apparently use the fructose bisphosphatase–phosphofructokinase substrate cycle (Fig. 11-2, steps *b* and *c*) to warm their flight muscles.[268] It probably helps to keep us warm, too.

5. Nuclear Magnetic Resonance, Isotopomer Analysis, and Modeling of Metabolism

As as been pointed out in Boxes 3-C and 17-C, the use of ^{13}C and other isotopic tracers together with NMR and mass spectroscopy have provided powerful tools for understanding the complex interrelationships among the various interlocking pathways of metabolism. In Box 17-C the application of ^{13}C NMR to the citric acid cycle was described. Similar approaches have been used to provide direct measurement of the glucose concentration in human brain (1.0 ± 0.1 mM; 4.7 ± 0.3 mM in plasma)[226] and to study gluconeogenesis[269–271] as well as fermentation.[271a] Similar investigations have been made using mass spectroscopy.[272] The metabolism of acetate through the glycoylate pathway in yeast has been observed by ^{13}C NMR.[200] Data obtained from such experiments are being used in attempts to model metabolism and to understand how flux rates through the various pathways are altered in response to varying conditions.[65,273–276]

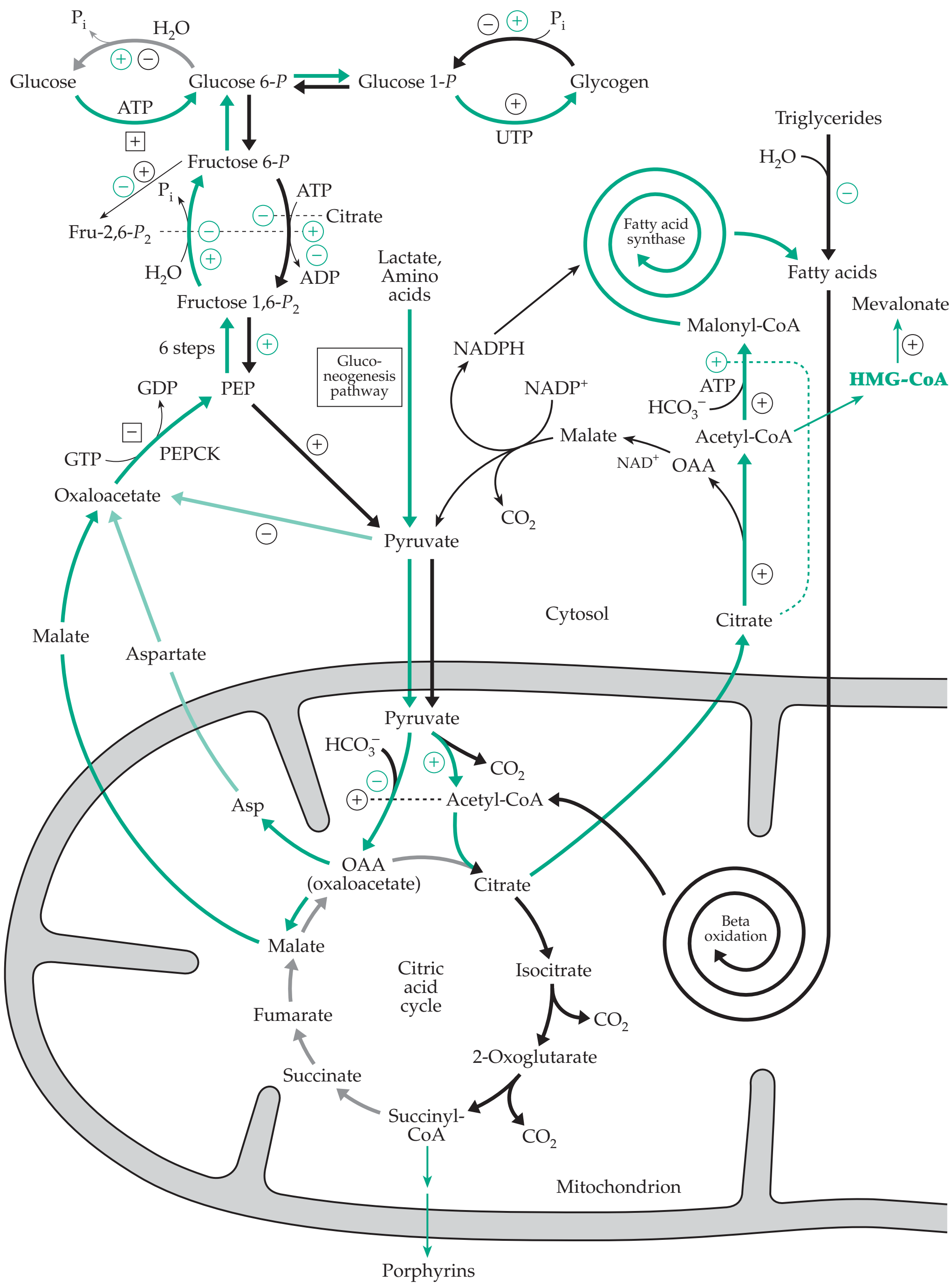


Figure 17-20 The interlocking pathways of glycolysis, gluconeogenesis, and fatty acid oxidation and synthesis with indications of some aspects of control in hepatic tissues. (⟶) Reactions of glycolysis, fatty acid degradation, and oxidation by the citric acid cycle. (⟶) Biosynthetic pathways. Some effects of insulin via indirect action on enzymes ⊕, ⊖, or on transcription ⊞, ⊟. Effects of glucagon ⊕, ⊖.

6. The Fasting State

During prolonged fasting, glycogen supplies are depleted throughout the body and fats become the principal fuels. Both glucose and pyruvate are in short supply. While the hydrolysis of lipids provides some glycerol (which is phosphorylated and oxidized to dihydroxyacetone-*P*), the quantity of glucose precursors formed in this way is limited. Since the animal body cannot reconvert acetyl-CoA to pyruvate, there is a continuing need for both glucose and pyruvate. The former is needed for biosynthetic processes, and the latter is a precursor of oxaloacetate, the regenerating substrate of the citric acid cycle. For this reason, during fasting the body readjusts its metabolism. As much as 75% of the glucose need of the brain can gradually be replaced by ketone bodies derived from the breakdown of fats (Section A,4).[277] Glucocorticoids (e.g., cortisol; Chapter 22) are released from the adrenal glands. By inducing enzyme synthesis, these hormones increase the amounts of a variety of enzymes within the cells of target organs such as the liver. Glucocorticoids also appear to increase the sensitivity of cell responses to cAMP and hence to hormones such as glucagon.[268]

The overall effects of glucocorticoids include an increased release of glucose from the liver (increased activity of glucose 6-phosphatase), an elevated blood glucose and liver glycogen, and a decreased synthesis of mucopolysaccharides. The reincorporation of amino acids released by protein degradation is inhibited and synthesis of enzymes degrading amino acids is enhanced. Among these enzymes are tyrosine and alanine aminotransferases, enzymes that initiate amino acid degradation which gives rise to the glucogenic precursors fumarate and pyruvate.

The inability of the animal body to form the glucose

BOX 17-F LACTIC ACIDEMIA AND OTHER DEFICIENCIES IN CARBOHYDRATE METABOLISM

The lactate concentration in blood can rise from its normal value of 1–2 mM to as much as 22 mM after very severe exercise such as sprinting, but it gradually returns to normal, requiring up to 6–8 h, less if mild exercise is continued. However, continuously high lactic acid levels are observed when enzymes of the gluconeogenic pathway are deficient or when oxidation of pyruvate is partially blocked.[a,b] Severe and often lethal deficiencies of the four key gluconeogenic enzymes pyruvate carboxylase, PEP carboxykinase, fructose 1,6-bisphosphatase, and glucose 6-phosphatase are known.[b] Pyruvate carboxylase deficiency may be caused by a defective carboxylase protein, by an absence of the enzyme that attaches biotin covalently to the three mitochondrial biotin-containing carboxylases (Chapter 14, Section C), or by defective transport of biotin from the gut into the blood. The latter types of deficiency can be treated successfully with 10 mg biotin per day.

Deficiency of pyruvate dehydrogenase is the most frequent cause of lactic acidemia.[a,c] Since this enzyme has several components (Fig. 15-15), a number of forms of the disease have been observed. Patients are benefitted somewhat by a high-fat, low-carbohydrate diet. Transient lactic acidemia may result from infections or from heart failure. One treatment is to administer dichloroacetate, which stimulates increased activity of pyruvate dehydrogenase, while action is also taken to correct the underlying illness.[d] Another problem arises if a lactate transporter is defective so that lactic acid accumulates in muscles.[e]

A different problem results from deficiency of enzymes of glycolysis such as phosphofructokinase (see Box 20-D), phosphoglycerate mutase, and pyruvate kinase. Lack of one isoenzyme of phosphoglycerate mutase in muscle leads to intolerance to strenuous exercise.[f] A deficiency in pyruvate kinase is one of the most common defects of glycolysis in erythrocytes and leads to a shortened erythrocyte lifetime and hereditary hemolytic anemia.[g]

Deficiency of the first enzyme of the pentose phosphate pathway, glucose 6-phosphate dehydrogenase, is widespread.[h] Its geographical distribution suggests that, like the sickle-cell trait, it confers some resistance to malaria. A partial deficiency of 6-phosphogluconolactonase (Eq. 17-12, step *b*) has also been detected within a family and may have contributed to the observed hemolytic anemia.[i]

[a] Robinson, B. H. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 1 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 1479–1499, McGraw-Hill, New York

[b] Robinson, B. H. (1982) *Trends Biochem. Sci.* **7**, 151–153

[c] McCartney, R. G., Sanderson, S. J., and Lindsay, J. G. (1997) *Biochemistry* **36**, 6819–6826

[d] Stacpoole, P. W., 17 other authors, and Dichloroacetate-Lactic Acidosis Study Group (1992) *N. Engl. J. Med.* **327**, 1564–1569

[e] Fishbein, W. N. (1986) *Science* **234**, 1254–1256

[f] DiMauro, S., Mirando, A. F., Khan, S., Gitlin, K., and Friedman, R. (1981) *Science* **212**, 1277–1279

[g] Tanaka, K. R., and Paglia, D. E. (1995) in *The Metabolic and Molecular Bases of Inherited Disease* (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 3485–3511, McGraw-Hill, New York

[h] Pandolfi, P. P., Sonati, F., Rivi, R., Mason, P., Grosveld, F., and Luzzatto, L. (1995) *EMBO J.* **14**, 5209–5215

[i] Beutler, E., Kuhl, W., and Gelbart, J. (1985) *Proc. Natl. Acad. Sci. U.S.A.* **82**, 3876–3878

precursors pyruvate or oxaloacetate from acetyl units is sometimes a cause of severe metabolic problems. Ketosis, which was discussed in Section A,4, develops when too much acetyl-CoA is produced and not efficiently oxidized in the citric acid cycle. Ketosis occurs during starvation, with fevers, and in insulin-dependent diabetes (see also Box 17-G). In cattle, whose metabolism is based much more on acetate than is ours, spontaneously developing ketosis is a frequent problem.

7. Lipogenesis

A high-carbohydrate meal leads to an elevated blood glucose concentration. The glycogen reserves within cells are filled. The ATP level rises, blocking the citric acid cycle, and citrate is exported from mitochondria (Fig. 17-20). Outside the mitochondria citrate is cleaved by the ATP-requiring citrate lyase (Eq. 14-37) to acetyl-CoA and oxaloacetate. The oxaloacetate can be reduced to malate and the latter oxidized with $NADP^+$ to pyruvate (Eq. 17-46), which can again enter the mitochondrion. In this manner acetyl groups are exported from the mitochondrion as acetyl-CoA which can be carboxylated, under the activating influence of citrate, to form malonyl-CoA, the precursor of fatty acids. The NADPH formed from oxidation of the malate provides part of the reducing equivalents needed for fatty acid synthesis. Additional NADPH is available from the pentose phosphate pathway. Thus, excess carbohydrate is readily converted into fat by our bodies. These reactions doubtless occur to some extent in most cells, but they are quantitatively

BOX 17-G DIABETES MELLITUS

The most prevalent metabolic problem affecting human beings is diabetes mellitus.[a–c] Out of a million people about 400 develop **type I** (or juvenile-onset) **insulin-dependent diabetes mellitus** (IDDM) between the ages of 8 and 12. Another 33,000 (over 3%) develop diabetes by age 40–50, and by the late 70s over 7% are affected. A propensity toward diabetes is partially hereditary, and recessive susceptibility genes are present in a high proportion of the population. The severity of the disease varies greatly. About half of the type I patients can be treated by diet alone, while the other half must receive regular insulin injections because of the atrophy of the insulin-producing cells of the pancreas. Type I diabetes sometimes develops very rapidly with only a few days of ravenous hunger and unquenchable thirst before the onset of ketoacidosis. Without proper care death can follow quickly. This suggested that a virus infection might cause the observed death of the insulin-secreting β cells of the pancreatic islets. However, the disease appears to be a direct result of an autoimmune response (Chapter 31). Antibodies directed against such proteins as insulin, glutamate decarboxylase,[d,e] and a tyrosine phosphatase[f] of the patient's own body are present in the blood. There may also be a direct attack on the β cells by T cells of the immune system (see Chapter 31).[g,h] The events that trigger such autoimmune attacks are not clear, but there is a strong correlation with susceptibility genes, in both human beings[i,j] and mice.[k,l]

Adults seldom develop type I diabetes but often suffer from **type II** or **non-insulin-dependent diabetes mellitus** (NIDDM). This is not a single disease but a syndrome with many causes. There is usually a marked decrease in sensitivity to insulin (referred to as **insulin resistance**) and poor uptake and utilization of glucose by muscles.[m] In rare cases this is a result of a mutation in the gene for the insulin molecule precursors (Eq. 10-8) or in the gene regulatory regions of the DNA.[n,o] Splicing of the mRNA[p] may be faulty or there may be defects in the structure or in the mechanisms of activation of the insulin receptors (Figs. 11-11 and 11-12).[q] The number of receptors may be too low or they may be degraded too fast to be effective. About 15% of persons with NIDDM have mutations in the insulin substrate protein IRS-1 (Chapter 11, Section G) but the significance is not clear.[m,r] Likewise, the causes of the loss of sensitivity of insulin receptors as well as other aspects of insulin resistance are still poorly understood.[s] In addition, prolonged high glucose concentrations result in decreased insulin synthesis or secretion, both of which are also complex processes. After synthesis the insulin hexamer is stored as granules of the hexamer $(\text{insulin})_6Zn_2$ (Fig. 7-18) in vesicles at low pH. For secretion to occur the vesicles must first dock at membrane sites and undergo exocytosis. The insulin dissolves, releasing the Zn^{2+}, and acts in the monomeric form.[t] Because the mechanisms of action of insulin are still not fully understood, it is difficult to interpret the results of the many studies of diabetes mellitus.

A striking symptom of diabetes is the high blood glucose level which may range from 8 to 60 mM. Lower values are more typical for mild diabetes because when the glucose concentration exceeds the renal threshold of ~8 mM the excess is secreted into the urine. Defective utilization of glucose seems to be tied to a failure of glucose to exert proper

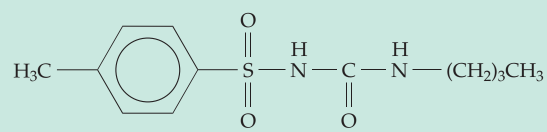
BOX 17-G DIABETES MELLITUS (continued)

feedback control. The result is that gluconeogenesis is increased with corresponding breakdown of proteins and amino acids. The liver glycogen is depleted and excess nitrogen from protein degradation appears in the urine. In IDDM diabetes the products of fatty acid degradation accumulate, leading to ketosis. The volume of urine is excessive and tissues are dehydrated.

Although the acute problems of diabetes, such as coma induced by ketoacidosis, can usually be avoided, it has not been possible to prevent long-term complications that include cataract formation and damage to the retina and kidneys. Most diabetics eventually become blind and half die within 15–20 years. Many individuals with NIDDM develop insulin-dependent diabetes in later life as a result of damage to the pancreatic β cells. The high glucose level in blood appears to be a major cause of these problems. The aldehyde form of glucose reacts with amino groups of proteins to form Schiff bases which undergo the Amadori rearrangement to form ketoamines (Eq. 4-8). The resulting modified proteins tend to form abnormal disulfide crosslinkages. Crosslinked aggregates of lens proteins may be a cause of cataract. The accumulating glucose-modified proteins may also induce autoimmune responses that lead to the long-term damage to kidneys and other organs. Another problem results from reduction of glucose to sorbitol (Box 20-A). Accumulation of sorbitol in the lens may cause osmotic swelling, another factor in the development of cataracts.[w,x] Excessive secretion of the 37-residue polypeptide **amylin**, which is synthesized in the β cells along with insulin, is another frequent complication of diabetes.[u,v] Amylin precipitates readily within islet cells to form **amyloid deposits** which are characteristic of NIDDM.

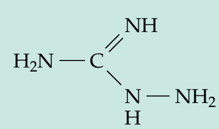
For many persons with diabetes regular injections of insulin are essential. Insulin was discovered in 1921 in Toronto by Banting and Best, with a controversial role being played by Professor J. J. R. Macleod, who shared the Nobel prize with Banting in 1923.[y,z] In 1922 the first young patients received pancreatic extracts and a new, prolonged life.[z–bb] Persons with IDDM are still dependent upon daily injections of insulin, but attempts are being made to treat the condition with transplanted cells from human cadavers.[cc] Animal insulins are suitable for most patients, but allergic reactions sometimes make **human insulin** essential. The human hormone, which differs from bovine insulin in three positions (Thr in human vs Ala in bovine at positions 8 of the A chain and 30 of the B chain and Ile vs Val at position 10 of the A chain), is now produced in bacteria using recombinant DNA. Nonenzymatic laboratory synthesis of insulin has also been achieved, but it is difficult to place the disulfide crosslinks in the proper positions. New approaches mimic the natural synthesis, in which the crosslinking takes place in proinsulin (Fig. 10-7).

NIDDM is strongly associated with obesity,[dd] and dieting and exercise often provide adequate control of blood glucose. Sulfonylurea drugs such as the following induce an increase in the number of insulin receptors formed and are also widely used in treatment of the condition.[ee,ff] These drugs bind to and inhibit ATP-sensitive K^+ channels in the β cell membranes. A defect in this sulfonylurea receptor has been associated with excessive insulin secretion



1-Butyl-*p*-tolylsulfonylurea

in infants.[17] New types of drugs are being tested.[gg–kk] These include inhibitors of aldose reductase,[ii] which forms sorbitol; compounds such as aminoguanidine, which inhibit formation of advanced products of glycation and newly discovered fungal metabolites that activate insulin receptors.[jj]



Aminoguanidine

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BOX 17-G (continued)

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most important in the liver, in fat cells of adipose tissue, and in mammary glands. The process is also facilitated by insulin, which promotes the activation of pyruvate dehydrogenase (Eq. 17-9) and of fatty acid synthase of adipocytes.[277a] Activity of fatty acid synthase seems to be regulated by the rate of transcription of its gene, which is controlled by a transcription factor designated either as **adipocyte determination and differentiation factor-1** (**ADD-1**) or **sterol regulatory element-binding protein-1c** (**SREBP-1c**). This protein (ADD-1/SREBP-1c) may be a general mediator of insulin action.[277b] The nuclear DNA-binding protein known as **peroxisome proliferator-activated receptor gamma** (**PPAR$_\gamma$**) is also involved in the control of insulin action, a conclusion based directly on discovery of mutations in persons with type II diabetes.[277c] A newly discovered hormone **resistin**, secreted by adipocytes, may also play a role.[277d] Another adipocyte hormone, **leptin**, impairs insulin action.[277e] Recent evidence suggests that both insulin and leptin may have direct effects on the brain which also influence blood glucose levels.[277f] Malonyl-CoA, which may also play a role in insulin secretion,[278,279] inhibits carnitine palmitoyltransferase I (CPT I; Fig. 17-2), slowing fatty acid catabolism.[280]

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Study Questions

1. Write out a complete step-by-step mechanism for the reactions by which citrate can be synthesized from pyruvate and then exported from mitochondria for use in the biosynthesis of fatty acids. Include a chemically reasonable mechanism for the action of ATP–citrate lyase, which catalyzes the following reaction:

   ATP + citrate + CoA-SH → Acetyl-CoA + oxaloacetate + ADP + P_i

   Show how this reaction can be incorporated into an ATP-driven cyclic pathway for generating NADPH from NADH.

2. Show which parts (if any) of the citric acid cycle are utilized in each of the following reactions and what, if any, additional enzymes are needed in each case.
   a. Oxidation of acetyl-CoA to CO_2
   b. Catabolism of glutamate to CO_2
   c. Biosynthesis of glutamate from pyruvate
   d. Formation of propionate from pyruvate

3. Here is a possible metabolic reaction for a fungus.

   L-Leucine + 2-oxoglutarate^{2-} + 2 $^1/_2$ O_2 → L-glutamate$^-$ + citrate^{3-} + H_2O + $2H^+$
   $\Delta G°'$ (pH 7) = –1026 kJ/mol

   Suggest a metabolic pathway for this reaction. Is it thermodynamically feasible?

4. It has been suggested that in *Escherichia coli* pyruvate may act as the regenerating substrate for a catalytic cycle by which glyoxylate, OHC–COO$^-$, is oxidized to CO_2. Key enzymes in this cycle are thought to be a 2-oxo-4-hydroxyglutarate aldolase and 2-oxoglutarate dehydrogenase. Propose a detailed pathway for this cycle.

5. Some bacteria use a "dicarboxylic acid cycle" to oxidize glyoxylate OHC–COO$^-$ to CO_2. The regenerating substrate for this cycle is acetyl-CoA. It is synthesized from glyoxylate by a complex pathway that begins with conversion of two molecules of glyoxylate to tartronic semialdehyde: $^-$OOC–CHOH–CHO. The latter is then dehydrogenated to D-glycerate.

   Write out a detailed scheme for the dicarboxylate cycle. Also indicate how glucose and other cell constituents can be formed from intermediates created in this biosynthetic pathway.

6. Some bacteria that lack the usual aldolase produce ethanol and lactic acid in a 1:1 molar ratio via the "heterolactic fermentation." Glucose is converted to ribulose 5-phosphate via the pentose phosphate pathway enzymes. A thiamin diphosphate-dependent "phosphoketolase" cleaves xylulose 5-phosphate in the presence of inorganic phosphate to acetyl phosphate and glyceraldehyde 3-phosphate.

   Propose a mechanism for the phosphoketolase reaction and write a balanced set of equations for the fermentation.

7. Bacteria of the genera *Aerobacter* and *Serratia* ferment glucose according to the following equation:

   1 $^1/_2$ Glucose → 2,3-butanediol + 3 CO_2 + H_2 + ethanol

   Write out a detailed pathway for the reactions. Use the pyruvate formate–lyase reaction. What yield of ATP do you expect per molecule of glucose fermented?

8. Some Clostridia ferment glucose as follows:

   2 Glucose + 2 H_2O → butyrate$^-$ + 2 acetate$^-$ + 4 CO_2 + 6 H_2 + 3 H^+
   $\Delta G°'$ (pH 7) = –479 kJ/ mol of butyrate formed

   Write out detailed pathways. How much ATP do you think can be formed per glucose molecule fermented?

9. Propionic acid bacteria use the following fermentation:

   1 $^1/_2$ Glucose → 2 propionate + acetate$^-$ + 3 H^+ + CO_2 + H_2O
   $\Delta G°'$ = –464 kJ/ mol of acetate formed

   Write out a detailed pathway for the reactions. How much ATP can be formed per molecule of glucose?

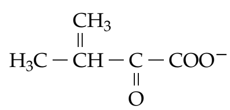
10. Consider the following reaction which can occur in the animal body:

    2 Lactate$^-$ ($C_3H_5O_3^-$) + CoASH + 2 NAD^+ → butyryl-CoA (C_4H_7O–S–CoA) + 2 CO_2 + 2 NADH + H_2O
    $\Delta G°'$ (pH 7) ~ 70 kJ/ mol of butyryl-CoA formed

    Outline the sequence of reactions involved in this transformation. Do you think that any ATP will either be used or generated in the reaction? Explain.

11. Leucine, an essential dietary constituent for human beings, is synthesized in many bacteria and plants using pyruvate as a starting material. Outline this pathway of metabolism and illustrate the chemical reaction mechanisms involved in each step.

12. Write a step-by-step sequence for all of the chemical reactions involved in the biosynthesis of L-leucine from 2-oxoisovalerate:



Notice that this compound contains one carbon atom less than leucine. Start by condensing 2-oxoisovalerate with acetyl-CoA in a reaction similar to that of citrate synthase. Use structural formulas. Show all intermediate structures and indicate what coenzymes are needed. Use curved arrows to indicate the flow of electrons in each step.

13. Some fungi synthesize lysine from 2-oxoglutarate by elongating the chain using a carbon atom derived from acetyl-CoA to form the 6-carbon 2-oxoadipate. The latter is converted by an ATP-dependent reduction to the ε-aldehyde. Write out reasonable mechanisms for the conversion of 2-oxoglutarate to the aldehyde. The latter is converted on to L-lysine by a non–PLP-dependent transamination via saccharopine (Chapter 24).

14. Outline the pathway for biosynthesis of L-leucine from glucose and NH_4^+ in autotrophic organisms. In addition, outline the pathways for degradation of leucine to CO_2, water, and NH_4^+ in the human body. For this overall pathway or "metabolic loop," mark the locations (one or more) at which each of the following processes occurs.
   a. Synthesis of a thioester by dehydrogenation
   b. Substrate-level phosphorylation
   c. Thiamin-dependent α condensation
   d. Oxoacid chain-elongation process
   e. Transamination
   f. Oxidative decarboxylation of an α-oxoacid
   g. Partial β oxidation
   h. Thiolytic cleavage
   i. Claisen condensation
   j. Biotin-dependent carboxylation

15. A photosynthesizing plant is exposed to $^{14}CO_2$. On which carbon atoms will the label first appear in glucose?

16. The Calvin–Benson cycle and the pentose phosphate pathway (Eq. 17-12) have many features in common but run in opposite directions. Since the synthesis of glucose from CO_2 requires energy, the energy expenditure for the two processes will obviously differ. Describe the points in each pathway where a Gibbs energy difference is used to drive the reaction in the desired direction.

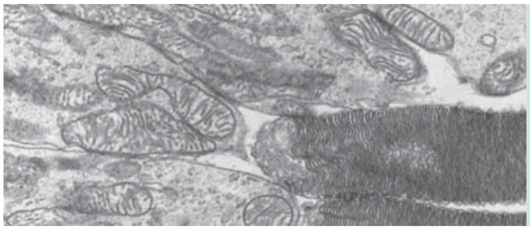
17. Draw the structure of ribulose 1(^{14}C), 5-bisphosphate. Enter an asterisk (*) next to carbon 1 to show that this position is ^{14}C-labeled.

   Draw the structures of the products of the ribulose bisphosphate carboxylase reaction, indicating the radioactive carbon position with an asterisk.

18. A wood-rotting fungus is able to convert glucose to oxalate approximately according to the following equation:

$$\text{Glucose} + 5\ O_2 \rightarrow 2\ \text{Oxalic acid}\ (C_2O_4H_2) + 2\ CO_2 + 4\ H_2O$$

   Propose a mechanism. See Munir *et al.*[281] for details.



Electrons flowing through the electron transport chains in the membranes of the mitochondria, (at the left) in this thin section through the retina of a kangaroo rat (*Dipodomys ordi*) generates ATP. The ATP provides power for the synthesis and functioning of the stacked photoreceptor disks seen at the right. The outer segment of each rod cell (See Fig. 23-40), which may be 15–20 mm in length, consists of these disks, whose membranes contain the photosensitive protein pigment rhodopsin. Absorption of light initiates an electrical excitation which is sent to the brain. Micrograph from Porter and Bonneville (1973) Fine Structure of Cells and Tissues, Lea and Febiger, Philadelphia, Pennsylvania. Courtesy of Mary Bonneville.

Contents

Electron Transport, Oxidative Phosphorylation, and Hydroxylation

18

In this chapter we will look at the processes by which reduced carriers such as NADH and $FADH_2$ are oxidized within cells. Most familiar to us, because it is used in the human body, is **aerobic respiration**. Hydrogen atoms of NADH, $FADH_2$, and other reduced carriers appear to be transferred through a chain of additional carriers of increasingly positive reduction potential and are finally combined with O_2 to form H_2O. In fact, the hydrogen nuclei move freely as protons (or sometimes as H^- ions); it is the *electrons* that are deliberately transferred. For this reason, the chain of carriers is often called the **electron transport chain**. It is also referred to as the **respiratory chain**.

Because far more energy is available to cells from oxidation of NADH and $FADH_2$ than can be obtained by fermentation, the chemistry of the electron transport chain and of the associated reactions of ATP synthesis assumes great importance. A central question becomes "How is ATP generated by flow of electrons through this series of carriers"? Not only is most of the ATP formed in aerobic and in some anaerobic organisms made by this process of **oxidative phosphorylation**, but the solar energy captured during photosynthesis is used to form ATP in a similar manner. The mechanism of ATP generation may also be intimately tied to the function of membranes in the transport of ions. In a converse manner, the mechanism of oxidative phosphorylation may be related to the utilization of ATP in providing energy for the contraction of muscles.

In some organisms, especially bacteria, energy may be obtained through oxidation of H_2, H_2S, CO, or Fe^{2+} rather than of the hydrogen atoms removed from organic substrates. Furthermore, some bacteria use **anaerobic respiration** in which NO_3^-, SO_4^{2-}, or CO_2 act as oxidants either of reduced carriers or of reduced inorganic substances. In the present chapter, we will consider these energy-yielding processes as well as the chemistry of reactions of oxygen that lead to incorporation of atoms from O_2 into organic compounds.

The oxidative processes of cells have been hard to study, largely because the enzymes responsible are located in or on cell membranes. In bacteria the sites of electron transport and oxidative phosphorylation are on the inside of the plasma membrane or on membranes of mesosomes. In eukaryotes they are found in the inner membranes of the mitochondria and, to a lesser extent, in the endoplasmic reticulum. For this reason we should probably start with a closer look at mitochondria, the "power plants of the cell."

A. The Architecture of the Mitochondrion

Mitochondria are present in all eukaryotic cells that use oxygen in respiration, but the number per cell and the form and size vary.[1–4] Certain tiny trypanosomes have just *one* mitochondrion but some oocytes have as many as 3×10^5. Mammalian cells typically contain several hundred mitochondria and liver cells[5] more than 1000. Mammalian sperm cells may contain 50–75 mitochondria,[6] but in some organisms only one very large helical mitochondrion, formed by the fusion of many individual mitochondria, wraps around the base of the tail. Typical mitochondria appear to be about the size of cells of *E. coli*. However, study of ultrathin serial sections of a single yeast cell by electron microscopy has shown that, under some growth conditions, all of the mitochondria are interconnected.[7]

In every case a mitochondrion is enclosed by two concentric membranes, an *outer* and an *inner* membrane, each ~5–7 nm thick (Figs. 18-1, 18-2). The inner membrane is folded to form the **cristae**. The number of cristae, the form of the cristae, and the relative amount of the internal **matrix** space are variable. In liver there is little inner membrane and a large matrix space, while in heart mitochondria there are more folds and a higher rate of oxidative phosphorylation. The enzymes catalyzing the tricarboxylic acid cycle are also unusually active in heart mitochondria. A typical heart mitochondrion has a volume of 0.55 μm^3; for every cubic micrometer of mitochondrial volume there are 89 μm^2 of inner mitochondrial membranes.[9]

Mitchondria can swell and contract, and forms other than that usually seen in osmium-fixed electron micrographs have been described. In some mitochondria the cristae are swollen, the matrix volume is much reduced, and the **intermembrane space** between the membranes is increased. Rapidly respiring mitochondria fixed for electron microscopy exhibit forms that have been referred to as "energized" and "energized-twisted."[10] The micrograph (Fig. 18-1) and drawing (Fig. 18-2) both show a significant amount of intermembrane space. However, electron micrographs of mitochondria from rapidly frozen aerobic tissues show almost none.[11] Recent studies by electron microscopic tomography show cristae with complex tubular structures. The accepted simple picture of mitochondrial structure (Fig. 18-1) is undergoing revision.[12–12b] The isolated mitochondria that biochemists have studied may be fragments of an interlinked **mitochondrial reticulum** that weaves its way through the cell.[12b] However, this reticulum may not be static but may break and reform. The accepted view that the mitochondrial matrix space is continuous with the internal space in the cristae is also the subject of doubts. Perhaps they are two different compartments.[12a]

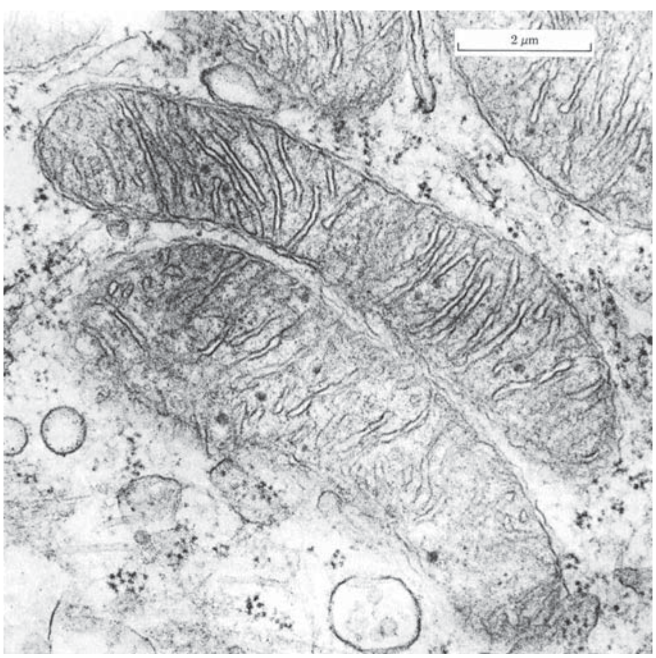


Figure 18-1 Thin section of mitochondria of a cultured kidney cell from a chicken embryo. The small, dark, dense granules within the mitochondria are probably calcium phosphate. Courtesy of Judie Walton.

1. The Mitochondrial Membranes and Compartments

The outer membranes of mitochondria can be removed from the inner membranes by osmotic rupture.[13] Analyses on separated membrane fractions show that *the outer membrane* is less dense (density ~1.1 g / cm^3) than the inner (density ~1.2 g / cm^3). It is highly permeable to most substances of molecular mass 10 kDa or less because of the presence of pores of ~2 nm diameter. These are formed by **mitochondrial porins**,[14–17] which are similar to the outer membrane porins of gram-negative bacteria (Fig. 8-20). The ratio of phospholipid to protein (~0.82 on a weight basis) is much higher than in the inner membrane. Extraction of the phospholipids by acetone destroys the membrane. Of the lipids present, there is a low content of cardiolipin, a high content of phosphatidylinositol and cholesterol, and no ubiquinone.

The inner membrane is impermeable to many substances. Neutral molecules of <150 Da can penetrate the membranes, but the permeability for all other materials including small ions such as H^+, K^+, Na^+, and Cl^- is tightly controlled. The ratio of phospholipid to protein in the inner membrane is ~0.27, and cardiolipin makes up ~20% of the phospholipid present. Cholesterol is absent. Ubiquinone and other components of the respiratory chain are all found in the inner membrane. Proteins account for 75% of the mass of the membrane.

Another characteristic of the inner mitochondrial membrane is the presence of projections on the inside surface, which faces the mitochondrial matrix. See Fig. 18-14. These spherical 85-kDa particles, discovered by Fernandez Moran in 1962 and attached to the membrane through a "stalk", display ATP-hydrolyzing (ATPase) activity. The latter was a clue that the knobs might participate in the *synthesis* of ATP during oxidative phosphorylation. In fact, they are now recognized as a complex of proteins called **coupling factor 1** (F_1) or **ATP synthase**.

In addition to bacterialike mitochondrial ribosomes and small circular molecules of DNA, mitochondria may contain variable numbers of dense granules of calcium phosphate, either $Ca_3(PO_4)_2$ or hydroxylapatite (Fig. 8-34),[4,18] as well as of phospholipoprotein.[4]

Quantitatively the major constituents of the matrix space are a large number of proteins. These account for about 56% by weight of the matrix material and exist in a state closer to that in a protein crystal than in a true solution.[19–20a] Mitochondrial membranes also contain proteins, both tightly bound relatively nonpolar intrinsic proteins and extrinsic proteins bound to the membrane surfaces. The other mitochondrial compartment, the intermembrane space, may normally be very small but it is still "home" for a few enzymes.

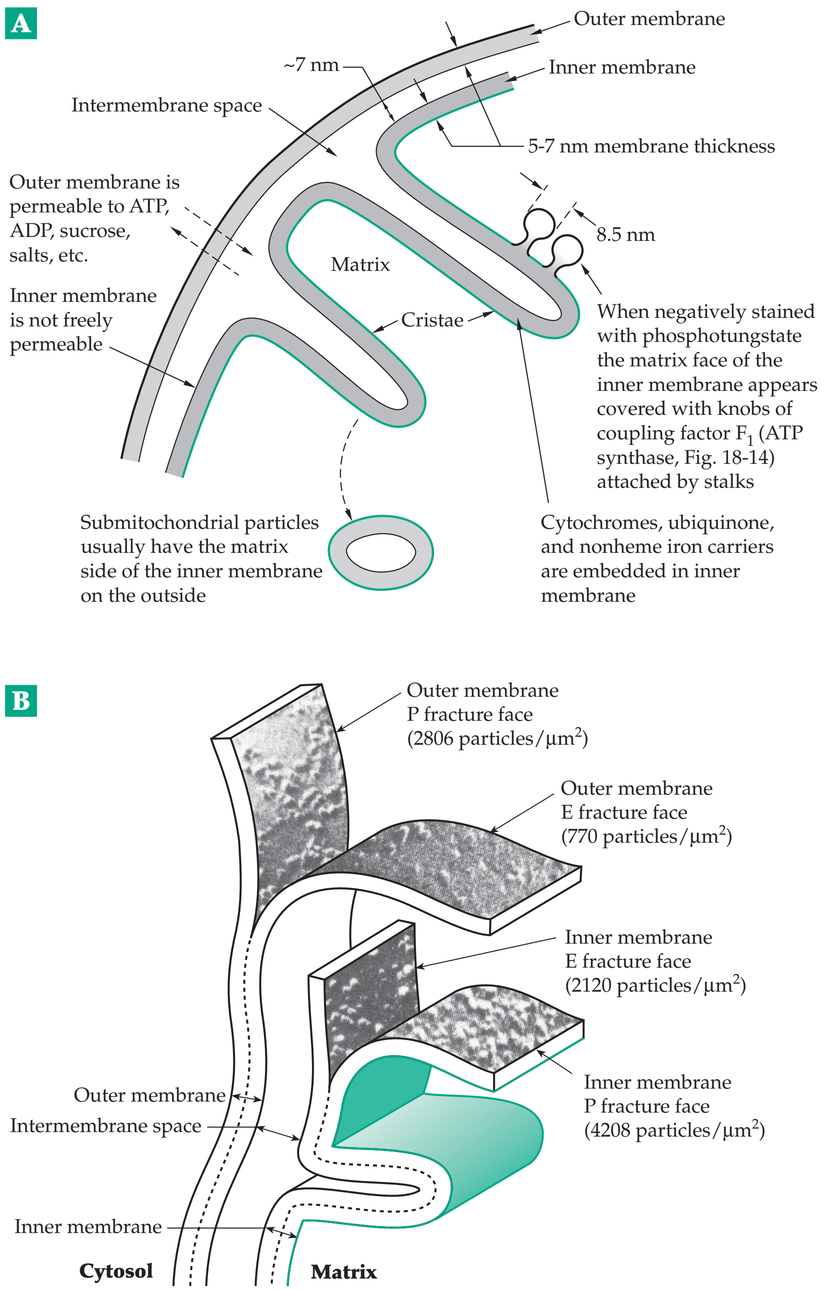


Figure 18-2 (A) Schematic diagram of mitochondrial structure. (B) Model showing organization of particles in mitochondrial membranes revealed by freeze-fracture electron microscopy. The characteristic structural features seen in the four half-membrane faces (EF and PF) that arise as a result of fracturing of the outer and inner membranes are shown. The four smooth membrane surfaces (ES and PS) are revealed by etching. From Packer.[8]

2. The Chemical Activities of Mitochondria

Mention of mitochondria usually brings to the mind of the biochemist the **citric acid cycle,** the **β oxidation pathway** of fatty acid metabolism, and **oxidative phosphorylation**. In addition to these major processes, many other chemical events also occur. Mitochondria concentrate Ca^{2+} ions and control the entrance and exit of Na^+, K^+, dicarboxylates, amino acids, ADP, P_i and ATP, and many other substances.[16] Thus, they exert regulatory functions both on catabolic and biosynthetic sequences. The glycine decarboxylase system (Fig. 15-20) is found in the mitochondrial matrix and is especially active in plant mitochondria (Fig. 23-37). Several cytochrome P450-dependent hydroxylation reactions, important to the biosynthesis and catabolism of steroid hormones and to the metabolism of vitamin D, take place within mitochondria. Mitochondria make only a few of their own proteins but take in several hundred other proteins from the cytoplasm as they grow and multiply.

Where within the mitochondria are specific enzymes localized? One approach to this question is to see how easily the enzymes can be dissociated from mitochondria. Some enzymes come out readily under hypotonic conditions. Some are released only upon sonic oscillation, suggesting that they are inside the matrix space. Others, including the cytochromes and the flavoproteins that act upon succinate and NADH, are so firmly embedded in the inner mitochondrial membranes that they can be dissociated only through the use of non-denaturing detergents.

Because the enzymes of the citric acid cycle[20a] (with the exception of succinate dehydrogenase) and β oxidation are present in the matrix, the reduced electron carriers must approach the inner membrane from the matrix side (the M side). Thus, the embedded enzymes designed to oxidize NADH, succinate, and other reduced substrates must be accessible from the matrix side. However, *sn*-glycerol 3-phosphate dehydrogenase, a flavoprotein, is accessible from the "outside" of the cytoplasmic (C side) of the inner membrane.[21] Fluorescent antibodies to cytochrome *c* bind only to the cytoplasmic (intermembrane) side of the inner membrane, but antibodies to cytochrome oxidase label both sides, which suggested that this protein complex spans the membrane.[22] However, oxidation of cytochrome *c* by cytochrome oxidase occurs only on the cytoplasmic surface.[22] Antibodies to the ATP synthase that makes up the knobs bind strictly to the matrix side.

The outer mitochondrial membrane contains monoamine oxidase, cytochrome b_5, fatty acyl-CoA synthase, and enzymes of cardiolipin synthesis[22a] as well as other proteins. Cardiolipin (diphosphatidylglycerol; Fig. 21-4) is found only in the inner mitochondrial membrane and in bacteria. It is functionally important for several mitochondrial enzymes including cytochrome oxidase and cytochome bc_1.[22a–c] It is also essential to photosynthetic membranes for which an exact role in an interaction between the lipid membrane and the associated protein has been revealed by crystallography.[22d] In other respects the composition of the inner mitochondrial membrane resembles that of the membranes of the endoplasmic reticulum. Isoenzyme III of adenylate kinase, a key enzyme involved in equilibrating ATP and AMP with ADP (Eq. 6-65), is one of the enzymes present in the intermembrane space. A number of other kinases, as well as sulfite oxidase, are also present between the membranes.[4]

As mentioned in Box 6-D, mitochondria sometimes take up calcium ions. The normal total concentration of Ca^{2+} is ~1 mM and that of free Ca^{2+} may be only ~0.1 µM.[22e,f] However, under some circumstances mitochondria accumulate large amounts of calcium, perhaps acting as a Ca^{2+} buffer.[22g,f] The so called ryanodine receptors (Fig. 19-21), prominent in the endoplasmic reticulum, have also been found in heart mitochondria, suggesting a function in control of calcium oscillations.[22i,j] On the other hand, accumulation of calcium by mitochondria may be pathological and the activation of Ca^{2+}-dependent proteases may be an initial step in apoptosis.[22h,22k]

TABLE 18-1
Catalog of Mitochondrial Genes[a]

Name and symbol	*Homo sapiens*	*Reclinomonas americana*	*Saccharomyces cerevisiae*	*Arabidopsis thaliana*
Ribosomal RNA				
s rRNA (small, 12s)	1	1	1	1
l rRNA (large, 16s)	1	1	1	1
5 S RNA		1		1
Transfer RNAs	22[b]	26	24	22
NADH dehydrogenase				
Subunits ND1–6, ND4L	7	12	0	9
Cytochrome *b*	1	1	1	1
Cytochrome oxidase				
Subunits I, II, III	3	3	3	3
ATP synthase				
Subunits 6, 8, others	2	5	3	4
Total protein coding genes	13	62	8	27
Total genes	37	92	35	53
Size of DNA (kbp)	16.596	69	75	367

[a] Data from Palmer, J. D. (1997) *Nature (London)* **387**, 454–455.[1]
[b] One for each amino acid of the genetic code but two each for serine and leucine.

3. The Mitochondrial Genome

Each mitochondrion contains several molecules of DNA (**mtDNA**), usually in a closed, circular form, as well as the ribosomes, tRNA molecules, and enzymes needed for protein synthesis.[1,23–26] With rare exceptions almost all of the mitochondrial DNA in a human cell is inherited from the mother.[6,26a] The size of the DNA circles varies from 16–19 kb in animals[27] to over 200 kb in many higher plants. Complete sequences of many mitochondrial DNAs are known.[28,28a] Among these are the 16,569 bp human mtDNA,[29] the 16,338 bp bovine mtDNA, the 16,896 bp mtDNA of the wallaroo *Macropus robustus*,[30] and the 17,533 bp mtDNA of the amphibian *Xenopus laevis*.[31,32] The sea urchin *Paracentotus lividus* has a smaller 15,697 bp genome. However, the order of the genes in this and other invertebrate mtDNA is different from that in mammalian mitochondria.[33] Protozoal mtDNAs vary in size from ~5900 bp for the parasitic malaria organism *Plasmodium falciparum*[34,35] to 41,591 bp for *Acanthamoeba castellani*[36] and 69,034 bp for the fresh water flagellate *Reclinomonas americana*.[26,37]

All of the mammalian mtDNAs are organized as shown in Fig. 18-3. The two strands of the DNA can be separated by virtue of their differing densities. The heavy (H) strand has a 5'→3' polarity in a counterclockwise direction in the map of Fig. 18-3, while the light (L) strand has a clockwise polarity. From the sequences 13 genes for specific proteins, 2 genes for ribosomal RNA molecules, and 22 genes for transfer RNAs have been identified. The genes are listed in Table 18-1 and have also been marked on the map in Fig. 18-3. The map also shows the tRNA genes, labeled with standard one-letter amino acid abbreviations, and the directions of transcription. Most of the protein genes are on the H-strand. One small region, the D-loop, contains an origin of replication and control signals for transcription (see Chapters 27 and 28).

The genes in mammalian mtDNA are closely packed with almost no nucleotides between them. However, the 19.5-kb mtDNA of *Drosophila* contains an (A+T)-rich region, which varies among species.[38] In the much larger 78-kb genome of yeast *Saccharomyces cerevisiae* many (A+T)-rich spacer regions are present.[39] The yeast mitochondrial genome also contains genes for several additional proteins. Mitochondria of *Reclinomonas americana* contain 97 genes, including those for 5S RNA, the RNA of ribonuclease P as well as a variety of protein coding genes. Perhaps this organism is primitive, resembling the original progenitor of eukaryotic life.[26] The mtDNA of trypanosomes is present in the large mitochondrion or kinetoplast as 40–50 "maxicircles" ~20–35 kb in size, together with 5000–10,000 "minicircles", each of 645–2500 bp (see Fig. 5-16). The latter encode **guide RNA** for use in RNA editing (Chapter 28). The large mitochondrial DNAs of higher plants, e.g., *Arabidopsis* (Table 18-1), also contain additional protein genes as well as large segments of DNA between the genes. The genome of the turnip (*Brassica campestris*) exists both as a 218-kb **master chromosome** and smaller 83- and 135-kb incomplete chromosomes, a pattern existing for most land plants.[40–42] The muskmelon contains 2500 kb of mitochondrial DNA (mtDNA). On the other hand, most mtDNA of the liverwort *Marchantia polymorpha* consists of 186-kb linear duplexes.[42a]

The compact size of the mammalian genome is dependent, in part, on alterations in the genetic code, as shown in Table 18-2, and a modification of tRNA structures that permits mitochondria to function with a maximum of 22 tRNAs (Chapter 28).[43–45] However, the more primitive *Reclinomonas* utilizes the standard genetic code in its mitochondria.[26] The mammalian mitochondrial genes contain no introns, but some yeast mitochondrial genes do. Furthermore, some of the introns contain long open reading frames. At least two of these genes-within-genes encode enzymes that excise the introns.

Why does mtDNA contain *any* protein genes, or why does mtDNA even exist? It seems remarkable that the cells of our bodies make the 100 or so extra proteins (encoded in the nucleus) needed for replication, transcription, amino acid activation, and mitochondrial ribosome formation and bring these into the mitochondria for the sole purpose of permitting the synthesis there of 13 proteins. The explanation is not evident. What are the 13 proteins?

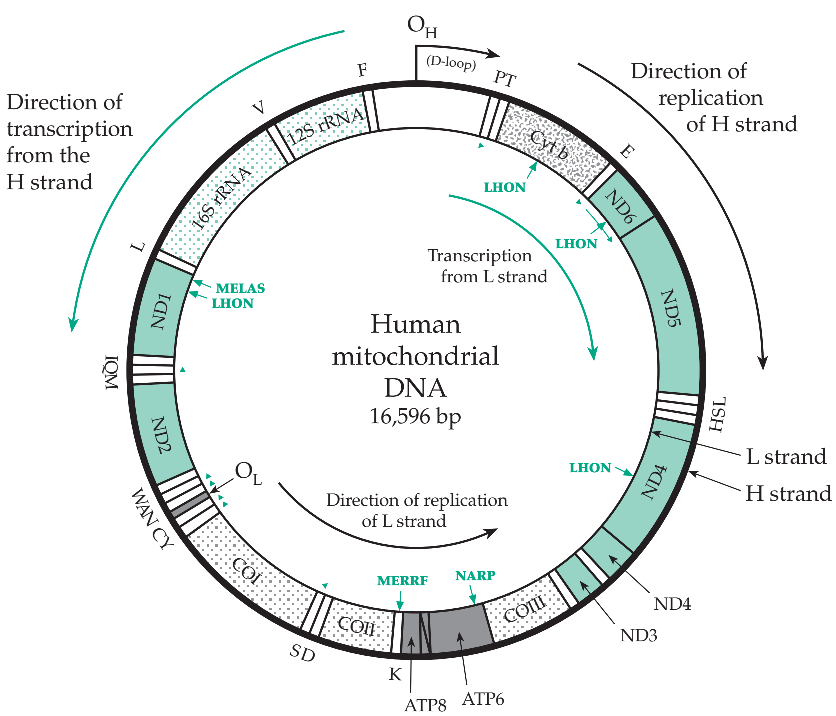


Figure 18-3 Genomic map of mammalian mitochondrial DNA. The stippled areas represent tRNA genes which are designated by the single-letter amino acid code; polarity is counterclockwise except for those marked by green arrow heads. All protein-coding genes are encoded on the H strand (with counterclockwise polarity), with the exception of ND6, which is encoded on the L strand. COI, COII, and COIII: cytochrome oxidase subunits I, II, and III; Cyt *b*: cytochrome *b*; ND: NADH dehydrogenase; ATP:ATP synthase. O_H and O_L: the origins of H and L strand replication, respectively. After Wallace[46] and Shoffner and Wallace.[45] Positions of a few of many known mutations that cause serious diseases are marked using abbrevations defined in Box 18-B.

TABLE 18-2
Alterations in the Genetic Code in the DNA of Animal Mitochondria

Codons	Nuclear DNA[a]	Mitochondrial DNA
AGA, ACG	Arg	Termination
AUA	Ile	Met
UGA	Termination	Trp

[a] See Table 5-5 for the other "standard" codons.

Three are the large functional subunits of cytochrome oxidase, one is cytochrome *b*, and seven are subunits of the NADH dehydrogenase system (Complex I). Two are subunits of ATP synthase. These are all vitally involved in the processes of electron transport and oxidative phosphorylation, but so are other proteins that are imported from the cytoplasm.

One gene in yeast mtDNA is especially puzzling. The *var 1* gene encodes a mitochondrial ribosomal protein, whose sequence varies with the strain of yeast. The gene is also involved in unusual recombinational events.[47] Another unusual aspect of yeast mitochondrial genetics is the frequent appearance of "petite" mutants, which grow on an agar surface as very small colonies. These have lost a large fraction of their mitochondrial DNA and, therefore, the ability to make ATP by oxidative phosphorylation. The remaining petite mtDNA may sometimes become integrated into nuclear DNA.[48] A few eukaryotes that have no aerobic metabolism also have no mitochondria.[49]

4. Growth and Development

Mitochondria arise by division and growth of preexisting mitochondria. Because they synthesize only a few proteins and RNA molecules, they must import many proteins and other materials from the cytoplasm. A mitochondrion contains at least 100 proteins that are encoded by nuclear genes.[50,50a] The mechanisms by which proteins are taken up by mitochondria are complex and varied. Many of the newly synthesized proteins carry, at the N terminus, presequences that contain **mitochondrial targeting signals**[51–53] (Chapter 10). These amino acid sequences often lead the protein to associate with receptor proteins on the outer mitochondrial membrane and subsequently to be taken up by the mitochondria. While the targeting sequences are usually at the N terminus of a polypeptide, they are quite often internal. The N-terminal sequences are usually removed by action of the **mitochondrial processing peptidase** (MPP) in the matrix, but internal targeting sequences are not removed.[52] Targeting of proteins to mitochondria may be assisted by **mRNA binding proteins** that guide appropriate mRNAs into the vicinity of mitochondria or other organelles.[53]

In addition to targeting signals, polypeptides destined for the inner mitochondrial membrane contain additional **topogenic signals** that direct the polypeptide to its destination. These topogenic signals are distinct from the targeting signals, which they sometimes follow. Topogenic signals are usually hydrophobic sequences, which may become transmembrane segments of the protein in its final location.[52,54] The uptake of many proteins by mitochondria requires the electrical potential that is usually present across the inner membrane (Section E). The fact that mitochondrial proteins usually have higher isoelectric points and carry more positive charges at neutral pH favors uptake.[55] In addition, chaperonins assist in

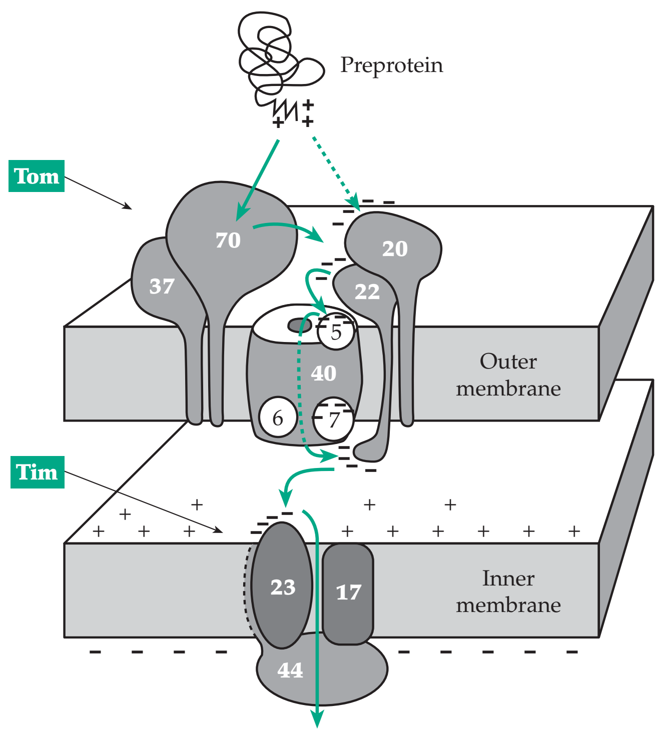


Figure 18-4 Schematic diagram of the protein transport machinery of mitochondrial membranes labeled according to the uniform nomenclature.[57] Subunits of outer membrane receptors and translocase are labeled Tom (translocase of outer membrane) and those of the inner membrane Tim (translocase of inner membrane). They are designated Tom70, etc., according to their sizes in kilodaltons (kDa). Preproteins are recognized by receptor Tom70•Tom 37 and / or by Tom22•Tom20. Clusters of negative charges on many components help guide the preprotein through the uptake pores in one or both membranes.[50,58] See Pfanner *et al.*,[57] Schatz,[50,50a] and Gabriel *et al.*[50b]

unfolding the protein to be taken up, assist in transport of some proteins,[50a] and may help the imported proteins to assemble into oligomeric structures.[51–53,56]

Protein uptake also requires a set of special proteins described as the **translocase of the outer mitochondrial membrane** (**Tom**) and **translocase of the mitochondrial inner membrane** (**Tim**). Subunits that form the receptor targets and transport pores are designated, according to their approximate molecular masses in kD as Tom70, Tim23, etc. (Fig. 18-4).[57] Preproteins are recognized by the receptor complexes Tom70 • Tom37 and / or Tom22 • Tom20 on the mitochondrial surface. They then enter the **general import pore** formed by Tom40, Tom6, and Tom7 with the assistance of a small integral membrane protein Tom5, which has a positively charged C-terminal membrane anchor segment and a negatively charged N-terminal portion that may bind to the positively charged mitochondrial targeting sequences.[50,59] A number of other translocase components, including Tom20 and Tom22 of the outer membrane and Tim23 of the inner membrane, also have acidic extramembranous domains.[58] This suggested an "acid chain" hypothesis according to which the targeting signal interacts consecutively with a series of acidic protein domains that help to guide it across the two membranes.[50a,58,59] A series of small proteins, Tim8, 9, 10, 12, 13, function in yeast mitochondria to mediate the uptake of metabolite transporters. A defect in the human nuclear gene for a protein that resembles Tim8 causes **deafness dystonia**, a recessive X-linked neurodegenerative disorder.[59a,b]

B. Electron Transport Chains

During the 1940s, when it had become clear that formation of ATP in mitochondria was coupled to electron transport, the first attempts to pick the system apart and understand the molecular mechanism began. This effort led to the identification and at least partial characterization of several flavoproteins, iron-sulfur centers, ubiquinones, and cytochromes, most of which have been described in Chapters 15 and 16. It also led to the picture of mitochondrial electron transport shown in Fig. 10-5 and which has been drawn in a modern form in Fig. 18-5.

1. The Composition of the Mitochondrial Electron Transport System

Because of the difficulty of isolating the electron transport chain from the rest of the mitochondrion, it is easiest to measure ratios of components (Table 18-3). Cytochromes a, a_3, b, c_1, and c vary from a 1:1 to a 3:1 ratio while flavins, ubiquinone, and nonheme iron occur in relatively larger amounts. The much larger amount of pyridine nucleotides is involved in carrying electrons from the various soluble dehydrogenases of the matrix to the immobile carriers in the inner membrane, while ubiquinone has a similar function within the lipid bilayer of mitochondrial membranes.

TABLE 18-3
Ratios of Components in the Electron Transport Chain of Mitochondria[a,b]

Electron carrier	Rat liver mitochondria	Beef heart mitochondria
Cytochrome a_3	1.0	1.1
Cytochrome a	1.0	1.1
Cytochrome b	1.0	1.0
Cytochrome c_1	0.63	0.33–0.51
Cytochrome c	0.78	0.66–0.85
Pyridine nucleotides	24	
Flavins	3	1
Ubiquinone	3–6	7
Copper		2.2
Nonheme iron		5.5

[a] From Wainio, W. W. (1970) *The Mammalian Mitochondrial Respiratory Chain*, Academic Press, New York, and references cited therein.
[b] Molecular ratios are given. Those for the cytochromes refer to the relative numbers of heme groups.

What are the molar concentrations of the electron carriers in mitochondrial membranes? In one experiment, cytochrome b was found in rat liver mitochondria to the extent of 0.28 μmol/g of protein. If we take a total mitochondrion as about 22% protein, the average concentration of the cytochrome would be ~0.06 mM. Since all the cytochromes are concentrated in the inner membranes, which may account for 10% or less of the volume of the mitochondrion, the concentration of cytochromes may approach 1 mM in these membranes. This is sufficient to ensure rapid reactions with substrates.

2. The Sequence of Electron Carriers

Many approaches have been used to deduce the sequence of carriers through which electron flow takes place (Fig. 18-5). In the first place, it seemed reasonable to suppose that the carriers should lie in order of increasing oxidation–reduction potential going from left to right of the figure. However, since the redox potentials existing in the mitochondria may be somewhat different from those in isolated enzyme preparations, this need not be strictly true.

The development by Chance of a dual wavelength spectrophotometer permitted easy observation of the state of oxidation or reduction of a given carrier within mitochondria.[60] This technique, together with the study of specific inhibitors (some of which are indicated in Fig. 18-5 and Table 18-4), allowed some electron transport sequences to be assigned. For example, blockage with **rotenone** and **amytal** prevented reduction of the cytochrome system by NADH but allowed reduction by succinate and by other substrates having their own flavoprotein components in the chain. Artificial electron acceptors, some of which are shown in Table 18-5, were used to bypass parts of the chain as indicated in Fig. 18-5.

Submitochondrial particles and complexes. Many methods have been employed to break mitochondrial membranes into submitochondrial particles that retain an ability to catalyze some of the reactions of the chain.[61] For example, the Keilin–Hartree preparation of heart muscle is obtained by homogenizing mitochondria and precipitation at low pH.[62] The resulting particles have a low cytochrome *c* content and do not carry out oxidative phosphorylation.

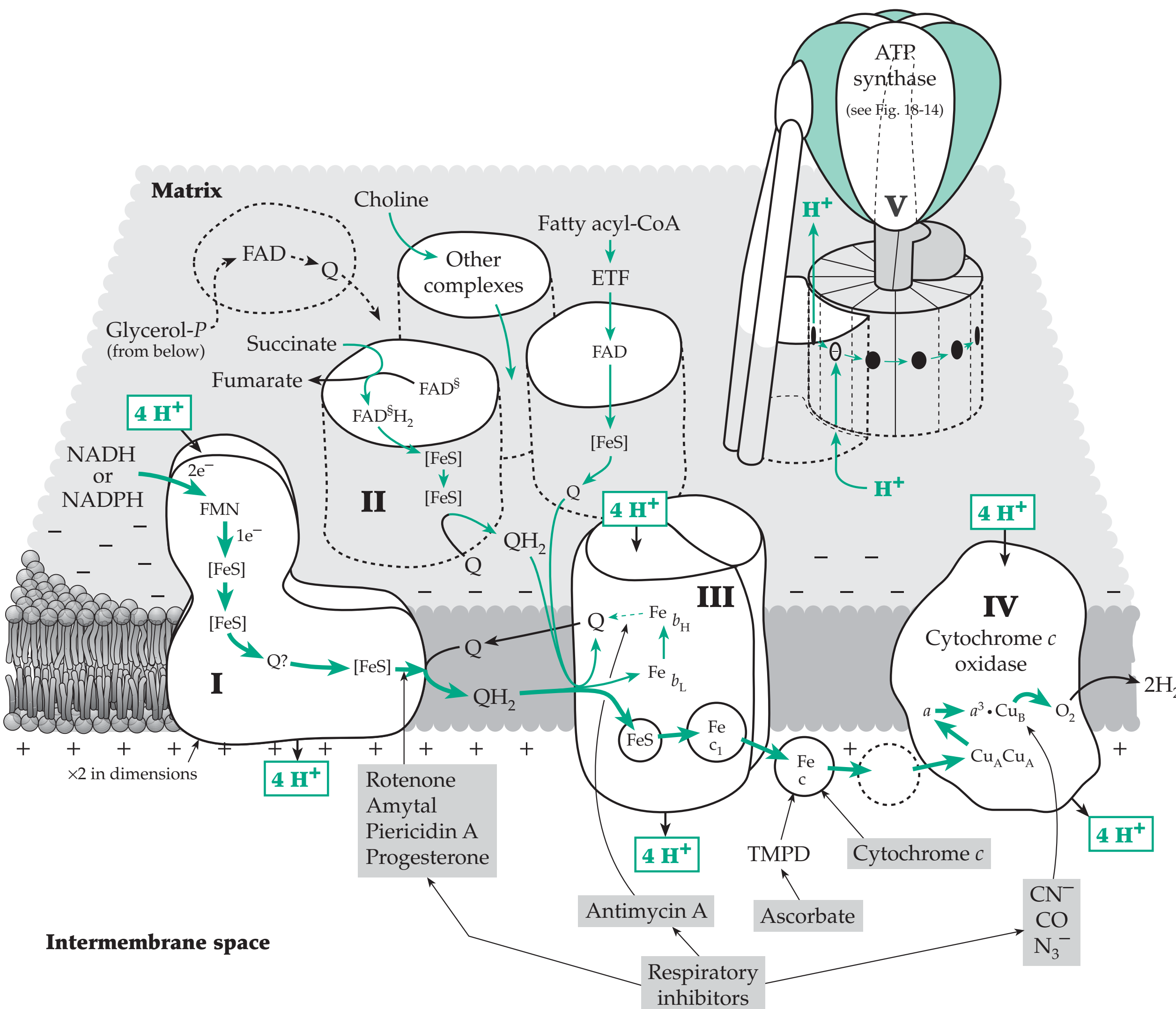


Figure 18-5 A current concept of the electron transport chain of mitochondria. Complexes I, III, and IV pass electrons from NADH or NADPH to O_2, one NADH or two electrons reducing one O to H_2O. This electron transport is coupled to the transfer of about 12 H^+ from the mitochondrial matrix to the intermembrane space. These protons flow back into the matrix through ATP synthase (V), four H^+ driving the synthesis of one ATP. Succinate, fatty acyl-CoA molecules, and other substrates are oxidized via complex II and similar complexes that reduce ubiquinone Q, the reduced form QH_2 carrying electrons to complex III. In some tissues of some organisms, glycerol phosphate is dehydrogenated by a complex that is accessible from the intermembrane space.

However, they do transport electrons and react with O_2. Other electron transport particles have been prepared by sonic oscillation. Under the electron microscope such particles appear to be small membranous vesicles resembling mitochondrial cristae.

Many detergents are strong denaturants of proteins, but some of them disrupt mitochondrial membranes without destroying enzymatic activity. A favorite is **digitonin** (Fig. 22-12), which causes disintegration of the outer membrane. The remaining fragments of inner membrane retain activity for oxidative phosphorylation. Such submitochondrial particles can be fractionated further by chemical treatments. Separate complexes can be obtained by treating the inner membranes with the nondenaturing detergent cholate (Fig. 22-10) and isolating the complexes by differential salt fractionation using ammonium sulfate. The isolated complexes I – IV catalyze reactions of four different portions of the electron transport process[63–65] as indicated in Eq. 18-1:

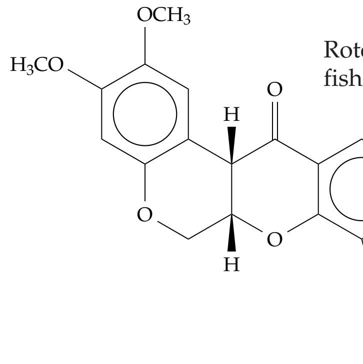
$$\begin{array}{l} \text{Succinate} \xrightarrow{\text{II}} \text{Q} \\ \text{NADH} \xrightarrow{\text{I}} \text{Q} \xrightarrow{\text{III}} \text{cytochrome } c \xrightarrow{\text{IV}} \text{O}_2 \end{array} \qquad (18\text{-}1)$$

These complexes are usually named as follows: I, **NADH-ubiquinone oxidoreductase**; II, **succinate-ubiquinone oxidoreductase**; III, **ubiquinol-cytochrome *c* oxidoreductase**; IV, **cytochrome *c* oxidase**. The designation **complex V** is sometimes applied to ATP synthase (Fig. 18-14). Chemical analysis of the electron transport complexes verified the probable location of some components in the intact chain. For example, a high iron content was found in both complexes I and II and copper in complex IV.

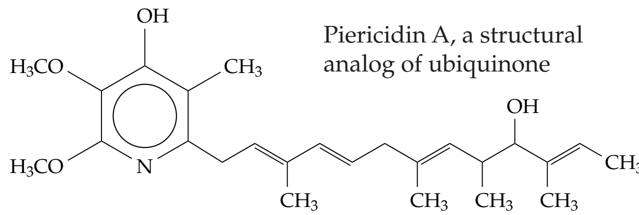
We now recognize not only that these complexes are discrete structural units but also that they are functional units. Complete X-ray crystallographic structures are available for complexes III and IV and for much of the ATP synthase complex. As is indicated in Fig. 18-5, complexes I – IV are linked by two soluble electron carriers, ubiquinone and cytochrome *c*.

The lipid-soluble ubiquinone (Q) is present in both bacterial and mitochondrial membranes in relatively large amounts compared to other electron carriers (Table 18-2). It seems to be located at a point of convergence of the NADH, succinate, glycerol phosphate, and choline branches of the electron transport chain. Ubiquinone plays a role somewhat like that of NADH, which carries electrons between dehydrogenases in the cytoplasm and from soluble dehydrogenases in the aqueous mitochondrial matrix to flavoproteins embedded in the membrane. Ubiquinone transfers electrons plus protons between proteins within the

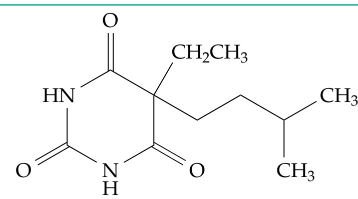
TABLE 18-4
Some Well-Known Respiratory Inhibitors[a]



Rotenone, an insecticide and fish poison from a plant root



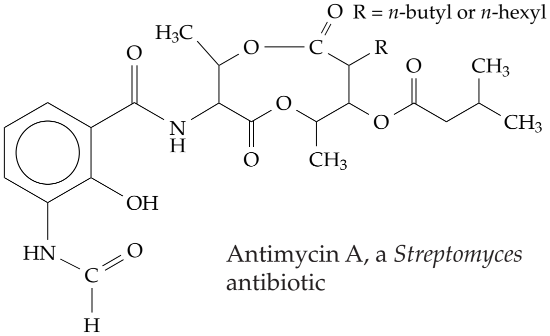
Piericidin A, a structural analog of ubiquinone



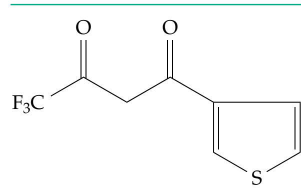
Amytal (amobarbital)

Progesterone (See Fig. 22-11)

R = *n*-butyl or *n*-hexyl



Antimycin A, a *Streptomyces* antibiotic



Thenoyltrifluoroacetone (TTB, 4,4,4-trifluoro-1-(2-thienyl)-1,3-butadione)

Cyanide	$^{-}C \equiv N$
Azide	$N \equiv N \equiv N^{-}$
Carbon monoxide	CO

[a] See Fig. 18-5 for sites of inhibition.

BOX 18-A HISTORICAL NOTES ON RESPIRATION

Animal respiration has been of serious interest to scientists since 1777, when Lavoisier concluded that foods undergo slow combustion within the body, supposedly in the blood. In 1803–1807, Spallanzani established for the first time that the tissues were the actual site of respiration, but his conclusions were largely ignored. In 1884, MacMunn discovered that cells contain the heme pigments, which are now known as **cytochromes**. However, the leading biochemists of the day dismissed the observations as experimental error, and it was not until the present century that serious study of the chemistry of biological oxidations began.[a,b]

Recognition that substrates are oxidized by **dehydrogenation** is usually attributed to H. Wieland. During the years 1912–1922 he showed that synthetic dyes, such as methylene blue, could be substituted for oxygen and would allow respiration of cells in the absence of O_2. Subsequent experiments (see Chapter 15) led to isolation of the soluble pyridine nucleotides and flavoproteins and to development of the concept of an electron transport chain.

Looking at the other end of the respiratory chain, Otto Warburg[c,d] noted in 1908 that all aerobic cells contain iron. Moreover, iron-containing charcoal prepared from blood catalyzed nonenzymatic oxidation of many substances, but iron-free charcoal prepared from cane sugar did not. Cyanide was found to inhibit tissue respiration at low concentrations similar to those needed to inhibit nonenzymatic catalysis by iron salts. On the basis of these investigations, Warburg proposed in 1925 that aerobic cells contain an iron-based *Atmungsferment* (respiration enzyme), which was later called **cytochrome oxidase**. It was inhibited by carbon monoxide.

Knowing that carbon monoxide complexes of hemes are dissociated by light, Warburg and Negelein, in 1928, determined the photochemical **action spectrum** (see Chapter 23) for reversal of the carbon monoxide inhibition of respiration of the yeast *Torula utilis*. The spectrum closely resembled the absorption spectrum of known heme derivatives (Fig. 16-7). Thus, it was proposed that O_2, as well as CO, combines with the iron of the heme group in the Atmungsferment.

Meanwhile, during 1919–1925, David Keilin, while peering through a microscope equipped with a spectroscopic ocular at thoracic muscles of flies and other insects, observed a pigment with four distinct absorption bands. At first he thought it was derived by some modification of hemoglobin, but when he found the same pigment in fresh baker's yeast, he recognized it as an important new substance. In his words:[e]

One day while I was examing a suspension of yeast freshly prepared from a few bits of compressed yeast shaken vigorously with a little water in a test-tube, I failed to find the characteristic four-banded absorption spectrum, but before I had time to remove the suspension from the field of vision of the microspectroscope the four absorption bands suddenly reappeared. This experiment was repeated time after time and always with the same result: the absorption bands disappeared on shaking the suspension with air and reappeared within a few seconds on standing.

I must admit that this first visual perception of an intracellular respiratory process was one of the most impressive spectacles I have witnessed in the course of my work. Now I have no doubt that cytochrome is not only widely distributed in nature and completely independent of haemoglobin, but that it is an intracellular respiratory pigment which is much more important than haemoglobin.

Keilin soon realized that three of the absorption bands, those at 604, 564, and 550 nm (*a*, *b*, and *c*), represented different pigments, while the one at 521 nm was common to all three. Keilin proposed the names cytochromes *a*, *b*, and *c*. The idea of an electron transport or respiratory chain followed[e] quickly as the flavin and pyridine nucleotide coenzymes were recognized to play their role at the dehydrogenase level. Hydrogen removed from substrates by these carriers could be used to oxidize reduced cytochromes. The latter would be oxidized by oxygen under the influence of cytochrome oxidase.

In 1929, Fiske and Subbarow,[d,f–h] curious about the occurrence of purine compounds in muscle extracts, discovered and characterized ATP. It was soon shown (largely through the work of Lundsgaard and Lohman)[f] that hydrolysis of ATP provided energy for muscular contraction. At about the same time, it was learned that synthesis of ATP accompanied glycolysis. That ATP could also be formed as a result of electron transport became clear following an observation of Engelhardt[h,i] in 1930, that methylene blue stimulated ATP synthesis by tissues.

The study of electron transport chains and of oxidative phosphorylation began in earnest after Kennedy and Lehninger,[j] in 1949, showed that mitochondria were the site not only of ATP synthesis but also of the operation of the citric acid cycle and fatty acid oxidation pathways. By 1959, Chance had introduced elegant new techniques of spectrophotometry that led to formulation of the electron

BOX 18-A (continued)

transport chain as follows:

$$\text{Substrate} \rightarrow \text{pyridine nucleotides} \rightarrow \text{flavoprotein} \rightarrow \text{cyt } b \rightarrow \text{cyt } c \rightarrow \text{cyt } a \rightarrow \text{cyt } a_3 \rightarrow O_2$$

Since that time, some new components have been added, notably the ubiquinones and iron-sulfur proteins, but the basic form proposed for the chain was correct.

[a] Kalckar, H. M. (1969) *Biological Phosphorylations*, Prentice-Hall, Englewood Cliffs, New Jersey

[b] Kalckar, H. M. (1991) *Ann. Rev. Biochem.* **60**, 1–37

[c] Edsall, J. T. (1979) *Science* **205**, 384–385

[d] Fiske, C. H., and Subbarow, Y. (1929) *Science* **70**, 381–382

[e] Keilin, D. (1966) *The History of Cell Respiration and Cytochrome*, Cambridge Univ. Press, London and New York

[f] Kalckar, H. (1980) *Trends Biochem. Sci.* **5**, 56–57

[g] Schlenk, F. (1987) *Trends Biochem. Sci.* **12**, 367–368

[h] Saraste, M. (1998) *Science* **283**, 1488–1493

[i] Slater, E. C. (1981) *Trends Biochem. Sci.* **6**, 226–227

[j] Talalay, P., and Lane, M. D. (1986) *Trends Biochem. Sci.* **11**, 356–358

membrane bilayer. Membranes also contain **ubiquinone-binding proteins**,[66,67] which probably hold the ubiquinone that is actively involved in electron transport. Perhaps some ubiquinone molecules function as fixed carriers. There is also uncertainty about the number of sites at which ubiquinone functions in the chain.

Mitochondrial electron transport in plants and fungi. Plant mitochondria resemble those of mammals in many ways, but they contain additional dehydrogenases and sometimes utilize alternative pathways of electron transport,[68–73] as do fungi.[74] Mitochondria are impermeable to NADH and NAD^+. Animal mitochondria have shuttle systems (see Fig. 18-16) for bringing the reducing equivalents of NADH into mitochondria and to the NADH dehydrogenase that faces the matrix side of the inner membrane. However, plant mitochondria also have an NADH dehydrogenase on the *outer* surface of the inner membrane (Fig. 18-6). This enzyme transfers electrons to ubiquinone, is not inhibited by rotenone (see Fig. 18-5), and also acts on NADPH. Inside the mitochondria a high-affinity NADH dehydrogenase resembles complex I of animal mitochondria and is inhibited by rotenone.[73] There is also a low-affinity NADH dehydrogenase, which is insensitive to rotenone. Some plant mitochondria respire slowly in the presence of cyanide. They utilize an **alternative oxidase** that replaces complex III and cytochome *c* oxidase and which is not inhibited by antimycin or by cyanide (Fig. 18-6).[68,71,75] It is especially active in thermogenic plant tissues (Box 18-C). A

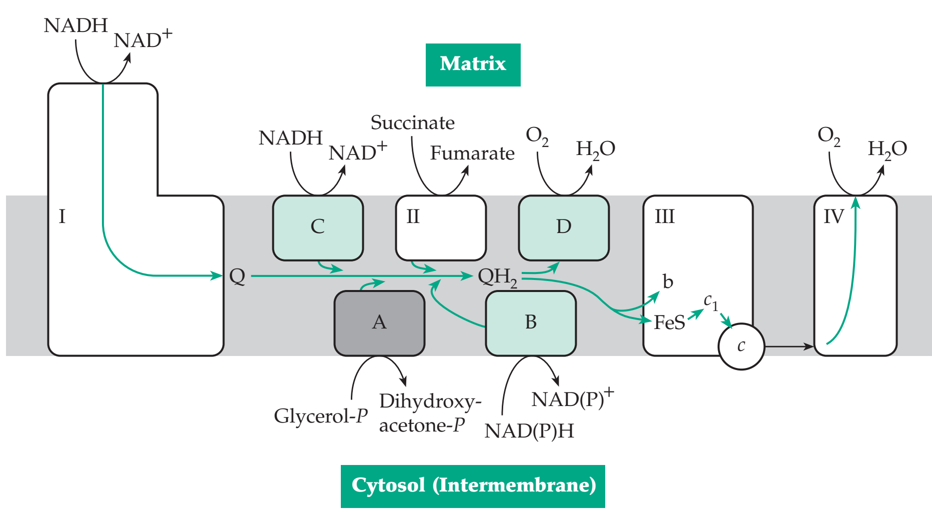


Figure 18-6 Schematic diagram of some mitochondrial dehydrogenase and oxidase complexes of plants and also the glycerol phosphate dehydrogenase of animals, which is embedded in the inner membrane. Complexes I–IV are also shown. (A) The glycerol phosphate dehydrogenase of some animal tissues. It is accessible from the intermembrane space on the cytosolic side. (B) The rotenone-insensitive NAD(P)H dehydrogenase of the external membrane surface of plants. (C) The rotenone-insensitive NADH dehydrogenase facing the matrix side in some plants. (D) The plant alternative oxidase. Ubiquinone, Q. The three green stippled dehydrogenases are not coupled to proton pumps or ATP synthesis. After Hoefnagel *et al.*[73]

BOX 18-B DEFECTS OF MITOCHONDRIAL DNA

A mutation in any of the 13 protein subunits, the 22 tRNAs, or the two rRNAs whose genes are carried in mitochondrial DNA may possibly cause disease. The 13 protein subunits are all involved in electron transport or oxidative phosphorylation. The syndromes resulting from mutations in mtDNA frequently affect oxidative phosphorylation (OXPHOS) causing what are often called "OXPHOS diseases."[a–g] Mitochondrial oxidative phosphorylation also depends upon ~100 proteins encoded in the nucleus. Therefore, OXPHOS diseases may result from defects in either mitochondrial or nuclear genes. The former are distinguished by the fact that they are inherited almost exclusively maternally. Most mitochondrial diseases are rare. However, mtDNA is subject to rapid mutation, and it is possible that accumulating mutants in mtDNA may be an important component of aging.[h–k]

The first recognition of mitochondrial disease came in 1959. A 30-year old Swedish woman was found to have an extremely high basal metabolic rate (180% of normal), a high caloric intake (>3000 kcal / day), and an enormous perspiration rate. She had developed these symptoms at age seven. Examination of her mitochondria revealed that electron transport and oxidative phosphorylation were very loosely coupled. This explains the symptoms. However, the disease (Luft disease) is extremely rare and the underlying cause isn't known.[i] Its recognition did focus attention on mitochondria, and by 1988, 120 different mtDNA defects had been described.[e,i]

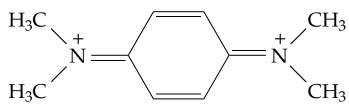
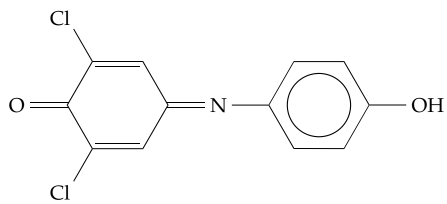
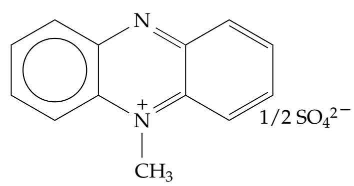
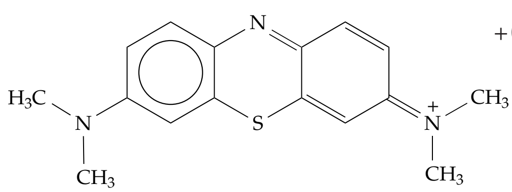
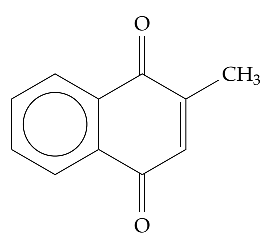
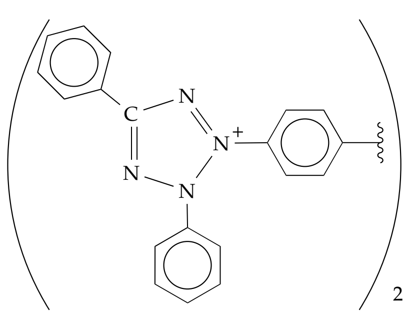
Some OXPHOS disorders, including Luft disease, result from mutations in nuclear DNA. A second group arise from point mutations in mtDNA and a third group involve deletions, often very large, in mtDNA. Persons with these deletions survive because they have both mutated and normal mtDNA, a condition of **heteroplasmy** of mtDNA. As these persons age their disease may become more severe because they lose many normal mitochondria.[d,e]

The names of mitochondrial diseases are often complex and usually are described by abbreviations. Here are a few of them: **LHON**, Lebers hereditary optical neuropathy; **MERRF**, myoclonic epilepsy and ragged-red-fiber disease; **MELAS**, mitochondrial encephalomyopathy, lactic acidosis, and strokelike episodes; **NARP**, neurological muscle weakness, ataxia, and retinitis pigmentosa; **Leigh disease — SNE**, subacute necrotizing encephalomyelopathy; **KSS**, Kearns–Sayre syndrome; **CPEO**, chronic progressive external ophthalmoplegia. LHON is a hereditary disease that often leads to sudden blindness from death of the optic nerve especially among males. Any one of several point mutations in subunits ND1, 2, 4, 5, and 6 of NADH dehydrogenase (complex I; Figs. 18-3 and 18-5), cytochrome *b* of complex II, or subunit I of cytochrome oxidase may cause this syndrome. Most frequent is an R340H mutation of the ND4 gene at position 11,778 of mtDNA (Fig. 18-3).[e,l,m] It may interfere with reduction of ubiquinone.[n] Mutations in the ND1 gene at position 3460 and in the ND6 gene at position 14484 or in the cytochrome *b* gene at position 15257 cause the same disease.[l] The most frequent (80–90%) cause of MERRF, which is characterized by epilepsy and by the appearance of ragged red fibers in stained sections of muscle, is an A → G substitution at position 8344 of mtDNA in the TψC loop (Fig. 5-30) of mitochondrial $tRNA^{Lys}$. A similar disease, MELAS, is accompanied by strokes (not seen in MERRF) and is caused in 80% of cases by an A → G substitution in the dihydrouridine loop (Fig. 5-30) of mitochondrial $tRNA^{Leu}$.[o] CPEO, Leigh disease, and KSS often result from large deletions of mtDNA.[p] NARP and related conditions have been associated with an L156R substitution in the ATPase 6 gene of ATP synthase.[q]

Can mitochondrial diseases be treated? Attempts are being made to improve the function of impaired mitochondria by adding large amounts of ubiquinone, vitamin K, thiamin, riboflavin, and succinate to the diet.[e] One report suggests that mitochondrial decay during aging can be reversed by administration of *N*-acetylcarnitine.[k]

[a] Palca, J. (1990) *Science* **249**, 1104–1105
[b] Capaldi, R. A. (1988) *Trends Biochem. Sci.* **13**, 144–148
[c] Darley-Usmar, V., ed. (1994) *Mitochondria: DNA, Proteins and Disease*, Portland Press, London
[d] Wallace, D. C. (1999) *Science* **283**, 1482–1488
[e] Shoffner, J. M., and Wallace, D. C. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 1 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 1535–1609, McGraw-Hill, New York
[f] Schon, E. A. (2000) *Trends Biochem. Sci.* **25**, 555–560
[g] Tyler, D. D. (1992) *The Mitochondrion in Health and Disease*, VCH Publ., New York
[h] Wallace, D. C. (1992) *Science* **256**, 628–632
[i] Luft, R. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 8731–8738
[j] Tanhauser, S. M., and Laipis, P. J. (1995) *J. Biol. Chem.* **270**, 24769–24775
[k] Shigenaga, M. K., Hagen, T. M., and Ames, B. N. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 10771–10778
[l] Hofhaus, G., Johns, D. R., Hurko, O., Attardi, G., and Chomyn, A. (1996) *J. Biol. Chem.* **271**, 13155–13161
[m] Brown, M. D., Trounce, I. A., Jun, A. S., Allen, J. C., and Wallace, D. C. (2000) *J. Biol. Chem.* **275**, 39831–39836
[n] Zickermann, V., Barquera, B., Wikström, M., and Finel, M. (1998) *Biochemistry* **37**, 11792–11796
[o] Hayashi, J.-I., Ohta, S., Kagawa, Y., Takai, D., Miyabayashi, S., Tada, K., Fukushima, H., Inui, K., Okada, S., Goto, Y., and Nonaka, I. (1994) *J. Biol. Chem.* **269**, 19060–19066
[p] Moraes, C. T., and 19 other authors. (1989) *N. Engl. J. Med.* **320**, 1293–1299
[q] Hartzog, P. E., and Cain, B. D. (1993) *J. Biol. Chem.* **268**, 12250–12252

TABLE 18-5
Some Artificial Electron Acceptors[a,b]

Compound	Structure	E°′(pH 7) 30°C
Ferricyanide	$Fe(CN)_6^{3-}$	+0.36 V (25°C)
Oxidized form of tetramethyl-*p*-phenylenediamine		+0.260 V
2,6-Dichlorophenol-indophenol (DCIP)		+0.217 V
Phenazine methosulfate (PMS)		+0.080 V
Ascorbate	(See Box 18-D)	+0.058 V
Methylene blue		+0.011 V
Menadione		+0.008 V (°25C)
Tetrazolium salts, e.g., "neotetrazolium chloride"		−0.125 V

[a] From Wainio, W. W. (1970) *The Mammalian Mitochondrial Respiratory Chain*, Academic Press, New York, pp. 106–111.
[b] See Fig. 18-5 for sites of action.

similar oxidase is present in trypanosomes.[72] Neither the rotenone-insensitive dehydrogenases nor the alternative oxidases are coupled to synthesis of ATP.

Electron transport chains of bacteria. The bacterial electron transport systems are similar to that of mitochondria but simpler. Bacteria also have a variety of alternative pathways that allow them to adapt to various food sources and environmental conditions.[76,77] The gram-negative soil bacterium *Paracoccus denitrificans*, which has been called "a free-living mitochondrion," has a mammalian-type respiratory system. Its complexes I–IV resemble those of animals and of fungi,[78–79] but *Paracoccus* has fewer subunits in each complex. Complex I of *E. coli* is also similar to that of our own bodies.[79–80] However, other major flavoprotein dehydrogenases in *E. coli* act on D-lactate and *sn*-3-glycerol phosphate.[81] Pyruvate is oxidized by a membrane-bound flavoprotein (Fig. 14-2). All of these enzymes pass electrons to ubiquinone-8 (Q_8).[82] Succinate dehydrogenase of *E. coli* resembles that of mitochondria,[83] and the ubiquinol oxidase of *Paracoccus* resembles complexes III + IV of mitochondria. It can be resolved into a three-subunit bc_1 complex, a three-subunit c_1aa_3 complex, and another 57-kDa peptide.[84] The last contains a 22-kDa cytochrome c_{552}, which is considerably larger than mitochondrial cytochrome *c*.

The cytochrome aa_3 terminal oxidase is produced constitutively, i.e., under all conditions. However, when cells are grown on succinate or H_2 another set of enzymes is produced with the *b*-type cytochrome o_3 as the terminal **quinol oxidase** (Eq. 18-2).[85]

$$\begin{array}{ccccc} & & H_2 & & \\ & & \downarrow & & \\ \text{Succinate} & \rightarrow & Q & \rightarrow & \text{cytochrome } b \\ & & & & \downarrow \\ & & & & \text{cytochrome } o_3 \\ & & & & \downarrow \\ & & & & O_2 \end{array} \quad (18\text{-}2)$$

Two terminal quinol oxidase systems, both related to cytochrome *c* oxidase, are utilized by *E. coli* to oxidize ubiquinol-8. When cultured at high oxygen tensions, cytochrome bo_3 (also called cytochrome *bo*) is the major oxidase. It utilizes heme *o* (Fig. 16-5) instead of heme *a*. However, at low oxygen tension, e.g., in the late logarithmic stage of growth, the second oxidase, cytochrome *bd*, is formed.[76,86–88a] It contains two molecules of the chlorin heme *d* (Fig. 16-5), which appear to be involved directly in binding O_2. This terminal oxidase system is present in many bacteria and can utilize either O_2 or nitrite as the oxidant. A simpler electron transport chain appears to be involved in the oxidation of pyruvate by *E. coli*. The flavoprotein pyruvate oxidase passes electrons to Q_8, whose reduced form can pass electrons directly to cytochrome *d*. Incorporation of these two pure protein complexes and ubiquinone-8 into phospholipid vesicles has given an active reconstituted chain.[82] Other bacteria utilize a variety of quinol oxidase systems, which contain various combinations of cytochromes: aa_3, caa_3, *cao*, bo_3, and ba_3.[88b,c]

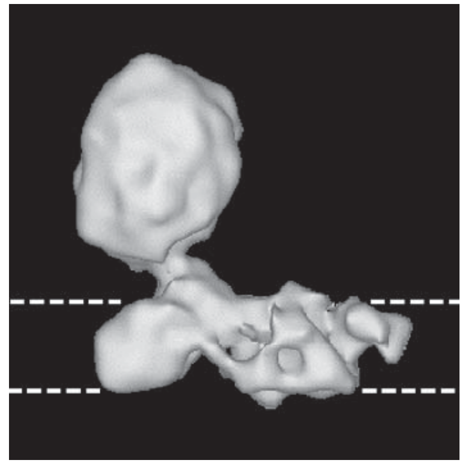


Figure 18-7 Three-dimensional image of bovine NADH-Ubiquinone oxidoreductase (complex I) reconstructed from individual images obtained by electron cyro-microscopy. The resolution is 2.2 nm. The upper portion projects into the mitochondrial matrix while the horizontal part lies within the membrane as indicated. Courtesy of N. Grigorieff.[90]

3. Structures and Functions of the Individual Complexes I – IV and Related Bacterial Assemblies

What are the structures of the individual electron transport complexes? What are the subunit compositions? What cofactors are present? How are electrons transferred? How are protons pumped? We will consider these questions for each of complexes I–IV, as found in both prokaryotes and eukaryotes.[88d,e]

Complex I, NADH-ubiquinone oxidoreductase. Complex I oxidizes NADH, which is generated within the mitochondrial matrix by many dehydrogenases. Among these are the pyruvate, 2-oxoglutarate, malate, and isocitrate dehydrogenases, which function in the tricarboxylic acid cycle; the β-oxoacyl-CoA dehydrogenase of the β oxidation system for fatty acids; and 2-hydroxybutyrate, glutamate, and proline dehydrogenases. All produce NADH, which reacts with the flavoprotein component of complex I. Whether from bacteria,[79] fungal mitochondria,[89] or mammalian mitochondria[89a,90] complex I exists as an L-shaped object, of which each of the two arms is ~23 nm long. One arm projects into the matrix while the other lies largely within the inner mitochondrial membrane (Fig. 18-7). The mitochondrial complex, which has a mass of ~1 MDa, has the same basic structure as the 530-kDa bacterial complex. However, the arms are thicker in the mitochondrial complex. Analysis of the denatured proteins by gel electrophoresis revealed at least 43 peptides.[78,90] Bound to some of these are the electron carriers FMN, Fe_2S_2, and Fe_4S_4 clusters, ubiquinone or other quinones, and perhaps additional unidentified cofactors.[79] Complex I from *E. coli* is smaller, containing only 14 subunits. These are encoded by a cluster of 14 genes, which can be directly related by their sequences to subunits of mitochondrial complex I and also to the corresponding genes of *Paracoccus denitrificans*.[80,91] Complex I of *Neurospora* contains at least 35 subunits.[89] The 14 subunits that are present both in bacteria and in mitochondria probably form the structural core of the complex. The other subunits thicken, strengthen, and rigidify the arms. Some of the "extra" subunits have enzymatic activities that are not directly related to electron transport. Among these are a 10-kDa prokaryotic type acyl carrier protein (ACP), which may be a relic of a bacterial fatty acid synthase, reflecting the endosymbiotic origin of mitochondria.[92] Also present is a 40-kDa NAD(P)H dependent reductase / isomerase, which may be involved in a biosynthetic process, e.g., synthesis of a yet unknown redox group.[79,92]

In all cases, FMN is apparently the immediate acceptor of electrons from NADH. From the results of extrusion of the Fe–S cores (Chapter 16) and EPR measurements it was concluded that there are three tetranuclear (Fe_4S_4) iron–sulfur centers and at least two binuclear (Fe_2S_2) centers[93,94] as well as bound ubiquinone.[95] Chemical analysis of iron and sulfide suggested up to eight Fe–S clusters per FMN, while gene sequences reveal potential sites for formation of six Fe_4S_4 clusters and two Fe_2S_2 clusters.[78] Treatment of complex I with such "chaotropic agents" as 2.5 M urea or 4 M sodium trichloroacetate followed by fractionation with ammonium sulfate[95] gave three fractions:

(1) A soluble NADH dehydrogenase consisting of a 51-kDa peptide that binds both the FMN and also one tetranuclear Fe – S cluster (designated N3) and a 24-kDa peptide that carries a binuclear Fe–S center designated N1b. (2) A 75-kDa peptide bearing two binuclear Fe–S centers, one of which is called N1a and also 47-, 30-, and 13-kDa peptides. One of these carries tetranuclear center N4. (3) A group of insoluble, relatively nonpolar proteins, one of which carries tetranuclear cluster N2. It may be the immediate donor of electrons to a ubiquinone held by a ubiquinone-binding protein designated QP-N. In bacteria seven of these are homologs of the seven NADH dehydrogenase subunits encoded by mtDNA (Fig. 18-3). A 49-kDa subunit of complex I in the yeast *Yarrowia lipolytica* is strikingly similar to the hydrogen reactive subunit of NiFe hydrogenases (Fig. 16-26).[95a] These proteins are thought to lie within the membrane arm and to form ~55 transmembrane α helices. Ubiquinone may also function as a carrier within complex I,[96,97] and there may be a new redox cofactor as well.[79] The following tentative sequence (Eq. 18-3) for electron transfer within complex I (with apparent $E^{\circ\prime}$ values of carriers) has been suggested. By equilibration with external redox systems, the redox potentials of these centers within the mitochondria have been estimated and are given (in V) in Eq. 18-3. The presence of a

$$\begin{array}{ccccccccccccccc} & & & & & & & & & & & & & \text{rotenone} & \\ \text{NADH} & \rightarrow & \text{FMN} & \rightarrow & \text{N1a} & \rightarrow & (\text{N3}, & \text{N1b}, & \text{N4}) & \rightarrow & \text{N2} & \rightarrow & \text{QP-N} & \rightarrow & \text{Q} \\ & & & & Fe_2S_2 & & Fe_4S_4 & Fe_2S_2 & Fe_4S_4 & & Fe_4S_4 & & & \ominus & \\ -0.32 & & & & -0.38 & & & -0.24 & & & -0.21\ \text{V} & & & & \end{array} \tag{18-3}$$

large fraction of the bound ubiquinone as a free radical suggests that the quinone functions as a one-electron acceptor rather than a two-electron acceptor. A characteristic of complex I is inhibition by rotenone or piericidin, both of which block electron transport at the site indicated in Fig. 18-5.

Complex II, succinate-ubiquinone oxidoreductase. Complex II, which carries electrons from succinate to ubiquinone, contains covalently linked 8^{α}-(*N*-histidyl)-FAD (Chapter 15) as well as Fe–S centers and one or more ubiquinone-binding sites. There are four subunits whose structures and properties have been highly conserved among mitochondria and bacteria and also in **fumarate reductases**. The latter function in the opposite direction during anaerobic respiration with fumarate as the terminal oxidant, both in bacteria[98–99a] and in parasitic helminths and other eukaryotes that can survive prolonged anaerobic conditions (Chapter 17, Section F,2).[100] Complex II from *E. coli* consists of 64-, 27-, 14-, and 13-kDa subunits, which are encoded by genes *sdhCDAB* of a single operon.[101–103] The two larger hydrophilic subunits associate to form the readily soluble succinate dehydrogenase. The 64-kDa subunit carries the covalently bound FAD while the 27-kDa subunit carries three Fe–S centers. The two small 13- and 14-kDa subunits form a hydrophobic anchor and contain a ubiquinol-binding site (QD-S)[103] as well as a heme that may bridge the two subunits[102] to form cytochrome b_{556}. The functions of the heme is uncertain. The soluble mammalian succinate dehydrogenase resembles closely that of *E. coli* and contains three Fe – S centers: binuclear S1 of $E^{\circ\prime}$ 0 V, and tetranuclear S2 and S3 of −0.25 to −0.40 and + 0.065 V, respectively. Center S3 appears to operate between the −2 and −1 states of Eq. 16-17 just as does the cluster in the *Chromatium* high potential iron protein. The function of the very low potential S2 is not certain, but the following sequence of electron transport involving S1 and S3 and the bound ubiquinone QD–S[66] has been proposed (Eq. 18-4).

$$\text{Succinate} \rightarrow (\text{FAD*}, \text{S1}) \rightarrow \text{S3} \rightarrow \text{QP–S} \rightarrow \text{Q} \tag{18-4}$$

In addition to complexes I and II several other membrane-associated FAD-containing dehydrogenase systems also send electrons to soluble ubiquinone. These include dehydrogenases for choline, *sn*-glycerol 3-phosphate, and the electron-transfer protein (ETF) of the fatty acyl-CoA β oxidation system (Fig. 18-5). The last also accepts electrons from dehydrogenases for sarcosine (*N*-methylglycine), dimethylglycine, and other substrates. The *sn*-glycerol 3-phosphate dehydrogenase is distinguished by its accessibility from the intermembrane (cytosolic) face of the inner mitochondrial membrane (Fig. 18-6).

Complex III (ubiquinol-cytochrome c oxidoreductase or cytochrome bc_1 complex). Mitochondrial complex III is a dimeric complex, each subunit of which contains 11 different subunits with a total molecular mass of ~240 kDa per monomer.[104–107] However, in many bacteria the complex consists of only three subunits, cytochrome *b*, cytochrome c_1, and the high potential (~0.3 V) Rieske iron-sulfur protein, which is discussed in Chapter 16, Section A,7. These three proteins are present in all bc_1 complexes. In eukaryotes the 379-residue cytochrome *b* is mitochondrially encoded. Although there is only one cytochrome *b* gene in the mtDNA, two forms of cytochrome *b* can be seen in absorption spectra: b_H (also called b_{562} or b_K) and lower potential b_L (also called b_{566} or b_T).[107a,b]

X-ray diffraction studies have revealed the complete 11-subunit structure of bovine bc_1 complex[104,106–107] as well as a nearly complete structure of the chicken bc_1 complex (Fig. 18-8).[105] The bovine complex contains 2166 amino acid residues per 248-kDa monomer and exists in crystals as a 496-kDa dimer and probably functions as a dimer.[106-107] The two hemes of cytochrome b are near the two sides of the membrane, and the Fe–S and cytochrome c_1 subunits are on the surface next to the intermembrane space (Fig. 18-8). On the matrix side (bottom in Fig. 18-8A) are two large ~440 residue "core" subunits that resemble subunits of the mitochondrial processing protease. They may be evolutionary relics of that enzyme.[106,108,108a] Mitochondrial cytochrome b_H has an $E°'$ value of +0.050 V, while that of b_L is −0.090 V at pH 7.[109] That of the Rieske Fe–S protein is + 0.28 V.[110]

The sequence of electron transport within complex III has been hard to determine in detail. For reasons discussed in Section C, the "Q-cycle" shown in Fig. 18-9 has been proposed.[111–114a] As is indicated in Fig. 18-9, complex II accepts electrons from QH_2 and passes them consecutively to the Fe–S protein, cytochrome c_1, and the external cytochrome c. However, half of the electrons are recycled through the two heme groups of cytochrome b, as is indicated in the figure and explained in the legend. The X-ray structure (Fig. 18-8) is consistent with this interpretation. Especially intriguing is the fact that the Fe_2S_2 cluster of the Rieske protein subunit has been observed in two or three different conformations.[105–107,114a-c] In Fig. 18-8C the structures of two conformations are superimposed. The position of the long helix at the right side is unchanged but the globular domain at the top can be tilted up to bring the Fe_2S_2 cluster close to the heme of cytochrome c_1, or down to bring the cluster close to heme b_L. Movement between these two positions is probably part of the catalytic cycle.[115]

The simpler cytochrome bc_1 complexes of bacteria such as *E. coli*,[102] *Paracoccus dentrificans*,[116] and the photosynthetic *Rhodobacter capsulatus*[117] all appear to function in a manner similar to that of the large mitochondrial complex. The bc_1 complex of *Bacillus subtilis* oxidizes reduced **menaquinone** (Fig. 15-24) rather than ubiquinol.[118] In chloroplasts of green plants photochemically reduced **plastoquinone** is oxidized by a similar complex of cytochrome b, c-type cytochrome f, and a Rieske Fe–S protein.[119–120a] This cytochrome b_6f complex delivers electrons to the copper protein plastocyanin (Fig. 23-18).

The electron acceptor for complex III is cytochrome c, which, unlike the other cytochromes, is water soluble and easily released from mitochondrial membranes. Nevertheless, it is usually present in a roughly 1:1 ratio with the fixed cytochromes, and it seems unlikely that it is as free to diffuse as are ubiquinone and NAD^+.[121,122] However, a small fraction of the cytochrome c may diffuse through the intermembrane space and accept electrons from cytochrome b_5, which is located in the outer membrane.[123] Cytochrome c forms a complex with cardiolipin (diphosphatidylglycerol), a characteristic component of the inner mitochondrial membrane.[124]

Complex IV. Cytochrome c oxidase (ubiquinol-cytochrome c oxidoreductase). Complex IV from mammalian mitochondria contains 13 subunits. All of them have been sequenced, and the three-dimensional structure of the complete complex is known (Fig. 18-10).[125–127] The simpler cytochrome c oxidase from *Paracoccus denitrificans* is similar but consists of only three subunits. These are homologous in sequence to those of the large subunits I, II, and III of the mitochondrial complex. The three-dimensional structure of the *Paracoccus* complex is also known. Its basic structure is nearly identical to that of the catalytic core of subunits I, II, and III of the mitochondrial complex (Fig. 18-10,A).[128] All three subunits have transmembrane helices. Subunit III seems to be structural in function, while subunits I and II contain the oxidoreductase centers: two hemes a (a and a_3) and two different copper centers, Cu_A (which contains two Cu^{2+}) and a third Cu^{2+} (Cu_B) which exists in an EPR-silent exchange coupled pair with a_3. Bound Mg^{2+} and Zn^{2+} are also present in the locations indicated in Fig. 18-10.

The Cu_A center has an unusual structure.[130–132] It was thought to be a single atom of copper until the three-dimensional structure revealed a dimetal center, whose structure follows. The Cu_B-cytochrome a_3 center is also unusual. A histidine ring is covalently attached to tyrosine.[133–135a] Like the tyrosine in the active site of galactose oxidase (Figs. 16-29, 16-30), which carries a covalently joined cysteine, that of cytochrome oxidase may be a site of tyrosyl radical formation.[135]

Cytochrome c oxidase accepts four electrons, one at a time from cytochrome c, and uses them to reduce O_2 to two H_2O. Electrons enter the oxidase via the Cu_A center and from there pass to the cytochrome a and on to the cytochrome a_3 – Cu_B center where the reduction of O_2 takes place. A possible sequence of steps in the catalytic cycle is given in Fig. 18-11. Reduction of O_2 to two H_2O requires four electrons and also four protons. An additional four protons are evidently pumped across the membrane for each catalytic cycle.[136–138] The overall reaction is:

$$4\ e^- + 8\ H^+_{IN} + O_2 \longrightarrow 2\ H_2O + 4\ H^+_{OUT} \qquad (18\text{-}5)$$

The reaction of O_2 with cytochrome c oxidase to form the oxygenated species A (Fig. 18-11) is very rapid, occurring with apparent lifetime τ (Eq. 9-5) of ~8–10 μs.[139] Study of such rapid reactions has depended upon a flow-flash technique developed by Greenwood and Gibson.[136,140,141] Fully reduced cytochrome oxidase is allowed to react with carbon monoxide, which binds to the iron in cytochrome a_3 just as does O_2. In fact, it was the spectroscopic observation that only half of the

Figure 18-8 Stereoscopic ribbon diagrams of the chicken bc_1 complex (A) The native dimer. The molecular twofold axis runs vertically between the two monomers. Quinones, phospholipids, and detergent molecules are not shown for clarity. The presumed membrane bilayer is represented by a gray band. (B) Isolated close-up view of the two conformations of the Rieske protein (top and long helix at right) in contact with cytochrome *b* (below), with associated heme groups and bound inhibitors, stigmatellin, and antimycin. The isolated heme of cytochrome c_1 (left, above) is also shown. (C) Structure of the intermembrane (external surface) domains of the chicken bc_1 complex. This is viewed from within the membrane, with the transmembrane helices truncated at roughly the membrane surface. Ball-and-stick models represent the heme group of cytochrome c_1, the Rieske iron–sulfur cluster, and the disulfide cysteines of subunit 8. SU, subunit; cyt, cytochrome. From Zhang *et al.*[105]

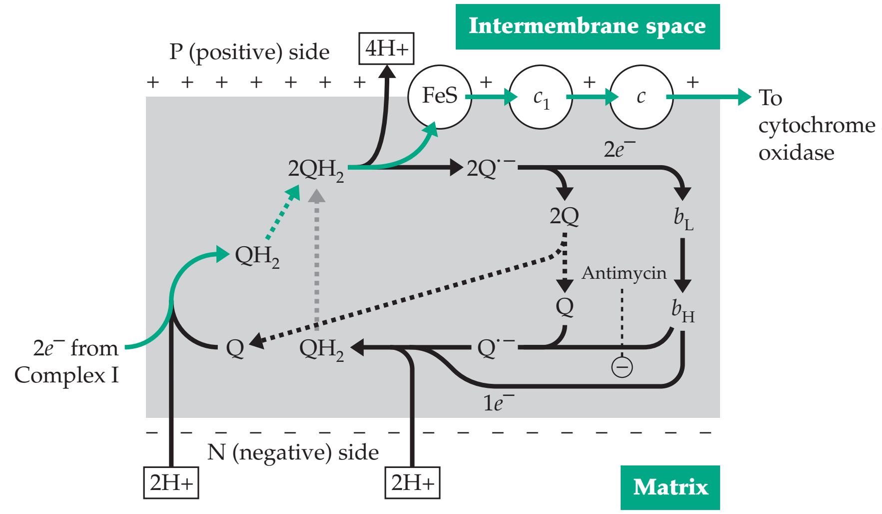
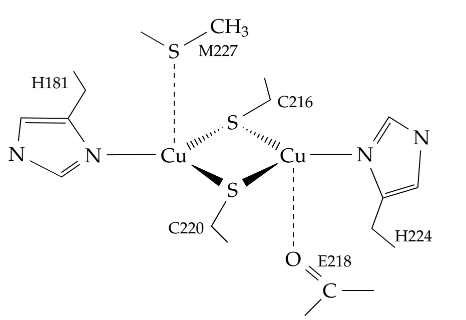
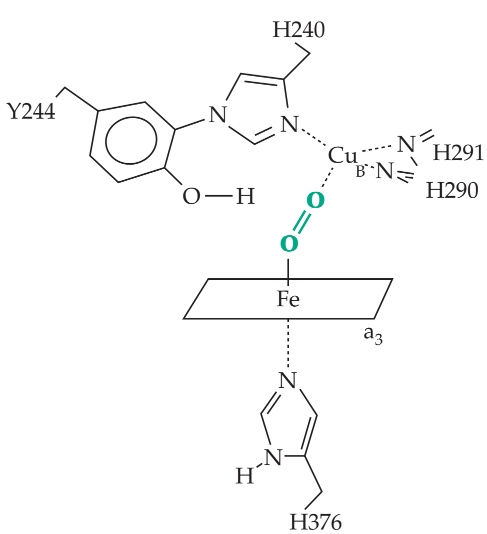


Figure 18-9 Proposed routes of electron transfer in mitochondrial complex III according to Peter Mitchell's Q cycle. Ubiquinone (Q) is reduced to QH_2 by complex I (left side of diagram) using two H^+ taken up from the matrix (leaving negative charges on the inner membrane surface). After diffusing across the bilayer (dashed line) the QH_2 is oxidized in the two steps with release of the two protons per QH_2 on the positive (P) side of the membrane. In the two-step oxidation via anionic radical $Q^{\cdot -}$ one electron flows via the Rieske Fe–S protein and the cytochrome c_1 heme to external cytochrome c. The other electron is transferred to heme b_L of cytochrome b, then across the membrane to heme b_H which now reduces Q to Q^-. A second QH_2 is dehydrogenated in the same fashion and the electron passed through the cytochrome b centers is used to reduce Q^- to QH_2 with uptake from the matrix of 2 H^+. The resulting QH_2 diffuses back across the membrane to function again while the other Q diffuses back to complex I. The net result is pumping of 4 H^+ per 2 e^- passed through the complex. Notice that in the orientation used in this figure the matrix is at the bottom, not the top as in Figs. 18-4 and 18-5.



The Cu_A center of cytochrome oxidase



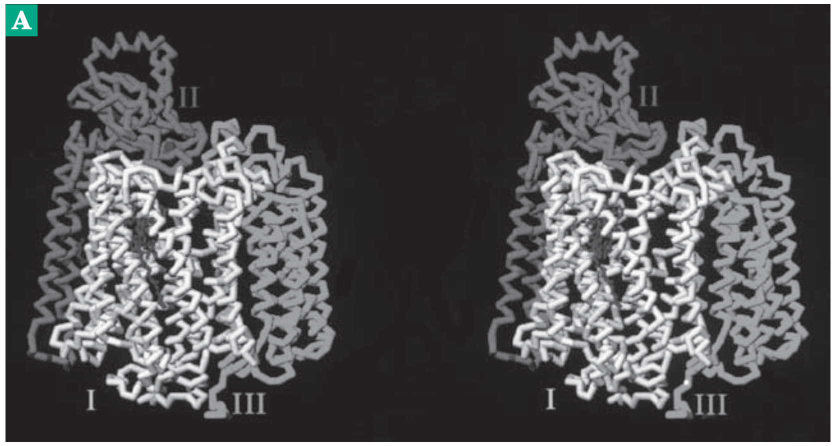
The $Cu_B \bullet A_3$ center of cytochrome oxidase

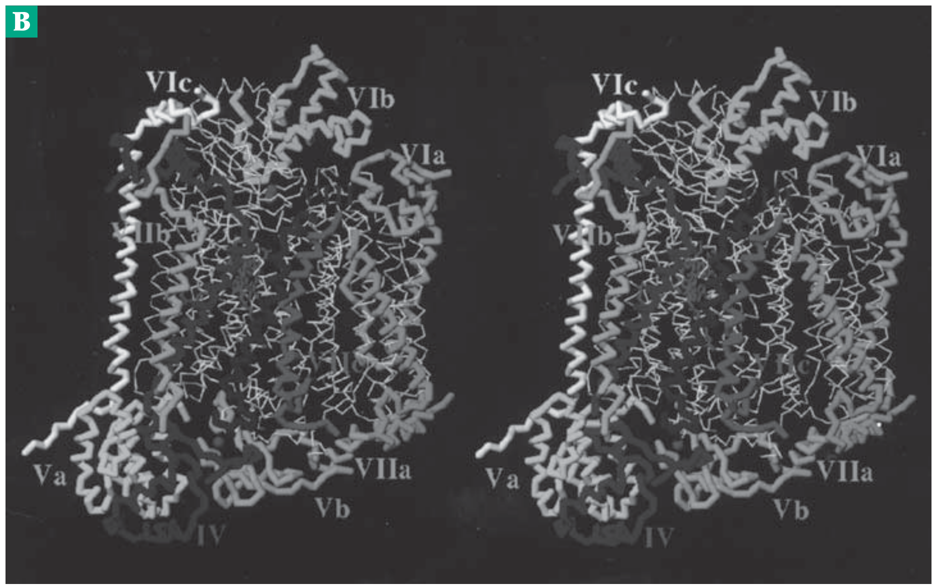
cytochrome *a* combined with CO that led Keilin to designate the reactive component a_3. This CO complex is mixed with O_2-containing buffer and irradiated with a laser pulse to release the CO and allow O_2 to react. The first rapid reaction observed is the binding of O_2 (step *c* in Fig. 18-11). Formation of a peroxy intermediate from the initial oxygenated form (A in Fig. 18-11) is very fast. The O–O bond of O_2 has already been cleaved in form P (Fig. 18-11), which has until recently been thought to be the peroxy intermediate. In fact, spectroscopic measurements indicate that form P contains an oxo-ferryl ion with the second oxygen of the original O_2 converted to an OH ion and probably coordinated with Cu_B.[136a,136c,142,142a–c] P may also contain an organic radical, perhaps formed from tyrosine 244 as indicated in Fig. 18-11.

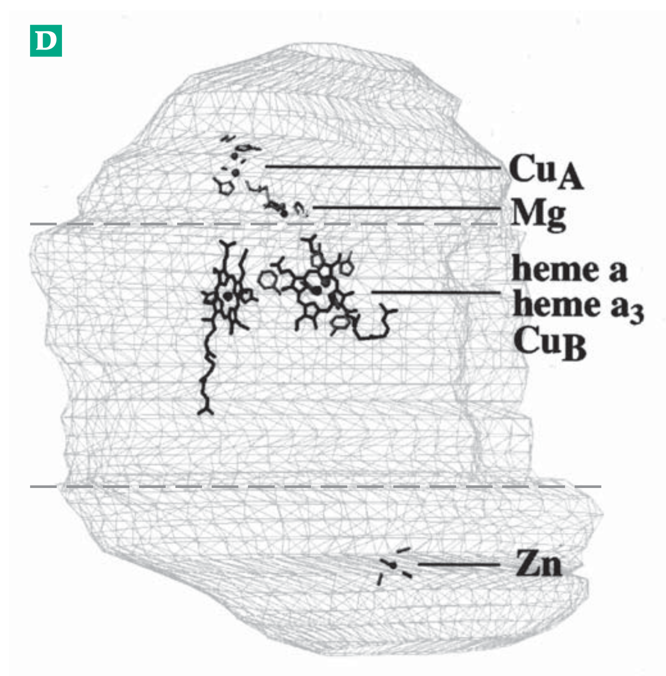
A second relaxation time of $\tau = 32–45\ \mu s$ has been assigned[139] to the conversion of the peroxide intermediate P to P′. A third relaxation time ($\tau = 100–140\ \mu s$) is associated with the oxidation of Cu_A by *a* (not shown in Fig. 18-11).[143] This electron transfer step limits the rate of step *f* of Fig. 18-11. Another reduction step with $\tau \sim 1.2$ ms is apparently associated with electron transfer in step *h*. This slowest step still allows a first-order reaction rate of ~800 s^{-1}.

When O_2 reacts with cytochrome *c* oxidase, it may be bound initially to either the a_3 iron or to Cu_B, but in the peroxy intermediate P it may bind to both atoms. Oxyferryl compound F (Fig. 8-11) as well as radical species, can also be formed by treatment of the oxidized

Figure 18-10 Structure of mitochondrial cytochrome *c* oxidase. (A) Stereoscopic C_α backbone trace for one monomeric complex of the core subunits I, II, and III. (B) Stereoscopic view showing all 13 subunits. The complete complex is a dimer of this structure. From Tsukihara *et al.*[125] (C) MolScript ribbon drawing of one monomeric unit. The horizontal lines are drawn at distances of ±1.0 and ± 2.0 nm from the center of the membrane bilayer as estimated from eight phospholipid molecules bound in the structure. From Wallin *et al.*[127] Courtesy of Arne Elofsson. (D) Schematic drawing of the same complex showing positions of the Cu_A dimetal center, bound Mg^{2+}, heme *a*, the bimetal heme a_3 - Cu_B center, and bound Zn^{2+}. The location of an 0.48-nm membrane bilayer is marked. From Tsukihara *et al.*[129] (A), (B), and (D) courtesy of Shinya Yoshikawa.







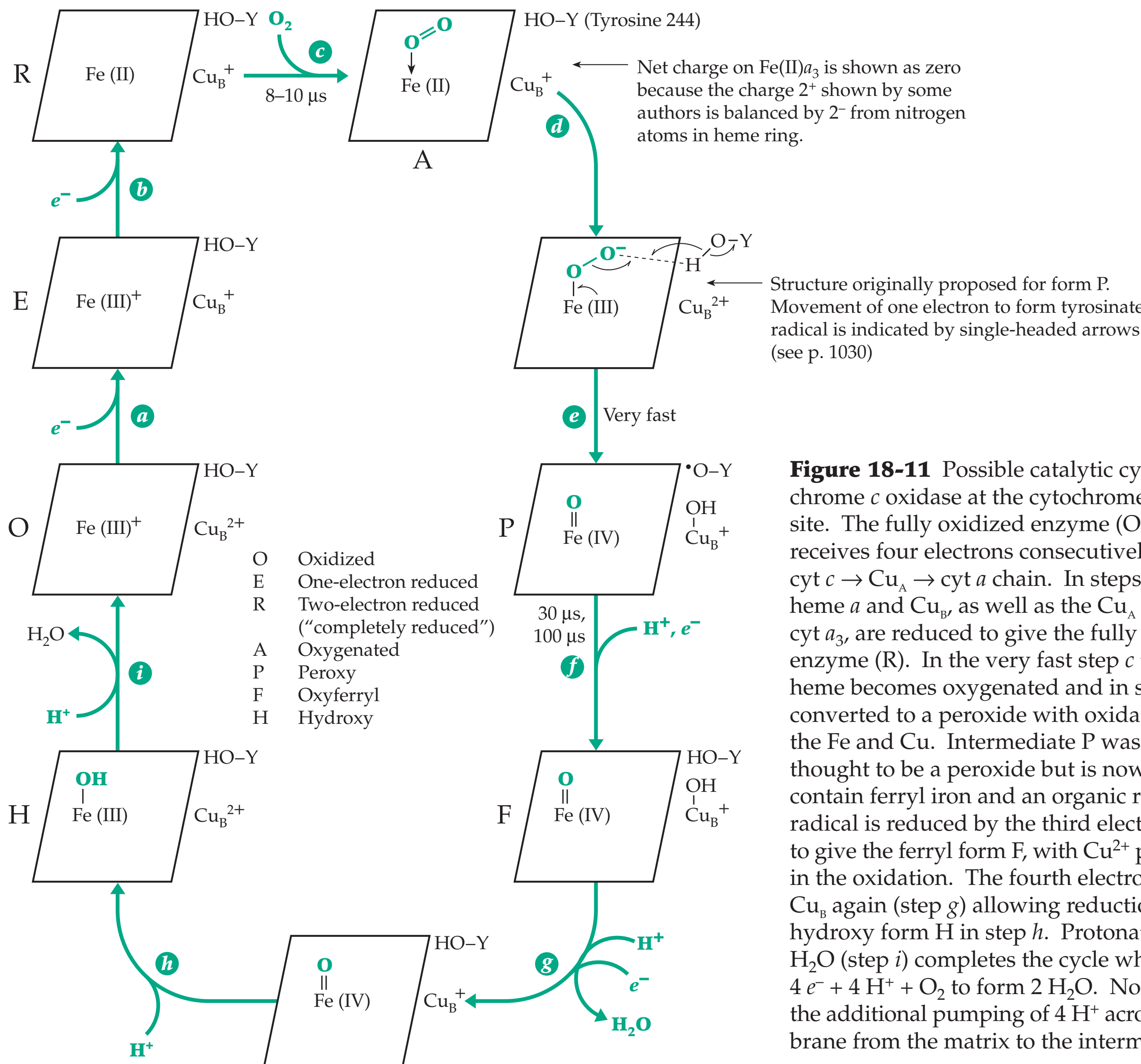


Figure 18-11 Possible catalytic cycle of cytochrome c oxidase at the cytochrome a_3 – Cu_B site. The fully oxidized enzyme (O; left center) receives four electrons consecutively from the cyt $c \rightarrow Cu_A \rightarrow$ cyt a chain. In steps a and b both heme a and Cu_B, as well as the Cu_A center and cyt a_3, are reduced to give the fully reduced enzyme (R). In the very fast step c the cyt a_3 heme becomes oxygenated and in step d is converted to a peroxide with oxidation of both the Fe and Cu. Intermediate P was formerly thought to be a peroxide but is now thought to contain ferryl iron and an organic radical. This radical is reduced by the third electron in step f to give the ferryl form F, with Cu^{2+} participating in the oxidation. The fourth electron reduces Cu_B again (step g) allowing reduction to the hydroxy form H in step h. Protonation to form H_2O (step i) completes the cycle which utilizes $4\ e^- + 4\ H^+ + O_2$ to form $2\ H_2O$. Not shown is the additional pumping of 4 H^+ across the membrane from the matrix to the intermembrane space.

enzyme O with hydrogen peroxide.[143a-144] Use of various inhibitors has also been important in studying this enzyme. Cyanide, azide, and sulfide ions, as well as carbon monoxide, are powerful inhibitors. Cyanide specifically binds to the Fe^{3+} form of cytochrome a_3 preventing its reduction,[145] while CO competes with O_2 for its binding site. A much-used reagent that modifies carboxyl groups in proteins, and which inhibits many proton translocating proteins, is dicyclohexyl carbodiimide (Eq. 3-10).[146] The step-by-step flow of electrons through cytochrome c oxidase seems quite well defined. However, one of the most important aspects is unclear. How is the pumping of protons across the membrane coupled to electron transport?[137,138,142,147,147a] Many recent studies have employed directed mutation of residues in all four subunits to locate possible proton pathways or channels.[148–152] Most ideas involve movement through hydrogen bonded chains (Eq. 9-94), which may include the carboxylate groups of the bound hemes.[153] Conformational changes may be essential to the gating of proton flow by electron transfers.[143]

The surface of the matrix side of cytochrome oxidase contains histidine and aspartate side chains close together. It has been suggested that they form a proton collecting antenna that contains groups basic enough to extract protons from the buffered matrix and guide them to a proton conduction pathway.[154] Calcium ions also affect proton flow.[153a,b] We will return to this topic in Section C,3 (p. 1040).

C. Oxidative Phosphorylation

During the 1940s when it had become clear that formation of ATP from ADP and inorganic phosphate was coupled to electron transport in mitochondria, intensive efforts were made to discover the molecular mechanisms. However, nature sometimes strongly resists attempts to pry out her secrets, and the situation which prevailed was aptly summarized by Ephraim Racker: "Anyone who is not confused about oxidative phosphorylation just doesn't understand the situation."[155] The confusion is only now being resolved.

1. The Stoichiometry (P/O Ratio) and Sites of Oxidative Phosphorylation

Synthesis of ATP *in vitro* by tissue homogenates was demonstrated in 1937 by Kalckar, who has written a historical account.[156] In 1941, Ochoa[157] obtained the first reliable measurement of the P/O ratio, the *number of moles of ATP generated per atom of oxygen utilized* in respiration. The P/O ratio is also equal to the number of moles of ATP formed for each pair of electrons passing through an electron transport chain. Ochoa established that for the oxidation of pyruvate to acetyl-CoA and CO_2, with two electrons passed down the mitochondrial electron transport chain, the P/O ratio was ~3. This value has since been confirmed many times.[158–160] However, experimental difficulties in measuring the P/O ratio are numerous.[161] Many errors have been made, even in recent years, and some investigators[162] have contended that this ratio is closer to 2.5 than to 3. One method for measuring the P/O ratio is based on the method of determining the amount of ATP used that is described in the legend to Fig. 15-2.

The experimental observation of a P/O ratio of ~3 for oxidation of pyruvate and other substrates that donate NADH to the electron transport chain led to the concept that there are *three sites for generation of ATP*. It was soon shown that the P/O ratio was only 2 for oxidation of succinate. This suggested that one of the sites (site I) is located between NADH and ubiquinone and precedes the diffusion of QH_2 formed in the succinate pathway to complex III.

In 1949, Lehninger used ascorbate plus tetramethylphenylene-diamine (TMPD, Table 18-4) to introduce electrons into the chain at cytochrome *c*. The sequence ascorbate → TMPD → cytochrome *c* was shown to occur nonenzymatically. Later, it became possible to use cytochrome *c* as an electron donor directly. In either case only one ATP was generated, as would be anticipated if only site III were found to the right of cytochrome *c*. Site I was further localized by Lardy, who used hexacyanoferrate (III) (ferricyanide) as an artificial oxidant to oxidize NADH in the presence of antimycin *a*. Again a P/O ratio of one was observed. Finally, in 1955, Slater showed that passage of electrons from succinate to cytochrome c also gave only one ATP, the one generated at site II. The concept of three sites of ATP formation became generally accepted. However, as we shall see, these sites are actually proton-pumping sites, and there may be more than three of them.

Respiratory control and uncoupling. With proper care relatively undamaged mitochondria can be isolated. Such mitochondria are said to be **tightly coupled**. By this we mean that electrons cannot pass through the electron transport chain without generation of ATP. If the concentration of ADP or of P_i becomes too low, both phosphorylation and respiration cease. This **respiratory control** by ADP and P_i is a property of undamaged mitochondria. It may seem surprising that damaged mitochondria or submitochondrial particles are often able to transfer electrons at a faster rate than do undamaged mitochondria. However, electron transfer in damaged mitochondria occurs *without synthesis of ATP* and with no slowdown as the ADP concentration drops. A related kind of **uncoupling** of electron transport from ATP synthesis is brought about by various lipophilic anions called **uncouplers**, the best known of which is **2,4-dinitrophenol**. Even before the phenomenon of uncoupling was discovered, it had been known that dinitrophenol substantially increased the respiration rates of animals. The compound had even been used (with some fatal results) in weight control pills. The chemical basis of uncoupling will be considered in Section D.

"States" of mitochondria and spectrophotometric observation. Chance and Williams defined five **states** of tightly coupled mitochondria[60,163]; of these, states 3 and 4 are most often mentioned. If no oxidizable substrate or ADP is added the mitochondria have a very low rate of oxygen uptake and are in state 1. If oxidizable substrate and ADP are added rapid O_2 uptake is observed, the rate depending upon the rate of flow of electrons through the electron transport chain. This is state 3. As respiration occurs the coupled phosphorylation converts ADP into ATP, exhausting the ADP. Respiration slows to a very low value and the mitochondria are in state 4. If the substrate is present in excess, addition of more ADP will return the mitochondria to state 3.

Chance and associates employed spectrophotometry on intact mitochondria or submitochondrial particles to investigate both the sequence of carriers and the sites of phosphorylation. Using the dual wavelength spectrophotometer, the light absorption at the absorption maximum (λ_{max}) of a particular component was followed relative to the absorption at some other reference wavelength (λ_{ref}). The principal wavelengths used are given in Table 18-6. From these measurements the state of oxidation or reduction of each one of the carriers could be observed in the various states and in the presence of inhibitors. The

TABLE 18-6
Wavelengths of Light Used to Measure States of Oxidation of Carriers in the Electron Transport Chain of Mitochondria[a]

Carrier	λ_{max} (nm)[b]	λ_{ref} (nm)
NADH	340	374
Flavins	465	510
Cytochromes		
b^{2+}	564(α)	575
	530(β)	
	430(γ)	
c_1^{2+}	534(α)	
	523(β)	
	418(γ)	
c^{2+}	550(α)	540
	521(β)	
	416(γ)	
a^{2+}	605(α)	630(590)
	450(γ)[a]	
a_3^{2+}	600(α)[a]	
	445(γ)	455

[a] After Chance, B. and Williams, G. R. (1955) *J. Biol. Chem*, **217**, 409–427; (1956) Adv. Enzymol. **17**, 409–427.
[b] The wavelengths used for each carrier in dual wavelength spectroscopy appear opposite each other in the two columns. Some positions of other absorption bands of cytochromes are also given.

experiments served to establish that electrons passing down the chain do indeed reside for a certain length of time on particular carriers. That is, in a given state each carrier exists in a defined ratio of oxidized to reduced forms ([ox] / [red]). Such a result would not be seen if the entire chain functioned in a cooperative manner with electrons passing from the beginning to the end in a single reaction. By observing changes in the ratio [ox] / [red] under different conditions, some localization of the three phosphorylation sites could be made. In one experiment antimycin *a* was added to block the chain ahead of cytochrome c_1. Then tightly coupled mitochondria were allowed to go into state 4 by depletion of ADP. Since the concentration of oxygen was high and cytochrome a_3 has a low K_m for O_2 (~3 μM) cytochrome a_3 was in a highly oxidized state. Cytochrome *a* was also observed to be oxidized, while cytochrome c_1 and *c* remained reduced. The presence of this **crossover point** suggested at the time that cytochrome *c* might be at or near one of the "energy conservation sites." Accounts of more recent experiments using the same approach are given by Wilson *et al*.[164]

2. Thermodynamics and Reverse Electron Flow

From Table 6-8 the value of $\Delta G'$ for oxidation of one mole of NADH by oxygen (1 atm) is −219 kJ. At a pressure of ~10^{-2} atm O_2 in tissues the value is −213 kJ. However, when the reaction is coupled to the synthesis of three molecules of ATP ($\Delta G'$ = +34.5 kJ mol^{-1}) the net Gibbs energy change for the overall reaction becomes $\Delta G'$ = −110 kJ mol^{-1}. This is still very negative. However, we must remember that the concentrations of ATP, ADP, and P_i can depart greatly from the 1:1:1 ratio implied by the $\Delta G'$ value.

An interesting experiment is to allow oxidative phosphorylation to proceed until the mitochondria reach state 4 and to measure the **phosphorylation state ratio R_p**, which equals the value of [ATP] / [ADP][P_i] that is attained. This mass action ratio, which has also been called the "phosphorylation ratio" or "phosphorylation potential" (see Chapter 6 and Eq. 6-29), often reaches values greater than 10^4–10^5 M^{-1} in the cytosol.[164] An extrapolated value for a zero rate of ATP hydrolysis of log R_p = 6.9 was estimated. This corresponds (Eq. 6-29) to an increase in group transfer potential (ΔG of hydrolysis of ATP) of 39 kJ/mol. It follows that the overall value of ΔG for oxidation of NADH in the coupled electron transport chain is less negative than is $\Delta G'$. If synthesis of three molecules of ATP is coupled to electron transport, the system should reach an equilibrium when $R_p = 10^{6.4}$ at 25°C, the difference in ΔG and $\Delta G'$ being 3RT ln Rp = 3 x 5.708 x 6.4 = 110 kJ mol^{-1}. This value of Rp is, within experimental error, the same as the maximum value observed.[165] There apparently is an almost true equilibrium among NADH, O_2, and the adenylate system if the P/O ratio is 3.

Within more restricted parts of the chain it is possible to have *reversed electron flow*. Consider the passage of electrons from NADH, partway through the chain, and back out to fumarate, the oxidized form of the succinate–fumarate couple. The Gibbs energy change $\Delta G'$ (pH 7) for oxidation of NADH by fumarate is −67.7 kJ mol^{-1}. In uncoupled mitochondria electron flow would always be from NADH to fumarate. However, in tightly coupled mitochondria, in which ATP is being generated at site I, the overall value of $\Delta G'$ becomes much less negative. If $R_p = 10^4$ M^{-1}, $\Delta G'$ for the coupled process becomes approximately zero (−67.7 + 68 kJ mol^{-1}). Electron flow can easily be reversed so that succinate reduces NAD^+. Such ATP-driven reverse flow occurs under some physiological conditions within mitochondria of living cells, and some anaerobic bacteria generate all of their NADH by reversed electron flow (see Section E).

Another experiment involving equilibration with the electron transport chain is to measure the "observed potential" of a carrier in the chain as a function of the concentrations of ATP, ADP, and P_i. The observed potential E is obtained by measuring log([ox] / [red]) and applying Eq. 18-6 in which $E°'$ is the known midpoint potential of the couple (Table 6-8) and n is the number of electrons required to reduce one molecule of the carrier. If the system is equilibrated with a "redox buffer" (Chapter 6), E can be fixed at a preselected value. For example, a 1:1 mixture of succinate and fumarate would fix E at +0.03 V while the couple 3-hydroxybutyrate-acetoacetate in a 1:1 ratio would fix it at $E°' = -0.266$ V. Consider the potential of cytochrome b_{562} (b_H), which has an $E°'$ value of 0.030 V. Substituting this in Eq. 18-7 and using $E = -0.266$ V (as obtained by equilibration with 3-hydroxybutyrate–acetoacetate), it is easy to calculate that at equilibrium the ratio [ox] / [red] for cytochrome b_{562} is about 10^{-5}. In other words, in the absence of O_2 this cytochrome will be kept almost completely in the reduced form in an uncoupled mitochondrion.

$$E = \frac{-\Delta G}{nF} = E°' + \frac{0.0592}{n}\log\frac{[\text{ox}]}{[\text{red}]}$$
$$= \text{observed potential of carrier} \quad (18\text{-}6)$$

TABLE 18-7
Electrode Potentials of Mitochondrial Electron Carriers and Gibbs Energy Changes Associated with Passage of Electrons[a]

	Electron carrier	$E°'$ (pH 7) isolated	$E°'$ (pH 7.2) in mitochondria	ΔG (kJ mol^{-1}) for 2 e^- flow to O_2 at 10^{-2} atm, carriers at pH 7
	NADH / NAD+	–0.320		–213
Group I	Flavoprotein		~ –0.30	
~ –0.30 V	Fe–S protein		~ –0.305	
	β-Hydroxybutyrate–acetoacetate	–0.266		–203
	Lactate–pyruvate	–0.185		–187
	Succinate-fumarate	0.031		–146
Group II	Flavoprotein		~ –0.045	
~ 0 V	Cytochrome b_T		–0.030	
	Cu		0.001	
	Fe–S protein		0.030	
	Cytochrome b_K		0.030	
	Ubiquinone	0.10	0.045	–132
	Cytochrome a_3 + ATP		0.155	
Group III	Cytochrome c_1		0.215	
	Cytochrome c	0.254	0.235	–102
	Cytochrome b_T + ATP		0.245	
	Cytochrome a	0.29	0.210	
	Cu		0.245	
	Fe–S protein		0.28	
Group IV	Cytochrome a_3		0.385	–77
	O_2 (10^{-2} atm)	0.785		0.00
	1 atm	0.815		

[a] Data from Wilson, D. F., Dutton, P. L., Erecinska, M., Lindsay, J. G., and Soto, N. (1972) *Acc. Chem. Res.* **5**, 234–241 and Wilson, D. F., Erecinska, M., and Sutoon, P. L. (1974) *Ann. Rev. Biophys. Bioeng.* **3**, 203–230.

However, if the electron transport between 3-hydroxybutyrate and cytochrome b_{562} is tightly coupled to the synthesis of one molecule of ATP, the observed potential of the carrier will be determined not only by the imposed potential E_i of the equilibrating system but also by the phosphorylation state ratio of the adenylate system (Eq. 18-7). Here $\Delta G'_{ATP}$ is the group transfer potential ($-\Delta G'$ of hydrolysis) of ATP at pH 7 (Table 6-6), and n' is the number of electrons passing through the chain required to synthesize one ATP. In the upper part of the equation n is the number of electrons required to reduce the carrier, namely one in the case of cytochrome b_{562}.

From Eq. 18-7 it is clear that in the presence of a high phosphorylation state ratio a significant fraction of cytochrome b_{562} may remain in the reduced form at equilibrium. Thus, if $R_p = 10^4$, if $E°'$ for cytochrome b_{562} is 0.030 V, if $n' = 2$, and the potential E is fixed at -0.25 V using the hydroxybutyrate–acetoacetate couple, we calculate, from Eq. 18-7, that the ratio [ox] / [red] for cytochrome b_{562} will be 1.75. Now, if R_p is varied the observed potential of the carrier should change as predicted by Eq. 18-7. This variation has been observed.[164] For a tenfold change in R_p the observed potential of cytochrome b_{562} changed by 0.030 V, just that predicted if $n' = 2$. On the other hand, the observed potential of cytochrome c varied by 0.059 V for every tenfold change in the ratio. This is just as expected if $n' = 2$, and if synthesis of two molecules of

$$\begin{aligned} E(\text{observed}) &= E^{\circ\prime} + \frac{0.0592}{n}\log_{10}\frac{[\text{ox}]}{[\text{red}]} \\ &= E_i + \frac{\Delta G'_{\text{ATP}}}{96.5n'} + \frac{RT}{n'F}\ln\frac{[\text{ATP}]}{[\text{ADP}][\text{P}_\text{i}]} \\ &= E_i + \frac{0.358}{n'} + \frac{0.0592}{n'}\log_{10} R_\text{P} \end{aligned} \tag{18-7}$$

ATP is coupled to the electron transport to cytochrome *c*. Thus, we have experimental evidence that when one-electron carriers such as the cytochromes are involved, the passage of *two electrons* is required to synthesize one molecule of ATP. Furthermore, from experiments of this type it was concluded that the sites of phosphorylation were localized in or related to complexes I, III, and IV.

Another kind of experiment is to equilibrate the electron transport chain with an external redox pair of known potential using *uncoupled* mitochondria. The value of $E^{\circ\prime}$ of a particular carrier can then be measured by observation of the ratio [ox] / [red] and applying Eq. 18-7. While changes in the equilibrating potential E will be reflected by changes in [ox] / [red] the value of $E^{\circ\prime}$ will remain constant. The $E^{\circ\prime}$ values of Fe–S proteins and copper atoms in the electron transport chain have been obtained by equilibrating mitochondria, then rapidly freezing them in liquid nitrogen, and observing the ratios [ox] / [red] by EPR at 77K (Table 18-7).

The values of $E^{\circ\prime}$ of the mitochondrial carriers fall into four **isopotential groups** at ~–0.30, ~0, ~+0.22, and ~+0.39 V (Table 18-7). When tightly coupled mitochondria are allowed to go into state 4 (low ADP, high ATP, O_2 present but low respiration rate), the observed potentials change. That of the lowest isopotential group (which includes NAD^+ / NADH) falls to ~–0.38 V, corresponding to a high state of reduction of the carriers to the left of the first phosphorylation site in Fig. 18-4. Groups 2 and 3 remain close to their midpoint potentials at ~–0.05 and +0.26 V. In this condition the potential difference between each successive group of carriers amounts to ~0.32 V, just enough to balance the formation of one molecule of ATP for each two electrons passed at a ratio $R_\text{p} \approx 10^4$ M^{-1} (Eq. 18-7).

Two cytochromes show exceptional behavior and appear twice in Table 18-7. The midpoint potential $E^{\circ\prime}$ of cytochrome b_{566} (b_L) changes from –0.030 V in the absence of ATP to +0.245 V in the presence of a high concentration of ATP. On the other hand, $E^{\circ\prime}$ for cytochrome a_3 drops from +0.385 to 0.155 V in the presence of ATP. These shifts in potential must be related to the coupling of electron transport to phosphorylation.

3. The Mechanism of Oxidative Phosphorylation

It was natural to compare mitochondrial ATP synthesis with substrate-level phosphorylations, in which high-energy intermediates are generated *by the passage of electrons through the substrates*. The best known example is oxidation of the aldehyde group of glyceraldehyde 3-phosphate to an acyl phosphate, which, after transfer of the phospho group to ADP, becomes a carboxylate group (Fig. 15-6). The Gibbs energy of oxidation of the aldehyde to the carboxylate group provides the energy for the synthesis of ATP. However, this reaction differs from mitochondrial electron transport in that *the product, 3-phosphoglycerate, is not reconverted to glyceraldehyde 3-phosphate*. Electron carriers of the respiratory chain must be regenerated in some cyclic process. Because of this, it was difficult to imagine practical mechanisms for oxidative phosphorylation that could be related to those of substrate level phosphorylation. Nevertheless, many efforts were made over a period of several decades to find such high-energy intermediates.

Search for chemical intermediates. An early hypothetical model, proposed by Lipmann,[166] is shown in Fig. 18-12. Here A, B, and C are three electron carriers in the electron transport chain. Carrier C is a better oxidizing agent than B or A. Carrier B has some special chemistry that permits it, in the reduced state, to react with group Y of a protein (step *b*) to form $Y–BH_2$. The latter, an unidentified adduct, is converted by oxidation with carrier C (step *c*) to a "high energy" oxidized form indicated as Y ~ B. Once the possibility of generating such an intermediate is conceded, it is easy to imagine plausible ways in which the energy of this intermediate could be transferred into forms with which we are already familiar. For example, another protein X could react (step *d*) to form X ~ Y in which the X ~ Y linkage could be a thioester, an acyl phosphate, or other high-energy form. Furthermore, it might not be necessary to have two proteins; X and Y could be different functional groups of the same protein. They might be nonprotein components, e.g., Y might be a phospholipid.

Generation of ATP by the remaining reactions (steps *e* and *f* of Fig. 18-12) is straightforward. For example, if X ~ Y were a thioester the reactions would be the reverse of Eq. 12-48. These reaction steps would also be responsible for observed exchange reactions, for example, the mitochondrially catalyzed exchange of inorganic phosphate ($H^{32}PO_4^{2-}$) into the terminal position of ATP. Mitochondria and submitochondrial particles also contain ATP-hydrolyzing **(ATPase)** activity, which is thought to depend upon the same machinery that synthesizes ATP in tightly coupled mitochondria. In the scheme of Fig. 18-12, ATPase

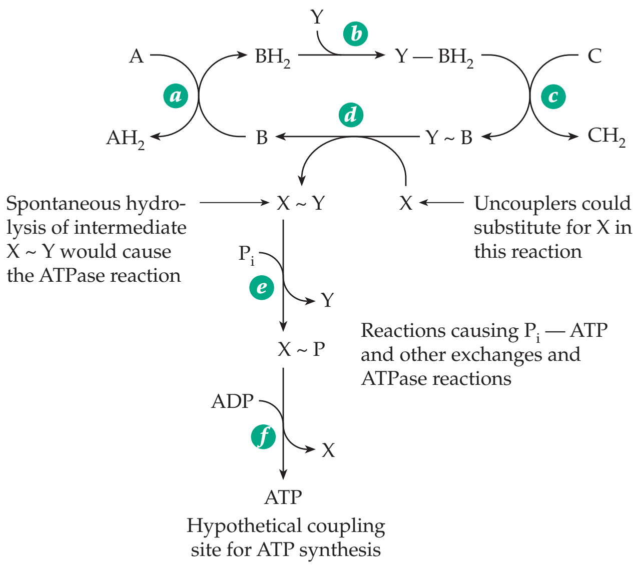


Figure 18-12 An early proposal for formation of ATP via "high-energy" chemical intermediates.

activity would be observed if hydrolysis of X ~ Y were to occur. Partial disruption of the system would lead to increased ATPase reactivity, as is observed. Uncouplers such as the dinitrophenolate ion or arsenate ion, acting as nucleophilic displacing groups, could substitute for a group such as X. Spontaneous breakdown of labile intermediates would permit oxidation to proceed unimpaired. Since there are three different sites of phosphorylation, we might expect to have three different enzymes of the type Y in the scheme of Fig. 18-12, but it would be necessary to have only one X.

In Lipmann's original scheme group Y was visualized as adding to a carbon–carbon double bond to initiate the sequence. Isotopic exchange reactions ruled out the possibility that either ADP or P_i might serve as Y, but it was attractive to think that a bound phosphate ion, e.g., in a phospholipid or coenzyme, could be involved. Y ~ B of Fig. 18-12 would be similar in reactivity to an acyl phosphate or thioester. However, whatever the nature of Y ~ B, part of group Y would be left attached to B after the transfer of Y to X. For example, if Y were Y'OH compound X ~ OY' would be formed, and the carrier would be left in step *d* in the form of B–OH. Elimination of a hydroxyl group would be required to regenerate B. Perhaps nature has shunned this mechanism because there is no easy way to accomplish such an elimination. Many variations on the scheme of Fig. 18-12 were proposed,[166] and some were discussed in the first edition of this textbook.[167] However, as attractive as these ideas may have seemed, *all attempts to identify discrete intermediates that might represent X ~ Y failed. Furthermore, most claims to have seen Y ~ B by any means have been disproved*.

Peter Mitchell's chemiosmotic theory. To account for the inability to identify high energy intermediates as well as the apparent necessity for an intact membrane, Peter Mitchell, in 1961, offered his **chemiosmotic theory** of oxidative phosphorylation.[168–175a] aThis theory also accounts for the existence of **energy-linked processes** such as the accumulation of cations by mitochondria. The principal features of the Mitchell theory are illustrated in Fig. 18-13. Mitchell proposed

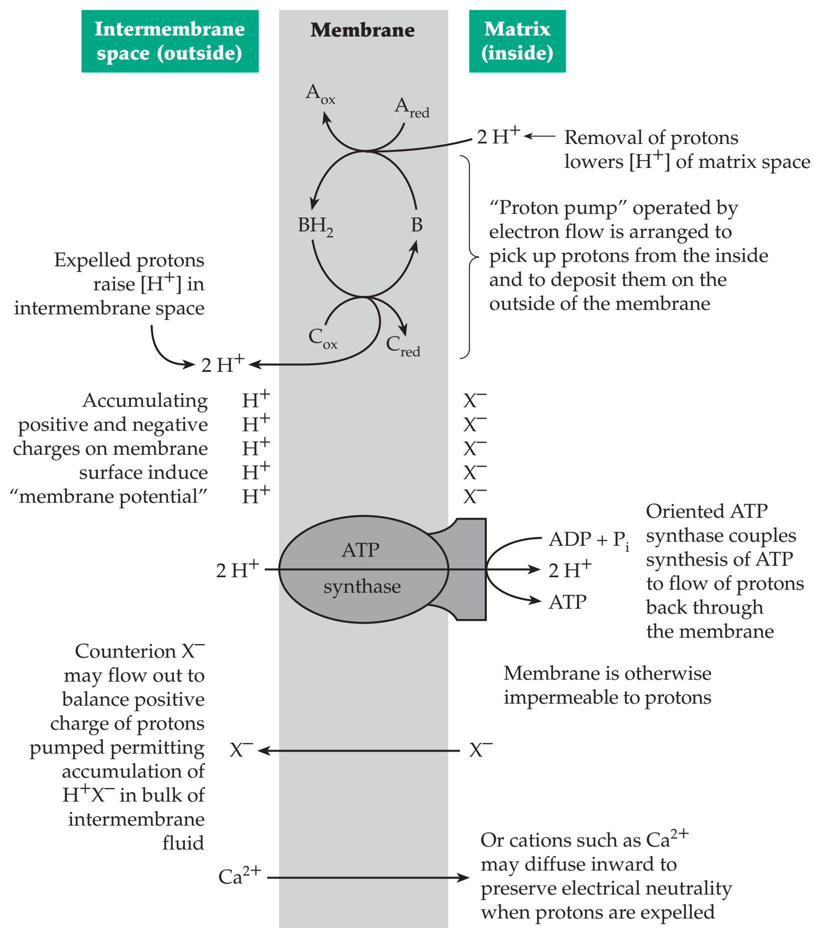


Figure 18-13 Principal features of Mitchell's chemiosmotic theory of oxidative phosphorylation.

that the inner membrane of the mitochondrion is a closed, proton-impermeable **coupling membrane**, which contains **proton pumps** operated by electron flow and which cause protons to be expelled through the membrane from the matrix space. As indicated in Fig. 18-13, an oxidized carrier B, upon reduction to BH_2, acquires two protons. These protons do not necessarily come from reduced carrier AH_2, and Mitchell proposed that they are picked up from the solvent on the matrix side of the membrane. Then, when BH_2 is reoxidized by carrier C, protons are released on the outside of the membrane. On the basis of existing data, *Mitchell assumed a stoichiometry of two protons expelled for each ATP synthesized*. It followed that there should be three different proton pumps in the electron transport chain corresponding to the three phosphorylation sites.

The postulated proton pumps would lead either to bulk accumulation of protons in the intermembrane space and cytoplasm, with a corresponding drop in pH, *or to an accumulation of protons along the membrane itself*. The latter would be expected if counterions X^- do not pass through the membrane with the protons. The result in such a case would be the development of a **membrane potential**, a phenomenon already well documented for nerve membranes (Chapter 8).

A fundamental postulate of the chemiosmotic theory is the presence of an oriented ATP synthase that utilizes the Gibbs energy difference of the proton gradient to drive the synthesis of ATP (Fig. 18-9). Since $\Delta G'$ (pH 7) for ATP synthesis is +34.5 kJ mol^{-1} and, if as was assumed by Mitchell, the passage of two protons through the ATP synthase is required to form one ATP, the necessary pH gradient (given by Eq. 6-25 or Eq. 18-9 with $E_m = 0$) would be 34.5/(2 x 5.708) = 3.0 pH units at 25°C. On the other hand, if the phosphorylation state ratio is ~10^4 M^{-1}, the pH difference would have to be 5 units. Most investigators now think that 4 H^+ per ATP are needed by the synthase. If so, a pH difference of 2.5 units would be adequate. Various experiments have shown that passage of electrons does induce a pH difference, and that an artifically induced pH difference across mitochondrial membranes leads to ATP synthesis. However, pH gradients of the required size have not been observed. Nevertheless, if the membrane were charged as indicated in Fig. 18-13, without accumulation of protons in the bulk medium, a membrane potential would be developed, and this could drive the ATP synthase, just as would a proton gradient.

The mitochondrial membrane potential E_m (or $\Delta\psi$) is the potential difference measured across a membrane relative to a reference electrode present in the surrounding solution.[176] For both mitochondria and bacteria E_m normally has a negative value. The Gibbs energy change $\Delta\psi_H+$ for transfer of one mole of H^+ from the inside of the mitochondrion to the outside, against the concentration and potential gradients, is given by Eq. 18-8. This equation follows directly from Eqs. 6-25

$$\begin{aligned}\Delta G_{H}+ &= 2.303\,RT\,\Delta pH - E_m F\\ &= 5.708\,\Delta pH - 96.5\,E_m \text{ kJ/mol at 25°C}\\ &\text{where } \Delta pH = pH\text{ (inside)} - pH\text{ (outside)}\end{aligned} \tag{18-8}$$

and 6-63 with n = 1. The same information is conveyed in Eq. 18-9, which was proposed by Mitchell for what he calls the **total protonic potential difference Δp**.

$$\begin{aligned}\Delta p\text{ (volts)} &= E_m\text{(volts)} - 2.303\,\frac{RT}{F}\,\Delta pH\\ \Delta p\text{ (mV)} &= E_m\text{(mV)} - 59.2\,\Delta pH \text{ at 25°C}\\ E_m &= \Delta\psi\end{aligned} \tag{18-9}$$

Mitchell was struck by the parallel between the force and flow of electrons, which we call electricity, and the force and flow of protons, which he named **proticity**.[174] This led one headline writer in *Nature*[177] to describe Mitchell as "a man driven by proticity," but if Mitchell is right, as seems to be the case, we are all driven by proticity! Mitchell also talked about **protonmotive** processes and referred to Δp as the **protonmotive force**. Although it is a potential rather than a force, this latter name is a popular designation for Δp.

The reader should be aware that considerable confusion exists with respect to names and definitions.[176] For example, the ΔG_H+ of Eq. 18-8 can also be called the **proton electrochemical potential** $\Delta\mu_H+$, which is analogous to the chemical potential μ of an ion (Eq. 6-24) and has units of kJ/mol (Eq. 18-10).

$$\begin{aligned}-\Delta G_{H}+ = \Delta\mu_{H}+ &= F\,\Delta p\\ &= 96.5\,\Delta p \text{ kJ/mol at 25°C}\end{aligned} \tag{18-10}$$

However, many authors use $\Delta\mu_H+$ as identical to the protonmotive force Δp.

From Eq. 18-9 or Eq. 18-10 it can be seen that a membrane potential E_m of −296 mV at 25°C would be equivalent to a 5.0 unit change in pH and would be sufficient, if coupled to ATP synthesis via 2 H^+, to raise R_p to 10^4 M^{-1}. Any combination of ΔpH and E_m providing Δp of −296 mV would also suffice. If the ratio H^+/ATP = 4, Δp of −148 mV would suffice.

The chemiosmotic hypothesis had the great virtue of predicting the following consequences which could be tested: (1) electron-transport driven proton pumps with defined stoichiometries and (2) a separate ATP synthase, which could be driven by a pH gradient or membrane potential. Mitchell's hypothesis was initially greeted with skepticism but it encouraged many people, including Mitchell and his associate Jennifer Moyle, to test these predictions, which were soon found to be correct.[178]

Observed values of E_m and pH. One of the problems[179] in testing Mitchell's ideas has been the difficulty of reliably measuring Δp. To evaluate the pH term in Eq. 18-10 measurements have been made with microelectrodes and indicator dyes. However, the most reliable approach has been to observe the distribution of weak acids and bases across the mitochondrial membrane.[180] This is usually done with a suspension of freshly isolated active mitochondria. The method has been applied widely using, for example, methylamine. A newer method employs an isotope exchange procedure to measure the pH-sensitive carbonic anhydrase activity naturally present in mitochondria.[181]

The measurement of E_m ($\Delta\psi$) is also difficult.[179] Three methods have been used: (1) measurement with microelectrodes; (2) observation of fluorescent probes; (3) distribution of permeant ions. Microelectrodes inserted into mitochondria[182] have failed to detect a significant value for E_m. Fluorescent probes are not very reliable,[179,183] leaving the distribution of permeant ions the method of choice. In this method a mitochondrial suspension is exposed to an ion that can cross the membrane but which is not pumped or subject to other influences that would affect its distribution. Under such conditions the ion will be distributed according to Eq. 18-11. The most commonly used ions are K^+, the same ion that is thought to reflect the membrane potential of nerve axons (Chapter 30), or Rb^+. To make the inner mitochondrial membrane permeable to K^+, valinomycin (Fig. 8-22) is added. The membrane potential, with n = 1 in Eq. 9-1, becomes:

$$E_m = -59.2\ ([K^+]_{inside} / [K^+]_{outside})\ \text{volts} \tag{18-11}$$

In these experiments respiring mitochondria are observed to take up the K^+ or Rb^+ to give a high ratio of K^+ inside to that outside and consequently a negative E_m. There are problems inherent in the method. The introduction of a high concentration of ion perturbs the membrane potential, and there are uncertainties concerning the contribution of the Donnan equilibrium (Eq. 8-5) to the observed ion distribution.[184]

In most instances, either for mitochondrial suspensions or whole bacteria, ΔpH is less negative than −0.5 unit making a contribution of, at most, −30 mV to Δp. The exception is found in the thylakoid membranes of chloroplasts (Chapter 23) in which protons are pumped into the thylakoid vesicles and in which the internal pH falls dramatically upon illumination of the chloroplasts.[185] The ΔpH reaches a value of −3.0 or more units and Δp is ~180 mV, while E_m remains ~0. Reported values of E_m for mitochondria and bacteria range from −100 to −168 mV and Δp from −140 to −230 mV.[172,179] Wilson concluded that E_m for actively respiring mitochondria, using malate or glutamate as substrates, attains maximum (negative) values of E_m = −130 mV and Δp = −160 mV.[179] However, Tedeschi and associates[183,184] argued that E_m is nearly zero for liver mitochondria and seldom becomes more negative than −60 mV for any mitochondria.

A crucially important finding is that submitochondrial particles or vesicles from broken chloroplasts will synthesize ATP from ADP and P_i, when an artificial pH gradient is imposed.[172,186] Isolated purified F_1F_0 ATPase from a thermophilic *Bacillus* has been coreconstituted into liposomes with the light-driven proton pump **bacteriorhodopsin** (Chapter 23). Illumination induced ATP synthesis.[187] These observations support Mitchell's proposal that the ATP synthase is both spatially separate from the electron carriers in the membrane and utilizes the protonmotive force to make ATP. Thus, the passage of protons from the outside of the mitochondria back in through the ATP synthase induces the formation of ATP. What is the stoichiometry of this process?

It is very difficult to measure the flux of protons across the membrane either out of the mitochondria into the cytoplasm or from the cytoplasm through the ATP synthase into the mitochondria. Therefore, estimates of the stoichiometry have often been indirect. One argument is based on thermodynamics. If Δp attains values no more negative than −160 mV and R_p within mitochondria reaches at least $10^4\ M^{-1}$, we must couple ΔG_H of −15.4 kJ/mol to ΔG of formation of ATP of +57.3 kJ/mol. To do this four H^+ must be translocated per ATP formed. Recent experimental measurements with chloroplast ATP synthase[188] also favor four H^+. It is often proposed that one of these protons is used to pump ADP into the mitochondria via the ATP-ADP exchange carrier (Section D). Furthermore, if R_p reaches $10^6\ M^{-1}$ in the cytoplasm, it must exceed $10^4\ M^{-1}$ in the mitochondrial matrix.

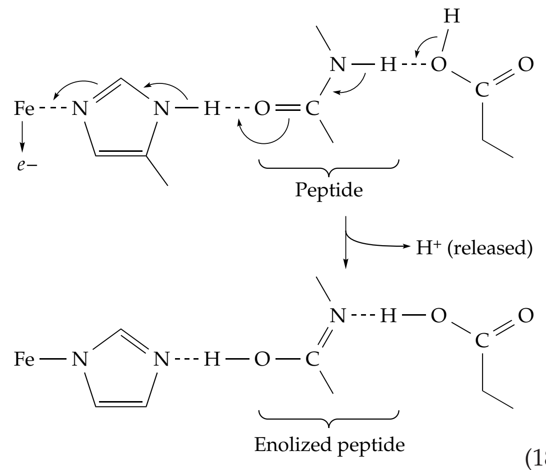
Proton pumps driven by electron transport. What is the nature of the proton-translocating pumps that link Δp with electron transport? In his earliest proposals Mitchell suggested that electron carriers, such as flavins and ubiquinones, each of which accepts two protons as well as two electrons upon reduction, could serve as the proton carriers. Each pump would consist of a pair of oxidoreductases. One, on the inside (matrix side) of the coupling membrane, would deliver two electrons (but no protons) to the carrier (B in Fig. 18-13). The two protons needed for the reduction would be taken from the solvent in the matrix. The second oxidoreductase would be located on the outside of the membrane and would accept two electrons from the reduced carrier (BH_2 in Fig. 18-13) leaving the two released protons on the outside of the membrane. To complete a "loop" that would allow the next carrier to be reduced, electrons would have to be transferred through fixed electron carriers embedded in the membrane from the reduced electron acceptor (C_{red} in Fig. 18-13) to the oxidized form of the oxidoreductase to be used as reductant for the next loop. These loops, located in complexes I and III of Fig. 18-5, would pump three protons per electron or six H^+/O. With a P/O ratio of three this would provide two H^+ per ATP formed. Mitchell regarded this stoichiometry as appropriate.

The flavin of NAD dehydrogenase was an obvious candidate for a carrier, as was ubiquinone. However, the third loop presented a problem. Mitchell's solution was the previously discussed **Q cycle**, which is shown in Fig. 18-9. This accomplishes the pumping in complex III of 2 H^+/e^-, the equivalent of two loops.[111] However, as we have seen, the magnitude of Δp suggests that 4 H^+, rather than 2 H^+, may be coupled to synthesis of one ATP. If this is true, mitochondria must pump 12 H^+/O rather than six when dehydrogenating NADH, or eight H^+/O when dehydrogenating succinate.

The stoichiometry of proton pumping was measured by Lehninger and associates using a fast-responding O_2 electrode and a glass pH electrode.[189,190] They observed an export of eight H^+/O for oxidation of succinate rat liver mitochondria in the presence of a permeant cation that would prevent the buildup of E_m, and four H^+/O (2 H^+/e^-) for the cytochrome oxidase system. These are equivalent to two H^+/e^- at each of sites II and III as is indicated in Fig. 18-4. Some others have found lower H^+/e^- ratios.

If two H^+/e^- are pumped out of mitochondria, where do we find the pumping sites? One possibility is that protons are pumped through the membrane by a **membrane Bohr effect**, so named for its similarity to the Bohr effect observed upon oxygenation of hemoglobin. In the latter case (Chapter 7), the pK_a values of certain imidazole and terminal amino groups are decreased when O_2 binds. This may result, in part, from an electrostatic effect of O_2 in inducing a partial positive charge in the heme. This partial charge may then cause a decrease in the pK_a values of nearby groups. Similarly, complete loss of an electronic charge from a heme group or an iron–sulfur protein in the electron transport chain would leave a positive charge, an electron "hole," which could induce a large change in the pK_a of a neighboring group. One manifestation of this phenomenon may be a strong pH dependence of the reduction potential (Eq. 16-19).

Protons that could logically be involved in a membrane Bohr effect are those present on imidazole rings coordinated to Fe or Cu in redox proteins. Removal of an electron from the metal ion could be accompanied by displacement of electrons within the imidazole, within a peptide group that is hydrogen-bonded to an imidazole, or within some other acidic group. A hypothetical example is illustrated in Eq. 18-12 in which a carboxyl group loses a proton when "handed" a second. If the transiently enolized peptide linkage formed in



(18-12)

this process is tautomerized back to its original state before the iron is reduced again, the proton originally present on the carboxyl group will be released. It is easy to imagine that a proton could then be "ferried" in (as in Eq. 9-96) from the opposite side of the membrane to reprotonate the imidazole group and complete the pumping process.

In view of the large number of metal-containing electron carriers in the mitochondrial chain, there are many possible locations for proton pumps. However, the presence of the three isopotential groups of Table 18-7 suggests that the pumps are clustered in complexes I-III as pictured in Fig. 18-5. One site of pumping is known to be in the cytochrome *c* oxidase complex. When reconstituted into phospholipid, the purified complex does pump protons in response to electron transport, H^+/e^- ratios of ~1 being observed.[136,137,147,191] As mentioned in Section B,3 a large amount of experimental effort has been devoted to identifying proton transport pathways in cytochrome *c* oxidase and also in the cytochrome bc_1 (complex II).[192] Proton pumping appears to be coupled to chemical changes occurring between intermediates P and F of Fig. 18-11, between F and O,[136,193] and possibly between O and R.[137,138] Mechanisms involving direct coupling of chemical changes at the A_3Cu_B center and at the Cu_A dimetal center have been proposed.[147,194]

How do protons move from the pumping sites to ATP synthase molecules? Since protons, as H_3O^+, are sufficiently mobile, ordinary diffusion may suffice. Because of the membrane potential they will tend to stay close to the membrane surface, perhaps being transported on phosphatidylethanolamine head groups (see Chapter 8). According to the view of R. J. P. Williams protons are not translocated across the entire membrane by the proton pumps, but flow through the proteins of the membrane to the ATP synthase.[195] There the protons induce the necessary conformational changes to cause ATP synthesis. A related idea is that transient high-energy intermediates generated by electron transport within membranes are proton-carrying conformational isomers. When an electron is removed from an electron-transporting metalloprotein, the resulting positively charged "hole" could be stable for some short time, while the protein diffused within the membrane until it encountered an F_0 protein of an ATP synthase. Then it might undergo an induced conformational change at the same time that it "handed" the Bohr effect proton of Eq. 18-14 to the F_0 protein and simultaneously induced a conformational change in that protein. The coupling of proton transport to conformational changes seems plausible, when we recall that the induction of conformational changes within proteins almost certainly involves rearrangement of hydrogen bonds.

A consequence of the chemiosmotic theory is that there is no need for an integral stoichiometry between protons pumped and ATP formed or for an integral P/O ratio. There are bound to be inefficiencies in coupling, and Δp is also used in ways other than synthesis of ATP.

4. ATP Synthase

In 1960, Racker and associates[196,197] discovered that the "knobs" or "little mushrooms" visible in negatively stained mitochondrial fragments or fragments of bacterial membranes possess ATP-hydrolyzing (**ATPase)** activity. Earlier the knob protein had been recognized as one of several **coupling factors** required for reconstitution of oxidative phosphorylation by submitochondrial particles.[197] Electron micrographs showed that the submitochondrial particles consist of closed vesicles derived from the mitochondrial cristae, and that the knobs (Fig. 18-14A) are on the *outside* of the vesicles. They can be shaken loose by ultrasonic oscillation with loss of phosphorylation and can be added back with restoration of phosphorylation. The knob protein became known as **coupling factor F_1**. Similar knobs present on the outside of the thylakoids became **CF_1** and those inside thermophilic bacteria **TF_1**. The ATPase activity of F_1 was a clue that *the knobs were really ATP synthase*. It also became clear that a portion of the ATP synthase is firmly embedded in the membranes. This part became known as **F_0**. Both the names F_1F_0 ATP synthase and F_1F_0 ATPase are applied to the complex, the two names describing different catalytic activities. The ATPase activity is usually not coupled to proton pumping but is a readily measurable property of the F_1 portion. In a well-coupled submitochondrial particle the ATPase activity will be coupled to proton transport and will represent a reversal of the ATP synthase activity.

The synthase structure. The F_1 complex has been isolated from *E. coli*,[204,205] other bacteria,[206,207] yeast,[208a,b] animal tissues,[199,209–211] and chloroplasts.[212–214] In every case it consists of five kinds of subunits with the stoichiometry $\alpha_3\beta_3\gamma\delta\varepsilon$.[214a,b] The F_0 complex of *E. coli* contains three subunits designated a,b, and c. All of these proteins are encoded in one gene cluster, the *unc* operon (named for uncoupled mutants), with the following order:

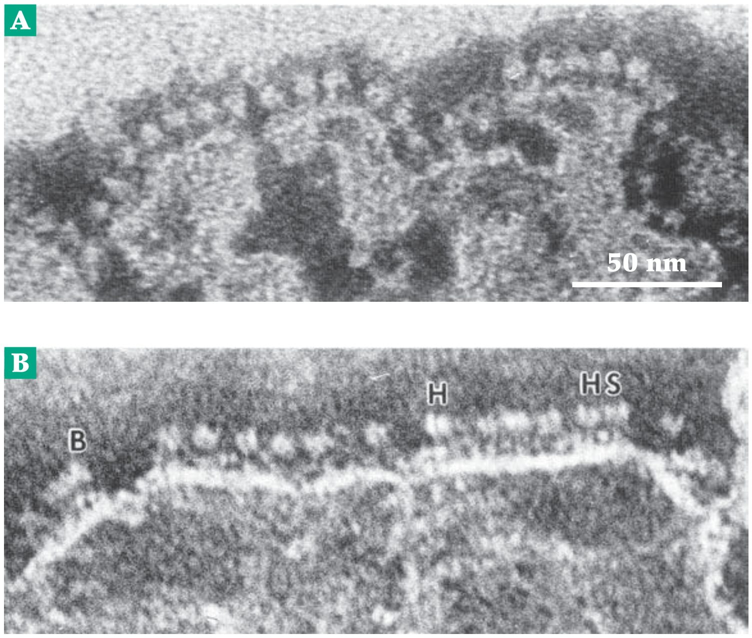
I	*B*	*E*	*F*	*H*	*A*	*G*	*D*	*C*	--- Gene symbol
i	a	c	b	δ	α	γ	β	ε	--- Subunit symbol

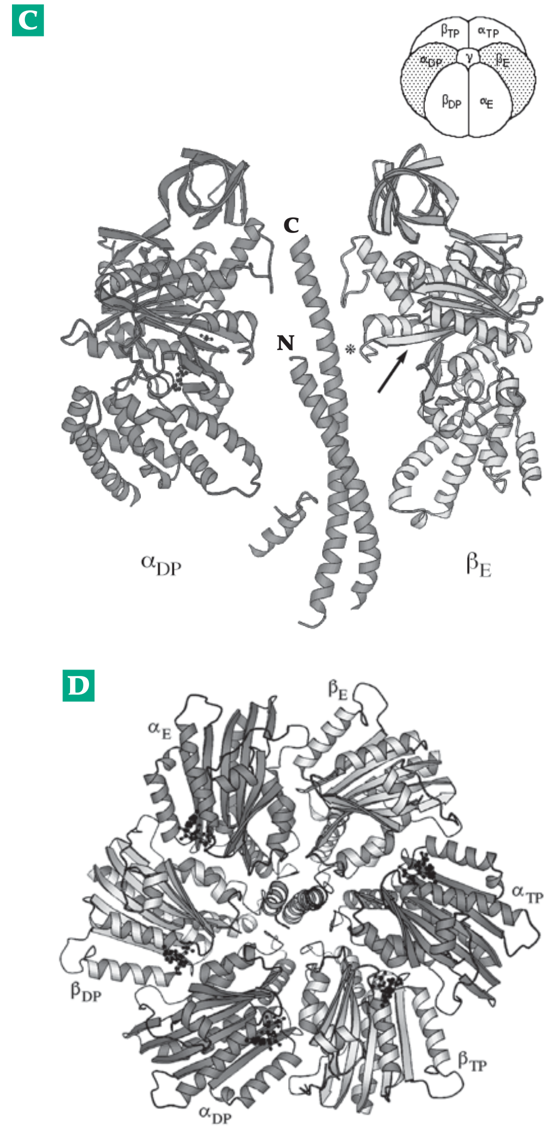
Here I is the regulatory gene (as in Fig. 28-1). The *E. coli* F_0 appears to have approximately the unusual stoichiometry $ab_2c_{9\text{-}11}$. This suggested the possibility that 12 c subunits form a ring with D_6 or D_{12} symmetry, the latter being illustrated in the structural proposal shown in Fig. 18-14E. However, crystallographic evidence suggests that there may be 10, not 12 subunits.[214c]

Mitochondrial ATP synthase of yeast contains at least 13 different kinds of subunits[208] and that of animals[215] 16, twice as many as in *E. coli*. Subunits α, β, γ, a, b, and c of the mitochondrial synthase correspond to those of *E. coli*. However, the mitochondrial homolog of *E. coli* δ is called the **oligomycin-sensitivity-conferring protein** (OSCP).[216–218] It makes the ATPase activity sensitive to oligomycin. The mitochondrial δ subunit corresponds to ε of *E. coli* or of chloroplasts.[217,219] Mitochondrial ε has no counterpart in bacteria.[209,220] In addition,[209,215] mitochondria contain subunits called d, e, f, g, A6L, F6, and IF_1, the last being an 84-residue inhibitor, a regulatory subunit.[221] The subunits of yeast ATP synthase correspond to those of the animal mitochondrial synthase but include one additional protein (**h**).[208a]

	F_1							F_0								
E. coli	α	β	γ	δ	ε			a	b	c						
Mitochondria	α	β	γ	OSCP	δ	ε	IF_1	a	b	c	d	e	f	g	A6L	F6

Six of the relatively large (50–57 kDa) α and β subunits associate to an $\alpha_3\beta_3$ complex that constitutes the knobs.[202,210] Chemical crosslinking, directed mutation, electron cryomicroscopy,[222,222a] and high-resolution X-ray diffraction measurements[199,207,211,223,224] have established that the α and β subunits alternate in a quasisymmetric cyclic head that contains active sites for ATP formation in the three β subunits (Fig. 18-14C–E). The α subunits also contain ATP-binding sites, but they are catalytically inactive, and their bound MgATP does not exchange readily with external ATP and can be replaced by the nonhydrolyzable AMP-PNP (Fig. 12-31) with retention of activity. The $\alpha_3\beta_3$ complex is associated with the F_0 part by a slender **central stalk**





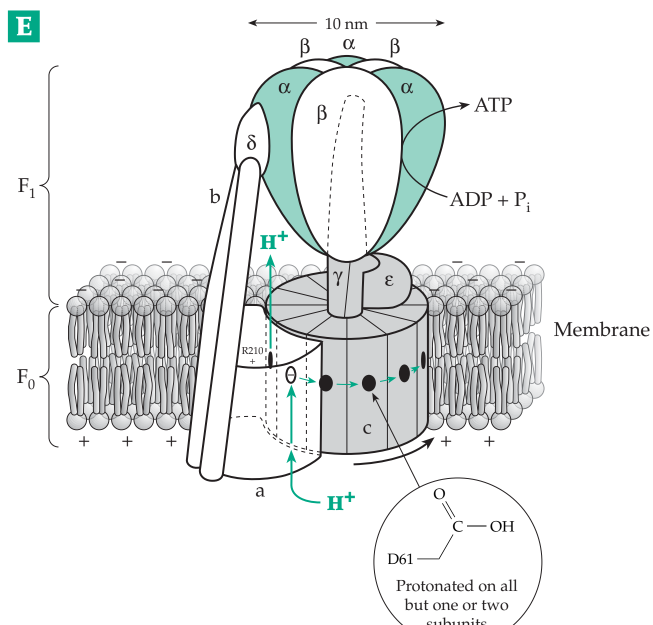


Figure 18-14 ATP synthase and vacuolar ATPase. (A) "Knobs" of ATP synthase on mitochondrial membranes negatively stained with phosphotungstate. (B) Vacuolar proton-pumping ATPase from an intact vacuolar membrane stained in the same way. Some images have been marked to indicate well-resolved head groups (H), stalks (S), and basal components (B). (A) and (B) are from Dschida and Bowman.[198] Courtesy of Barry J. Bowman. (C) Ribbon model of the atomic structure of the F_1 part of bovine heart mitochondrial ATP synthase. This section through the knob, which is drawn schematially at the upper right, shows one α subunit, containing bound ADP, α_{DP}, to the left and one empty β subunit, β_E, to the right. In the center are the N and C termini of subunit γ. The arrow points to a disruption in the β sheet of the subunit structure in the β_E conformation. The asterisk marks a loop that would collide with the C-terminal part of subunit γ if the latter were rotated. (D) View of the F_1 ATP synthase from the membrane side. The section shown contains the nucleotide binding sites. Subunits with empty sites are labeled α_E and β_E. Those with bound ADP are labeled α_{DP} and β_{DP}, and those with bound ATP α_{TP} and β_{TP}. Deviation from perfect threefold symmetry can be seen in this view. (C) and (D) are from Abrahams *et al.*[199] Courtesy of John E. Walker. (E) A recent model of the *E. coli* ATP synthase. The $\alpha_3\beta_3$ head contains the ATP-synthesizing sites in the β subunits. The $\alpha_3\beta_3$ complex and also subunits a, b, and δ remain stationary and form the stator for a protic engine, whose rotor consists of 12 c subunits and attached γ and ε subunits. Rotation is induced by the membrane potential or difference in H^+ activity on the two sides of the membrane. The carboxylate group of Asp 61 in each c subunit must be protonated to –COOH in order for it to move away from the entrance channel in subunit a. The presence of the positively charged Arg 210 near the exit channel in subunit a induces release of the proton when the c subunit has rotated almost 360°. According to this picture, 12 protons are required for one rotation with synthesis of three molecules of ATP. After Elston *et al.*[200] and Zhou *et al.*[201] See also Junge *et al.*[202] and Engelbrecht and Junge.[203]

or **shaft**. Much effort has gone into establishing the subunit composition of the shaft and the F_0 parts of the structure. As is indicated in Fig. 18-14E, subunits γ and ε of the *E. coli* enzyme are both part of the central shaft.[219,225,226] The same is true for the mitochondrial complex, in which the δ subunit corresponds to bacterial ε.[227] The role of this subunit is uncertain. It is part of the shaft but is able to undergo conformational alterations that can permit its C-terminal portion to interact either with F_0 or with the $\alpha_3\beta_3$ head.[227,227a] The unique ε subunit of mitochondrial ATPase appears also to be part of the shaft.[220]

The most prominent component of the central shaft is the 270-residue subunit γ, which associates loosely with the $\alpha_3\beta_3$ head complex but more tightly with F_0. About 40 residues at the N terminus and 60 at the C terminus form an α-helical coiled coil, which is visible in Fig. 18-14E[199,211] and which protrudes into the central cavity of the $\alpha_3\beta_3$ complex. Because it is asymmetric, the γ subunit apparently acts as a rotating camshaft to physically alter the α and β subunits in a cyclic manner. Asymmetries are visible in Fig. 18-14D.[211] The central part of subunit γ forms a more globular structure, which bonds with the c subunits of F_0.[205] Exact structures are not yet clear.

The δ subunit of *E. coli* ATP synthase (OSCP of mitochondria) was long regarded as part of the central stalk. However, more recent results indicate that it is found in a **second stalk**, which joins the $\alpha_3\beta_3$ complex to F_0. The central stalk rotates, relative to the second stalk. The second stalk may be regarded as stationary and part of a **stator** for a protic engine.[228,229] This stalk has been identified[230] in electron micrographs of chloroplast F_1F_0 and by crosslinking studies. As is depicted in Fig. 18-14E, a major portion of the second stalk is formed by two molecules of subunit b. Recent results indicate that bacterial subunit δ (mitochondrial OSCP) extends further up than is shown in Fig. 18-14, and together with subunit F6 may form a cap at the top of the $\alpha_3\beta_3$ head.[230a, 230b]

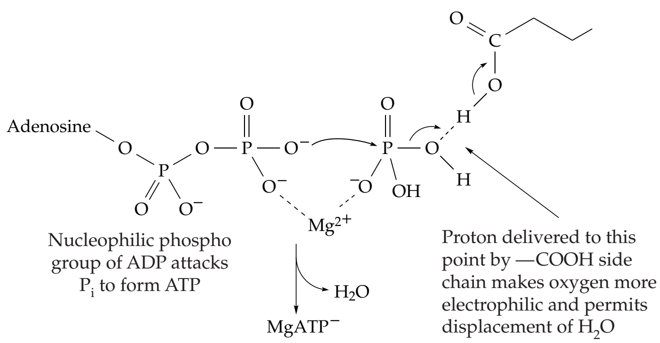
The F_0 portion of bacterial ATP synthase, which is embedded in the membrane, consists of one 271-residue subunit a, an integral membrane protein probably with five transmembrane helices,[231,232] two 156-residue b subunits, and ~ twelve 79-residue c subunits. The c subunit is a proteolipid, insoluble in water but soluble in some organic solvents. The structure of monomeric c in chloroform:methanol:H_2O (4:4:1) solution has been determined by NMR spectroscopy. It is a hairpin consisting of two antiparallel α helices.[233] Twelve of the c subunits are thought to assemble into a ring with both the N and C termini of the subunit chains in the periplasmic (or intermembrane) face of the membrane.[234,235] The ratio of c to a subunits has been difficult to measure but has been estimated as 9–12. The fact that both genetically fused c_2 dimers and c_3 trimers form function F_0 suggested that they assemble to a c_{12} ring as shown in Fig. 18-14E.[236] However, the recent crystallographic results that revealed a C_{10} ring[214c] raise questions about stoichiometry.

Since ATP synthesis takes place in F_1, it has long been thought that the F_0 part of the ATP synthase contains a "proton channel," which leads from the inside of the mitochondria to the F_1 assemblage.[146] Such a channel would probably not be an open pore but a chain of hydrogen-bonded groups, perhaps leading through the interior of the protein and able to transfer protons via icelike conduction. One residue in the c subunit, Asp 61, which lies in the center of the second of the predicted transmembrane helices, is critical for proton transport.[236a] Natural or artificial mutants at this position (e.g., D61G or D61N) do not transport protons. This carboxyl group also has an unusually high reactivity and specificity toward the protein-modifying reagent dicyclohexylcarbodiimide (DCCD; see Eq. 3-10).[146,237] Modification of a single c subunit with DCCD blocks the proton conductance.

An interesting mutation is replacement of alanine 62 of the c subunit with serine. This mutant will support ATP synthase using Li^+ instead of H^+.[237] Certain alkylophilic bacteria, such as *Propionigenium modestum*, have an ATP synthase that utilizes the membrane potential and a flow of Na^+ ions rather than protons through the c subunits.[238–240c] The sodium transport requires glutamate 65, which fulfills the same role as D61 in *E. coli*, and also Q32 and S66. Study of mutants revealed that the polar side chains of all three of these residues bind Na^+, that E65 and S66 are needed to bind Li^+, and that only E65 is needed for function with H^+.

The a subunit is also essential for proton translocation.[231,232,241] Structural work on this extremely hydrophobic protein has been difficult, but many mutant forms have been studied. Arginine 210 is essential as are E219 and H245. However, if Q252 is mutated to glutamate, E219 is no longer essential.[241] One of the OXPHOS diseases (NARP; Box 18-B) is a result of a leucine-to-arginine mutation in human subunit a.[241a] The b subunit is an elongated dimer, largely of α-helical structure.[242,243] Its hydrophobic N terminus is embedded in the membrane,[229,244] while the hydrophilic C-terminal region interacts with subunit σ of F_1, in the stator structure (Fig. 18-14E). Some of the F_0 subunits (d, e, f, g, A6L) may form a collar around the lower end of the central stalk.[210a,b]

How is ATP made? No covalent intermediates have been identified, and isotopic exchange studies indicate a direct dehydration of ADP and P_i to form bound ATP.[245] For the nucleophilic terminal phospho group of ADP to generate a high-energy linkage directly by attack on the phosphorus atom of P_i an OH^- ion must be eliminated (Eq. 18-13). This is not a probable reaction at pH 7, but it would be reasonable at low pH. Thus, one function of the oriented ATP synthase might be to deliver one or more protons flowing in from F_0 specifically to the oxygen that is to be eliminated (Eq. 18-13). As we have seen (Section 2), on thermodynamic grounds 3–4 protons would probably be needed. Perhaps they could be positioned nearby to exert a large electrostatic effect, or they could assist in releasing the ATP formed from the synthase by inducing a conformational change. However, it isn't clear how protons could be directed to the proper spots.



(18-13)

Paul Boyer's binding change mechanism. Boyer and associates suggested that ATP synthesis occurs rapidly and reversibly in a closed active site of the ATP synthase in an environment that is essentially anhydrous. ATP would then be released by an energy-dependent conformational change in the protein.[245–249] Oxygen isotope exchange studies verified that a rapid interconversion of bound ADP, P_i, and ATP does occur. Studies of soluble ATP synthase, which is necessarily uncoupled from electron transport or proton flow, shows that ATP is exceedingly tightly bound to F_1 as expected by Boyer's mechanism.[248] According to his **conformational coupling** idea, protons flowing across the membrane into the ATP synthase would in some way induce the conformational change necessary for release of ATP.

The idea of conformational coupling of ATP synthesis and electron transport is especially attractive when we recall that ATP is used in muscle to carry out mechanical work. Here we have the hydrolysis of ATP coupled to motion in the protein components of the muscle. It seems reasonable that ATP should be formed as a result of motion induced in the protein components of the ATPase. Support for this analogy has come from close structural similarities of the F_1 ATPase β subunits and of the active site of ATP cleavage in the muscle protein myosin (Chapter 19).

A simple version of Boyer's binding change mechanism is shown in Figure 18-15. The three F_1 β subunits are depicted in three different conformations. In O the active site is open, in T it is closed, and if ATP is present in the active site it is tightly bound. In the low affinity L conformation ligands are bound weakly. In step *a* MgADP and P_i enter the L site while MgATP is still present in the T site. In step *b* a protonic-energy-dependent step causes synchronous conformational changes in all of the subunits. The tight site opens and MgATP is free to leave. At the same time MgADP and P_i in the T site are converted spontaneously to tightly bound ATP. The MgATP is in reversible equilibrium with MgADP + P_i, which must be bound less tightly than is MgATP. That is, the high positive value of $\Delta G'$ for formation of ATP must be balanced by a corresponding negative $\Delta G'$ for a conformational or electronic reorganization of the protein in the T conformation. Opening of the active site in step *b* of Fig. 18-15 will have a high positive $\Delta G'$ unless it is coupled to proton flow throught F_0. Of three sites in the subunits, one binds MgATP very tightly (K_d ~ 0.1 μM) while the other sites bind less tightly (K_d ~ 20 μM).[250,251] However, it has been very difficult to establish binding constants or K_m values for the ATPase reaction.[248] Each of the three β sites probably, in turn, becomes the high-affinity site, consistent with an ATP synthase mechanism involving protein conformational changes.

Rotational catalysis. Boyer suggested that there is a cyclic rotation in the conformations of the three β subunits of the ATP synthase, and that this might involve rotation about the stalk. By 1984, it had been shown that bacterial flagella are rotated by a protonic motor (Chapter 19), and a protic rotor for ATP synthase had been proposed by Cox *et al.*[252] and others.[245] However, the b subunits were thought to be in the central stalk.[222] More recently chemical crosslinking experiments,[201,253] as well as electron microscopy, confirmed the conclusion that an intact stator structure must also be present as in Fig. 18-14E.[202] The necessary second stalk is visible in CF_1F_0 ATPase of chloroplasts[230] and also in the related vacuolar ATPase, a proton or Na^+ pump from a clostridium.[254] See also Section 5. Another technique, **polarized absorption recovery after photobleaching**, was applied after labeling of Cys 322, the penultimate residue at the C terminus of the γ subunit with the dye eosin. After photobleaching with a laser beam the polarization of the light absorption by the dye molecule relaxed because of rotation. Relaxation was observed when ATP was added but not with ADPPNP.[202,255,256]

The most compelling experiments were performed by Noji *et al.*[202,257–260] They prepared the $\alpha_3\beta_3\gamma$ subcomplex of ATPase from a thermophilic bacterium. The complex was produced in *E. coli* cells from the cloned genes allowing for some "engineering" of the proteins. A ten-histidine "tag" was added at the N termini of the β subunits so that the complex could be "glued" to a microscope coverslip coated with a nickel complex with a high affinity for the His tags. The γ subunit shafts protrude upward as shown in Fig. 18-16. The γ subunit was mutated to replace its

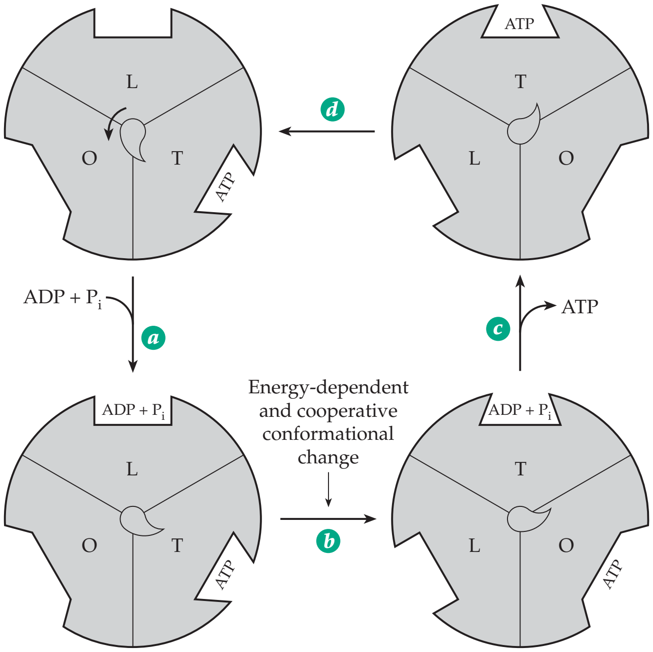


Figure 18-15 Boyer's binding change mechanism for ATP synthase in a simple form. After Boyer[245] but modified to include a central camshaft which may drive a cyclic alteration in conformations of the subunits. The small "pointer" on this shaft is not to be imagined as real but is only an indicator of rotation with induced conformational changes. The rotation could occur in 120° steps rather than the smaller steps suggested here.

only cysteine by serine and to introduce a cysteine in place of Ser 107 of the stalk region of γ. The new cysteine was biotinylated and attached to streptavidin (see Box 14-B) which was also attached to a fluorescently labeled actin filament (Fig. 7-10) ~1–3 µm in length as shown in Fig. 18-16. The actin fiber rotated in a counterclockwise direction when ATP was added but did not rotate with AMPPNP. At low ATP concentrations the rotation could be seen to occur in discrete 120° steps.[258,261,262] Each 120° step seems to consist of ~90° and ~30° substeps, each requiring a fraction of a millisecond.[262a] The ATPase appears to be acting as a **stepper motor**, hydrolysis of a single ATP turning the shaft 120°. Rotation at a rate of ~14 revolutions per second would require the hydrolysis of ~42 ATP per second. If the motor were attached to the F_0 part it would presumably pump four (or perhaps three) H^+ across a membrane for each ATP hydrolyzed. Acting in reverse, it would make ATP. A modification of the experiment of Fig. 18-16 was used to demonstrate that the c subunits also rotate with respect to the $\alpha_3\beta_3$ head.[262b] Other experiments support rotation of the c ring relative to subunit a.[262c,d]

Still to be answered are important questions. How does ATP hydrolysis turn the shaft? Are four H^+ pumped for each step, or are there smaller single proton substeps? Is the simple picture in Fig. 18-15 correct or, as proposed by some investigators,[263–265] must all three β subunits be occupied for maximum catalytic activity?[266] How is the coupling of H^+ transport to mechanical motion accomplished?[267,267a–d]

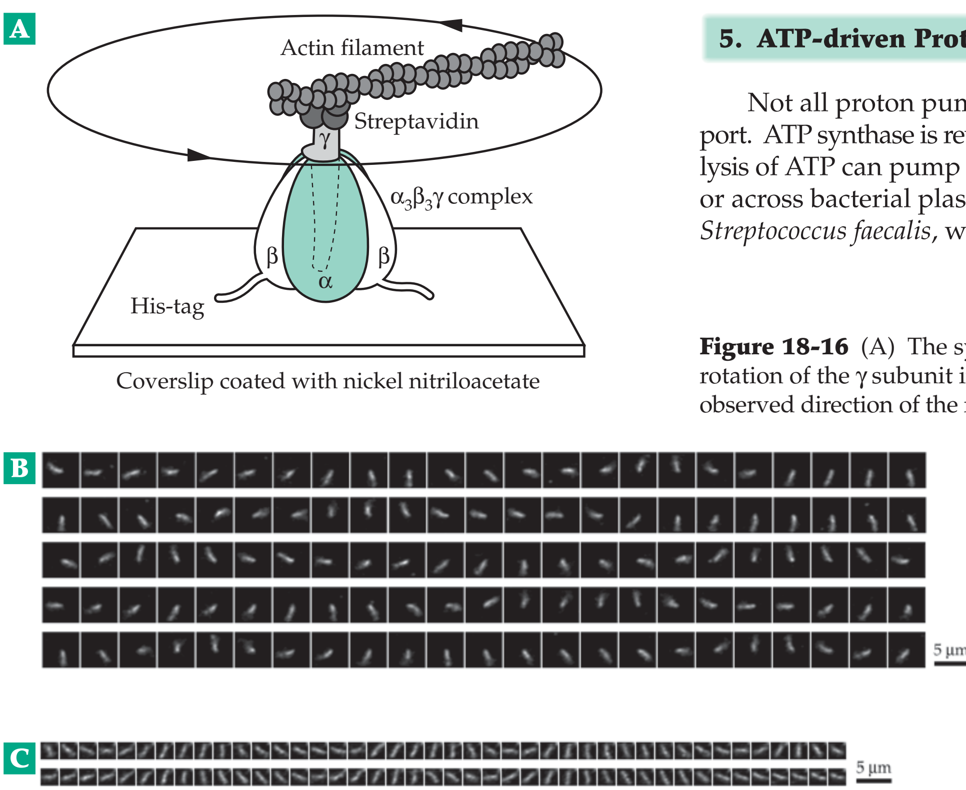


Figure 18-16 (A) The system used for observation of the rotation of the γ subunit in the $\alpha_3\beta_3\gamma$ subcomplex. The observed direction of the rotation of the γ subunit is indicated by the arrows. (B) Sequential images of a rotating actin filament attached as in (A). (C) Similar images obtained with the axis of rotation near the middle of the filament. The images correspond to the view from the top in (A). Total length of the filament, 2.4 µm; rotary rate, 1.3 revolutions per second; time interval between images, 33 ms. From Noji *et al.*[258]

5. ATP-driven Proton Pumps

Not all proton pumps are driven by electron transport. ATP synthase is reversible, and if Δp is low, hydrolysis of ATP can pump protons out of mitochondria or across bacterial plasma membranes.[268] Cells of *Streptococcus faecalis*, which have no respiratory chain and form ATP by glycolysis, use an F_1F_0 ATP synthase complex to pump protons out to help regulate cytoplasmic pH.[268] Similar **vacuolar (V-type) H^+-ATPases** or V_1V_0 ATPases pump protons into vacuoles, Golgi and secretory vesicles, coated vesicles, and lysosomes[198,269-270] in every known type of eukaryotic cell.[271,272] These proton pumps are similar in appearance (Fig. 18-14,B) and in structure to F_1F_0 ATPases.[272a–c] The 65- to 77-kDa A subunits and 55- to 60-kDa B subunits are larger than the corresponding F_1F_0 α and β subunits. Accessory 40-, 39-, and 33-kDa subunits are also present in V_1. The V_0 portion appears to contain a hexamer of a 16-kDa proteolipid together with 110- and 21-kDa subunits.[271] V-type ATPases are also found in archaebacteria[271,273] and also in some clostridia[254] and other eubacteria.[273a] A type of proton pump, the **V-PPase**, uses hydrolysis of inorganic pyrophosphate as a source of energy.[274] It has been found in plants, in some phototrophic bacteria, and in acidic calcium storage vesicles (acidocalcisomes) of trypanosomes.[274a]

Other ATP-dependent proton pumps are present in the plasma membranes of yeast and other fungi[274b] and also in the acid-secreting parietal cells of the stomach (Fig. 18-17). The H^+-ATPase of *Neurospora* pumps H^+ from the cytoplasm without a counterion. It is electrogenic.[275,275a] However, the gastric H^+,K^+-ATPase exchanges H_3O^+ for K^+ and cleaves ATP with formation of a phosphoenzyme.[276] It belongs to the family of P-type ion pumps that includes the mammalian Na^+,K^+-ATPase (Fig. 8-25) and Ca^{2+}-ATPase (Fig. 8-26). These are discussed in Chapter 8. The H^+,K^+-ATPases, which are widely distributed within eukaryotes, are also similar, both in sequence and in the fact that a phospho group is transferred from ATP onto a carbox-ylate group of an aspartic acid residue in the protein. All of them, including a Mg-ATPase of *Salmonella*, are two-subunit proteins. A large catalytic α subunit contains the site of phosphorylation as well as the ATP- and ion-binding sites. It associates noncovalently with the smaller heavily glycosylated β subunit.[276–278] For example, the rabbit H^+,K^+-ATPase consists of a 1035-residue α chain which has ten transmembrane segments and a 290-residue β chain with a single transmembrane helix and seven N-linked glycosylation sites.[278]

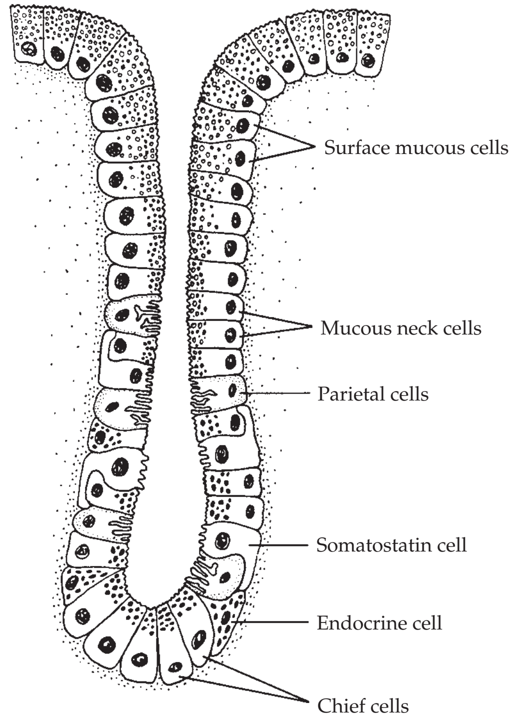
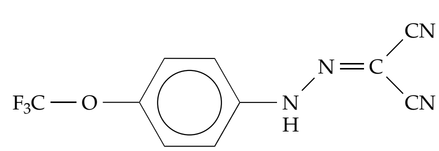


Figure 18-17 Schematic diagram of an acid-producing oxyntic gland of the stomach. The normal human stomach contains about 10^9 parietal (oxyntic) cells located in the walls of these glands. From Wolfe and Soll.[279] Modified from Ito. These glands also produce mucus, whose role in protecting the stomach lining from the high acidity is uncertain.[280]

6. Uncouplers and Energy-linked Processes in Mitochondria

Many compounds that uncouple electron transport from phosphorylation, like 2,4-dinitrophenol, are weak acids. Their anions are nucleophiles. According to the scheme of Fig. 18-12, they could degrade a high energy intermediate, such as Y ~ B, by a nucleophilic attack on Y to give an inactive but rapidly hydrolyzed derivative of Y. On the other hand, according to Mitchell's hypothesis uncouplers facilitate the transport of protons back into the mitochondria thereby destroying Δp. The fact that the anions of the uncouplers are large, often aromatic, and therefore soluble in the lipid bilayer supports this interpretation; the protonated uncouplers can diffuse into the mitochondria and the anion can diffuse back out. Mitochondria can also be uncoupled by a combination of ionophores, e.g., a mixture of valinomycin (Fig. 8-22), which carries K^+ into the mitochondria, plus nigericin, which catalyzes an exchange of K^+ (out) for H^+ (in).[172]

The uncoupler carbonyl cyanide *p*-(trifluoromethoxy)phenylhydrazone (FCCP) and related compounds are widely used in biochemical studies. Their action can be explained only partially by increased proton conduction.

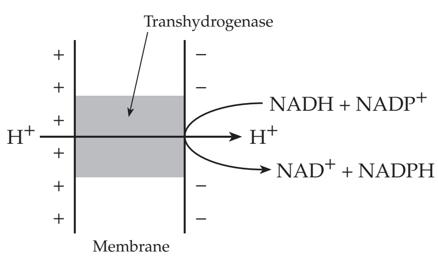


Carbonyl cyanide *p*-(trifluoromethoxy)phenylhydrazone (FCCP)

Uncoupling is sometimes important to an organism. The generation of heat by uncoupling is discussed in Box 18-C. The fungus *Bipolaris maydis* caused a crisis in maize production when it induced pore formation in mitochondrial membranes of a special strain used in production of hybrid seeds.[281,282]

Synthesis of ATP by mitochondria is inhibited by oligomycin, which binds to the OSCP subunit of ATP synthase. On the other hand, there are processes that require energy from electron transport and that are not inhibited by oligomycin. These **energy-linked processes** include the transport of many ions across the mitochondrial membrane (Section E) and reverse electron flow from succinate to NAD^+ (Section C,2). Dinitrophenol and many other uncouplers block the reactions, but oligomycin has no effect. This fact can be rationalized by the Mitchell hypothesis if we assume that Δp can drive these processes.

Another energy-linked process is the **transhydrogenase** reaction by which NADH reduces $NADP^+$ to form NADPH. In the cytoplasm various other reactions are used to generate NADPH (Chapter 17, Section I), but within mitochondria a membrane-bound transhydrogenase has this function.[283–286a] It couples the transhydrogenation reaction to the transport of one (or possibly more than one) proton back into the mitochondria (Eq. 18-14). A value of Δp of −180 mV could increase the ratio of $[NADPH] / [NADP^+]$ within mitochondria by a factor of as much as 1000.



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Transhydrogenases function in a similar way within bacteria. Whether from *E. coli*, photosynthetic bacteria, or bovine mitochondria, transhydrogenases have similar structures.[285] Two 510-residue α subunits associate with two 462-residue β subunits to form an $\alpha_2\beta_2$ tetramer with 10−14 predicted transmembrane helices. The α subunits contain separate NAD(H) and NADP(H) binding sites. A conformational change appears to be associated with the binding or release of the $NADP^+$ or NADPH.[287]

D. Transport and Exchange across Mitochondrial Membranes

Like the external plasma membrane of cells, the inner mitochondrial membrane is selective. Some nonionized materials pass through readily but the transport of ionic substances, including the anions of the dicarboxylic and tricarboxylic acids, is restricted. In some cases energy-dependent active transport is involved but in others one anion passes inward in exchange for another anion passing outward. In either case specific translocating carrier proteins are needed.

Solutes enter mitochondria through pores in thousands of molecules of the **voltage-gated anion-selective channel VDAC**, also known as mitochondrial porin.[15,16,288,289] In the absence of a membrane potential these pores allow free diffusion to molecules up to ~1.2 kDa in mass and may selectively permit passage of anions of 3- to 5-kDa mass. However, a membrane potential greater than ~20 mV causes the pores to close. NADH also decreases permeability. In the closed state the outer membrane becomes almost impermeant to ATP.[289,290]

An example of energy-dependent transport is the uptake of Ca^{2+} by mitochondria. As indicated in the lower part of Fig. 18-13, there are two possibilities for preservation of electrical neutrality according to the chemiosmotic theory. Counterions X^- may flow out to balance the protons discharged on the outside. On the other hand, if a cation such as Ca^{2+} flows inward to balance the two protons flowing outward, neutrality will be preserved and the mitochondrion will accumulate calcium ions. Experimentally such accumulations via a **calcium uniporter**[4] are observed to accompany electron transport. In the presence of a suitable ionophore energy-dependent accumulation of potassium ions also takes place.[291] In contrast, an **electroneutral exchange** of one Ca^{2+} for two Na^+ is mediated by a Na^+–Ca^{2+} exchanger.[292,293] It permits Ca^{2+} to leave mitochondria. A controversial role of mitochondria in accumulating Ca^{2+} postulates a special **rapid uptake mode** of exchange (see p. 1049).[294]

It is thought that glutamate enters mitochondria as the monoanion Glu^- in exchange for the dianion of aspartate Asp^{2-}. Like the uptake of Ca^{2+} this exchange is driven by Δp. Since a membrane potential can be created by this exchange in the absence of Δp, the process is electrogenic.[4] In contrast, an **electroneutral** exchange of malate^{2-} and 2-oxoglutarate^{2-} occurs by means of carriers that are not energy-linked.[295,296] This dicarboxylate transporter is only one of 35 structurally related mitochondrial carriers identified in the complete genome of yeast.[296,297] Another is the **tricarboxylic transporter** (citrate transport protein) which exchanges the dianionic form of citrate for malate, succinate, isocitrate, phosphoenolpyruvate, etc.[298,298a,b]

The important **adenine nucleotide carrier** takes ADP into the mitochondrial matrix for phosphorylation in a 1:1 ratio with ATP that is exported into the cytoplasm.[299-300b] This is one of the major rate-determining processes in respiration. It has been widely accepted that the carrier is electrogenic,[300]

BOX 18-C USING METABOLISM TO GENERATE HEAT: THERMOGENIC TISSUES

A secondary but important role of metabolism in warm-blooded animals is to generate heat. The heat evolved from ordinary metabolism is often sufficient, and an animal can control its temperature by regulating the heat exchange with the environment. Shivering also generates heat and is used from birth by pigs.[a] However, this is insufficient for many newborn animals, for most small mammals of all ages, and for animals warming up after hibernation. The need for additional heat appears to be met by **brown fat**, a tissue which contrasts strikingly with the more abundant white adipose tissue. Brown fat contains an unusually high concentration of blood vessels, many mitochondria with densely packed cristae, and a high ratio of cytochrome *c* oxidase to ATP synthase. Also present are a large number of sympathetic nerve connections, which are also related to efficient generation of heat. Newborn humans have a small amount of brown fat, and in newborn rabbits it accounts for 5–6% of the body weight.[b–d] It is especially abundant in species born without fur and in hibernating animals. Swordfish also have a large mass of brown adipose tissue that protects their brains from rapid cooling when traveling into cold water.[a]

The properties of brown fat pose an interesting biochemical question. Is the energy available from electron transport in the mitochondria dissipated as heat because ATP synthesis is uncoupled from electron transport? Or does ATP synthesis take place but the resulting ATP is hydrolyzed wastefully through the action of ATPases? Part of the answer came from the discovery that mitochondria of brown fat cells synthesize a 32-kDa **uncoupling protein** (UCP1 or thermogenin). It is incorporated into the inner mitochondrial membranes where it may account for 10–15% of total protein.[d–f] This protein, which is a member of a family of mitochondrial membrane metabolic carriers (Table 18-8), provides a "short-circuit" that allows the protonmotive force to be dissipated rapidly, perhaps by a flow of protons out through the uncoupling protein.[g–i] Synthesis of the uncoupling protein is induced by exposure to cold, but when an animal is warm the uncoupling action is inhibited.

The uncoupling protein resembles the ATP/ADP and phosphate *anion* carriers (Table 18-8),[g,i] which all have similar sizes and function as homodimers. Each monomeric subunit has a triply repeated ~100-residue sequence, each repeat forming two transmembrane helices. Most mitochondrial transporters carry anions, and UCP1 will transport Cl^-.[h,i] However, the relationship of chloride transport to its real function is unclear. Does the protein transport H^+ into, or does it carry HO^- out from, the mitochondrial matrix?[g,h] Another possibility is that a fatty acid anion binds H^+ on the intermembrane surface and carries it across into the matrix as an unionized fatty acid. The fatty acid anion could then pass back out using the anion transporter function and assisted by the membrane potential.[h,i]

The uncoupling protein is affected by several control mechanisms. It is inhibited by nucleotides such as GDP, GTP, ADP, and ATP which may bind at a site corresponding to that occupied by ATP or ADP in the ADP/ATP carrier.[i] Uncoupling is stimulated by noradrenaline,[f] which causes a rapid increase in heat production by brown fat tissues, apparently via activation of adenylate cyclase. Uncoupling is also stimulated by fatty acids.[j] Recently UCP1 and related uncoupling proteins have been found to require both fatty acids and **ubiquinone** for activity.[jj,jk]

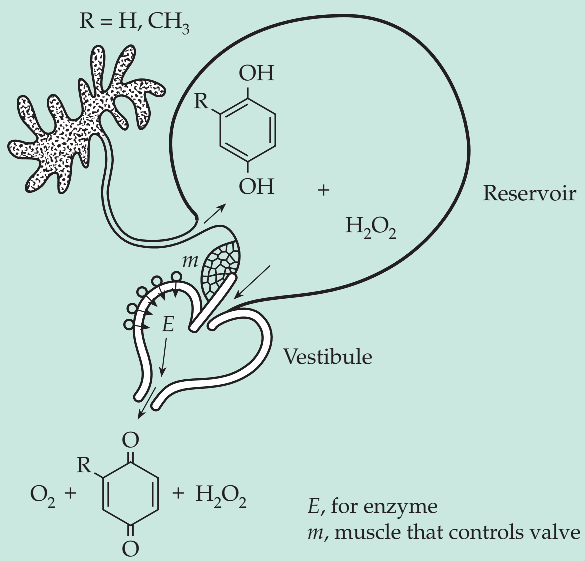
It has been suggested that brown adipose tissue may also function to convert excess dietary fat into heat and thereby to resist obesity.[k–m] Mice lacking the gene for the mitochondrial uncoupling protein are cold-sensitive but not obese. However, other proteins, homologous to UCP1, have been discovered. They may partially compensate for the loss.[m,h]

The bombardier beetle generates a hot, quinone-containing defensive discharge, which is sprayed in a pulsed jet from a special reaction chamber at a temperature of 100°C.[n–p] The reaction mixture of 25% hydrogen peroxide and 10% hydroquinone plus methylhydroquinone is stored in a reservoir as shown in the accompanying figure and reacts with explosive force when it comes into contact with catalase and peroxidases in the reaction chamber. The synthesis and storage of 25% H_2O_2 poses interesting biochemical questions!

Some plant tissues are thermogenic. For example, the spadix (or inflorescence, a sheathed floral spike) of the skunk cabbage *Symplocarpus foetidus* can maintain a 15–35°C higher temperature than that of the surrounding air.[q] The voodoo lily in a single day heats the upper end of its long spadex to a temperature 22°C above ambient, volatilizing a foul smelling mixture of indoles and amines.[r,s] This is accomplished using the alternative oxidase system[s] (Box D in Fig. 18-6). The lotus flower maintains a temperature of 30–35°C, while the ambient temperature may vary from 10–30°C.[t] While the volatilization of insect attractants may be the primary role for thermogenesis in plants, the warm flowers may also offer an important reward to insect pollinators. Beetles and bees require thoracic temperatures above 30°C to initiate flight and, therefore,

BOX 18-C (continued)

benefit from the warm flowers.[t] While in flight bees vary their metabolic heat production by altering their rate of flight, hovering, and other changes in physical activity.[u]



Reservoir and reaction vessel of the bombardier beetle. From D. J. Aneshansley, *et al*.[n]

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bringing in ADP^{3-} and exporting ATP^{4-} in an exchange driven by Δp. However, an electroneutral exchange, e.g., of ADP^{3-} for ATP^{3-}, may also be possible. The carrier is an ~300-residue 32-kDa protein, which is specifically inhibited by the plant glycoside **atractyloside** or the fungal antibiotic **bongkrekate**. The carrier is associated with bound cardiolipin.[301] This one transporter accounts for ~10% of all of the mitochondrial protein.[302,303]

A separate dimeric carrier allows P_i to enter, probably as $H_2PO_4^-$.[304-305a] This ion enters mitochondria in an electroneutral fashion, either in exchange for OH^- or by cotransport with H^+. A less important carrier[306] exchanges HPO_4^{2-} for malate^{2-}. Several other transporters help to exchange organic and inorganic ions. One of them allows pyruvate to enter mitochondria in exchange for OH^- or by cotransport with H^+. Some of the identified carriers are listed in Table 18-8. As discussed in Chapter 8, exchange carriers are also important in plasma membranes of organisms from bacteria to human beings. For example, many metabolites enter cells by cotransport with Na^+ using the energy of the Na^+ gradient set up across the membrane by the Na^+,K^+ pump.

Under some circumstances the inner membrane develops one or more types of large-permeability pore. An increase in Ca^{2+} may induce opening of an unselective pore which allows rapid uptake of Ca^{2+}.[294,307,307a] A general anion-specific channel may be involved in volume homeostasis of mitochondria.[308]

Mitochondria are not permeable to NADH. However, reactions of glycolysis and other dehydrogenations in the cytoplasm quickly reduce available NAD^+ to NADH. For aerobic metabolism to occur, the "reducing equivalents" from the NADH must be transferred into the mitochondria. Fungi and green plants have solved the problem by providing *two* NADH dehydrogenases embedded in the inner mitochondrial membranes (Fig. 18-6). One faces the matrix space and oxidizes the NADH produced in the matrix while the second faces outward to the intermembrane space and is able to oxidize the NADH formed in the cytoplasm. In animals the reducing equivalents from NADH enter the mitochondria indirectly. There are several mechanisms, and more than one may function simultaneously in a tissue.

In insect flight muscle, as well as in many mammalian tissues, NADH reduces dihydroxyacetone phosphate. The resulting *sn*-3-glycerol *P* passes through the permeable outer membrane of the mitochondria, where it is reoxidized to dihydroxyacetone phosphate by the FAD-containing glycerol-phosphate dehydrogenase embedded in the outer surface of the inner membrane (Figs. 18-5, 18-6). The dihydroxyacetone can then be returned to the cytoplasm. The overall effect of this **glycerol-phosphate shuttle** (Fig. 18-18A) is to provide for mitochondrial oxidation of NADH produced in the cytoplasm. In heart and liver the same function is served by a more complex **malate–aspartate shuttle** (Fig. 18-18B).[309] Reduction of oxaloacetate to malate by NADH, transfer of malate into mitochondria, and reoxidation with NAD^+ accomplishes the transfer of reduction equivalents into the mitochondria. Mitochondrial membranes are not very permeable to oxaloacetate. It returns to the cytoplasm mainly via transamination to aspartate, which leaves the mitochondria together with 2-oxoglutarate. At the same time glutamate enters the mitochondria in exchange for aspartate. The 2-oxoglutarate presumably exchanges with the entering malate as is indicated in Fig. 18-18B. The export of aspartate may be energy-linked as a result of the use of an electrogenic carrier that exchanges glutamate$^-$ + H^+ entering mitochondria for aspartate$^-$ leaving the mitochondria. Thus, Δp may help to expel aspartate from mitochondria and to drive the shuttle.

The glycerol-phosphate shuttle, because it depends upon a mitochondrial flavoprotein, provides ~ 2 ATP per electron pair (P / O = 2), whereas the malate–aspartate shuttle may provide a higher yield of ATP. The glycerol-phosphate shuttle is essentially irreversible, but the reactions of the malate–aspartate shuttle can be reversed and utilized in gluconeogenesis (Chapter 17).

TABLE 18-8
Some Mitochondrial Membrane Transporters[a]

Ion Diffusing In	Ion Diffusing Out	Comment[b]
ADP^{3-}	ATP^{3-}	
or ADP^{3-}	ATP^{4-}	Electrogenic symport
$H_2PO_4^-$ + H^+		Electroneutral symport
or $H_2PO_4^-$	OH^-	
HPO_4^{2-}	Malate^{2-}	
Malate^{2-}	2-Oxoglutarate^{2-}	
Glutamate^{2-} + H^+	Aspartate^{2-}	
Glutamate$^-$	OH^-	
Pyruvate$^-$	OH^-	
or Pyruvate$^-$ + H^+		Electroneutral symport
Citrate^{3-} + H^+	Malate^{2-}	
Ornithine$^+$	H^+	
Acylcarnitine	Carnitine	
2 Na^+	Ca^{2+}	
H^+	K^+	
H^+	Na^+	
General transporters		
VDAC (porin)		Outer membrane
Large anion pores		Inner membrane

[a] From Nicholls and Ferguson[172] and Tyler[4].
[b] Unless indicated otherwise the transporters are *antiporters* that catalyze an electroneutral ion exchange.

E. Energy from Inorganic Reactions

Some bacteria obtain all of their energy from inorganic reactions. These **chemolithotrophs** usually have a metabolism that is similar to that of heterotrophic organisms, but they also have the capacity to obtain all of their energy from an inorganic reaction. In order to synthesize carbon compounds they must be able to fix CO_2 either via the reductive pentose phosphate cycle or in some other way. The chloroplasts of green plants, using energy from sunlight, supply the organism with both ATP and the reducing agent NADPH (Chapter 17). In a similar way the lithotrophic bacteria obtain both energy and reducing materials from inorganic reactions.

Chemolithotrophic organisms often grow slowly, making study of their metabolism difficult.[310] Nevertheless, these bacteria usually use electron transport chains similar to those of mitochondria. ATP is formed by oxidative phosphorylation, the amount formed per electron pair depending upon the number of proton-pumping sites in the chain. This, in turn, depends upon the electrode potentials of the reactions involved. For example, H_2, when oxidized by O_2, leads to passage of electrons through the entire electron transport chain with synthesis of ~3 molecules of ATP per electron pair. On the other hand, oxidation by O_2 of nitrite, for which $E°'$ (pH 7) = +0.42 V, can make use only of the site III part of the chain. Not only is the yield of ATP less than in the oxidation of H_2 but also there is another problem. Whereas reduced pyridine

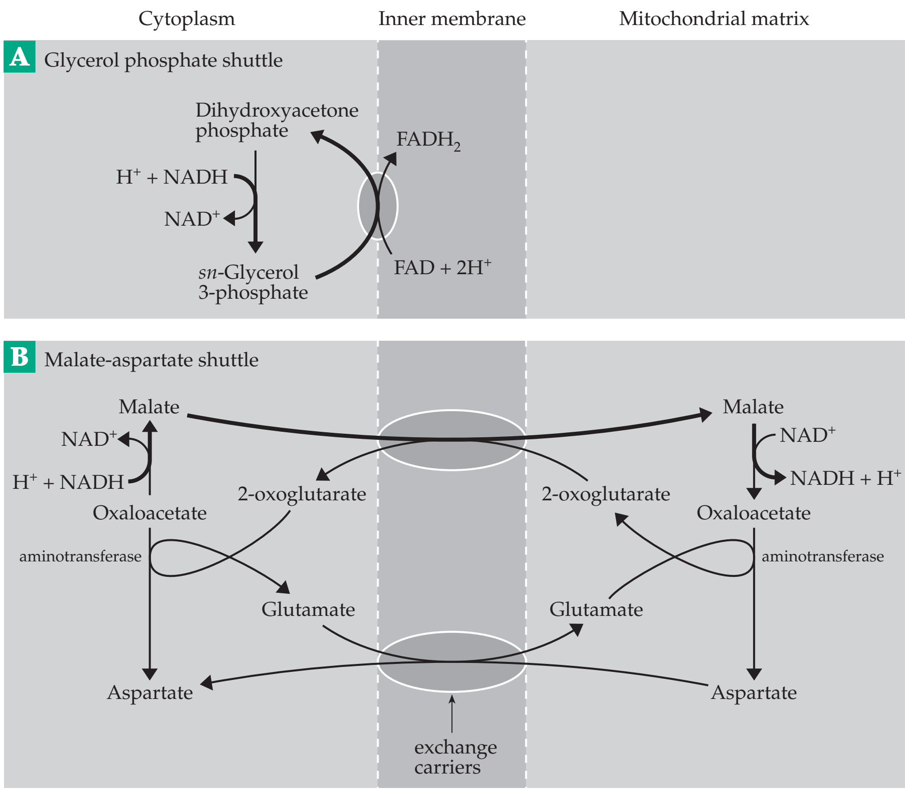


Figure 18-18 (A) The glycerol-phosphate shuttle and (B) the malate–aspartate shuttle for transport from cytoplasmic NADH into mitochondria. The heavy arrows trace the pathway of the electrons (as 2H) transported.

nucleotides needed for biosynthesis can be generated readily from H_2, nitrite is not a strong enough reducing agent to reduce NAD^+ to NADH. The only way that reducing agents can be formed in cells utilizing oxidation of nitrite as an energy source is via *reverse electron flow* driven by hydrolysis of ATP or by Δp. Such reverse electron flow is a common process for many chemolithotrophic organisms.

Let us consider the inorganic reactions in two groups: (1) oxidation of reduced inorganic compounds by O_2 and (2) oxidation reactions in which an inorganic oxidant, such as nitrate or sulfate, substitutes for O_2. The latter reactions are often referred to as **anaerobic respiration**.

1. Reduced Inorganic Compounds as Substrates for Respiration

The hydrogen-oxidizing bacteria. Species from several genera including *Hydrogenomonas*, *Pseudomonas*, and *Alcaligenes* oxidize H_2 with oxygen:

$$H_2 + 1/2\ O_2 \rightarrow H_2O$$
$$\Delta G° = -237.2 \text{ kJ mol}^{-1} \quad (18\text{-}15)$$

Some can also oxidize carbon monoxide:

$$CO + 1/2\ O_2 \rightarrow CO_2$$
$$\Delta G° = -257.1 \text{ kJ mol}^{-1} \quad (18\text{-}16)$$

The hydrogen bacteria can also oxidize organic compounds using straightforward metabolic pathways. The key enzyme is a membrane-bound nickel-containing hydrogenase (Fig. 16-26), which delivers electrons from H_2 into the electron transport chain.[310a] A second soluble hydrogenase (sometimes called **hydrogen dehydrogenase**) transfers electrons to $NADP^+$ to form NADPH for use in the reductive pentose phosphate cycle and for other biosynthetic purposes.

Nitrifying bacteria. Two genera of soil bacteria oxidize ammonium ion to nitrite and nitrate (Eqs. 18-17 and 18-18).[311]

$$NH_4^+ + 3/2\ O_2 \rightarrow NO_2^- + 2\ H^+ + H_2O\ (\textit{Nitrosomonas})$$
$$\Delta G'\ (\text{pH } 7) = -272 \text{ kJ mol}^{-1} \quad (18\text{-}17)$$

$$NO_2^- + 1/2\ O_2 \rightarrow NO_3^-\ (\textit{Nitrobacter})$$
$$\Delta G'\ (\text{pH } 7) = -76 \text{ kJ mol}^{-1} \quad (18\text{-}18)$$

The importance of these reactions to the energy metabolism of the bacteria was recognized in 1895 by Winogradsky, who first proposed the concept of chemiautotrophy. Because the nitrifying bacteria grow slowly (generation time ~10–12 h) it has been hard to get enough cells for biochemical studies and progress has been slow. The reaction catalyzed by *Nitrosomonas* (Eq. 18-17) is the more complex; it occurs in two or more stages and is catalyzed by two enzymes as illustrated in Fig. 18-19. The presence of hydrazine blocks oxidation of hydroxylamine (NH_2OH) in step *b* and permits that intermediate to accumulate. The oxidation of ammonium ion by O_2 to hydroxylamine (step *a*) is endergonic with $\Delta G'$ (pH 7) = 16 kJ mol^{-1} and is incapable of providing energy to the cell. It occurs by a hydroxylation mechanism (see Section G). On the other hand, the oxidation of hydroxylamine to nitrite by O_2 in step *b* is highly exergonic with $\Delta G'$ (pH 7) = –228 kJ mol^{-1}. The hydroxylamine oxidoreductase that catalyzes this reaction is a trimer of 63-kDa subunits, each containing seven *c*-type hemes and an unusual heme P450, which is critical to the enzyme's function[312–314a] and which is covalently linked to a tyrosine as well as to two cysteines.

The electrode potentials for the two- and four-electron oxidation steps are indicated in Fig. 18-19. It is apparent that step *b* can feed four electrons into the electron transport system at about the potential of ubiquinone. Two electrons are needed to provide a cosubstrate (Section G) for the ammonia monooxygenase and two could be passed on to the terminal cytochrome aa_3 oxidase. The stoichiometry of proton pumping in complexes III and IV is uncertain, but if it is assumed to be as shown in Fig. 18-19 and similar to that in Fig. 18-5, there will be ~13 protons available to be passed through ATP synthase to generate ~3 ATP per NH_3 oxidized. However, to generate NADH for reductive biosynthesis *Nitrosomonas* must send some electrons to NADH dehydrogenase (complex I) using reverse electron transport, a process that depends upon Δp to drive the reaction via a flow of protons through the NADH dehydrogenase from the periplasm back into the bacterial cytoplasm (Fig. 18-19).

Nitrobacter depends upon a simpler energy-yielding reaction (Eq. 18-18) with a relatively small Gibbs energy decrease. The two-electron oxidation delivers electrons to the electron transport chain at $E°'$ = +0.42 V. The third oxygen in NO_3^- originates from H_2O, rather than from O_2 as might be suggested by Eq. 18-18.[316,317] It is reasonable to anticipate that a single molecule of ATP should be formed for each pair of electrons reacting with O_2. However, *Nitrobacter* contains a confusing array of different cytochromes in its membranes.[311] Some of the ATP generated by passage of electrons from nitrite to oxygen must be used to drive a reverse flow of electrons through both a bc_1-type complex and NADH dehydrogenases. This generates reduced pyridine nucleotides required for biosynthetic reactions (Fig. 18-20).

An interesting feature of the structure of *Nitrobacter* is the presence of several double-layered membranes which completely envelop the interior of the cell. Nitrite entering the cell is oxidized on these membranes and cannot penetrate to the interior, where it might have toxic effects.

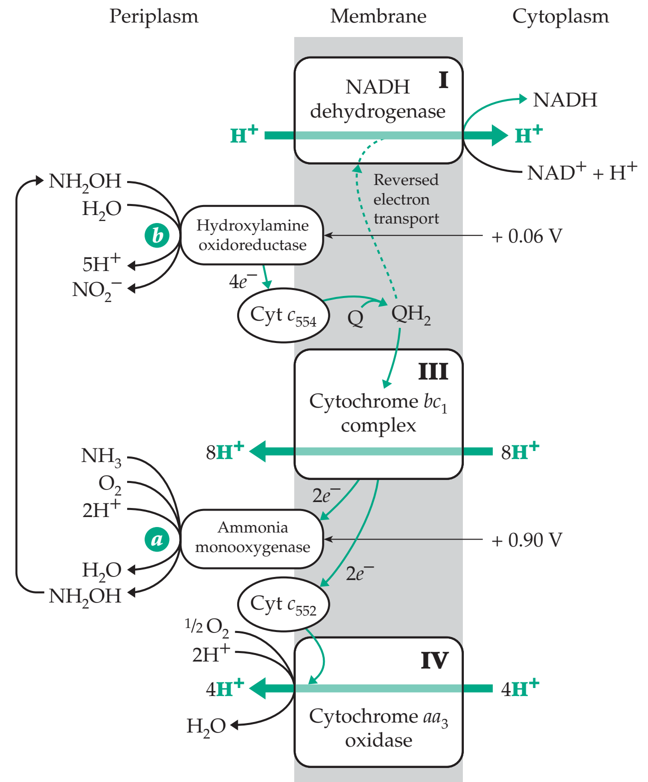


Figure 18-19 The ammonia oxidation system of the bacterium *Nitrosomonas*. Oxidation of ammonium ion (as free NH_3) according to Eq. 18-17 is catalyzed by two enzymes. The location of ammonia monooxygenase (step *a*) is uncertain but hydroxylamine oxidoreductase (step *b*) is periplasmic. The membrane components resemble complexes I, III, and IV of the mitochondrial respiratory chain (Fig. 18-5) and are assumed to have similar proton pumps. Solid green lines trace the flow of electrons in the energy-producing reactions. This includes flow of electrons to the ammonia monoxygenase. Complexes III and IV pump protons out but complex I catalyzes reverse electron transport for a fraction of the electrons from hydroxylamine oxidoreductase to NAD^+. Modified from Blaut and Gottschalk.[315]

The sulfur-oxidizing bacteria. Anaerobic conditions prevail in marine sediments, in poorly stirred swamps, and around hydrothermal vents at the bottom of the sea. Sulfate-reducing bacteria form high concentrations (up to mM) of H_2S (in equilibrium with HS^- and S^{2-})[318–320] This provides the substrate for bacteria of the genus *Thiobacillus*, which are able to oxidize sulfide, elemental sulfur, thiosulfate, and sulfite to sulfate and live where the aerobic and anaerobic regions meet.[311,321–323] Most of these small gram-negative

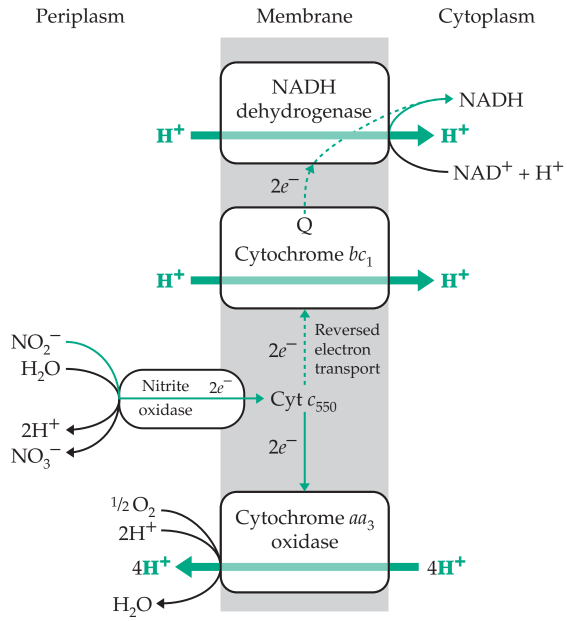
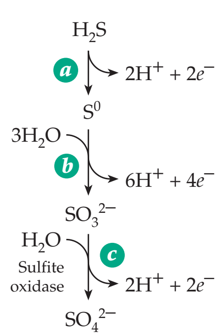


Figure 18-20 Electron transport system for oxidation of the nitrite ion to the nitrate ion by *Nitrobacter*. Only one site of proton pumping for oxidative phosphorylation is available. Generation of NADH for biosynthesis requires two stages of reverse electron transport.

organisms, found in water and soil, are able to grow in a simple salt medium containing an oxidizable sulfur compound and CO_2. One complexity in understanding their energy-yielding reactions is the tendency of sulfur to form chain molecules. Thus, when sulfide is oxidized, it is not clear that it is necessarily converted to monoatomic elemental sulfur as indicated in Eq. 18-19. Elemental sulfur (S_8^0) often precipitates. In *Beggiotoa*, another sulfide-oxidizing bacterium, sulfur is often seen as small globules within the cells. Fibrous sulfur precipitates are often abundant in the sulfide-rich layers of ponds, lakes, and oceans.[318]



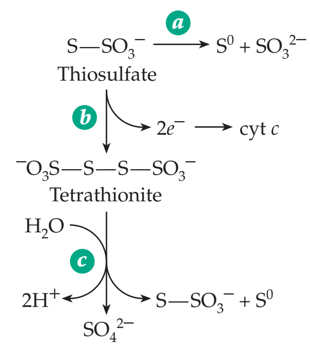
(18-19)

The reactions of Eq. 18-19 occur in the periplasmic space of some species.[315,323a,324] Steps *a* and *b* of Eq. 18-19 are catalyzed by a 67-kDa sulfide dehydrogenase in the periplasm of a purple photosynthetic bacterium.[324] The enzyme consists of a 21-kDa subunit containing two cytochrome *c*-like hemes, presumably the site of binding of S^{2-}, and a larger 46-kDa FAD-containing flavoprotein resembling glutathione reductase.[324] The molybdenum-containing sulfite oxidase (Fig. 16-32), which is found in the intermembrane space of mitochondria, may be present in the periplasmic space of these bacteria. However, there is also an intracellular pathway for sulfite oxidation (see Eq. 18-22).

The sulfide-rich layers inhabited by the sulfur oxidizers also contain thiosulfate, $S_2O_3^{2-}$. It may arise, in part, from reaction of glutathione with elemental sulfur:

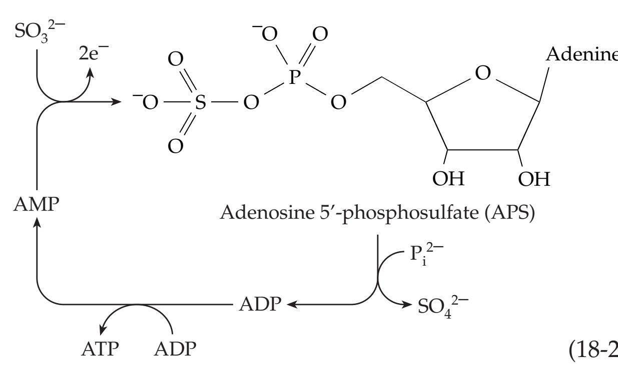
$$GSH + S_8^0 \rightarrow G{-}S{-}S_8{-}H \tag{18-20}$$

The linear polysulfide obtained by this reaction may be oxidized, the sulfur atoms being removed from the chain either one at a time to form sulfite or two at a time to form thiosulfate.[322,322a] Thiosulfate is oxidized by all species, the major pathway beginning with cleavage to S^0 and SO_3^{2-} (Eq. 18-21, step *a*). At high thiosulfate concentrations some may be oxidized to tetrathionate (step *b*), which is hydrolyzed to sulfate (step *c*).



(18-21)

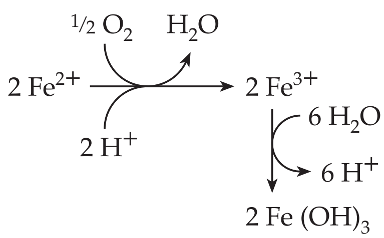
Oxidation of sulfite to sulfate within cells occurs by a pathway through **adenosine 5'-phosphosulfate** (**APS**, adenylyl sulfate). Oxidation via APS (Eq. 18-22) provides a means of substrate-level phosphorylation,



(18-22)

the only one known among chemolithotrophic bacteria. No matter which of the two pathways of sulfite oxidation is used, thiobacilli also obtain energy via electron transport. With a value of $E°'$ (pH 7) of −0.454 V [$E°'$ (pH 2) = −0.158 V] for the sulfate–sulfite couple an abundance of energy may be obtained. Oxidation of sulfite to sulfate produces hydrogen ions. Indeed, pH 2 is optimal for the growth of *Thiobacillus thiooxidans*, and the bacterium withstands 5% sulfuric acid.[322]

The "iron bacterium" *Thiobacillus ferrooxidans* obtains energy from the oxidation of Fe^{2+} to Fe^{3+} with subsequent precipitation of ferric hydroxide (Eq. 18-23). However, it has been recognized recently that a previously unknown species of Archaea is much more important than *T. ferrooxydans* in catalysis of this reaction.[324a]



(18-23)

Since the reduction potential for the Fe(II) / Fe(III) couple is + 0.77 V at pH 7, the energy obtainable in this reaction is small. These bacteria always oxidize reduced sulfur compounds, too. Especially interesting is their oxidation of **pyrite**, ferrous sulfide (Eq. 18-24). The Gibbs energy change was calculated from published data[325] using a value of $G_f°$ for $Fe(OH)_3$ of

$$2\,FeS + 4\tfrac{1}{2}\,O_2 + 5\,H_2O \rightarrow 2\,Fe(OH)_3 + 2\,SO_4^{2-} + 4\,H^+$$
$$\Delta G'\ (\text{pH } 2) = -1340\ \text{kJ} \qquad (18\text{-}24)$$

−688 kJ mol^{-1} estimated from its solubility product. Because sulfuric acid is generated in this reaction, a serious water pollution problem is created by the bacteria living in abandoned mines. Water running out of the mines often has a pH of 2.3 or less.[326]

Various invertebrates live in S^{2-}-containing waters. Among these is a clam that has symbiotic sulfur-oxidizing bacteria living in its gills. The clam tissues apparently carry out the first step in oxidation of the sulfide.[327] Among the animals living near sulfide-rich thermal vents in the ocean floor are giant 1-meter-long tube worms. Both a protective outer tube and symbiotic sulfide-oxidizing bacteria protect them from toxic sulfides.[319,320]

2. Anaerobic Respiration

Nitrate as an electron acceptor. The use of nitrate as an alternative oxidant to O_2 is widespread among bacteria. For example, *E. coli* can subsist anaerobically by reducing nitrate to nitrite (Eq. 18-25).[311,328] The respiratory (dissimilatory) nitrate reductase that

$$NO_3^- + 2\,H^+ + 2e^- \rightarrow NO_2^- + H_2O$$
$$E°'\ (\text{pH7}) = +0.421\ \text{V} \qquad (18\text{-}25)$$

catalyzes the reaction is a large three-subunit molybdenum-containing protein. The enzyme is present in the plasma membrane, and electrons flow from ubiquinone through as many as six heme and Fe–S centers to the molybdenum atom.[328-329] A second molybdoenzyme, formate dehydrogenase (discussed in Chapter 16), appears to be closely associated with nitrate reductase. Formate is about as strong a reducing agent as NADH (Table 6-8) and is a preferred electron donor for the reduction of NO_3^- (Eq. 18-26).[329a,b] Since cytochrome *c* oxidase of the electron transport chain is bypassed, one less ATP is formed than when O_2 is the oxidant. Nitrate is the oxidant preferred by bacteria grown under anaerobic conditions. The

$$HCOO^- + NO_3^- + H^+ \rightarrow NO_2^- + CO_2 + H_2O$$
$$\Delta G'\ (\text{pH } 7) = 165\ \text{kJ mol}^{-1} \qquad (18\text{-}26)$$

presence of NO_3^- induces the synthesis of nitrate reductase and represses the synthesis of alternative enzymes such as **fumarate reductase**,[330,331] which reduces fumarate to succinate (see also p. 1027). On the other hand, if NO_3^- is absent and fumarate, which can be formed from pyruvate, is present, synthesis of fumarate reductase is induced. Although it is a much weaker oxidant than is nitrate ($E^{o'}$ = 0.031 V), fumarate is still able to oxidize H_2 or NADH with oxidative phosphorylation. Like the related succinate dehydrogenase, fumarate reductase of *E. coli* is a flavoprotein with associated Fe / S centers. It contains covalently linked FAD and Fe_2S_2, Fe_4S_4, and three-Fe iron–sulfur centers.[332] In some bacteria a soluble periplasmic cytochrome c_3 carried out the fumarate reduction step.[332a] Trimethylamine *N*-oxide[330,333] or dimethylsulfoxide (DMSO; Eq. 16-62)[334,335] can also serve as alternative oxidants for anaerobic respiration using appropriate molybdemum-containing reductases (Chapter 16).

Reduction of nitrite: denitrification. The nitrite formed in Eq. 18-25 is usually reduced further to ammonium ions (Eq. 18-27). The reaction may not be important to the energy metabolism of the bacteria, but it provides NH_4^+ for biosynthesis. This six-electron reduction is catalyzed by a hexaheme protein containing six *c*-type hemes bound to a single 63-kDa polypeptide chain.[336,337]

$$NO_2^- + 8\,H^+ + 6\,e^- \rightarrow 2\,H_2O + NH_4^+ \qquad (18\text{-}27)$$

Several types of **denitrifying bacteria**[315,338–340] use either nitrate or nitrite ions as oxidants and reduce nitrite to N_2. A typical reaction for *Micrococcus denitrificans* is oxidation of H_2 by nitrate (Eq. 18-28). *Thiobacillus denitrificans*, like other thiobacilli, can oxidize

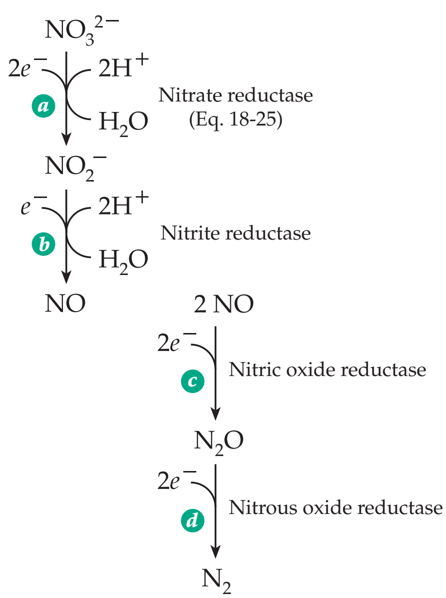
$$5\,H_2 + 2\,NO_3^- + 2\,H^+ \rightarrow N_2 + 6\,H_2O$$
$$\Delta G' \text{ (pH 7)} = -561 \text{ kJ mol}^{-1} \text{ of nitrate reduced or } -224 \text{ kJ mol}^{-1} \text{ of } H_2 \text{ oxidized} \quad (18\text{-}28)$$

sulfur as well as H_2S or thiosulfate using nitrate as the oxidant (Eq. 18-29):

$$5\,S + 6\,NO_3^- + 2\,H_2O \rightarrow 5\,SO_4^{2-} + 3\,N_2 + 4\,H^+$$
$$\Delta G' \text{ (pH 7)} = -455 \text{ kJ mol}^{-1} \text{ of nitrate reduced or } -546 \text{ kJ mol}^{-1} \text{ of S oxidized} \quad (18\text{-}29)$$

The reactions begin with reduction of nitrate to nitrite (Eq. 18-25) and continue with further reduction of nitrite to nitric oxide, **NO**; nitrous oxide, **N_2O**; and dinitrogen, **N_2**. A probable arrangement of the four enzymes needed for the reactions of Eq. 18-30 in *Paracoccus denitrificans* is shown in Fig. 18-21. See also pp. 884, 885.

Two types of dissimilatory nitrite reductases catalyze step *b* of Eq. 18-30. Some bacteria use a copper-containing enzyme, which contains a type 1 (blue) copper bound to a β barrel domain of one subunit and a type 2 copper at the catalytic center. The type 1 copper is thought to receive electrons from the small copper-containing carrier pseudoazurin (Chapter 16).[341–342b]



(18-30)

More prevalent is **cytochrome *cd*$_1$** nitrite reductase.[340,343–346] The water-soluble periplasmic enzyme is a homodimer of ~60-kDa subunits, each containing a *c*-type heme in a small N-terminal domain and **heme *d*$_1$**, a ferric dioxoisobacteriochlorin (Fig. 16-6). The

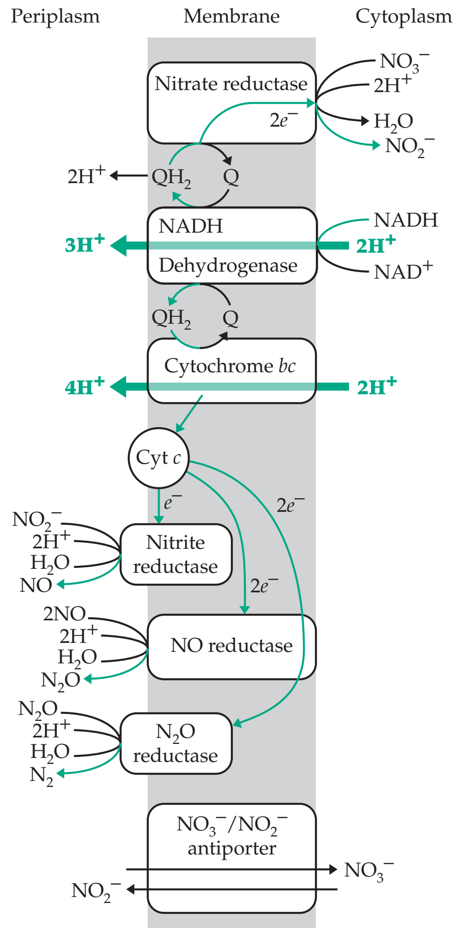


Figure 18-21 Organization of the nitrate reduction system in the outer membrane of the bacterium *Paracoccus dinitrificans* as outlined by Blaut and Gottschalk.[315] The equations are not balanced as shown but will be balanced if two NO_3^- ions are reduced to N_2 by five molecules of NADH (see also Eq. 18-28). Although this will also require seven protons, about 20 additional H^+ will be pumped to provide for ATP synthesis.

latter is present in the central channel of an eight-bladed β-sheet propeller[345-346g] similar to that in Fig. 15-23A. The heme d_1 is unusual in having its Fe atom ligated by a tyrosine hydroxyl oxygen, which may be displaced to allow binding of NO_3^-. The electron required for the reduction is presumably transferred from the electron transfer chain in the membrane to the heme *d*, via the heme *c* group.[347] Cytochrome cd_1 nitrite reductases have an unexpected second enzymatic activity. They catalyze the four-electron reduction of O_2 to H_2O, as does cytochrome *c* oxidase. However, the rate is much slower than that of nitrite reduction.[340,348]

The enzyme catalyzing the third step of Eq. 18-30 (step *c*), **nitric oxide reductase**, is an unstable membrane-bound protein cytochrome *bc* complex.[349,350]

It has been isolated as a two-subunit protein, but genetic evidence suggests the presence of additional subunits.[350] The small subunit is a cytochrome *c*, while the larger subunit is predicted to bind two protohemes as well as a nonheme iron center. This protein also shows sequence homology with cytochrome *c* oxidase. It contains no copper, but it has been suggested that a heme *b*–nonheme Fe center similar to the heme *a*–Cu_B center of cytochrome *c* oxidase may be present. It may be the site at which the nitrogen atoms of two molecules of NO are joined to form N_2O.[350,351] A different kind of NO reductase is utilized by the denitrifying fungus *Fusarium oxysporum*. It is a cytochrome P450 but with an unusually low redox potential (–0.307 V). This **cytochrome P450$_{nor}$** does not react with O_2 (as in Eq. 18-57) but binds NO to its heme Fe^{3+}, reduces the complex with two electrons from NADH, then reacts with a second molecule of NO to give N_2O and H_2O.[352]

Reduction of N_2O to N_2 by bacteria (Eq. 18-30, step *d*) is catalyzed by the copper-containing nitrous oxide reductase. The purple enzyme is a dimer of 66-kDa subunits, each containing four atoms of Cu.[353] It has spectroscopic properties similar to those of cytochome *c* oxidase and a dinuclear copper–thiolate center similar to that of Cu_A in cytochrome *c* oxidase (p. 1030). The nature of the active site is uncertain.[354]

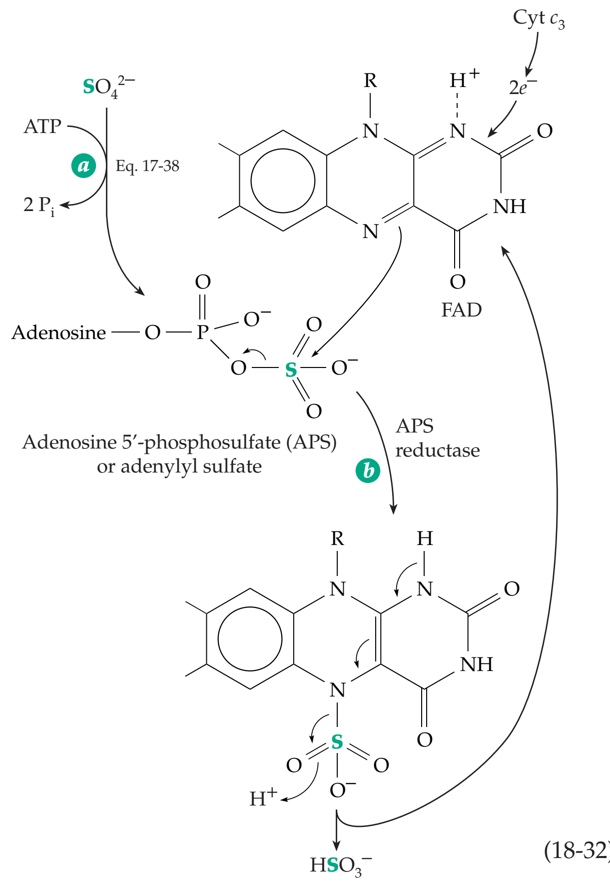
Sulfate-reducing and sulfur-reducing bacteria. A few obligate anaerobes obtain energy by using sulfate ion as an oxidant.[355–356a] For example, *Desulfovibrio desulfuricans* catalyzes a rapid oxidation of H_2 with reduction of sulfate to H_2S (Eq. 18-31).

$$4\ H_2 + SO_4^{2-} + 2\ H^+ \rightarrow H_2S + 4\ H_2O$$
$$\Delta G'\ (\text{pH } 7) = -154\ \text{kJ mol}^{-1} \text{ of sulfate reduced} \qquad (18\text{-}31)$$

While this may seem like an esoteric biological process, the reaction is quantitatively significant. For example, it has been estimated that within the Great Salt Lake basin bacteria release sulfur as H_2S in an amount of 10^4 metric tons (10^7 kg) per year.[357]

The reduction potential for sulfate is extremely low ($E°'$, pH 7 = –0.454 V), and organisms are not known to reduce it directly to sulfite. Rather, a molecule of ATP is utilized to form adenosine 5'-phosphosulfate (APS) through the action of **ATP sulfurylase** (ATP:sulfate adenylyltransferase, Eq. 17-38).[358,359] APS is then reduced by cytochrome c_3 (Eq. 18-32, step *b*). The 13-kDa low-potential ($E°'$, pH 7 = 0.21 V) cytochrome c_3 contains four heme groups (Figure 16-8C) and is found in high concentration in sulfate-reducing bacteria.[360,361] Some of these bacteria have larger polyheme cytochromes *c*.[361a] For example, *Desulfovibrio vulgaris* forms a 514-residue protein carrying 16 hemes organized as four cytochrome c_3-like domains.[362] Each heme in cytochrome c_3 has a distinct redox potential within the range –0.20 to –0.40 V.[361–363]

APS is reduced (Eq. 18-32, step *b*) by **APS reductase**, a 220-kDa iron–sulfur protein containing FAD and several Fe–S clusters. An intermediate in the reaction may be the adduct of sulfite with FAD, which may be formed as in Eq. 18-32. The initial step in this hypothetical mechanism is displacement on sulfur by a strong nucleophile generated by transfer of electrons from reduced ferredoxin to cytochrome c_3 to the flavin.[364]



(18-32)

Bisulfite produced according to Eq. 18-32 is reduced further by a **sulfite reductase,** which is thought to receive electrons from flavodoxin, cyt c_3, and a hydrogenase. ATP synthesis is coupled to the reduction. Sulfite reductases generally contain both siroheme and Fe_4S_4 clusters (Fig. 16-19). They appear to be able to carry out the 6-electron reduction to sulfide without accumulation of intermediates.[365,366] However, in contrast to the assimilatory sulfite reductases present in many organisms, the dissimilatory nitrite reductases of sulfur-reducing bacteria may also release some thiosulfate $S_2O_3^{2-}$.[367] A possible role of menaquinone (vitamin K_2), present in large amounts in *Desulfovibrio*, has been suggested.[311] Although *Desulfovibrio* can obtain their energy from Eq. 18-31, they are not true autotrophs and must utilize compounds such as acetate together with CO_2 as a carbon source.

Some thermophilic archaeobacteria are able to live with CO_2 as their sole source of carbon and reduction of elemental sulfur with H_2 (Eq. 18-33) as their sole source of energy.[368,369]

$$H_2 + S^0 \rightarrow H_2S \qquad \Delta G° = -27.4 \text{ kJ mol}^{-1} \tag{18-33}$$

This is remarkable in view of the small standard Gibbs energy decrease. Some species of the archaeobacterium *Sulfolobus* are able either to live aerobically oxidizing sulfide to sulfate with O_2 (Eq. 18-22) or to live anaerobically using reduction of sulfur by Eq. 18-33 as their source of energy.[369]

The sulfate-reducing bacterium *Desulfovibrio sulfodismutans* carries out what can be described as "inorganic fermentations" which combine the oxidation of compounds such as sulfite or thiosulfate (as observed for sulfur-oxidizing bacteria; Eq. 18-22), with reduction of the same compounds (Eq. 18-34).[370,370a] Dismutation of $S_2O_3^{2-}$ plus H_2O to form SO_4^{2-} and H_2S also occurs but with a less negative Gibbs energy change.

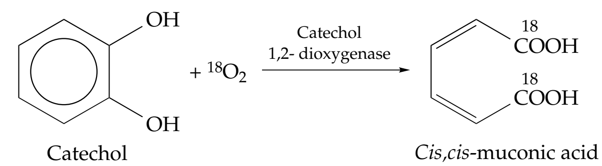
$$4\ SO_3^{2-} + H^+ \rightarrow 3\ SO_4^{2-} + HS^- \qquad \Delta G' \text{ (pH 7)} = -185 \text{ kJ/mol} \tag{18-34}$$

A strain of *Pseudomonas* obtains all of its energy by reducing sulfate using phosphite, which is oxidized to phosphate.[370b]

Methane bacteria. The methane-producing bacteria (Chapter 15) are also classified as chemiautotrophic organisms. While they can utilize substances such as methanol and acetic acid, they can also reduce CO_2 to methane and water using H_2 (Fig. 15-22). The electron transport is from hydrogenase, perhaps through ferredoxin to formate dehydrogenase and via the deazaflavin F_{420} and $NADP^+$ to the methanopterin-dependent dehydrogenases that carry out the stepwise reduction of formate to methyl groups (Fig. 16-28). Generation of ATP probably involves proton pumps, perhaps in internal coupling membranes .[315,371]

F. Oxygenases and Hydroxylases

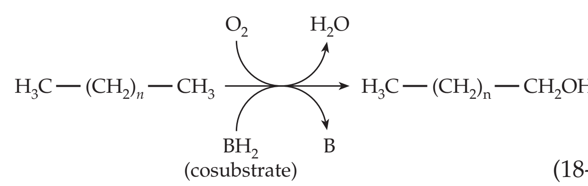
For many years the idea of dehydrogenation dominated thinking about biological oxidation. Many scientists assumed that the oxygen found in organic substances always came from water, e.g., by addition of water to a double bond followed by dehydrogenation of the resulting alcohol. Nevertheless, it was observed that small amounts of O_2 were essential, even to anaerobically growing cells.[372] In 1955, Hayaishi and Mason independently demonstrated that ^{18}O was sometimes incorporated into organic compounds directly from $^{18}O_2$ as in Eq. 18-35. Today a bewildering variety of **oxygenases** are



(18-35)

known to function in forming such essential metabolites as sterols, prostaglandins, and active derivatives of vitamin D. Oxygenases are also needed in the catabolism of many substances, often acting on nonpolar groups that cannot be attacked readily by other types of enzyme.[372]

Oxygenases are classified either as **dioxygenases** or as **monooxygenases**. The monooxygenases are also called mixed function oxidases or **hydroxylases**. Dioxygenases catalyze incorporation of two atoms of oxygen as in Eq. 18-35, but monooxygenases incorporate only one atom. The other oxygen atom from the O_2 is converted to water. A typical monooxygenase-catalyzed reaction is the hydroxylation of an alkane to an alcohol (Eq. 18-36).



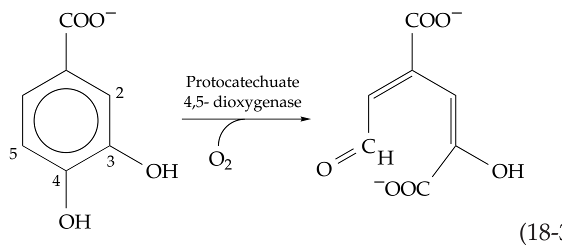
(18-36)

A characteristic of the monooxygenases is that an additional reduced substrate, a **cosubstrate** (BH_2 in Eq. 18-36), is usually required to reduce the second atom of the O_2 molecule to H_2O.

Since O_2 exists in a "triplet" state with two unpaired electrons, it reacts rapidly only with transition metal ions or with organic radicals (Chapter 16). For this reason, most oxygenases contain a transition metal ion, usually of iron or copper, or contain a cofactor, such as FAD, that can easily form a radical or act on a cosubstrate or substrate to form a free radical.

1. Dioxygenases

Among the best known of the oxygenases that incorporate both atoms of O_2 into the product are those that participate in the biological degradation of aromatic compounds by cleaving double bonds at positions between two OH groups as in Eq. 18-35 or adjacent to one OH group of an *ortho* or *para* hydroxyl pair.[373] A much studied example is **protocatechuate 3,4-dioxygenase**,[373–375] which cleaves its substrate between the two OH groups (*intradiol cleavage*) as in Eq. 18-35. A different enzyme, **protocatechuate 4,5-dioxygenase**,[357] cleaves the same substrate next to just one of the two OH groups (*extradiol cleavage*; Eq. 18-37) to form the aldehyde α-hydroxy-δ-carboxymuconic semialdehyde. Another extradiol cleaving enzyme, **protocatechuate 2,3-dioxygenase**, acts on the same substrate. Many other dioxygenases attack related substrates.[376–380] Intradiol-cleaving enzymes are usually iron-tyrosinate proteins (Chapter 16) in which the



(18-37)

iron is present in the Fe(III) oxidation state and remains in this state throughout the catalytic cycle.[375] The enzymes usually have two subunits and no organic prosthetic groups. For example, a protocatechuate 3,4-dioxygenase from *Pseudomonas* has the composition $(\alpha\beta Fe)_{12}$ with subunit masses of 23 (α) and 26.5 (β) kDa. The iron is held in the active site cleft between the α and β subunits by Tyr 408, Tyr 447, His 460, and His 462 of the β subunit and a water molecule.[375] These enzymes and many other oxygenases probably assist the substrate in forming radicals that can react with O_2 to form organic peroxides. Some plausible intermediate species are pictured in Fig. 18-22. The reactions are depicted as occurring in two-electron steps. However, O_2 is a diradical, and it is likely that the Fe^{3+}, which is initially coordinated to both phenolate groups of the ionized substrate, assists in forming an organic free radical that reacts with O_2.

Extradiol dioxygenases have single Fe^{2+} ions in their active sites. The O_2 probably binds to the Fe^{2+} and may be converted transiently to an Fe^{3+}-superoxide complex which adds to the substrate. Some extradiol dioxygenases require an Fe_2S_2 ferredoxin to reduce any Fe^{3+}-enzyme that is formed as a side reaction back to the Fe^{2+} state.[381] Possible intermediates are given in Fig. 18-22 (left side) with two-electron steps used to save space and to avoid giving uncertain details about free radical intermediates. Formation of the organic radical is facilitated by the iron atom, which may be coordinated initially to both phenolate groups of the ionized substrate. The peroxide intermediates, for both types of dioxygenases, may react and be converted to various final products by several mechanisms.[382]

Tryptophan dioxygenase (indoleamine 2,3-dioxygenase)[383] is a heme protein which catalyzes the reaction of Eq. 18-38. The oxygen atoms designated by

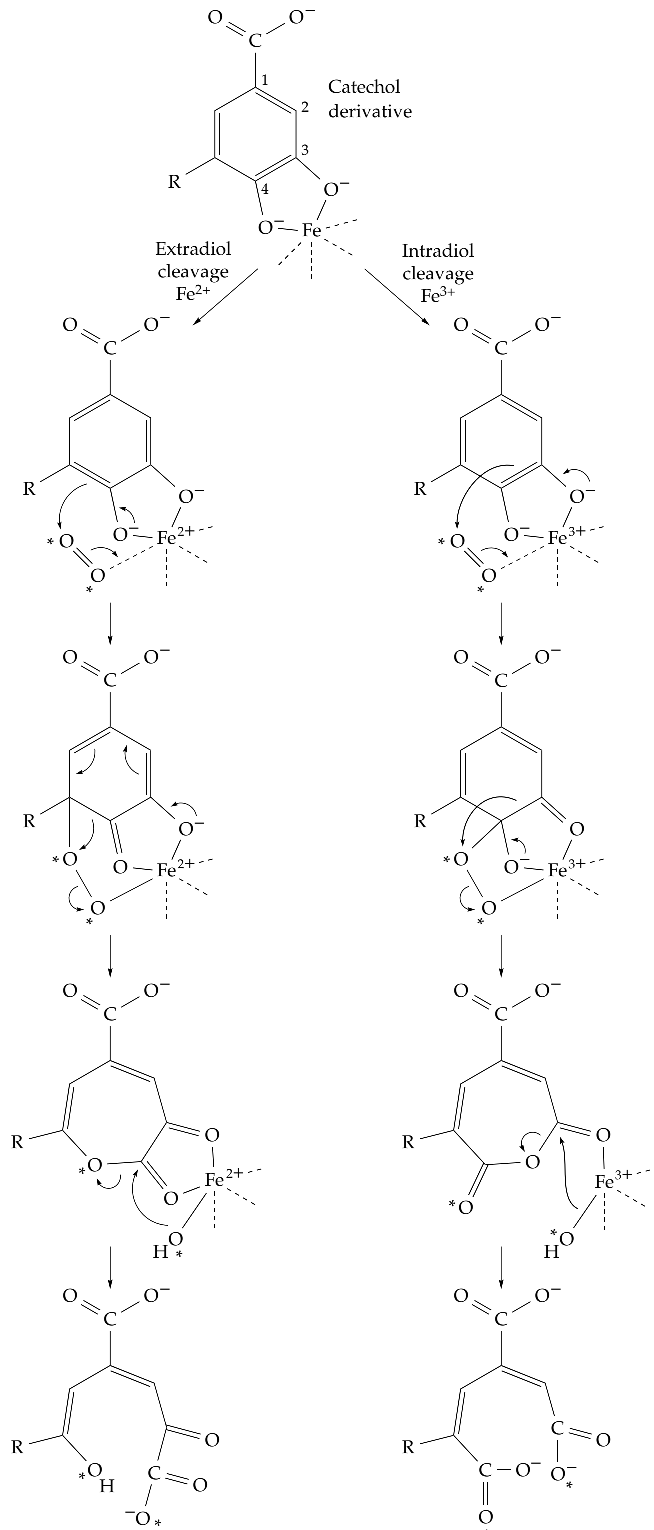
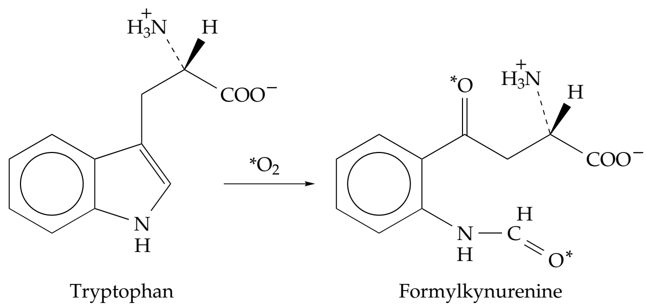


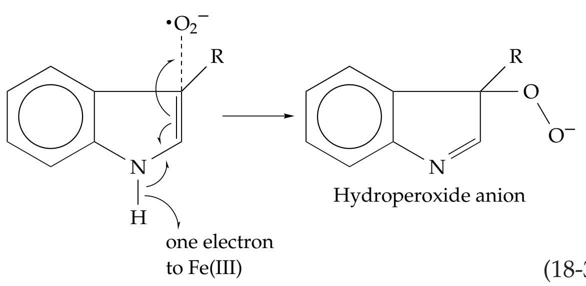
Figure 18-22 Some possible intermediates in the action of extradiol (left) and intradiol (right) aromatic dioxygenases. Although the steps depict the flow of pairs of electrons during the formation and reaction of peroxide intermediates, the mechanisms probably involve free radicals whose formation is initiated by O_2. The asterisks show how two atoms of labeled oxygen can be incorporated into final products. After Ohlendorf *et al.*[374]

the asterisks are derived from O_2. Again, the first step is probably the formation of a complex between Fe(II) and O_2, but tryptophan must also be present before this can occur. At 5°C the enzyme, tryptophan, and O_2 combine to give an altered spectrum reminiscent of that of compound III of peroxidase (Fig. 16-14). This oxygenated complex may, perhaps, then be converted to a complex of Fe(II) and superoxide ion.



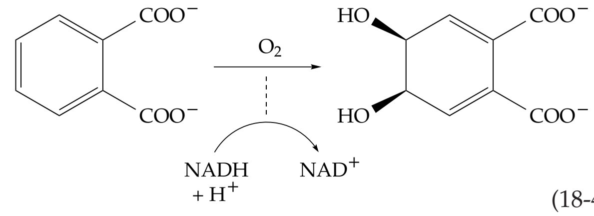
(18-38)

There is much evidence, including inhibition by superoxide dismutase and stimulation by added potassium superoxide,[384] that the superoxide anion radical is the species that attacks the substrate (Eq. 18-39). In this reaction one electron is returned to the Fe(III) form of the enzyme to regenerate the original Fe(II) form. Subsequent reaction of the hydroperoxide anion would give the observed products.



(18-39)

Some dioxygenases require a cosubstrate. For example, **phthalate dioxygenase**[385] converts phthalate to a *cis*-dihydroxy derivative with NADH as the cosubstrate (Eq. 18-40). Similar double hydroxylation reactions catalyzed by soil bacteria are known for benzene, benzoate,[386] toluene, naphthalene, and several other aromatic compounds.[386a] The formation of the *cis*-glycols is usually followed by dehydrogenation or oxidative decarboxylation by NAD^+ to give a catechol, whose ring is then opened by another dioxygenase reaction (Chapter 25). An elimination of Cl^- follows dioxygenase action on *p*-chlorophenylacetate and produces 3,4-dihydroxyphenylacetate as a product. Pthalate dioxygenases consist of two subunits. The 50-kDa dioxygenase subunits receive electrons from reductase subunits that contain a Rieske-type Fe–S



(18-40)

centers and bound FMN.[387] The dioxygenase also contains an Fe_2S_2 center, and electrons flow from NADH to FMN and through the two Fe–S centers to the Fe^{2+} of the active site.[387–388]

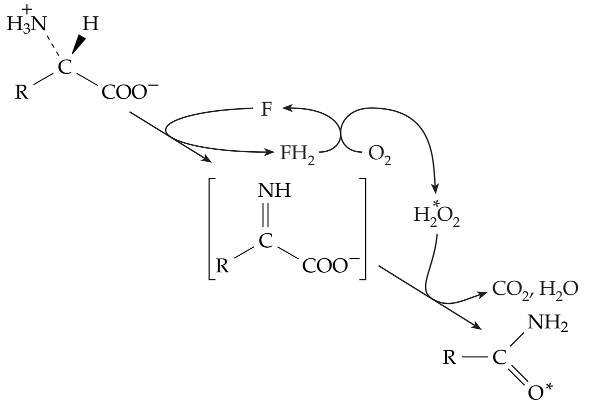
Lipoxygenases catalyze oxidation of polyunsaturated fatty acids in plant lipids. Within animal tissues the lipoxygenase-catalyzed reaction of arachidonic acid with O_2 is the first step in formation of **leukotrienes** and other mediators of inflammation. These reactions are discussed in Chapter 21.

2. Monooxygenases

Two classes of monooxygenases are known. Those requiring a cosubstrate (BH_2 of Eq. 18-36) in addition to the substrate to be hydroxylated are known as **external monooxygenases**. In the other group, the **internal monooxygenases**, some portion of the substrate being hydroxylated also serves as the cosubstrate. Many internal monooxygenases contain flavin cofactors and are devoid of metal ions.

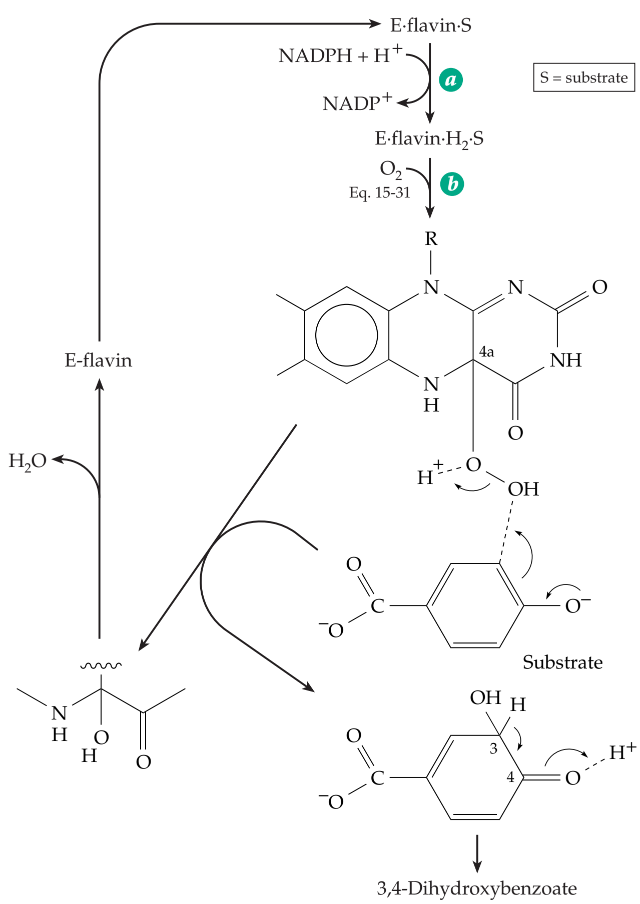
Flavin-containing monooxygenases. One group of flavin-dependent monooxygenases form H_2O_2 by reaction of O_2 with the reduced flavin and use the H_2O_2 to hydroxylate 2-oxoacids. An example is **lactate monooxygenase,** which apparently dehydrogenates lactate to pyruvate and then oxidatively decarboxylates the pyruvate to acetate with H_2O_2 (Eq. 15-36). One atom of oxygen from O_2 is incorporated into the acetate formed.[389,390] In a similar manner, the FAD-containing bacterial **lysine monooxygenase** probably catalyzes the sequence of reactions shown in Eq. 18-41.[391] When native lysine monooxygenase was treated with sulfhydryl-blocking reagents the resulting modified enzyme produced a 2-oxoacid, ammonia, and H_2O_2, just the products predicted from the hydrolytic decomposition of the bracketed intermediate of Eq. 18-41. Similar bacterial enzymes act on tryptophan and phenylalanine.[392]

NADPH can serve as a cosubstrate of flavoprotein monooxygenase by first reducing the flavin, after which the reduced flavin can react with O_2 to generate the hydroxylating reagent.[393] An example is the bacterial **4-hydroxybenzoate hydroxylase** which forms 3,4-dihydroxybenzoate.[394] The 43-kDa protein consists of three domains, the large FAD-binding domain being folded in nearly the same way as that of glutathione reductase (Fig. 15-10). The 4-hydroxybenzoate binds



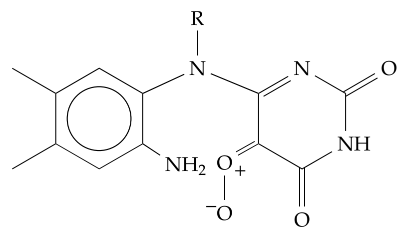
(18-41)

first into a deep cleft below the N-5 edge of the isoalloxazine ring of the FAD; then the NADH binds. Spectroscopic studies have shown the existence of at least three intermediates. The first of these has been identified as the 4a-peroxide whose formation (Eq. 15-31) is discussed in Chapter 15. The third intermediate is the corresponding 4a-hydroxyl compound. The substrate hydroxylation must occur in a reaction with the flavin peroxide, presumably with the phenolate anion form of the substrate (Eq. 18-42).[395] The initial hydroxylation product is tautomerized to form the product 3,4-dihydroxybenzoate.

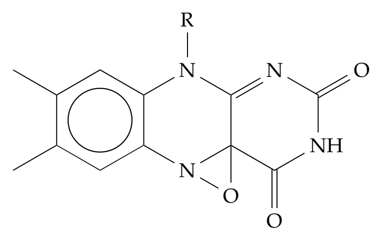


(18-42)

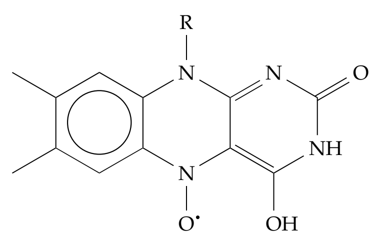
According to this mechanism, one of the two oxygen atoms in the hydroperoxide reacts with the aromatic substrate, perhaps as OH^+ or as a superoxide radical. A variety of mechanisms for activating the flavin peroxide to give a more potent hydroxylating reagent have been proposed. These include opening of the central ring of the flavin to give a carbonyl oxide intermediate which could transfer an oxygen atom to the substrate,[396] elimination of H_2O to form an **oxaziridine,**[397] or rearrangement to a **nitroxyl radical**.[398] Any of these might be an active electrophilic hydroxylating reagent. However, X-ray structural studies suggest that conformational changes isolate the substrate–FAD-enzyme complex from the medium stabilizing the 4a peroxide via hydrogen bonding[399–400] in close proximity to the substrate. Reaction could occur by the simple mechanism of Eq. 18-42, a mechanism also supported by ^{19}F NMR studies with fluorinated substrate analogs[401] and other investigations.[401a,b]



Carbonyl oxide

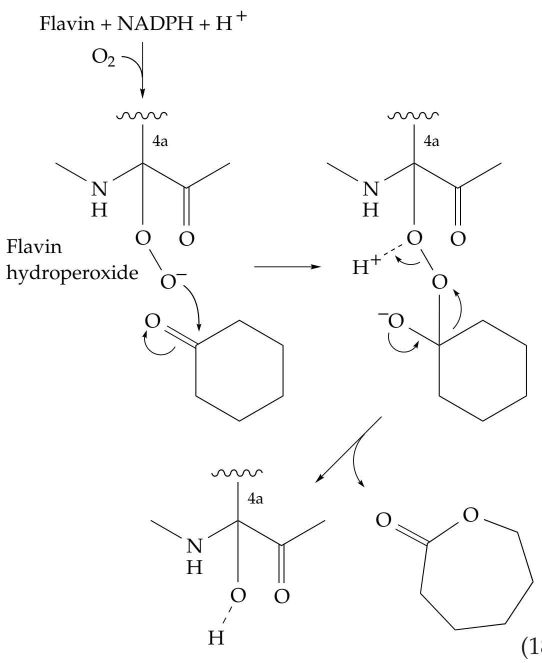


An oxaziridine



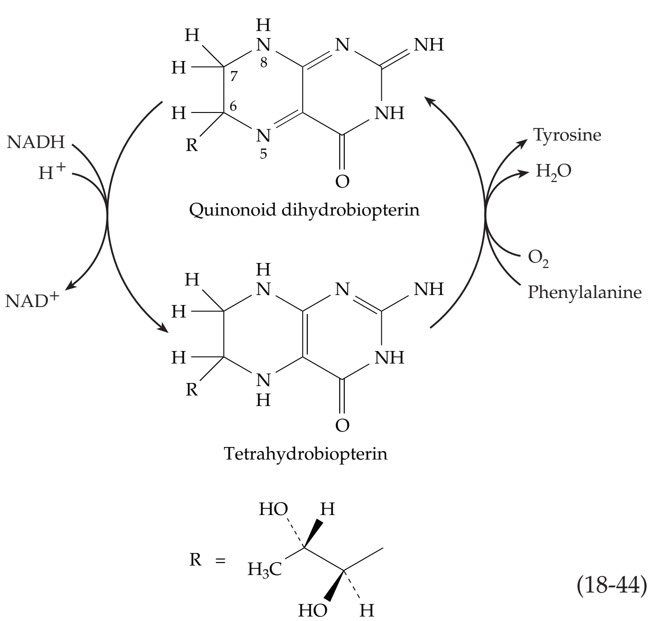
A nitroxyl radical

Related flavin hydroxylases act at nucleophilic positions on a variety of molecules[393,402] including phenol,[403] salicylate,[404] anthranilate,[405] *p*-cresol,[406] 4-hydroxyphenylacetate,[407,408] and 4-aminobenzoate.[409] Various microsomal flavin hydroxylases are also known.[410] Flavin peroxide intermediates are also able to hydroxylate some electrophiles.[411] For example, the bacterial **cyclohexanone oxygenase** catalyzes the ketone to lactone conversion of Eq. 18-43.[411a] The mechanism presumably involves the nucleophilic attack of the flavin hydroperoxide on the carbonyl group of the substrate followed by rearrangement. This parallels the Baeyer–Villiger rearrangement that results from treatment of ketones with peracids.[393] Cyclohexanone oxygenase also catalyzes a variety of other reactions,[412] including conversion of sulfides to sulfoxides.



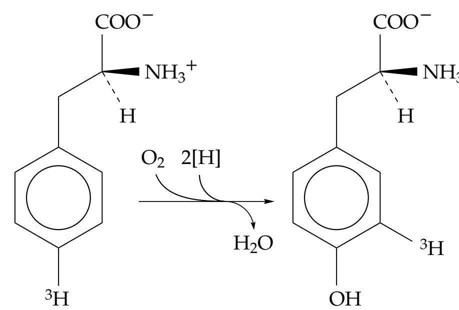
(18-43)

Reduced pteridines as cosubstrates. A dihydro form of biopterin (Fig. 15-17) serves as a cosubstrate, that is reduced by NADPH (Eq. 18-44) in hydroxylases that act on phenylalanine, tyrosine, and tryptophan.



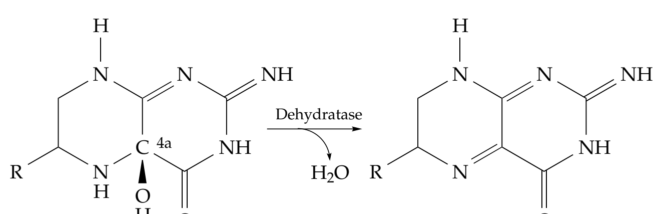
(18-44)

The tetrahydrobiopterin formed in this reaction is similar in structure to a reduced flavin. The mechanism of its interaction with O_2 could reasonably be the same as that of 4-hydroxybenzoate hydroxylase. However, **phenylalanine hydroxylase**, which catalyzes the formation of tyrosine (Eq. 18-45), a dimer of 451-residue subunits, contains one Fe per subunit,[413–415a] whereas flavin monooxygenases are devoid of iron. **Tyrosine hydroxylase**[416–419a] and **tryptophan hydroxylase**[420] have very similar properties. All three enzymes contain regulatory, catalytic, and tetramerization domains as well as a common Fe-binding motif in their active sites.[413,421,421a]



(18-45)

The role of the iron atom in these enzymes must be to accept an oxygen atom from the flavin peroxide, perhaps forming a reactive ferryl ion and transferring the oxygen atom to the substrate, e.g., as do cytochromes P450 (see Eq. 18-57). The 4*a*-hydroxytetrahydrobiopterin, expected as an intermediate if the mechanism parallels that of Eq. 18-42, has been identified by its ultraviolet absorption spectrum.[422] A ring-opened intermediate has also been ruled out for phenylalanine hydroxylase.[423] However, the 4*a* –OH adduct has been observed by ^{13}C-NMR spectroscopy. Its absolute configuration is 4*a*(S) and the observation of an ^{18}O-induced shift in the ^{13}C resonance of the 4*a*-carbon atom[424] confirms the origin of this oxygen from $^{18}O_2$ (see Eq. 18-42). A "stimulator protein" needed for rapid reaction of phenylalanine hydroxylase has been identified as a **4*a*-carbinolamine dehydratase** (Eq. 18-46).[425–426] This protein also has an unexpected function as part of a complex with transcription factor HNF1 which is found in nuclei of liver cells.[425a,426]



(18-46)

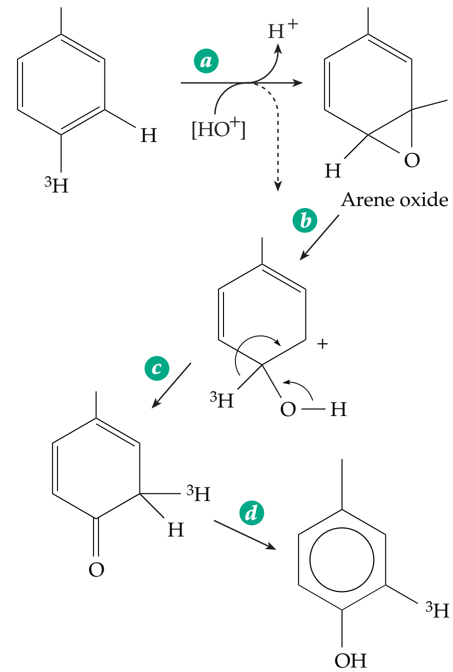
Dihydrobiopterin can exist as a number of isomers. The quinonoid form shown in Eqs. 18-44 and 18-46 is a tautomer of 7,8-dihydrobiopterin, the form generated by dihydrofolate reductase (Chapter 15). A pyridine nucleotide-dependent **dihydropteridine reductase**[427–429] catalyzes the left-hand reaction of Eq. 18-44.

The hereditary absence of phenylalanine hydroxylase, which is found principally in the liver, is the cause of the biochemical defect **phenylketonuria** (Chapter 25, Section B).[430,430a] Especially important in the metabolism of the brain are tyrosine hydroxylase, which converts tyrosine to 3,4-dihydroxyphenylalanine, the rate-limiting step in biosynthesis of the catecholamines (Chapter 25), and tryptophan hydroxylase, which catalyzes formation of 5-hydroxytryptophan, the first step in synthesis of the neurotransmitter 5-hydroxytryptamine (Chapter 25). All three of the pterin-dependent hydroxylases are under complex regulatory control.[431,432] For example, tyrosine hydroxylase is acted on by at least four kinases with phosphorylation occurring at several sites.[431,433,433a] The kinases are responsive to nerve growth factor and epidermal growth factor,[434] cAMP,[435] Ca^{2+} + calmodulin, and Ca^{2+} + phospholipid (protein kinase C).[436] The hydroxylase is inhibited by its endproducts, the catecholamines,[435] and its activity is also affected by the availability of tetrahydrobiopterin.[436]

Hydroxylation-induced migration. A general result of enzymatic hydroxylation of aromatic compounds is the intramolecular migration of a hydrogen atom or of a substituent atom or group as is shown for the ^3H atom in Eq. 18-45.[437] Dubbed the NIH shift (because the workers discovering it were in a National Institutes of Health laboratory), the migration tells us something about possible mechanisms of hydroxylation. In Eq. 18-45 a tritium atom has shifted in response to the entering of the hydroxyl group. The migration can be visualized as resulting from electrophilic attack on the aromatic system, e.g., by an oxygen atom from Fe(N)=O or by OH^+ (Eq. 18-47).

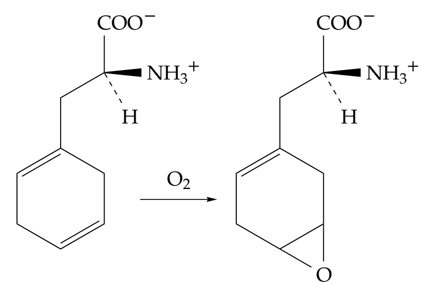
Such an attack could lead in step *a* either to an **epoxide** (**arene oxide**) or directly to a carbocation as shown in Eq. 18-47. Arene oxides can be converted, via the carbocation step *b*, to end products in which the NIH shift has occurred.[438] The fact that phenylalanine hydroxylase also catalyzes the conversion of the special substrate shown in Eq. 18-48 to a stable epoxide, which cannot readily undergo ring opening, also supports this mechanism.

Operation of the NIH shift can cause migration of a large substituent as is illustrated by the hydroxylation of 4-hydroxyphenylpyruvate (Eq. 18-49), a key step in the catabolism of tyrosine (Chapter 25). Human 4-hydroxyphenylpyruvate dioxygenase is a dimer of 43-kDa subunits.[439] A similar enzyme from *Pseudomonas* is a 150-kDa tetrameric iron-tyrosinate protein, which must be maintained in the reduced Fe(II) state for catalytic activity.[440] Although this enzyme is a



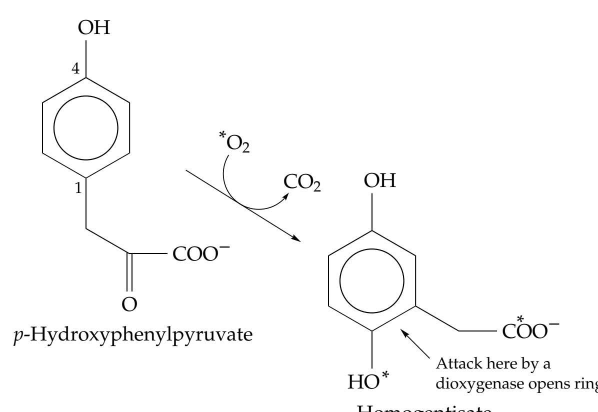
(18-47)

dioxygenase, it is probably related in its mechanism of action to the 2-oxoglutarate-dependent monooxygenases discussed in the next section (Eqs. 18-51, 18-52). It probably uses the oxoacid side chain of the substrate



(18-48)

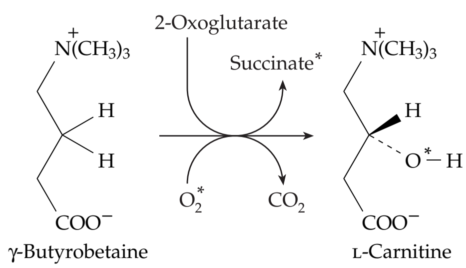
to generate a reactive oxygen intermediate such as Fe(IV)=O by the decarboxylative mechanism of Eqs. 18-50 and 18-51. The iron-bound oxygen attacks C1 of



(18-49)

the aromatic ring, the electron-donating *p*-hydroxyl group assisting. This generates a hydroxylated carbocation of the type shown in Eq. 18-47 in which the whole two-carbon side chain undergoes the NIH shift.

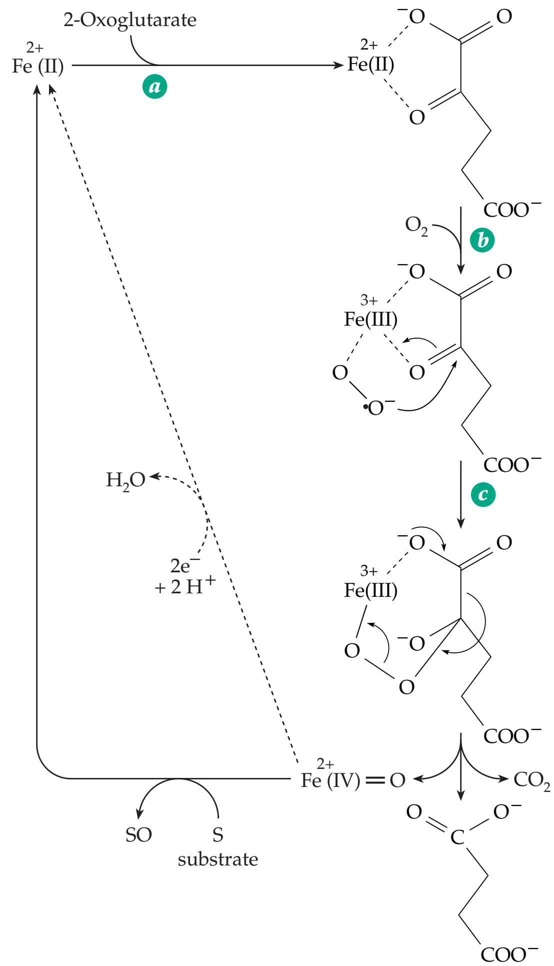
2-Oxoglutarate as a decarboxylating cosubstrate. Several oxygenases accept hydrogen atoms from 2-oxoglutarate, which is decarboxylated in the process to form succinate. Among these are enzymes catalyzing hydroxylation of residues of proline in both the 3- and 4-positions (Eq. 8-6)[441–444] and of lysine in the 5-position (Eq. 8-7)[445,446] in the collagen precursor **procollagen**. The hydroxylation of prolyl residues also takes place within the cell walls of plants.[447] Similar enzymes hydroxylate the β-carbon of aspartyl or asparaginyl side chains in EGF domains (Table 7-3) of proteins.[441] Thymine[448] and taurine[449,449a] are acted on by related dioxygenases. A bacterial oxygenase initiates the degradation of the herbicide 2,4-dichlorophenoxyacetic acid (2,4-D) using another 2-oxoglutarate-dependent hydroxylase.[450,451] In the human body a similar enzyme hydroxylates γ-butyrobetaine to form carnitine (Eq. 18-50).[452] All of



(18-50)

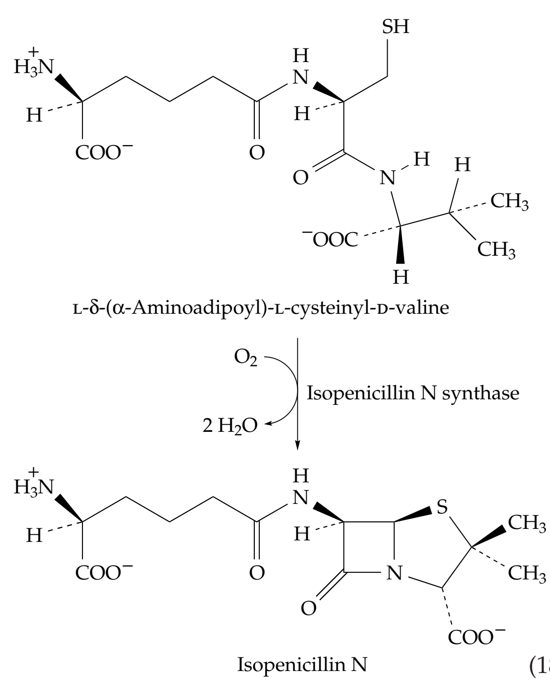
these enzymes contain iron and require ascorbate, whose function is apparently to prevent the oxidation of the iron to the Fe(III) state.

When $^{18}O_2$ is used for the hydroxylation of γ-butyrobetaine (Eq. 18-51), one atom of ^{18}O is found in the carnitine and one in succinate. The reaction is stereospecific and occurs with retention of configuration at C-3, the *pro*-R hydrogen being replaced by OH while the *pro*-S hydrogen stays.[453] Under some conditions these enzymes decarboxylate 2-oxoglutarate in the absence of a hydroxylatable substrate, the iron being oxidized to Fe^{3+} and ascorbate being consumed stoichiometrically.[454] A plausible mechanism (Eq. 18-51) involves formation of an Fe(II)–O_2 complex, conversion to $Fe(III)^+ \,{}^\bullet O_2^-$, and addition of the superoxide ion to 2-oxoglutarate to form an adduct.[451,455] Decarboxylation of this adduct could generate the oxidizing reagent, perhaps Fe(IV)=O. In the absence of substrate S the ferryl iron could be reconverted to Fe(II) by a suitable reductant such as ascorbate. In the absence of ascorbate the Fe(IV) might be reduced to a catalytically inert Fe(III) form.



(18-51)

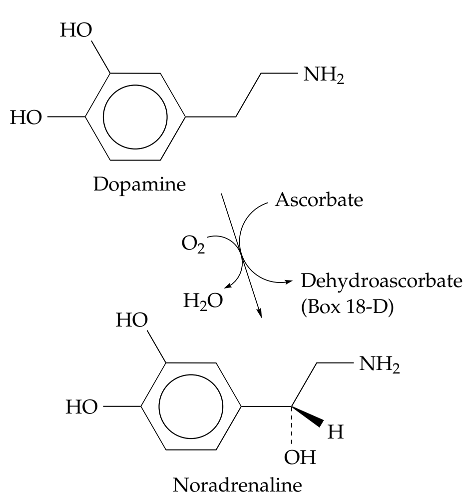
An unusual oxygenase with a single Fe^{2+} ion in its active site closes the four-membered ring in the biosynthesis of penicillins (Eq. 18-52). It transfers four



(18-52)

hydrogen atoms from its dipeptide substrate to form two molecules of water and the product isopenicillin N.[456,457] Sequence comparison revealed several regions including the Fe-binding sites that are homologous with the oxoacid-dependent oxygenases. A postulated mechanism for **isopenicillin N synthase** involves formation of an Fe^{3+} superoxide anion complex as in Eq. 18-51. However, instead of attack on an oxoacid as in Eq. 18-51, it removes a hydrogen from the substrate to initiate the reaction sequence.[457] Other related oxygenases include **aminocyclopropane-1-carboxylate oxidase** (Eq. 24-35); **deacetoxycephalosporin C synthase**,[457a] an enzyme that converts penicillins to cephalosporins (Box 20-G); and **clavaminate synthase**,[458,459] an enzyme needed for synthesis of the β-lactamase inhibitor clavulanic acid, and **clavaminate synthase**.[458,459] This 2-oxoglutarate-dependent oxygenase catalyzes three separate reactions in the synthesis of the clinically important β-lactamase inhibitor clavulanic acid. The first step is similar to that in Eq. 18-50. The second is an oxidative cyclization and the third a desaturation reaction.

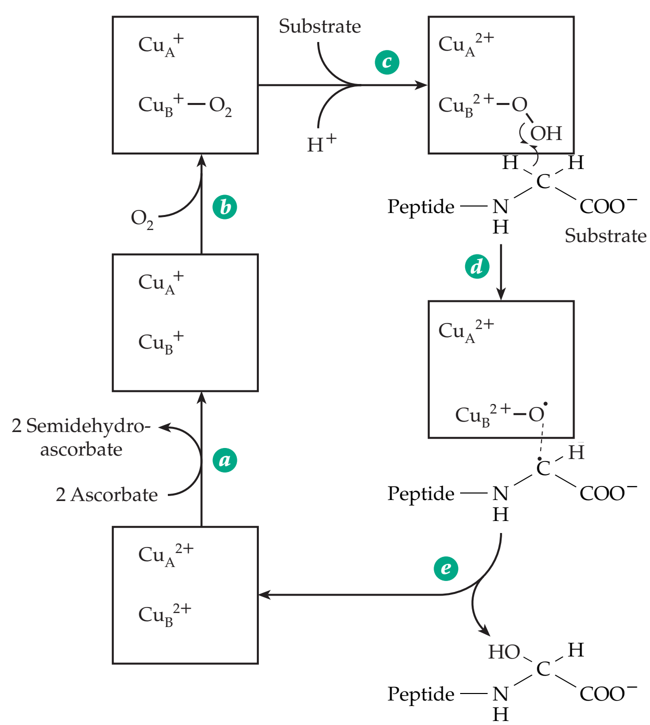
Copper-containing hydroxylases. Many Fe(II)-containing hydroxylases require a reducing agent to maintain the iron in the reduced state, and ascorbate is often especially effective. In addition, ascorbate is apparently a true cosubstrate for the copper-containing **dopamine β-hydroxylase**, an enzyme required in the synthesis of noradrenaline according to Eq. 18-53. This reaction takes place in neurons of the brain and in the adrenal gland, a tissue long known as especially rich in ascorbic acid. The reaction requires two molecules of ascorbate, which are converted in two one-electron steps to **semidehydroascorbate**.[460] Both the structure of this free radical and that of the fully oxidized form of vitamin C, **dehydroascorbic acid**, are shown in Box 18-D. Dopamine β-hydroxylase is a 290-kDa tetramer, consisting of a pair of identical disulfide-crosslinked homodimers, which contains two Cu ions per subunit.[461]



(18-53)

A similar copper-dependent hydroxylase constitutes the N-terminal domain of the **peptidylglycine α-amidating enzyme** (Eq. 10-11). This bifunctional enzyme hydroxylates C-terminal glycines in a group of neuropeptide hormones and other secreted peptides. The second functional domain of the enzyme cleaves the hydroxylated glycine to form a C-terminal amide group and glyoxylate.[462–464b] The three-dimensional structure of a 314-residue catalytic core of the hydroxylase domain is known.[463] Because of similar sequences and other properties, the structures of this enzyme and of dopamine β-hydroxylase are thought to be similar. The hydroxylase domain of the α-amidating enzyme is folded into two eight-stranded antiparallel jelly-roll motifs, each of which binds one of the two copper ions. Both coppers can exist in a Cu(II) state and be reduced by ascorbate to Cu(I). One Cu (Cu_A) is held by three imidazole groups and is thought to be the site of interaction with ascorbate. The other copper, Cu_B, which is 1.1 nm away from Cu_A, is held by two imidazoles. The substrate binds adjacent to Cu_B.[463]

The reaction cycle of these enzymes begins with reduction of both coppers from Cu(II) to Cu(I) (Eq. 18-54, step *a*). Both O_2 and substrate bind (steps *b* and *c*, but not necessarily in this order). The O_2 bound to Cu_B is reduced to a peroxide anion that remains bound to Cu_B. Both Cu_A and Cu_B donate one electron, both being oxidized to Cu(II). These changes are also included in step *c* of Eq. 18-54. One proposal is that the resulting peroxide is cleaved homolytically while removing the *pro-S* hydrogen of the glycyl residue.



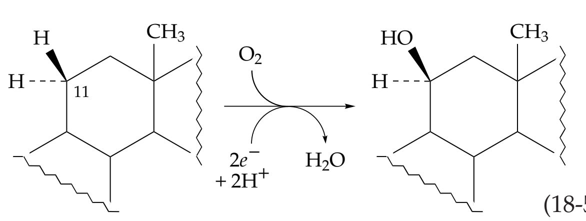
(18-54)

The resulting glycyl radical couples with the oxygen radical that is bound to Cu_B (step *e*).

A variety of other copper hydroxylases are known. For example, **tyrosinase**, which contains a binuclear copper center, catalyzes both hydroxylation of phenols and aromatic amines and dehydrogenation of the resulting catechols or *o*-aminophenols (Eq. 16-57). As in hemocyanin, the O_2 is thought to be reduced to a peroxide which bridges between the two copper atoms. Methane-oxidizing bacteria, such as *Methylococcus capsulatus,* oxidize methane to methanol to initiate its metabolism. They do this with a copper-containing membrane-embedded monooxygenase whose active site is thought to contain a trinuclear copper center. Again a bridging peroxide may be formed and may insert an oxygen atom into the substrate.[465,466] The same bacteria produce a soluble methane monooxygenase containing a binuclear iron center.

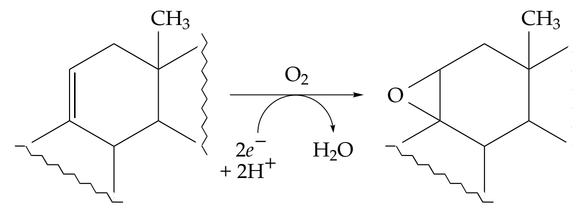
Hydroxylation with cytochrome P450. An important family of heme-containing hydroxylases, found in most organisms from bacteria to human beings, are the cytochromes P450. The name comes from the fact that in their reduced forms these enzymes form a complex with CO that absorbs at 450 nm. In soil bacteria cytochromes P450 attack compounds of almost any structure. In the adrenal gland they participate in steroid metabolism,[467,468] and in the liver microsomal cytochromes P450 attack drugs, carcinogens, and other xenobiotics (foreign compounds).[469–471] They convert cholesterol to bile acids[472] and convert vitamin D,[473] prostaglandins, and many other metabolites to more soluble and often biologically more active forms. In plants cytochromes P450 participate in hydroxylation of fatty acids at many positions.[474] They play a major role in the biosynthetic phenylpropanoid pathway (Fig. 25-8) and in lignin synthesis.[475] More than 700 distinct isoenzyme forms have been described.[476,476a]

Cytochromes P450 are monooxygenases whose cosubstrates, often NADH or NADPH, deliver electrons to the active center heme via a separate flavoprotein and often via an iron-sulfur protein as well.[476a,b] A typical reaction (Eq. 18-55) is the 11 β-hydroxylation of a steroid, an essential step in the biosynthesis of steroid hormones (Fig. 22-11). The hydroxyl group is introduced without inversion of configuration. The same enzyme converts unsaturated derivatives to epoxides (Eq. 18-56), while other cytochromes P450



(18-55)

epoxidize olefins.[477] Epoxide hydrolases, which act by a mechanism related to haloalkane dehalogenase (Fig. 12-1), convert the epoxides to diols.[478] Cytochromes P450 are able to catalyze a bewildering array of other



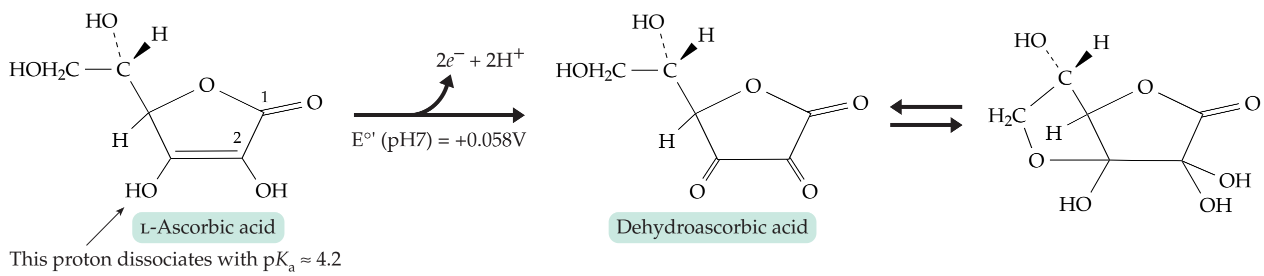
(18-56)

reactions[479–481] as well. Most of these, such as conversion of amines and thioesters to *N*- or *S*-oxides, also involve transfer of an oxygen atom to the substrate. Others, such as the reduction of epoxides, *N*-oxides, or nitro compounds, are electron-transfer reactions.

Several different cytochromes P450 are present in mammalian livers.[470] All are bound to membranes of the endoplasmic reticulum and are difficult to solubilize. Biosynthesis of additional forms is induced by such agents as phenobarbital,[470] 3-methylcholanthrene,[469] dioxin,[482] and ethanol.[483] These substances may cause as much as a 20-fold increase in P450 activity. Another family of cytochrome P450 enzymes is present in mitochondria.[483a] A large number of cytochrome P450 genes have been cloned and sequenced. Although they are closely related, each cytochrome P450 has its own gene. There are at least ten families of known P450 genes and the total number of these enzymes in mammals may be as high as 200. Microorganisms, from bacteria to yeast, produce many other cytochromes P450.

Microsomal cytochromes P450 receive electrons from an **NADPH-cytochrome P450** reductase, a large 77-kDa protein that contains one molecule each of FAD and FMN.[484–485a] It is probably the FAD which accepts electrons from NADPH and the FMN which passes them on to the heme of cytochrome P450. Cytochrome b_5 is also reduced by this enzyme,[486] and some cytochromes P450 may accept one electron directly from the flavin of the reductase and the second electron via cytochrome b_5. However, most bacterial and mitochondrial cytochromes P450 accept electrons only from small iron–sulfur proteins. Those of the adrenal gland receive electrons from the 12-kDa **adrenodoxin**.[487,488] This small protein of the ferredoxin class contains one Fe_2S_2 cluster and is, therefore, able to transfer electrons one at a time from the FAD-containing NADPH-adrenodoxin reductase[489] to the cytochrome P450. The camphor 5-monoxygenase from *Pseudomonas putida* consists of three components: an FAD-containing reductase, the Fe_2S_2 cluster-containin **putidaredoxin**,[489a] and cytochrome $P450_{cam}$.[490] Some other bacterial

BOX 18-D VITAMIN C: ASCORBIC ACID



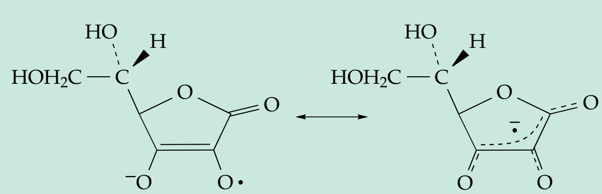
Hemorrhages of skin, gums, and joints were warnings that death was near for ancient sea voyagers stricken with **scurvy**. It was recognized by the year 1700 that the disease could be prevented by eating citrus fruit, but it was 200 years before efforts to isolate vitamin C were made. Ascorbic acid was obtained in crystalline form in 1927,[a–e] and by 1933 the structure had been established. Only a few vertebrates, among them human beings, monkeys, guinea pigs, and some fishes, require ascorbic acid in the diet; most species are able to make it themselves. Compared to that of other vitamins, the nutritional requirement is large.[e] Ten milligrams per day prevents scurvy, but subclinical deficiency, as judged by fragility of small capillaries in the skin, is present at that level of intake. "Official" recommendations for vitamin C intake have ranged from 30 to 70 mg / day. A more recent study[f] suggests 200 mg / day, a recommendation that is controversial.[g]

The biological functions of vitamin C appear to be related principally to its well-established reducing properties and easy one-electron oxidation to a free radical or two-electron reduction to **dehydroascorbic acid**. The latter is in equilibrium with the hydrated hemiacetal shown at the beginning of this box as well as with other chemical species.[h–j] Vitamin C is a weak acid which also has metal complexing properties.

Ascorbate, the anion of ascorbic acid, tends to be concentrated in certain types of animal tissues and may reach 3 mM or more in leukocytes, in tissues of eyes and lungs, in pituitary, adrenal, and parotid glands,[k,l] and in gametes.[m] Uptake into vesicles of the endoplasmic reticulum may occur via glucose transporters.[n] Ascorbate concentrations are even higher in plants and may exceed 10 mM in chloroplasts.[o] In animals the blood plasma ascorbate level of 20–100 µM is tightly controlled.[p,q] Cells take up ascorbate but any excess is excreted rapidly in the urine.[q] Both in plasma and within cells most vitamin C exists as the reduced form, ascorbate. When it is formed, the oxidized dehydroascorbate is reduced back to ascorbate or is degraded. The lactone ring is readily hydrolyzed to 2,3-dioxogulonic acid, which can undergo decarboxylation and oxidative degradation, one product being oxalate (see Fig. 20-2).[r] Tissues may also contain smaller amounts of L-ascorbic acid 2-sulfate, a compound originally discovered in brine shrimp. It is more stable than free ascorbate and may be hydrolyzed to ascorbate in tissues.[s]

In the chromaffin cells of the adrenal glands and in the neurons that synthesize catecholamines as neurotransmitters, ascorbate functions as a cosubstrate for dopamine β-hydroxylase (Eq. 18-53).[t,u] In fibroblasts it is required by the prolyl and lysyl hydroxylases and in hepatocytes by homogentisate dioxygenase (Eq. 18-49). Any effect of ascorbic acid in preventing colds may be a result of increased hydroxylation of procollagen and an associated stimulation of procollagen secretion.[v] High levels of ascorbate in guinea pigs lead to more rapid healing of wounds.[w] An important function of ascorbate in the pituitary and probably in other endocrine glands is in the α-amidation of peptides (Eq. 10-11).[x,y] Together with Fe(II) and O_2 ascorbate is a powerful nonenzymatic hydroxylating reagent for aromatic compounds. Like hydroxylases, the reagent attacks nucleophilic sites, e.g., converting phenylalanine to tyrosine. Oxygen atoms from $^{18}O_2$ are incorporated into the hydroxylated products. While H_2O_2 is formed in the reaction mixture, it cannot replace ascorbate. The relationship of this system to biochemical functions of ascorbate is not clear. An unusual function for vitamin C has been proposed for certain sponges that are able to etch crystalline quartz (SiO_2) particles from sand or rocks.[z]

Ascorbate is a major antioxidant, protecting cells and tissues from damage by free radicals, peroxides, and other metabolites of O_2.[p,r,aa,bb] It is chemically suited to react with many biologically important radicals and is present in high enough concentrations to be effective. It probably functions in cooperation with glutathione (Box 11-B),[cc] α-tocopherol (Fig. 15-24),[dd] and lipoic acid.[ee] Ascorbate can react with radicals in one-election transfer reactions to give the monodehydroascorbate radical[aa]:



BOX 18-D (continued)

Two ascorbate radicals can react with each other in a disproportionation reaction to give ascorbate plus dehydroascorbate. However, most cells can reduce the radicals more directly. In many plants this is accomplished by NADH + H^+ using a flavoprotein **monodehydroascorbate reductase**.[o] Animal cells may also utilize NADH or may reduce dehydroascorbate with reduced glutathione.[cc,ff] Plant cells also contain a very active blue copper ascorbate oxidase (Chapter 16, Section D,5), which catalyzes the opposite reaction, formation of dehydroascorbate.[gg] A heme ascorbate oxidase has been purified from a fungus.[hh] Action of these enzymes initiates an oxidative degradation of ascorbate, perhaps through the pathway of Fig. 20-2.

Ascorbate can also serve as a signal. In cultured cells, which are usually deficient in vitamin C, addition of ascorbate causes an enhanced response to added iron, inducing synthesis of the iron storage protein ferritin.[ii] Ascorbate indirectly stimulates transcription of procollagen genes[v] and decreases secretion of insulin by the pancreas.[jj] However, since its concentration in blood is quite constant this effect is not likely to cause a problem for a person taking an excess of vitamin C.

Should we take extra vitamin C to protect us from oxygen radicals and slow down aging? Linus Pauling, who recommended an intake of 0.25–10 g / day, maintained that ascorbic acid also has a specific beneficial effect in preventing or ameliorating symptoms of the common cold.[j,kk] However, critics point out that unrecognized hazards may exist in high doses of this seemingly innocuous compound. Ascorbic acid has antioxidant properties, but it also promotes the generation of free radicals in the presence of Fe(III) ions, and it is conceivable that too much may be a bad thing.[ll] Catabolism to oxalate may promote formation of calcium oxalate kidney stones. Under some conditions products of dehydroascorbic acid breakdown may accumulate in the lens and contribute to cataract formation.[i,mm,nn] However, dehydroascorbate, or its decomposition products, apparently *protects* low-density lipoproteins against oxidative damage.[bb] Pauling pointed out that nonhuman primates synthesize within their bodies many grams of ascorbic acid daily, and that there is little evidence for toxicity. Pauling's claim that advanced cancer patients are benefited by very high (10 g daily) doses of vitamin C has been controversial, and some studies have failed to substantiate the claim.[oo,pp]

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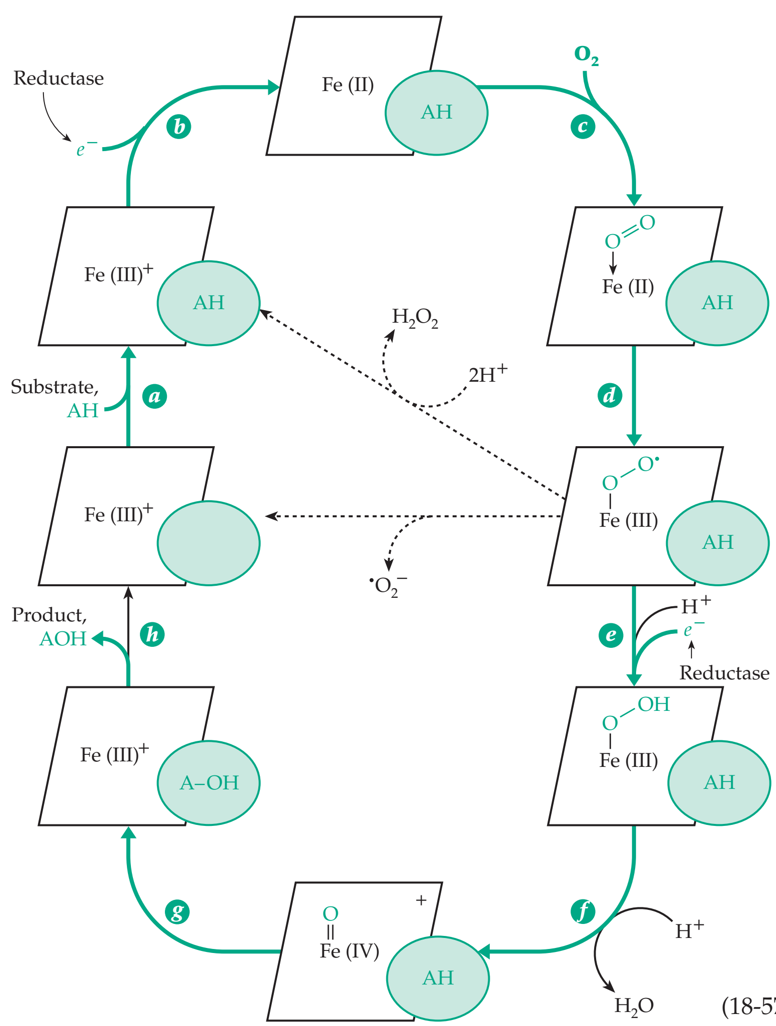
cytochrome P450s, such as a soluble fatty acid hydroxylase from *Bacillus megaterium*, have reductase domains with tightly bound FMN and FAD bound to the same polypeptide chain as is the heme.[491]

All cytochromes P450 appear to have at their active sites a molecule of heme with a thiolate anion as an axial ligand in the fifth position (Fig. 18-23). These relatively large heme proteins of ~ 45- to 55-kDa mass may consist of as many as 490 residues. Only a few three-dimensional structures are known,[490,492–494] and among these there are significant differences. However, on the basis of a large amount of experimental effort[487,495,496] it appears that all cytochromes P450 act by basically similar mechanisms.[474,496a,b,497] As indicated in Eq. 18-57, the substrate AH binds to the protein near the heme, which must be in the Fe(III) form. An electron delivered from the reductase then reduces the iron to the Fe(II) state (Eq. 18-57, step *b*). Then O_2 combines with the iron, the initial oxygenated complex formed in step *c* being converted to an Fe(III)-superoxide complex (Eq. 18-57, step *d*). Subsequent events are less certain.[497a] Most often a second electron is transferred in from the reductase (Eq. 18-57, step *e*) to give a peroxide complex of Fe(III), which is then converted in step *f* to a ferryl iron form, as in the action of peroxidases (Fig. 16-14). This requires transfer of two H^+ into the active site. The ferryl Fe(IV)=O donates its oxygen atom to the substrate regenerating the Fe(III) form of the heme (step *g*) and releasing the product (step *h*).

Microsomal cytochromes P450 often form hydrogen peroxide as a side product. This may arise directly from the Fe–O–O$^-$ intermediate shown in Eq. 18-57. Some cytochromes P450 use this reaction in reverse to carry out hydroxylation utilizing peroxides instead of O_2 (Eq. 18-58).

$$AH + ROOH \rightarrow AOH + ROH \qquad (18\text{-}58)$$

Cytochromes P450 often convert drugs or other foreign compounds to forms that are more readily excreted.[499] However, the result is not always beneficial. For example, 3-methylcholanthrene, a strong inducer of cytochrome P450, is converted to a powerful carcinogen by the hydroxylation reaction.[500] See also Box 18-E.



(18-57)

Other iron-containing oxygenases. Hydroxylases with properties similar to those of cytochrome P450 but containing nonheme iron catalyze ω-oxidation of alkanes and fatty acids in certain bacteria, e.g., *Pseudomonas oleovarans*. A flavoprotein rubredoxin reductase, is also required.[501] The methylotrophs *Methylococcus* and *Methylosinus* hydroxylate methane using as cosubstrate NADH or NADPH (Eq. 18-59). A soluble complex consists of 38-kDa reductase containing FAD and an Fe_2S_2

$$CH_4 + O_2 + NADH + H^+ \rightarrow CH_3OH + H_2O + NAD^+ \qquad (18\text{-}59)$$

center, a small 15-kDa component, and a 245-kDa hydroxylase with an $(\alpha\beta\gamma)_2$ composition and a three-dimensional structure[502–503] similar to that of ribonucleotide reductase (Chapter 16, Section A,9). Each large α subunit contains a diiron center similar to that shown in Fig. 16-20C. It is likely that O_2 binds between the two iron atoms in the Fe(II) oxidation state and, oxidizing both irons to Fe(III), is converted to a bridging peroxide group as shown in Eq. 18-60. In this intermediate, in which the two metals are held rigidly by the surrounding ligands including a bridging carboxylate side chain, the O–O bond may be broken as in Eq. 18-60, steps *a* and *b*, to generate an Fe(IV)–O$^\bullet$ radical that may

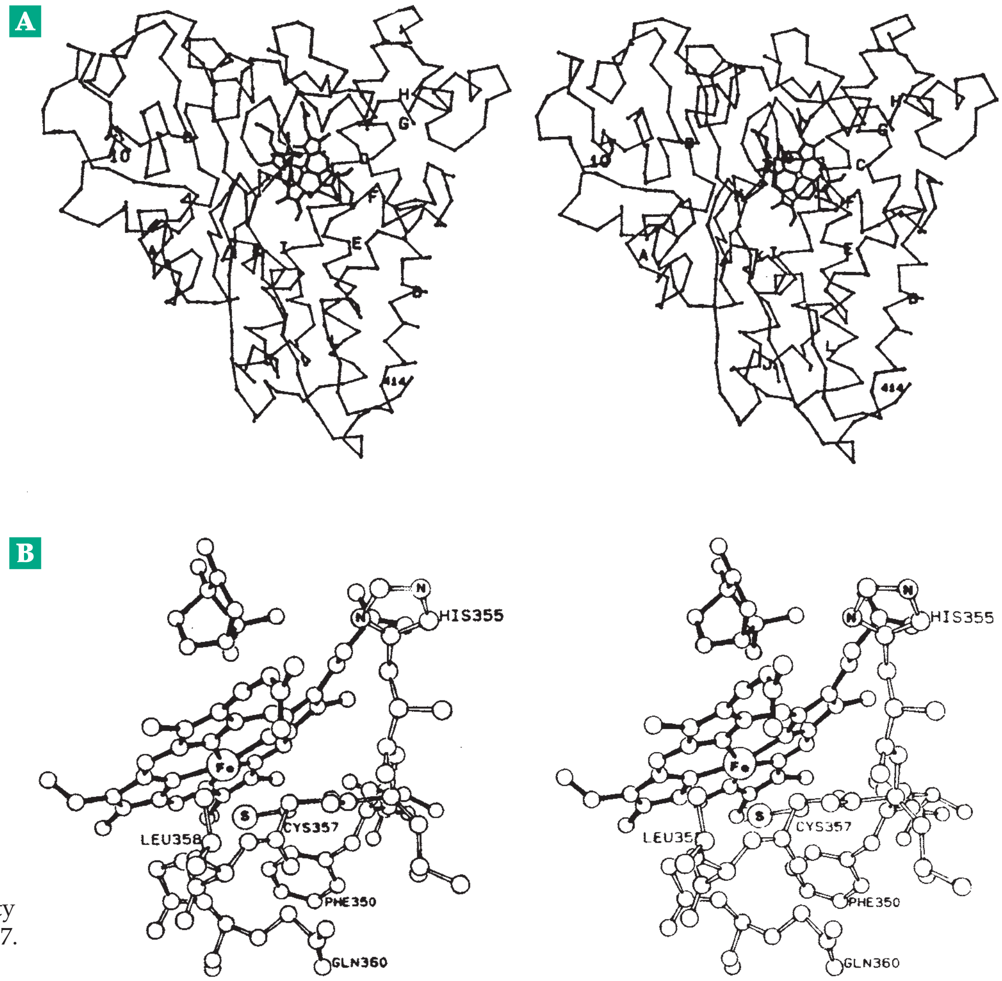
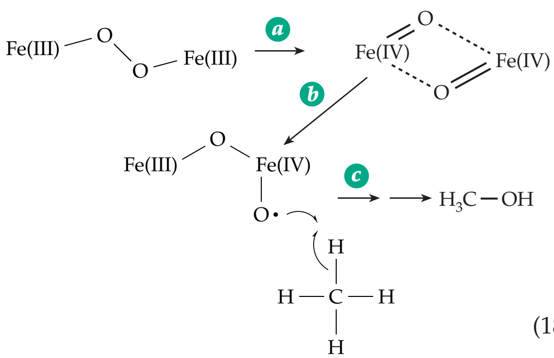


Figure 18-23 (A) Stereoscopic α-carbon backbone model of cytochrome $P450_{cam}$ showing the locations of the heme and of the bound camphor molecule. (B) View in the immediate vicinity of the thiolate ligand from Cys 357. From Poulos *et al.*[498]

remove a hydrogen atom from the substrate (step *c*) and undergo subsequent reaction steps analogous to those in the cytochrome P450 reaction cycle.[504–506]



(18-60)

A group of related bacterial enzymes hydroxylate alkanes,[507] toluene,[508] phenol,[509] and other substrates.[509a] Eukaryotic fatty acid desaturases (Fig. 16-20B) belong to the same family.[508] Some bacteria use cytochrome P450 or other oxygenase to add an oxygen atom to an alkene to form an epoxide. For example, propylene may be converted to either *R* or *S* epoxypropane which may be hydrolyzed, rearranged by a coenzyme M-dependent reaction, and converted to acetoacetate, which can be used as an energy source.[509a,b]

Nitric oxide and NO synthases. Nitric oxide (NO) is a reactive free radical whose formula is often written as $^{\bullet}NO$ to recognize this characteristic. However, NO is not only a toxic and sometimes dangerous metabolite but also an important hormone with functions in the circulatory system, the immune system, and the brain.[510–512] The hormonal effects of NO are discussed in Chapter 30, but it is appropriate here to mention a few reactions. Nitric oxide reacts rapidly with O_2 to form nitrite (Eq. 18-61).

$$4\ NO + O_2 + 2\ H_2O \rightarrow 4\ NO_2^- + 4\ H^+ \qquad (18\text{-}61)$$

It also combines very rapidly with superoxide anion radical to form **peroxynitrite** (Eq. 18-62).[513] This is another reactive oxidant which, because of its relatively high pK_a of 6.8, is partially protonated and able to diffuse through phospholipids within cells.[514,515]

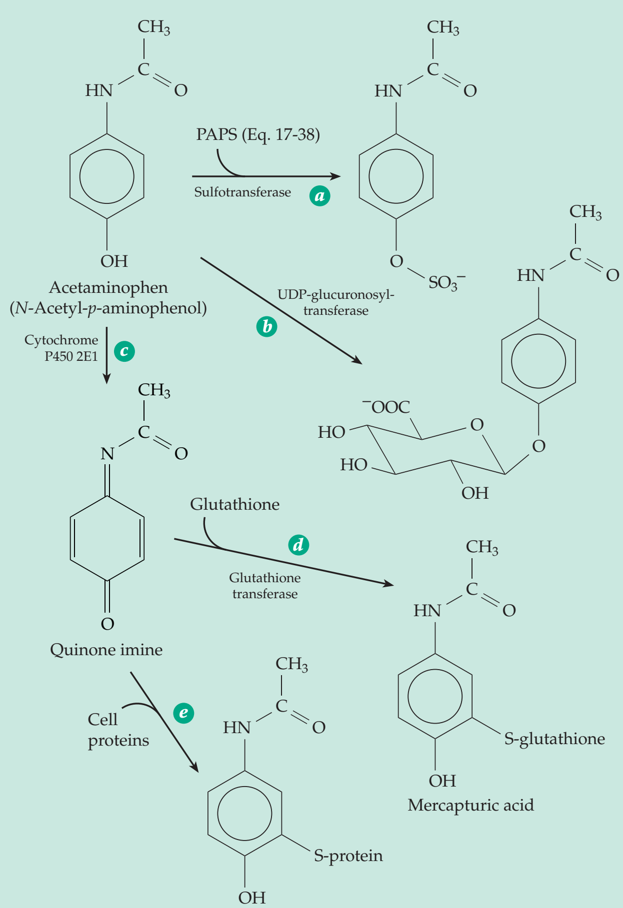
BOX 18-E THE TOXICITY OF ACETAMINOPHEN

Most drugs, as well as toxins and other xenobiotic compounds, enter the body through membranes of the gastrointestinal tract, lungs, or skin. Drugs are frequently toxic if they accumulate in the body. They are often rather hydrophobic and are normally converted to more polar, water-soluble substances before elimination from the body. Two major types of reaction take place, usually in the liver. These are illustrated in the accompanying scheme for acetaminophen (*N*-acetyl-*p*-aminophenol), a widely used analgesic and antipyretic (fever relieving) nonprescription drug sold under a variety of trade names: (1) A large water-soluble group such as sulfate[a] or glucuronate is transferred onto the drug by a nucleophilic displacement reaction (steps *a* and *b* of scheme). (2) Oxidation, demethylation, and other alterations are catalyzed by one or more of the nearly 300 cytochrome P450 monooxygenases present in the liver (step *c*). Oxidation products may be detoxified by glutathione S-transferases, step *d* (see also Box 11-B).[b,c,cc]

These reactions protect the body from the accumulation of many compounds but in some cases can cause serious problems. The best known of these involves acetaminophen. Its oxidation by cytochrome P450 2E1 or by prostaglandin H synthase[d] yields a highly reactive quinone imine which reacts with cell proteins.[e] Since the cytochrome P450 oxidation can occur in two steps, a reactive intermediate radical is also created.[c,f] At least 20 drug-labeled proteins arising in this way have been identified.[c] Both addition of thiol groups of proteins to the quinone imine (step *e* of scheme) and oxidation of protein thiols occur.[g] Mitochondria suffer severe damage,[h] some of which is related to induction of Ca^{2+} release.[i]

Acetaminophen is ordinarily safe at the recommended dosages, but large amounts exhaust the reserve of glutathione and may cause fatal liver damage. By 1989, more than 1000 cases of accidental or intentional (suicide) overdoses had been reported with many deaths. Prompt oral or intravenous administration of *N*-acetylcysteine over a 72-hour period promotes synthesis of glutathione and is an effective antidote.[j]

Similar problems exist for many other drugs. Both acetaminophen and phenacetin, its ethyl ether derivative, may cause kidney damage after many years of use.[k,l] Metabolism of phenacetin and several other drugs varies among individuals. Effective detoxification may not occur in individuals lacking certain isoenzyme forms of cytochrome P450.[m] Use of the anticancer drugs daunomycin (daunorubicin; Figs. 5-22 and 5-23) and adriamycin is limited by severe cardiac toxicity arising from free radicals generated during oxidation of the drugs.[n] These are only a few examples of the problems with drugs, pesticides, plasticizers, etc.



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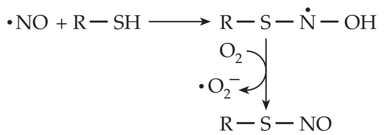
$$^{\bullet}NO + {^{\bullet}O_2^-} \rightarrow O{=}N{-}O{-}O^- \qquad (18\text{-}62)$$

NO binds to the iron atoms in accessible heme groups such as those of hemoglobin[516] and of guanylate cylases,[517,518] and in some Fe – S proteins such as aconitase. Some blood-sucking insects utilize salivary heme proteins called **nitrophorins** to carry NO into host tissues where it activates guanylate cyclase causing vasodilation. Nitrophorins also bind histamine and inhibit blood coagulation, which assists feeding.[518a,b] In the presence of a suitable oxidant such as O_2, nitric oxide reacts with thiol groups of proteins and small molecules to give ***S*-nitrosothiols** (Eq. 18-63).[511,519-520]



S-Nitrosothiol (18-63)

However, the physiological mechanisms of formation of these *S*-nitroso compounds is not clear.[520–521] One mechanism may involve conversion by O_2 to nitrous anhydride.[511,522]

$$2\,NO + O_2 \rightarrow ON{-}O{-}NO \qquad (18\text{-}64)$$

Nucleophilic attack on this compound by $-SH$, $-NH_2$, and other nucleophiles would yield *S*-nitroso and *N*-nitroso compounds with release of nitrous oxide N_2O.

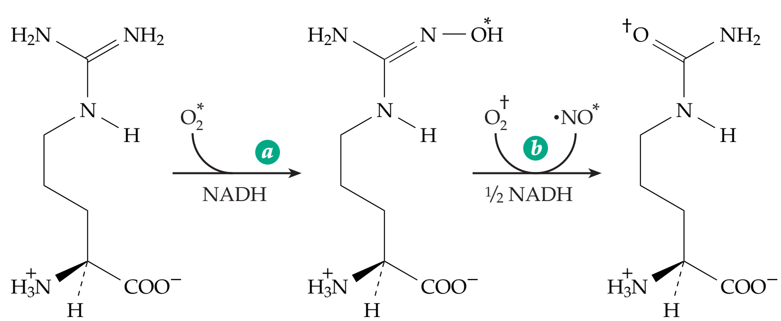
The relatively stable *S*-nitrosothiols derived from glutathione, cysteine, and proteins such as hemoglobin[516] may be important storage and transport forms of NO. If so, mechanisms of release of NO are important. A simple homolytic cleavage of R–S–NO to NO plus a thiyl radical R–S$^{\bullet}$ has often been assumed. However, rapid cleavage requires catalysis by a transition metal ion or reaction with reducing agents such as ascorbate or other thiols.[523,524] *S*-Nitrosothiols may also give rise to nitrosonium (NO^+) or nitroxy (NO^-) ions.[524]

NO synthases are oxygenases that carry out a two-step oxidation of L-arginine to L-citrulline with production of NO. In the first step, a normal monooxygenase reaction, L-N^{γ}-hydroxyarginine is formed (Eq. 18-65, step *a*). In the second step (Eq. 18-65, step *b*) NO is formed in a three-electron oxidation. In this equation the symbols * and † indicate positions of incorporation of labeled O_2 atoms in the intermediate and final products.

The human body contains three types of nitric oxide synthase known as **neuronal** (nNOS or NOS1), **inducible** (iNOS or NOS2), and **endothelial** (eNOS or NOS3).[511a–f] These enzymes have a broad distribution within various tissues, but NOS1 is especially active in neurons and NOS3 in endothelial cells. The inducible NOS2 originally discovered in macrophages is transcriptionally regulated. When these phagocytic cells are at rest the activity of NOS2 is very low, but it becomes highly active after induction by cytokines or by the lipopolysaccharides of bacterial cell membranes.[525,526] Both NOS1 and NOS2 are constitutively expressed but require calcium ions, which bind to a calmodulin domain of the protein. The inducible NOS2 doesn't require added Ca^{2+} but does contain the calmodulin domain. NOS3 carries an N-terminal myristoyl group as well as cysteines that may be palmitoylated. It is located in caveolae of plasma membranes and in Golgi complexes.[527,528]

Nucleotide sequences revealed a close similarity of NO synthases to cytochrome P450 reductase.[525,529,530] Study of spectra suggested that NO synthases, in which the heme is held by a thiolate sulfur of a cysteine side chain, might be specialized cytochromes P450.[531] However, although the heme in NO synthases is bound by a thiolate group the protein fold is unlike that of cytochromes P450.[532,533] The NO synthases all share a three-component structure.[534] In the NOS2 from macrophages residues 1–489 form the catalytic oxygenase domain, residues 499–530 bind calmodulin, and residues 531–1144 form the reductase domain.[535] The last contains a binding site for NADPH as well as bound FMN and FAD. The reductase domain structure can be modeled after that of cytochrome P450 reductase.[533] Electrons from NADPH are apparently transferred to FAD, then to FMN, and then to the catalytic site.[525]

The least understood aspect of NO synthases is the requirement for tetrahydrobiopterin, BH_4, the same coenzyme required by the other pterin-dependent monooxygenases (Eq. 18-44). The presence of this coenzyme in the reduced BH_4 form is essential for step *a* of Eq. 18-65 but not for step *b*. This suggests that in step *a* an organic peroxide might be generated by BH_4 and used to form an oxo-iron hydroxylating reagent.

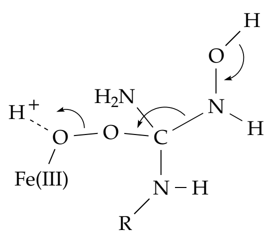


L-Arginine L-N^{γ}-hydroxyarginine L-Citrulline

(18-65)

However, there is no evidence for the expected quinonoid dihydropterin, and the three-dimensional structure suggested that BH_4 plays a structural role in mediating essential conformational changes.[532,535a,b] Nevertheless, newer data indicate a role in electron transfer.[535c]

Step *b* of Eq. 18-65 is an unusual three-electron oxidation, which requires only one electron to be delivered from NADPH by the reductase domain. Hydrogen peroxide can replace O_2 in this step.[536] A good possibility is that a peroxo or superoxide complex of the heme in the Fe(III) state adds to the hydroxyguanidine group. For example, the following structure could arise from addition of Fe(III)–O–O⁻:

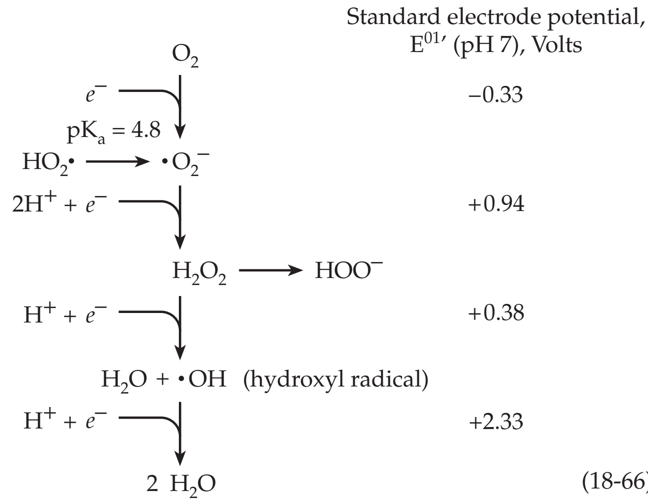


Breakup as indicated by the arrows on this structure would give Fe(III)–OH, citrulline, and O=N–H, **nitroxyl**. This is one electron (e^- + H^+) more reduced than •NO. Perhaps the adduct forms from Fe(III)–O–O•. On the other hand, there is evidence that NO synthases may produce nitroxyl or nitroxyl ion NO^- as the initial product.[537–538] NO and other products such as N_2O and NO_2^- may arise rapidly in subsequent reactions. Nitrite is a major oxidation product of NO in tissues.[538a] The chemistry of NO in biological systems is complex and not yet fully understood. See also pp. 1754, 1755.

G. Biological Effects of Reduced Oxygen Compounds

Although molecular oxygen is essential to the aerobic mode of life, it is toxic at high pressures. Oxidative damage from O_2 appears to be an important cause of aging and also contributes to the development of cancer. Reduced forms of oxygen such as superoxide, hydrogen peroxide, and hydroxyl radicals are apparently involved in this toxicity.[539,540] The same agents are deliberately used by phagocytic cells such as the neutrophils (polymorphonuclear leukocytes) to kill invading bacteria or fungi and to destroy malignant cells.[541]

The reactions shown with vertical arrows in Eq. 18-66 can give rise to the reduced oxygen compounds. The corresponding standard redox potential at pH 7 for each is also given.[539,542–544] As indicated by the low value of the redox potential for the O_2 / $O_2•^-$ couple, the formation of superoxide by reduction of O_2 is spontaneous only for strongly reducing one-electron donors. Superoxide ion is a strong reductant, but at the same time a powerful one-electron oxidant, as is indicated by the high electrode potential of the $O_2•^-$/ H_2O_2 couple.



(18-66)

1. The Respiratory Burst of Neutrophils

Some 25 x 10^9 neutrophils circulate in an individual's blood, and an equal number move along the surfaces of red blood cells. Invading microorganisms are engulfed after they are identified by the immune system as foreign. Phagocytosis is accompanied by a rapid many-fold rise in the rate of oxygen uptake as well as an increased glucose metabolism. One purpose of this **respiratory burst**[545–548] is the production of reduced oxygen compounds that kill the ingested microorganisms. In the very serious **chronic granulomatous disease** the normal respiratory burst does not occur, and bacteria are not killed.[549] The respiratory burst seems to be triggered not by phagocytosis itself, but by stimulation of the neutrophil by chemotactic formylated peptides such as formyl-Met-Leu-Phe[550] and less rapidly by other agonists such as phorbol esters.

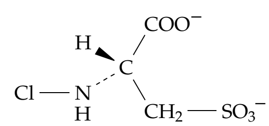
The initial product of the respiratory burst appears to be superoxide ion O_2^-. It is formed by an **NADPH oxidase,** which transports electrons from NADPH to O_2 probably via a flavoprotein and cytochrome b_{558}. Either the flavin or the cytochrome b_{558} must donate one electron to O_2 to form the superoxide anion.

$$\text{NADPH} \rightarrow \text{FAD} \rightarrow \text{cyt } b_{558} \rightarrow O_2 \qquad (18\text{-}67)$$

Flavocytochrome b_{558} (also called b_{-245}) has the unusually low redox potential of –0.245 V. It exists in phagocytic cells as a heterodimer of membrane-associated subunits p22-*phox* and gp91-*phox* where *phox* indicates phagocytic oxidase. The larger 91-kDa subunit contains two heme groups as well as one FAD and the presumed NADPH binding site.[548,551–552a] The mechanism of interaction with O_2 is unclear. Unlike hemoglobin but like other cytochromes *b*, cytochrome b_{558} does not form a complex with CO.[553] NADPH oxidase also requires two cytosolic components p47-*phox* and p67-*phox*. In resting phagocytes they reside in the cytosol as a 240-kDa complex with a third component, p40-*phox*, which may serve as an inhibitor.[548,554] Upon activation of the phagocyte in response to chemotactic signals the cytosolic components undergo phosphorylation at several sites,[554] and protein p47-*phox* and p67-*phox* move to the membrane and bind to and with the assistance of the small G protein Rac[552a,554a] activate flavocytochrome b_{558}. Phosphorylation of p47-*phox* may be especially important.[555]

In the X-chromosome-linked type of chronic granulomatous disease flavocytochrome b_{558} is absent or deficient, usually because of mutation in gp91-*phox*.[556,557] In an autosomal recessive form the superoxide-forming oxidase system is not activated properly. In some patients protein kinase C fails to phosphorylate p47-*phox*.[556,558] Less severe symptoms arise from deficiences in myeloperoxidase, chloroperoxidase (Chapter 16), glucose 6-phosphate dehydrogenase, glutathione synthetase, and glutathione reductase. The importance of these enzymes can be appreciated by examination of Fig. 18-24, which illustrates the relationship of several enzymatic reactions to the formation of superoxide anion and related compounds. Not only neutrophils but monocytes, macrophages, **natural killer cells** (NK cells), and other phagocytes apparently use similar chemistry in attacking ingested cells (Chapter 31).[559] Superoxide-producing NADH oxidases have also been found in nonphagocytic cells in various tissues.[559a]

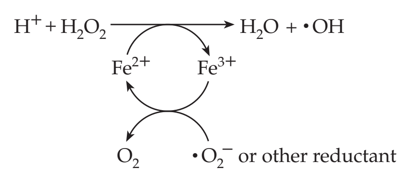
What kills the ingested bacteria and other microorganisms? Although superoxide anion is relatively unreactive, its protonated form HO_2•, is very reactive. Since its pK_a is 4.8, there will be small amounts present even at neutral pH. Some of the $^•O_2^-$ may react with NO to form peroxynitrite (Eq. 18-62).[559b] Peroxynitrite, in turn, can react with the ubitquitous CO_2 to give $^•CO_3^-$ and $^•NO_2$ radicals.[559c] Peroxynitrite anion also reacts with metalloenzyme centers[559d] and causes nitration and oxidation of aromatic residues in proteins.[559d,e] However, neutrophils contain active superoxide dismutases, and most of the superoxide that is formed is converted quickly to O_2 and H_2O_2. The latter may diffuse into the phagosomes as well as into the extracellular space. The H_2O_2 itself is toxic, but longer lived, more toxic oxidants are also formed. Reaction of H_2O_2 with **myeloperoxidase** (Chapter 16) produces hypochlorous acid, **(HOCl;** Eqs. 16-12, Fig. 18-24) and **chloramines** such as NH_2Cl, RNHCl, and $RNCl_2$. An important intracellular chloramine may be that of taurine.



Chloramine formed from taurine

Human neutrophils use HOCl formed by myeloperoxidase to oxidize α-amino acids such as tyrosine to reactive aldehydes that form adducts with –SH, $–NH_2$, imidazole, and other nucleophilic groups.[560] They also contain NO synthases, which form NO, peroxynitrite (Fig. 18-24), and nitrite.[561,562]

Hydroxyl radicals •OH, which attack proteins, nucleic acids, and a large variety of other cellular constituents, may also be formed. Although too reactive to diffuse far, they can be generated from H_2O_2 by Eq. 18-68. This reaction involves catalysis by Fe ions as shown.[562a]



(18-68)

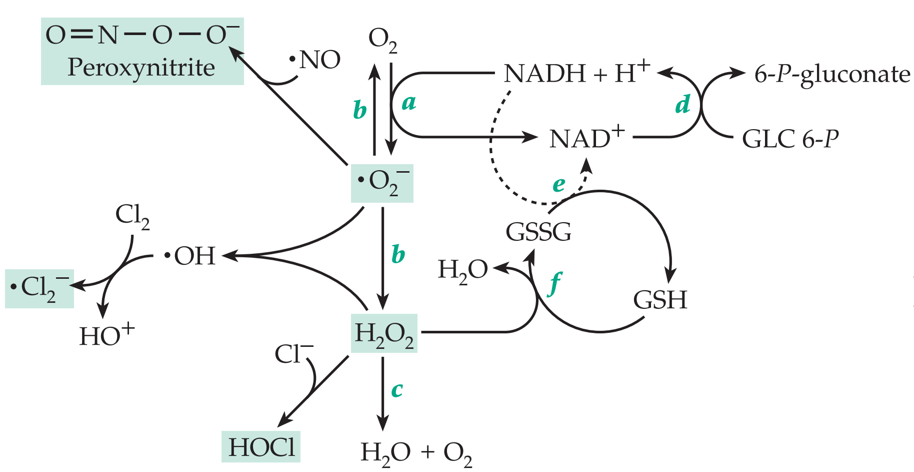


Figure 18-24 Some reactions by which superoxide anions, hydrogen peroxide and related compounds are generated by neutrophils and to a lesser extent by other cells: (*a*) NADPH oxidase, (*b*) superoxide dismutase, (*c*) catalase, (*d*) glucose-6-phosphate dehydrogenase, (*e*) glutathione reductase, (*f*) glutathione peroxidase. Abbrevations: GSH, glutathione; GSSB, oxidized glutathione.

Because Fe^{3+} is present in such low concentrations, there is uncertainty as to the biological significance of this reaction.[563] However, other iron compounds may function in place of Fe^{3+} and Fe^{2+} in Eq. 18-68.[564] A mixture of ferrous salts and H_2O_2 (Fenton reagent) has long been recognized as a powerful oxidizing mixture, which generates •OH or compounds of similar reactivity.[564–567] Ascorbate, and various other compounds, can also serve as the reductant in Eq. 18-68.[568]

Eosinophils, whose presence is stimulated by parasitic infections, have a peroxidase which acts preferentially on Br^- to form HOBr.[569] This compound can react with H_2O_2 more efficiently than does HOCl (Eq. 16-16) to form the very reactive **singlet oxygen.**[570] Singlet oxygen can also be generated from H_2O_2 and $•O_2$ by Eq. 18-69[571] and also photochemically.[572]

$$•O_2^- + H_2O_2 \rightarrow {}^1O_2({}^1\Delta_g) + OH + OH^- \qquad (18\text{-}69)$$

Additional killing mechanisms used by phagocytes include acidification of the phagocytic lysosomes with the aid of a proton pump[573] and formation of toxic peptides. For example, bovine neutrophils produce the bacteriocidal peptide RLCRIVVIRVCR which has a disulfide crosslinkage between the two cysteine residues.[574] Microorganisms have their own defenses against the oxidative attack by phagocytes. Some bacteria have very active superoxide dismutases. The protozoan *Leishmania* produces an acid phosphatase that shuts down the production of superoxide of the host cells in response to activating peptides.[575]

A respiratory burst accompanies fertilization of sea urchin eggs.[576,577] In this case, the burst appears to produce H_2O_2 as the major or sole product and is accompanied by release of **ovoperoxidase** from cortical granules. This enzyme uses H_2O_2 to generate **dityrosine crosslinkages** between tyrosine side chains during formation of the fertilization membrane. Defensive respiratory bursts are also employed by plant cells.[578,579] See also Box 18-B.

2. Oxidative Damage to Tissues

Superoxide anion radicals are formed not only in phagocytes but also as an accidental by-product of the action of many flavoproteins,[580,581] heme enzymes, and other transition metal-containing proteins. An example is xanthine oxidase. It is synthesized as xanthine dehydrogenase which is able to use NAD^+ as an oxidant, but upon aging, some is converted into the $^•O_2$-utilizing xanthine oxidase (Chapter 16). This occurs extensively during ischemia. When oxygen is readmitted to a tissue in which this conversion of xanthine dehydrogenase to xanthine oxidase has occurred, severe oxidative injury may occur.[582] In animals the intravenous administration of superoxide dismutase or pretreatment with the xanthine oxidase inhibitor **allopurinol** (Chapter 25) prevents much of the damage, suggesting that superoxide is the culprit.

Hydrogen peroxide is also generated within cells[583] by flavoproteins and metalloenzymes and by the action of superoxide dismutase on $^•O_2^-$. Since H_2O_2 is a small uncharged molecule, it can diffuse out of cells and into other cells readily. If it reacts with Fe(II), it can be converted within cells to $^•OH$ radicals according to Eq. 18-68. Such radicals and others have been detected upon readmission of oxygen to ischemic animal hearts.[584,585] It has also been suggested that NADH may react with Fe(III) compounds in the same way as does O_2^- in Eq. 18-68 to provide a mechanism for producing hydroxyl radicals from H_2O_2.[539] Nitric oxide, formed by the various NO synthases in the cytosol and in mitochondria[586] and by some cytochromes P450,[587] is almost ubiquitous and can also lead to formation of peroxynitrate (Eq. 18-62). Thus, the whole range of reduced oxygen compounds depicted in Eq. 18-24 are present in small amounts throughout cells.

There is little doubt that these compounds cause extensive damage to DNA, proteins, lipids, and other cell constituents.[539,540,563,588] For example, one base in 150,000 in nuclear DNA is apparently converted from guanine to 8-hydroxyguanine presumably as a result of attack by oxygen radicals.[589] In mitochondrial DNA one base in 8000 undergoes this alteration. This may be a result of the high rate of oxygen metabolism in mitochondria and may also reflect the lack of histones and the relatively inefficient repair of DNA within mitochondria. Proteins undergo chain cleavage, crosslinking, and numerous side chain modification reactions.[588] Dissolved O_2 can react directly with exposed glycyl residues in protein backbones to create glycyl radicals which may lead to chain cleavage as in Eq. 15-39.[588a] Iron–sulfur clusters, such as the Fe_4S_4 center of aconitase (Fig. 13-4), are especially sensitive to attack by superoxide anions.[588a–c] "Free iron" released from the Fe–S cluster may catalyze formation of additional damaging radicals.

Antioxidant systems. Cells have numerous defenses against oxidative damage.[563,590,591] Both within cells and in extracellular fluids superoxide dismutase (Eq. 16-27) decomposes superoxide to O_2 and H_2O_2. The H_2O_2 is then broken down by catalase (Eq. 16-8) to O_2 and H_2O. In higher animals the selenoenzyme glutathione peroxidase (Chapter 16) provides another route for decomposition of H_2O_2 and lipid peroxides of membranes. The oxidized glutathione formed is reduced by NADPH. The system has a critical role within erythrocytes (Box 15-H). In chloroplasts an analogous system utilizes ascorbate peroxidase, ascorbate, and glutathione to break down peroxides.[592]

$$NADPH + H^+ \rightarrow \text{glutathione} \rightarrow \text{ascorbate} \rightarrow H_2O_2 \quad (18\text{-}70)$$

Ascorbate,[593–594] glutathione, NADPH,[594a] and tocopherols (Box 15-G)[595] all act as scavengers of free radicals such as O_2^-, $^{\bullet}OH$ and $ROO^{\bullet}$, $^{\bullet}CO_3^-$, and of singlet oxygen. Antioxidant protection is needed in extracellular fluids as well as within cells. In addition to glutathione and ascorbate, bilirubin,[596] uric acid,[597] melatonin,[598,598a] circulating superoxide dismutase, and the copper protein ceruloplasmin (Chapter 16) all act as antioxidants. Methionine residues of proteins may have a similar function.[599] Various proteins and small chelating compounds such as citrate tie up Fe^{3+} preventing it from promoting radical formation. Tocopherols, ubiquinols, and lipoic acid[600,600a–c] protect membranes. Beta carotene (Fig. 22-5), another lipid-soluble antioxidant, is the most effective quencher of singlet O_2 that is known. Even nitric oxide, usually regarded as toxic, sometimes acts as an antioxidant.[601] Trehalose protects plants against oxidative damage.[601a]

An increasing number of proteins are being recognized as protectants against oxidative damage. The exposed $-SH$ and $-SCH_3$ groups of cysteine and methionine residues in proteins may function as appropriately located scavengers which may donate electrons to destroy free radicals or react with superoxide ions to become sulfonated. The thioredoxin (Box 15-C) and glutathione (Eq. 18-70) systems, in turn, reduce the protein radicals formed in this way.[601b–d] Methionine sulfoxide, both free and in polypeptides, is reduced by **methionine sulfoxide reductase** in organisms from bacteria to humans.[601e–g] Biotin, together with biotin sulfoxide reductase,[601h] may provide another antioxidant system. Some bacteria utilize glutathione-independent **alkylperoxide reductases** to scavenge organic peroxides.[601i] while mammals accomplish the same result with **peroxiredoxins** and with thioredoxin.[601j] Many other proteins will doubtless be found to participate in defense against oxidative damage. Oxygen is always present and its reactions in our bodies are essential. Generation of damaging reduced oxygen compounds and radicals is inevitable. Evolution will select in favor of many proteins that have been modified to minimize the damage.

Antioxidant enzymes do not always protect us. There was great excitement when it was found that victims of a hereditary form of the terrible neurological disease **amyotrophic lateral sclerosis** (**ALS**; see also Chapter 30) carry a defective gene for Cu / Zn-superoxide dismutase (SOD; Eq. 16-27).[602–603b] This discovery seemed to support the idea that superoxide anions in the brain were killing neurons. However, it now appears that in some cases of ALS the defective SOD is *too active*, producing an excess of H_2O_2, which damages neurons.

Transcriptional regulation of antioxidant proteins. Certain proteins with easily accessible Fe–S clusters, e.g., aconitase, are readily inactivated by oxidants such as peroxynitrite.[540,604] At least two proteins of this type function as transcription factors in *E. coli*. These are known as **SoxR** and **OxyR**. The SoxR protein is sensitive to superoxide anion, which carries out a one-electron oxidation on its Fe_2S_2 centers.[540,605–607] In its oxidized form SoxR is a transcriptional regulator that controls 30–40 genes, among them several that are directly related to "**oxidative stress.**"[608] These include genes for manganese SOD, glucose-6-phosphate dehydrogenase, a DNA repair nuclease, and aconitase (to replace the inactivated enzyme). The OxyR protein, which responds to elevated $[H_2O_2]$, is activated upon oxidation of a pair of nearby $-SH$ groups to form a disulfide bridge.[607,609] It controls genes for catalase, glutathione reductase, an alkyl hydroperoxide reductase,[610,610a] and many others. Similar transcriptional controls in yeast result in responses to low doses of H_2O_2 by at least 167 different proteins.[608] Animal mitochondria also participate in sensing oxidant levels.[611,612] (See also Chapter 28, Section C,6.)

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Study Questions

1. Reticulocytes (immature red blood cells) contain mitochondria that are capable of both aerobic and anaerobic oxidation of glucose. In an experiment using these cells, incubated in oxygenated Krebs–Ringer solution with 10 mM glucose, the addition of antimycin A produced the following changes in metabolite concentration after 15 min (From Ghosh, A. K. and Sloviter, H. A. (1973) *J. Biol. Chem.* **248**, 3035–3040). Interpret the observed changes in ATP, ADP, and AMP concentrations (see tabulation). Express the concentration of each component after addition of antimycin as a percentage of that before addition. Then plot the resulting figures for each compound in the sequence found in glycolysis, i.e., label the *X* axis as follows:

G6P F6P FDP TP etc.

Metabolite	Abbreviation	Concentrations (mmol/1 of cells) Before addition of antimycin	After addition of antimycin
Glucose 6-phosphate	G6P	460	124
Fructose 6-phosphate	F6P	150	30
Fructose 1,6-bisphosphate	FBP	8	33
Triose phosphates	TP	18	59
3-Phosphoglycerate	3PGA	45	106
2-Phosphoglycerate	2PGA	26	19
Phosphoenolpyruvate	PEP	46	34
Pyruvate	Pyr	126	315
Lactate	Lac	1125	8750
ATP		2500	1720
ADP		280	855
AMP		36	206

2. The following problem can be solved using standard reduction potentials (Table 6-8). Use $E°'$ (pH 7) values for NAD^+, enzyme-bound FAD, and fumarate of −0.32, 0.0, and −0.03 volts, respectively. Values of numerical constants are given in Table 6-1.

   a) Derive an equation relating the equilibrium constant for a reaction, K_{eq}, to differences in E_0'.

   b) Calculate the numerical values of K_{eq} for the reactions

   $$\text{Succinate} + NAD^+ \rightarrow \text{Fumarate} + NADH + H^+$$

   $$\text{Succinate} + FAD \rightarrow \text{Fumarate} + FADH_2$$

   at pH 7 and 25°C. The values should be calculated for succinate and the oxidant in the numerator.

3. Compare the catalytic cycles of the following enzymes:

   Peroxidase
   Cytochrome *c* oxidase
   Cytochrome P450

4. What chemical properties are especially important for the following compounds in the electron transport complexes of mitochondria?

   FAD or FMN
   Ubiquinone (coenzyme Q)
   Cytochrome *c*

5. Describe the operation of the F_1F_0ATP synthase of mitochondrial membranes.

6. In studies of mitochondrial function the following stoichiometric ratios have been measured.

   a) The P/O ratio: number of molecules of ATP formed for each atom of oxygen (as O_2) taken up by isolated mitochondria under specified conditions.
   b) The ratio of H^+ ions translocated across a mitochondrial inner membrane to the molecules of ATP formed.
   c) The ratio of H^+ ions pumped out of a mitochondrion to the number of molecules of ATP formed.

   Discuss the experimental difficulties in such measurements. How do uncertainties affect conclusions about the mechanism of ATP synthase? Are the ratios in (b) and (c) above necessarily equal? Explain.

7. Compare P/O ratios observed for mitochondrial respiration with the following substrates and conditions:

Study Questions

a) Oxidation of NADH by O_2.
b) Oxidation of succinate by O_2.
c) Dehydrogenation of ascorbate by O_2.

How would the ratio of ATP formed to the number of electrons passing from NADH through the respiratory chain differ for these three oxidants: O_2, fumarate, nitrite?

8. What is the mitochondrial glycerol phosphate shuttle? Is it utilized by plant cells? Explain.

9. What chemical reactions are included in these two important components of the nitrogen cycle (see also Fig. 24-1)?

   Nitrification
   Denitrification

10. What is the difference between a dioxygenase and a monooxygenase? What is meant by a cosubstrate for a monooxygenase?

11. The enzyme *p*-hydroxybenzoate hydroxylase utilizes a cosubstrate together with O_2 to form 3,4-dihydroxybenzoate. Indicate the mechanisms by which the bound FAD cofactor participates in the reaction.

12. What pterin-dependent hydroxylation reactions are important to the human body? Point out similarities and differences between flavin and pterin hydroxylase mechanisms.

13. Describe the basic properties of nitric oxide synthases (NOSs) and their varied functions in the body. What are the three different types of NOS? In what ways do they differ?

14. List several compounds that cause oxidative stress in cells and describe some chemical and physiological characteristics of each.

15. Propylene glycol is metabolized by several aerobic bacteria to acetoacetate, which can be catabolized as an energy source (see references 509a and 509b). The first step is conversion to an epoxide which reacts further in coenzyme M-dependent and CO_2-dependent reactions to form acetoacetate. Can you propose chemical mechanisms?

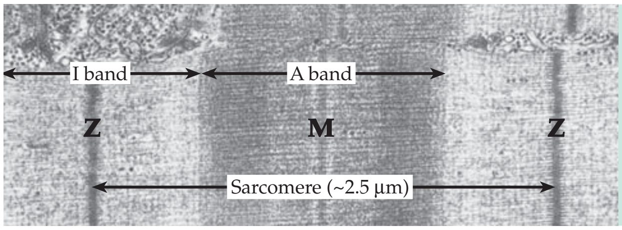
16. A group of slow-growing denitrifying bacteria obtain energy by oxidizing ammonium ions anaerobically with nitrite ions.[613,614]

    $$NO_2^- + NH_4^+ \rightarrow N_2 + 2\,H_2O$$

    Intermediate metabolites are hydroxylamine (H_2NOH) and hydrazine (N_2H_4). The reaction takes place within internal vesicles known as **anammoxosomes**. Unusual cyclobutane- and cyclohexane-based lipids in their membranes are thought to partially prevent the escape of the toxic intermediates from the anammoxosomes.[614]

    Four protons may move from the cytoplasm into the vesicles for each ammonium ion oxidized. Can you write a reaction sequence? What is the Gibbs energy change for the reaction? How is ATP generated? See p. 1052.

Notes



Electron micrograph of a thin longitudinal section of a myofibril from pig muscle. The basic contractile unit is the **sarcomere**, which extends from one Z line to the next. Thin **actin** filaments are anchored at the M lines and the thick **myosin** filaments at the Z lines. The (anisotropic) A bands are regions of overlap of interdigitated thick and thin filaments, while the I ("isotropic") bands are devoid of thick filaments. The ATP-driven contraction of muscle involves sliding of the interdigitated filaments and shortening of the sarcomere to ~1.8 µm. Micrograph courtesy of Marvin Stromer

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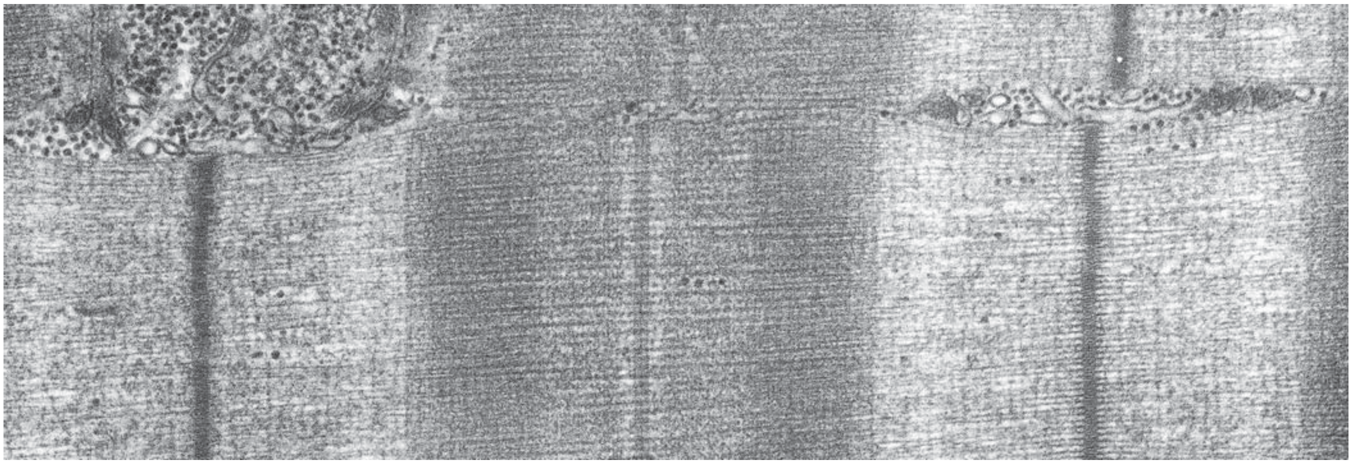
The Chemistry of Movement 19



The swimming of bacteria, the flowing motion of the ameba, the rapid contraction of voluntary muscles, and the slower movements of organelles and cytoplasm within cells all depend upon transduction of chemical energy into mechanical work.

A. Motility of Bacteria

The smallest organs of propulsion are the bacterial flagella (Figs. 1-1, 1-3), and we have been able to unravel some of the mystery of movement by looking at them. When a cell of *E. coli* or *Salmonella* swims smoothly, each flagellum forms a left-handed superhelix with an ~2.3 µm pitch. Rotation of these "propellors" at rates of 100–200 revolutions / s (100–200 Hz) or more[1,2] in a counterclockwise direction, as viewed from the distal end of the flagellum, drives the bacterium forward in a straight line.[3–8] Several flagella rotate side-by-side as a bundle.[4] The observed velocities of 20–60 µm / s are remarkably high in comparison with the dimensions of the bacteria. Also remarkable is the fact that a cell may travel straight for a few seconds, but then tumble aimlessly for about 0.1 s before again moving in a straight line in a different direction. The tumbling occurs when the flagellum reverses its direction of rotation and also changes from a left-handed to a right-handed superhelix, which has just half the previous pitch.

Such behavior raised many questions. What causes reversal of direction of the propellor? Why do the bacteria tumble? How does a bacterium "decide" when to tumble? How is the flagellum changed from a left-handed to a right-handed superhelix? How does this behavior help the bacterium to find food? Most intriguing of all, what kind of motor powers the flagella? The answers are complex, more than 50 genes being needed to specify the proteins required for assembly and operation of the motility system of *E. coli* or *Salmonella typhimurium*.[9]

1. The Structure and Properties of Bacterial Flagella

Twenty or more structural proteins are present from the base to the tip of a complete bacterial flagellum. However, over most of their length the long thin shafts (Figs. 1-1, 19-1) are composed of subunits of single proteins called **flagellins**. Flagellin molecules have a high content of hydrophobic amino acids and, in *Salmonella*, contain one residue of the unusual N^{ε}-methyllysine. The subunits are arranged in a helix of outside diameter ~20 nm in which they also form 11 nearly longitudinal rows or **protofilaments**.[10–12a] Each subunit gives rise to one of the projections seen in the stereoscopic view in Fig. 19-1B. The flagella usually appear under the electron microscope to be supercoiled (Fig 19-1C–E) with a long "wavelength" (pitch) of ~2.5 µm. The supercoiled structure is essential for function, and mutant bacteria with straight flagella are nonmotile. Under some conditions and with some mutant flagellins, straight flagella, of the type shown in Fig. 19-1B, are formed. There is a central hole which is surrounded by what appears to be inner and outer tubes with interconnecting "spokes." However, all of the 494-residue flagellin subunits presumably have identical conformations, and each subunit contributes to both inner and outer tubes as well as to the outer projections. **Basal bodies** (Fig. 19-2) anchor the flagella to the cell wall and plasma membrane and contain the protic motors (Fig. 19-3) that drive the flagella.[14–16]

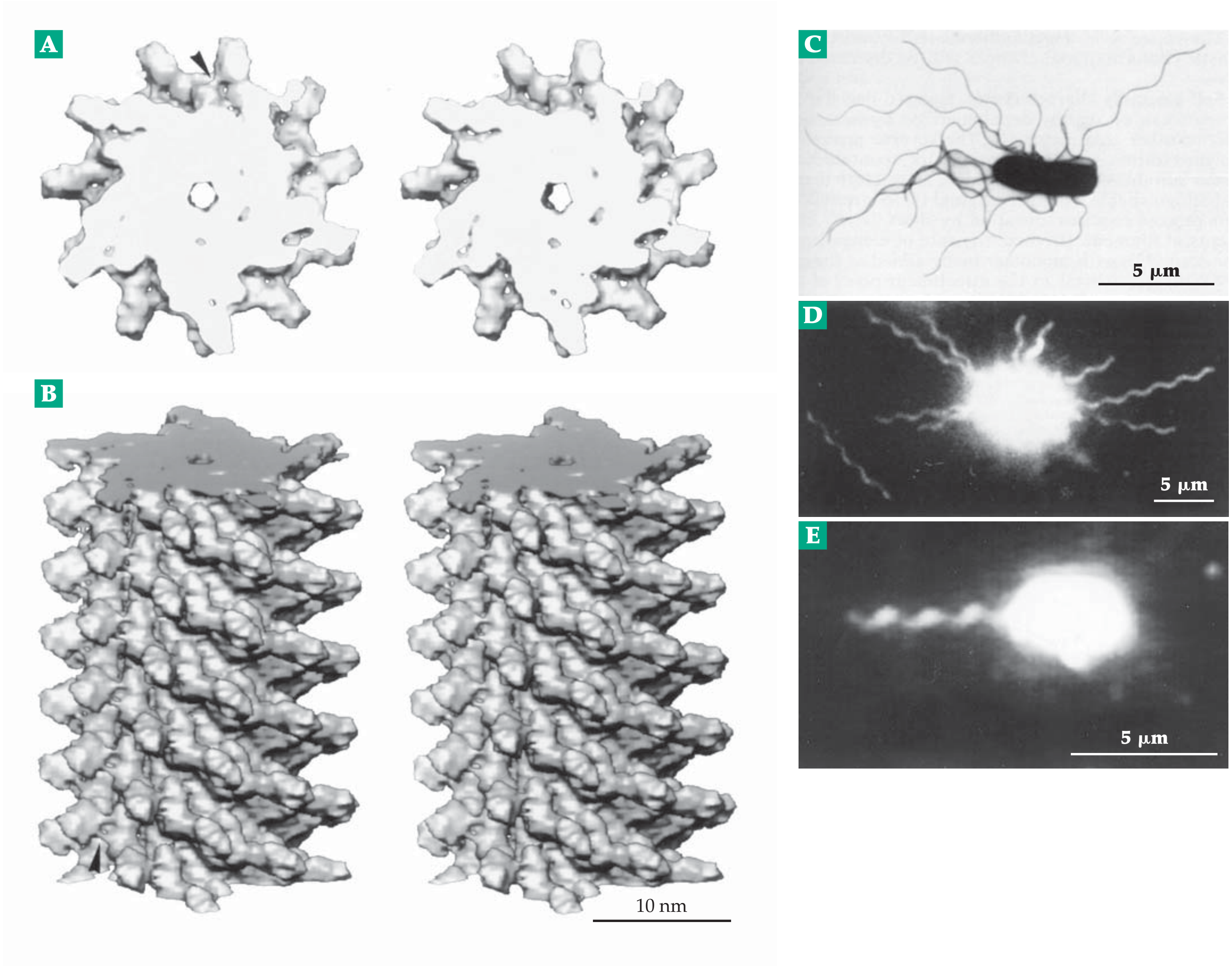


Figure 19-1 (A) Axial view of a 5-nm thick cross-section of the flagellar filament shown in (B). The 11 subunits form two turns of the one-start helix. (B) Stereoscopic oblique view of a 30-nm long section of a flagellum of *Salmonella typhimurium*. This is a straight flagellum from a nonmotile strain of bacteria. The structure was determined to a resolution of 0.9 nm by electron cryomicroscopy. From Mimori *et al.*[11] Courtesy of Keiichi Namba. (C) Electron micrograph of a cell of *S. typhimurium* showing peritrichous (all-around) distribution of flagella. Courtesy of S. Aizawa.[3] (D) Dark-field light micrograph of a flagellated cell of *S. typhimurium* with flagella dispersed during tumbling (see text). Courtesy of R. M. Macnab.[3] (E) Image of a cell of *Vibrio alginolyticus* obtained with dark-field illumination showing the single polar flagellum.[13] Because the illumination was strong, the size of the cell body and the thickness of the flagellum in the image appear large. Courtesy of Michio Homma.

Quasiequivalence. There are two distinct types of straight flagella: one (R) in which the protofilaments have a right-handed twist (as in Fig. 19-1) and the other (L) in which the protofilaments have a left-handed twist. These arise from two different conformations of the subunit proteins. Native supercoiled flagella contain a mixture of flagellins in the R- and L-states with all subunits in a given protofilament being in the same state. The supercoiling of the filament cannot be explained by stacking of identical subunits but is thought to arise because of an asymmetric distribution of protofilaments in a given state around the filament.[17–19a] Here, as with the icosahedral viruses (Chapter 7), quasiequivalence permits formation of a structure that would be impossible with full equivalence of subunits. The corkscrew shape of the flagellum is essential to the conversion of the motor's torque into a forward thrust.[18] Certain mutants of *Salmonella* have "curly" flagella with a superhelix of one-half the normal pitch. The presence of *p*-fluorophenylalanine in the growth medium also produces curly flagella, and normal flagella can be transformed to curly ones by a suitable change of pH. More important for biological function, the transformation from normal to curly also appears to take place during the tumbling of bacteria associated with chemotaxis.[17]

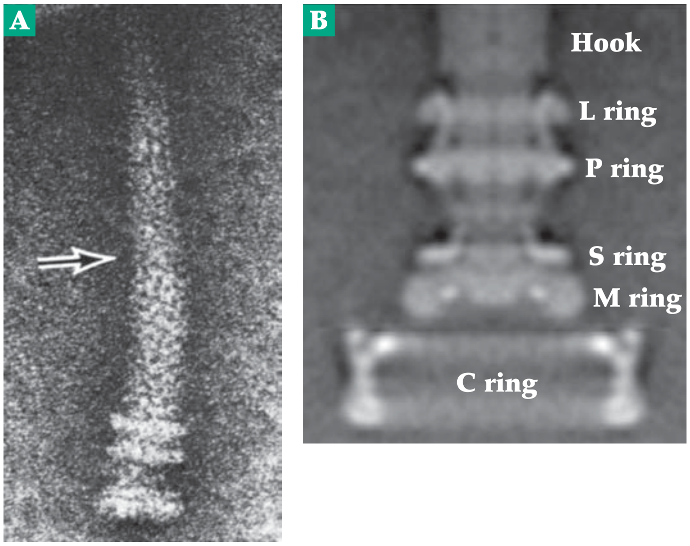


Figure 19-2 (A) Electron micrograph of a flagellum from *E. coli* stained with uranyl acetate. The M-and S-rings are seen at the end. Above them are the P-ring, thought to connect to the peptidoglycan layer, and the L-ring, thought to connect to the outer membrane or lipopolysaccharide layer (see Fig. 8-28). An arrow marks the junction between hook and thinner filament. From DePamphilis and Adler.[14] The hook is often bent to form an elbow. (B) Average of ~100 electron micrographs of frozen-hydrated preparations of basal bodies showing the cytoplasmic C-ring (see Fig. 19-3) extending from the thickened M-ring. From DeRosier.[16]

Growth of flagella. Iino added *p*-fluorophenylalanine to a suspension of bacteria, whose flagella had been broken off at various distances from the body.[20] Curly ends appeared as the flagella grew out. Unlike the growth of hairs on our bodies, the flagella grew from the outer ends. Because no free flagellin was found in the surrounding medium, it was concluded that the flagellin monomers are synthesized within the bacterium, then pass out, perhaps in a partially unfolded form, through the 2- to 3-nm diameter hole[10,12] in the flagella, and bind at the ends.[21] Flagella of *Salmonella* grow at the rate of 1 µm in 2–3 min initially, then more slowly until they attain a length of ~15 µm. More recent studies have provided details. The hook region (Fig. 19-3) grows first to a length of ~55 nm by addition to the basal-body rod of ~140 subunits of protein **FlgE**. During growth a **hook cap** formed from subunits **FlgD** prevents the FlgE subunits from passing out into the medium.[22,23] Hook subunits are added beneath the cap, moving the cap outward. Hook growth is terminated by protein **FlgK** (also called hook-associated protein Hap1). This protein displaces the hook cap and initiates growth of the main filament.[24] The first 10–20 subunits added are those of the FlgK (Fig. 19-3). These are followed by 10–20 subunits of **FlgL** (Hap3), a modified flagellin whose mechanical properties can accomodate the stress induced in the flagella by their rotation.[25] FlgJ is also needed for rod formation.[25a]

Growth of the flagellum to a length of up to 20 µm continues with subunits of **FliC** that are added at the tip, which is now covered by a dodecamer of the **cap protein FliD** (HAP2).[24,26,26a,b] Its 5-fold rotational symmetry means that this "star-cap" does not form a perfect plug for the 11-fold screw-symmetry of the flagellum, a fact that may be important in allowing new flagellin subunits to add at the growing tip. If the cap protein is missing, as in some *FliD* mutants, a large amount of flagellin leaks into the medium.[24]

Still unclear is how the protein synthesis that is taking place on the ribosomes in the bacterial cytoplasm is controlled and linked to "export machinery" at the base of the flagellin. As indicated in Fig. 19-3, the genetically identified proteins FlhA, FliH, and FliI are involved in the process that sends the correct flagellin subunits through the growing flagellum at the appropriate time. FliI contains an ATPase domain.[26c] FliS protein may be an export chaperone.[26d]

2. Rotation of Flagella

A variety of experiments showed that the flagellum is a rigid propellor that is rotated by a "motor" at the base. For example, a bacterium, artificially linked by means of antibodies to a short stub of a flagellum of another bacterium, can be rotated by the second bacterium. Rotation of cells tethered to a cover slip has also been observed. Although it is impossible to see individual flagella on live bacteria directly, bundles of flagella and even single filaments (Fig. 19-1C) can be viewed by dark-field light microscopy.[8,29] Normal flagella appear to have a left-handed helical form, but curly *Salmonella* flagella, which have a superhelix of one-half the normal pitch, form a right-handed helix.[5] Normal bacteria swim in straight lines but periodically "tumble" before swimming in a new random direction. This behavior is part of the system of **chemotaxis** by which the organism moves toward a food supply.[30] Curly mutants tumble continuously. When bacteria tumble the flagella change from normal to curly. The pitch is reversed and shortened. A proposed mechanism for the change of pitch involves propagation of cooperative conformational changes down additional

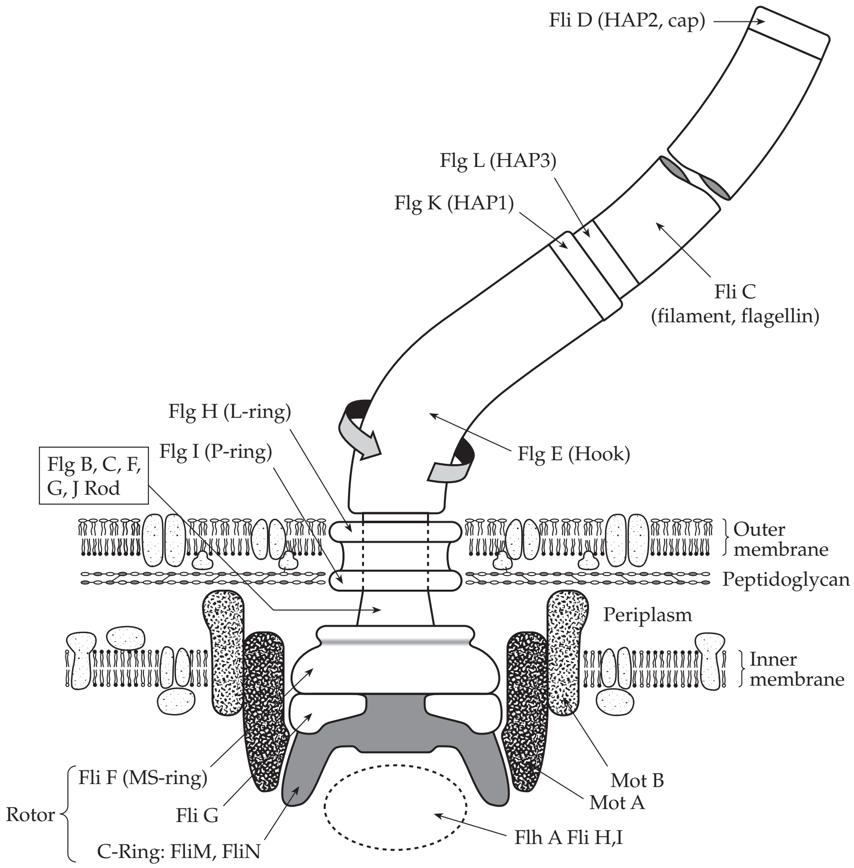


Figure 19-3 Schematic drawing of bacterial flagellar motor. Based on drawings of Berg,[27] Zhou and Blair,[28] and Elston and Oster.[1]

rows of flagellin subunits.[31]

There are no muscle-type proteins in the flagella. By incubating flagellated bacteria with penicillin and then lysing them osmotically, Eisenbach and Adler obtained cell envelopes whose flagella would rotate in a counterclockwise fashion if a suitable artificial electron donor was added.[32] This and other evidence showed that ATP is not needed. Rather, the torque developed is proportional to the **protonmotive force** and, under some circumstances, to ΔpH alone. It is the flow of protons from the external medium into the cytoplasm that drives the flagella.[8] Movement of *E. coli* cells in a capillary tube can also be powered by an external voltage.[33] In alkalophilic strains of *Bacillus* and some *Vibrio* species a sodium ion gradient will substitute.[13] Several hundred protons or Na^+ ions must pass through the motor per revolution.[8] Some estimates, based on energy balance,[29] are over 1000. However, Na^+-dependent rotation at velocities of up to 1700 Hz has been reported for the polar flagellum of *Vibrio alginolyticus*. It is difficult to understand how the bacterium could support the flow of 1000 Na^+ per revolution to drive the flagellum.[2]

What kind of protic motor can be imagined for bacterial flagella? Electron microscopy reveals that the flagellar hook is attached to a rod that passes through the cell wall and is, in turn, attached to a thin disc, the **M-ring** (or MS-ring), which is embedded in the cytoplasmic membrane both for gram-positive and gram-negative bacteria (Fig. 19-3). Two additional rings are present above the M-ring of flagella from gram-negative bacteria. The P-ring interacts with the peptidoglycan layer, and the L-ring contacts the outer membrane (lipopolysaccharide; Fig. 19-3). A logical possibility is that the M-ring, which lies within the plasma membrane, is the rotor, and a ring of surrounding protein subunits is the stator for the motor (Fig. 19-4).[34,35] Glagolev and Skulachev suggested in 1978 that attraction between $-COO^-$ and $-NH_3^+$ groups provides the force for movement.[34] Protons passing down an H^+-conducting pathway from the outer surface could convert $-NH_2$ groups to $-NH_3^+$, which would then be attracted to the $-COO^-$ groups on the stator subunits. When these two oppositely charged groups meet, a proton could be transferred from $-NH_3^+$ to $-COO^-$ destroying the electrostatic attraction. At the same time, movement of the M-ring would bring the next $-NH_2$ group to the H^+-conducting pathway from the outside. The $-COOH$ of the stator would now lose its proton through a conducting pathway to the inside of the bacterium, the proximity of the new $-NH_3^+$ assisting in this proton release. Since that time, other models based on electrostatic interactions have been advanced.[1,29,36]

Approximately 40 genes are required for assembly of the flagella, but mutations in only five motility genes have produced bacteria with intact flagella that do not rotate. Among these genes are *motA*, *mot B*, *FliG*, *FliM*, and *FliN*.[16,29,37,37a] Infection with a lambda transducing bacteriophage carrying functional *motB* genes restores motility to *motB* mutants by inducing synthesis of the motB protein. Block and Berg observed rotation of single bacteria tethered to a coverslip by their flagella. As the synthesis of the motB protein increased,the flagellar rotation rate increased in as many as 16 steps. This suggested that as many as 16 subunits of the *motB* gene product may contribute to the operation of the motor.[38] Later studies suggest eight subunits[39] rather than 16.

Both the M-ring and the thin S-ring, which lies directly above it and is now usually referred to as the MS-ring, are formed from ~20–25 subunits of the 61-kDa **FliF** protein.[39] Both the MotA and MotB proteins are embedded in the inner bacterial membrane and appear to form a circular array of "studs" around the M-ring.[16] MotA has a large cytosolic domain as well as four predicted trans-membrane helices[40] while MotB has a large periplasmic domain and probably binds to the peptidoglycan.[37,41,41a] The MotA and MotB proteins, which bind to each other, are thought to form the ~8 functional units in the stator of the motor.[37] Proteins FliG, FliM, and FliN are evidently parts of the rotor assembly. FliM and FliN form an additional ring, the cytoplasmic or **C-ring**, which had been difficult to see in early electron microscopy. As many as 40 of each of these subunits may be present in the ring.[42,43] A ring of FliG subunits joins the C-ring to the MS-ring (Fig. 19-3). FliE is also a part of the basal body.[25a]

From study of mutants it has been concluded that three charged residues of FliG, R279, D286, and D287 are directly involved in generation of torque by the motor.[44] Side chains of these residues may interact with the cytoplasmic domains of MotA and MotB. Residues R90 and E98 of MotA may be involved in controlling proton flow through the motor units.[28,44] The two prolines P173 and P122 are also essential for torque generation.[28]

There are obvious similarities between the flagellar motors and the protic turbines of ATP synthases (Fig. 18-14), but there are also substantial differences. It apparently takes about 12 protons for one revolution of the ATP synthase but about 1000, or ~125 per motor unit, for rotation of a bacterial flagellum. Elston and Oster propose an ion turbine more complex than that of ATP synthase. They suggest that the rotor might contain about 60 slanted rows of positively charged groups spaced as shown in Fig. 19-4. The motor is reversible, i.e., it can rotate in either direction. One possibility is that the subunits alter their conformations cooperatively in such a way that the slant of the rows of charged groups is reversed. Other possibilities for altering the constellation of charges via conformational changes can be imagined.[1] See also Thomas *et al.*[44a]

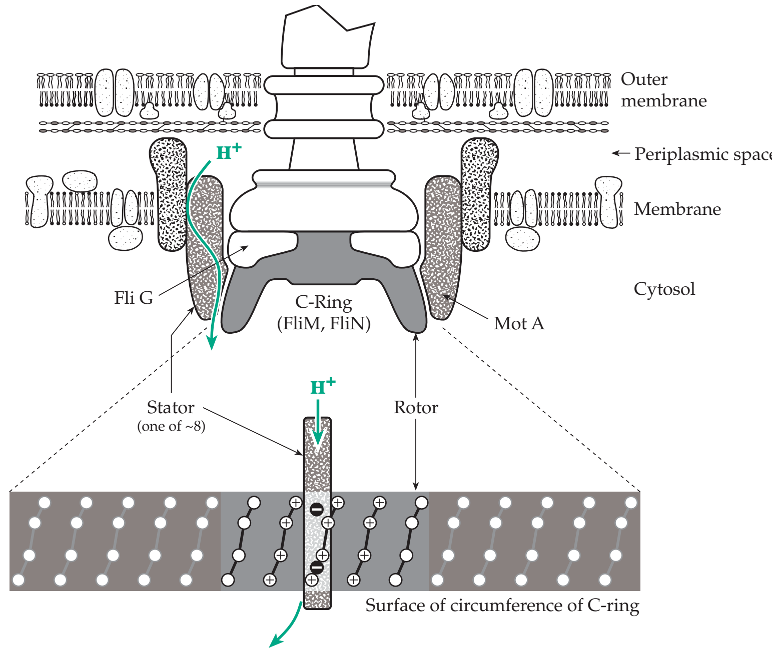


Figure 19-4 Schematic drawing of a hypothetical configuration of rotor and one stator unit in a flagellar motor as proposed by Elston and Oster.[1] The rotor can hold up to 60 positive charges provided by protons flowing from the periplasm through the stator motor units that surround the C-ring and hopping from one site to the next along the slanted lines. The rotor is composed of 15 repeating units, each able to accommodate four protons. Negative charges on the stator units are 0.5 nm from the rotor charges at their closest approach. For details see the original paper.

3. Chemotaxis

The flagellar motor is reversible, and in response to some signal from the bacterium it will turn in the opposite direction. At the same time, the flagellin subunits and those of the hook undergo conformational changes that change the superhelical twist. Perhaps synchronous conformational changes in the M-ring also are associated with the change in direction of rotation and are induced by interaction with a **switch complex** that lies below the M-ring. This consists of proteins FliG, FliM, and FliN.[44b] Mutations in any one of these proteins lead to the following four phenotypes: absence of flagella, paralyzed flagella, or flagella with the switch biased toward clockwise or toward counterclockwise rotation.[45]

What signals a change in direction of rotation? The answer lies in the attraction of bacteria to compounds that they can metabolize. Bacteria will swim toward such compounds but away from repellent substances, a response known as **chemotaxis**. Cells of *E. coli* swim toward higher concentrations of L-serine (but not of D-serine), of L-aspartate, or of D-ribose.

Phenol and Ni^{2+} ions are repellent.[46–48] By what mechanism can a minuscule prokaryotic cell sense a concentration gradient? It is known that the plasma membrane contains receptor proteins, whose response is linked to control of the flagella. Since the dimensions of a bacterium are so small, it would probably be impossible for them to sense the difference in concentration between one end and the other end of the cell. The chemotatic response apparently results from the fact that a bacterium swims for a relatively long time without tumbling when it senses that the concentration of the attractant is increasing with *time*. When it swims in the opposite direction and the concentration of attractant decreases, it tumbles sooner.[49]

Koshland[47] proposed that as the membrane receptors become increasingly occupied with the attractant molecule, the rate of formation v_f of some compound X, within the membrane or within the bacterium, is increased (Eq. 19-1). When [X] rises higher than a threshold level, tumbling is induced. At the same time, X is destroyed at a velocity of v_d.

$$\xrightarrow{v_f} [X] \xrightarrow{v_d} \qquad (19\text{-}1)$$

Subsequently, a readjustment of v_f and v_d occurs such that the concentration of X falls to its normal steady state level. X would act directly on the flagellar motor.

The receptors for L-serine[50–51a] and L-aspartate[52,53] are 60-kDa proteins encoded by genes *tsr* and *tar* in *Salmonella* or *E. coli*.[46,54] These proteins span the inner plasma membrane of the bacteria as shown in Figs. 11-8 and 19-5. The functioning of the receptor has been discussed in some detail in Chapter 11. However, there is still much that is not understood. The symmetric head, whose structure is known (Fig. 11-8), has two binding sites, but the aspartate receptor binds only one aspartate tightly. There is substantial evidence that suggests a piston-type sliding of one helix toward the cytoplasm as part of the signaling mechanism.[54a] While the flagella are distributed around the cell, the receptors appear to be clustered at the cell poles.[55]

Proteins encoded by genes *cheA*, *cheW*, *cheY* and *cheZ*, *cheB*, and *ChR* are all involved in controlling chemotaxis.[48,56] Their functions are indicated in the scheme of Fig. 19-5. All of the corresponding protein products have been isolated and purified, and the whole chemotaxis system has been reconstituted in phospholipid vesicles.[57] Gene *CheA* encodes a 73-kDa protein kinase, which binds as a dimer to the cytoplasmic domains of the related aspartate, serine, and ribose/galactose receptors with the aid of a coupling protein, cheW (Fig. 19-5). A great deal of effort has been expended in trying to understand how binding of an attractant molecule to the periplasmic domain of the receptor can affect the activity of the CheA kinase, but the explanation is unclear. There is a consensus that a small but distinct conformational alteration is transmitted through the receptor.[58–61a] An apparently α-helical region containing methylation sites (Fig. 19-5) appears to be critically involved in the signaling, responding not only to occupancy of the receptor site but also to intracellular pH and temperature and to methylation. Mutation of the buried Gly 278 found in this region to branched hydrophobic amino acids, such as Val or leucine, locks the receptor in state with a superactivated CheA kinase, while *substitution* of Gly 278 with aspartate leaves the kinase *inactive*.[61] Occupancy of the normal receptor site with ligand (aspartate, serine, etc.,) dramatically decreases the kinase activity.

The CheA protein is an autokinase which, upon activation by the receptor, becomes phosphorylated on N^{ε} of the imidazole ring of His 48. It then transfers this phospho group from His 48 to the carboxylate of Asp 57 of the 654-residue protein **CheY**, which is known as the **response regulator**.[62–65d] The unregulated flagellum rotates counterclockwise (CCW). *Phospho-CheY* (CheY-*P*, which qualifies as X in Eq. 19-1) *carries the message to the flagellar motor to turn clockwise (CW).* This is apparently accomplished through the binding of CheY-*P* to the N-terminal portion of protein FliM. This presumably induces a conformational change, which is propagated to FliG and to all of the proteins of the rotor and flagellar rod, hook, and filament.[45,65,66,66a] The flagella fly apart, and the bacterium tumbles and heads randomly in a new direction.

Tumbling occurs most often when receptors are unoccupied, and the bacteria change directions often, as if lost. However, if a receptor is occupied by an attractant, the activity of CheY is decreased and less CheY-*P* will be made. The carboxyl phosphate linkage in this compound is labile and readily hydrolyzed, a process hastened by the phosphatase **CheZ**.[67–69] Consequently, in the presence of a high enough attractant concentration the tumbling frequency is decreased, CCW flagellar rotation occurs, and the bacterium swims smoothly for a relatively long time.

There are still other important factors. Occupancy of the receptor by a ligand makes the receptor protein itself a substrate for the chemotaxis-specific methyltransferase encoded by the *cheR* gene.[62,70,71] This enzyme transfers methyl groups from *S*-adenosylmethionine to specific glutamate side chains of the receptor to form methyl esters. In the aspartate receptor there are four such glutamate residues in a large cytoplasmic domain that includes the C terminus. Two of these glutamates are initially glutamines and can undergo methylation only if they are deaminated first.[72] An esterase encoded by the *cheB* gene[72] removes the methyl ester groupings as methanol.

The action of the CheR methyltransferase is apparently unregulated, but the esterase activity of CheB is controlled by the phosphorylation state of the

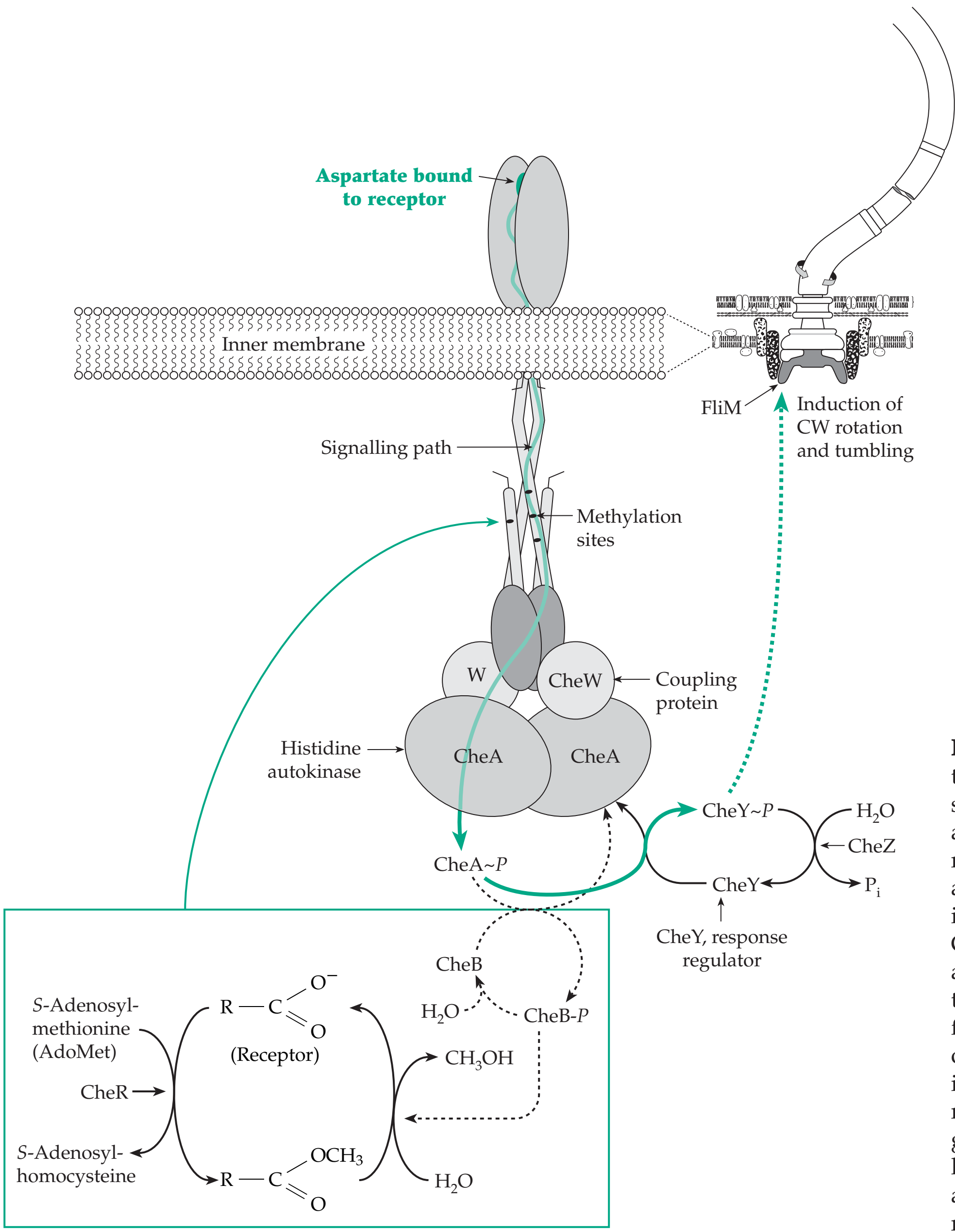


Figure 19-5 Schematic representation of an important chemotactic system of *E. coli*, *S. typhimurium*, and other bacteria. The transmembrane receptor activates the autokinase CheA, which transfers its phospho group to proteins CheY and CheB to form CheY-*P* and CheB-*P*. CheY-*P* regulates the direction of rotation of the flagella, which are distributed over the bacterial surface. CheR is a methyltransferase which methylates glutamate carboxyl groups in the receptor and modulates the CheA activity. CheZ is a phosphatase and CheB-*P* a methylesterase.

autokinase CheA. CheB competes with CheY (Fig. 19-5), and CheB-*P* is the active form of the esterase. After a chemotactic stimulus the level of CheA-*P* falls and so does the activity of the methylesterase. The number of methyl groups per receptor rises making the CheA kinase more active and opposing the decrease in kinase activity caused by receptor occupancy. The system is now less sensitive to the attractant; the bacterium has *adapted* to a higher attractant concentration.[62,73,73a] It tumbles more often unless the attractant concentration rises; if it is headed toward food tumbling is still inhibited. If it is headed away from the attractant the levels of both CheY-*P* and ChB-*P* rise. A high level of fumarate within the cell also acts on the switch–motor complex and favors CW rotation.[74]

For some bacterial attractants such as D-galactose, D-ribose, maltose, and dipeptides[75] the corresponding binding proteins,[38,76] which are required for the sugar uptake (e.g., Fig. 4-18A), are also necessary for chemotaxis. The occupied binding proteins apparently react with membrane-bound receptors to trigger the chemotactic response. The aspartate receptor (*tar* gene product) appears also to be the receptor for the maltose-binding protein complex,[47] and both the aspartate and the serine receptor (*tsr* gene product) also mediate thermotaxis and pH taxis.[77,77a] Clusters of identical receptors may function cooperatively to provide high sensitivity and dynamic range.[77b]

B. Muscle

There is probably no biological phenomenon that has excited more interest among biochemists than the movement caused by the contractile fibers of muscles. Unlike the motion of bacterial flagella, the movement of muscle is directly dependent on the hydrolysis of ATP as its source of energy. Several types of muscle exist within our bodies. **Striated** (striped) **skeletal muscles** act under voluntary control. Closely related are the **involuntary striated heart muscles,** while **smooth involuntary muscles** constitute a third type. Further distinctions are made between fast-twitch and slow-twitch **fibers**. **Fast-twitch fibers** have short isometric contraction times, high maximal velocities for shortening, and high rates of ATP hydrolysis. They occur predominately in white muscle. Because of the absence of the strong oxidative metabolism found in red muscles, fast-twitch fibers fatigue rapidly. Although red muscle sometimes contains fast-twitch fibers, it more often consists of **slow-twitch fibers,** which have a longer contraction time, low shortening velocity, and low ATPase. They are more resistant to fatigue[56] than fast-twitch fibers.[78] Embryonic muscle contains fast-twitch fibers as well as embryonic forms which contract slowly.[79] Some organisms contain specialized types of muscle. For example, the asynchronous flight muscles of certain insects cause the wings to beat at rates of 100–1000 Hz. In these muscles nerve impulses are used only to start and to stop the action; otherwise the cycle of contraction and relaxation continues automatically.[80] The adductor muscles, which close the shells of oysters and clams, can sustain large tensions for long periods of time with little expenditure of energy. This is accomplished by a **catch mechanism**.[81]

1. The Structural Organization of Striated Muscle

Skeletal muscles consist of bundles of long **muscle fibers,** which are *single cells* of diameter 10–100 µm formed by the fusion of many embryonic cells. The lengths are typically 2–3 cm in mammals but may sometimes be as great as 50 cm. Each fiber contains up to 100–200 nuclei. Typical cell organelles are present but are often given special names. Thus, the plasma membrane (plasmalemma) of muscle fibers is called the **sarcolemma**. The cytoplasm is **sarcoplasm,** and mitochondria may be called **sarcosomes**. The major characteristic of muscle is the presence of the contractile **myofibrils**, organized bundles of proteins 1–2 µm in diameter and not separated by membranes from the cytoplasm. Since they occupy most of the cytoplasm, a substantial number of myofibrils are present in each muscle fiber.

In the light microscope cross striations with a repeating distance of ~2.5 µm can be seen in the myofibrils (Figs. 19-6 and 19-7). The space between two of the dense **Z-discs** (Z lines) defines the **sarcomere**, the basic contractile unit. In the center of the sarcomere is a dense **A-band** (anisotropic band). The name refers to the intense birefringence of the band when viewed with plane polarized light. Straddling the Z-discs are less dense **I-bands** (the abbreviation stands for isotropic, a misnomer, for although the bands lack birefringence, they are not isotropic). Weakly staining **M-lines** (usually visible only with an electron microscope) mark the centers of the A-bands and of the sarcomeres.

The fine structure of the sarcomere was a mystery until 1953, when H. E. Huxley, examining thin sections of skeletal muscle with the electron microscope, discovered a remarkably regular array of interdigitated protein filaments.[82,83] **Thick filaments**, 12–16 nm in diameter and ~1.6 µm long, are packed in a hexagonal array on 40- to 50-nm centers throughout the A-bands (Fig. 19-6B). Between these thick filaments are **thin filaments** only 8 nm in diameter and extending from the Z-line for a length of ~1.0 µm. When contracted muscle was examined, it was found that the I-bands had become so thin that they had nearly disappeared and that the amount of overlap between the thick and the thin filaments had increased. This indicated that contraction had consisted of the sliding movement of the thick and thin filaments with respect to each other.[84] In skeletal muscle the sarcomere shortens to a length of ~2 µm, but in insect flight muscle a much smaller shortening occurs repetitively at a very high rate.

2. Muscle Proteins and Their Structures

The myofibrillar proteins make up 50–60% of the total protein of muscle cells. Insoluble at low ionic strengths, these proteins dissolve when the ionic strength exceeds ~0.3 and can be extracted with salt solutions. Analysis of isolated mammalian myofibrils[86] shows that nine proteins account for 96% or more of the protein; **myosin**, which constitutes the bulk of the thick filaments, accounts for 43% and **actin**, the principal component of the thin filaments, 22%.

Actin and the thin filaments. There are at least six forms of actin in adult mammalian tissues: α-cardiac, α-skeletal muscle, α- and γ-smooth muscle, β- and γ-cytoplasmic.[87–89] All of them are closely homologous, e.g., the 42-kDa α-skeletal muscle actin differs in only 4 of 375 residues from the α-cardiac form and only in 6 residues from the γ-smooth form. In almost all organisms actins contain one residue of N^{δ}-methylhistidine at position 73.[87,88,90] Actin is an unusual protein in that

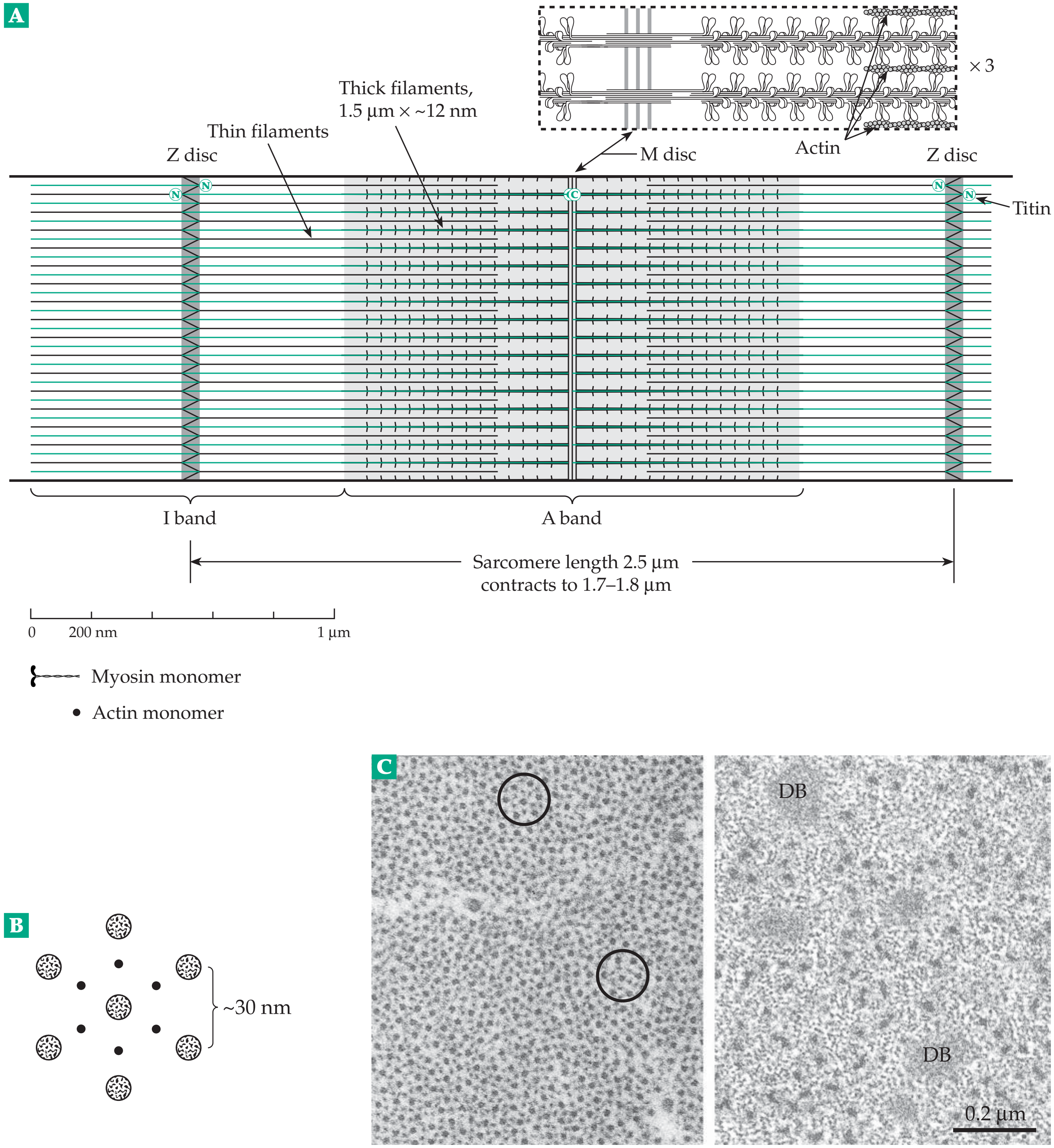


Figure 19-6 (A) The structure of a typical sarcomere of skeletal muscle. The longitudinal section depicted corresponds to that of the electron micrograph, Fig. 19-7A. The titin molecules in their probable positions are colored green. The heads of only a fraction of the myosin molecules are shown protruding toward the thin actin filaments with which they interact. A magnified section at the top is after Spudich.[85] It shows the interactions of the myosin heads with the thin filaments at the right-hand edge. (B) A sketch showing the arrangement of thick and thin filaments as seen in a transverse section of a striated muscle fiber. (C) Left: electron micrograph of a transverse section of a glycerated rabbit psoas muscle. The hexagonal arrangement of six thin filaments around one thick filament can be seen in the center of the circle. Six other thick filaments form a larger concentric circle as in (B). Right: transverse section of a smooth muscle fiber. Notice the irregular arrangement of thick and thin filaments. Filaments of intermediate diameter are also present, as are dense bodies (DB). The latter are characteristically present in smooth muscle.

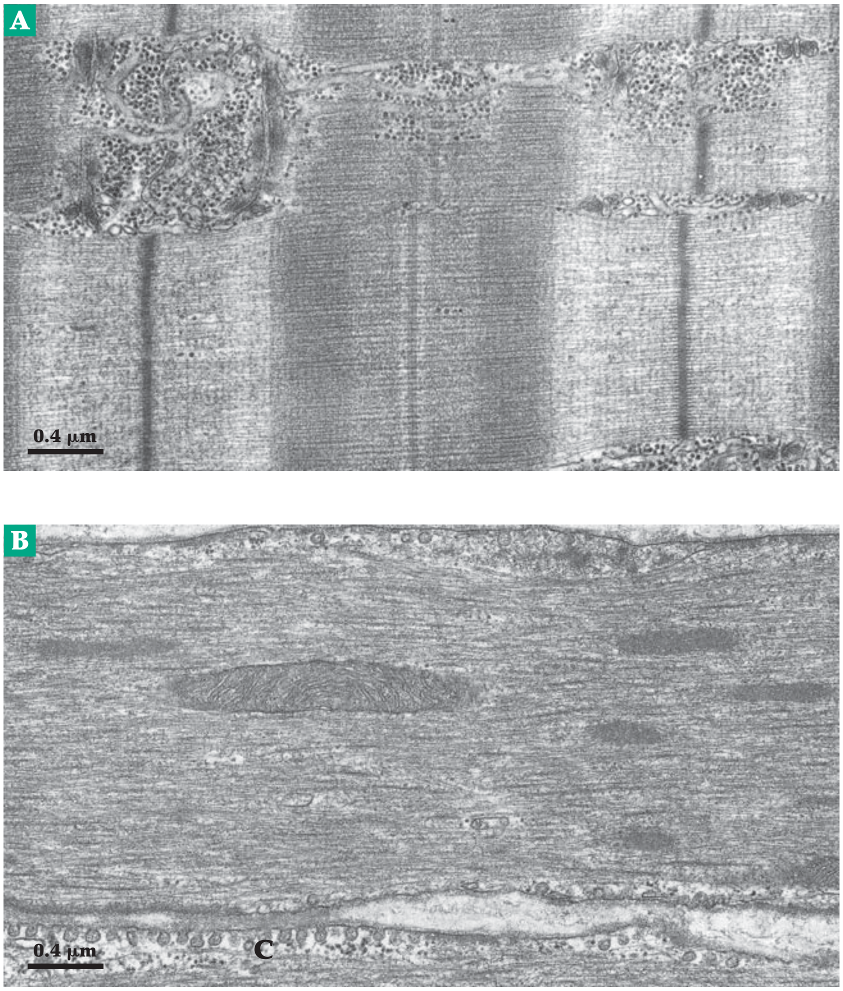


Figure 19-7 (A) Electron micrograph of a longitudinal section of a mammalian skeletal muscle (pig biceps muscle). The tissue was doubly fixed, first with formaldehyde and glutaraldehyde, then with osmium tetroxide. It was then stained with uranyl acetate and lead citrate. The section shows a white muscle fiber containing few mitochondria and narrow Z-lines. The Z-discs (marked Z), M-line, A- and I-bands, and thick and thin filaments can all be seen clearly. The periodicity of ~40 nm along the thin filaments corresponds to the length of the tropomyosin molecules, and the cross striation is thought to represent bound tropomyosin and troponin. The numerous dense particles in the upper part of the micrograph are glycogen granules, while the horizontal membranous structures are longitudinal tubules of the sarcoplasmic reticulum (endoplasmic reticulum). These come into close apposition to the T tubules leading from the surface of the muscle fiber. The T tubules (T) are visible in longitudinal section at the upper left of the micrograph on both sides of the Z-line and in cross-section in the upper right-hand corner. There a T tubule is seen lying between two lateral cisternae of the sarcoplasmic reticulum. (B) Longitudinal section of smooth muscle (chicken gizzard) fixed as in A. Thick filaments (Th), which are considerably thicker than those in striated muscle and less regular, can be seen throughout the section. They are surrounded by many thin filaments, which are often joined to dense bodies (DB). A mitochondrion (Mi) is seen in the center of the micrograph, and at the lower edge is a boundary between two adjacent cells. Notice the caveolae (C), which are present in large numbers in the plasma membrane and which are extremely active in smooth muscle. Micrographs courtesy of Marvin Stromer.

it can exist in both a filamentous and a soluble state. The interconversion between them is of great physiological importance. Actin filaments dissolve in a low ionic strength medium containing ATP to give the soluble, monomeric **G-actin**. Each G-actin monomer usually contains one molecule of bound ATP and a calcium ion.

Because of its tendency to polymerize, G-actin has been difficult to crystallize. However, it forms crystalline complexes with several other proteins, e.g., deoxyribonuclease I,[91] a fragment of gelsolin, and profilin,[92] which block polymerization and it has recently been crystallized as the free ADP complex.[92a] The three-dimensional structure of the actin is nearly the same in all cases. The molecule folds into four domains, the ATP binding site being buried in a deep cleft. The atomic structure (Fig. 7-10) resembles that of hexokinase, of glycerol kinase, and of an ATP-binding domain of a chaperonin of the Hsp 70 family.[90] As with the kinases, actin can exist in a closed and more open conformations, one of which is seen in the profilin complex. Addition of 1 mM Mg^{2+} or 0.1 M KCl to a solution of G-actin leads to spontaneous transformation into filaments of **F-actin** (Figs. 7-10 and 19-9) each containing 340–380 actin monomers and resembling the thin filaments of muscle.[93-94a] The ATP bound in the F-actin filament is hydrolyzed within ~100 s to ADP and P_i. However, the hydrolysis is not as rapid as polymerization so that a "cap" of ATP-containing monomers may be found at each end of the filament.[94,95,95a] There is a striking similarity to the binding of nucleotides to microtubule subunits (Fig. 7-33) and in the contractile tail sheath of bacteriophage (Box 7-C).

The two ends of the F-actin filaments have different surfaces of the monomer exposed and grow at different rates. This has been demonstrated by allowing the myosin fragment called heavy meromyosin (HMM; see Fig. 19-10) to bind to (or "decorate") an actin filament. The myosin heads bind at an angle, all pointed in one direction. This gives a “pointed” appearance to one end and a “barbed” appearance at the other. When monomeric actin is added to such an HMM-decorated F-actin filament the barbed ends grow much faster than the pointed ends.[94,96] In the intact sarcomere the ends that become pointed when decorated are free, while the opposite barbed ends of the filaments are attached at the Z-line (Fig. 19-6A). The existence in the cytoplasm of proteins that “cap” the fast-growing end of actin filaments thereby preventing further growth[96,97] suggests that cap proteins may be present at the ends of the thin filaments of the myofibrils.

Titin and nebulin. The third most abundant protein (10%), titin (also called **connectin**),[98–100a] is one of the largest of known proteins. Titin cDNA from human cardiac muscle encodes a 26,926-residue chain. Several tissue-specific isoforms of the protein are created by alternative mRNA splicing.[101] A single titin molecule stretches ~1200 nm from the Z-disc, where the N terminus is bound, to the M-line, where the C-terminal domain is attached (Fig. 19-8A). Thoughout much of the A-band titin binds to the thick filament and appears to be part of a scaffold for maintenance of the sarcomere structure. The I-band portion of titin has elastic properties that allow it to lengthen greatly or to shorten as the sarcomere changes length.[98,100,102]

Under the electron microscope titin appears as a flexible beaded string ~4 nm in diameter. Most of the molecule is made up of repetitive domains of two types. In human cardiac titin there are 132 folded domains that resemble type III fibronectin repeats and 112 immunoglobulin-like domains.[98] In a “PEVK region,” between residues 163 and 2174, 70% of the residues are Pro, Glu, Val, or Lys. The titin molecule may be organized as polyproline helices in this elastic region.[102a] At the C terminus of titin 800 residues, including a Ser / Thr protein kinase domain, are found within the M-line.

Another very large protein, **nebulin** (3% of the total protein),[103] appears to be stretched alongside the thin filaments. In the electron microscope it appears as a flexible, beaded string ~4 nm in diameter. Ninety-seven percent of the 6669-residue human nebulin is organized as 185 copies of an ~ 35-residue module.[104,105] Nebulin has a proline residue at about every 35th position, possibly corresponding in length to the pitch of the actin helix (Fig. 7-10). At the C terminus is an SH3 domain (see Fig. 11-14), which is preceded by a 120-residue segment rich in potential phosphorylation sites.[106] This part of the peptide chain is anchored in the Z-discs (Fig. 19-8B, C). The three extreme N-terminal modules of nebulin bind to tropomodulin, which caps the pointed ends of thin filments.[106a] Avian cardiac muscle contains a much shorter 100-kDa protein called **nebulette**, which resembles the C-terminal parts of nebulin. Nebulin has been described as encoding a blueprint for thin filament architecture.[99,103]

Proteins of the M-line and Z-disc. The M-line region contains the structural protein **myomesin**, which binds to both titin and myosin and holds the two together.[107] Fast skeletal and cardiac fibers also contain another **M-protein**, which may bridge between myosin filaments. Both the C-terminal region of nebulin and the N termini of pairs of titin molecules meet in the Z-disc, where they are bound into a lattice containing **α-actinin**[98,108-109b] and other proteins (Fig. 19-8B). The dimeric α-actinin, a member of the spectrin family, has a subunit mass of ~97 kDa.[109a] Found primarily in the Z-discs, it is also present in nonmuscle cells in stress fibers and at other locations in the cytoskeleton (Chapter 7). It may anchor actin filaments to various structures outside of the sarcomere.[110] In the dense Z-disc of insect flight muscle a regular hexagonal lattice of α-actinin[111] and a large (500–700 kDa) modular protein called **kettin**[112,112a,b] bind the thin filaments of opposite polarity together.

The **C-protein** (thick filaments), myomesin (M-line protein), and α-actinin (Z-line protein)[110,113,114] each provide 2% of the protein in the myofibril. Less than 1% each of 11 or more other proteins may also be present within the sarcomere.[86,115] Several of these, including the cytoskeletal proteins **desmin** and **vimentin**, and **synemin** surround the Z-discs.[116,116a]

The regulatory proteins troponin and tropomyosin. These two proteins are also associated with the filaments, each one contributing ~5% to the total protein of myofibrils. Tropomyosin is an elongated α-helical coiled-coil molecule, each molecule of which associates with seven actin subunits of an actin filament. Troponin consists of three subunits known as troponins C, T, and I. The elongated troponin T binds to tropomyosin. Troponin I is an inhibitor of the interaction of myosin and actin necessary for muscle contraction. Troponin C, a member of the calmodulin family (Fig. 6-8), binds Ca^{2+} and induces conformational changes that relieve the inhibition and allow contraction to occur. Nebulin is also thought to bind to tropomyosin. A possible arrangement of one of the tropomyosin–troponin–nebulin complexes that lie along the length of the thin filaments is shown schematically and as a three-dimensional model in Figs. 19-8C and D. These proteins are discussed further in Section 4. Figure 19-9 shows a model of the thin filaments with tropomyosin coiled-coil molecules on each side. The troponin subunits are not shown.

Myosins. There are 15 distinct families of proteins within the myosin superfamily.[117–120] They vary greatly in size, but all of them bind and hydrolyze ATP, and all bind to actin. Most have C-terminal tails. At their N

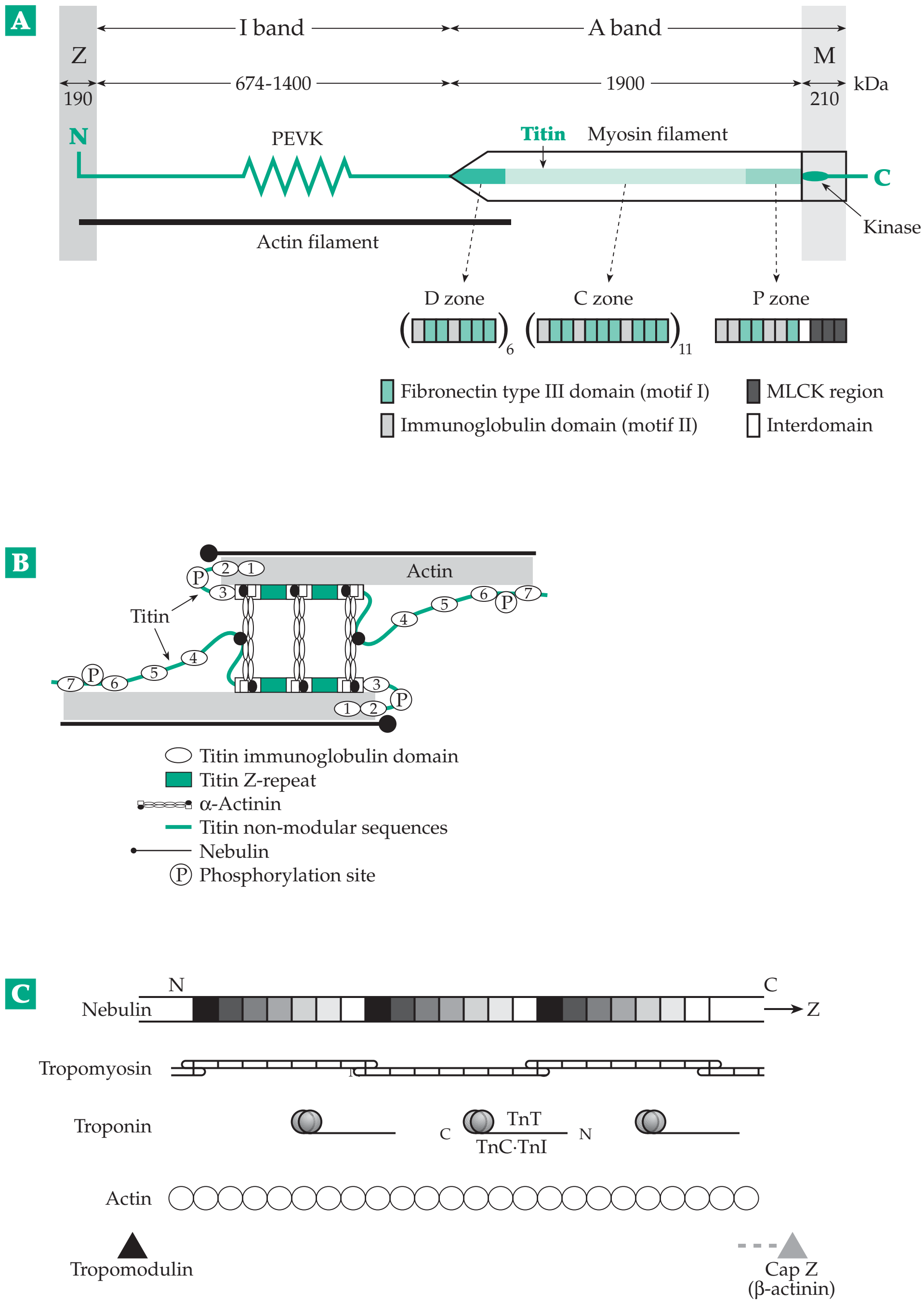


Figure 19-8 (A) Schematic drawing showing one molecule of titin (connectin) in a half sarcomere and its relationship to thick myosin filaments and thin actin filaments. The complex repeat patterns of fibronectin type III, immunoglobulin, in the three zones D, C, and P are also indicated.[98] See Maruyama.[98,98a] (B) Schematic drawing of the molecular structure of the sarcomere Z-disc. Titin, which is thought to parallel the thin filaments through the I-band, consists of various modules that are numbered from the N termini. In the Z-disc titin binds to α-actin, shown here as three vertical rods, and also to actin or actin-binding proteins. The SH3 domain (shown as a sphere) of nebulin and the N terminus of titin may interact. Regulatory phosphorylation sites are marked P. From Young *et al.*[108] Courtesy of Mathias Gautel. (C) Hypothetical model of a composite regulatory complex containing nebulin, tropomyosin, and troponin on the thin filaments of a skeletal muscle sarcomere. Each seven-module nebulin super-repeat (squares with graded shading) binds one tropomyosin, possible through the seven charge clusters along the length of each tropomyosin, and one troponin complex (shaded spheres with a tail). This complex consists of TnT, TnI, and TnC in orientations indicated by the N and C termini. Each nebulin super-repeat binds to seven actin monomers (open circles) along the thin filament. **Tropomodulin** caps the pointed ends of actin filaments and Cap Z, the “barbed ends.” From Wang *et al.*[103]

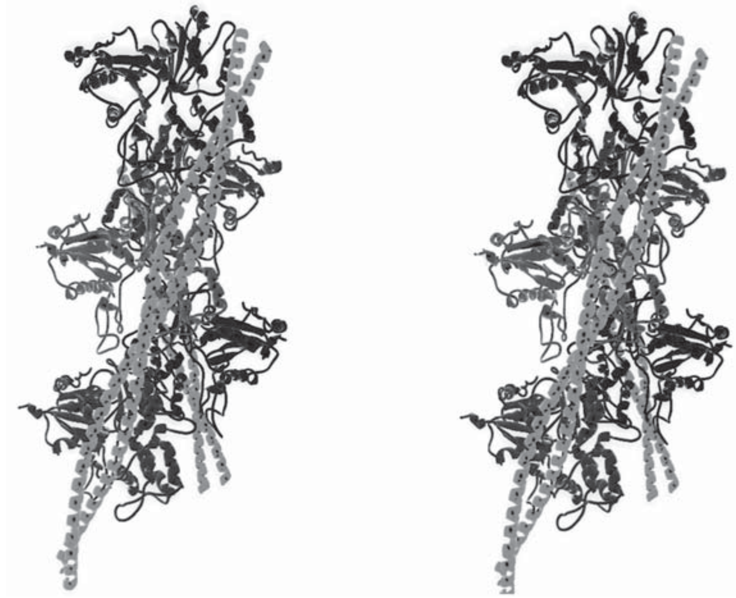


Figure 19-9 Stereoscopic ribbon drawing of the proposed structure of a thin actin filament with tropomyosin coiled-coils bound on opposing sides.[124] Five actin nomomers are assembled in the structure as is also illustrated in Fig. 7-10. From Lorenz *et al*.[125] Courtesy of Michael Lorenz.

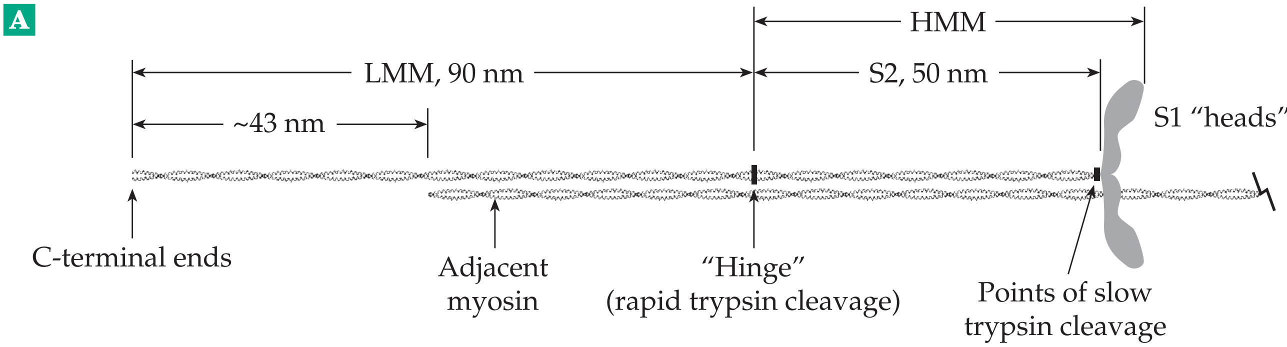
termini are one or two globular heads, which contain the catalytic centers in which ATP hydrolysis occurs. Sizes vary from 93 kDa for a myosin with a very short tail from *Toxoplasma*[118] to over 300 kDa. Myosins I, found in ameboid organisms and also in our own bodies (for example in the microvilli of the brush border of intestinal epithelial cells), are small single-headed molecules.[117,121] Myosins II are the "conventional" myosins of myofibrils and are often referred to simply as myosin. However, each of the three muscle types (skeletal, cardiac, and smooth) has its own kind of myosin II.[121a–c] Likewise, at least six different genes have been identified for the light chains of the myosin heads.[122] Fast and slow muscle as well as embryonic muscle have their own light chains. Each myosin II molecule consists of two identical ~230-kDa **heavy chains,** which are largely α-helical, together with two pairs of smaller 16- to 21-kDa **light chains**. Human skeletal muscle heavy chains contain 1938 residues of which the first ~850 are folded into pear-shaped heads, which contain the catalytic sites involved in harnessing ATP cleavage to movement. Following proline 850 nearly all of the remaining 1088 residues form an α-helical coiled-coil rod of dimensions ~160 x 2 nm (Fig. 19-10) in which the two chains coil around each other. The two heavy chains are parallel, each having its N terminus in one of the two heads and its C terminus bound in the shaft of the thick filament.

Myosins II from other sources have similar structures. For example, analysis of the DNA sequence for a heavy chain gene from the nematode *Caenorhabditis* showed that the protein contains 1966 residues, 1095 of which contain an amino acid sequence appropriate for a 160-nm long coiled coil.[123] There are no prolines within this sequence, which lies between Pro 850 and Pro 1944. Although there are many bands containing positively and negatively charged side chains along the myosin rod, the interactions between the two coiled helices are largely nonpolar. In *Drosophila* 15 different heavy-chain isoforms are created by splicing of a single mRNA.[123a]

While the C-terminal portions of the two parallel myosin heavy chains form a rod, the N-terminal portions fold into two separate heads. Each head also contains two smaller 16- to 21-kDa peptide chains which belong to the calmodulin family. One of these, the **essential light chain**, is tightly bound to the heavy chain. The second, the **regulatory light chain**, is able to bind Ca^{2+} and is less tightly bound to the rest of the head. A short treatment with trypsin or papain cuts the myosin molecule into two pieces. The tail end gives rise to **light meromyosin (LMM)**, a molecule ~90 nm in length. The remainder of the molecule including the heads is designated **heavy meromysin (HMM)**. A longer trypsin treatment leads to cleavage of HMM into one ~62-kDa **S2** fragment 40 nm long, and two ~130-kDa **S1** fragments, each of the latter representing one of the two heads (Fig. 19-10).

The junction of the head and tail portions of myosin appears rigid in Fig. 19-10A. However, there must be considerable conformational flexibility and perhaps some uncoiling of the helices to allow the two heads to interact with a single thin filament as is observed by electron microscopy.[126,128] There also appears to be a hinge between the S2 and LMM segments (Figs. 19-10A and 19-14).

The thick filaments. Dissociated myosin molecules can be induced to aggregate into rods similar to the thick filaments of muscle.[129] Since the filaments



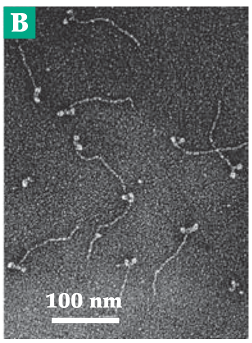


Figure 19-10 (A) An approximate scale drawing of the myosin molecule. The "hinge" is a region that is rapidly attacked by trypsin to yield the light and heavy meromyosins (LMM and HMM). Total length ~160 nm, molecular mass, 470 kDa; two ~200-kDa heavy chains; two pairs of 16- to 21-kDa light chains; heads: ~15 x 4 x 3 nm. (B) Electron micrograph of rabbit myosin monomers that became dissociated from thick filaments in the presence of ATP, fixed and shadowed with platinum.[127] Courtesy of Tsuyoshi Katoh.

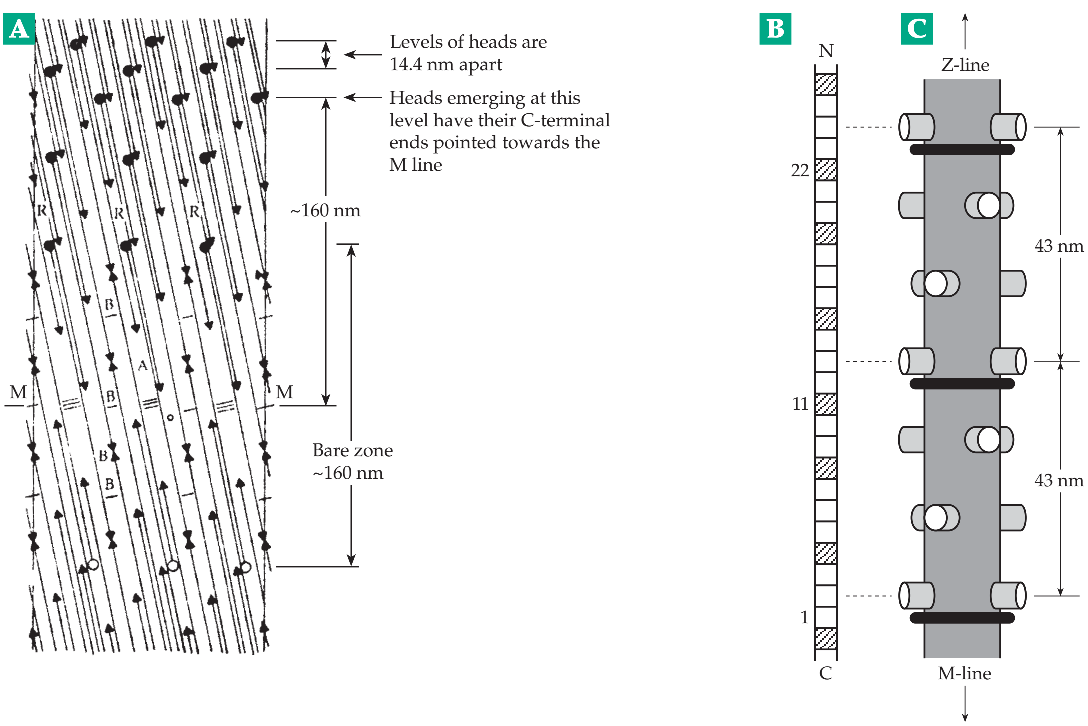


Figure 19-11 (A) Radial projection illustrating packing of myosin rods as suggested by Squire[130] for thick filaments of vertebrate skeletal muscle. The region of the bare zone at the M-line is shown. The filled circles represent the head ends of the myosin molecules and the arrowheads represent the other end of the rod, i.e., the end of the LMM portion. Antiparallel molecules interacting with overlaps of 43 and 130 nm are shown joined by single and triple cross-lines, respectively. Positions where two arrowheads meet are positions of end-to-end butting. O is an "up" molecule (thin lines) and A a "down" molecule (thick lines). The molecules move from the core at the C-terminal end to the filament surface at the head end. The levels marked B may be the levels of attachment of M-bridge material to the myosin filament. The level M-M is the center of the M-line and of the whole filament. The lateral scale is exaggerated more than threefold. (B) A segment of titin showing the 43-nm 11-domain super-repeat. (C) Model of a segment of a thick filament showing the 43-nm repeat, the C-protein, also bound at 43 nm intervals.[99] (B) and (C) Courtesy of John Trinick.

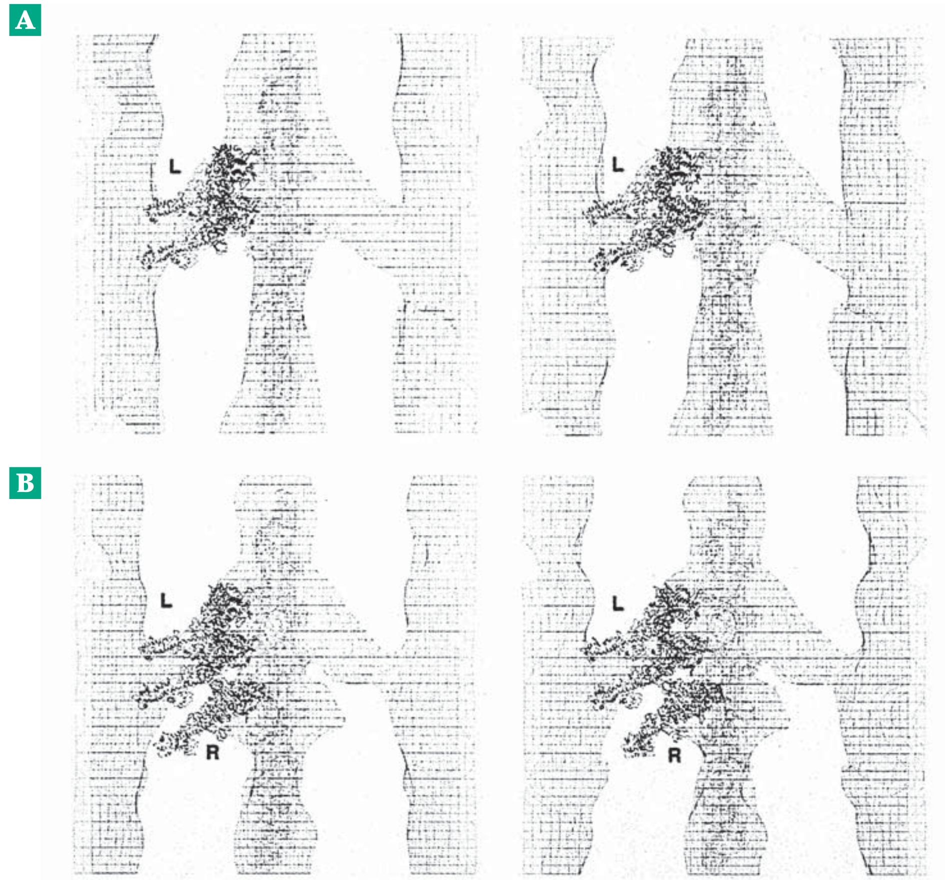


Figure 19-12 (A) Stereoscopic views of computer-assisted reconstructions of images of myosin heads attached to an F-actin filament centered between two thick filaments. Atomic structures of actin (Fig. 7-10) and of myosin heads (Fig. 19-15) have been built into the reconstructed images obtained by electron microscopy. (A) With the nonhydrolyzable ATP analog ATPPNP bound in the active sites. (B) Rigor. Two myosin heads are apparently bound to a single actin filament in (A). If they belong to the same myosin molecule the two C-terminal ends must be pulled together from the location shown here. In (B) a third head is attached, presumably from another myosin rod. This configuration is often seen in rigor. From Winkler *et al.*[134] Courtesy of K. A. Taylor.

have a diameter of ~14 to 20 nm, a large number of the thin 2-nm myosin molecules must be packed together. Electron microscopy reveals the presence of the heads projecting from the thick filaments at intervals of ~43.5 nm. However, there is a bare zone centered on the M-line, a fact that suggests tail-to-tail aggregation of the myosin monomers at the M-line in the centers of the thick filaments (see magnified section of Fig. 19-6, A). A helical packing arrangement involving about 300 myosin molecules (up to 30 rods in a single cross section) in close packing with a small central open core has been proposed for skeletal muscle myosin[130,131] and is illustrated in Fig. 19-11A,B. There are approximately three heads per turn of the helix, each group of three heads spaced 14.3 nm from the preceding one along the thick filament. It is apparently the zones of positive and negative charge, which are especially prevalent in the LMM segment toward the C termini, that lock the successive myosin molecules into this 14.3-nm spacing.[116,132] Titin also binds to the LMM segment of the myosin rod,[99,133] and its 11-domain super repeat of IgG-like and fibronectin-like modules are also 43 nm in length.[98,101] There are typically 47–49 of these super repeats in titin, and if each fits to a turn of the helix, as shown in Fig. 19-11B, there would be 147 myosin molecules in one-half of the thick filament.

Not all muscles have the thick filament structure of Fig. 19-11. In the tarantula muscle, which has a particularly well-ordered structure, there are four myosins per turn.[135,135a] Figure 19-13 shows a reconstruction of scallop myosin which has a 7-fold rotational symmetry. The thick filaments often contain

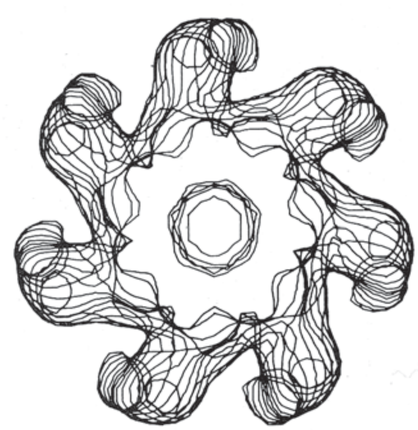


Figure 19-13 Superimposed sections for the 14 nm thickness of a computer-assisted reconstruction of the myosin filaments of the scallop adductor muscle. From Vibert and Craig.[137]

other proteins in addition to myosin. Thus, skeletal muscle contains the C-protein in a series of helical bands along the thick rod.[135b,c] In nematodes, molluscs, and insects the thick filament has a cylindrical core of **paramyosin**, another protein with a structure resembling that of the myosin rod. A minor component of *Drosophila* myosin, the **myosin rod protein**, lacks heads but is transcribed from the myosin heavy chain gene.[136]

3. Actomyosin and Muscle Contraction

That actin and myosin are jointly responsible for contraction was demonstrated long before the fine structure of the myofibril became known. In about 1929, ATP was recognized as the energy source for muscle contraction, but it was not until 10 years later that Engelhardt and Ljubimowa showed that isolated myosin preparations catalyzed the hydrolysis of ATP.[138] Szent-Györgi[139,140] showed that a combination of the two proteins actin (discovered by F. Straub[141]) and myosin was required for Mg^{2+}-stimulated ATP hydrolysis (ATPase activity). He called this combination **actomyosin**.

Under the electron microscope the myosin heads can sometimes be seen to be attached to the nearby thin actin filaments as **crossbridges**. When skeletal muscle is relaxed (not activated by a nerve impulse), the crossbridges are not attached, and the muscle can be stretched readily. The thin filaments are free to move past the thick filaments, and the muscle has some of the properties of a weak rubber band. However, when the muscle is activated and under tension, the crossbridges form more frequently. When ATP is exhausted (e.g., after death) muscle enters the state of **rigor** in which the crossbridges can be seen by electron microscopy to be almost all attached to thin filaments, accounting for the complete immobility of muscle in rigor (Figs. 19-12, 14).[134]

In rigor the crossbridges are almost all firmly attached to the thin actin filaments, making an approximately 45° angle to the actin filaments.[142–144] However, the addition of ATP causes their instantaneous release and the relaxation of the muscle fiber. In contrast, activation by a nerve impulse, with associated release of calcium ions (Section B,4), causes the thin filaments to slide between the thick filaments with shortening of the muscle. An activated muscle shortens if a low tension is applied to the muscle, but at a higher tension it maintains a constant length. Because the maximum tension developed is proportional to the length of overlap between the thick and thin filaments, it was natural to identify the individual crossbridges as the active centers for generation of the force needed for contraction.

The rowing hypothesis. H. E. Huxley[145,146] and A. F. Huxley and R. M. Simmons[147] independently proposed that during contraction the myosin heads attach themslves to the thin actin filaments. The hydrolysis of ATP is then coupled to the generation of a tension that causes the thick and thin filaments to be pulled past each other. The heads then release themselves and become attached at new locations on the actin filament. Repetition of this process leads to the sliding motion of the filaments (Fig. 19-14). The evidence in favor of this "rowing" or "swinging bridge" hypothesis was initially based largely on electron microscopy. For example, contracting muscle was frozen rapidly and fixed for microscopy in the frozen state.[148] Relaxed muscle shows no attached crossbridges, but contracting muscle has many. However,

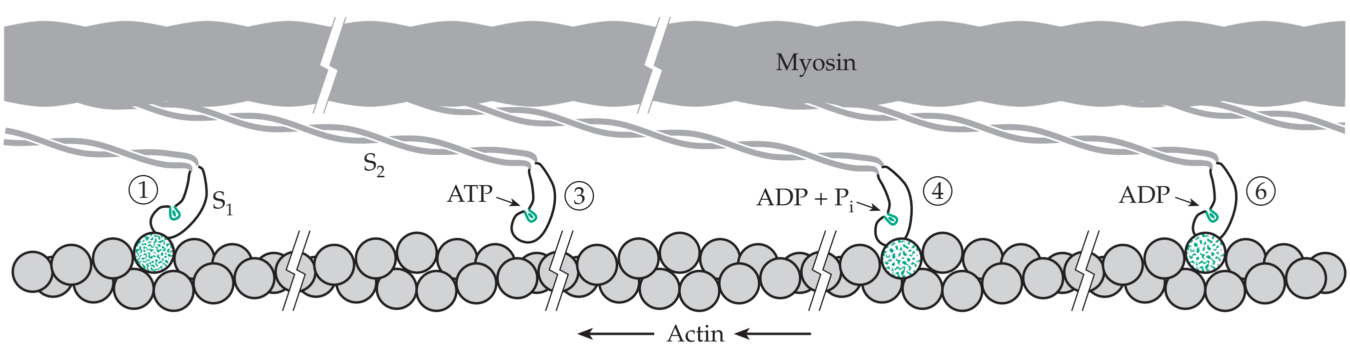


Figure 19-14 A model for the coupling of ATP hydrolysis to force production in muscle based on proposals of H. E. Huxley, and A. F. Huxley and Simmons. The power stroke is depicted here as a rotation of the crossbridge from a 90° to a 45° configuration. Four representative stages are shown: (1) the rigor complex, (3) the dissociated myosin ATP complex, (4) the actomyosin ADP pre-power stroke state in which the actin–myosin band has reformed but with a different actin subunit, which may be distant from that in (1), and (6) the actomyosin ADP post-power stroke state. Force production and contraction result from crossbridges passing cyclically through the steps depicted from left to right. Numbering of the stages corresponds approximately to that in Fig. 19-18. After H. Huxley.[146]

their appearance was distinct from that seen in rigor. The model was also supported by indirect physical methods.

An impressive demonstration that myosin heads do move along the actin filaments was provided by Sheetz and Spudich, who found that myosin-coated fluorescent beads ~0.7 µm in diameter will move along actin filaments from cells of the alga *Nitella* in an ATP-dependent fashion at velocities similar to those required in muscle.[149] The myosin heads literally glide along the thick cables of parallel actin filaments present in these algae.

Why two heads? The actin filament is a two-start helix, and it is natural to ask whether the two myosin heads bind to just one or simultaneously to both of the actin strands. Most evidence supports a 1:1 interaction of a single head with just one strand of actin. However, the other actin strand may associate with heads from a different thick filament. Another question concerns the role of the pairs of myosin heads. Could the two heads bind sequentially to the actin and exert their pull in a fixed sequence? In the reconstruction of the actomyosin complex in rigor (Fig. 19-12B) two different images are seen for the crossbridges. This suggests the existence of two different conformations for the attached myosin heads. Similar images for smooth muscle heavy meromyosin in its inactive (resting) dephosphorylated state (see p. 1116) show the two heads in very different orientations with one binding to the other of the pair and blocking its movement.[121b] Perhaps one head is tightly bound at the end of the power stroke while the other is at a different stage of the catalytic cycle. Nevertheless, single-headed myosin from *Acanthamoeba* will propell organelles along actin filaments,[150] and actin filaments will slide across a surface coated with single-headed myosin formed by controlled proteolysis.[151] The additional interactions seen in rigor may be peculiar to that state.

Structure of the myosin heads. Myosin and myosin fragments can be isolated in large quantities, but they have been difficult to crystallize. However, Rayment and coworkers purified S1 heads cleaved from chicken myosin by papain and subjected them to reductive methylation (using a dimethylamine–borane complex; see also Eq. 3-34). With most of the lysine side chain amino groups converted to dimethylamine groups, high-quality crystals were obtained, and a structure was determined by X-ray diffraction.[152] Since that time various forms of both modified and unmodified myosin heads from several species have been studied by X-ray crystallography.[153–160] Especially clear results were obtained with unmodified myosin from the ameba *Dictyostelium discoideum*. The head structure, shown in Fig. 19-11, includes a 95-kDa piece of the heavy chain and both light chains. A clearer picture of the neck region containing the light chains was provided by the structure of the "regulatory domain" of scallop myosin.[161] Unlike mammalian or avian myosins, molluscan myosins are regulated by binding of Ca^{2+} to a site in the essential light chain, but the structures are similar to those in Figs. 19-10 and 19-15.

Cleavage of the ~850-residue S1 heads with trypsin yields mainly three large fragments that correspond to structural domains of the intact protein as shown in Fig. 19-15. They are known as the 25-kDa (N-terminal), 50-kDa, and 20-kDa fragments, and for myosin from *D. discoideum* correspond to residues 1 to 204, 216 to 626, and 647 to 843, respectively. The ATP-binding site is in a deep cleft between the 20-kDa and 50-kDa

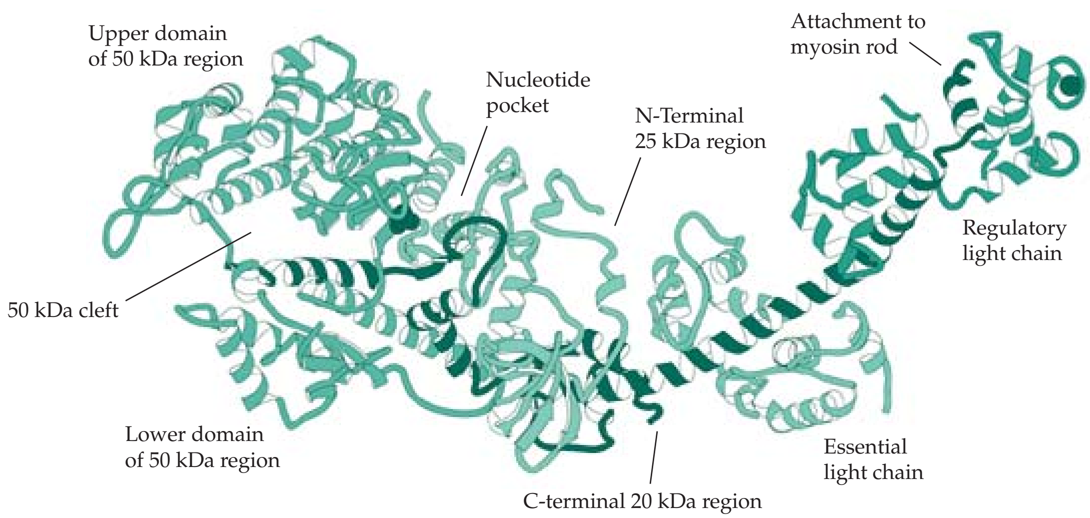


Figure 19-15 Ribbon representation of chicken skeletal myosin subfragment-1 showing the major domains and tryptic fragments. Prepared with the program MolScript. From Rayment.[157]

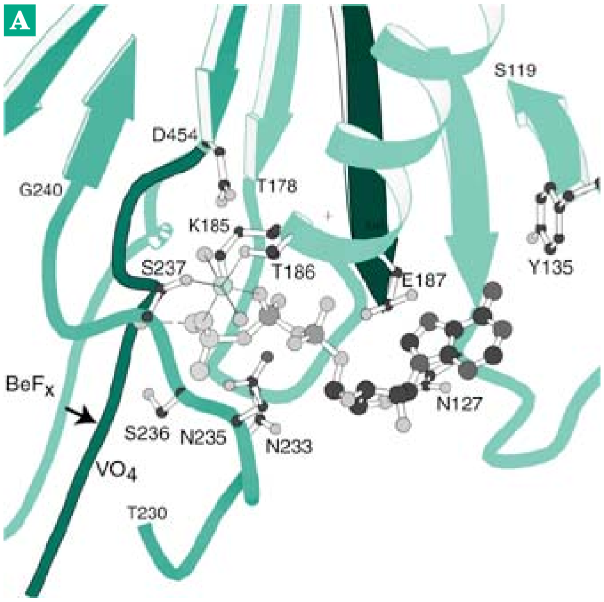
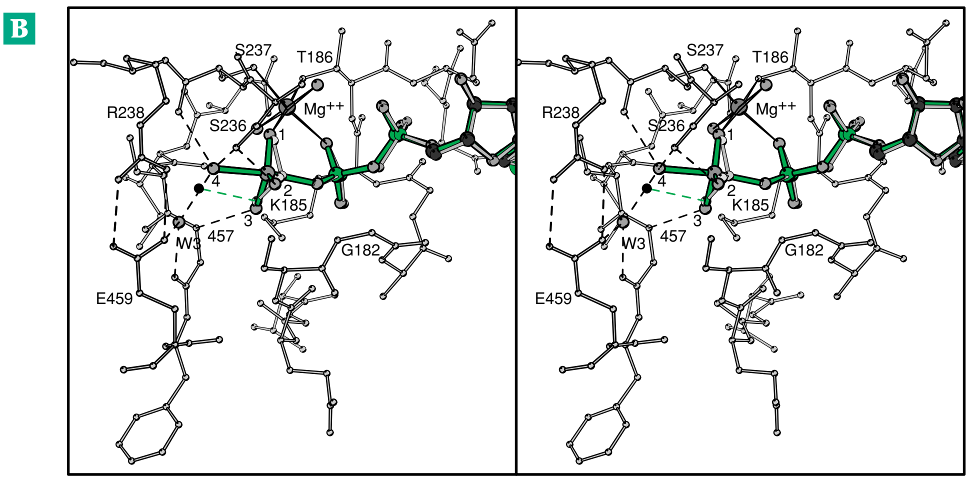


Figure 19-16 (A) The nucleotide binding site of myosin with MgADP•BeF_x bound in a conformation thought to mimic that of ATP prior to hydrolysis. The β-sheet strands are contributed by both the 25-kDa and 50-kDa domains. The P-loop lies between T178 and E187. The conserved N233 to G240 loop, which also contributes important ATP-binding residues, comes from the 50-kDa region. (B) Stereoscopic view of the γ-phospho group binding pocket with the bound MgADP•VO_4 (vanadate) complex. The coordinated Mg^{2+} and associated water molecules are seen clearly. Courtesy of Ivan Rayment.[157]



regions. Figure 19-16 illustrates the binding of an ATP analog, the beryllium fluoride complex of MgADP, in the active site. As can be seen, the ATP binds to loops at the C termini of the β strands of the 8-stranded β sheet from the 25-kDa domain. The conserved P-loop (Chapter 12, E), which lies between T178 and E187, curls around the α and β phospho groups, and has the sequence G(179)ESGAGKT. A second conserved loop N(233)SNSSR-G(240) from the 50-kDa domain contributes to the binding of ATP.

The actin-binding region of the myosin head is formed largely by the 50-kDa segment, which is split by a deep cleft into two separate domains (Fig. 19-15), both of which are thought to participate in binding to actin. A surface loop (loop 1) near the ATP-binding site at the junction of the 25- and 50-kDa regions affects the kinetic properties of myosin, probably by influencing product release. A second loop (loop 2, residues 626–647) at the junction of the 50- and 20-kDa regions interact with actin. Loop 2 contains a GKK sequence whose positive charges may interact with negative charges in the N-terminal part of actin.[162–164]

The C-terminal fragment of myosin contains a globular domain that interacts with both the 20-kDa and 50-kDa regions and contains an α-helical neck that connects to the helix of the coiled-coil myosin rod. This helical region is surrounded by the two myosin light chains (Fig. 19-15).[157] A pair of reactive thiol groups (from C697 and C707) in the globular domain are near the active site. Crosslinking of these cysteines by an –S–S– bridge has been utilized to trap nucleotide analogs in the active site.[165]

How does actin bind? The actin monomer consists of four subdomains, 1, 2, 3, and 4 numbered from the N terminus (Fig. 7-10). The negatively charged N-terminal region of actin contains the sequence

$$\mathrm{D-E-D-\overset{4}{E}\text{- - -}\overset{24}{D}-\overset{25}{D}.}$$

It may interact with loop 1 of myosin, which contains five lysines. However, to form a strong interaction with the myosin head a conformational change must occur in the myosin. A change may also occur in actin. Modeling suggests that a large nonpolar contact region involves actin residues A144, I341, I345, L349, and F352 and myosin residues P529, M530, I535, M541, F542, and P543. A conformational change in actin, which might involve largely the highly conserved actin subdomain 2, may also be required for tight interaction.[142,166–168]

Kinesins and other molecular motors. Before considering further how the myosin motor may work, we should look briefly at the **kinesins**, a different group of motor molecules,[168a] which transport various cellular materials along microtubule "rails." They also participate in organization of the mitotic spindle and other microtubule-dependent activities.[168a,b,c] See Section C,2 for further discussion. More than 90 members of the family have been identified. Kinesin heads have much shorter necks than do the myosin heads. A myosin head is made up of ~850 residues, but the motor domain of a kinesin contains only ~345. Like myosin, the 950- to 980-residue kinesins have a long coiled-coil C- terminal region that forms a "neck" of ~50 residues, a "stalk" of ~190 and ~330 residue segments with a Pro / Gly-rich hinge between them, and an ~45 residue "tail."[169–171]

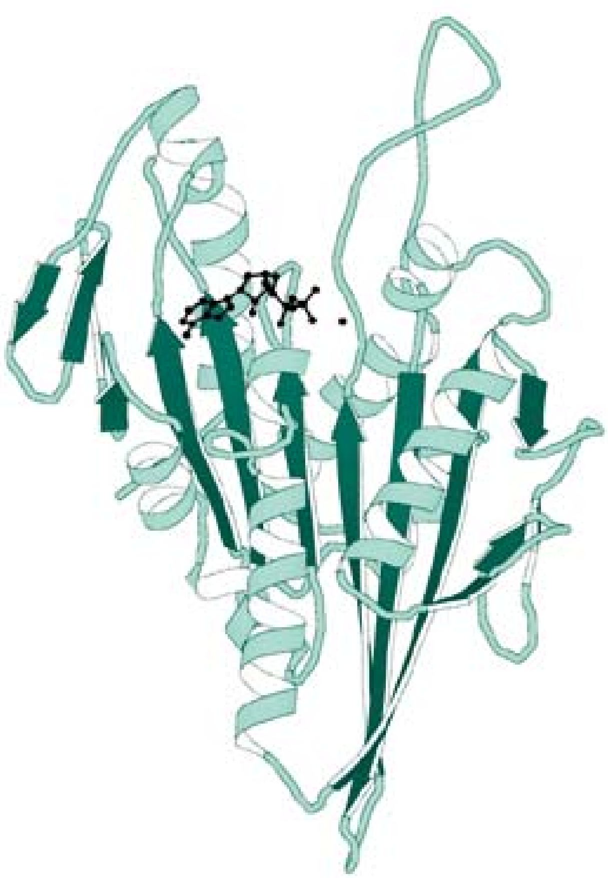


Figure 19-17 Ribbon drawing of human kinesin with bound Mg•ADP. From Gulick *et al.*[174] Courtesy of Ivan Rayment and Andy Gulick.

Crystal structures are known for motor domains of human kinesin[172] and of a kinesin from rat brain.[169,173] The structures of one of six yeast kinesins,[174] a protein called **Kar3**, and also of a *Drosophila* motor molecule designated **Ncd** have also been determined.[175] The last was identified through study of a *Drosophila* mutant called non-claret disjunctional (Ncd). The motor domains of various members of the kinesin family show ~40% sequence identity and very close structural identity (Fig. 19-17).[174] Although the sequences are different from those of the myosin heads or of G proteins, the folding pattern in the core structures is similar in all cases. An 8-stranded β sheet is flanked by three α helices on each side and a P-loop crosses over the ATP-binding site as in Fig. 19-16. Further similarity is found in the active site structures, which, for a monomeric kinesin KIF1A,[174a] have been determined both with bound ADP and with a nonhydrolyzable analog of ATP.[174b,174c] Although there is little similarity in amino acid sequences the structures in the catalytic core are clearly related to each other, to those of dimeric kinesins,[174d] to those of myosins, and to those of the GTP-hydrolyzing G proteins.

A puzzling discovery was that the motor domain of kinesin, which binds primarily to the β subunits of tubulin (Fig. 7-34) and moves toward the fast growing *plus* end of the microtubule,[176] is located at the N terminus of the kinesin molecule, just as is myosin. However, the Ncd and Kar3 motor domains are at the C-terminal ends of their peptide chains and move their "cargos" toward the *minus* ends of microtubules.[174] Nevertheless, the structures of all the kinesin heads are conserved as are the basic chemical mechanisms. The differences in directional preference are determined by a short length of peptide chain between the motor domain and the neck, which allows quite different geometric arrangements when bound to microtubules.[173,177] Like Ncd, myosin VI motor domains also move "backwards" toward the pointed (minus) ends of actin filaments.[178–179a]

Other major differences between kinesins and myosin II heads involve kinetics[180,181] and processivity.[173] Dimeric kinesin is a **processive** molecule. It moves rapidly along microtubules in 8-nm steps but remains attached.[182,182a] Myosins V and VI are also processive[183–183e] but myosin II is not. It binds, pulls on actin, and then releases it. The many myosin heads interacting with each actin filament accomplish muscle contraction with a high velocity in spite of the short time of attachment. Ncd and Kar3 are also nonprocessive and slower than the *plus* end-oriented kinesins.[184]

The ATPase cycles of actomyosin and of the kinesins. The properties of the protein assemblies found in muscle have been described in elegant details, but the most important question has not been fully answered. How can the muscle machinery use the Gibbs energy of hydrolysis of ATP to do mechanical work? Some insight has been obtained by studying the ATPase activity of isolated myosin heads (S1) alone or together with actin. Results of numerous studies of ATP binding, hydrolysis, and release of products using fast reaction techniques[185–191] and cryoenzymology[191a] are summarized in Fig. 19-18. In resting muscle the myosin heads swing freely in the ~20-nm space between the thick and thin filaments. However, in activated muscle some heads are bound tightly to actin as if in rigor (complex A•M in Fig. 19-18). When ATP is added MgATP binds into the active site of the myosin (Fig. 19-18, step *a*) inducing a conformational change to form A•M*•ATP in which the bond between actin and myosin is weakened greatly, while that between myosin and ATP is strengthened. The complex dissociates (step *b*) to give free actin and (M*•ATP), which accumulates at −15°C. However, at higher temperatures the bound ATP is hydrolyzed rapidly (step *c*) to a form M**•ADP•P_i in which the ATP has been cleaved to ADP + P_i but in which the split products remain bound at the active site.[116,192,192a,b] All of these reactions are reversible. That is, the split products can recombine to form ATP. This fact suggests that most of the Gibbs energy of hydrolysis of the ATP must be stored, possibly through a conformational change in the myosin head or through tighter bonding to ATP. As long as calcium ions are absent, there is only a slow release of the bound ADP and P_i and replacement with fresh ATP takes place. Thus, myosin alone shows a very weak ATPase activity.

On the other hand, in activated muscle the head with the split ATP products will bind to actin (step *d*), probably at a new position. The crossbridges that form appear to be attached almost at right angles to the thin filaments. In step *e*, P_i is released following a conformational alteration that is thought to open a "back door" to allow escape of the phosphate ion.[193] In the final two steps (*f* and *g*) the stored energy in the myosin head (or in the actin) is used to bring about another conformational change that alters the angle of attachment of myosin head to the thin filament from ~90° to ~45°. At least some indication of such a change can be observed directly by electron microscopy.[144] Such a change in angle is sufficient to cause the actin filament to move ~10 nm with respect to the thick filaments to complete the movement cycle (Fig. 19-18), if the head is hinged at the correct place. However, the existence of at least four different conformational states suggests a more complex sequence.[193a,193b] Examination of the three-dimensional structures available also suggest a complex sequence of alterations in structure and geometry. X-ray crystallographic structures of myosin heads, in states thought to correspond to states 1 and 3 of Figs. 19-14 and 19-18, are also in agreement on an ~10 (5−12) nm movement of the lever arm.[194,195] Six states of the actomyosin complex are depicted in Fig. 19-18, but a complete kinetic analysis requires at least eight and possibly 12 states.[196,197]

Observing single molecules. A major advance in the study of molecular motors has been the development of ways to observe and study single macromolecules. The methods make use of **optical traps** (optical "tweezers") that can hold a very small (~1 µm diameter) polystyrene or silica bead near the waist of a laser beam focused through a microscope objective.[198–202] In one experimental design an F-actin filament is stretched between two beads, held in a pair of optical traps. The filament is pulled taut and lowered onto a stationary silica bead to which a few myosin HMM fragments have been attached (Fig. 19-19). If ATP is present, short transient movements along the filament are detected by observation of displacements of one of the beads when the actin filament contacts HMM heads. An average lateral displacement of 11 nm was observed. Each HMM head exerted a force of 3−4 pN, a value consistent with expectations for the swinging bridge model.[200] From the duration of a single displacement (≤7 ms) and an estimated k_{cat} for ATP hydrolysis of 10 s^{-1}, the fraction of time that the head was attached during one catalytic cycle of the head was therefore only 0.07. This ratio, which is called the **duty ratio**, is low for actomyosin. However, many myosin heads bind to each actin filament in a muscle. Each head exerts its pull for a short time. but the actin is never totally unattached.[203] Similar measurements with smooth muscle revealed similar displacements but with a 10-fold slower sliding velocity and a 4-fold increase in the duty ratio. This may perhaps account for an observed 3-fold increase in force as compared with skeletal muscle.[204,204a]

Other single molecule techniques involve direct observation of motor molecules or of S1 myosin fragments tagged with highly fluorescent labels.[205,206] All measurements of single molecule movement are subject to many errors. Brownian motion of the beads makes measurements difficult.[207] Not all results are in agreement, and some are difficult to understand.[207a] Most investigators agree that there is a step size of ~4−10 nm. Kitamura *et al.* found 5.3 nm as the average.[206] However, they also reported the puzzling observation that some single S1 molecules moved 11−30 nm in two to five rapid successive steps during the time of hydrolysis of a single molecule at ATP. They suggested that some of the energy of ATP hydrolysis may be stored in S1 or in the actin filament and be released in multiple steps. Veigel *et al.*[208] observed that a brush border myosin I from chicks produced ~ 6 nm movements, each of which was followed by an additional ~ 5.5 nm step within ~5 ms. They attribute these steps to two stages in the power stroke, e.g., to steps *f* and *g* of Fig. 19-18. A value of ~10nm was reported recently by Piazzesi *et al*.[208a] Myosin V moves along actin filaments with very large 36-nm steps.[208b]

Motion of kinesin heads has been observed by movement of microtubules over biotinylated kinesin fixed to a steptavidin-coated surface,[209] by direct observation of fluorescent kinesin moving along microtubules,[171] and by optical trap interferometry.[210] Kinesin heads move by 8-nm steps, evidently the exact length

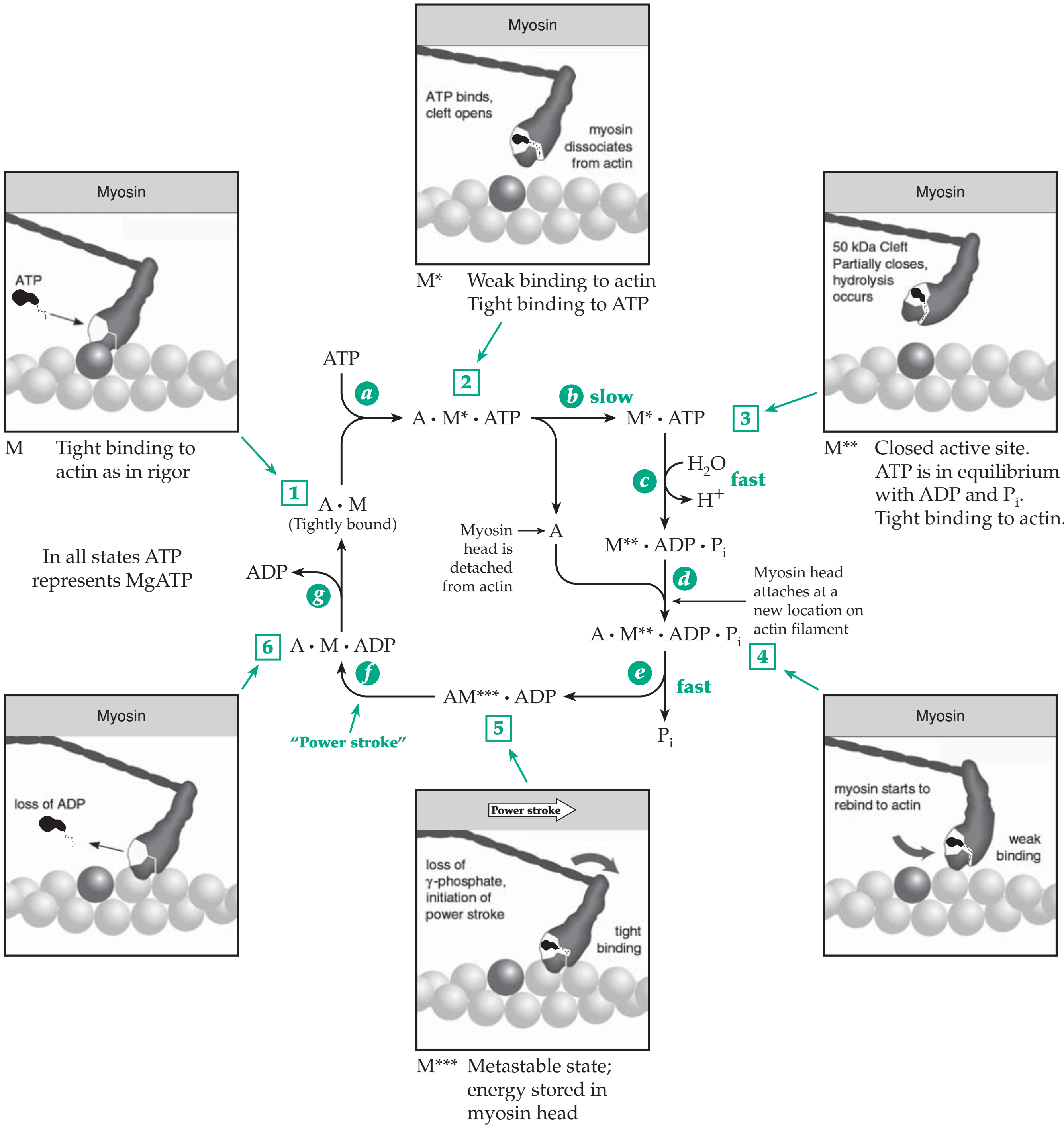


Figure 19-18 Simplified view of the ATP hydrolysis cycle for actomyosin. A similar cycle can be written for kinesins and dyneins. Here A stands for fibrous actin and M, M*, M**, and M*** for four different conformations of the myosin heads. As indicated by the numbers in squares, four of the six states of actomyosin shown here can also be correlated with those in Fig. 19-14.

of an αβ tubulin dimer in the microtubule structure (Fig. 7-33). One ATP is apparently hydrolyzed for each 8-nm step. However, shorter steps of ~5 nm have sometimes been seen.[211,212]

The movement is processive, kinesin motors typically taking 100 steps before dissociating from the microtubule.[201,212a] Kinesin is bound to the microtubule continuously. Its duty ratio is nearly 1.0 (the same is true for the bacterial flagellar motor; Fig. 19-4).[212b] However, single kinesin heads, which lack the coiled-coil neck region, have a duty ratio of <0.45. The movement is nonprocessive.[213] The Ncd motor is also nonprocessive.[214–216] As mentioned previously, the Ncd and kinesin motor domains are at opposite ends of the peptide chain, and the motors move in opposite directions along microtubules.[217,218] The critical difference between the two motor molecules was found in the neck domains, which gave rise to differing symmetrics in the two heads.[219] The latter are shown in Fig. 19-20, in which they have been docked onto the tubulin protofilament structure. One head, both of ncd and of kinesin, occupies a similar position on the microtubule, but the other head points toward the microtubular plus end for kinesin but toward the minus end for Ncd. Cryoelectron microscopy also supports this interpretation.[220]

Still not fully understood is the processive action of kinesin.[221–224] It is often assumed that this protein moves in a hand-over-hand fashion with the two heads binding alternatively to the microtubule. Some substantial reorganization of the peptide chain in the hinge region at the end of the neck is presumably involved.[173] An alternative "inchworm" mechanism has been suggested.[220a]

Thinking about chemical mechanisms. We have now examined the active sites of kinases that cleave ATP (Chapter 12), ATPases that pump ions by cleaving ATP, ATP synthases that form ATP from ADP and P_i (Chapter 18), and GTP hydrolyzing enzymes that cause movement and shape changes that control metabolic processes (Chapter 11). It is striking that the active site regions where the ATP or GTP bind have such a highly conserved structure.[225] This suggests that the secret of movement can be found in the very strong interactions of the nucleotides and their split products with the proteins. In every case there is at least one tight binding or closed conformation in which a large number of hydrogen bonds and ionic interactions bind the nucleotide. This is shown for a kinase in Fig. 12-32 and for myosin in Fig. 19-16. During the actomyosin reaction several conformational changes must occur. Not only does the affinity for the bound nucleotide vary, but also the binding of actin to myosin can be strong, as in the nucleotide-free state or

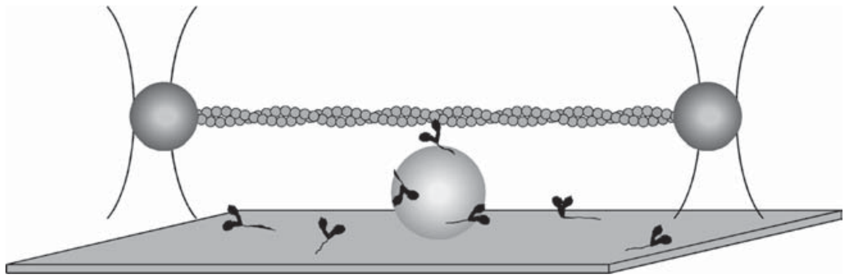


Figure 19-19 Schematic drawing (not to scale) illustrating the use of two optical traps that are focused on beads attached to a single actin filament. The filament is lowered onto a stationary silica bead sparsely coated with HMM fragments of myosin. In the presence of ATP the myosin heads bind transiently for a few milliseconds to the actin, moving it in one direction and displacing the beads from their positions in the optical traps. An image of one of the beads is projected onto photodiode detectors capable of measuring small displacements. The displacing force can also be recorded. For details of the experiments and of the optical traps and measuring devices see Finer *et al.*[200] Courtesy of J. A. Spudich.

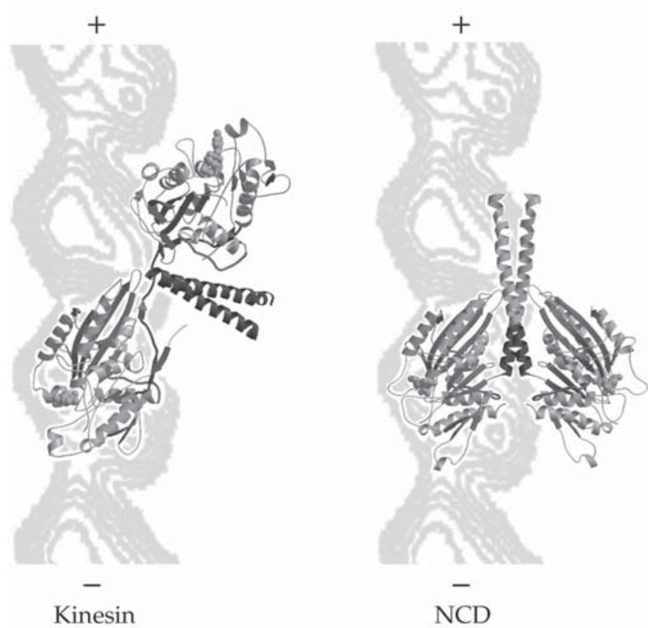
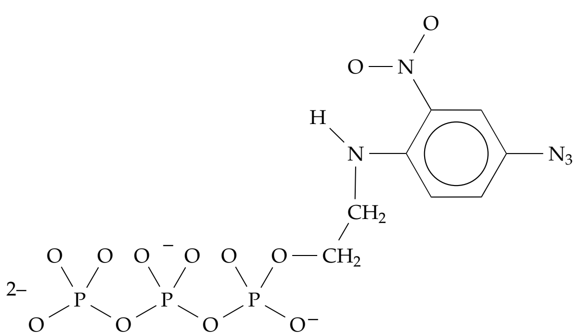


Figure 19-20 Model showing the ncd and kinesin dimer structures docked onto a tubulin protofilament. The bound ncd and kinesin heads are positioned similarly. Because of the distinct architectures of the kinesin and ncd necks, the unbound kinesin head points toward the plus end, whereas the unbound ncd head is tilted toward the minus end of the protofilament. From Sablin *et al.*[219] Courtesy of Ronald Vale.

in the presence of bound ADP. Binding is weak when ATP or the split product ADP + P_i are in the active site.

To understand these differences we should look at the structure of ATP itself. The triphosphate group has many negative charges repelling each other. What must happen to allow the binding of ATP to myosin to break the actin-myosin bond? The electrostatic attraction of these phospho groups for active site groups is doubtless one cause of the observed conformational changes. Could it be that electrostatic repulsion, via a proton shuttle mechanism, is also induced at the right point in the actin–myosin interface? Many studies with analogs of ATP have contributed to our understanding. Neither the purine nor the ribose ring of ATP is absolutely essential. The compound 2-[(4-azido-2-nitrophenyl) amino] diphosphate (NANDP) and related nonnucleotide analogs[165,196,226] support muscle contraction and relaxation in the same

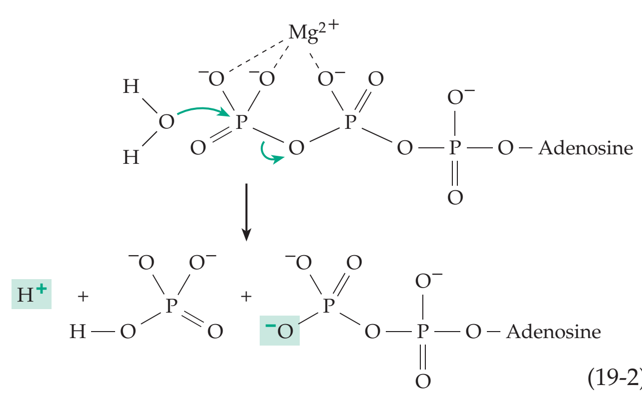


NANTP

way as does ATP. An analog with a rigid five-membered ring, 2',3'-dideoxydidehydro-ATP, is also active.[227] A comparison of kinetic data and X-ray structural data supports the proposal that the ATP must be bound in the conformation shown for MgADP•BeF_x in Fig. 19-16A.[196] When the two SH groups of C697 and C707 of the myosin head are crosslinked by various reagents,[227a] this NANDP analog can be trapped at the active site. Because of the presence of its azide ($-N_3$) group the trapped compound can serve as a photoaffinity label, attaching itself to a tryptophan side chain upon activation with visible light (Eq. 23-27).

How can cleavage of ATP to bound ADP + P_i create a metastable high-energy state of the myosin head ready to hold onto and pull the actin chain? This may be compared with the inverse problem of generating ATP in oxidative phosphorylation, in which ADP and P_i coexist in equilibrium with ATP in a closed active site (Fig. 18-14). Comparison should also be made with the GTP-hydrolyzing G proteins (Fig. 11-7).[227b,c] During hydrolysis of GTP by the Ras protein binding to the protein induces a shift of negative charge from the γ- to β-oxygens of GTP facilitating bond cleavage as in Eq. 19-2. G proteins also couple substrate hydrolysis to mechanical motion. We should also think about the fact that when ATP is cleaved within myosin there will necessarily be a flow of electrical charge from the water to the ADP (Eq. 19-2). This will be followed by some accomodation of the protein to the new charge constellation. As we have seen previously, movement of protons is often the key to conformational changes. In this case, the initial change must be to create a high-energy state of myosin which, following loss of the orthophosphate ion, can cause the major conformational change that swings the lever arm of the myosin. The conformational changes may occur in several steps in which the packing of groups within the myosin head is always tight in some places and rather loose in others. Movement within the head is being observed not only by X-ray crystallography but by **fluorescence resonance energy transfer** (FRET; Chapter 23)[227d] and by the newer **luminescence resonance energy transfer** (LRET). For example, a terbium chelate of azide-ATP was photochemically bound in the active site, and a fluorescent dye was attached to Cys 108 in the regulatory light chain. The terbium ion was irradiated, and fluorescence of the dye was observed. Distance changes, measured in the absence and presence of ATP, were consistent with the swinging arm model.[228] Dyes have been attached to –SH groups engineered into various locations in the myosin molecule to permit other distance measurements.[229,230] In another elegant application of the FRET technique the **green fluorescent protein** of *Aequoria* (Box 23-A) was fused to the C terminus of the motor domain of myosin giving a fluorescent lever arm. Energy transfer to blue fluorescent protein fused to the N terminus of the S1 head was measured. The distance between these was estimated by the FRET technique and was also consistent with expectations for the “rowing model.”[231]

The “rowing model” is generally accepted, but other quite different processes have been proposed to account for the elementary cycle of muscle contraction. Muscle contracts nearly *isovolumetrically*; thus, anything that expands the sarcomere will cause a contraction. It has been suggested that the hydrolysis of ATP deposits negatively charged phospho groups on the actin filaments, and that the electrostatic repulsion is responsible for



(19-2)

BOX 19-A HEREDITARY DISEASES OF MUSCLE

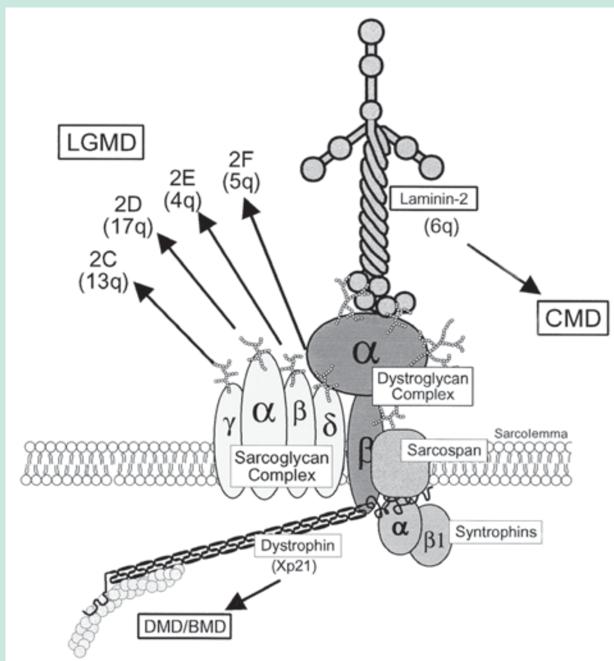
Considering the numerous specialized proteins in muscle it is not surprising that many rare hereditary muscle diseases are known. The most frequent and most studied of these is **Duchenne muscular dystrophy**. An X-linked disease of boys, it may not be recognized until two to three years of age, but victims are usually in a wheelchair by age 12 and die around age 20. Individual muscle fibers disintegrate, die, and are replaced by fibrous or fatty tissue.[a–d] The disease strikes about 1 out of 3500 boys born. The less serious **Becker muscular dystrophy** arises from defects in the same gene but affects only 1 in 30,000 males, some of whom have a normal life span. Because of its frequency and the knowledge that the gene must lie in the X-chromosome, an intensive search for the gene was made. It was found in 1986 after a five-year search.[a,e] This was the first attempt to find a faulty gene whose protein product was totally unknown. The project, which relied upon finding restriction fragment polymorphisms (Chapter 26) that could serve as markers in the genome, made use of the DNA from patients with a range of related diseases. The very rare female patients in whom the faulty gene had been translocated from the X-chromosome to an autosome also provided markers. DNA probes obtained from a young man with a large X-chromosome deletion that included genes related to retinitis pigmentosa and several other diseases provided additional markers. The result was a triumph of "reverse genetics" which has since been applied to the location of many other disease genes.[e]

The muscular dystrophy gene may be the largest human gene. It consists of 2.3 million base pairs, which include 79 exons which encode a huge 427 kDa protein named **dystrophin**. The protein consists of four main domains.[a,f,g] The N-terminal domain binds to actin and is homologous to β-actinin. The central domain is an elongated rod resembling spectrin. It contains repetitive coiled-coil segments and four hinge regions. The third domain is rich in cysteine and binds Ca^{2+}, while the fourth domain has a structure that is shared by several other proteins of the dystrophin family. Dystrophin is quantitatively a minor protein of muscle. It forms part of the cytoskeleton, lying adjacent to the sarcolemma (cell membrane) along with β-spectrin and vinculin (see Fig. 8-16).

While one end of the dystrophin molecule binds to actin filaments, the C-terminal domain associates with several additional proteins to form a **dystrophin–glycoprotein complex** (see figure).[f,h–k] Dystrophin is linked directly to the membrane-spanning protein **β-dystroglycan,** which in the outer membrane surfaces associates with a glycoprotein **α-dystroglycan**. The latter binds to laminin-2 (Fig. 8-33), a protein that binds the cell to the basal lamina. Four other membrane-spanning proteins, α-, β-, γ-, and δ-**sarcoglycans**, are among additional members of the complex.[k–m]

Patients with Duchenne muscular dystrophy are deficient not only in dystrophin but also in the dystroglycan and sarcoglycan proteins.[f,n] Evidently, dystrophin is needed for formation of the complex which plays an essential role in muscle. In both types of X-linked muscular dystrophy there are individuals with a wide range of point mutations, frame-shift mutations, and deletions in the dystrophin gene.[d] The essential function of dystrophin and associated proteins is uncertain but may be related to the linkage from actin filaments through the membrane to laminin. Individuals with Becker muscular dystrophy also have defects in dystrophin, but the protein is partially functional. Some other muscular dystrophies are caused by defects in the autosomal genes of any of the four sarcoglycan subunits.[j,k,o] or in that of laminin α1 chain.[p,q] The arrows in the accompanying drawing indicate chromosome locations of the sarcoglycan subunits, which are sites for mutations causing **limb girdle muscular dystrophy**.[k]

Dystrophin, shorter isoforms, and related proteins are found in many tissues including the brain.[s] One related protein, **utrophin** (dystrophin-related



Schematic model of the dystrophin–glycoprotein complex. Courtesy of Kevin P. Campbell. See Lim and Campbell[r] Abbreviations: LGMD, Limb Girdle muscular dystrophy; CMD, congenital muscular dystrophy; DMD / BMD, Duchenne / Becker muscular dystrophies.

BOX 19-A (continued)

protein 1), is present in the neuromuscular junction of adult skeletal muscle. One approach to therapy of Duchenne muscular dystrophy is to stimulate a higher level of expression of the utrophin gene.[t] Because the dystrophin gene is so large treatment by gene transfer is not practical, but transfer of parts of the gene may be. Myoblast transfer has not been successful, but new approaches will be devised.[d]

Myotonic dystrophy is a generalized adult-onset disorder with muscular spasms, weakness, and many other symptoms.[u–y] It is one of the **triple-repeat diseases** (Table 26-4). The affected gene encodes a protein kinase of unknown function. The corresponding mRNA transcript has ~2400 nucleotides. The gene has a CTG repeat $(CTG)_n$ near the 3'-end with $n < 30$ normally. For persons with the mildest cases of myotonic dystrophy n may be over 50 while in severe cases it may be as high as 2000. As in other triple-repeat diseases the repeat number tends to increase in successive generations of people as does the severity of the disease.[x]

For some individuals, muscular dystrophy causes no obvious damage to skeletal muscle but affects the heart producing a severe **cardiomyopathy,** and persons with Duchenne muscular dystrophy often die from heart failure. Heart failure from other causes, some hereditary, is a major medical problem, especially among older persons. Hereditary forms are often autosomal dominant traits that may cause sudden death in young persons. At least seven genes for cardiac sarcomeric proteins including actin,[z] myosin, both heavy and light chains,[aa–dd] three subunits of troponin,[ee] tropomyosin, and protein C (p. 1104) may all carry mutations that cause cardiomyopathy.

A hereditary disease common in Japan results from a defect in migration of neurons and is associated with brain malformation as well as muscular dystrophy.[ff] In **nemaline myopathy** a defect in nebulin leads to progressive weakness and often to death in infancy. A characteristic is the appearance of "nemaline bodies" or thickened Z-discs containing Z-disc proteins.[gg] Some hereditary diseases involve nonmuscle myosins. Among these is **Usher syndrome**, the commonest cause of deaf-blindness. The disease, which results from a defect in the myosin VIA gene, typically causes impairment of hearing and retinitis pigmentosa (Chapter 23).[hh]

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lateral expansion of the sarcomere.[232] Still other ideas have been advanced.[233,234]

4. Control of Muscle Contraction

Skeletal muscle must be able to rest without excessive cleavage of ATP but able to act rapidly with a high expenditure of energy upon nervous excitation. Even a simple physical activity requires that a person's muscles individually contract and relax in rapid response to nerve impulses from the brain. To allow for this control the endoplasmic reticulum (**sarcoplasmic reticulum, SR**) of striated muscle fibers is organized in a striking regular manner.[235–237] Interconnecting tubules run longitudinally through the fibers among the bundles of contractile elements. At regular intervals they come in close contact with infoldings of the outer cell membrane (the **T system** of membranes, Fig. 19-21; see also Fig. 19-7A). A nerve impulse enters the muscle fiber through the neuromuscular junctions and travels along the sarcolemma and into the T tubules. At the points of close contact the signal is transmitted to the longitudinal tubules of the sarcoplasmic reticulum, which contain a high concentration of calcium ions.

Calcium ions in muscle. A nerve signal arriving at a muscle causes a sudden release of the calcium ions into the cytoplasm from cisternae of the sarcoplasmic reticulum (SR) that are located adjacent to the T-tubules. Diffusion of the Ca^{2+} into the myofibrils initiates contraction. In smooth muscle the signals do not come directly from the nervous system but involve hormonal regulation.[238] Again, calcium ions play a major role, which is also discussed in Chapter 6, Section E, and in Box 6-D. Muscle contains a large store of readily available Ca^{2+} in lateral cisternae of the SR. The free intracellular Ca^{2+} concentration is kept low by a very active ATP-dependent calcium ion pump (Fig. 8-26), which is embedded in the membranes of the SR.[238a] Within the vesicles Ca^{2+} is held loosely by the ~63-kDa protein **calsequestrin,** which binds as many as 50 calcium ions per molecule. When the cytoplasmic concentration of free Ca^{2+} falls below ~10^{-6} M, contraction ceases. In fast-contracting skeletal muscles the Ca^{2+}-binding protein **parvalbumin** (Fig. 6-7) may

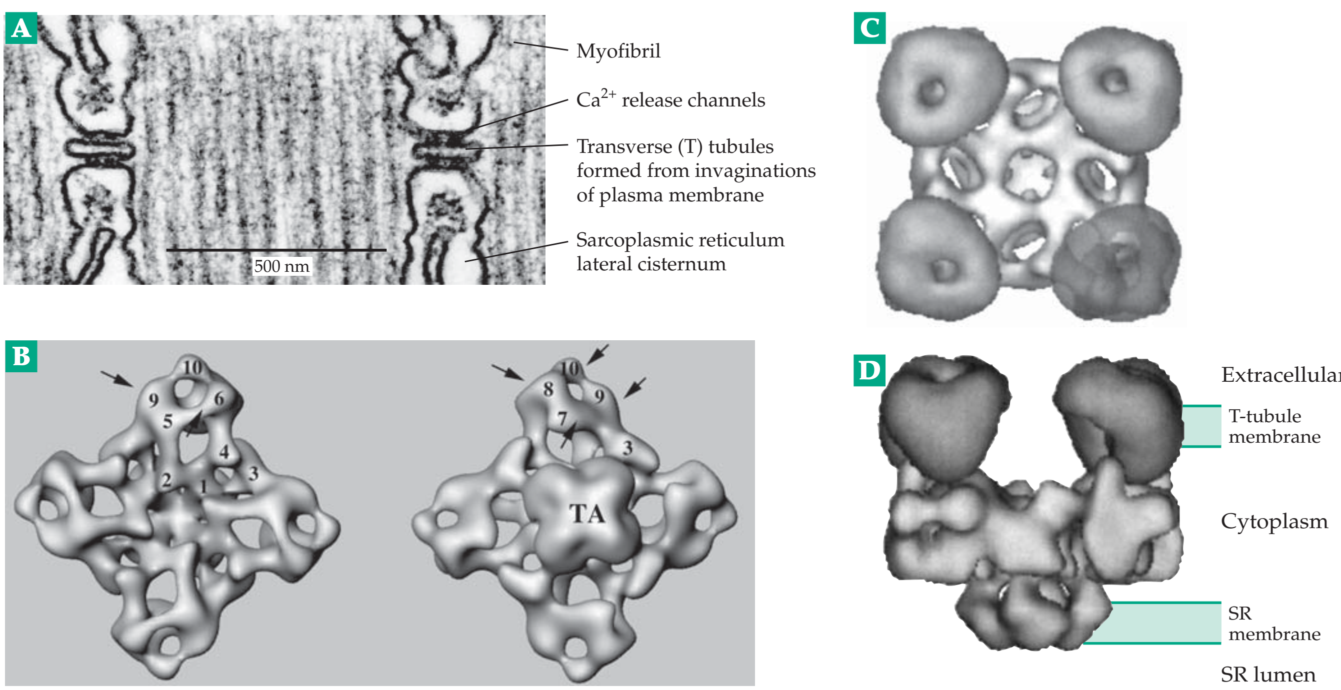
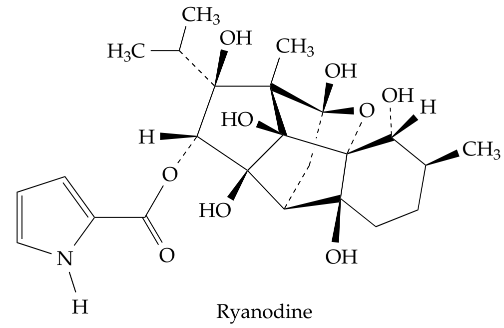


Figure 19-21 (A) Electron micrograph showing two transverse tubules (T-tubules) that are formed by infolding of the plasma membrane. They wrap around a skeletal muscle fiber and carry nerve impulses to all parts of the fiber. From Alberts *et al.*[237] Courtesy of Clara Franzini-Armstrong. (B) Three-dimensional surface representation of the calcium release channels known as ryanodine receptors, type RyR1 based on cryoelectron microscopy and image reconstruction at a resolution of 4.1 nm. The image to the left shows the surface that would face the cytoplasm while that to the right shows the surface that would interact with the sarcoplasmic reticulum, TA representing the transmembrane portion. Notice the fourfold symmetry of the particle, which is composed of four 565-kDa subunits. From Sharma *et al.*[239] Courtesy of Manjuli Rani Sharma. (C), (D) Model showing proposed arrangement of ryanodine receptors and dihydropyridine receptors (round) in the T tubule and SR membranes. From Serysheva *et al.*[245a]

assist in rapid removal of free Ca^{2+} from the cytoplasm. Contraction is activated when Ca^{2+} is released from the SR through the **calcium release channels**,[240–244] which are often called **ryanodine receptors**. The name arises from their sensitivity to the insecticidal plant alkaloid ryanodine, which at low concentration (≤60 µM) causes the channels to open, but a high concentration closes the channels.[243] These calcium release channels consist of tetramers of ~5000-residue proteins. The bulk of the 565-kDa polypeptides are on the cytosolic surface of the SR membranes, where they form a complex "foot" structure (Fig. 19-21B) that spans the ~12 nm gap between the SR vesicles and the T-tubule membrane.[239,240] Ryanodine receptor function is modulated by NO, which apparently binds to –SH groups within the Ca^{2+} channel.[243a,243b] Some ryanodine receptors are activated by cyclic ADP-ribose (cADPR, p. 564).[243c] Some have an oxidoreductase-like structural domain.[243d]

The release channels open in response to an incompletely characterized linkage to the **voltage sensor** that is present in the T-tubule membrane and is known as the **dihydropyridine receptor**.[240,245] This too, is a Ca^{2+} channel, which opens in response to arrival of an **action potential** (nerve impulse; see Chapter 30) that move along the T-tubule membrane. Because the



Ryanodine

action potential arrives almost simultaneously throughout the T-tubules of the muscles, the dihydropyridine receptors all open together. It isn't clear whether the linkage to the calcium channels is via stimulation from released Ca^{2+} passing from the dihydropyridine receptor to the surfaces of the feet of the release channels, or is a result of depolarization of the T-tubule membrane, or involves direct mechanically linked conformational changes.[240,245] The close cooperation of the Ca^{2+} release channel and the voltage sensor is reflected in their close proximity. In the sarcoplasmic reticulum every second release channel is adjacent to a voltage sensor in the opposing T-tubule membrane.[240,245a] The essential nature of the voltage

BOX 19-B MALIGNANT HYPERTHERMIA AND STRESS-PRONE PIGS

Very rarely during surgery the temperature of a patient suddenly starts to rise uncontrollably. Even when heroic measures are taken, sudden death may follow within minutes. This **malignant hyperthermia syndrome** is often associated with administration of halogenated anesthetics such as a widely used mixture of halothane (2-bromo-2-chloro-1,1,1-trifluoroethane) and succinylcholine.[a–d] There is often no warning that the patient is abnormally sensitive to anesthetic. However, development of an antidote together with increased alertness to the problem has greatly decreased the death rate. Nevertheless, severe damage to nerves and kidneys may still occur.[c] Biochemical investigation of the hyperthermia syndrome has been facilitated by the discovery of a similar condition that is prevalent among certain breeds of pigs. Such "stress-prone" pigs are likely to die suddenly of hyperthermia induced by some stress such as shipment to market. The sharp rise in temperature with muscles going into a state of rigor is accompanied by a dramatic lowering of the ATP content of the muscles.

The problem, both in pigs and in humans susceptible to malignant hyperthermia, was found in the Ca^{2+} release channels (ryanodine receptors). Study of inheritance in human families together with genetic studies in pigs led to the finding that the stress-prone pigs have cysteine replacing arginine 615 in the Ca^{2+} channel protein. This modification appears to facilitate opening of the channels but to inhibit their closing.[e] A similar mutation has been found in some human families in which the condition has been recognized. However, there is probably more than one site of mutation in humans.[c,f] Similar mutations in the nematode *C. elegans* are being investigated with the hope of shedding light both on the problem of hyperthermia and on the functioning of the Ca^{2+} release channels.[g]

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sensor is revealed by a lethal mutation (**muscular disgenesis**) in mice. Animals with this autosomal recessive trait generate normal action potentials in the sarcolemma but Ca^{2+} is not released and no muscular contraction occurs. They lack a 170-kDa dihydropyridine-binding subunit of the sensor.[246]

Some aspects of regulation by calcium ions are poorly understood. The frequent observations of oscillations in [Ca^{2+}] in cells is described in Box 6-D. Another phenomenom is the observation of Ca^{2+} "sparks," detected with fluorescent dyes and observation by confocal microscopy.[247] These small puffs of Ca^{2+} have been seen in cardiac muscle[247] and in a somewhat different form in smooth muscle.[248] They may represent the release of Ca^{2+} from a single release channel or a small cluster of channels. When the calcium release channels open, Ca^{2+} ions flow from the cisternae of the SR into the cytoplasm, where they activate both the troponin–tropomyosin system and also the Ca^{2+}-calmodulin-dependent **light chain protein kinase,** which acts on the light chains of the myosin head. These light chains resemble calmodulin in their Ca^{2+}-binding properties. The function of light chain phosphorylation of skeletal muscle myosin is uncertain but it is very important in smooth muscle.[248a]

The regulatory complex of tropomyosin and troponin is attached to the actin filaments as indicated in Fig. 19-8D and also in Fig. 19-9. The latter shows a model at near atomic resolution but without side chains on the tropomyosin and without the troponin components. When the regulatory proteins are completely removed from the fibrils, contraction occurs until the ATP is exhausted. However, in the presence of the regulatory proteins and in the absence of calcium, both contraction and hydrolysis of ATP are blocked. Tropomyosin (Tm) is a helical coiled-coil dimer, a 40-nm rigid rod, in which the two 284-residue 33-kDa monomers have a parallel orientation (Fig. 19-9)[249] as in the myosin tail. However, an 8– to 9– residue overlap at the ends may permit end-to-end association of the Tm molecules bound to the actin filament. As with other muscle proteins there are several isoforms,[250,251] whose distribution differs in skeletal and smooth muscle and in platelets. The elongated Tm rods appear to fit into the grooves between the two strands of actin monomers in the actin filament.[252–254] In resting muscle the Tm is thought to bind to actin near the site at which the S1 portion of the myosin binds. As a consequence, the Tm rod may block the attachment of the myosin heads to actin and prevent actin-stimulated hydrolysis of ATP. The 40-nm Tm rod can contact about seven actin subunits at once (Fig. 19-9). Thus, one Tm–troponin complex controls seven actin subunits synchronously.

Troponin (Tn) consists of three polypeptides (TnC, TnI, TnT) that range in mass from 18 to 37 kDa. The complex binds both to Tm and to actin.[255,256] Peptide TnT binds tightly to Tm and is thought to link the TnI•TnC complex to Tm.[256,257] TnI interacts with actin and inhibits ATPase activity in the absence of Ca^{2+}.[258–261] It may work with the other two peptides to keep the Tm in the proper position to inhibit ATP hydrolysis. TnC binds calcium ions. This ~160-residue protein has a folding pattern almost identical to that of calmodulin (Fig. 6-8) with four Ca^{2+}-binding domains arranged in two pairs at the ends of a long 9-turn helix. When Ca^{2+} binds to TnC, a conformational change occurs[258,259,262-265] (p. 313). This induces changes in the troponin•tropomyosin•actin complex, releases the inhibition of actomyosin ATPase, and allows contraction to occur.[265a] In the heart additional effects are exerted by β-adrenergic stimulation, which induces phosphorylation of two sites on TnI by the action of the cAMP-dependent protein kinase PKA. Dephosphorylation by protein phosphatase 2A completes a regulatory cycle in which the doubly phosphorylated TnI has a decreased sensitivity to calcium ions.[266] Cardiac muscle also contains a specialized protein called **phospholamban**. An oligomer of 52-residue subunits, it controls the calcium ion pump in response to β-adrenergic stimulation. Unphosphorylated phospholamban inhibits the Ca ATPase, keeping [Ca^{2+}] high in the cytoplasm. Phosphorylation of phospholamban by cAMP and/or calmodulin-dependent protein kinase activates the Ca^{2+} pump,[267–268a] removing Ca^{2+} and ending contraction.

X-ray diffraction and electron microscopy in the 1970s suggested that when calcium binds to troponin the tropomyosin moves through an angle of ~ 20° away from S1, uncovering the active site for the myosin–ATP–actin interactions.[252,253] Tropomyosin could be envisioned as rolling along the surface of the actin, uncovering sites on seven actin molecules at once. Side-chain knobs protruding from the tropomyosin like teeth on a submicroscopic gear might engage complementary holes in the actin. At the same time a set of magnesium ion bridges between zones of negative charge on tropomyosin and actin could hold the two proteins together. This proposal has been difficult to test. Although the older image reconstruction is regarded as unreliable, recent work still supports this **steric blocking** model.[255,269-270c] Image reconstruction from electron micrographs of thin filaments shows that, in the presence of Ca^{2+}, the tropomyosin does move 25° away from the position in low [Ca^{2+}]. However, instead of two states of the thin filament, "on" and "off," there may be at least three, which have been called "blocked," "closed," and "open."[255,269,271,271a] The closed state may be attained in rigor.[269] In addition, the possibility that changes in the conformation of actin as well as of myosin occur during the contraction cycle must be considered.[255]

Smooth muscle. The primary regulation of smooth muscle contraction occurs via phosphorylation of the Ser 19 –OH group in the 20-kDa regulatory light chains of each myosin head.[121b,160,272–274] The phosphorylated form is active, participating in the contraction process. Removal of Ca^{2+} by the calcium pump and dephosphorylation of the light chains by a protein phosphatase[275] restores the muscle to a resting state. The N-terminal part of the myosin light chain kinase binds to actin, while the catalytic domain is in the center of the protein. The C-terminal part binds to myosin, and this binding also has an activating effect.[276]

Another protein, **caldesmon**, binds to smooth muscle actin and blocks actomyosin ATPase.[271,277–278a] It is present in smooth muscle in a ratio of actin:tropomyosin:caldesmon of ~14:2:1. Inhibition can be reversed by Ca^{2+}, but there is no agreement on the function of caldesmon.[277] It is an elongated ~756-residue protein with N-terminal domain, which binds to myosin, and a C-terminal domain, which binds to actin, separated by a long helix.[278] Caldesmon may be a substitute for troponin in a tropomyosin-type regulatory system, or it may promote actomyosin assembly. Another possibility is that it functions in a **latch state**, an energy-economic state of smooth muscle at low levels of ATP hydrolysis.[278,279] In molluscan muscles Ca^{2+} binds to a myosin light chain and activates contraction directly. Some molluscan smooth muscles (**catch muscles**) also have a latch state, which enables these animals to maintain muscular tension for long periods of time, e.g., holding their shells tightly closed, with little expenditure of energy.[279a] Catch muscles contain myosin plus a second protein, **catchin**, which is formed as a result of alternative mRNA splicing. Catchin contains an N-terminal sequence that may undergo phosphorylation as part of a regulatory mechanism.[280] However, recent experiments indicate that twitchin (see next paragraph), rather than catchin, is essential to the catch state and is regulated by phosphorylation.[280a] Regulation of the large groups of unconventional myosins is poorly understood. Phosphorylation of groups on the myosin heavy chains is involved in ameba myosins and others.[281]

An unexpected aspect of regulation was discovered from study of the 40 or more genes of *Caenorhabditis elegans* needed for assembly and function of muscle. The mutants designated *unc-22* showed a constant twitch arising from the muscles in the nematode's body. The gene was cloned using transposon tagging (Chapter 27) and was found to encode a mammoth 753-kDa 6839-residue protein which has been named **twitchin**.[282–285] Twitchin resembles titin (Fig. 19-8) and like titin has a protein kinase domain, which is normally inhibited by the end of its peptide chain, which folds over the active site of the kinase. Perhaps the protein kinase activities of twitchin, titin, and related proteins[285] are required in assembly of the sarcomere.

5. The Phosphagens

ATP provides the immediate source of energy for muscles but its concentration is only ~5 mM. As discussed in Chapter 6, Section D, **phosphagens**, such as **creatine phosphate**, are also present and may

$$\text{Creatine-}P + \text{ADP} \longrightarrow \text{Creatine} + \text{ATP}$$

attain a concentration of 20 mM in mammalian muscle. This provides a reserve of high-energy phospho groups and keeps the adenylate system buffered at a high phosphorylation state ratio[286] (see Eq. 6-67).

The concentration of both ATP and creatine phosphate as well as their rates of interconversion can be monitored by ^{31}P NMR within living muscles (Figs. 6-4 and 19-22). Phospho groups were observed to be

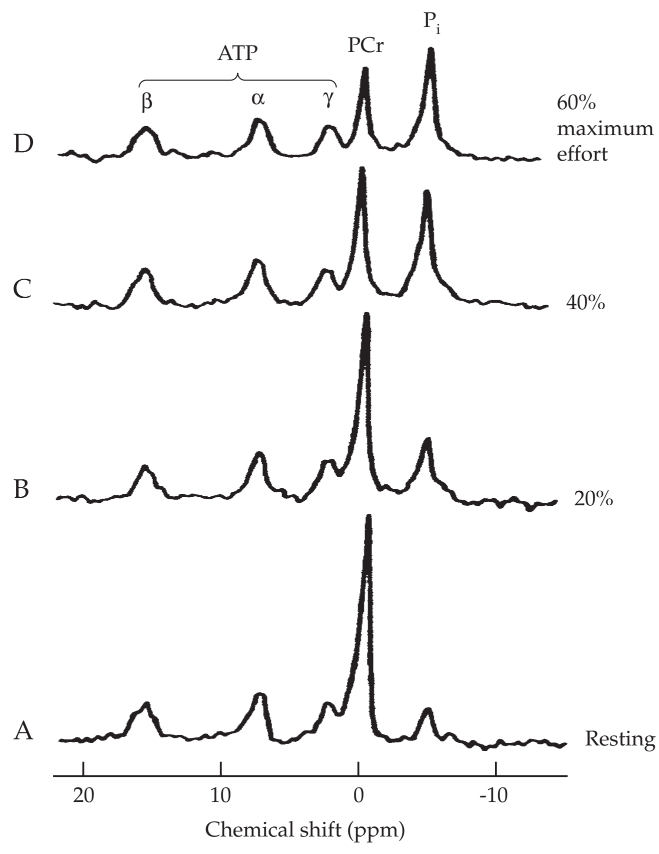

Figure 19-22 Phosphorus-31 magnetic resonance spectra of wrist flexor muscles of the forearm of a trained long-distance runner at rest and during contraction at three different levels of exercise. Ergometer measurements indicating the percent of initial maximum strength (% max) were recorded over each 6-min period. Spectra were obtained during the last 3 min of each period. Times of spectral data collection: A, resting; B, 4–6 min; C, 10–12 min; and D, 16–18 min. The pH ranged from 6.9 to 7.0. From Park *et al*.[288]

transferred from creatine phosphate to ADP to form ATP with a flux of 13 mmol kg^{-1}s^{-1} in rat legs.[287] The reverse reaction must occur at about the same rate because little cleavage of ATP to P_i was observed in the anaesthetized rats. Use of surface coils has permitted direct observation of the operation of this shuttle system in human muscle (Fig. 19-22)[288] as well as in animal hearts (see Chapter 6). Only a fraction of the total creatine present within cells participates in the shuttle, however.[289]

C. Motion in Nonmuscle Cells

At one time actin and myosin were thought to be present only in muscles, but we know now that both actin proteins of the myosin family are present in all eukaryotic cells. Ameboid movement, the motion of cilia and flagella, and movement of materials along microtubules within cells also depend upon proteins of this group.

1. Actin-Based Motility

Ameboid movements of protozoa and of cells from higher organisms, the ruffling movements of cell membranes, phagocystosis, and the cytoplasmic streaming characteristic of many plant cells[289a] have all been traced to actin filaments or actin cables rather

TABLE 19-1
Some Actin-Binding Proteins

Function	Name
Bind and stabilize monomeric actin	Profilin[a,b,c,d,e,f,g,h]
	ADF/Cofilin[i,j,h,k]
	Thymosin[f,l]
Cap actin filament ends	CapZ[m,n]
barbed end	Fragmin[o,p]
pointed end	β-Actin[q]
	Tropomodulin[r]
	Arp2/3, a complex of seven polypeptides
Sever or dissociate actin filament	Gelsolin[s,t,u,v,w]
	Depactin
	Profilin[d,e]
	ADF/cofilin[h,k,x,y]
Crosslink actin filaments or monomers	
Tight bundles	Villin[c,z,aa]
Loose bundles	α-Actinin[bb]
Spectrin[bb]	
	Fascin[cc]
	MARCKS[a]
Network	Filamin[bb,c]
	Gelactins
Bind actin filaments to membrane	Talin[dd]
	"ERM" proteins[ee,ff]

[a] Aderem, A. (1992) *Trends Biochem. Sci.* **17**, 438–443
[b] Mannherz, H. G. (1992) *J. Biol. Chem.* **267**, 11661–11664
[c] Way, M., and Weeds, A. (1990) *Nature (London)* **344**, 292–293
[d] Eads, J. C., Mahoney, N. M., Vorobiev, S., Bresnick, A. R., Wen, K.-K., Rubenstein, P. A., Haarer, B. K., and Almo, S. C. (1998) *Biochemistry* **37**, 11171–11181
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[i] Lappalainen, P., Fedorov, E. V., Fedorov, A. A., Almo, S. C., and Drubin, D. G. (1997) *EMBO J.* **16**, 5520–5530
[j] Rosenblatt, J., and Mitchison, T. J. (1998) *Nature (London)* **393**, 739–740
[k] Chen, H., Bernstein, B. W., and Bamburg, J. R. (2000) *Trends Biochem. Sci.* **25**, 19–23
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[p] Khaitlina, S., and Hinssen, H. (1997) *Biophys. J.* **73**, 929–937
[q] See main text
[r] Gregorio, C. C., Weber, A., Bondad, M., Pennise, C. R., and Fowler, V. M. (1995) *Nature (London)* **377**, 83–86
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[t] De Corte, V., Demol, H., Goethals, M., Van Damme, J., Gettemans, J., and Vandekerckhove, J. (1999) *Protein Sci.* **8**, 234–241
[u] McGough, A., Chiu, W., and Way, M. (1998) *Biophys. J.* **74**, 764–772
[v] Sun, H. Q., Yamamoto, M., Mejillano, M., and Yin, H. L. (1999) *J. Biol. Chem.* **274**, 33179–33182
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[x] Carlier, M.-F., Ressad, F., and Pantaloni, D. (1999) *J. Biol. Chem.* **274**, 33827–33830
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[z] Markus, M. A., Matsudaira, P., and Wagner, G. (1997) *Protein Sci.* **6**, 1197–1209
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[bb] Matsudaira, P. (1991) *Trends Biochem. Sci.* **16**, 87–92
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[ff] Tsukita, S., and Yonemura, S. (1999) *J. Biol. Chem.* **274**, 34507–34510

than to microtubules.[289b] Actin is one of the most abundant proteins in all eukaryotes. Its network of filaments is especially dense in the lamellipodia of cell edges, in microvilli, and in the specialized stereocilia and acrosomal processes (see also pp. 369-370).[289c] Actin filaments and cables are often formed rapidly and dissolve quickly. When actin filaments grow, the monomeric subunits with bound ATP are added most rapidly at the "barbed end" and dissociate from the filament at the "pointed end" (see Section B,2).[94,290] The rate of growth may be ~20–200 nm / s, which requires the addition of 10–100 subunits / s.[291] Various actin-binding proteins control the growth and stability of the filaments. The actin-related proteins **Arp2** and **Arp3**, as a complex Arp2/3, together with recently recognized **formins**.[291a], provide nuclei for rapid growth of new actin filaments as branches near the barbed ends.[290,292–293c]

Growth of the barbed ends of actin filaments is stimulated by phosphoinositides and by members of the Rho family of G proteins (p. 559)[293d] through interaction with proteins of the **WASp** group.[293b,d,e] The name WAS comes from the immune deficiency disorder Wiskott–Aldrich Syndrome. Yet another family, the **Ena/VASP** proteins, are also implicated in actin dynamics. They tend to localize at focal adhesions and edges of lamellipodia.[293e,f] Profilin (Table 19-1) stabilizes a pool of monomeric actin when the barbed ends of actin filaments are capped. However, it catalyzes both the addition of actin monomers to uncapped barbed ends and rapid dissociation of subunits from pointed ends, leading to increased **treadmilling**.[294,295] Actin-severing proteins such as the **actin depolymerizing factor** (**ADF** or **cofilin**, Table 19-1) promote breakdown of the filaments.[296–297a] Treadmilling in the actin filaments of the lamellipods of crawling cells or pseudopods of amebas provides a motive force for many cells[291,298–299] ranging from those of *Dictyostelium* to human leukocytes. A series of proteins known as the **ezrin**, **radixin**, **moesin** (**ERM**) group attach actin to integral membrane proteins (Fig. 8-17)[292,300,301] and may interact directly with membrane lipids.[301a,b] Bound ATP in the actin subunits is essential for polymerization, and excess ATP together with crosslinking proteins stabilize the filaments. However, when the bound ATP near the pointed ends is hydrolyzed to ADP the filaments become unstable and treadmilling is enhanced. Thus, as in skeletal muscle, ATP provides the energy for movement.

Bacteria also contain filamentous proteins that resemble F-actin and which may be utilized for cell-shape determination.[301c] Actin-based motility is used by some bacteria and other pathogens during invasion of host cells (Box 19-C). It is employed by sperm cells of *Ascaris* and of *C. elegans*, which crawl by an ameboid movement that utilizes treadmilling of filaments formed from a motile sperm protein, which does not resemble actin.[302,302a] Cells are propelled on a glass surface at rates up to ~1 μm / s.

Various nonmuscle forms of myosin also interact with actin without formation of the myofibrils of muscle.[299] In most higher organisms nonmuscle myosins often consist of two ~200-kDa subunits plus two pairs of light chains of ~17 and 24 kDa each. These may form bipolar aggregates, which may bind to pairs of actin filaments to cause relative movement of two parts of a cell.[303] Movement depending upon the cytoskeleton is complicated by the presence of a bewildering array of actin-binding proteins, some of which are listed in Table 19-1.

2. Transport along Microtubules by Kinesin and Dynein

Many materials are carried out from the cell bodies of neurons along microtubules in the axons, which in the human body may be as long as 1 m. The rates of this **fast axonal transport** in neurons may be as high as 5 μm / s or 0.43 m / day. The system depends upon ATP and kinesin (Fig. 19-17) and permits small vesicles to be moved along single microtubules.[304–305b] Movement is from the minus end toward the plus end of the microtubule as defined in Figs. 7-33, 7-34. Slower **retrograde axonal transport** carries vesicles from the synapses at the ends of the axons (Fig. 30-8) back toward the cell body. This retrograde transport depends upon the complex motor molecule **cytoplasmic dynein** which moves materials from the plus end of the microtubule toward the minus end.[305,305c] In addition to these movements, as mentioned in Chapter 7, microtubules often grow in length rapidly or dissociate into their tubulin subunits. Growth occurs at one end by addition of tubulin subunits with their bound GTP. The fast growing *plus*-ends of the microtubules are usually oriented toward the cell periphery, while the *minus*-ends are embedded in the **centrosome** or **microtubule-organizing center** (p. 372).[306] Just as with actin, in which bound ATP is hydrolyzed to ADP, the bound GTP in the β-tubulin subunits of microtubules is hydrolyzed to GDP[307–310] decreasing the stability of the microtubules, a phenomenon described as **dynamic instability**. Various **microtubule-associated proteins** (MAPs) have strong effects on this phenomenon.[311] The MAPs are often regulated by phosphorylation–dephosphorylation cycles involving serine / threonine kinases. Microtubules also undergo posttranslational alterations not seen in other proteins. These include addition or removal of tyrosine at the C terminus.[312] Polyglycyl groups containing 3–34 glycine residues may be bound covalently to γ-carboxyl groups of glutamate side chains in both α- and β-tubulins.[312,313] This stabilizes the microtubules and is important to the long-lived microtubules of the axonemes of flagella and

BOX 19-C ACTIN-BASED MOTILITY AND BACTERIAL INVASION

Listeria monocytogenes is a dangerous food-borne bacterium that has become a major problem in the United States. This is one of the best understood *intracellular* pathogens. It is able to enter cells, escape from phagocytic vesicles, spread from cell to cell, and cross intestinal, blood–brain, and placental barriers.[a–c] Within cells these bacteria move using actin-based motility. Actin subunits polymerize at one end of a bacterium leaving a "comet tail" of crosslinked fibrous actin behind (see micrographs). Actin polymerization occurs directly behind the bacteria with subunits of monomeric actin adding to the fast growing "barbed end" (see Section B,2) of the actin strands. Growth has been described as a "Brownian ratchet."[c,d] Continual Brownian movement opens up spaces behind the bacteria, spaces that are immediately filled by new actin subunits. This provides a propulsive force adequate to move the bacteria ahead at velocities of about 0.3 µm / s.

Polymerization of actin is induced by interaction of a dimer of a 610-residue bacterial protein **ActA** with proteins of the host cell.[a,e–h] ActA is a composite protein with an N-terminal region that protrudes from the bacterial cell, a central region of proline-rich repeats that appear to be essential for recognition by host cells, and a C-terminal hydrophobic membrane anchor. There are also regions of close sequence similarity to the human actin-binding proteins vinculin and zyxin. The number of host proteins needed in addition to monomeric actin are:[i,j] the two actin-related proteins, **Arp2** and **Arp3**, which stimulate actin polymerization and branching;[h] **ADF/ cofilin**, which increases the rates, both of growth at the barbed ends and dissociation from the pointed ends of the filaments; and **Cap Z**, which caps barbed ends (Table 19-1). The need for ADF/ cofilin and Cap Z seems paradoxical. Cap Z may cap mostly older and slower growing filaments, restricting rapid filament assembly to the region closest to the bacterium. The need for ADF/ cofilin is unclear.[i] Growth rates are also enhanced by the human protein called **VASP** (vasodilator-stimulated phosphoprotein). The proline-rich region of the bacterial ActA may bind to VASP to initiate polymerization.[g] Both profilin (Table 19-1) and the crosslinking protein α-actinin also stimulate comet tail growth. Myosin does not participate in actin-based motility, but the hydrolysis of ATP drives the process through its link to actin polymerization.[i]

Although *Listeria* has been studied most, actin-based motility is employed by other pathogens as well, e.g., *Shigella flexneri* (the dysentery bacterium),[k] *Rickettsia*,[l] and vaccinia virus.[l] Although enteropathogenic *E. coli* do not use this method of movement, they induce accumulation of actin beneath the bacteria. They also promote formation of actin-rich adherent pseudopods and highly organized cytoskeletal structures that presumably assist the bacteria in entering a cell.[m]

a Cossart, P., and Lecuit, M. (1998) *EMBO J.* **17**, 3797–3806
b Sechi, A. S., Wehland, J., and Small, J. V. (1997) *J. Cell Biol.* **137**, 155–167
c Pantaloni, D., Le Clainche, C., and Carlier, M.-F. (2001) *Science* **292**, 1502–1506
d Mogilner, A., and Oster, G. (1996) *Biophys. J.* **71**, 3030–3045
e Mourrain, P., Lasa, I., Gautreau, A., Gouin, E., Pugsley, A., and Cossart, P. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 10034–10039
f Southwick, F. S., and Purich, D. L. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 5168–5172
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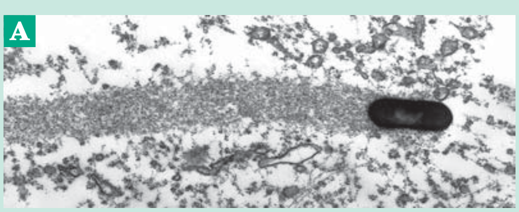
h Welch, M. D., Rosenblatt, J., Skoble, J., Portnoy, D. A., and Mitchison, T. J. (1998) *Science* **281**, 105–108
i Loisel, T. P., Boujemaa, R., Pantaloni, D., and Carlier, M.-F. (1999) *Nature (London)* **401**, 613–616
j Machesky, L. M., and Cooper, J. A. (1999) *Nature (London)* **401**, 542–543
k Bourdet-Sicard, R., Rüdiger, M., Jockusch, B. M., Gounon, P., Sansonetti, P. J., and Tran Van Nhieu, G. (1999) *EMBO J.* **18**, 5853–5862
l Cudmore, S., Cossart, P., Griffiths, G., and Way, M. (1995) *Nature (London)* **378**, 636–638
m Rosenshine, I., Ruschkowski, S., Stein, M., Reinscheid, D. J., Mills, S. D., and Finlay, B. B. (1996) *EMBO J.* **15**, 2613–2624



(A) *Listeria* cell with "comet tail" of cross-linked actin filaments. From Kocks *et al*. (1992) *Cell* **68**, 521–531. Courtesy of Pascale Cossart.

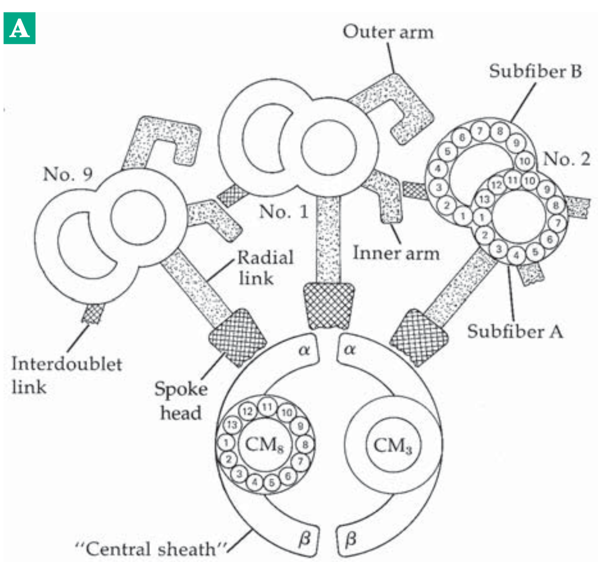
(B) Enlarged section of a thin comet tail of high resolution showing the actin filaments. From Sechi *et al*.[b] Courtesy of Antonio Sechi.

cilia. Polyglutamyl groups of 6–7 glutamates are also often added.[314]

Both dynein and several kinesins act as motors for formation of the spindle and for movement of chromosomes toward the minus ends of spindle microtubules during mitosis and meiosis (Fig. 26-11).[314a] In the genome of *Saccharomyces cerevisiae* there is only one dynein gene, but genes for six different kinesin-type motor molecules are present.[315] In higher organisms there may be even more genes for kinesins but there is apparently only one dynein in most species.[316] **Axonemal dyneins** drive the motion of eukaryotic flagella and cilia. As with the cytoplasmic dyneins a complete molecule consists of two or three heavy chains with molecular mass ~520 kDa, some localized in the dynein tail, and several lighter chains.[305a,317–321] Like myosin dynein is an ATPase.

3. Eukaryotic Cilia and Flagella

The motion of eukaryotic flagella (Fig. 1-8) involves a sliding of the microtubular filaments somewhat analogous to the sliding of muscle filaments.[305,322–325] Sliding between the outer doublet microtubules (Fig. 19-23) via their inner and outer arms (dynein compounds) is thought to provide the characteristic bending waves.[325a,b] The movement is powered by dynein and ATP hydrolysis. Force and displacement measurements made by optical trapping nanometry suggest that the characteristic rhythmic beating of flagella results from an oscillatory property of the dynein.[326] The extremely complex structure of flagella is illustrated in Fig. 19-23. About 250 individual axonemal proteins have been detected in flagella of the alga *Chlamydomonas* (Fig. 1-11),[327] and a large number of mutants with various defects in their flagella have been isolated. The radial spokes (Fig. 19-23) alone contain 17 different proteins. These spokes protrude at ~29-nm intervals while the dynein molecules lie between pairs of the outer microtubule doublets at ~24-nm intervals. The dynein "arms" protrude from the "A" microtubule of each outer doublet and make contact with the incomplete "B" microtubule of the next doublet (Fig. 19-12). Although the shapes of the molecules are quite different, the basic chemistry of the ATPase activity of the dynein–microtubule system resembles that of actomyosin. However, the complexity of the dynein arms,[328]



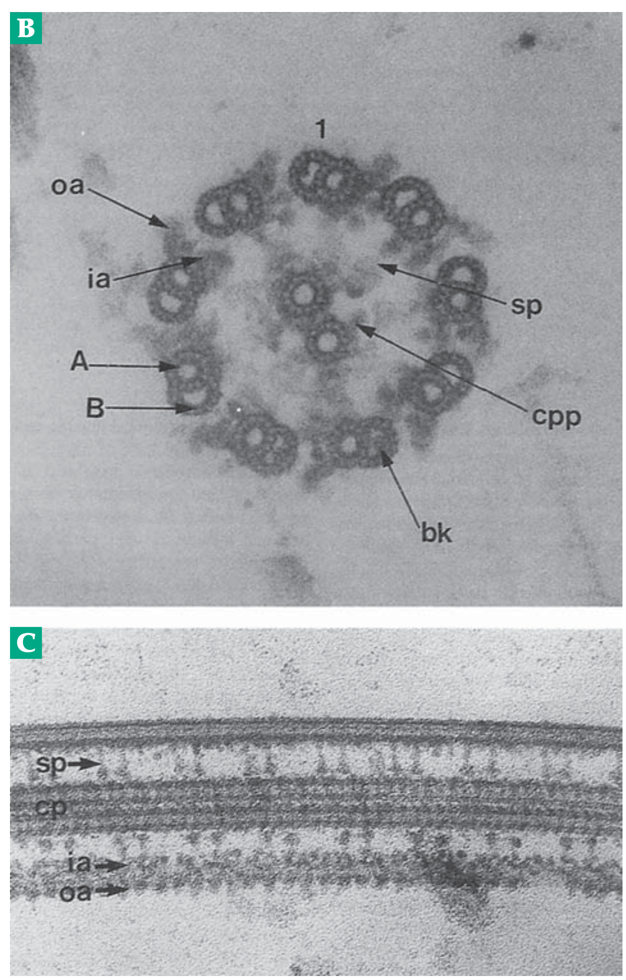


Figure 19-23 (A) Diagram of a cross-sectional view of the outer portion of a lamellibranch gill cilium. This has the 9+2 axoneme structure as shown in Fig. 1-8 and in (B). The viewing direction is from base to tip. From M. A. Sleigh.[329] (B, C) Thin-section electron micrographs of transverse (B) and longitudinal (C) sections of wild-type *Chlamydomonas* axonemes. In transverse section labels A and B mark A and B subtubules of microtubule doublets; oa, ia, outer and inner dynein arms, respectively; sp, spokes; cpp, central pair projections; bk, beaks. From Smith and Sale.[329a]

which exist in two types, inner and outer, suggests a complex contraction cycle.

4. Chemotaxis

As described in Box 11-C, the ameboid cells of the slime mold *Dictyostelium discoideum* are attracted to nutrients such as folic acid during their growth stage. Later, as the cells undergo developmental changes they become attracted by pulses of cyclic AMP.[330] Occupancy of 7-helix receptors for cAMP on the outer plasma membrane appears to induce methylation of both proteins and phospholipids and a rise in cytosolic Ca^{2+} and changes in the cytoskeleton that result in preferential extension of the actin-rich pseudopods toward the chemoattractant.[331]

In a similar manner, human ameboid leukocytes are attracted to sites of inflammation by various **chemotactic factors**.[332] These include the 74-residue cleavage product C5a formed from the fifth component of complement (Chapter 31),[333] various **lymphokines** (Chapter 31) secreted by lymphocytes, and peptides such as VGSE and AGSE, as well as larger peptides released by mast cells, basophils, or stimulated monocytes[334] and oxoicosenoids.[335] Polymorphonuclear leukocytes, upon engulfing sodium urate crystals in gouty joints, release an 8.4-kDa chemotactic protein which may cause a damaging response in this arthritic condition. Leukotriene B is a potent chemotactic agent as are a series of specific bacterial products, formylated peptides such as *N*-formyl-MLF.[332,336]

Neutrophils, monocytes, macrophages, eosinophils, basophils, and polymorphonuclear leukocytes are all affected by several or all of these factors. Binding to specific receptors results in a variety of changes in the cells. These include alterations in membrane potential, cyclic nucleotide levels, and ion fluxes (Na^+, K^+, Ca^{2+}) as well as increased methylation of specific proteins. A reorganization of microtubules and actin fibrils occurs, probably in response to an altered gradient of Ca^{2+}. The morphology of the cells changes, and they begin immediately to crawl toward the chemoattractants. It appears that these ameboid cells detect a gradient of attractant concentration between one end of the cell and the other, even though the anticipated difference may amount to only 0.1% of the total.[337,338]

5. Other Forms of Movement

Movement is characteristic of life and is caused not only by motor proteins but by various springs and ratchets which may be energized in a number of ways.[339] A striking example, which any one with a microscope and some fresh pond water can observe, is contraction of the stalk of protozoa of the genus *Vorticella*. Apparently first reported in 1676 by Leeuwenhoek the organism's 2–3 mm-long stalk contracts into a coiled spring (see p. 1 and also p. 281) when the animal is disturbed. Application of calcium ions causes contraction within a few milliseconds to ~40% of the original length. The process reverses slowly after a few seconds. Contraction is caused by a spring-like organ the **spasmoneme**, which is a bundle of short 2 nm-diameter fibrils inside the stalk. The fibrils are thought to be weakly crosslinked and held in the extended state by electrostatic repulsion between the negatively charged rods. Addition of Ca^{2+} neutralizes the charges permitting an entropy-driven collapse of the fibers.[339]

Another remarkable example is extension of the acrosomal process from a sperm cell of the horseshoe crab *Limulus polyphemus* at fertilization. A bundle of actin filaments in a crystalline state lies coiled around the base of the nucleus. At fertilization the bundle uncoils and slides through a tunnel in the nucleus to form a 60 μm-long acrosomal process within a few seconds. The uncoiled bundle is also crystalline. The coiled bundle is apparently overtwisted and an actin crosslinking protein **scruin** mediates the conformational alteration that takes place.[339] A somewhat related process may be involved in contraction of bacteriophage tails (pp. 363, 364)

Some bacteria glide with a twitching movement induced by rapid retraction of pili.[340] Another type of movement involves the pinching off of vesicles, e.g., of clathrin-coated pits (Fig. 8-27). This is a GTP-driven process that requires a mechanoenzyme called **dynamin**.[341,342]

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Study Questions

1. Describe briefly major aspects of the structure, properties, locations, and functions of each of the following proteins of skeletal muscle.

Actin	Tropomyosin
Myosin	Troponin
Titin	Myomesin
Nebulin	Desmin
α-Actinin	Vimentin
C-protein	

2. Describe the generally accepted sliding filament model of muscle contraction. List some uncertainties in this description.

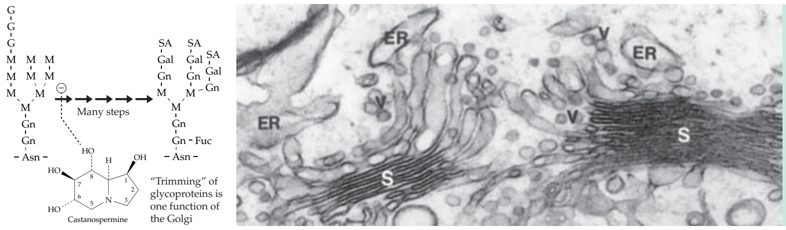
3. Compare mechanisms that regulate contraction in skeletal muscle and in smooth muscle.

4. Compare myosin with kinesins and dyneins. What features do they have in common? What differences can you describe?

5. Compare the properties of actin in skeletal muscle and in nonmuscle cells. What is meant by "treadmilling?" What is "actin-based motility?"

6. The human genome contains more than 100 genes for proteins of the kinesin superfamily. Why?

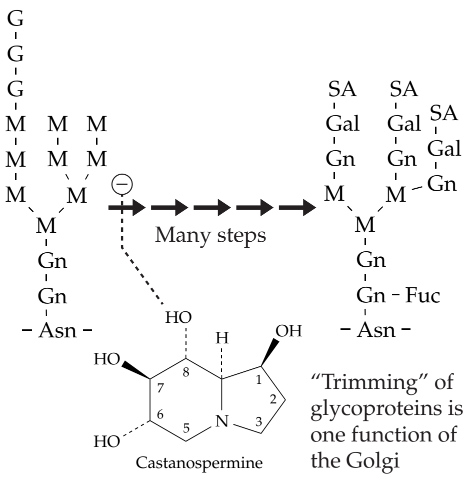
7. Describe some of the major diseases that involve muscle proteins.



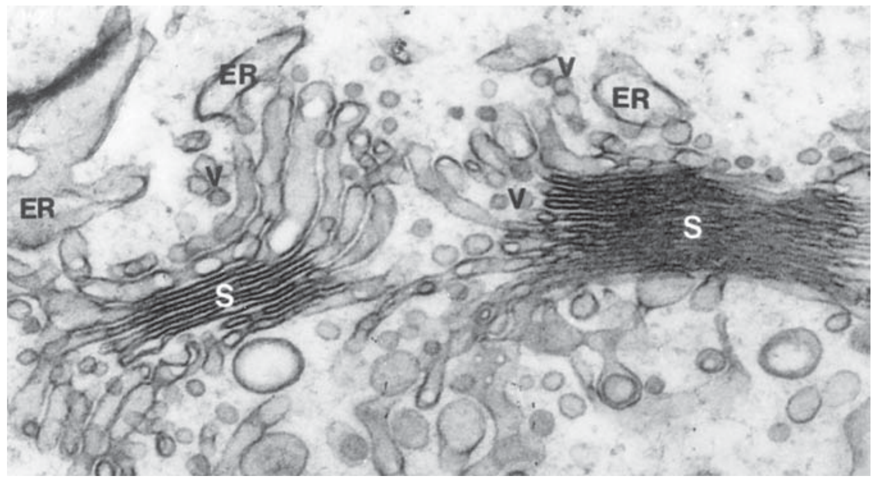
The branched oligosaccharides of glycoprotein surfaces are formed on asparagine side chains of selected cell surface proteins. The oligosaccharide at the left is formed in the ER and is transferred intact (Fig. 20-6) to an acceptor asparagine. It is then trimmed by removal of glucose and mannose units and residues of glucosamine, galactose, and fucose are added as in Fig. 20-7. These reactions begin in the ER and continue in the Golgi apparatus (right). See also Fig. 20-8.

Contents

Some Pathways of Carbohydrate Metabolism 20



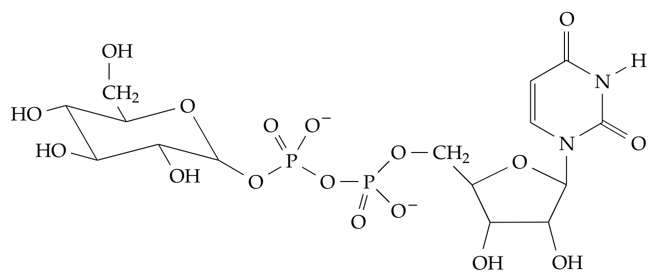
"Trimming" of glycoproteins is one function of the Golgi



The general principles of biosynthesis, as well as the pathways of formation of major carbohydrate and lipid precursors, are considered in Chapter 17. Also described are the processes of gluconeogenesis, the synthesis of glucose 6-phosphate and fructose 6-phosphate from free glucose, and typical polymerization pathways for formation of polysaccharides. In this chapter, additional aspects of the metabolism of monosaccharides, oligosaccharides, polysaccharides, glycoproteins, and glycolipids are considered. These are metabolic transformations that affect the physical properties of cell surfaces and body fluids. They are essential to signaling between cells, to establishment of the immunological identity of individuals, and to the development of strong cell wall materials. Some of the differences in carbohydrates found in bacteria, fungi, green plants, and mammals are considered.

A. Interconversions of Monosaccharides

Chemical interconversions between compounds are easiest at the level of oxidation of carbohydrates. Consequently, many reactions by which one sugar can be changed into another are known. Most of the transformations take place in the "sugar nucleotide derivatives" (see also Eq. 17-56). The first of this group of compounds to be recognized was **uridine diphosphate glucose** (UDPG), which was discovered around 1950 by L. F. Leloir[1,2] during his investigation of the metabolism of galactose 1-*P*. The fact that interconversions of hexoses take place largely at the sugar nucleotide level was unknown at the time. Leloir's studies led to the characterization of both UDP-glucose and UDP-galactose.



Uridine diphosphate D-glucose

Figure 20-1 summarizes pathways by which glucose 6-phosphate or fructose 6-phosphate can be converted into many of the other sugars found in living things. Galactose and mannose can also be interconverted with the other sugars. A kinase forms **mannose 6-phosphate** which equilibrates with fructose 6-phosphate. Galactokinase converts free galactose to **galactose 1-phosphate**, which can be isomerized to glucose 1-phosphate by the reactions of Eq. 20-1. Fructose, an important human dietary constituent[3] derived largely by hydrolysis of sucrose, can also be formed in tissues via the **sorbitol pathway**[4] (Box 20-A). Fructose can be phosphorylated to fructose 1-phosphate by liver **fructokinase**. We have no mutase able to convert fructose 1-*P* to fructose 6-*P*, but a special aldolase cleaves fructose 1-*P* to dihydroxyacetone phosphate and free glyceraldehyde. Lack of this aldolase leads to occasionally observed cases of fructose intolerance.[5,6] The glyceraldehyde formed from fructose can be metabolized by reduction to glycerol followed by phosphorylation (glycerol kinase) and reoxidation to dihydroxyacetone phosphate. Some phosphorylation of fructose 1-*P* to fructose 1,6-P_2 apparently also occurs.[7] Interconversion of ribose 5-*P* and other sugar phosphates is a central part of the pentose phosphate pathway (Fig. 17-8). Free ribose can be phosphorylated by a **ribokinase**.[8]

Oxidation of UDP-glucose in two steps[9,9a] by NAD+ yields **UDP-glucuronic** acid, which can be epimerized to **UDP-galacturonic** acid. Likewise (see bottom of Fig. 20-1), **guanosine disphosphate-mannose** (GDP-mannose) is oxidized to **GDP-mannuronic acid**, which undergoes 4-epimerization to **GDP-guluronic acid**. Looking again at the top of the scheme, notice that UDP-D-glucuronic acid may be epimerized at the 5 position to **UDP-L-iduronic acid**. However, the iduronic acid residues in dermatan sulfate arise by inversion at C-5 of D-glucuronic acid residues in the polymer.[10] The mechanism of these reactions, like that of the decarboxylation of UDP-glucuronic acid to UDP-xylose (near the top of Fig. 20-1), apparently have not been well investigated.

Notice that glucuronic acid is abbreviated GlcA, in accord with IUB recommendations. However, many authors use GlcUA, ManUA, etc., for the uronic acids.

1. The Metabolism of Galactose

The reactions of galactose have attracted biochemists' interest because of the occurrence of the rare (30 cases / million births) hereditary disorder **galactosemia**. When this defect is present, the body cannot transform galactose into glucose metabolites but reduces it to the sugar alcohol **galactitol** or oxidizes it to **galactonate**, both products being excreted in the urine. Unfortunately, severe gastrointestinal troubles often appear within a few days or weeks of birth. Growth is slow and cataracts develop in the eyes, probably as a result of the accumulating galactitol. Death may come quickly from liver damage. Fortunately, galactose-free diets can be prepared for young infants, and if the disease is diagnosed promptly the most serious damage can be avoided. However, it has not been possible to prevent long-term effects that include speech difficulties, learning disabilities, and ovarian dysfunction.[1,11]

In some less seriously affected galactosemic patients **galactokinase** (Eq. 20-1, step *a*) is absent, but it is more often **galactose-1-phosphate uridyltransferase** (Eq. 20-1, step *b*) that is missing or inactive.[12–15a] This enzyme transforms galactose 1-*P* to UDP-galactose by displacing glucose 1-*P* from UDP-glucose. The UDP-galactose is then isomerized by the NAD+-dependent

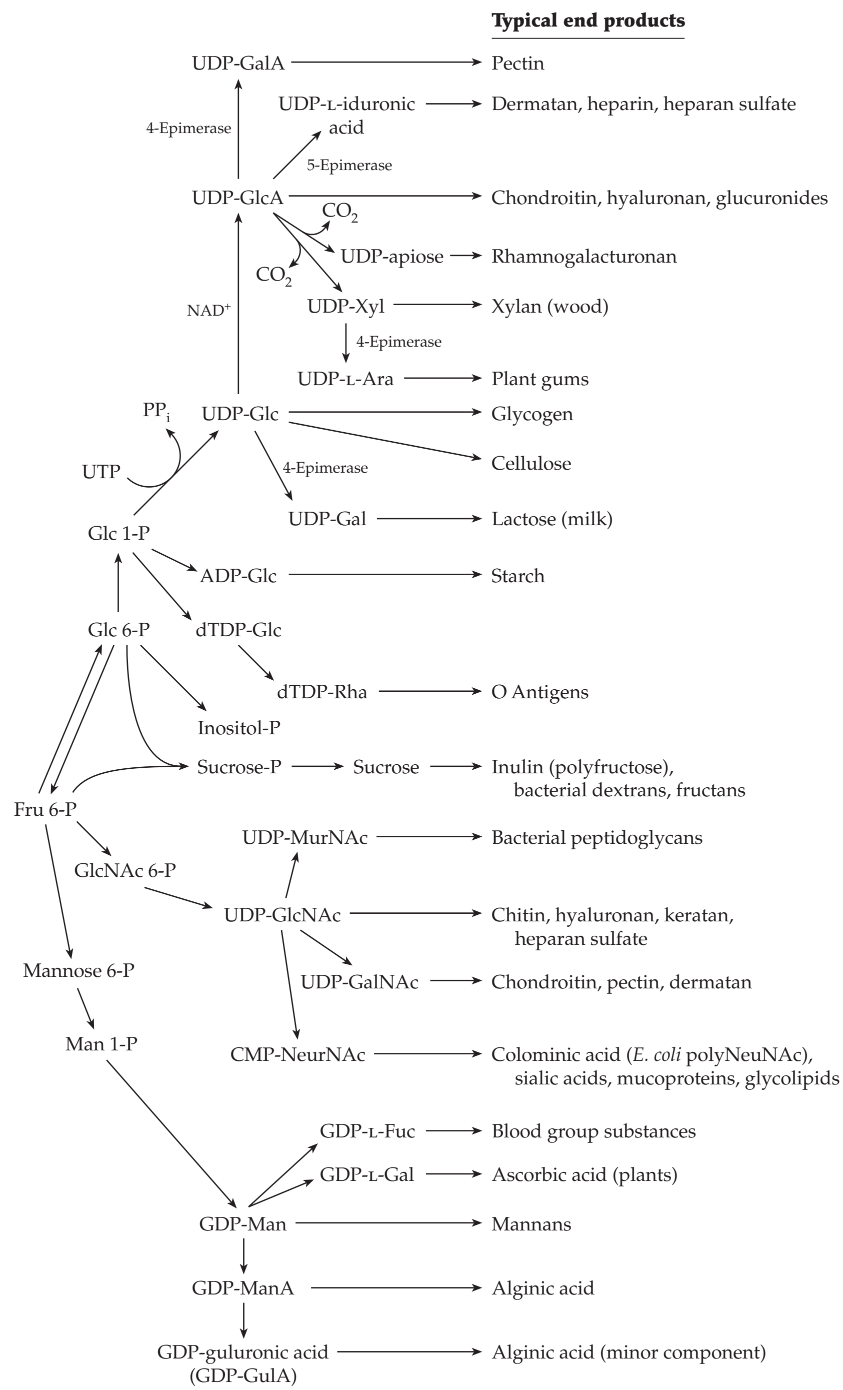
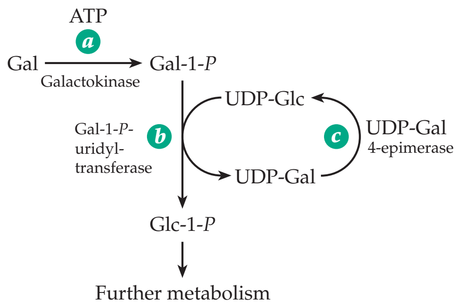


Figure 20-1 Some routes of interconversion of monosaccharides and of polymerization of the activated glycosyl units.



(20-1)

UDP-Gal 4-epimerase[16–16b] (Eq. 20-1, step *c*; see also Eq. 15-13 and accompanying discussion). Absence of this enzyme also causes galactosemia.[11] The overall effect of the reactions of Eq. 20-1 is to transform galactose into glucose 1-*P*. At the same time, the 4-epimerase can operate in the reverse direction to convert UDP-glucose to UDP-galactose, when the latter is needed for biosynthesis (Fig. 20-1).

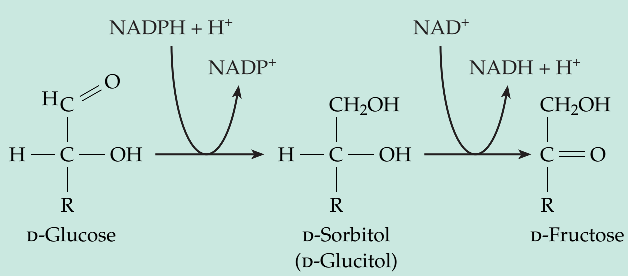
Another enzyme important to galactose metabolism, at least in *E. coli*, is **galactose mutarotase**.[17] Cleavage of lactose by β-galactosidase produces β-D-galactose which must be converted to the α-anomer by the mutarotase before it can be acted upon by galactokinase. Galactose is present in most glycoproteins and glycolipids in the pyranose ring form. However, in bacterial O-antigens, in cell walls of mycobacteria and fungi, and in some protozoa glactose occurs in the furanose form. The precursor is UDP-Gal*f*, which is formed from UDP-Gal*p* by **UDP-Gal*p* mutase**.[17a]

2. Inositol

Related to the monosaccharides is the hexahydroxycyclohexane ***myo*-inositol** (Eq. 20-2). This **cyclitol**, which is apparently present universally within cells (Fig. 11-9), can be formed from glucose-6-*P* according to Eq. 20-2 using a synthase that contains bound

BOX 20-A FRUCTOSE FOR SPERM CELLS VIA THE POLYOL PATHWAY

An interesting example of the way in which the high [NADPH]/[NADP$^+$] and [NAD$^+$]/[NADH] ratios in cells can be used to advantage is found in the metabolism of sperm cells. Whereas D-glucose is the commonest sugar used as an energy source by mammalian cells, spermatozoa use principally D-fructose, a sugar that is not readily metabolized by cells of surrounding tissues.[a–c] Fructose, which is present in human semen at a concentration of 12 mM, is made from glucose by cells of the seminal vesicle by reduction with NADPH to the sugar alcohol D-sorbitol, which in turn is oxidized in the 2 position by NAD$^+$. The combination of high [NADPH]/[NADP$^+$] and high [NAD$^+$]/[NADH] ratio is sufficient to shift the equilibrium far toward fructose formation.[d]



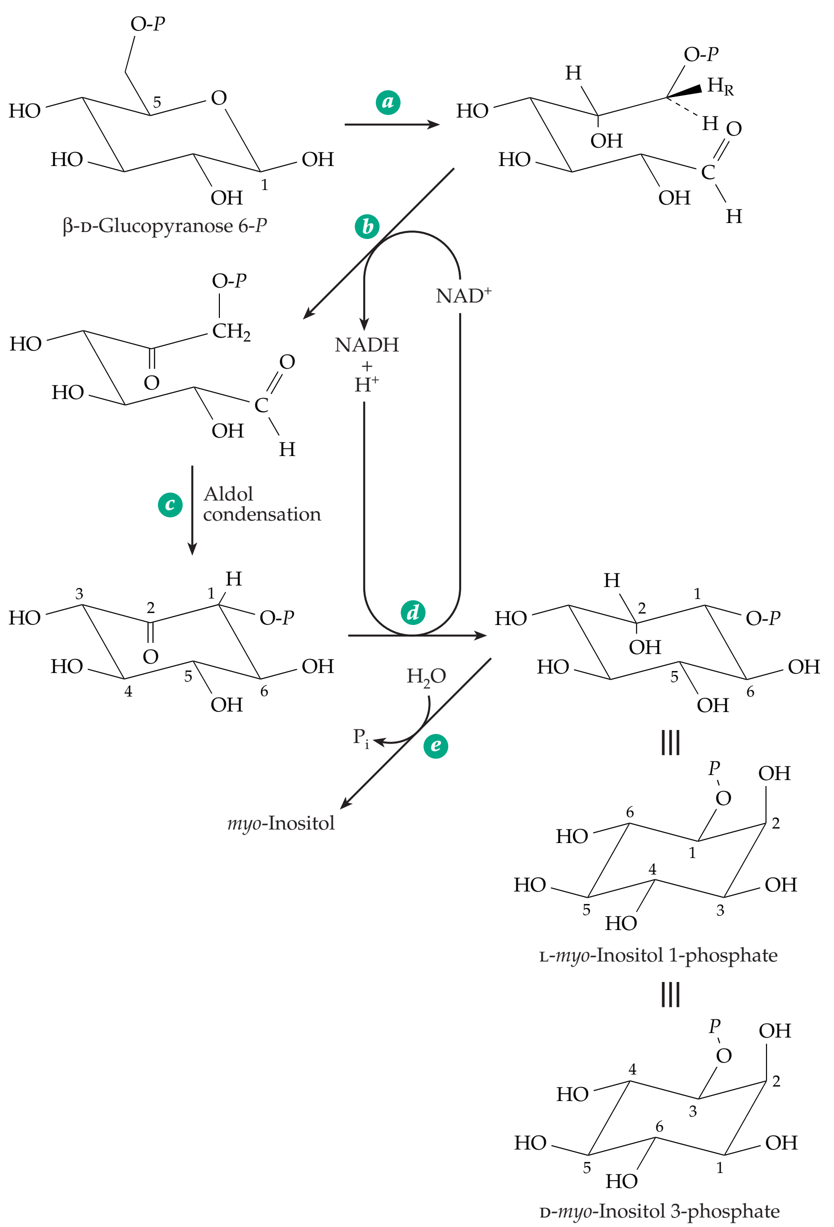
The polyol pathway is an active bypass of the dominant glycolysis pathway in many organisms.[e] Sorbitol and other polyols such as glycerol, erythritol, threitol, and ribitol serve as cryoprotectants in plants, insects, and other organisms.[f] Sorbitol is also an important osmolyte in some organisms (see Box 20-C). On the other hand, accumulation of sorbitol in lenses of diabetic individuals has often been blamed for development of cataract. However, doubts have been raised about this conclusion. The polyol pathway is more active than normal in diabetes, and there is evidence that the increased flow in this pathway may lead to an increase in oxidative damage to the lens. This may result, in part, from the depletion of NADPH needed for reduction of oxidized glutathione in the antioxidant system.[g] Aldose reductase inhibitors, which reduce the rate of sorbitol formation, decrease cataract formation. However, the reason for this is not yet clear.[h]

[a] McGilvery, R. W. (1970) *Biochemistry, A Functional Approach*, Saunders, Philadelphia, Pennsylvania (pp. 631–632)
[b] Hers, H. G. (1960) *Biochim. Biophys. Acta.* **37**, 127–
[c] Gitzelmann, R., Steinmann, B., and Van den Berghe, G. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 1 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 905–935, McGraw-Hill, New York
[d] Prendergast, F. G., Veneziale, C. M., and Deering, N. G. (1975) *J. Biol. Chem.* **250**, 1282–1289
[e] Luque, T., Hjelmqvist, L., Marfany, G., Danielsson, O., El-Ahmad, M., Persson, B., Jörnvall, H., and Gonzàlez-Duarte, R. (1998) *J. Biol. Chem.* **273**, 34293–34301
[f] Podlasek, C. A., and Serianni, A. S. (1994) *J. Biol. Chem.* **269**, 2521–2528
[g] Lee, A. Y. W., and Chung, S. S. M. (1999) *FASEB J.* **13**, 23–30
[h] Srivastava, S., Watowich, S. J., Petrash, J. M., Srivastava, S. K., and Bhatnagar, A. (1999) *Biochemistry* **38**, 42–54

NAD+. In addition to the two redox steps (Eq. 20-2, *b* and *d*), this enzyme catalyzes both the conversion of the β anomer of glucose 6-*P* to the open-chain aldehyde form and the internal aldol condensation of Eq. 20-2, step *c*.[18-19b] The pro-*R* hydrogen at C-6 of glucose 1-*P* is lost in step *b* while the pro-*S* hydrogen is retained.[20] The ring numbering system is different for glucose and for the inositols, C-5 of glucose 1-*P* becoming C-2 of L-*myo*-inositol. Since *myo*-inositol contains a plane of symmetry D- and L- forms are identical. However, they are numbered differently (Eq. 20-2). The phosphoinositides and inositol polyphosphates are customarily numbered as derivatives of D-*myo*-inositol.

Synthesis of inositol by animals is limited and *myo*-inositol is sometimes classified as a vitamin. Mice grow poorly and lose some of their hair if deprived of dietary inositol. Various phosphate esters of inositol occur in nature. For example, large amounts of the hexaphosphate (**phytic acid**) are present in grains, usually as the calcium or mixed Ca^{2+}–Mg^{2+} salts known as **phytin**. The two apical cells of the 28-cell larvae of mesozoa (Fig. 1-12A) contain enough magnesium phytate in granular form to account for up to half of the weight of the larvae.[21] Inositol pentaphosphate is an allosteric activator for hemoglobin in birds and turtles (p. 358). Di-*myo*-inositol-1,1'-phosphate is an osmolyte in some hypothermophilic archaea.[19a] Inositol is a component of **galactinol**, the β glycoside of D-galactose with inositol (Eq. 20-15). Galactinol, as well as free inositol, circulates in human blood and in plants and may be a precursor of cell wall polysaccharides. However, in our own bodies the greatest importance for inositol doubtless lies in the inositol-containing phospholipids known as **phosphoinositides** (Figs. 8-2, 11-9, 21-5). Their function in generation of "second messengers" for various hormones is dealt with in Chapters 11 and 21.

A person typically ingests daily about one gram of inositol, some in the free form, some as phosphoinositides, and some as phytin. As much as four grams of inositol per day may be synthesized in the kidneys.[22] Breast milk is rich in inositol and dietary supplementation with inositol has increased survival of premature infants with respiratory distress syndrome.[22] The action of insulin is reported to be improved by administration of D-*chiro*-inositol (p. 998) to women with polycystic ovary syndrome.[22a]



(20-2)

3. D-Glucuronic Acid, Ascorbic Acid, and Xylitol

In bacteria, as well as in animal kidneys,[23] inositol may be converted to D-glucuronic acid (Fig. 20-1) with the aid of an oxygenase. Free glucuronic acid may also be formed by animals from glucose or from UDP-glucose (Fig. 20-2). Within the animal body glucuronic acid can be reduced with NADH (Fig. 20-2, step *a*) to yield **L-gulonic acid**, an aldonic acid that could also be formed by oxidation at the aldehyde end of the sugar **gulose**. Because C-6 of the glucuronic acid has become C-1 of gulonic acid, the latter belongs to the L family of

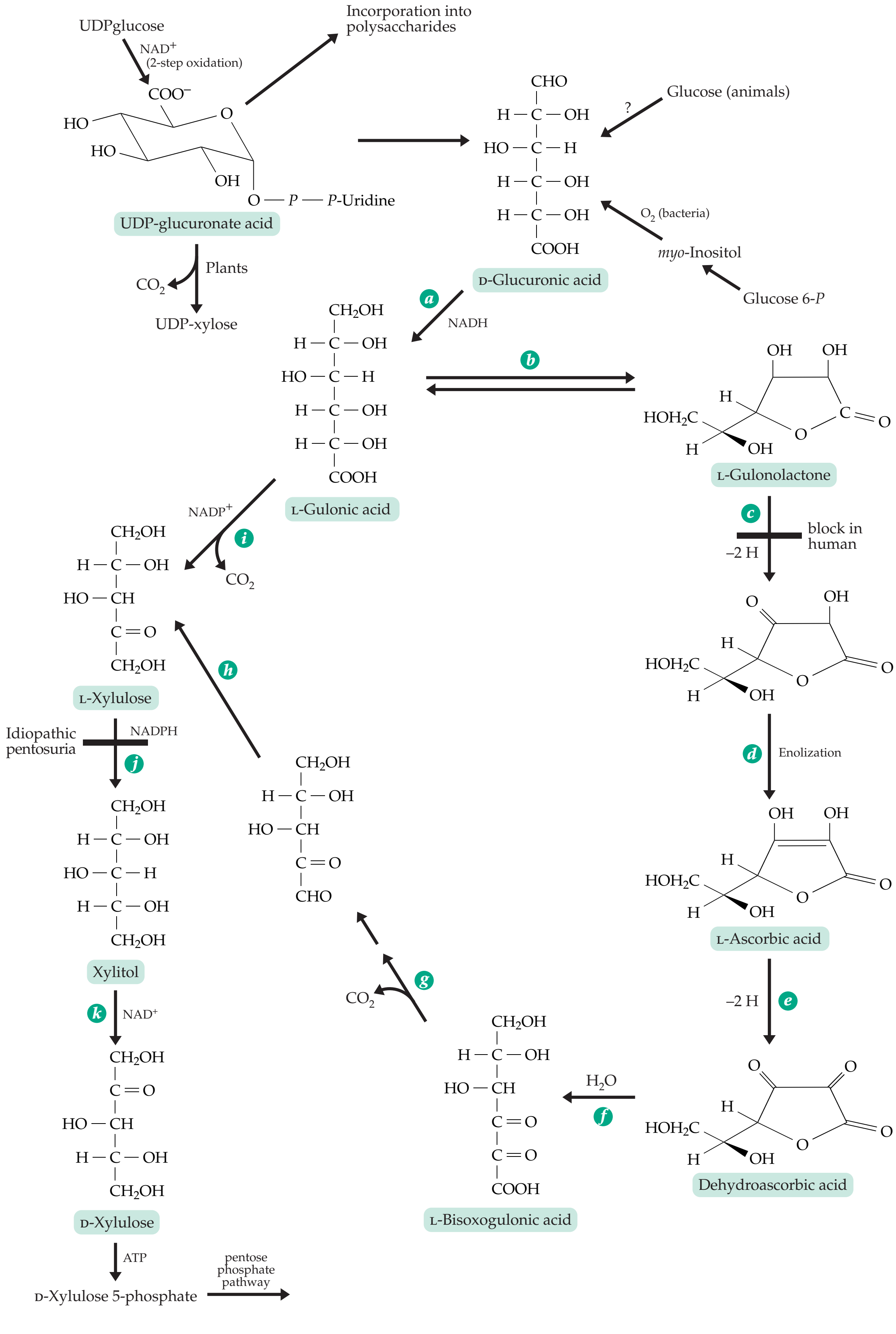
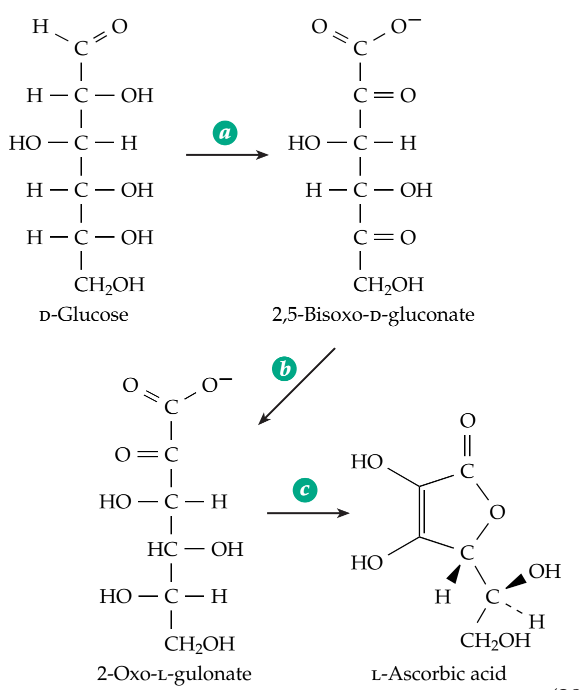


Figure 20-2 Some pathways of metabolism of D-glucuronic acid and of ascorbic acid, vitamin C.

sugars. Gulonic acid can be converted to a cyclic lactone (step *b*) which, in a two-step process involving dehydrogenation and enolization (steps *c* and *d*), is converted to **L-ascorbic acid**. This occurs in most higher animals.[24] However, the dehydrogenation step is lacking in human beings and other primates, in the guinea pig, and in a few other species. One might say that we and the guinea pig have a genetic defect at this point which obliges us to eat relatively large quantities of plant materials to satisfy our bodily needs for ascorbic acid (see Box 18-D). Gulonolactone oxidase is one of the enzymes containing covalently bound 8α-(N^1-histidyl)riboflavin.[25] The defective human gene for this enzyme has been identified, isolated, and sequenced. It is found to have accumulated a large number of mutations, which have rendered it inactive and now only a pseudogene.[26] Mice with an inactivated gulonolactone oxidase have a dietary requirement for vitamin C similar to that of humans. They suffer severe vascular damage on diets marginal in ascorbic acid.[26a] Even in rodents Na^+-dependent ascorbic acid transporters are present in metabolically active tissues to bring the vitamin from the blood into cells.[26b]

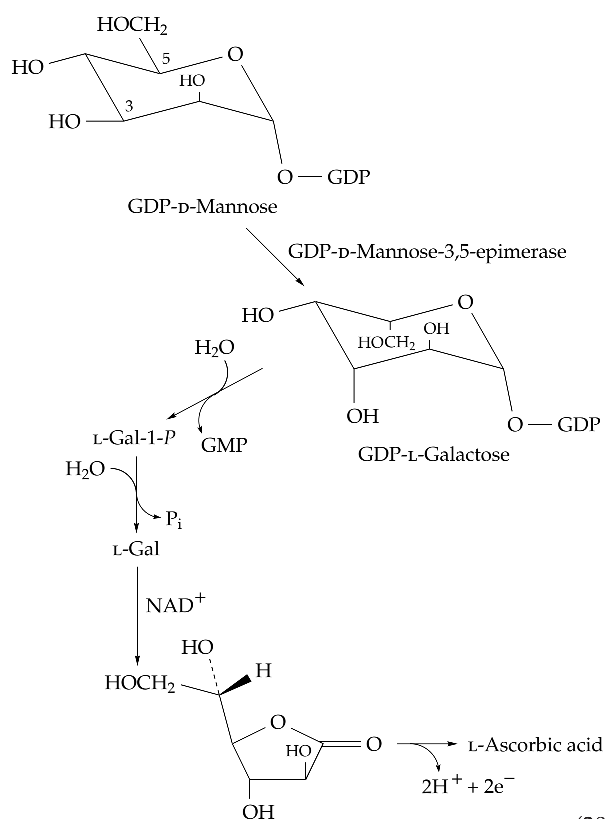
A clever bit of genetic engineering has permitted the conversion of D-glucose to 2-oxo-L-gulonate in the enzymatic sequence of Eq. 20-3, *a*, *b*.



(20-3)

The bacterium *Erwinia herbicola* naturally has the ability to oxidize glucose to 2,5-bisoxo-D-gluconate (Eq. 20-3, step *a*) but cannot carry out the next step, the stereospecific reduction to 2-oxo-L-gulonate. However, a gene encoding a suitable reductase was isolated from a genomic library from *Corynebacterium*. The cloned gene was fused to an *E. coli trp* promoter (see Chapter 28) and was introduced in a multicopy plasmid into *E. herbicola*. The resultant organism can carry out both steps *a* and *b* of Eq. 20-3 leaving only step *c*, a nonenzymatic acid-catalyzed reaction, to complete an efficient synthesis of vitamin C from glucose.[27]

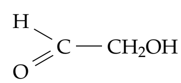
Higher plants make large amounts of L-ascorbate, which in leaves may account for 10% of the soluble carbohydrate content.[28] However, the pathway of synthesis differs from that in Fig. 20-2. Both D-mannose and L-galactose are efficient precursors. The pathway in Eq. 20-4, which starts with GDP-D-mannose and utilizes known enzymatic processes, has been suggested.[28,29] The GDP-D-mannose-3,5-epimerase is a well documented but poorly understood enzyme. Multistep mechanisms related to that of UDP-glucose 4-epimerase (Eqs. 20-1, 15-14) can be envisioned.



(20-4)

Ascorbic acid is readily oxidized to dehydroascorbic acid (Box 18-D; Fig. 20-2, step *e*), which may be hydrolyzed to L-bisoxogulonic acid (step *f*). The latter, after decarboxylation and reduction, is converted to L-xylulose (steps *g* and *h*), a compound that can also be formed by a standard oxidation and decarboxylation sequence on L-gulonic acid (step *i*). Reduction of xylulose to xylitol and oxidation of the latter with NAD^+ (steps *j* and *k*) produces D-xylulose, which can be phosphorylated with ATP and enter the pentose phosphate pathways. A metabolic variation produces a condition called **idiopathic pentosuria**. Affected individuals cannot reduce xylulose to xylitol and, hence, excrete large amounts of the pentose into the urine, especially if the diet is rich in glucuronic acid. The "defect" seems to be harmless, but the sugar in the urine can cause the condition to be mistaken for diabetes mellitus.[30]

Xylitol is as sweet as sucrose and has been used as a food additive. Because it does not induce formation of dental plaque, it is used as a replacement for sucrose in chewing gum. It appeared to be an ideal sugar substitute for diabetics. However, despite the fact that it is already naturally present in the body, ingestion of large amounts of xylitol causes bladder tumors as well as oxalate stones in rats and mice. Its use has, therefore, been largely discontinued. A possible source of the problem may lie in the conversion by fructokinase of some of the xylitol to D-xylulose 1-*P*, which can be cleaved by the xylulose 1-*P* aldolase to dihydroxyacetone *P* and glycolaldehyde.

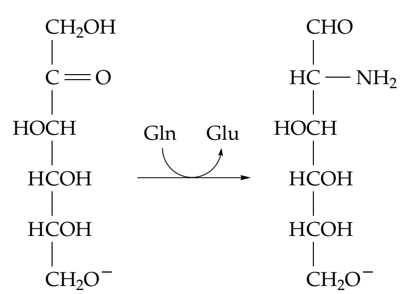


Glycolaldehyde

The latter can be oxidized to oxalate and may also be carcinogenic. As indicated in the upper left corner of Fig. 20-2, UDP-glucuronate can be decarboxylated to UDP-xylose.

4. Transformations of Fructose 6-Phosphate

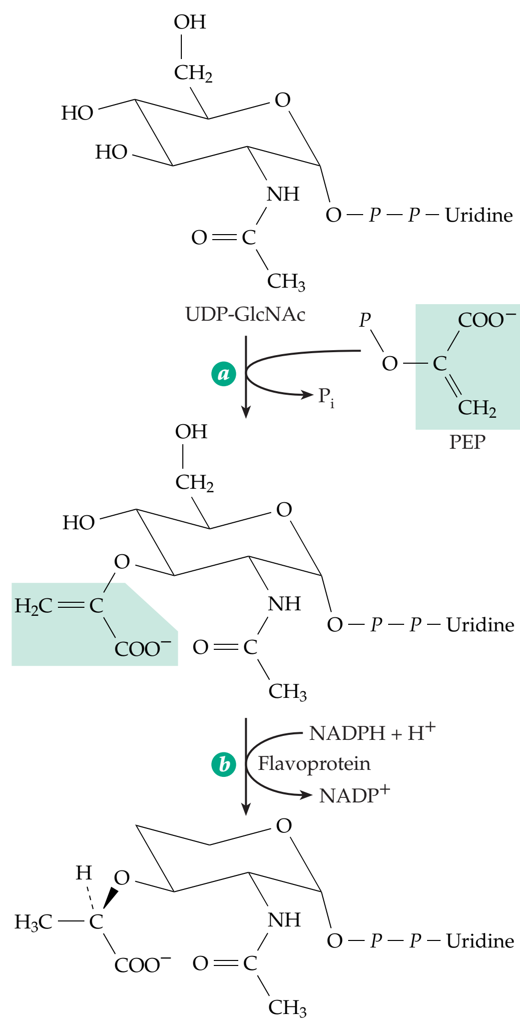
Biosynthesis of **D-glucosamine 6-phosphate** is accomplished by reaction of fructose 6-*P* with glutamine (Eq. 20-5):



Fructose 6-*P* D-Glucosamine 6-*P* (20-5)

Glutamine is one of the principal combined forms of ammonia that is transported throughout the body (Chapter 24). Glucosamine 6-phosphate synthase, which catalyzes the reaction of Eq. 20-5, is an amidotransferase of the N-terminal nucleophile hydrolase superfamily (Chapter 12).[31] It hydrolyzes the amide linkage of glutamine. The released ammonia presumably reacts with the carbonyl group of fructose 6-*P* to form an imine,[32–34a] which then undergoes a reaction analogous to that catalyzed by sugar isomerases.[35] The resulting D-glucosamine 6-*P* is acetylated on its amino group by transfer of an acetyl group,[36] and a mutase moves the phospho group to form *N*-acetylglucosamine 1-*P*. In *E. coli* acetylation occurs on GlcN 1-*P* and is catalyzed by a bifunctional enzyme that also has mutase activity.[37-37b] The resulting *N*-acetylglucosamine 1-*P* is converted to UDP-*N*-acetylglucosamine (UDP-GlcNAc) with cleavage of UTP to inorganic pyrophosphate as in the synthesis of UDP-glucose (Eq. 17-56). Cells of *E. coli* are also able to catabolize glucosamine 6-phosphate. A **deaminase,** with many properties similar to those of GlcN 6-*P* synthase, catalyzes a reaction resembling the reverse of Eq. 20-5 but releasing NH_3.[38,39]

One of the compounds formed from UDP-GlcNAc is **UDP-*N*-acetylmuramic acid**. The initial step in its synthesis is an unusual type of displacement reaction on the α-carbon of PEP by the 3-hydroxyl group of the sugar (Eq. 20-6, step *a*).[40-41c] Inorganic phosphate is



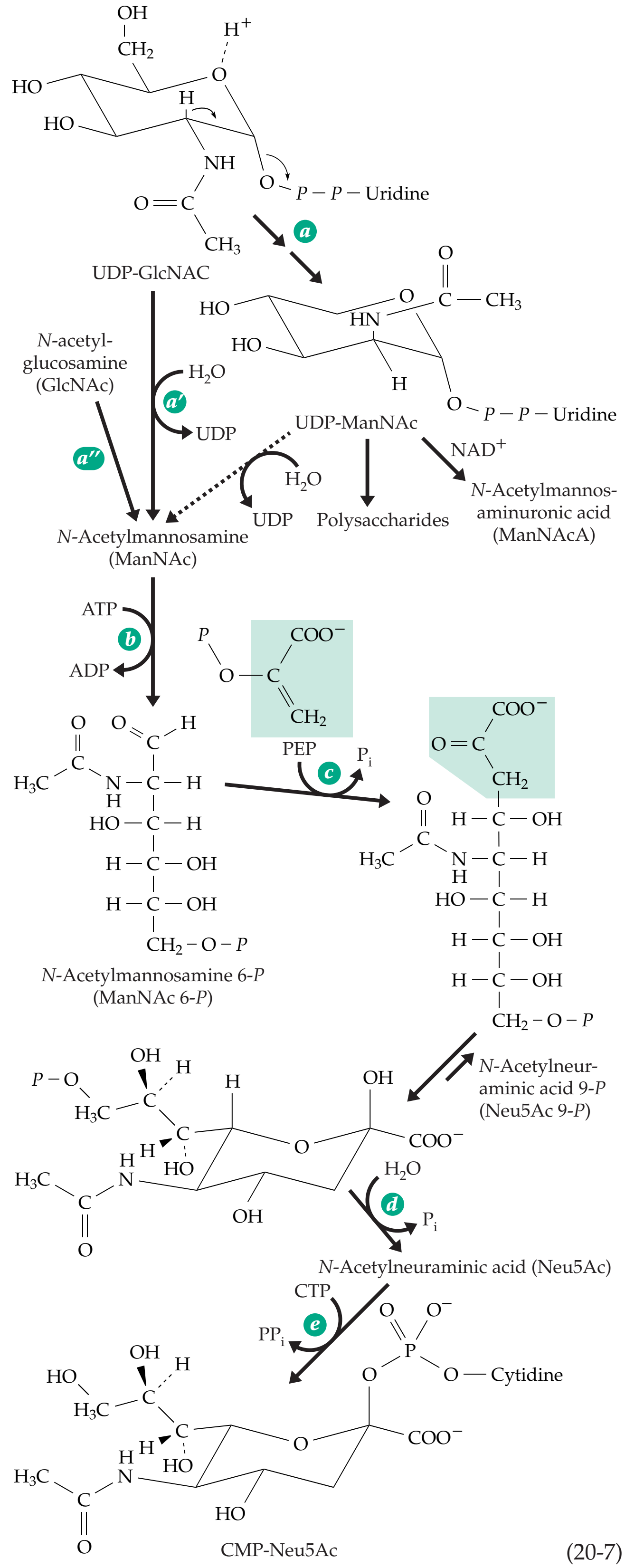
UDP-*N*-acetylmuramic acid
(UDP-MurNAc)

(20-6)

displaced with formation of an enolpyruvyl derivative of UDP-GlcNAc. This derivative is then reduced by NADPH (Eq. 20-6, step *b*).[42-43a] A second sugar nucleotide formed from UDP-GlcNAc is **UDP-*N*-acetylgalactosamine** (UDP-GalNAc), which may be created by the same 4-epimerase that generates UDP-Gal (Eq. 20-1).[44] Some animal tissues such as kidney and liver also have a **GalNAc kinase** that may salvage, for reuse, GalNAc that arises from the degradation of complex polysaccharides.[44] Bacteria may dehydrogenate UDP-GalNAc to UDP-*N*-acetylgalactosaminuric acid (UDP-GalNAcA).[44a]

UDP-GlcNAc can be converted to UDP-*N*-acetylmannosamine (UDP-ManNAc) with concurrent elimination of UDP (Eq. 20-7).[45–47b] This unusual epimerization occurs without creation of an adjacent carbonyl group that would activate the 2-H for removal as a proton. As indicated by the small arrows in Eq. 20-7, step *a′*, the UDP is evidently eliminated. In a bacterial enzyme it remains in the E–S complex and is returned after a conformational change involving the acetamido group. This allows the transient C1–C2 double bond to be protonated from the opposite side (Eq. 20-7, step *a*).[47] In bacteria the UDP-ManNAc may be dehydrogenated to UDP-*N*-acetylmannos-aminuronic acid (ManNAcA). Both ManNAc and ManNAcA are components of bacterial capsules.[47]

In mammals the epimerase (Eq. 20-7, step *a′*) probably utilizes a similar chemical mechanism but eliminates UDP and replaces it with HO^- to give free *N*-acetylmannosamine, which is then phosphorylated on the 6-hydroxyl (Eq. 20-7, step *b*). ManNAc may also be formed from free GlcNAc by another 2-epimerase (step a′′).[47c,d]

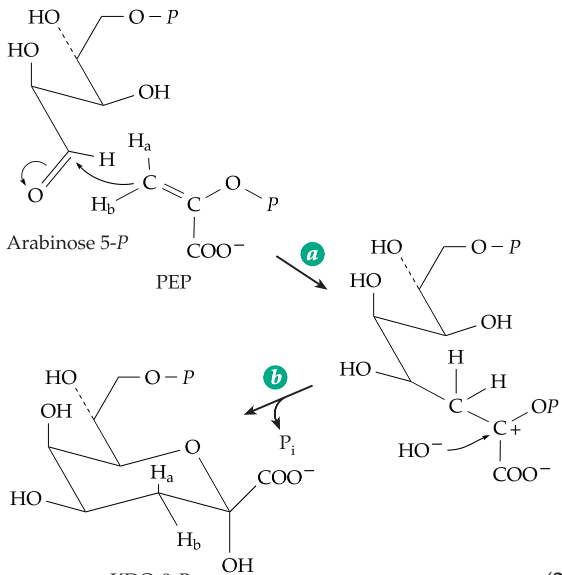


(20-7)

5. Extending a Sugar Chain with Phosphoenolpyruvate (PEP)

The six-carbon chain of ManNAc 6-*P* can be extended by three carbon atoms using an aldol-type condensation with a three-carbon fragment from PEP (Eq. 20-7, step *c*) to give ***N*-acetylneuraminic acid** (sialic acid).[48] The nine-carbon chain of this molecule can cyclize to form a pair of anomers with 6-membered rings as shown in Eq. 20-7. In a similar manner, arabinose 5-*P* is converted to the 8-carbon **3-deoxy-D-*manno*-octulosonic acid** (**KDO**) (Fig. 4-15), a component of the lipopolysaccharide of gram-negative bacteria (Fig. 8-30), and D-Erythrose 4-*P* is converted to 3-deoxy-D-*arabino*-heptulosonate 7-*P*, the first metabolite in the shikimate pathway of aromatic synthesis (Fig. 25-1).[48a] The arabinose-*P* used for KDO synthesis is formed by isomerization of D-ribulose 5-*P* from the pentose phosphate pathway, and erythrose 4-*P* arises from the same pathway.

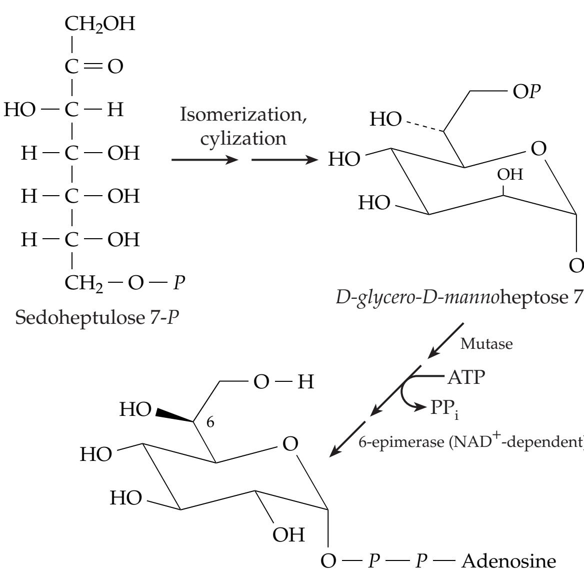
The mechanism of the aldol condenstion that forms these sugars is somewhat unexpected. A reactive enolate anion can be formed from PEP by hydrolytic attack on the phospho group with cleavage of the O–P bond. However, in reactions such as step *a* of



(20-8)

Eq. 20-6, step *c* of Eq. 20-7,and also in EPSP synthase (Eq. 25-4) the initial condensation does not involve O–P cleavage. NMR studies of the action of KDO synthase reveal that the C–O bond of PEP is cleaved as is indicated in Eq. 20-8.[49-52b] The *si* face of PEP faces the *re* face of the carbonyl group of the sugar phosphate. A carbanionic center is generated at C-3 of PEP with possible participation of the phosphate oxygen as well as electrostatic stabilization of the carbocation formed in step *a*. Ring closure (step *b*) occurs with loss of P_i. The immediate product of the aldol condensation, in Eq. 20-7, is *N*-acetylneuraminic acid 9-phosphate, which is cleaved through phosphatase action (step *d*) and is activated to the CMP derivative by reaction with CTP (Eq. 20-7,step *e*).[52c] Further alterations may occur. For example, CMP-Neu5Ac is hydroxylated to form CMP-*N*-glycolylneuraminic acid.[53] Furthermore, an additional type of sialic acid, 2-oxo-3-deoxy-D-*glycero*-D-*galacto*-nononic acid (**KDN**), has been found in human developmentally regulated glycoproteins and also in many other organisms.[54-55a] It has an –OH group in the 5-position rather than the acetamido group of the other sialic acids. Like NeuNAc it is activated by reaction with CTP forming CMP-KDN. These activated monosaccharides differ from most others in being derivatives of a CMP rather than of CDP. More than 40 different naturally occuring variations of sialic acid have been identified.[55b]

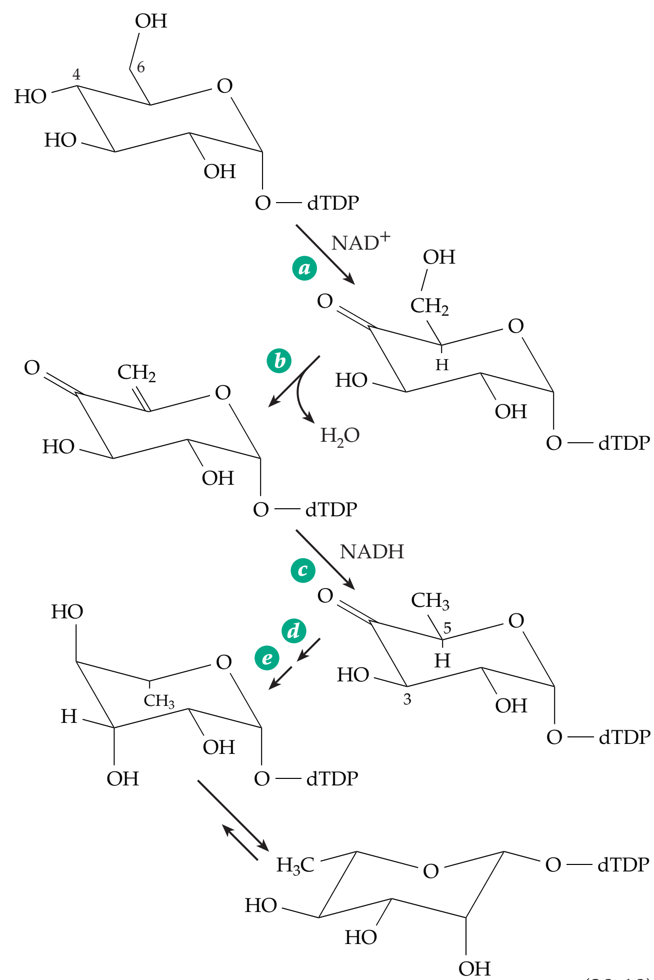
In a similar fashion, KDO is converted to the β-linked **CMP-KDO**,[56-56b] which is incorporated into lipid A as shown in Fig. 20-10. The ADP derivative of the ***L-glycero-D-manno*-heptose** (Fig. 4-15), which is also present in the lipopolysaccharide of gram-negative bacteria, is formed from sedoheptulose 7-*P* in a five-step process (Eq. 20-9).[57-58b]



(20-9)

6. Synthesis of Deoxy Sugars

Metabolism of sugars often involves dehydration to α,β-unsaturated carbonyl compounds. An example is the formation of 2-oxo-3-deoxy derivatives of sugar acids (Eq. 14-36). Sometimes a carbonyl group is



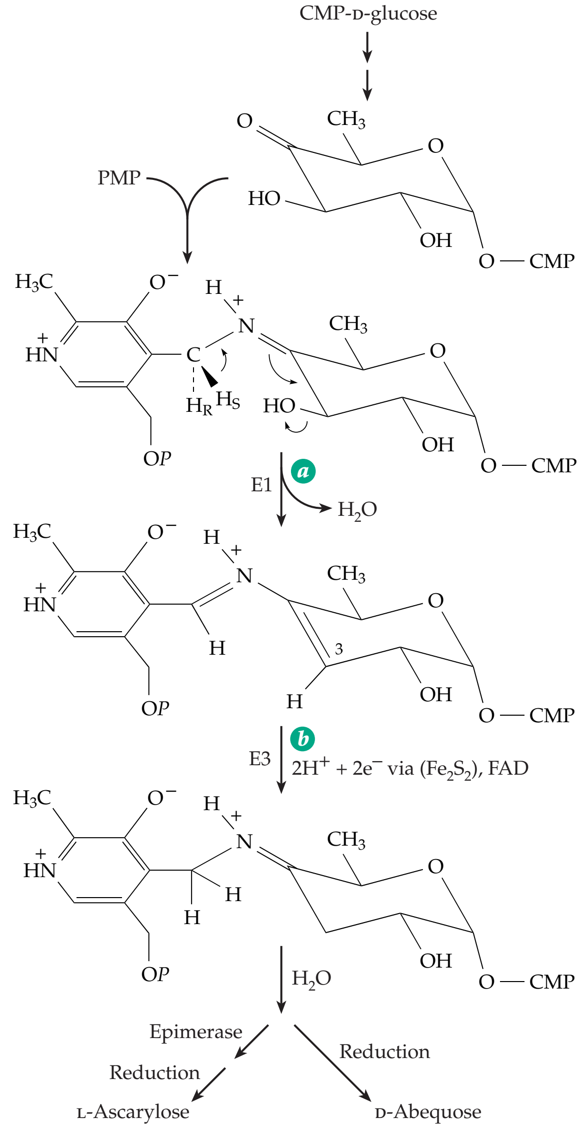
L-Rhamnose (20-10)

created by oxidation of an –OH group, apparently for the sole purpose of promoting dehydration. For example, the biosynthesis of L-rhamnose from D-glucose is a multistep process (Eq. 20-10) that takes place while the sugars are attached to deoxythymidine diphosphate.[59,59a,b] Introduction of the carbonyl group by dehydrogenation with tightly bound NAD^+ (Eq. 20-10, step *a*) is followed by dehydration (step *b*).[59c,d] To complete the sequence, the double bond formed by dehydration is reduced (step *c*) by the NADH produced in step *a*. A separate enzyme, a 3,5-epimerase catalyzes inversion at both C-3 and C-5 (step *d*).[59e] Finally, a third enzyme is needed for a second reduction (step *e*) using NADPH.[59f] The biosynthesis of **GDP-L-fucose** from GDP-D-mannose occurs by a parallel sequence.[60-61b]

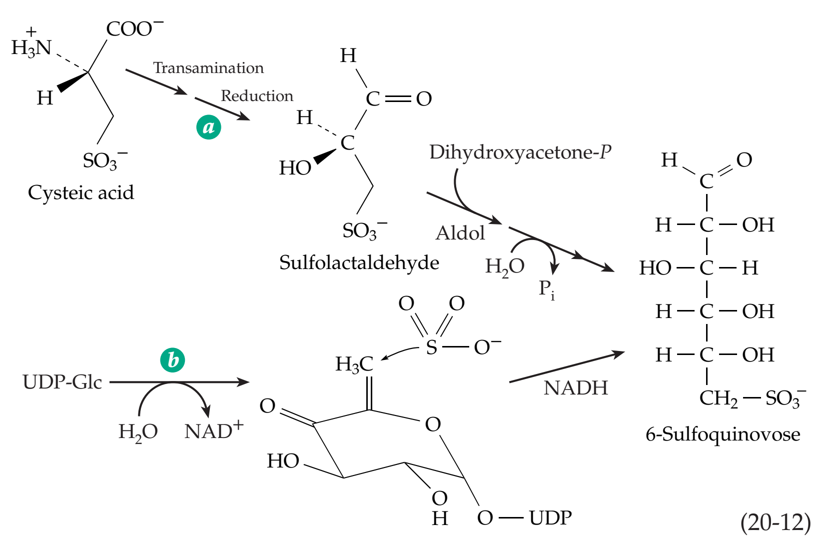
The metabolism of free L-fucose (6-deoxy-L-galactose), which is present in the diet and is also generated by degradation of glycoproteins, resembles the Entner–Doudoroff pathway of glucose metabolism (Eq. 17-18). Similar degradative pathways act on D-arabinose and L-galactose.[60]

Bacterial surface polysaccharides contain a variety of dideoxy sugars. The four 3,6-dideoxy sugars **D-paratose** (3,6-dideoxy-D-glucose), **D-abequose** (3,6-dideoxy-D-galactose), **D-tyvelose** (3,6-dideoxy-D-mannose), and **L-ascarylose** (3,6-dideoxy-L-mannose), whose structures are shown in Fig. 4-15, arise from CDP-glucose.[60a] This substrate is first converted, in reactions parallel to the first three steps of Eq. 20-10, to 4-oxo-6-deoxy-CDP-glucose which reacts in two steps with pyridoxamine 5'-phosphate (PMP) and NADH (Eq. 20-11). This unusual reaction[62–65] is catalyzed by a two-enzyme complex. The first component, E1, catalyzes the formation of a Schiff base of the substrate with PMP and a transamination, which also accomplishes dehydration, to give an unsaturated sugar ring (Eq. 20-11, step *a*). The protein also contains an Fe_2S_2 center suggesting a possible one-electron transfer. The second component, E_3, contains both an Fe_2S_2 plant type ferredoxin center and bound FAD.[65] Observation by EPR spectroscopy revealed accumulation of an organic free radical[64] that may be an intermediate in step *b* of Eq. 20-11. Hydrolysis, epimerization at C-5, and reduction yields L-ascarylose. A similar reaction sequence without the last epimerization would yield D-abequose. CDP-D-tyvelose arises by C-2 epimerization of CDP-D-paratose.[65a] Other unusual sugars[66–68] are formed from intermediates in Eq. 20-11. One is a **3-amino-3,4,6-trideoxyhexose** in which the amino group has been provided by transamination[67] (see also Box 20-B).

The unusual sulfur-containing sugar **6-sulfoquinovose** is present in

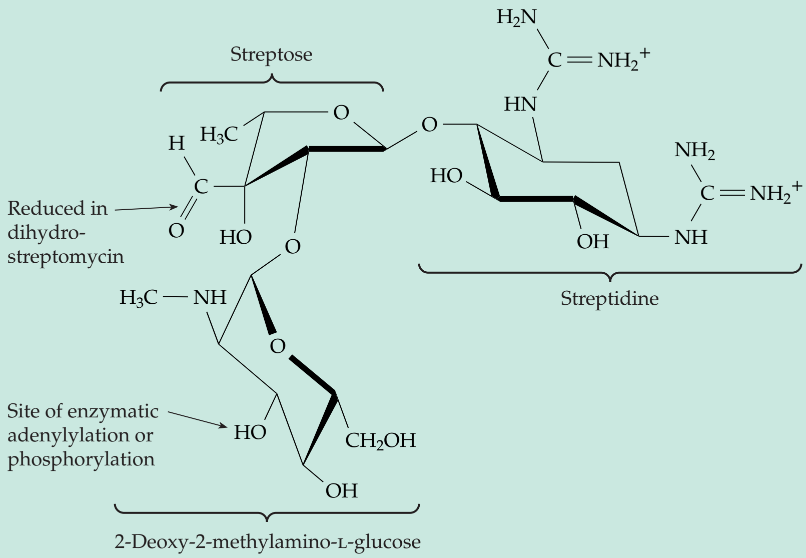


(20-11)



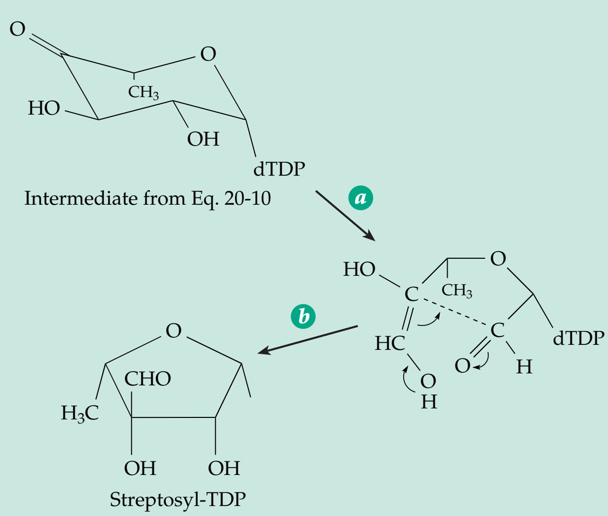
(20-12)

BOX 20-B THE BIOSYNTHESIS OF STREPTOMYCIN



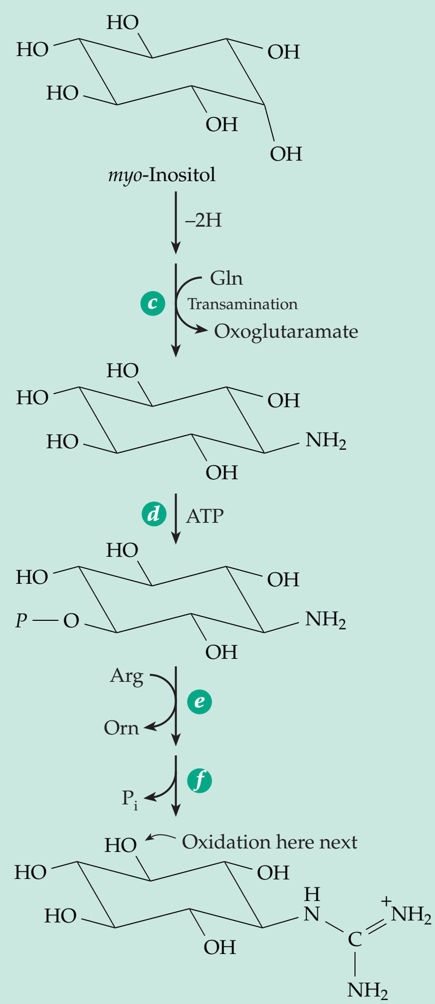
Streptomycin

Streptomycin, the kanamycins, neomycins, and gentamycins form a family of medically important **aminoglycoside antibiotics**.[a] They are all water-soluble basic carbohydrates containing three or four unusual sugar rings. D-Glucose is a precursor of streptomycin, all three rings being derived from it. While the route of biosynthesis of 2-deoxy-2-methylamino-L-glucose is not entirely clear, the pathways to L-streptose and streptidine, the other two rings, have been characterized.[b–d] The starting material for streptidine synthesis is a nucleoside diphosphate sugar, which is an intermediate in the synthesis of L-rhamnose (Eq. 20-10). The carbon–carbon chain undergoes an aldol cleavage as shown in step *a* of the following equation:



The ring-open product is written here as an enediol, which is able to recyclize in an aldol condensation (step *b*) to form a five-membered ring with a branch at C-3. The L-streptosyl nucleoside diphosphate formed in this way serves as the donor of streptose to streptomycin.

The basic cyclitol streptidine is derived from *myo*-inositol, which has been formed from glucose 6-*P* (Eq. 20-2). The guanidino groups are introduced by oxidation of the appropriate hydroxyl group to a carbonyl group followed by transamination from a specific amino donor. In the first step, illustrated by the following equation, glutamine is the amino donor for the transamination, the oxoacid product being α-oxoglutaramic acid.



BOX 20-B THE BIOSYNTHESIS OF STREPTOMYCIN (continued)

The amino group on the ring now receives an amidine group, which is transferred from arginine by nucleophilic displacement[e] in a reaction resembling that in the synthesis of urea (see Fig. 24-10, step *h*). However, there is first a phosphorylation at the 2 position. After the amidine transfer has occurred to form the guanidino group, the phospho group is hydrolyzed off by a phosphatase. This is another phosphorylation–dephosphorylation sequence (p. 977) designed to drive the reaction to completion in the desired direction. The second guanidino group is introduced in an analogous way by oxidation at the 3 position followed by transamination, this time with the amino group being donated by alanine. Again, a phosphorylation is followed by transfer of an amidine group from arginine. The final hydrolytic removal of the phospho group (which this time is added at C-6) does not occur until the two other sugar rings have been transferred on from nucleoside diphosphate precursors to form streptomycin phosphate.

As with other antibiotics,[f–i] streptomycin is subject to inactivation by enzymes encoded by genetic resistance factors (Chapter 26). Among these are enzymes that transfer phospho groups or adenylyl groups onto streptomycin at the site indicated by the arrow in the structure.[j,k] Thus, dephosphorylation at one site generates the active antibiotic as the final step in the biosynthesis, while phosphorylation at another site inactives the antibiotic.

[a] Benveniste, R., and Davies, J. (1973) *Ann. Rev. Biochem.* **42**, 471–506
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[c] Walker, J. B., and Skorvaga, M. (1973) *J. Biol. Chem.* **248**, 2441–2446
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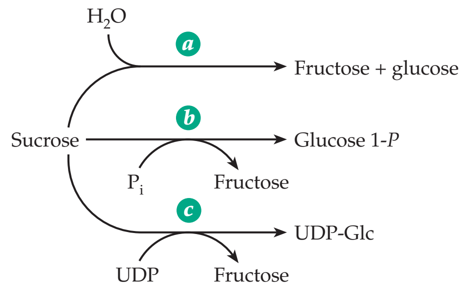
the sulfolipid of chloroplasts (p. 387).[69] A possible biosynthetic sequence begins with transamination of cysteic acid to 3-sulfopyruvate, reduction of the latter to sulfolactaldehyde, and aldol condensation with dihydroxyacetone-*P* as indicated in Eq. 20-12a.[70] See also Eq. 24-47 and Fig. 4-4. However, biosynthesis in chloroplasts appears to start with action of a 4,6-dehydratase on UDP-glucose followed by addition of sulfite and reduction (Eq. 20-12*b*).[70a,b] The sulfite is formed by reduction of sulfate via adenylyl sulfate (Fig. 24-25). However, biosynthesis in chloroplasts appears to start with action of a 4,6-dehydratase on UDP-glucose followed by addition of sulfite and reduction (Eq. 20-12*b*).[70a,b] The sulfite is formed by reduction of sulfate via adenylyl sulfate (Fig. 24-25).

B. Synthesis and Utilization of Oligosaccharides

Our most common food sugar **sucrose** is formed in all green plants and nowhere else. It is made both in the chloroplasts and in the vicinity of other starch deposits. It serves both as a transport sugar and, dissolved within vacuoles, as an energy store. Sucrose is very soluble in water and is chemically inert because the hemiacetal groups of both sugar rings are blocked. However, sucrose is thermodynamically reactive, the glucosyl group having a group transfer potential of 29.3 kJ mol^{-1}. It is extremely sensitive toward hydrolysis catalyzed by acid. Transport of sugar in the form of a disaccharide provides an advantage to plants in that the disaccharide has a lower osmotic pressure than would the same amount of sugar in monosaccharide form.

Biosynthesis of sucrose[71,71a] utilizes both UDP-glucose and fructose 6-*P* (Eq. 20-13). Reaction of UDP-glucose with fructose can also occur to give sucrose directly.[72] Because this reaction is reversible, sucrose serves as a source of UDP-glucose for synthesis of cellulose and other polysaccharides in plants. Metabolism of sucrose in the animal body begins with the action of **sucrase** (invertase), which hydrolyzes the disaccharide to fructose and glucose (Eq. 20-14, step *a*). The same enzyme is also found in higher plants and fungi. Mammalian sucrase is one of several carbohydrases that are anchored to the external surfaces of the microvilli of the small intestines. Sucrose is bound

$$\text{UDP–Glc} \xrightarrow[\text{Fru-6–}P]{} \text{Sucrose 6–}P \xrightarrow[\searrow\,\text{P}_\text{i}]{\text{H}_2\text{O}} \text{Sucrose} \qquad (20\text{-}13)$$



(20-14)

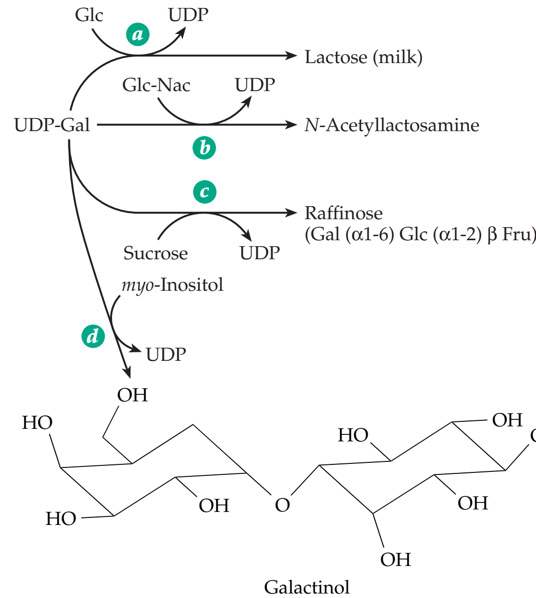
tightly but noncovalently to **isomaltase,** which hydrolyzes the α-1,6-linked isomaltose and related oligosaccharides. A nonpolar N-terminal segment of the isomaltase anchors the pair of enzymes to the microvillus membrane. The two-protein complex arises naturally because the two enzymes are synthesized as a single polypeptide, which is cleaved by intestinal proteases.[73,74]

Because of the relatively high group transfer potential of either the glucosyl or fructosyl parts, sucrose is a substrate for glucosyltransferases such as sucrose phosphorylase (Eq. 20-14, step *b*; see also Eq. 12-7 and associated discussion). In certain bacteria this reaction makes available the activated glucose 1-*P* which may enter catabolic pathways directly. Cleavage of sucrose for biosynthetic purposes can occur by reaction 20-14, step *c*, which yields UDP-glucose in a single step.

A disaccharide with many of the same properties as sucrose is **trehalose,** which consists of two α-glucopyranose units in 1,1 linkage (p. 168). The biosynthetic pathway from UDP-glucose and glucose 6-*P* parallels that for synthesis of sucrose (Eq. 20-13).[75,76] In *E. coli* the genes for the needed glucosyltransferase and phosphatase are part of a single operon. Its transcription is controlled in part by glucose-mediated catabolite repression (Chapter 28) and also by a repressor of the Lac family.[76,76a,77] The repressor is allosterically activated by trehalose 6-*P*, the intermediate in the synthesis. Trehalose formation in bacteria, fungi, plants, and microscopic animals is strongly induced during conditions of high osmolality (see Box 20-C).[77] Both trehalose and maltose can also be taken up via an ABC type transporter (p. 417).[77a,b]

Lactose, the characteristic sugar of milk, is formed by transfer of a galactosyl unit from UDP-galactose directly to glucose (Eq. 20-15, reaction *a*). The similar transfer of a galactosyl unit to *N*-acetylglucosamine to form *N*-acetyllactosamine (Eq. 20-15, reaction *b*) occurs in many animal tissues. An interesting regulatory mechanism is involved. The transferase catalyzing Eq. 20-15, reaction *b*, forms a complex with **α-lactalbumin** to become **lactose synthase**,[78–80b] the enzyme that catalyzes reaction *a*. Lactalbumin was identified as a milk constituent long before its role as a regulatory protein was recognized.

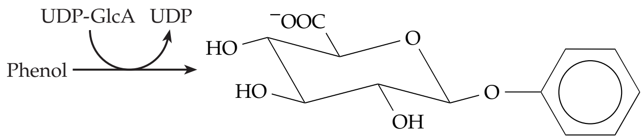
A very common biochemical problem is intolerance to lactose.[81] This results from the inability of the intestinal mucosa to make enough **lactase** to hydrolyze the sugar to its monosaccharide components galactose and glucose. Among most of the peoples of the earth only infants have a high lactase level, and the use of milk as a food for adults often leads to a severe diarrhea. The same is true for most animals. In fact, baby seals and walruses, which drink lactose-free milk, become very ill if fed cow's milk.



(20-15)

The plant trisaccharide **raffinose** arises from UDP-galactose by transfer of a galactosyl unit onto the 6-hydroxyl of the glucose ring of sucrose (Eq. 20-15, reaction *c*). Transfer of a galactosyl unit onto *myo*-inositol (Eq. 20-15, reaction *d*) produces **galactinol**, whose occurrence is widespread within the plant kingdom. Galactinol, in turn, can serve as a donor of activated galactosyl groups. Thus, many plants contain **stachyose** and higher homologs, all of which are formed by transfer of additional α-D-galactosyl units onto the 6-hydroxyl of the galactose unit of raffinose. These sugars appear to serve as antifreeze agents in the plants. The concentration of stachyose in soy beans can be as high as that of sucrose. Some seeds, e.g., those of maize, are coated with a glassy sugar mixture of sucrose and raffinose in a ratio of ~3:1.[81a]

Besides the oligosaccharides, living organisms form a great variety of glycosides that contain nonsugar components. Among these are the **glucuronides** (glucosiduronides), excretion products found in urine and derived by displacement of UDP from UDP-glucuronic acid by such compounds as phenol, benzoic acid, and sterols.[81b,c] Phenol is converted to phenyl glucuronide (Eq. 20-16), while benzoic acid (also excreted in part as hippuric acid, Box 10-A) yields an ester by the same type of displacement reaction. Many other aromatic or aliphatic compounds containing –OH, –SH, $-NH_2$, or $-COO^-$ groups also form glucuronides.[82]

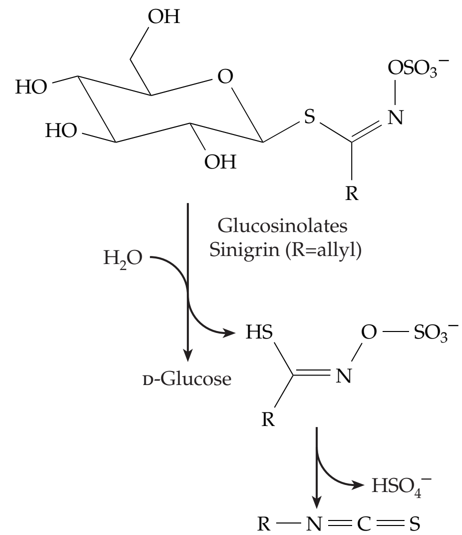


Phenyl glucuronide

(20-16)

Among these is bilirubin (Fig. 24-24). UDP-glucuronosyltransferases responsible for their synthesis are present in liver microsomes.

Among the many glycosides and glycosylamines made by plants are the anthocyanin and flavonoid pigments of flowers (Box 21-E), cyanogenic glycosides such as amygdalin (Box 25-B), and antibiotics (e.g., see Box 20-B).[83,84] Some are characteristic of certain families of plants. For example, more than 100 β-thioglucosides known as **glucosinolides** are found in the Cruciferae (cabbages, mustard, rapeseed). The compounds impart the distinctive flavors and aromas of the plants. However, some are toxic and may cause goiter or liver damage. The enzyme **myrosinase** hydrolyzes these compounds releasing isothiocyanates, thiocyanates, and nitriles (Eq. 20-17).[85-86a] L-Ascorbate acts as a cofactor for this enzyme, evidently providing a catalytic base.[86a]



(20-17)

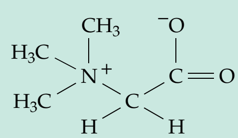
BOX 20-C OSMOTIC ADAPTATION

Bacteria, plants of many kinds, and a variety of other organisms are forced to adapt to conditions of variable osmotic pressure.[a,b] For example, plants must resist drought, and some must adapt to increased salinity. Some organisms live in saturated brine ~6 M in NaCl.[c] The **osmotic pressure** Π in dilute aqueous solutions is proportional to the total molar concentration of solute particles, c_s, as follows.

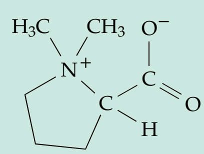
$$\Pi = RTc_s \quad \text{where } R \text{ is the gas constant and } T \text{ the Kelvin temperature}$$

At higher solute concentrations c_s must be replaced by the "effective molar concentration," which is called **osmolarity** (OsM) and has units of molarity (see Record *et al.* for discussion).[b] An osmolarity difference across a membrane of 0.04 OsM results in a turgor pressure of ~ 1 atmosphere. To adapt to changes in the environmental osmolarity organisms must alter their internal solute concentrations.

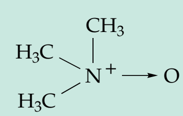
Cells of *E. coli* can adapt to at least 100-fold changes in osmolarity. Because of the porosity of the bacterial outer membrane the osmolarity of the periplasmic space is normally the same as that of the external medium. However, the inner membrane is freely permeable only to water and a few solutes such as glycerol.[b] The bacterial cells avoid loss of water when the external osmolarity is high by accumulating K^+ together with anions such as glutamate⁻ and nonionic **osmoprotectants** such as trehalose, sucrose, and oligosaccharides. *E. coli* cells will also take up other osmoprotectants such as **glycine betaine**, dimethylglycine, choline, proline, and proline betaine. Some methanogens accumulate N^{ε}-acetyl-β-lysine as well as glycine betaine.[d]



Betaine (glycine betaine)



Proline betaine



Trimethylamine *N*-oxide

Functioning in a somewhat different way in *E. coli* are 6- to 12-residue **periplasmic membrane-derived oligosaccharides**. These are β-1,2- and β-1,6-linked glucans covalently linked to *sn*-1-phosphoglycerol, phosphoethanolamine, or succinate (see Fig. 8-28).[b,e,f] They accumulate in the periplasm when cells are placed in a medium of low osmolarity. The resulting increased turgor in the periplasm is thought to buffer the cytoplasm against the loss of external osmolarity and to protect the periplasmic space from being eliminated by expansion of the plasma membrane. Related cyclic glucans, which are attached to *sn*-1-phosphoglycerol or *O*-succinyl ester residues, are accumulated by rhizobia.[g]

C. Synthesis and Degradation of Polysaccharides

Polysaccharides are all formed by transfer of glycosyl groups onto initiating molecules or onto growing polymer chains. The initiating molecule is usually a glycoprotein. However, let us direct our attention first to the growth of polysaccharide chains. The glycosyl are transferred by the action of glycosyltransferases from substrates such as UDP-glucose, other sugar nucleotides, and sometimes sucrose. The glycosyltransferases act by mechanisms discussed in Chapter 12 and are usually specific with respect both to substrate structure and to the type of linkage formed.

1. Glycogen and Starch

The bushlike glycogen molecules grow at their numerous nonreducing ends by the transfer of glucosyl units from UDP-glucose (Eq. 17-56)[87,87a] or in bacteria from ADP-glucose.[88–90] Utilization of glycogen by the cell involves removal of glucose units as glucose 1-*P* by the action of glycogen phosphorylase. The combination of growth and degradation from the same chain ends provides a means of rapidly storing and utilizing glucose units. The synthesis and breakdown of glycogen in mammalian muscle (Fig. 11-4) involves one of the first studied[91] and best known metabolic control systems. Various aspects have been discussed in Chapters 11, 12, and 17. The mechanism[92] and regulatory features[93–96b] have been described. An important recent development is the observation of glycogen concentrations in human muscles *in vivo* with ^{13}C NMR. This can be coupled with observation of glucose 6-*P* by ^{31}P NMR. The concentration of the latter is ~ 1 mM but increases after intense exercise.[94]

Glycogen phosphorylase and glycogen synthase alone are insufficient to synthesize and degrade glycogen. Synthesis also requires the action of the **branching enzyme** amylo-(1,4 → 1,6-transglycosylase,[97]

BOX 20-C (continued)

Fungi, green algae, and higher plants more often accumulate glycerol,[h,i] sorbitol, sucrose,[j] trehalose,[k] or proline.[a,l,m] These compounds are all "compatible solutes" which tend not to disrupt cellular structure.[n] Betaines and proline are especially widely used by a variety or organisms. How is it then that some desert rodents, some fishes, and other creatures accumulate **urea**, a well-known protein denaturant? The answer is that they also accumulate methylamine or trimethylamine *N*-oxide in an approximately 2:1 ratio of urea to amine. The mixture of compounds is compatible, the stabilizing effects of the amines offsetting the destabilizing effect of urea.[c,o]

Adaptation to changes in osmotic pressure involves sensing and signaling pathways that have been partially elucidated for *E. coli*[p] and yeasts.[i,q] Major changes in structure and metabolism may result. For example, in *E. coli* the outer membrane porin OmpF (Fig. 8-20) is replaced by OmpC (osmoporin), which has a smaller pore.[r]

A "resurrection plant" that normally contains an unusual 2-octulose converts this sugar almost entirely into sucrose when desiccated. This is one of a small group of plants that are able to withstand severe desiccation but can, within a few hours, reverse the changes when rehydrated.[j]

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an enzyme with dual specificity. After the chain ends attain a length of about ten glucose units, the branching enzyme attacks a 1,4-glycosidic linkage somewhere in the chain. Acting much as does a hydrolase, it forms a glycosyl enzyme or a stabilized carbocation intermediate. The enzyme does not release the severed chain fragment but transfers it to another nearby site on the glycogen molecule. There the enzyme rejoins the bound oligosaccharide chain that it carries to a free 6-hydroxyl group of the glycogen creating a new branch attached in α-1,6-linkage. Degradation of glycogen requires **debranching** after the long nonreducing ends of the polysaccharide have been shortened until only four glycosyl residues remain at each branch point. This is accomplished by **amylo-1,6-glucosidase / 4- α-glucanotransferase**. This 165-kDa bifunctional enzyme transfers a trisaccharide unit from each branch end to the main chain and also removes hydrolytically the last glucosyl residue at each branch point.[98-99a]

How are new glycogen molecules made? There is some evidence that a 37-kDa protein primer **glycogenin** is needed to initiate their formation.[100-101a] Thus, glycogen synthesis may be analogous to that of the glycosaminoglycans considered in Section D,1. Muscle glycogenin is a self-glycosylating protein, which catalyzes attachment of ~7 to 11 glucose units in α–1,4 linkage to the hydroxyl group of Tyr 194. The glucose units are added one at a time and when the chain is long enough it becomes a substrate for glycogen synthase.[100,102] The role of glycogenin in liver has been harder to demonstrate,[103] but a second glycogenin gene, which is expressed in liver, has been identified.[104] Genes for several glycogenins or glycogenin-like proteins have been identified in yeast, *Caenorhabditis elegans*, and *Arabidopsis*.[101a,105]

In contrast to animals, bacteria such as *E. coli* synthesize glycogen via ADP-glucose rather than UDP-glucose.[88] ADP-glucose is also the glucosyl donor for synthesis of starch in plants. The first step in the biosynthesis (Eq. 20-18) is catalyzed by the enzyme ADP-glucose pyrophosphorylase (named for the reverse reaction).

$$\text{Glucose 1-}P + \text{ATP} \rightarrow \text{ADP-glucose} + \text{PP}_i \qquad (20\text{-}18)$$

In bacteria this enzyme is usually inhibited by AMP and ADP and activated by glycolytic intermediates such as fructose 1,6-P_2, fructose 6-P, or pyruvate. In higher plants, green algae, and cyanobacteria the enzyme is usually activated by 3-phosphoglycerate, a product of photosynthetic CO_2 fixation, and is inhibited by inorganic phosphate (P_i).[106–108].

In eukaryotic plants starch is deposited within chloroplasts or in the cytoplasm as granules (Fig. 4-6) in a specifically differentiated and physically fragile plastid, the **amyloplast**.[108–110] Within the granules the starch is deposited in layers ~ 9 nm in thickness. About two-thirds of the thickness consists of nearly crystalline arrays, probably of double helical amylopectin side chains (Figs. 4-7, 4-8, 20-3) with "amorphous" segments between the layers.[111–114] In maize there are at least five starch synthases, one of which forms the straight chain amylose.[115–117] There are also at least three branching enzymes[118] and two or three debranching enzymes.[119,120] As in the synthesis of glycogen the molecules of amylopectin may grow at the many nonreducing ends. A current model, which is related to the broom-like cluster model of French (Fig. 4-7), is shown in Fig. 20-3. The branches are thought to arise, in part, by transglycosylation within the double helical strands. After branching the two chains remain in a double helix but the cut chain can now grow. Only double helical parts of strands pack well in the crystalline layer. A recent suggestion is that debranching enzymes then trim the molecule, removing single-stranded regions.[112]

The location (within the granule) of amylose, which makes up 15–30% by weight of many starches,[121] is uncertain. It may fill in the amorphous layers. It may be cut and provide primer pieces for new amylopectin molecules.[122,122a] Another possibility is that it grows by an insertion mechanism such as that portrayed for cellulose in Fig. 20-5 and is extruded inward from the membrane of the amyloplast. This mechanism might explain a puzzling question about starch. The branched amylopectin presumably grows in much the same way as does glycogen. A branching enzyme transfers part of the growing glycan chain to the $-CH_2-OH$ group of

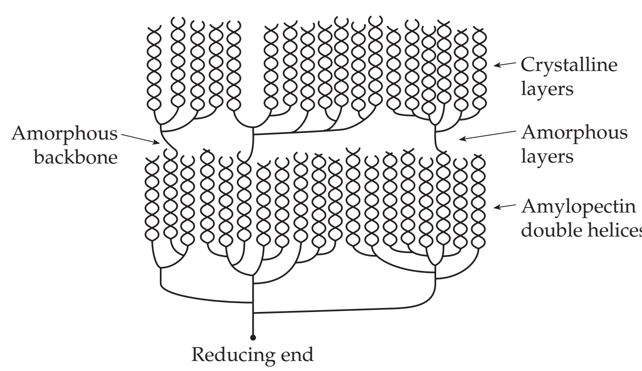

Figure 20-3 Proposed structure of a molecule of amylopectin in a starch granule. The highly branched molecule lies within 9 nm thick layers, about 2 / 3 of which contains parallel double helices of the kind shown in Fig. 4-8 in a semicrystalline array. The branches are concentrated in the amorphous region.[113,114,121] Some starch granules contain no amylose, but it may constitute up to 30% by weight of the starch. It may be found in part in the amorphous bands and in part intertwined with the amylopectin.[122]

BOX 20-D GENETIC DISEASES OF GLYCOGEN METABOLISM

In 1951, B. McArdle described a patient who developed pain and stiffness in muscles after moderate exericse.[a] Surprisingly, this person completely lacked muscle glycogen phosphorylase. Since that time several hundred others have been found with the same defect. Glycogen accumulates in muscle tissue in this disease, one of the several types of **glycogen storage disease**.[b] Severe exercise is damaging, but steady moderate exercise can be tolerated. Until the time of McArdle's discovery, it was assumed that glycogen was synthesized by reversal of the phosphorylase reaction. No hint of the UDP-glucose pathway had appeared, and it was, therefore, not obvious how glycogen could accumulate in the muscles of these patients. Leloir's discovery of UDP-glucose at about the same time provided the answer. Persons with McArdle syndrome are greatly benefitted by a high-protein diet, presumably because amino acids such as alanine and glutamine are converted efficiently to glucose and because branched-chain amino acids may serve as a direct source of muscle energy.[c,d]

Several other rare heritable diseases also lead to accumulation of glycogen because of some block in its breakdown through the glycolysis pathway. The enzyme deficiencies include those of muscle phosphofructokinase,[e] liver phosphorylase kinase, liver phosphorylase, and liver glucose-6-phosphatase. In the last case, glycogen accumulates because the liver stores cannot be released to the blood as free glucose.[b,f,g] This is a dangerous disease because blood glucose concentrations may fall too low at night. The prognosis improved greatly when methods were devised for providing the body with a continuous supply of glucose. The simplest treatment is ingestion of uncooked cornstarch which is digested slowly.[b,h] In one of the storage diseases the branching enzyme of glycogen synthesis is lacking, and glycogen is formed with unusually long outer branches. In another the debranching enzyme is lacking, and only the outer branches of glycogen can be removed readily.[i]

The most serious of the storage diseases involve none of the enzymes mentioned above. Pompe disease is a fatal generalized glycogen storage disease in which a lysosomal α-1,4-glucosidase is lacking. This observation suggested the existence of a new and essential pathway of degradation of glycogen to free glucose in the lysosomes. A few cases of glycogen synthase deficiency have been reported. Little or no glycogen is stored in muscle or liver, and patients must eat at regular intervals to prevent hypoglycemia. Severe diseases in which glycogen synthesis is impaired include deficiencies of the gluconeogenic enzymes pyruvate carboxylase and PEP carboxykinase.

The following tabulation includes deficiencies of glycogen metabolism, glycolysis, and gluconeogenesis.[a] Glycogen storage diseases are often designated as Types I–V and these terms are included.

Deficiency	Organ	Severity
Glycogen phosphorylase (Type V), McArdle disease	Muscle	Moderate, late onset
Glycogen phosphorylase	Liver	Very mild
Phosphorylase kinase	Liver	Very mild
Debranching enzyme (Type III)	Liver	Mild
Lysosomal α-glucosidase (Type II)		Lethal, infant and adult form
Phosphofructokinase	Muscle	Moderate, late onset
Phosphoglycerate mutase[j]	Muscle	Moderate
Pyruvate carboxylase		Lethal
PEP carboxykinase		Lethal
Fructose-1,6-bisphosphatase	Muscle	Severe
Glycogen synthase	Liver	Mild
Branching enzyme (Type IV)		Lethal, liver transplantation
Glucose-6-phosphatase (Type I)		Severe if untreated

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a glucose unit in an adjacent polysaccharide chain that lies parallel to the first, possibly in a double helix. Since amylose and amylopectin are intimately intermixed in the starch granules, it seems strange that the branching enzyme never transfers a branch to molecules of the straight-chain amylose. However, if the linear amylose chains are oriented in the opposite direction from the amylopectin chains, the nonreducing ends of the amylose molecules would be located toward the center of the starch granule. Growth could occur by an insertion mechanism at the reducing ends and the ends could move out continually with the amyloplast membrane as the granule grows.[123] Recent evidence from ^{14}C labeling indicates that both amylose and amylopectin too may grow by insertion at the reducing end of glucose units from ADP-glucose.[123a,b] Branching could occur to give the structure of Fig. 20-3. Starch synthesis in leaves occurs by day but at night the starch is degraded by amylases, α-glucosidases, and starch phosphorylase. Both the starch synthases and catabolic enzymes are present within the amyloplasts where they may be associated with regulatory proteins of the 14-3-3 class.[122a]

Digestion of dietary glycogen and starch in the human body begins with the salivary and pancreatic amylases, which cleave α-1,4 linkages at random. It continues with a **glucoamylase** found in the brush border membranes of the small intestine where it occurs as a complex with **maltase**.[74] Carbohydrases are discussed in Chapter 12, Section B.

2. Cellulose, Chitin, and Related Glycans

Cellulose synthases transfer glucosyl units from UDP-glucose, while chitin synthases utilize UDP-*N*-acetylglucosamine. Not only green plants but some fungi and a few bacteria form cellulose. The ameba *Dictyostelium discoideum* also coats its spores with cellulose.[124] Electron microscopic investigations suggest that both in bacteria[125] and in plants[126] multienzyme aggregates located at the plasma membrane synthesize many polymer chains side by side to generate hydrogen-bonded microfibrils which are extruded through the membrane. Both green plants and fungi also form important β-1,3-linked glycans.

The bacterial cellulose synthase from *Acetobacter xylinum* can be solubilized with detergents, and the resulting enzyme generates characteristic 1.7 nm cellulose fibrils (Fig. 20-4) from UDP-glucose.[125,127–129] These are similar, but not identical, to the fibrils of cellulose I produced by intact bacteria.[125,130] Each native fibril appears as a left-handed helix which may contain about nine parallel chains in a crystalline array. Three of these helices appear to coil together (Fig. 20-4) to form a larger 3.7-nm left-handed helical fibril. Similar fibrils are formed by plants. In both

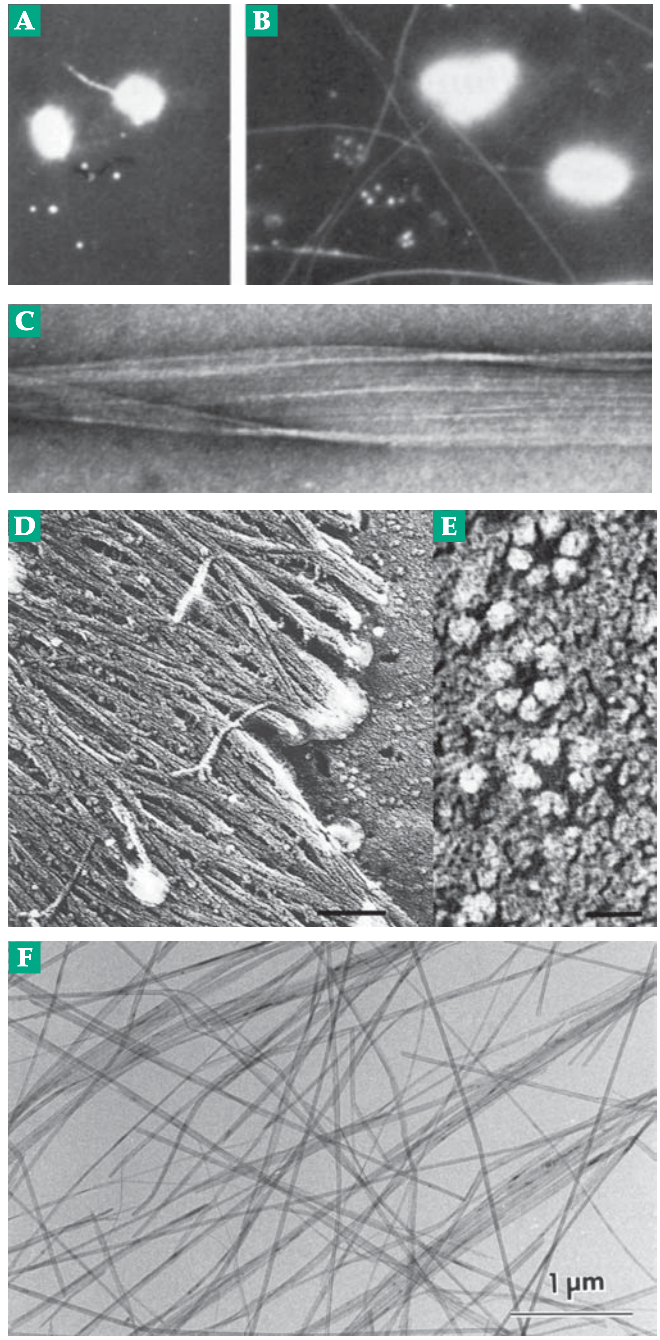


Figure 20-4 Cellulose microfibrils being formed by *Acetobacter xylinum*.[127] (A) Dark-field light micrograph after five minutes of cellulose production (x 1250). (B) After 15 minutes a pellicle of cellulose fibers is forming (x 2000) (C) Negatively stained cellulose ribbon. At the right the subdivision into microfibrils is visible. Courtesy of R. Malcolm Brown, Jr. (D) Cellulose microfibrils overlaying the plasma membrane in the secondary cell wall of a tracheary element of *Zinnia elegans*. Bar = 100 nm. (E) Rosettes in the plasma membrane underlying the cellulose-rich secondary cell wall thickening in *A. elegans.* Bar = 30 nm. (D) and (E) from Haigler and Blanton.[132] Courtesy of Candace H. Haigler. (F) Chitin microfibrils purified from protective tubes of the tube-worm *Lamellibrachia satsuma*.[137] Courtesy of Junji Sugiyama.

bacteria and plants the cellulose I fibrils that are formed are highly crystalline, contain parallel polysaccharide chains (Fig. 4-5), and have the tensile strength of steel. Electron micrographs show that the cell envelope of *A. xylinum* contains 5–80 pores, through which the cellulose is extruded, lying along the long axis of the cell.[129] The biosynthetic enzymes are probably bound to the plasma membrane. Similar, but more labile, cellulose synthases are present in green plants.[131] In *Arabidopsis* there are ten genes. The encoded cellulose synthases appear to be organized as rosettes on some cell surfaces (Fig. 20-4E).[131a-133a] The rosettes may be assembled to provide parallel synthesis of ~36 individual cellulose chains needed to form a fibril.[131a]

Because of the insolubility of cellulose fibrils it has been difficult to determine whether they grow from the reducing ends or the nonreducing ends of the chains. From silver staining of reducing ends and micro electron diffraction of cellulose fibrils attached to bacteria,[134] Koyama *et al.* concluded that the reducing ends are extruded from cells. New glucosyl rings would be added at the *nonreducing ends*, which remain attached noncovalently to the cells.[134] From amino acid sequence similarities it was also concluded that the same is true for *Arabidopsis*.[131,133] A single cellulose chain has a twofold screw axis, each residue being rotated 180° from the preceding residue (Fig. 4-5). It was postulated that two synthases act cooperatively to add cellobiose units. Another suggestion is that sitosterol β-glucoside acts in some fashion as a primer for cellulose synthesis in plants.[133b]

An insertion mechanism for synthesis of cellulose. Using ^{14}C "pulse and chase" labeling Han and Robyt found that new glucosyl units are added at the *reducing ends* of cellulose chains formed by cell membrane preparations from *A. xylinum*.[135] This conclusion is in accord with the generalization that extracellular polysaccharides made by bacteria usually grow from the reducing end by an insertion mechanism that depends upon a polyprenyl alcohol present in the cell membrane.[136] This lipid alcohol, often the C_{55}

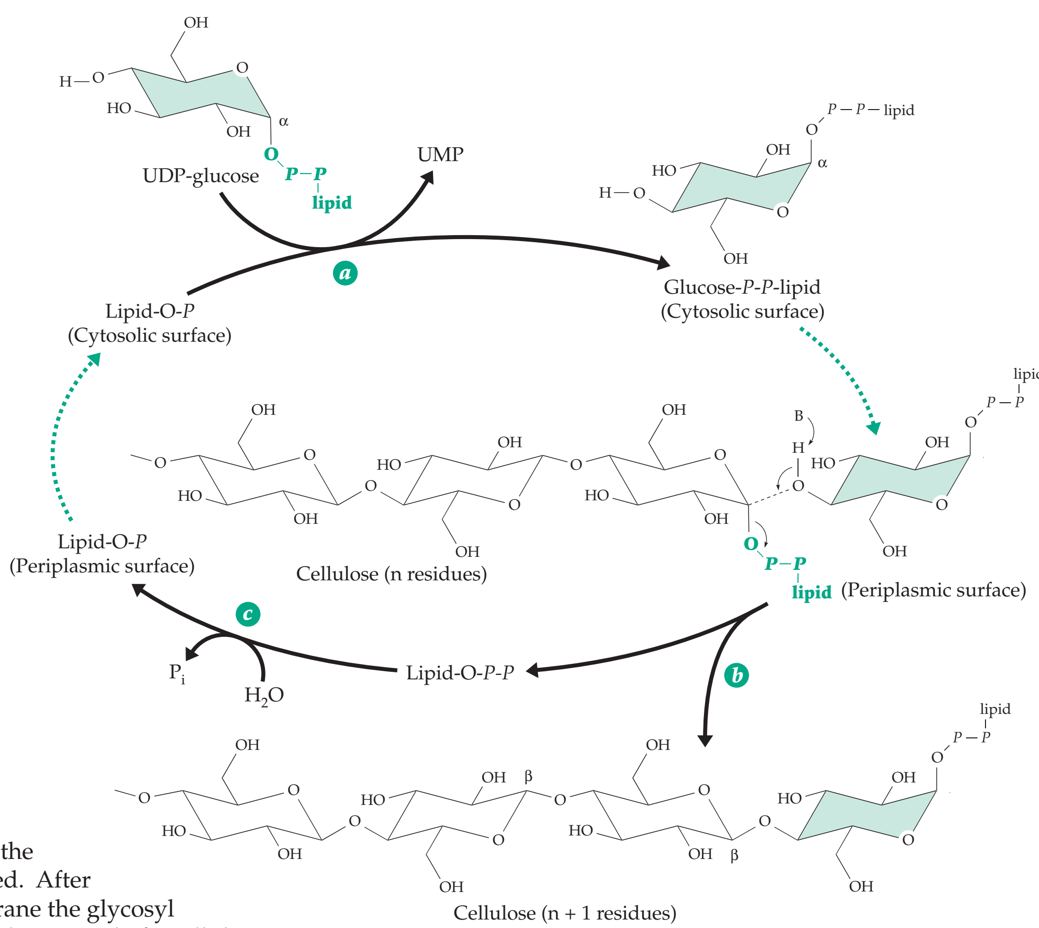
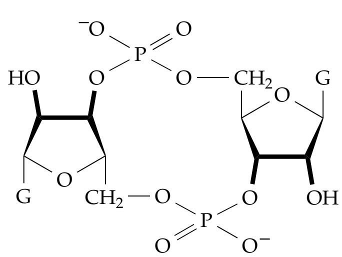


Figure 20-5 Proposed insertion mechanism for biosynthesis of cellulose. Three enzymatic steps are involved: a nucleophilic displacement reaction of a lipid phosphate on UDP-glucose yields a glucosyl diphosphate lipid in which the α-glycosyl linkage is retained. After passage through the membrane the glycosyl group is inserted into the reducing end of a cellulose chain, which is covalently attached by a pyrophosphate linkage to another lipid. The first lipid diphosphate is released and is hydrolyzed (step *c*) to the monophosphate, which crosses the membrane to complete the cycle. After Han and Robyt.[135] As throughout this book *P* represents the phospho group $-PO_3H$. The H may be replaced by groups which may contain oxygen atoms. This explains why an O is included in Lipid-O-*P* but no O is shown between the *P*'s in -O-*P*-*P*.

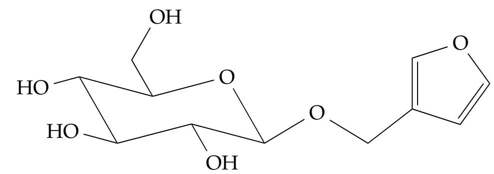
bactoprenol, reacts with UDP-glucose (or other glycosyl donor) to give a lipopyrophospho-glucose (step *a*, Fig. 20-5). The α linkage of the UDP-glucose is retained in this compound. The growing cellulose chain is attached at the reducing end by a similar linkage to a second lipid molecule. Then, in a displacement on the anomeric carbon of the first glucosyl residue of the cellulose chain, the new glucosyl unit is inserted with inversion of the α linkage to β. In step *c* the pyrophosphate linkage of the lipid diphosphate is hydrolyzed to regenerate the lipid monophosphate and to drive the reaction toward completion. Two of the steps in the cycle involve transport across the bacterial membrane. The first involves the lipid –O-*P*-*P*-glucose and the second the lipid monophosphate. This type of insertion mechanism is a common feature of polyprenol phosphate-dependent synthetic cycles for extracellular polysaccharides (Figs. 20-6, 20-9 and Eq. 20-20). However, further verification is needed for cellulose synthesis.

Regulation of cellulose synthesis in bacteria depends on allosteric activator of cellulose synthase, **cyclic diguanylate** (c-di-GMP), and a Ca^{2+}-activated phosphodiesterase that degrades the activator.[129,138–139a] Sucrose is the major transport form of glucose in plants. Synthesis of both cellulose and starch is reduced in mutant forms of maize deficient in sucrose synthase (Eq. 20-13). This synthase, acting in the reverse direction, forms UDP-Glc from sucrose.[140,141] The enzymatic degradation of cellulose is an important biological reaction, which is limited to certain bacteria, to fungi, and to organisms such as termites that obtain cellulases from symbiotic bacteria or by ingesting fungi.[142] These enzymes are discussed in Chapter 12, Section B,6. Genetic engineering methods now offer the prospect of designing efficient cellulose-digesting yeasts[143] that may be used to produce useful fermentation products from cellulose wastes.



Cyclic diguanylate

Callose and other β-1,3-linked glycans. Attempts to produce cellulose from UDP-Glc using enzymes of isolated plasma membranes from higher plants have usually yielded the β-1,3-linked glucan (callose) instead. This is a characteristic polysaccharide of plant wounds which, as healing occurs, is degraded and replaced by cellulose.[140,144] Callose formation is induced by a specific activator β-furfuryl-β-glucoside, and callose synthase is virtually inactive unless both the activator and Ca^{2+} are present.[144]



β-Furfuryl-β-glucoside

Beta-1,3-linked glycans are major components of the complex layered cell wall of yeasts and other fungi. In the fission yeast *Saccharomyces pombe* ~55% of the cell wall carbohydrate consists of β-1,3-linked glucan with some β-1,4-linked branches, ~ 28% is α-1,3-linked glucan, ~6% is α-1,6-linked glucan, and ~0.5% is chitin. There are two carbohydrate layers, the outer one appearing amorphous. The inner layer contains interwoven fibrils of both α-1,3-linked and α-1,4-linked glucans and holds the shape of the cell. The β-1,3 glucan synthase is localized on the inner side of the cell membrane and is activated by GTP and a small subunit of the Rho family of G proteins.[145]

Plants synthesize 1,3-β-glucanases that hydrolyze the glycans of fungal cell walls. Synthesis is induced by wounding as a defense reaction (see Box 20-E). These glycanases also function in the removal of callose.[146]

Chitin. Like cellulose synthase, fungal chitin synthases are present in the plasma membrane and extrude microfibrils of chitin to the outside.[147–150] In the fungus *Mucor* the majority of the chitin synthesized later has its *N*-acetyl groups removed hydrolytically to form the deacetylated polymer **chitosan**.[151,152] Chitin is also a major component of insect exoskeletons. For this reason, chitin synthase is an appropriate target enzyme for design of synthetic insecticides.[153]

Chitin hydrolyzing enzymes are formed by fungi and in marine bacteria.[154] Chitinases are also present in plant vacuoles, where they participate in defense against fungi and other pathogens[155] (Box 20-E). More recently a chitinase has been identified in human activated macrophages.[156] Another unanticipated discovery was that a developmental gene designated *DG*42, from *Xenopus*, has a sequence similar to that of the *NodC* gene. The latter encodes a synthase for chitin oligosaccharides (Nod factors) that serve as nodulation factors in *Rhizobia* (Chapter 24). The enzyme is synthesized for only a short time during early embryonic development.[157] The significance of this discovery is not yet clear. Synthesis of both the bacterial Nod factors and chitin oligosaccharides in zebrafish embryos occurs by transfer of GlcNAc residues from UDP-GlcNAc at the *nonreducing ends* of the chains.[158] Whether the same is true of chitin in fungi or arthropods remains uncertain.

Cell walls of plants. The thick walls of higher plant cells (Figs. 1-7, 4-14, and 20-4D) provide strength and rigidity to plants and, at the same time, allow rapid elongation during periods of growth.[159–163a] Northcote[164] likened the wall structure to glass fiber-reinforced plastic (fiber glass). Thus, the cell wall contains microfibrils of cellulose and other polysaccharides embedded in a matrix, also largely polysaccharide. The **primary cell wall** laid down in green plants during early stages of growth contains loosely interwoven cellulose fibrils ~10 nm in diameter and with an ~4 nm crystalline center. The cellulose in these fibrils has a degree of polymerization of 8000–12,000 glucose units. As the plant cell matures, a secondary cell wall is laid down on the inside of the primary wall. This contains many layers of closely packed microfibrils, alternate layers often being laid down at different angles to one another (Fig. 20-4D). The microfibrils in green plants are most often cellulose but may contain other polysaccharides as well. Some algae are rich in fibrils of xylan and mannan.

The materials present in the matrix phase vary with the growth period of the plant. During initial phases **pectin** (polygalacturonic acid derivatives) predominate but later xylans and a variety of other polysaccharides known as **crosslinking glycans** (or hemicelluloses) appear. Primary cell wall constituents of dicotyledons include **xyloglucans** (linear glucan chains with xylose, galactose, and fucose units in

BOX 20-E OLIGOSACCHARIDES IN DEFENSIVE AND OTHER RESPONSES OF PLANTS

Plants that are attacked by bacteria, fungi, or arthropods respond by synthesizing broad-spectrum antibiotics called **phytoalexins**,[a,b] by strengthening their cell walls with lignin and hydroxyproline-rich proteins called **extensins**,[c] and by making **protease inhibitors** and other proteins that help to block the chemical attack.[d] These plant responses seem to be initiated by the release from an invading organism of **elicitors**, which are often small oligosaccharide fragments, sometimes called **oligosaccharins**.[e] These include β-1,6-linked glucans that carry β-1,3-linked branches as well as chitin and chitosan oligomers, derived from fungal cell walls.[f] Other elicitors include galacturonic acid oligomers released from damaged plant cell walls,[g] metabolites such as arachidonic acid and glutathione,[h] and bacterial toxins.[i] Any of these may serve as signals to plants to take defensive measures.

Phytoalexins are often isoflavonoid derivatives (Box 21-E). Their synthesis, like that of lignin, occur via 4-coumarate (4-hydroxycinnamate, Fig. 25-8). The ligase which forms the thioester of 4-coumarate with coenzyme A is one of the **pathogenesis-related proteins** whose synthesis is induced.[j] A second induced enzyme is chalcone synthase, which condenses three acetyl units onto 4-coumaroyl-CoA as shown in Box 21-E. Its induction by elicitors acting on bean cells requires only five minutes.[h] Another rapidly induced gene is that of cinnamoyl alcohol dehydrogenase,[k] essential to lignin synthesis. Other proteins formed in response to infections include **chitinases** that are able to attack invading fungi[l,m] as well as the protease inhibitors. Their synthesis is induced via derivatives of **jasmonate**, a product of the octadecenoic acid pathway (Eq. 21-18).[a] As yet, little is known about the mechanism by which elicitors induce the defensive responses, but the presence of receptors, of phosphorylation, and of release of second messengers have been suggested.[d]

Lipooligosaccharides known as Nod factors (p. 1365) are another group of signaling molecules. These chitin-related *N*-acylated oligomers of *N*-acetylglucosamine (GlcNAc) do not defend against infection but invite infection of roots of legumes by appropriate species of *Rhizobia*[n–p] leading to formation of nitrogen-fixing root nodules.

[a] Blechert, S., Brodschelm, W., Hölder, S., Kammerer, L., Kutchan, T. M., Mueller, M. J., Xia, Z.-Q., and Zenk, M. H. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 4099–4105

[b] Ebel, J., and Grisebach, H. (1988) *Trends Biochem. Sci.* **13**, 23–27

[c] Kieliszewski, M. J., O'Neill, M., Leykam, J., and Orlando, R. (1995) *J. Biol. Chem.* **270**, 2541–2549

[d] Ryan, C. A. (1988) *Biochemistry* **27**, 8879–8883

[e] Ryan, C. A. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 1–2

[f] Baureithel, K., Felix, G., and Boller, T. (1994) *J. Biol. Chem.* **269**, 17931–17938

[g] Reymond, P., Grünberger, S., Paul, K., Müller, M., and Farmer, E. E. (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 4145–4149

[h] Dron, M., Clouse, S. D., Dixon, R. A., Lawton, M. A., and Lamb, C. J. (1988) *Proc. Natl. Acad. Sci. U.S.A.* **85**, 6738–6742

[i] Bidwai, A. P., and Takemoto, J. Y. (1987) *Proc. Natl. Acad. Sci. U.S.A.* **84**, 6755–6759

[j] Douglas, C., Hoffmann, H., Schulz, W., and Hahlbrock, K. (1987) *EMBO J.* **6**, 1189–1195

[k] Walter, M. H., Grima-Pettenati, J., Grand, C., Boudet, A. M., and Lamb, C. J. (1988) *Proc. Natl. Acad. Sci. U.S.A.* **85**, 5546–5550

[l] Legrand, M., Kauffmann, S., Geoffroy, P., and Fritig, B. (1987) *Proc. Natl. Acad. Sci. U.S.A.* **84**, 6750–6754

[m] Payne, G., Ahl, P., Moyer, M., Harper, A., Beck, J., Meins, F., Jr., and Ryals, J. (1990) *Proc. Natl. Acad. Sci. U.S.A.* **87**, 98–102

[n] Cedergren, R. A., Lee, J., Ross, K. L., and Hollingsworth, R. I. (1995) *Biochemistry* **34**, 4467–4477

[o] Jabbouri, S., Relic, B., Hanin, M., Kamalaprija, P., Burger, U., Promé, D., Promé, J. C., and Broughton, W. J. (1998) *J. Biol. Chem.* **273**, 12047–12055

[p] Dénarié, J., Debellé, F., and Promé, J.-C. (1996) *Ann. Rev. Biochem.* **65**, 503–535

branches),[164a] other crosslinking glycans, and galacturonic acid-rich pectic materials.[163a] The xyloglycans, which comprise 20% of the cell wall in some plants, have a backbone of α-1,4-linked glucose units with numerous α-1,6-linked xylose rings, some of which carry attached L-arabinose, galactose, or fucose. The structures, which vary from species to species, are organized as repeating blocks with a continuous glucan backbone. Another crosslinking glycan is **glucuronoarabinoxylan**. The backbone is β-1,4-linked xylose. Less abundant glucomannans, galactomannans, and galactoglucomannans, with β-1,4-linked mannan backbone structures, are also present in most angiosperms.[163a]

Pectins form a porous gel on the inside surface of plant cell walls.[163a,164a] A major component is a **homogalacturonan**, which consists of α-1,4-linked galacturonic acid (GalA). A second is rhamnogalacturonan I, an alternating polymer of (2-L-Rhaα1 → 4GalAα →) units. The most interesting pectin component is **rhamnogalacturonan II**, one of the less abundant constuents of pectin. It is obtained by hydrolytic cleavage of pectin by a polygalacturonidase. Before such release it forms parts (hairy regions) of pectin molecules that are largely homogalacturonans (in smooth regions). A rhamnogalacturonan II segment consists of 11 different monomer units.[164b–f] Attached to the polygalacturonic acid backbone are four oligosaccharides, consisting of rhamnose, galactose and fucose as well as some unusual sugars (see structure in Box 20-E). This polysaccharide is apparently present in all higher plants and is unusually stable, accumulating, for example, in red wine.[164e] It contains two residues of the branched chain sugar **apiose**, one of which is a site of crosslinking by boron (Box 20-F). A borate diol ester linkage binds two molecules of the pectin together as a dimer, perhaps controlling the porosity of the pectin gel. All of the complex cell wall polysaccharides bind, probably through multiple hydrogen bonds, to the cellulose microfibrils (Fig. 4-14). The resulting structures are illustrated in drawings of Carpita and McCann,[163a] which are more current than is Fig. 4-14. The cellulose plus crosslinking glycans form one network in the cell wall. The pectic substances form a second independent network. Some covalent crosslinking occurs, but most interactions are noncovalent.[163a] The site of biosynthesis of pectins and hemicelluloses is probably Golgi vesicles which pass to the outside via exocytosis. However, the cellulose fibrils as well as the chitin in fungi are apparently extruded from the plasma membrane.

Although the principal cell wall components of plants are carbohydrates, proteins account for 5–10% of the mass.[165] Predominant among these are glycoprotein **extensins**. Like collagen, they are rich in 4-hydroxyproline which is glycosylated with arabinose oligosaccharides and galactose (p. 181). Other hydroxyproline-containing proteins with the characteristic sequence (hydroxyproline)$_4$-Ser are also found, e.g., in soybean cell walls.[166] Some plant cell walls contain glycine-rich structural proteins. One in the petunia consists of 67% glycine residues.[167] During advanced stages of formation, as the walls harden into wood, large amounts of **lignins** are laid down in some plant cells. These chemically resistant phenylpropanoid polymers contain many crosslinked aromatic rings (Fig. 25-9).[163a]

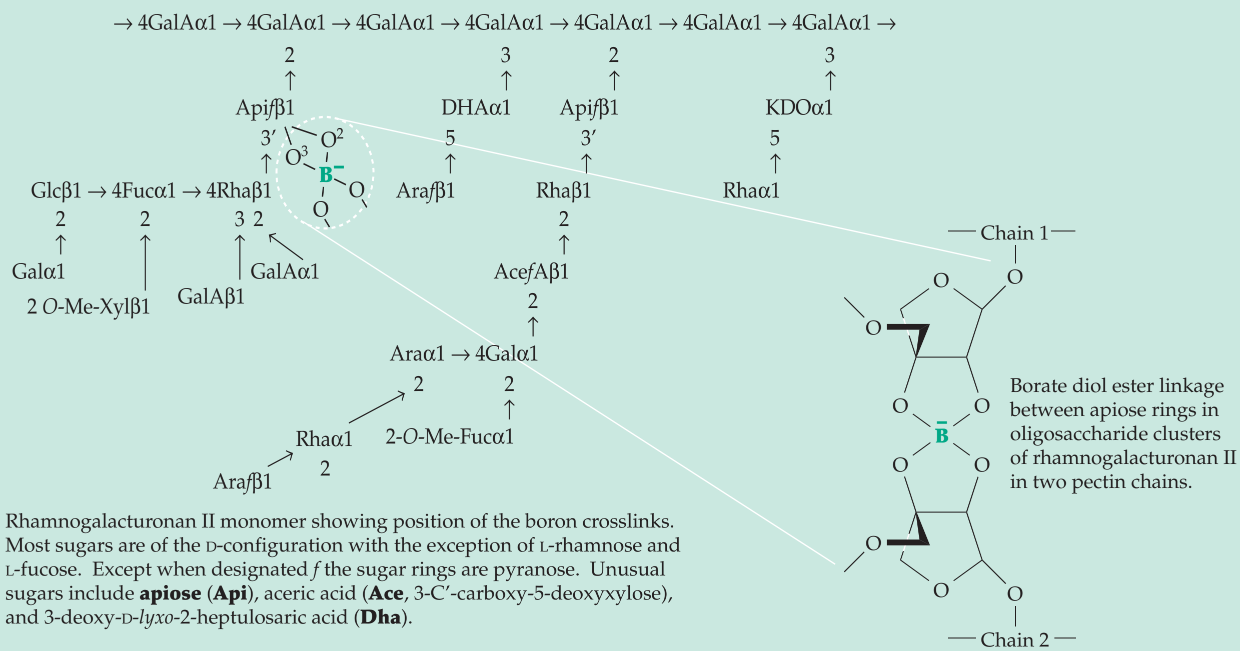
A remarkable aspect of primary plant cell walls is their ability to be elongated extremely rapidly during growth. While the driving force for cell expansion is thought to be the development of pressure within the cell, the manner in which the wall expands is closely regulated. After a certain point in development, elongation occurs in one direction only and under the influence of plant hormones. Most striking is the effect of the **gibberellins** (Eq. 22-5), which cause very rapid elongation. Elongation of plant cell walls may depend to some extent upon chemical cleavage and reforming of crosslinking polysaccharides. However, the cellulose fibrils probably remain intact and slide past each other.[161] A curious effect, which is mediated by proteins called **expansins**,[133a,168] is the ability of plant tissues to extend rapidly when incubated in a mildly acidic buffer of pH <5.5. Expansins are also involved in ripening of fruit. They may disrupt noncovalent bonding between cellulose fibrils and the hemicelluloses.[169,170] The β-expansins of grasses are allergens found in grass pollens.[133a,168] The borate diol ester linkages in the pectin may also facilitate expansion.

3. Patterns in Polysaccharide Structures

How can the many complex polysaccharides found in nature be synthesized? Are there genetically determined patterns? How are these controlled? The answer can be found in the *specificities* of the hundreds of known *glycosyltransferases*[171,172] and in the *patterns of expression of the genes* for transferases and other proteins. As a consequence, a great variety of structurally varied polysaccharide structures arise, especially on cell surfaces. The structures are not random but depend upon the assortment of glycosyltransferases available at the particular stage of development in a tissue. The numerous possibilities can account for much of the variation observed between species, between tissues, and also among individuals.

The simplest pattern is the growth of straight-chain homopolysaccharides such as amylose, cellulose, and chitin. The glycosyltransferases must recognize both the glycosyl donor, e.g., ADP-glucose, UDP-glucose, and also the correct end of the growing polymer, always adding the same monomer unit.

BOX 20-F WHAT DOES BORON DO?



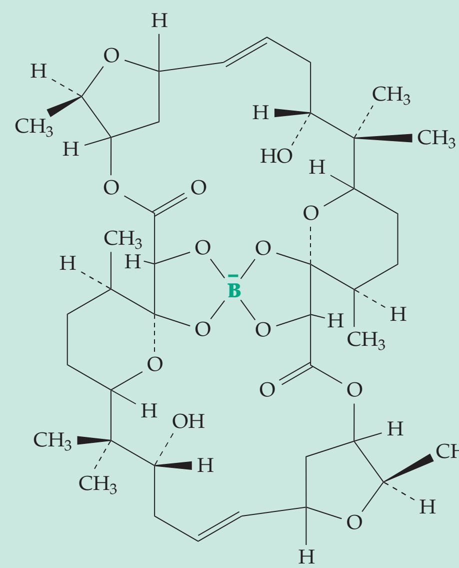
Rhamnogalacturonan II monomer showing position of the boron crosslinks. Most sugars are of the D-configuration with the exception of L-rhamnose and L-fucose. Except when designated *f* the sugar rings are pyranose. Unusual sugars include **apiose** (**Api**), aceric acid (**Ace**, 3-C′-carboxy-5-deoxyxylose), and 3-deoxy-D-*lyxo*-2-heptulosaric acid (**Dha**).

Borate diol ester linkage between apiose rings in oligosaccharide clusters of rhamnogalacturonan II in two pectin chains.

For 75 years or more it has been known that boron is essential for growth of green plants.[a,b] In its absence root tips fail to elongate normally, and synthesis of DNA and RNA is inhibited. Boron in the form of boric acid, $B(OH)_3$, is absorbed from soil. Although deficiency is rare it causes disintegration of tissues in such diseases as "heart rot" of beets and "drought spot" of apples. The biochemical role has been obscure, but is usually thought to involve formation of borate esters with sugar rings or other molecules with adjacent pairs of –OH groups (as in the accompanying structures). A regulatory role involving the plant hormones auxin, gibberelic acid, and cytokinin has also been suggested.

Diatoms also require boron, which is incorporated into the silicon-rich cell walls.[a] Some strains of *Streptomyces griseus* produce boron-containing macrolide antibiotics such as **aplasmomycin** (right).[c] Recently a function in plant cell walls has been identified (see also main text) as crosslinking of rhamnogalacturonan portions of pectin chains by borate diol ester linkages as illustrated.

It was long thought that boron was not required by human beings, but more recent studies suggest that we may need ~30 µg / day.[d] The possible functions are uncertain. Animals deprived of boron show effects on bone, kidney, and brain as well as a relationship to the metabolism of calcium, copper, and nitrogen. Nielson proposed a signaling function, perhaps via phosphoinositides, in animals.[b]

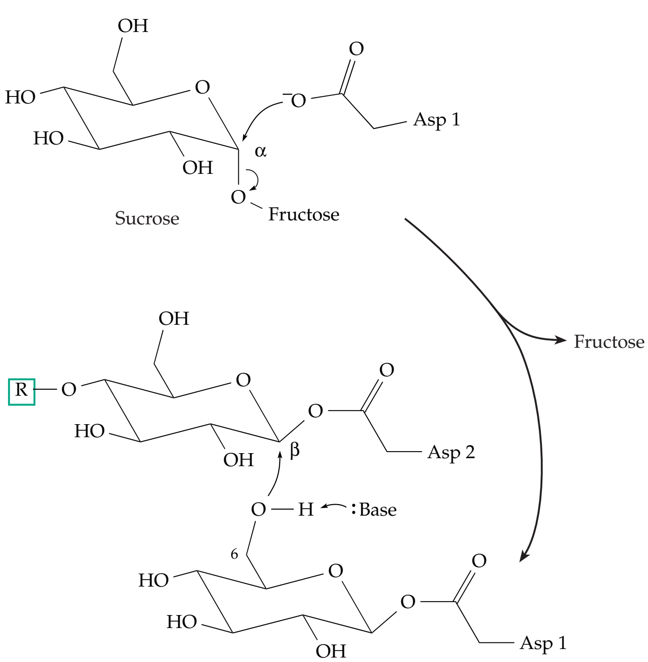


Aplasmomycin, a boron-containing antibiotic

[a] Salisbury, F. B., and Ross, C. W. (1992) *Plant Physiology*, 4th ed., Wadsworth, Belmont, California
[b] Nielsen, F. H. (1991) *FASEB J.* **5**, 2661–2667
[c] Lee, J. J., Dewick, P. M., Gorst-Allman, C. P., Spreafico, F., Kowal, C., Chang, C.-J., McInnes, A. G., Walter, J. A., Keller, P. J., and Floss, H. G. (1987) *J. Am. Chem. Soc.* **109**, 5426–5432
[d] Nielsen, F. H. (1999) in *Modern Nutrition in Health and Disease*, 9th ed. (Shils, M. E., Olson, J. A., Shike, M., and Ross, A. C., eds), pp. 283–303, Williams & Wilkins, Baltimore, Maryland

In contrast, hyaluronan and the polysaccharide chains of **glycosaminoglycans** (Fig. 4-11) have an alternating pattern. For a hyaluronan chain growing at the reducing end, one active site of hyaluronan synthase must be specific for UDP-GlcNAc and transfer the sugar unit only to the end of a glucuronic acid ring. A second active site must be specific for UDP-glucuronic acid but attach it only to the end of an acetylglucosamine unit.[172a,172b] There is still uncertainty about the direction of growth of hyaluronan.[173–175] Some hyaluronan synthases are lipid-dependent and their mechanism may resemble that proposed for cellulose synthesis (Fig. 20-5).

Dextrans. Some polysaccharides, such as the bacterial dextrans, are synthesized outside of cells by the action of secreted enzymes. An enzyme of this type, **dextran sucrase** of *Leuconostoc* and *Streptococcus*, adds glucosyl units at the *reducing* ends of the dextran chains (p. 174). Sucrose is the direct donor of the glucosyl groups, which are added by an insertion mechanism.[121,176–178] However, it is not dependent upon a membrane lipid as is that of Fig. 20-5. The glucosyl groups are transferred from sucrose to one of a pair of carboxylate groups of aspartate side chains in the active site.[179,180] If both carboxylates are glucosylated, a dextran chain can be initiated by insertion of one glucosyl group into the second (Eq. 20-19). The dextran grows alternating binding sites between the two carboxylates. Chain growth can be terminated by reaction with a sugar or oligosaccharide that fits into the active site and acts in place of the glucosyl group attached to Asp 1 as pictured in Eq. 20-19. The α-1,3-linked branches can be formed when a 3-OH group of

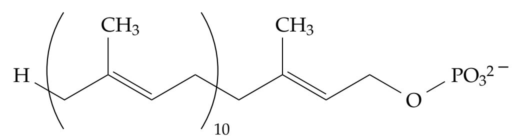


R = H or growing α–1,6-linked chain

(20-19)

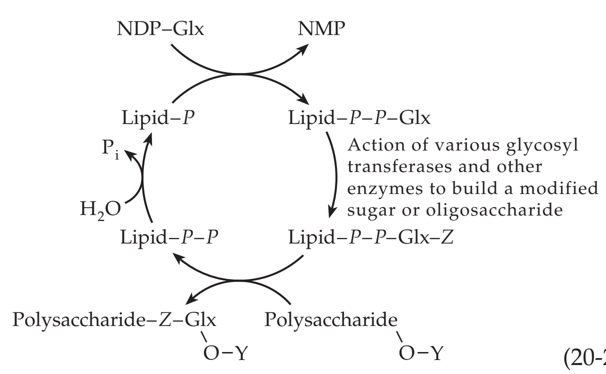
a second dextran chain enters the catalytic site, serving as the glycosyl acceptor. See Robyt for a detailed discussion of synthesis of dextrans and related polysaccharides such as **alternan** and the α-1,3-linked **mutan** (p. 175).[121,176] Some bacteria form β-2,6-linked **fructans** by a similar mechanism, with glucose being released by displacement on C2 of sucrose.[181] Fructans are also formed in green plants, apparently from reaction of two molecules of sucrose with release of glucose to form the trisaccharide Fru*f*β2 → 1Fru*f*β2–1α-Glc*p*, which then transfers a fructosyl group to the growing chain.

Lipid-dependent synthesis of polysaccharides. Insertion of monomer units at the base of a chain is a major mechanism of polymerization that is utilized for synthesis not only of polysaccharides but also of proteins (Chapter 29). For most carbohydrates the synthesis is dependent upon a polyprenyl lipid alcohol. In bacteria this is often the 55-carbon **undecaprenol** or **bactoprenol**,[136] which functions as a phosphate ester:



Undecaprenol phosphate

It serves as a membrane anchor for the growing polysaccharide. We have already discussed one example in the hypothetical cellulose synthase mechanism of Fig. 20-5. For some polysaccharides the mechanism is better established. The synthetic cycles all resemble that of Fig. 20-5 and can be generalized as in Eq. 20-20. Here NDP-Glx is a suitable nucleotide disphosphate derivative of sugar Glx, and Z-Glx is the repeating unit of the polysaccharide formed by the action of glycosyltransferases and other enzymes.



(20-20)

For example, the biosynthesis of alginate involves GDP-mannuronic acid (GDP-ManA) as NDP-Glx, bactoprenol as the lipid, and a glycosyltransferase that inserts a second mannuronate residue (as Z).

An additional transferase that uses acetyl-CoA as a substrate sometimes acetylates one mannuronate unit. The disaccharide units are then inserted into the growing chain. An additional modification, which occurs after polymerization, is random C5 epimerization of unacetylated D-mannuronate residues to L-guluronate.[136,182] Formation of alginate is of medical interest because infections by alginate-forming bacteria are a major cause of respiratory problems in cystic fibrosis.[182]

Sometimes an oligosaccharide assembled on the polyprenol phosphate represents a substantial block in assembly of a repeating polymer. For example, the xanthan gum (p. 179) produced by the bacterium *Xanthomonas campestris* is formed by several successive glycosyl transfers to bactoprenol-*P*-*P*-Glc. A second glucose is transferred onto the first from UDP-Glc, forming a pair of glucosyl groups in β-1,4 linkage. Mannose is then transferred from GDP-Man and joined in an α-1,3 linkage to the first GDP-Man to form a branch point. A glucuronate residue is then transferred from UDP-GlcA and another mannose from GDP-Man. The last mannose is modified by reaction with PEP to form a ketal (Eq. 4-9). The product of this assembly is the following lipid-bound oligosaccharide block.

$$\begin{array}{ccccc} \text{Pyruvate} & & \text{Acetyl} & \text{Glc}\beta 1 & \\ \| & & \downarrow & \downarrow & \\ & & 6 & 4 & \\ \text{Man}\beta 1 \rightarrow & 4\ \text{GlcA}\beta 1 \rightarrow & 2\ \text{Man}\alpha 1 \rightarrow & 3\ \text{Glc}\beta 1 \rightarrow & \text{O-}P\text{-}P\text{-lipid} \end{array}$$

This is inserted into the growing polysaccharide using the free 4-OH on the second glucose to link the units in a cellulose type chain. The twelve separate genes needed for synthesis of xanthan gum are contained in a 16-kb segment of the *X. campestris* genome.[136] Lipid-bound intermediates are also involved in synthesis of peptidoglycans (Fig. 20-9) and in the assembly of bacterial O-antigens (Fig. 8-30). Both of these also yield "block polymers."

D. Proteoglycans and Glycoproteins

The glycoproteins contain oligosaccharides attached to the protein either through *O*-glycosidic linkages with hydroxyl groups of side chains of serine, threonine, hydroxyproline, or hydroxylysine (*O*-linked) or via glycosylaminyl linkages to asparagine side chains (*N*-linked). The "core proteins" of the proteoglycans carry long polysaccharide chains, which are usually *O*-linked and are usually described as glycosaminoglycans.

1. Glycosaminoglycans

Synthesis of the alternating polysaccharide hyaluronan has been discussed in Section C,3 and may occur by an insertion mechanism. However, other glycosaminoglycans (sulfate esters of **chondroitin**, **dermatan**, **keratan**, **heparan**, and **heparin**) grow at their nonreducing ends.[183,184] Their synthesis is usually initiated by the hydroxyl group of serine or threonine side chains at special locations within several secreted proteins.[185] These proteins are synthesized in the rough ER and then move to the Golgi. Addition of the first sugar ring begins in the ER with transfer of single xylosyl residues to the initiating –OH groups.[186–190b] This reaction is catalyzed by the first of a group of special glycosyltransferases of high specificity that form the special terminal units (Chapter 4, Section D,1), that anchor the alternating polysaccharide represented here as $(\text{X–Y})_n$:

$$(\text{X} \rightarrow \text{Y})_n \rightarrow 4\ \text{GlcA}\beta 1 \rightarrow 3\ \text{Gal}\beta 1 \rightarrow 3\ \text{Gal}\beta 1 \rightarrow 4\ \text{Xyl}\beta 1 \rightarrow \text{O-Ser / Thr}$$

After transfer of the xylosyl residue from UDP-xylose to the –OH group in the protein,[190a] a second enzyme with proper specificity transfers a galactosyl group from UDP-galactose, joining it in β-1,4 linkage. A third enzyme transfers another galactosyl group onto the first one in β-1,3 linkage. A fourth enzyme, with a specificity different from that used in creating the main chain, then transfers a glucuronosyl group from UDP-glucuronic acid onto the chain terminus to complete the terminal unit.[190c] Then two more enzymes transfer the alternating units in sequence to form the repeating polymer with lengths of up to 100 or more monosaccharide residues. The sequence $(\text{X–Y})_n$ in the preceding formula is:

$$(\rightarrow 4\ \text{GlcA}\beta 1 \rightarrow 3\ \text{GalNAc}\beta 1 \rightarrow)_n$$
for chondroitin[172a,191,192]

$$(\rightarrow 4\ \text{GlcA}\beta 1 \rightarrow 4\ \text{GlcNAc}\alpha 1 \rightarrow)_n$$
for heparan sulfate and heparin

$$(\rightarrow 3\ \text{Gal}\beta 1 \rightarrow 4\ \text{GlcNAc}\beta 1 \rightarrow)_n$$
for dermatan sulfate

Subsequent modifications of the polymers involve extensive formation of *O*-sulfate esters,[190a,193–197] *N*-deacetylation and *N*-sulfation,[198,199] and epimerization at C5.[10] In some tissues almost all GluA is epimerized.[200] The modifications are especially extensive in dermatan, heparan sulfates, and heparin (see also p. 177).[196,201–203b] The modifications are not random and follow a defined order. *N*-Deacetylation must precede *N*-sulfation, and *O*-sulfation is initiated only after *N*-sulfation of the entire chain is complete. The modifications occur within the Golgi (see Fig. 20-7) but not all of the glycosyltransferases, PAPS (3'-phosphoadenosine 5'-phosphosulfate)-dependent sulfotransferases, and epimerases are present within a single compartment. Nevertheless, an entire glycosaminoglycan chain can be synthesized within 1–3 min.[189]

The completed polymers are modified uniformly. There are clusters of sulfo groups with unusual structures in chondroitin from squid and shark cartilages[204,205] and fucosylated chondroitin from echinoderms.[206] Similar modifications are present less extensively in vertebrates. One of the best known modifications forms the unique pentasaccharide sequence shown in Fig. 4-13, which is essential to the anticoagulant activity of heparin. This sequence has been synthesized in the laboratory as have related longer heparin chains. A sequence about 17 residues in length containing an improved synthetic version of the unique pentasaccharide binds tightly to both thrombin and antithrombin (Chapter 12, Section C,9).[207,208] This opens the door to the development of improved substitutes for the medically important heparin. Heparan sulfate chains are found on proteoglycans throughout the body, but the highly modified heparin does not circulate in the blood. It is largely sequestered in cytoplasmic granules within mast cells and is released as needed.[208a] Heparin binds to many different proteins (p. 177). Among them is the glycoprotein selenoprotein P (p. 824), which may impart antioxidant properties to the extracellular matrix.[208b]

Although glycosaminoglycans are most often attached to *O*-linked terminal units, chondroitin sulfate chains can also be synthesized with *N*-linked oligosaccharides attached to various glycoproteins serving as initiators.[209] At least one form of keratan sulfate, found in the cornea, is linked to its initiator protein via GlcNAc-Man to *N*-linked oligosaccharides of the type present in many glycoproteins (Section D).

At least 25 different proteins that are secreted into the extracellular spaces of the mammalian body carry glycosaminoglycan chains.[183,210,211] Most of these proteins can be described as (1) **small leucine-rich proteoglycans** with 36- to 42-kDa protein cores and (2) **large modular proteoglycans** whose protein cores have molecular masses of 40 to 500 kDa.[210] The most studied of the second group is **aggrecan**, a major component of cartilage. This 220-kDa protein carries ~100 chondroitin chains, each averaging about 100 monosaccharide residues and ~100 negative charges from the carboxylate and sulfate groups. Aggrecan has three highly conserved globular domains near the N and C termini.[212–213a] The G1 domain near the N terminus is a **lectin** (p. 186), which, together with a small link protein that is structurally similar to the G1 domain, binds to a decasaccharide unit of hyaluronan. One hyaluronan molecule of 500- to1000-kDa mass (~2500–5000 residues) may bind 100 aggrecan and link molecules to form an ~200,000-kDa particle such as that shown in Fig. 4-16. These enormous highly negatively charged molecules, together with associated counterions, draw in water and preserve osmotic balance. It is these molecules that keep our joints mobile and which deteriorate by proteolytic degradation in the common **osteoarthritis**.[214,215] The keratan sulfate content of cartilage varies with age, and the level in serum and in synovial fluid is increased in osteoarthritis.[215] Keratan sulfate is also found in the cornea and the brain. Its content is dramatically decreased in the cerebral cortex of patients with Alzheimer disease.[216]

Other modular core proteins[210] include **versican** of blood vessels and skin,[210,213a,217,217a] **neurocam** and **brevican** of brain, **perlecan** of basement membranes,[218] **agrin** of neuromuscular junctions, and **testican** of seminal fluid. However, several of these have a broader distribution than is indicated in the foregoing description. The sizes vary from 44 kDa for testican to greater than 400 kDa for perlecan. The numbers of glycosaminoglycan chains are smaller than for aggrecan, varying from 1 to 30. Another of the chondroitin sulfate-bearing core protein is **appican**, a protein found in brain and one of the splicing variants of the amyloid precursor protein that gives rise to amyloid deposits in Alzheimer disease (Chapter 30).[217a,b]

The core proteins of the leucine-rich proteoglycans have characteristic horseshoe shapes and are constructed from ~28-residue repeats, each containing a β turn and an α helix. The three-dimensional structures are doubtless similar to that of a ribonuclease inhibitor of known structure which contains 15 tandem repeats.[219,220] A major function of these proteoglycans seems to be to interact with collagen fibrils, which have distinct proteoglyan-binding sites,[221] and also with fibronectin.[222] The small leucine-rich proteoglycans have names such as **biglycan**, **decorin**,[222a] **fibromodulin**, **lumican**, **keratoglycan**, **chondroadherin**, **osteoglycin**, and **osteoadherin**.[219,223–223c] The distribution varies with the tissue and the stage of development. For example, biglycan may function in early bone formation; decorin, which has a high affinity for type I collagen, disappears from bone tissue as mineralization takes place. Osteoadherin is found in mature osteoblasts.[223] **Phosphocan**, another brain proteoglycan, has an unusually high content (about one residue per mole) of L-isoaspartyl residues (see Box 12-A).[224]

Proteoglycans bind to a variety of different proteins and polysaccharides. For example, the large extracellular matrix protein **tenascin**, which is important to adhesion, cell migration, and proliferation, binds to chondroitin sulfate proteoglycans such as neurocan.[225] **Syndecan**, a transmembrane proteoglycan, carries both chondroitin and heparan sulfate chains, enabling it to interact with a variety of proteins that mediate cell–matrix adhesion.[185]

The ability of dissociated cells of sponges to aggregate with cells only of a like type (p. 29) depends upon large extracellular proteoglycans. That of *Microciona prolifera* appears to be an aggregate of about three hundred 35-kDa core protein molecules with equal masses of attached carbohydrate. This **aggregation factor** has a total mass of ~2 x 10^4 kDa.[226,227] It apparently interacts specifically, in the presence of Ca^{2+} ions, with a 210-kDa cell matrix protein to hold cells of the same species together.[227]

2. *O*-Linked Oligosaccharides

A variety of different oligosaccharides are attached to hydroxyl groups on appropriate residues of serine, threonine, hydroxylysine, and hydroxyproline in many different proteins (Chapter 4). Such oligosaccharides are present on external cell surfaces, on secreted proteins, and on some proteins of the cytosol and the nucleus.[228–231b] The rules that determine which –OH groups are to become glycosylated are not yet clear.[232] Glycosylation occurs in the ER, and, just as during synthesis of the long carbohydrate chains of proteoglycans, the sugar rings are added directly to an –OH group, either of the protein or of the growing oligosaccharide. The first glycosyl group transferred is most often **GalNAc** for external and secreted proteins[233] but more often **GlcNAc** for cytosolic and nuclear proteins.[228,231,233a–c] Glycosylation of protein –OH groups can occur on either the luminal or cytosolic faces of the ER membranes.[234] The external *O*-linked glycoproteins often have large clusters of oligosaccharides attached to –OH groups of serine or threonine, but cytosolic proteins may carry only a small number of small oligosaccharides.

Of great importance are the **blood group determinants** which are discussed in Box 4-C. The ABO determinants are found at the nonreducing ends of *O*-linked oligosaccharides. Conserved Ser/Thr sites in the epidermal growth factor domains (Table 7-3) of various proteins carry ***O*-linked fucose**.[235]

The secreted **mucins** are unique in having clusters of large numbers of oligosaccharides linked by ***N*-acetylgalactosamine** to serine or threonine of the polypeptide.[236] The following core structures predominate.[237] These may be lengthened or further branched by the particular variety of glycosyltransferases present in a tissue and by their specificities.[233] The human genome contains at least nine mucin genes.[238] The large apomucins contain central domains with tandem repeats rich in Ser, Thr, Gly, and Ala and flanked at the ends by cysteine-rich domains.[239] For example, porcine submaxillary mucins are encoded by a gene with at least three alleles that encode 90, 125, or 133 repeats. The polypeptide may contain as many as 13,288 residues. N-terminal cysteine-rich regions are involved in dimer formation.[240]

core 1 $\quad \text{Gal}\beta1 \rightarrow 3\ \text{GalNAc} - \text{Ser/Thr}$

core 2
$$\begin{array}{rcl}\text{Gal}\beta1 \rightarrow 3 & \text{GalNAc} & - \text{Ser/Thr} - \text{Ser/Thr}\\ & 6 & \\ & \uparrow & \\ & \text{GalNAc}\beta1 & \end{array}$$

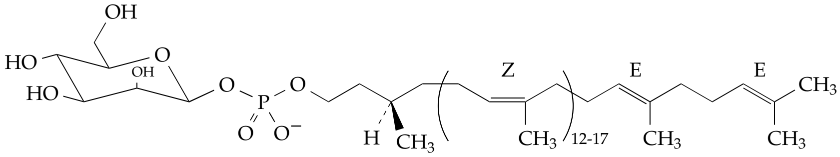
core 3 $\quad \text{GlcNAc}\beta1 \rightarrow 3\ \text{GalNAc} - \text{Ser/Thr}$

core 4
$$\begin{array}{rcl}\text{Gal}\beta1 \rightarrow 4\ \text{GlcNAc}\beta1 \rightarrow 6 & \text{GalNAc} & - \text{Ser/Thr}\\ & 3 & \\ & \uparrow & \\ & \text{GlcNAc}\beta1 & \end{array}$$

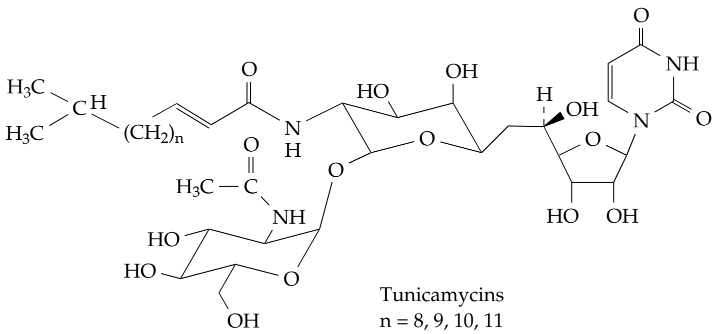
3. Assembly of *N*-Linked Oligosaccharides on Glycosyl Carrier Lipids

In eukaryotes the biosynthesis of the *N*-linked oligosaccharides of glycoproteins depends upon the polyprenyl alcohols known as **dolichols**, which are present in membranes of the endoplasmic reticulum. They contain 16–20 prenyl units, of which the one bearing the OH group is completely saturated as a result of the action of an NADPH-dependent reductase on the unsaturated precursor.[241] The predominant dolichol in mammalian cells contains 19 prenyl units. The structure of its mannosyl phosphate ester, one of the intermediates in the oligosaccharide synthesis, is illustrated below. The fully extended 95-carbon dolichol has a length of almost 10 nm, four times greater than that of oleic acid and twice the thickness of the nonpolar membrane bilayer core. The need for this great length is not clear nor is it clear why the first prenyl unit must be saturated for good acceptor activity.

The assembly of the oligosaccharides that will become linked to Asn residues in proteins occurs on the phosphate head of dolichol-*P*. The process begins on the cytoplasmic face of the membrane and within the lumen of the rough or smooth ER and continues within cisternae of the Golgi apparatus.[234,242–245] The initial transfer of GlcNAc-*P* to dolichol-*P* (Fig. 20-6, step *a*) appears to occur on the cytoplasmic face of the ER and is specifically inhibited by **tunicamycin**.[246,247] As the first "committed reaction" of *N*-glycosylation, it is regulated by a variety of hormonal and other factors.[248,249] The reaction takes place cotranslationally as the still unfolded peptide chain leaves the ribosome.[242] The oligosaccharide, still attached to the dolichol, continues to grow on the cytosolic surface of the ER membrane by transfer of GlcNAc and five residues of mannose from their sugar nucleotide forms (Fig. 20-6, steps *b* and *c*).[249a] The intermediate Dol-*P*-*P*-$\text{GlcNAc}_2\text{Man}_5$ crosses the membrane bilayer (Fig. 20-6, step *d*), after which mannosyl and glucosyl units are added (steps *e* and *f*). These sugars are carried across the membrane while attached to dolichol.



Man-*P*-dolichol



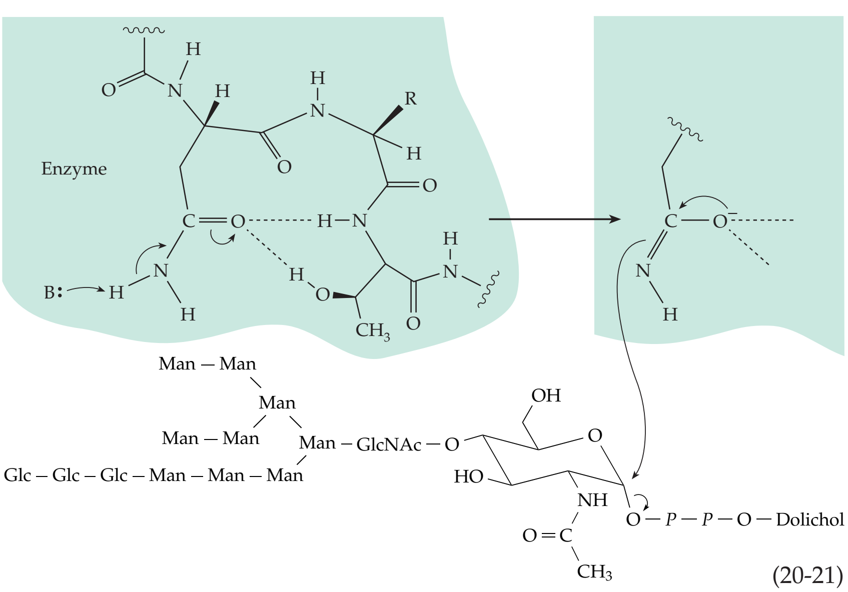
Tunicamycins
n = 8, 9, 10, 11

The completed 14-residue branched oligosaccharide $Glc_3Man_9GlcNAc_2$, with the structure indicated in Fig. 20-6, is then transferred to a suitable asparagine side chain (step *g*). This may be on a newly synthesized protein or on a still-growing polypeptide chain that is being extruded through the membrane into the lumenal space of the rough ER (Eq. 20-21; Fig. 20-6). The glycosylation site is often at the sequence Asn-X-Ser(Thr), which is likely to be present at a beta bend in the folded protein. Bends of the type illustrated in Eq. 20-21 and stabilized by the asparagine side chain are apparently favored.[250] In such a bend the –OH of the serine or threonine helps to polarize the amide group of the Asn side chain, perhaps enolizing it and generating a nucleophilic center that can participate in a displacement reaction[250,251] as indicated in Eq. 20-21. The **oligosaccharyltransferase** that catalyzes the reaction is a multisubunit protein. As many as eight different subunits have been reported for the enzyme in yeast. Genes for at least five of these are essential.[250–252b] One subunit serves to recognize suitable glycosylation sites.

Trimming of glycoprotein oligosaccharides. After transfer to glycoproteins the newly synthesized oligosaccharides undergo trimming, the hydrolytic removal of some of the sugar units, followed by addition of new sugar units to create the finished glycoproteins. The initial glycosylation process ensures that the glycoproteins remain in the lumen of the ER or within vesicles or cisternae separated from the cytoplasm. The subsequent processing appears to allow the cell to **sort** the proteins. Some remain attached to membranes and take up residence within ER, Golgi, or plasma membrane. Others are passed outward into transfer vesicles, Golgi, and secretion vesicles. A third group enter **lysosomes**. A series of specific inhibitors of trimming reactions, some of whose structures are shown in Fig. 20-7, has provided important insights.[253–255] Use of these inhibitors, together with immunochemical methods and study of yeast mutants,[250,252,256] is enabling us to learn many details of glycoprotein biosynthesis.

Whereas the formation of dolichol-linked oligosaccharides occurs in an identical manner in virtually all eukaryotic cells, trimming is highly variable as is the addition of new monosaccharide units.[257–257b] The major pathway for mammalian glycoproteins is shown in Fig. 20-7. Specific hydrolases in the ER remove all of the glucosyl units and one to three mannosyl units.[258] Removal of additional mannosyl residues occurs in the cis Golgi, to give the pentasaccharide core $Man_3GlcNAc_2$ which is common to all of the complex N-linked oligosaccharides. However, partial trimming without additional glycosylation produces some "high mannose" oligosaccharides.[258a] Removal of glucose may be necessary to permit some glycoproteins to leave the ER.

Sulfate groups and in some cases fatty acyl groups[259] may also be added. The exact composition of the oligosaccharides may depend upon the condition of the cell and may be altered in response to external influences.[257] Oligosaccharides attached to proteins that remain in the ER membranes may undergo



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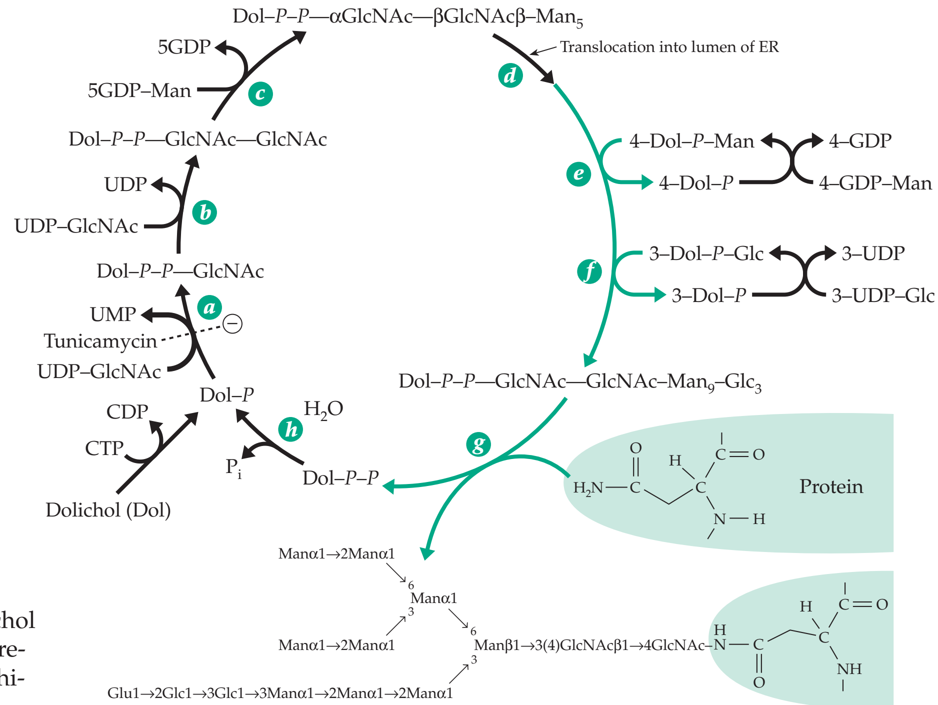


Figure 20-6 Biosynthesis of the dolichol diphosphate-linked oligosaccharide precursor to glycoproteins. The site of inhibition by tunicamycin is indicated.

very little trimming. However, the three glucosyl residues are usually removed by the glucosidases present in the rough ER. Some plant glycoproteins of the high-mannose type undergo no further processing.

Extensions and terminal elaborations. Even before trimming is completed, addition of new residues begins within the medial cisternae.[260] In mammalian Golgi, *N*-acetylglucosamine is added first. Galactose, sialic acid, and often fucose are then transferred from their activated forms to create such elaborate oligosaccharides as that shown in Fig. 4-17. As many as 500 glycosyltransferases, having different specificities for glycosyl donor and glucosyl acceptor, may be involved.[245] Extensions of the basic oligosaccharide structure often contain polylactosamine chains, branches with fucosyl residues, and sulfate groups.[245,261] More than 14 sialyltransferases place sialic acid residues, often in terminal positions, on these cell surface oligosaccharides.[262,263]

The cell wall of the yeast *Saccharomyces* is rich in **mannoproteins** that contain 50–90% mannose.[264] The ~250-residue mannan chains consist of an α-1,6-linked backbone with mono-, di-, and tri-mannosyl branches. These are attached to the same core structure as that of mammalian oligosaccharides. All of the core structures are formed in a similar way.[258,265] The mannoproteins may serve as a "filler" to occupy spaces in a cell wall constructed from β-1,3- and β-1,6-linked glycans and chitin. All of the four components, including the mannoproteins, are covalently linked together.[266] As was emphasized in Chapter 4 (pp. 186–188) glycoproteins serve many needs in biological recognition. The N-linked oligosaccharides play a major role in both animals and plants.[266a–c] Use of mass spectrometry, new automated methods of oligosaccharide synthesis,[266d] and development of new synthetic inhibitors[266c] are all contributing to current studies of what is commonly called "glycobiology."

The perplexing Golgi apparatus. First observed by Camillo Golgi[267,268] in 1898, the stacked membranes, now referred to simply as Golgi, remain somewhat mysterious.[268–271] There are at least three functionally distinct sets of Golgi cisternae, the **cis** (nearest the nucleus), **medial**, and **trans**. An additional series of tubules referred to as the **trans Golgi network** lies between the Golgi and the cell surface and may be the site at which lysosomal enzymes are sorted from proteins to be secreted.[260,272] Immunochemical staining directed toward specific glycosidases and glycosyl transferases suggested that the trimming reactions of glycoproteins start in the ER and continue as the proteins pass outward successively from one compartment of the Golgi to the next (Fig. 20-8). This has been the conventional view since the 1970s. The movement of the glycoproteins between compartments is thought to take place in small vesicles using a rather elaborate system of specialized proteins. Some of these coat the vesicles[273,274] while others target the vesicles to specific locations, e.g., the lysosomes[275] or the plasma membranes where they may be secreted.[273,277] A host of regulatory G-proteins assist these complex processes and drive them via hydolysis of GTP.[271,278]

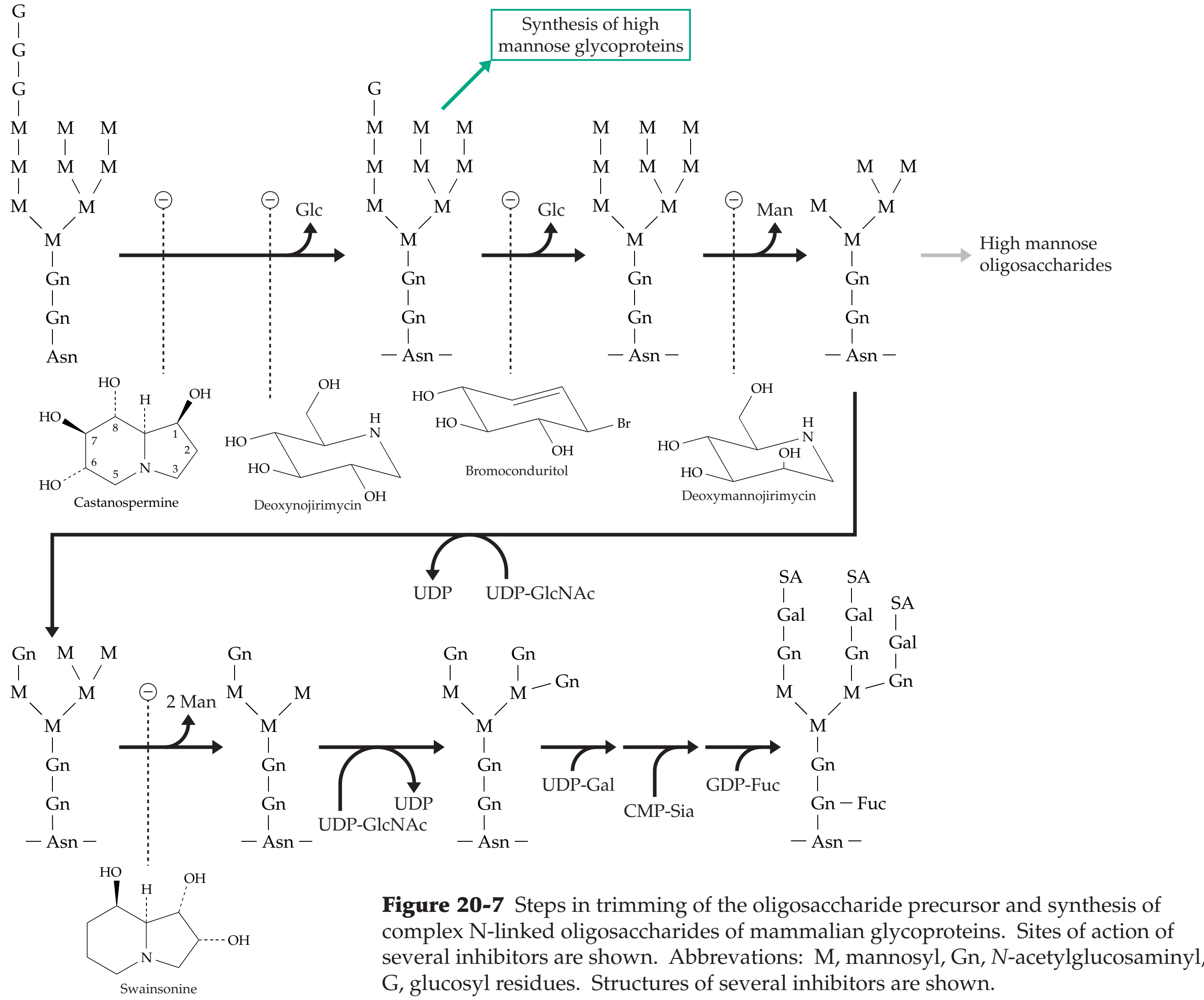


Figure 20-7 Steps in trimming of the oligosaccharide precursor and synthesis of complex N-linked oligosaccharides of mammalian glycoproteins. Sites of action of several inhibitors are shown. Abbrevations: M, mannosyl, Gn, *N*-acetylglucosaminyl, G, glucosyl residues. Structures of several inhibitors are shown.

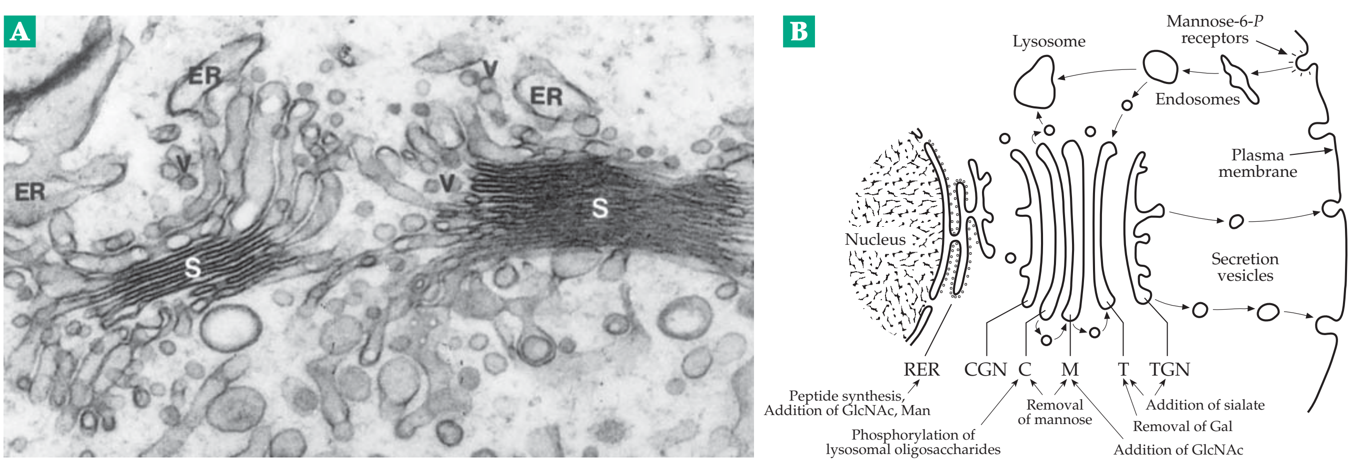
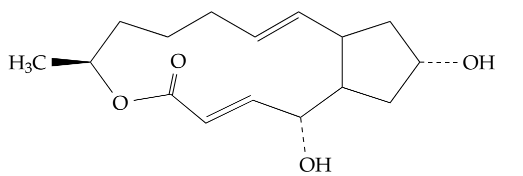


Figure 20-8 (A) Electron micrograph showing a transverse section through part of the Golgi apparatus of an early spermatid. Cisternae of the ER, Golgi stacks (S), and vesicles (V) can be seen. Curved arrows point to associated tubules. Magnification X45,000.[276] Courtesy of Y. Clermont. (B) Scheme showing functions of endoplasmic reticulum, transfer vesicles, Golgi apparatus, and secretion vesicles in the metabolism of glycoproteins.

While most proteins synthesized in the ER follow the exocytotic pathway through the Golgi, some are retained in the ER and some that pass on through the Golgi are returned to the ER.[279] In fact, such **retrograde transport** can carry some proteins taken up by endocytosis through the plasma membrane and through the Golgi to the ER where they undergo *N*-glycosylation. Retrograde transport is essential for recycling of plasma membrane proteins and lipids. The forward flow of glycoproteins and membrane components from the ER to the Golgi can be blocked by the fungal macrocyclic lactone **brefeldin A**. In cells treated with this drug, which inactivates a small



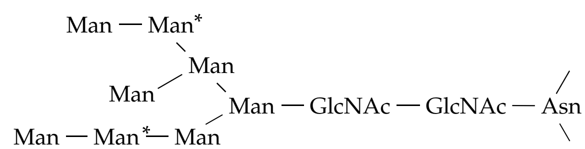
Brefeldin A

CTP-binding protein,[280] the Golgi apparatus is almost completely resorbed into the ER by retrograde transport. Proteins remaining in the ER undergo increased *O*-glycosylation as well as unusual types of *N*-glycosylation.

Although the conventional view of flow through the Golgi is generally accepted, it is difficult to distinguish it from an alternative explanation: *The Golgi compartments may move outward continuously while retrograde transport occurs via the observed vesicles*.[268,272] Some evidence for this **cisternal maturation model** has been known for many years but was widely regarded as reflecting unusual exceptions to the conventional model. In fact, both views could be partially correct; vesicular transport may function in both directions.[280a] High-resolution tomographic images are also altering our view of the Golgi.[280b]

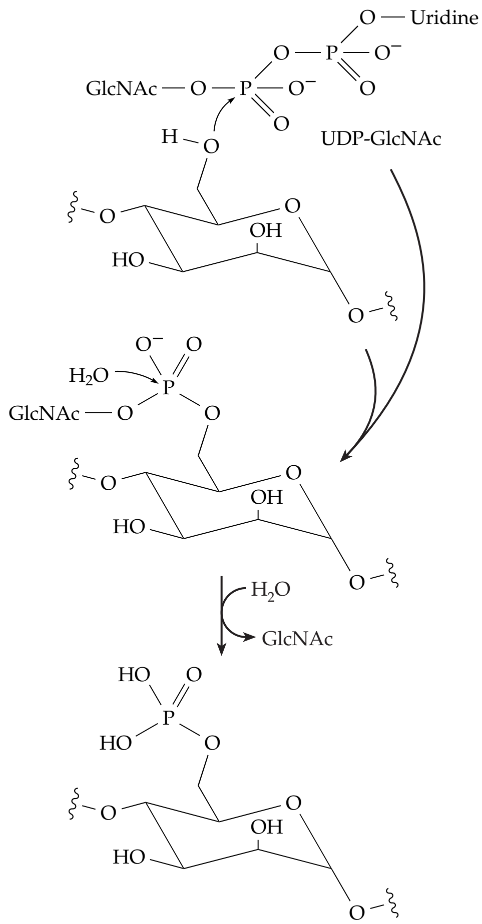
The proteins of Golgi membranes are largely integral membrane proteins and peripheral proteins associated with the cytosolic face. Some of the integral membrane proteins are the oligosaccharide-modifying enzymes, which protrude into the Golgi lumen.[280c,280d] Many other proteins participate in transport,[280d–f] docking, membrane fusion,[280g,h] and acidification of Golgi compartments.[280i] Many of the first studies of vesicular transport were conducted with synaptic vesicles and are considered in Chapter 30 (see Fig. 30-20). Other aspects of membrane fusion and transport are discussed in Chapter 8. A group of specialized Golgi proteins, the **golgins**, are also present. They are designated golgin-84, -95, -160, -245, and -376 (giantin or macrogolgin) and were identified initially as human **autoantigens** (Chapter 31), appearing in the blood of persons with autoimmune disorders such as Sjögren's syndrome.[281,282] Another protein of molecular mass ~130 kDa, and which appears to be specific to the trans-Golgi network, has been found in human serum of patients with renal vasculitis.[283]

Lysosomal enzymes. Various **sorting signals** are encoded within proteins. These include the previously mentioned C-terminal KDEL (mammals) or HDEL (yeast) amino acid sequence, which serves as a retrieval signal for return of proteins from the Golgi to the ER (p. 521).[279,284] Other sorting signals are provided by the varied structures of the oligosaccharides attached to glycoproteins. These sugar clusters convey important biological information, which is "decoded" in the animal body by interaction with various **lectins** that serve as receptors.[285] This often leads to endocytosis of the glycoprotein. An example is provided by the more than 50 proteins that are destined to become lysosomal enzymes and which undergo phosphorylation of 6-OH groups of the mannosyl residue marked by asterisks on the following structure. This is an *N*-linked oligosaccharide that has been partially trimmed. The phosphorylation is accomplished in



two steps by enzymes present in the cis Golgi compartment (Eq. 20-22) An ***N*-acetylglucosaminylphosphotransferase** transfers phospho-GlcNAc units from UDP-GlcNAc onto the 6-OH groups of mannosyl residues. These must be recognized by the phosphotransferase as appropriate.[286,287] Then a hydrolase cleaves off GlcNAc.

The proteins carrying the mannose 6-phosphate groups bind to one of two different **mannose 6-*P* receptors** present in the Golgi membranes and are subsequently transported in clathrin-coated vesicles to endosomes where the low pH causes the proteins to dissociate from the receptors, which may be recycled.[288–290a] The hydrolytic enzymes are repackaged in lysosomes. The same mannose 6-*P* receptors also appear on the external surface of the plasma membrane allowing many types of cells to take up lysosomal enzymes that have "escaped" from the cell. These proteins, too, are transported to the lysosomes. The mannose 6-*P* receptors have a dual function, for they also remove insulinlike growth factor from the circulation, carrying it to the lysosomes for degradation.[287,290] Most Man 6-*P* groups are removed from proteins once they reach the lysosomes but this may not always be true.[291] Not all lysosomal proteins are recruited by the mannose 6-*P* receptors. Some lysosomal membrane proteins are sorted by other mechanisms.[292]

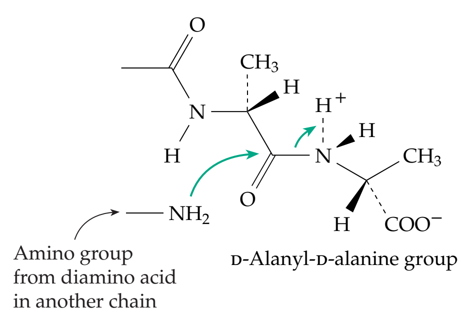


(20-22)

The hepatic asialoglycoprotein (Gal) receptor. A variety of proteins are taken out of circulation in the blood by the hepatocytes of the liver. Serum glycoproteins bearing sialic acid at the ends of their oligosaccharides (see Fig. 20-7) have relatively long lives, but if the sialic acid is removed by hydrolysis, the exposed galactosyl residues are recognized by the multisubunit **asialoglycoprotein receptor**.[293–295] The bound proteins are then internalized rapidly via the coated pit pathway and are degraded in the lysosomes. Other receptors, including those that recognize transferrin, low-density lipoprotein, α_2 macroglobulin, and T lymphocyte antigens, also depend upon interaction with oligosaccharides.[296]

E. Biosynthesis of Bacterial Cell Walls

The outer surfaces of bacteria are rich in specialized polysaccharides. These are often synthesized while attached to lipid membrane anchors as indicated in a general way in Eq. 20-20.[136,296a] One of the specific biosynthetic cycles (Fig. 20-9) that depends upon undecaprenol phosphate is the formation of the **peptidoglycan** (murein) layer (Fig. 8-29) of both gram-negative and gram-positive bacterial cell walls. Synthesis begins with attachment of L-alanine to the OH of the lactyl group of UDP-*N*-acetylmuramic acid in a typical ATP-requiring process (Fig. 20-9, step *a*).[297] Next D-glutamic acid, *meso*-diaminopimelate (Fig. 8-29), or L-lysine, and D-alanyl-D-alanine are joined in sequence, each in another ATP-requiring step.[298–301d] The entire unit assembled in this way is transferred to undecaprenol phosphate with creation of a pyrophosphate linkage (step *e*). An *N*-acetylglucosamine unit is added by action of another transferase (step *f*), and in an ATP-requiring process ammonia is sometimes added to cap the free α-carboxyl group of the D-glutamyl residue (step *g*). In *Staphylococcus aureus* and related gram-positive bacteria five glycyl units are also added, each from a molecule of glycyl-tRNA (green arrows in Fig. 20-9). The completely assembled repeating unit, together with the connecting peptide chain needed in the crosslinking reaction, is transferred onto the growing chain (step *h*). As in formation of dextrans, growth is by insertion of the repeating unit at the reducing end of the chain. The polyprenyl diphosphate is released, and the cycle is completed by the action of a pyrophosphatase (step *i*). This step is blocked by **bacitracin**, an antibiotic which forms an unreactive complex with the polyprenyl diphosphate carrier. Completion of the peptidoglycan requires crosslinking. This is accomplished by displacement of the terminal D-alanine of the pentapeptide by attack by the $-NH_2$ group of the diaminopimelate or lysine or other diamino acid (see also Fig. 8-29).[301e]



Because the peptidoglycan layer must resist swelling of the bacteria in media of low osmolarity, it must be strong and must enclose the entire bacterium. At the same time the bacterium must be able to grow in size and also to divide. For these reasons bacteria must continuously not only synthesize peptidoglycan but also degrade it.[302,303] The latter is accomplished by hydrolytic cleavage using cell wall enzymes to the ***N*-acetylglucosamine-anhydro-*N*-acetylmuramate-tripeptide** (GlcNAc-1,6-anhydro-MurNAc-L-Ala-D-Glu-A_2pm) fragment.[304,305] A hydrolase cuts the peptide bridge.[305a] This process is probably essential to formation of new growing points for expansion of the murein layer. Most of the peptide fragments that are released in the periplasm are transported back into the cytosol.

The anhydroMurNAc is removed, and new UDP-MurNAc and D-Ala-D-Ala units are added salvaging the tripeptide unit. The repaired UDP-MurNA-pentapeptide can then reenter the biosynthetic pathway (Fig. 20-9).

The O-antigens and lipid A. A cluster of sugar units of specific structure makes up the repeating unit of the "O-antigen" of *Salmonella*. The many structural variations in this surface polysaccharide account for the over two thousand serotypes of *Salmonella* (p. 180).[121,306] As is illustrated in Fig. 8-30, the O-antigen is a repeating block polymer that is attached to a complex lipopolysaccharide "core" and a hydrophobic membrane anchor known as lipid A (Figs. 8-28 and 8-30).[307–308a] Lipid A and the attached core and O-antigen are synthesized inside the bacterial cell by enzymes found in the cytoplasmic membrane.[309] The complete lipopolysaccharide units are then translocated from the inner membrane to the outer membrane of the bacteria. The synthesis of the O-antigen is understood best. Consider the following group E3 antigen, where Abe is abequose (Fig. 4-15) and Rha is rhamnose:

$$\begin{array}{l} \quad\ \text{Abe}\alpha 1 \\ \qquad\ \downarrow \\ \qquad\ 4 \\ (\rightarrow 6\,\text{Gal}\beta 1 \rightarrow 6\,\text{Man}1 \rightarrow 4\,\text{Rha}1 \rightarrow)_n \end{array}$$

Assembly of this repeating unit begins with the transfer of a *phosphogalactosyl* unit from UDP-Gal to the phospho group of the lipid carrier undecaprenol phosphate. The basic reaction cycle is much like that in Fig. 20-9 for assembly of a peptidoglycan.

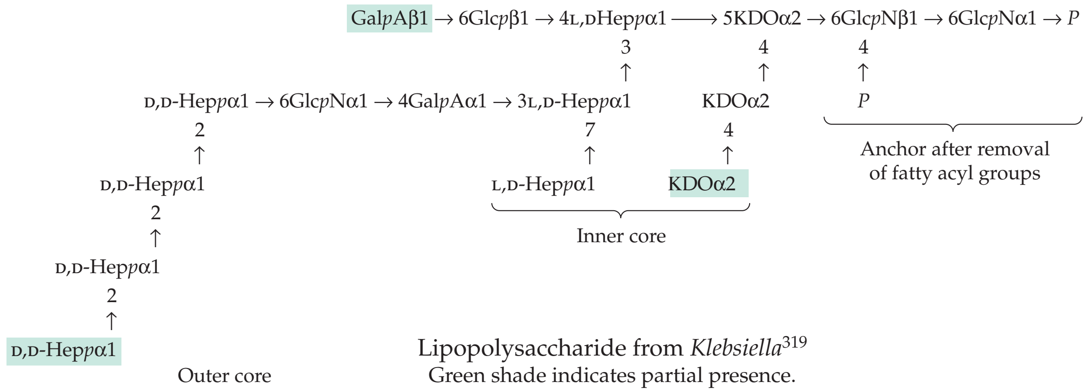
The oligosaccharide repeating unit of the O-antigen is constructed by the consecutive transferring action of three more transferases. For the antigen shown above, one enzyme transfers a rhamnosyl unit, another a mannosyl unit, and another an abequosyl unit from the appropriate sugar nucleotides. Then the entire growing O-antigen chain, which is attached to a second molecule of undecaprenol diphosphate, is transferred onto the end of the newly assembled oligosaccharide unit. In effect, the newly formed oligosaccharide is inserted at the reducing end of the growing chain just as in Fig. 20-9. Elongation continues by the transfer of the entire chain onto yet another tetrasaccharide unit. As each oligosaccharide unit is added, an undecaprenol diphosphate unit is released and a phosphatase cleaves off the terminal phospho group to regenerate the original undecaprenol phosphate carrier. When the O-antigen is long enough, it is attached to the rest of the lipopolysaccharide.

The lipid A anchor is also based on a carbohydrate skeleton. Its assembly in *E. coli*,[307] which requires nine enzymes, is depicted in Fig. 20-10. *N*-Acetylglucosamine 6-*P* is acylated at the 3-position[310] and after deacetylation[311] at the 2-position. As shown in this figure, acylation is accomplished by transfer of hydroxymyristoyl groups from acyl carrier protein (ACP). Two molecules of the resulting UDP-2,3-diacyl-GlcN are then joined via the reactions shown to give the acylated disaccharide precursor to lipid A. Stepwise transfer of KDO, L-*glycero*-D-*manno*-heptose, and other monosaccharide units from the appropriate sugar nucleotides and further acylation follows (Fig. 20-10). The assembled O-antigen chain is transferred from undecaprenol diphosphate onto the lipopolysaccharide core. This apparently occurs on the periplasmic surface of the plasma membrane. If so, the core lipid domain must be flipped across the plasma membrane before the O-antigen chain is attached.[312] Less is known about the transport of the completed lipopolysaccharide across the periplasmic space and into the outer membrane.

The core structures of the lipopolysaccharides vary from one species to another or even from one strain of bacteria to another. All three domains (lipid A, core, and O-antigen) contribute to the antigenic properties of the bacterial surface[313] and to the virulence of the organism.[313,314] Nitrogen-fixing strains of *Rhizobium* require their own peculiar lipopolysaccharides for successful symbiosis with a host plant.[315] However, there are some features common to most lipopolysaccharides. Two to three residues of KDO are usually attached to the acylated diglucosamine anchor, and these are often followed by 3 – 4 heptose rings.[316–318]

The structure of the inner core regions of a typical lipopolysaccharide from *E. coli* is indicated in Fig. 8-30. The complete structure of the lipopolysaccharide from a strain of *Klebsiella* is shown at the top of the next page.[319] Here L, D-Hep*p* is D-*manno*-heptopyranose and D, D-Hep*p* is D-*glycero*-D-*manno*-heptose. As in this case, the outer core often contains several different hexoses. The lipopolysaccharide of *Neisseria meningititis* has sialic acid at the outer end.[320] However, the major virulence factor for this organism, which is a leading cause of bacterial meningitis in young children, is a capsule of poly(ribosyl)ribitol phosphate that surrounds the cell.[321] *Haemophilus influenzae*, a common cause of ear infections and meningitis in children, has no *O*-antigen but a more highly branched core oligosaccharide than is present in *E. coli*.[321a] *Legionella* has its own variations.[321b]

Gram-positive bacteria. Although their outer coatings are extremely varied, all gram-positive bacteria have a peptidoglycan similar to that of gram-negative bacteria but often containing the intercalated pentaglycine bridge indicated in Fig. 8-29. However, the peptidoglycan of gram-positive bacteria is 20–50 nm thick, as much as ten times thicker than that of *E. coli*. Furthermore, the peptidoglycan is intertwined with the anionic polymers known as teichoic acids and



Lipopolysaccharide from *Klebsiella*[319]
Green shade indicates partial presence.

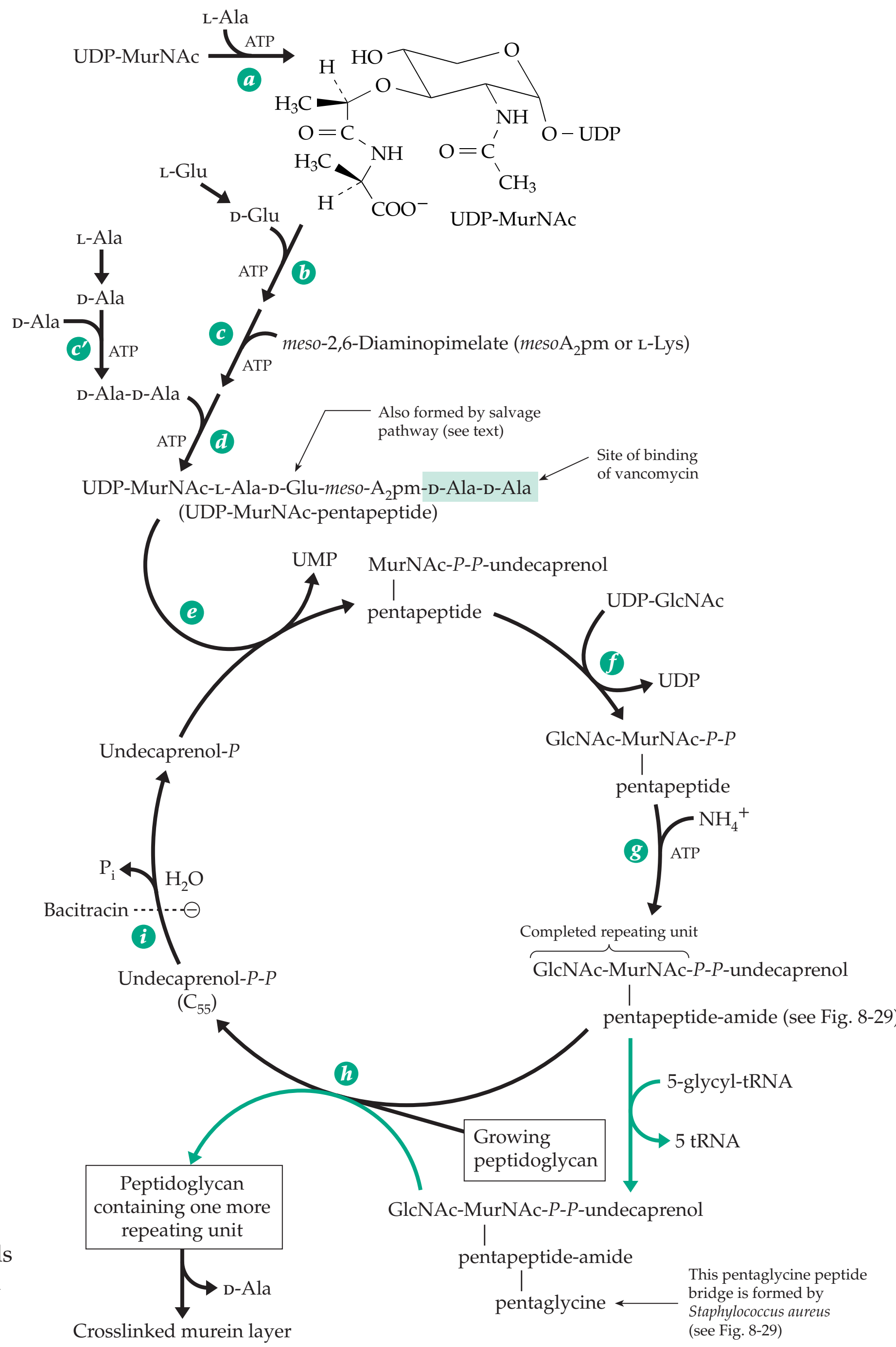


Figure 20-9 Biosynthesis of bacterial peptidoglycans. See Fig. 8-29 for details of the peptidoglycan structures. Green arrows show alternative route used by gram-positive bacteria.

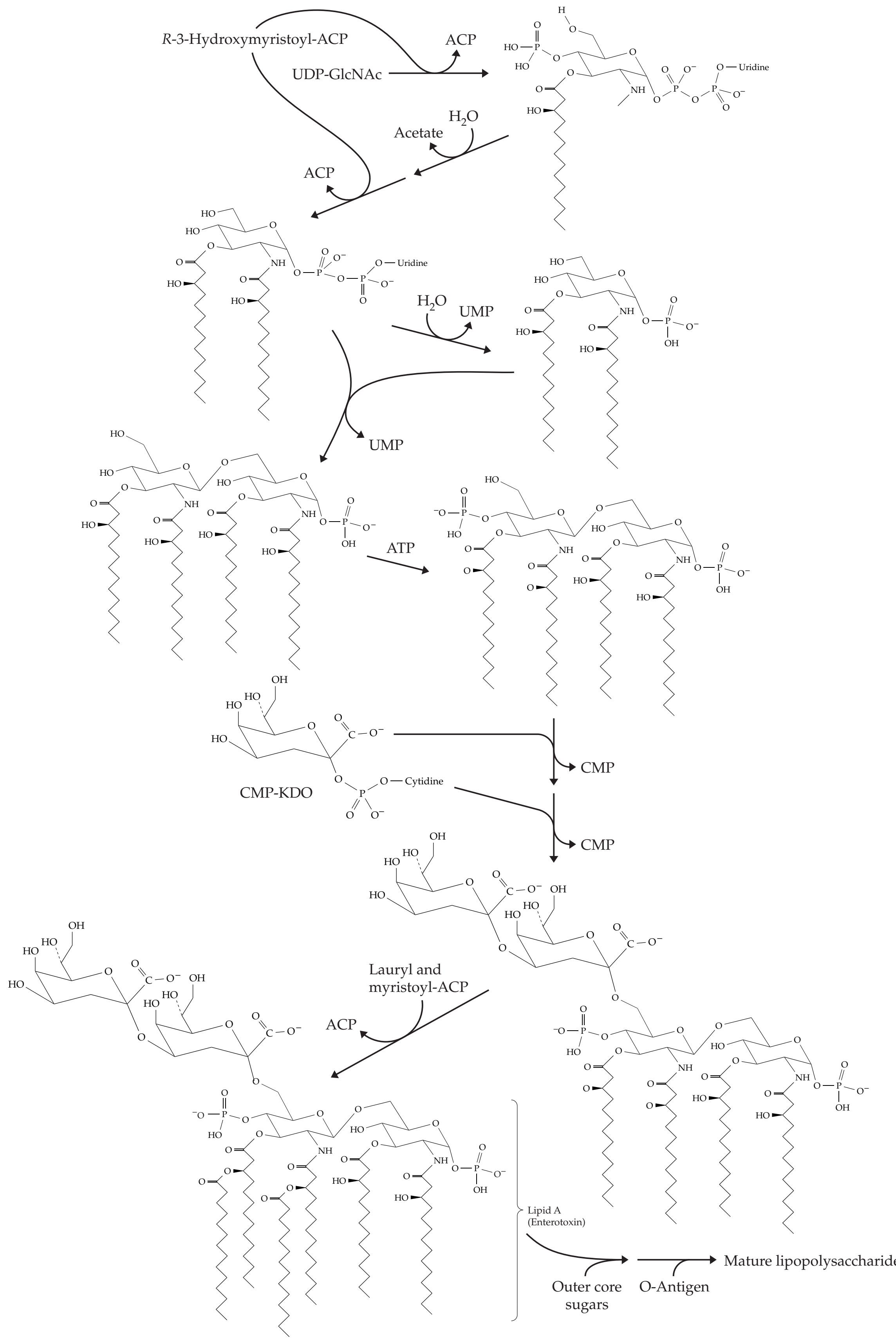
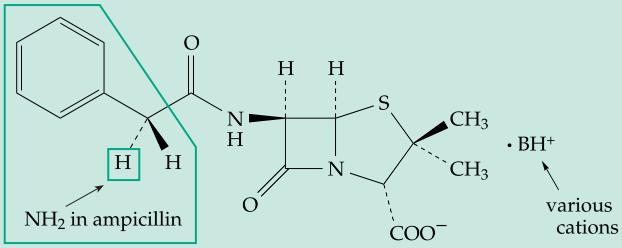


Figure 20-10 Proposed biosynthetic route for synthesis of lipid A and the mature lipopolysaccharide of the *E. coli* cell wall. After C. R. H. Raetz *et al.*[307]

BOX 20-G PENICILLINS AND RELATED ANTIBIOTICS



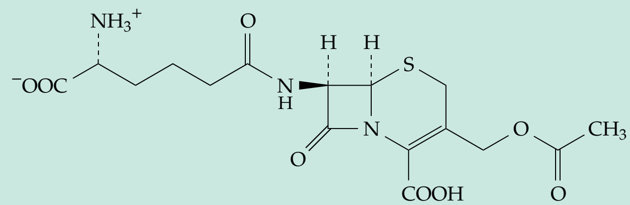
Penicillin G
(Benzylpenicillin)

Many organisms produce chemical substances that are toxic to other organisms. Some plants secrete from their roots or leaves compounds that block the growth of other plants. More familiar to us are the medicinal antibiotics produced by fungi and bacteria. The growth inhibition of one kind of organism upon another was well known in the last century, e.g., as reported by Tyndall[a,b] in 1876. The beginning of modern interest in the phenomenon is usually attributed to Alexander Fleming, who, in 1928, noticed the inhibition of growth of staphylococci by *Penicillium notatum*. His observation led directly to the isolation of penicillin, which was first used on a human patient in 1930. The early history as well as the subsequent purification, characterization, synthesis, and development as the first major antibiotic has been recorded in numerous books and articles.[c–i] During the same time period, Rene Dubos isolated the peptide antibiotics **gramicidin** and **tyrocidine**.[j] A few years later **actinomycin** (Box 28-A) and **streptomycin** (Box 20-B) were isolated from soil actinomycetes (streptomyces) by Waksman, who coined the name **antibiotic** for these compounds. Streptomycin was effective against tuberculosis, a finding that helped to stimulate an intensive search for additional antibacterial substances. Since that time, new antibiotics have been discovered at the rate of more than 50 a year. More than 100 are in commercial production.

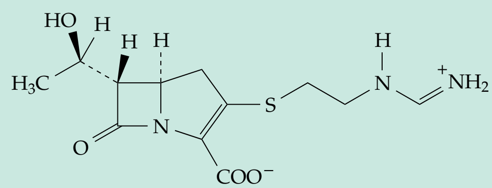
Major classes of antibiotics include more than 200 peptides such as the gramicidins, bacitracin, tyrocidines and valinomycin (Fig. 8-22)[k]; more than 150 **penicillins, cephalosporins,** and related compounds; **tetracyclines** (Fig. 21-10); the **macrolides**, large ring lactones such as the **erythromycins** (Fig. 21-11); and the **polyene** antibiotics (Fig. 21-10).

Penicillin was the first antibiotic to find practical use in medicine. Commercial production began in the early 1940s and benzylpenicillin (penicillin G), one of several natural penicillins that differ in the R group boxed in the structure above, became one of the most important of all drugs. Most effective against gram-positive bacteria, at higher concentrations it also attacks gram-negative bacteria including *E. coli*. The widely used semisynthetic penicillin **ampicillin** (R = D-α-aminobenzyl) attacks both gram-negative and gram-positive organisms. It shares with penicillin extremely low toxicity but some danger of allergic reactions. Other semisynthetic penicillins are resistant to β-lactamases, enzymes produced by penicillin-resistant bacteria which cleave the four-membered β-lactam ring of natural penicillins and inactivate them.

Closely related to penicillin is the antibiotic **cephalosporin C**. It contains a D-α-aminoadipoyl side chain, which can be replaced to form various semisynthetic cephalosporins. **Carbapenems** have similar structures but with CH_2 replacing S and often a different chirality in the lactam ring.



Cephalosporin C



Imipenem, a carbapenem antibiotic of last resort

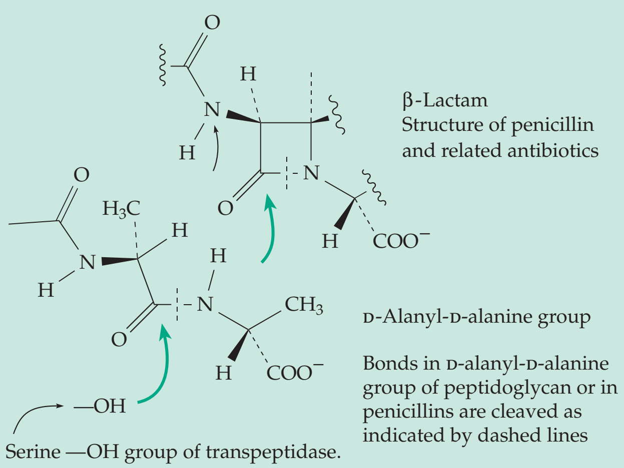
These and other related β-lactams are medically important antibacterial drugs whose numbers are increasing as a result of new isolations, synthetic modifications, and utilization of purified biosynthetic enzymes.[c,l,m]

How do antibiotics act? Some, like penicillin, block specific enzymes. Peptide antibiotics often form complexes with metal ions (Fig. 8-22) and disrupt the control of ion permeability in bacterial membranes. Polyene antibiotics interfere with proton and ion transport in fungal membranes. Tetracyclines and many other antibiotics interfere directly with protein synthesis (Box 29-B). Others intercalate into DNA molecules (Fig. 5-23; Box 28-A). There is no single mode of action. The search for suitable antibiotics for human use consists in finding compounds highly toxic to infective organisms but with low toxicity to human cells.

Penicillin kills only growing bacteria by preventing proper crosslinking of the peptidoglycan

BOX 20-G (continued)

layer of their cell walls. An amino group from a diamino acid in one peptide chain of the peptidoglycan displaces a D-alanine group in a transpeptidation (acyltransferase) reaction. The transpeptidase is also a hydrolase, a DD-carboxypeptidase. Penicillins are structural analogs of D-alanyl-D-alanine and bind to the active site of the transpeptidase.[l,n–p] The β-lactam ring of penicillins is unstable, making penicillins powerful acylating agents. The transpeptidase apparently acts by a double displacement mechanism, and the initial attack of a nucleophilic serine hydroxyl group of the enzyme on penicillin bound at the active site leads to formation of an inactivated, penicillinoylated enzyme.[q,r] More than one protein in a bacterium is derivatized by penicillin.[s] Therefore, more than one site of action may be involved in the killing of bacterial cells.



Several classes of **β-lactamases**, often encoded in transmissible plasmids, have spread worldwide rapidly among bacteria, seriously decreasing the effectivenss of penicillins and other β-lactam antibiotics.[t–y] Most β-lactamases (classes A and C) contain an active site serine and are thought to have evolved from the DD transpeptidases, but the B type[y] has a catalytic Zn^{2+}. The latter, as well as a recently discovered type A enzyme,[z] hydrolyze imipenem, currently one of the antibiotics of last resort used to treat infections by penicillin-resistant bacteria. Some β-lactam antibiotics are also powerful inhibitors of β-lactamases.[u,aa,bb] These antibiotics may also have uses in inhibition of serine proteases[cc,dd] such as elastase. Some antibiotic-resistant staphylococci produce an extra penicillin-binding protein that protects them from beta lactams.[ee] Because of antibiotic resistance the isolation of antibiotics from mixed populations of microbes from soil, swamps, and lakes continues. Renewed efforts are being made to find new targets for antibacterial drugs and to synthesize new compounds in what will evidently be a never-ending battle. We also need better antibiotics against fungi and protozoa.

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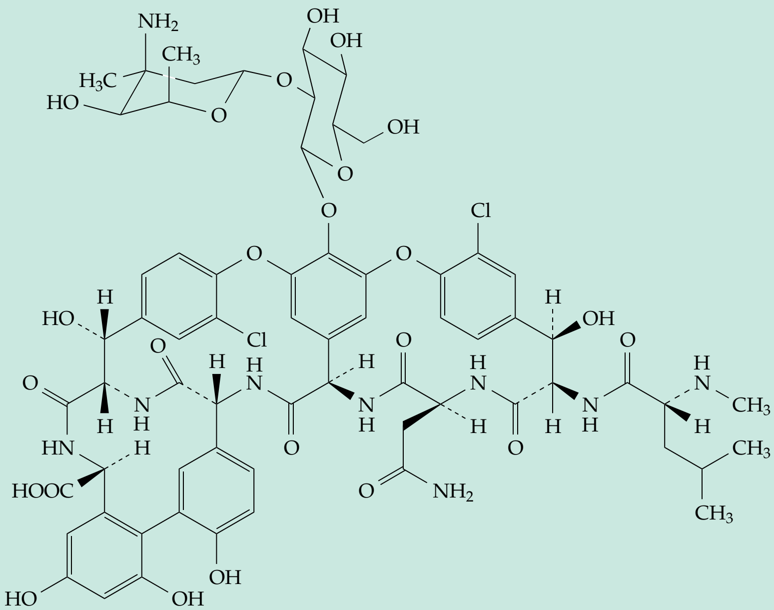
BOX 20-H ANTIBIOTIC RESISTANCE AND VANCOMYCIN



Vancomycin

As antibiotics came into widespread use, an unanticipated problem arose in the rapid development of resistance by bacteria. The problem was made acute by the fact that resistance genes are easily transferred from one bacterium to another by the infectious R-factor plasmids.[a–d] Since resistance genes for many different antibiotics may be carried on the same plasmid, "super bacteria," resistant to a large variety of antibiotics, have developed, often in hospitals.

The problem has reached the crisis stage, perhaps most acutely for tuberculosis. Drug-resistant *Mycobacteria tuberculosis* have emerged, especially, in patients being treated for HIV infection (see Box 21-C). Mechanisms of resistance often involve inactivation of the antibiotics. Aminoglycosides such as streptomycin, spectinomycin, and kanamycin (Box 20-B) are inactivated by en-zymes catalyzing phosphorylation or adenylylation of hydroxyl groups on the sugar rings.[e–g] Penicillin and related antibiotics are inactivated by β-lactamases (Box 20-G). Chloramphenicol (Fig. 25-10) is inactivated by acylation on one or both of the hydroxyl groups.

What is the origin of the drug resistance factors? Why do genes for inactivation of such unusual molecules as the antibiotics exist widely in nature? Apparently the precursor to drug resistance genes fulfill normal biosynthetic roles in nature. An antibiotic-containing environment, such as is found naturally in soil, leads to selection of mutants of such genes with drug-inactivating properties. Never-theless, it is not entirely clear why drug resistance factors have appeared so promptly in the population. Overuse of antibiotics in treating minor infections is one apparent cause.[h,i] Another is probably the widespread use on farms.[j] A nationwide effort to decrease the use of erythromycin in Finland had a very favorable effect in decreasing the incidence of erythromycin-resistant group A streptocci.[h]

Because of the rapid development of resistance, continuous efforts are made to alter antibiotics by semisynthesis (see Box 20-G) and to identify new targets for antibiotics or for synthetic antibacterial compounds.[d] An example is provided by the discovery of vancomycin. Like the penicillins, this antibiotic interferes with bacterial cell wall

teichuronic acids (p. 431). Both proteins and neutral polysaccharides, sometimes covalently bound, may also be present.[303] Like peptidoglycans, teichoic and teichuronic acids are assembled on undecaprenyl phosphates[303] or on molecules of diacylglycerol.[322] Either may serve as an anchor. A "linkage unit" may be formed by transfer of several glycosyl rings onto an anchor unit. For example, in synthesis of ribitol teichoic acid sugar rings are transferred from UDP-GlcNAc, UDP-ManNAC, and CDP-Gal to form the following linkage unit:

$$\text{Lipid}-P-P-\text{GlcNAc}-\text{ManNAc}-(P\text{-Gal})_{2,3}$$

Then many ribitol phosphate units are added by transfer from CDP-ribitol. Finally, the chain is capped by transfer of a glucose from UDP-Glc. Lipoteichoic acids often carry covalently linked D-alanine in ester linkage, altering the net electrical charge on the cell surface.[322a] The completed teichoic acid may then be transferred to a peptidoglycan, releasing the lipid phosphate for reuse.[303] Glycerol teichoic acid may be formed in a similar fashion.[322] Teichuronic acids arise by alternate transfers of *P*-GalNAc from UDP-GalNAc and of GlcA from UDP-GlcA.[303]

Gram-positive bacteria often carry surface proteins that interact with host tissues in establishing human infections. Protein A of *Staphylococcus* is a well-known

BOX 20-H (continued)

synthesis but does so by binding tightly to the D-alanyl-D-alanine termini of peptidoglycans that are involved in crosslinking (Fig. 8-29, Fig. 20-9).[k,l] Like penicillin, vancomycin prevents crosslinking but is unaffected by β-lactamases. Initially bacteria seemed unable to develop resistance to vancomycin, and this antibiotic was for 25 years the drug of choice for β-lactam resistant streptococci or staphylococci. However, during this period bacteria carrying a plasmid with nine genes on the transposon Tn1546 (see Fig. 27-30) developed resistance to vancomycin and were spread worldwide.[l] Vancomycin-resistant bacteria are able to sense the presence of the antibiotic and to synthesize an altered **D-alanine:D-alanine ligase**, the enzyme that joins two D-alanine molecules in an ATP-dependent reaction to form the D-alanyl-D-alanine needed to permit peptidoglycan crosslinking (Fig. 20-9, step *c′*). The altered enzyme adds D-lactate rather than D-alanine providing an – OH terminus in place of $-NH_3^+$. This prevents the binding of vancomycin[l,m] and allows crosslinking of the peptidoglycan via depsipeptide bonds. Another gene in the transposon encodes an oxoacid reductase which supplies the D-lactate.[l] A different resistant strain synthesizes a D-Ala-D-Ser ligase.[n] High-level vancomycin resistance is not attained unless the bacteria also synthesize a D-alanyl-D-alanine dipeptidase.[o]

The D-Ala-D-Ala ligase does provide yet another attractive target for drug design.[p,q] Still another is the D-Ala-D-Ala adding enzyme (Fig. 20-8, step *d*; encoded by the *E. coli MurF* gene).[r,s] Strategies for combatting vancomycin resistance include synthesis of new analogs of the antibiotic[t,u] and simultaneous administration of small molecules that catalyze cleavage of the D-Ala-D-lactate bond formed in cell wall precursors of resistant bacteria.[v]

Another possibility is to use **bacteriophages** directly as antibacterial medicines. This approach was introduced as early as 1919 and has enjoyed considerable success. It is now regarded as a promising alternative to the use of antibiotics in many instances.[w]

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example. After synthesis in the cytoplasm, it enters the secretory pathway. An N-terminal hydrophobic leader sequence and a 35-residue C-terminal sorting signal guide it to the correct destination. There a free amino group of an unlinked pentaglycyl group of the peptidoglycan carries out a transamidation reaction with an LPXTG sequence in the proteins, cutting the chain between the threonine and glycine residues, and anchoring the protein A to the peptidoglycan.[323]

Group A streptococci, which are serious human pathogens, form α-helical coiled-coil threads whose C termini are anchored in the cell membrane. They protrude through the peptidoglycan layers and provide a hairlike layer around the bacteria. A variable region at the N termini provides many antigens, some of which escape the host's immune system allowing infection to develop.[324] Group B streptococci form carbohydrate antigens linked to teichoic acid.[325] Streptococci, which are normally present in the mouth, utilize their carbohydrate surfaces as receptors for adhesion, allowing them to participate in formation of dental plaque.[326]

Cell walls of mycobacteria are composed of a peptidoglycan with covalently attached galactan chains. Branched chains of **arabinan**, a polymer of the furanose ring form of arabinose with covalently attached **mycolic acids**, are glycosidically linked to the galactan.[327] Shorter **glycopeptidolipids**, containing modified glucose and rhamnose rings as well as fatty acids, contribute to the complexity of mycobacterial surfaces.[328]

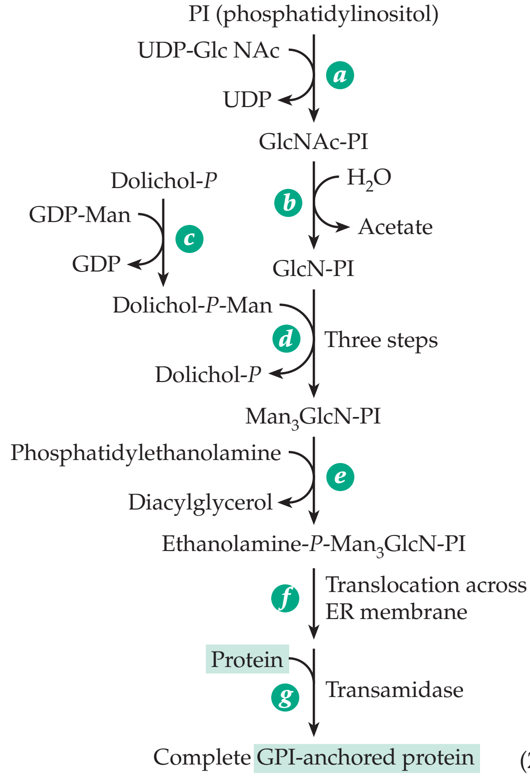
These examples describe only a small sample of the great diversity of cell coats found in the prokaryotic world. Some bacteria also provide themselves with additional protection in the form of external sheaths of crystalline arrays of proteins known as S-layers.[329]

F. Biosynthesis of Eukaryotic Glycolipids

Glycolipids may be thought of as membrane lipids bearing external oligosaccharides. In this sense, they are similar to glycoproteins both in location and in biological significance. Like the glycoproteins, glycolipids are synthesized in the ER, then transported into the Golgi apparatus and eventually outward to join the outer surface of the plasma membrane. Some glycolipids are attached to proteins by covalent linkage.

1. Glycophosphatidylinositol (GPI) Anchors

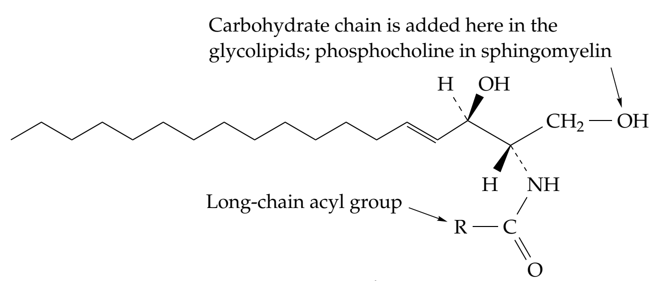
More than 100 different human proteins are attached to phosphatidylinositol anchors of the type shown in Fig. 8-13.[329a] Similar anchors are prevalent in yeast and in protozoa including *Leishmania* and *Trypanosoma*,[330–334b] and *Plasmodium*[334c] where they often bind major surface proteins to the plasma membrane.[335] They are also found in mycobacteria.[336] The structures of the hydrophobic anchor ends are all similar.[337] Two or three fatty acyl groups hold the molecule to the bilayer. Variations are found in the attached glycan portion, both in the number of sugar rings and in the structures of the covalently attached phosphoethanol-amine groups.[337–339] A typical assembly pathway is shown in Eq. 20-23. The first step (step *a* in Eq. 20-23), the transfer of an *N*-acetylglucosamine residue to phosphatidylinositol, is surprisingly complex, requiring at least three proteins.[340] The hydrolytic deacetylation (step *b*) helps to drive the synthetic process. Step *c* provides dolichol-*P*-mannose for the GPI anchors as well as for glycoproteins. The phosphoethanolamine part of the structure is added from phosphatidylethanolamine, apparently via direct nucleophilic displacement.[333] In this way the C terminus of the protein forms an amide linkage with the $-NH_2$ group of ethanolamine in the GPI anchor. Another unexpected finding was that this completed anchor unit undergoes "remodeling" during which the fatty acyl chains of the original phosphatidylinositol are replaced by other fatty acids.[339,339a]



(20-23)

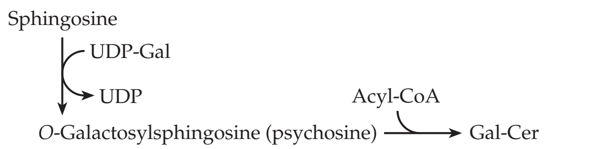
2. Cerebrosides and Gangliosides

These two groups of glycolipids are derived from the *N*-acylated sphingolipids known as **ceramides**. Some biosynthetic pathways from sphingosine to these substances are indicated in Fig. 20-11. Acyl, glycosyl, and sulfo groups are transferred from appropriate derivatives of CoA, CDP, UDP, CMP, and from PAPS to form more than 40 different gangliosides.[341,342] The biosynthesis of a sphingomyelin is also shown in this scheme but is discussed in Chapter 21.



A ceramide

Each biosynthetic step in Fig. 20-11 is catalyzed by a specific transferase. Most of these enzymes are present in membranes of the ER and the Golgi.[343–346] Furthermore, the sequence by which the transferases act may not always be fixed, and a complete biosynthetic scheme would be far more complex than is shown in the figure. For example, one alternative sequence is the synthesis of galactosyl ceramide by transfer of galactose to sphingosine followed by acylation (Eq. 20-24). However, the pathway shown in Fig. 20-11 is probably more important.



(20-24)

This "Hurler corrective factor" was identified as an **α-L-iduronidase**. In the **Hunter syndrome** (MPS II) dermatan sulfate and heparan sulfate accumulate. The missing enzyme is a **sulfatase** for 2-sulfated iduronate residues.[354,355] The diagram at the bottom of the page illustrates the need for both of these enzymes as well as three others in the degradation of dermatan.[352,356,357]

The **Sanfilippo disease** type A (MPS III) corrective factor is a heparan *N*-sulfatase. However, as is true for many other metabolic diseases, the same symptoms

G. The Intracellular Breakdown of Polysaccharides and Glycolipids

The attention of biochemists has been drawn to the importance of pathways of degradation of complex polysaccharides through the existence of at least 35 inherited **lysosomal storage diseases**.[347–351] In many of these diseases one of the 40-odd lysosomal hydrolases is defective or absent.

1. Mucopolysaccharidoses

There are at least seven mucopolysaccharidoses (Table 20-1) in which glycosaminoglycans such as hyaluronic acid accumulate to abnormal levels in tissues and may be excreted in the urine. The diseases cause severe skeletal defects; varying degrees of mental retardation; and early death from liver, kidney, or cardiovascular problems. As in other lysosomal diseases, undegraded material is stored in intracellular inclusions lined by a single membrane. Various tissues are affected to different degrees, and the diseases tend to progress with time.

First described in 1919 by Hurler, **mucopolysaccharidosis I** (MPS I, the **Hurler syndrome**) leads to accumulation of partially degraded dermatan and heparan sulfates (Fig. 4-11).[347,352,353] A standard procedure in the study of diseases of this type is to culture fibroblasts from a skin biopsy. Such cells cultured from patients with the Hurler syndrome accumulate the polysaccharide, but when fibroblasts from a normal person are cultured in the same vessel the defect is "corrected." It was shown that a protein secreted by the normal fibroblasts is taken up by the defective fibroblasts, permitting them to complete the degradation of the stored polysaccharide.

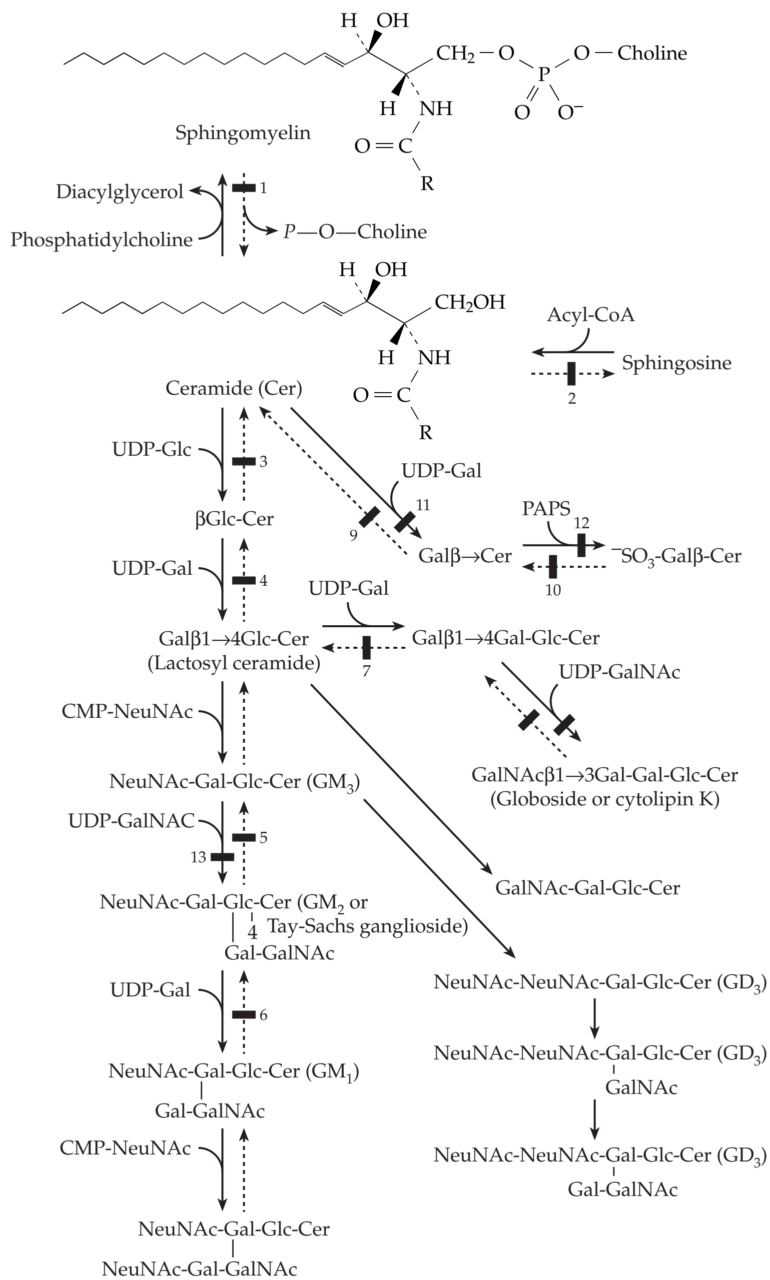


Figure 20-11 Biosynthesis and catabolism of glycosphingolipids. The heavy bars indicate metabolic blocks in known diseases.

may arise from several causes. Thus, Sanfilippo diseases B and D arise from lack of an *N*-acetylglucosaminidase and of a sulfatase for GlcNac-6-sulfate, respectively.[354] In Sanfilippo disease C the missing or defective enzyme is an acetyl transferase that transfers an acetyl group from acetyl-CoA onto the amino groups of glucosamine residues in heparan sulfate fragments. All four of these enzymes are needed to degrade the glucosamine-uronic acid pairs of heparan. The *N*-sulfate groups must be removed by the *N*-sulfatase. The free amino groups formed must then be acetylated before the *N*-acetylglucosaminidase can cut off the GlcNac groups. Remov-al of the 6-sulfate groups requires the fourth enzyme. Completion of the degradation also requires both β-glucuronidase and α-L-iduronidase. Another lysosomal enzyme deficiency, which is most prevalent in Finland, is the absence of **aspartylglucosaminidase**, an N-terminal nucleophile hydrolase (Chapter 12, Section C,3) that cleaves glucosamine from aspartate side chains to which oligosaccharides were attached in glycoproteins.[358–360]

Some hereditary diseases are characterized by lack of two or more lysosomal enzymes. In **I-cell disease** (mucolipidosis II), which resembles the Hurler syndrome, at least ten enzymes are absent or are present at much reduced levels.[350,361] The biochemical defect is the absence from the Golgi cisternae of the *N*-acetylglucosaminyl phosphotransferase that transfers *P*-GlcNAc units from UDP-GlcNAc onto mannose residues (Eq. 20-22) of glycoproteins marked for use in lysosomes.

2. Sphingolipidoses

There are at least ten lysosomal storage diseases, known as sphingolipidoses, that involve the metabolism of the glycolipids. Their biochemical bases are indicated in Fig. 20-11 and in Table 20-1. **Gaucher disease**[362–365] is a result of an autosomal recessive trait that permits glucosyl ceramide to accumulate in macrophages. The liver and spleen are seriously damaged, the latter becoming enlarged to four or five times normal size in the adult form of the disease. In the more severe juvenile form mental retardation occurs. By 1965, it was established that cerebroside is synthesized at a normal rate in the individuals affected, but that a lysosomal hydrolase was missing. This blocked the catabolic pathway indicated by dashed arrows in Fig. 20-11 (block No. 3 in the figure). In many patients a single base change causing a Leu → Pro substitution accounts for the defect. In **Fabry disease** an X-linked gene that provides for removal of galactosyl residues from cerebrosides is defective.[350] This leads to accumulation of the triglycosylceramide whose degradation is blocked at point 7 in Fig. 20-11.

The best known and the commonest sphingolipidosis is **Tay–Sachs disease**.[366–368] Several hundred cases have been reported since it was first described in 1881. A terrible disease, it is accompanied by mental deterioration, blindness, paralysis, dementia, and death by the age of three. About 15 children a year are born in North America with this condition, and the world figure must be 5–7 times this. The defect is in the α subunit of the β-hexosaminidase A (point 7 in Fig. 20-10)[366,366a] with accumulation of ganglioside G_{M2}. Somewhat less severe forms of the disease are caused by different mutations in the same gene[369] or in a protein activator. **Sandhoff disease**, which resembles Tay–Sachs disease, is caused by a defect in the β subunit, which is present in both β-hexosaminidases A and B.[368] Mutant "knockout" mice that produce only ganglioside GM_3 as the major ganglioside in their central nervous system die suddenly from seizures if they hear a loud sound. This provides further evidence of the essential nature of these components of nerve membranes.[369a]

3. Causes of Lysosomal Diseases

The descriptions given here have been simplified. For many lysosomal diseases there are mild and severe forms and infantile or juvenile forms to be contrasted with adult forms. Some of the enzymes exist as multiple isozymes. An enzyme may be completely lacking or

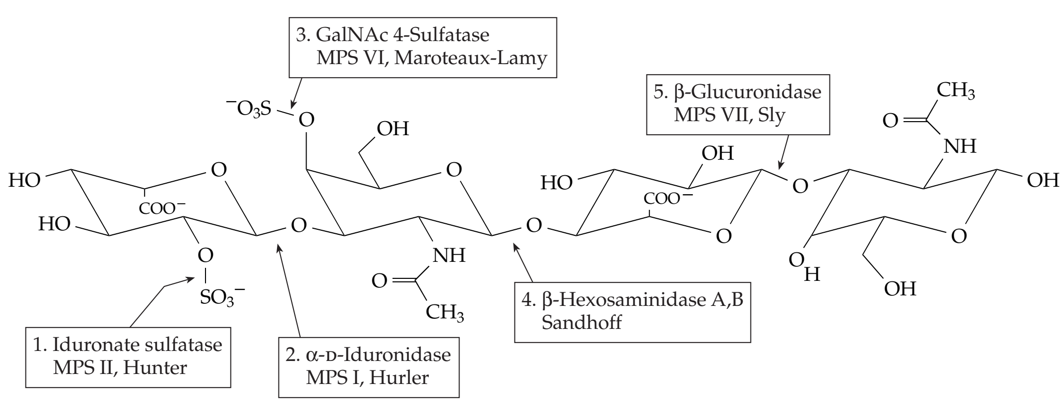


TABLE 20-1
Lysosomal Storage Diseases: Sphingolipidoses and Mucopolysaccharidoses[a]

No. in Fig. 20-11	Name	Defective enzyme
1.	Niemann–Pick disease[b]	Sphingomyelinase
2.	Farber disease (lipogranulomatosis)	Ceramidase
3.	Gaucher disease[c]	β-Glucocerebrosidase
4.	Lactosyl ceramidosis	β-Galactosyl hydrolase
5.	Tay–Sachs disease[c]	β-Hexosaminidase A
6.	G_{M1} gangliosidosis[d]	β-Galactosidase
7.	Fabry disease[c]	α-Galactosidase
8.	Sandhoff disease[e]	β-Hexosaminidases A and B
9.	Globoid cell leukodystrophy	Galactocerebrosidase
10.	Metachromatic leukodystrophy	Arylsulfatase A
13.	Hematoside (G_{M3}) accumulation	G_{M3}-*N*-acetylgalactosaminyltransferase
	Pompe disease[f]	α-Glucosidase
	Hurler syndrome (MPS I)[c]	α-L-Iduronidase
	Hunter syndrome (MPS II)[c]	Iduronate 2-sulfate sulfatase
	Sanfilippo disease[c,g]	
	Type A (MPS III)	Heparan *N*-sulfatase
	Type B	*N*-Acetylglucosaminidase
	Type C	Acetyl-CoA: α-glucosaminide *N*-acetyltransferase
	Type D	GlcNAc-6-sulfate sulfatase
	Maroteaux–Lamy syndrome (MPS VI)[g]	Arylsulfatase B
	Sly syndrome (MPS VII)[g,h]	β-Glucuronidase
	Aspartylglycosaminuria[i]	Aspartylglucosaminidase
	Mannosidosis	β-Mannosidase
	Fucosidosis	α-L-Fucosidase
	Mucolipidosis	α-*N*-Acetylneuraminidase
	Sialidosis[j]	

[a] A general reference is Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds. (1995) *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 1, McGraw-Hill, New York (pp. 2427–2879)
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[f] See Box 20-D
[g] Neufeld, E. F., and Muenzer, J. (1995) in *The Metabolic and Molecular Bases of Inherited Disease*, 7th ed., Vol. 2 (Scriver, C. R., Beaudet, A. L., Sly, W. S., and Valle, D., eds), pp. 2465 – 2494, McGraw-Hill, New York
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may be low in concentration. The causes of the deficiencies may include total absence of the gene, absence of the appropriate mRNA, impaired conversion of a proenzyme to active enzyme, rapid degradation of a precursor or of the enzyme itself, incorrect transport of the enzyme precursor to its proper destination, presence of mutations that inactivate the enzyme, or absence of protective proteins. Several lysosomal hydrolases require auxiliary **activator proteins** that allow them to react with membrane-bound substrates.[350,351,365,370]

4. Can Lysosomal Diseases Be Treated?

There has been some success in using enzyme replacement therapy for lysosomal deficiency diseases.[347,371-371b] One approach makes use of the fact that the mannose-6-*P* receptors of the plasma membrane take up suitably marked proteins and transfer them into lysosomes (Section C,2). The missing enzyme might simply be injected into the patient's bloodstream from which it could be taken up into the lysosomes.[372] This carries a risk of allergic reaction, and it may be safer to attempt microencapsulation of the enzyme, perhaps in ghosts from the patient's own erythrocytes.[347] A second approach, which has had limited success, is transplantation of an organ[371] or of bone marrow[373] from a donor with a normal gene for the missing enzyme. This is dangerous and is little used at present. However, new hope is offered by the possibility of transferring a gene for the missing enzyme into some of the patient's cells. For example, the cloned gene for the transferase missing in Gaucher disease has been transferred into cultured cells from Gaucher disease patients with apparent correction of the defect.[374] Long-term correction of the Hurler syndrome in bone marrow cells also provides hope for an effective therapy involving gene transfer into a patient's own bone marrow cells[375] or transplantation of selected hematopoietic cells.[375a]

In the cases of Gaucher disease and Fabry disease, it is hoped that treatment of infants and young children may prevent brain damage. However, in Tay–Sachs disease the primary sites of accumulation of the ganglioside GM_2 are the ganglion and glial cells of the brain. Because of the "blood–brain barrier" and the severity of the damage it seems less likely that the disease can be treated successfully.

The approach presently used most often consists of identifying carriers of highly undesirable genetic traits and offering genetic counseling. For example, if both parents are carriers the risk of bearing a child with Tay–Sachs disease is one in four. Women who have borne a previous child with the disease usually have the genetic status of the fetus checked by **amniocentesis**. A sample of the amniotic fluid surrounding the fetus is withdrawn during the 16th to 18th week of pregnancy. The fluid contains fibroblasts that have become detached from the surface of the fetus. These cells are cultured for 2–3 weeks to provide enough cells for a reliable assay of the appropriate enzymes. Such tests for a variety of defects are becoming faster and more sensitive as new techniques are applied.[376] In the case of Tay–Sachs disease, most women who have one child with the disease choose abortion if a subsequent fetus has the disease.

The diseases considered here affect only a small fraction of the problems in the catabolism of body constituents. On the other hand, fewer cases are on record of deficiencies in biosynthetic pathways. These are more often absolutely lethal and lead to early spontaneous abortion. However, blockages in the biosynthesis of cerebrosides are known in the special strains of mice known as Jimpy, Quaking, and msd (myelin synthesis deficient).[377,378] The transferases (points 11 and 12 of Fig. 20-11) are not absent but are of low activity. The mice have distinct neurological defects and poor myelination of nerves in the brain. A human ailment involving impaired conversion of GM_3 to GM_2 (with accumulation of the former; point 13 of Fig. 20-11) has been reported. Excessive synthesis of sialic acid causes the rare human **sialuria.**[379] This is apparently a result of a failure in proper feedback inhibition.

Animals suffer many of the same metabolic diseases as humans. Among these are a large number of lysosomal deficiency diseases.[380] Their availability means that new methods of treating the diseases may, in many cases, be tried first on animals. For example, enzyme replacement therapy for the Hurler syndrome is being tested in dogs.[381] Bone marrow transplantation for human **α-mannosidosis** is being tested in cats with a similar disease.[382] Mice with a hereditary deficiency of β-glucuronidase are being treated by gene transfer from normal humans.[357]

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Study Questions

1. Constitution of cell surface oligosaccharides or polysaccharides includes the following:
   - D-Glucose
   - D-Mannose
   - D-Galactose
   - L-Arabinose
   - L-Fucose
   - D-Glucuronic acid
   - D-Neuraminic acid

   Outline pathways for biosynthesis of these compounds from glucose.

2. Decarboxylation steps are required for synthesis of UDP-xylulose and UDP-apiose (Fig. 21-1). Propose chemical mechanisms for these reactions.

3. How do animals and plants differ with respect to transport and storage of glucose?

4. Comment on unresolved questions about the biosynthesis of cellulose, amylose, and amylopectin. What glycosyl carrier groups are required?

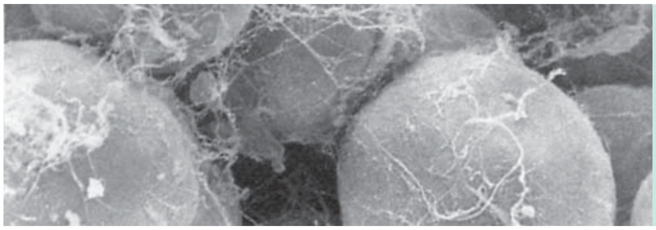
5. Most 5- and 6-membered sugars are found in nature as pyranose ring forms. Why is ribose in RNA in the furanose ring form?

6. If the ratio $[NAD^+]/[NADH]$ in a cell were 500 and the ratio $[NADP^+]/[NADPH]$ were 0.002, what concentrations of fructose and sorbitol would be in equilibrium with 0.1 mM glucose? See Box 20-A and Table 6-4.

7. Write a balanced equation for reaction of boric acid (H_3BO_3) with two sugar rings to give a borate diol ester linkage (Box 20-E).

8. Describe in general terms the process by which *N*-linked oligosaccharides are synthesized and attached to proteins. What are the functions of the ER and the Golgi?

9. What, if any, restrictions do you think should be applied to the use of antibiotics on farms?



Two large fat-filled adipose cells are seen in the foreground of this scanning electron micrograph. They are part of a larger cluster of cells from rat tissue. Delicate strands of connective tissue fibers intertwine the cells and hold them together. While most of the connective tissue substance has been washed away during preparation of the specimen, the remnants give a realistic impression of the soft, loose nature of the intercellular material. From Porter and Bonneville (1973) *Fine Structure of Cells and Tissues*, Lea and Febiger, Philadelphia, Pennsylvania. Courtesy of Mary Bonneville.

Contents

Specific Aspects of Lipid Metabolism

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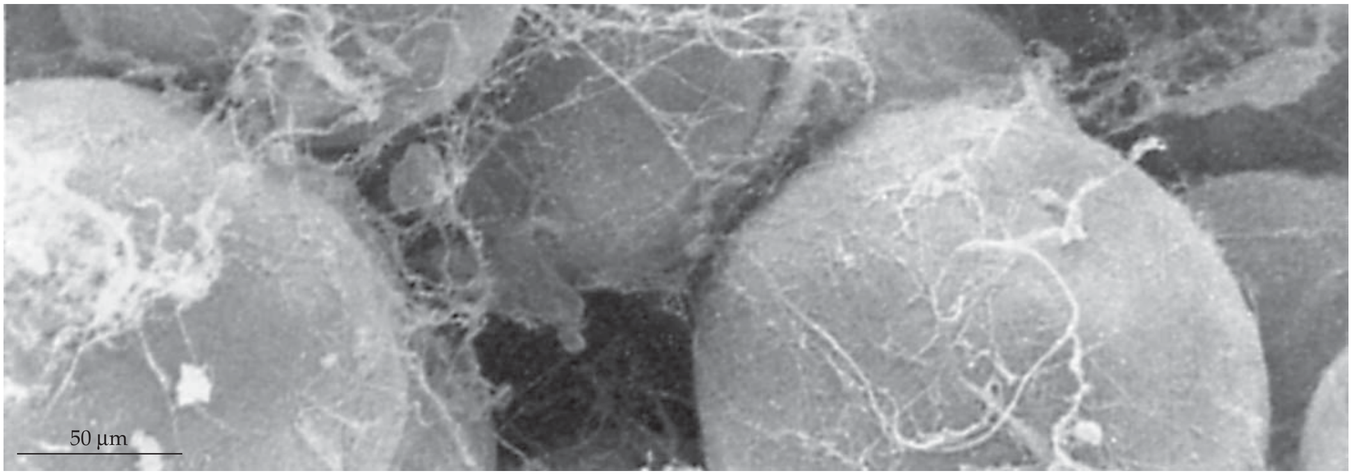

The basic pathways for both synthesis of fatty acids and for their β-oxidation (Fig. 17-1) have been described in Chapter 17. However, there are many variations to these pathways, and additional sets of enzymes are needed to synthesize the complex array of lipids present in most organisms. We will consider these details in this chapter. Like most other organisms, human beings are able to synthesize triacylglycerols (triglycerides), phospholipids, and glycolipids needed for cell membranes. Glucose can serve as the starting material. However, dietary lipids are also a major source. For this reason, we will start with a discussion of the digestion and uptake of lipids and of the distribution by way of the bloodstream of ingested lipids and of lipids synthesized in the liver or in other tissues.

A. Digestion, Synthesis, and Distribution of Triacylglycerols in the Human Body

Digestion of triglycerides begins in the stomach with emulsification and partial digestion by gastric lipase. Within the small intestine the ~100-residue protein called **colipase**[1–2a] binds to the surface of the fat droplets and provides an attachment site for the 449-residue **pancreatic lipase**. This Ca^{2+}-dependent serine esterase cleaves each triglyceride to two molecules of fatty acid and one of a 2-monoacylglycerol.[3–5] These products are emulsified by bile salts (Fig. 22-10) and are then taken up by the cells of the intestinal lining. The fatty acids are converted to acyl-CoA esters which transfer their acyl groups to the monoacylglycerols to resynthesize the triacylglycerols.[6] The latter are incorporated into the very large lipoprotein particles called **chylomicrons** (Table 21-1) and enter the bloodstream via the lymphatic system (Fig. 21-1).[7] Free fatty acids are also transported as complexes with serum albumin.

Synthesis of lipids from carbohydrates is an efficient process, which occurs largely in the liver and also in intestinal epithelial cells.[6] The newly synthesized triacylglycerols, together with smaller amounts of phospholipids and cholesterol, combine with specific **apolipoproteins**, which are also synthesized in the liver, to form **very low density lipoprotein (VLDL)** particles which are secreted into the blood stream. They transport the newly formed triacylglycerols from the liver to other body cells including the adipocytes, which store excess fat (Fig. 21-1).

1. Plasma Lipoproteins

The small particles of plasma lipoprotein, which carry triacylglycerols, can be separated according to their buoyant densities by centrifugation. They have been classified into five groups of increasing density but smaller size as **chylomicrons,** very low density lipoproteins **(VLDL)**, intermediate density lipoproteins (**IDL**), low density **(LDL),** and high density lipoproteins **(HDL)** (Table 21-1 and Fig. 21-2). Each lipoprotein particle contains one or more apolipoproteins (Table 21-2), whose sizes vary from the enormous 4536-residue apoB-100 to apoC-II and apoC-III, each of which contains just 79 residues[7a] and the 57-residue apoC-I.[7b]

The larger lipoproteins are spherical micelles containing a core of triacylglycerols and esters of cholesterol surrounded by a 2- to 3-nm-thick layer consisting of phospholipid, free cholesterol, and the apolipoprotein components.[8] The size of the lipoprotein particles also varies from a 200- to 500-nm diameter for chylomicrons to as little as 5 nm for the smallest HDL particles. The difference in volume is more impressive. If, as has been estimated,[9] a 22-nm diameter LDL particle contains about 2000 cholesterol and cholesteryl ester molecules and 800 phospholipids, a small HDL particle of 7-nm diameter will have room for only about 60 molecules of cholesterol and 90 of phospholipid, while a chylomicron may carry 10 million molecules of triacylglycerol. HDL particles are quite heterogeneous. As is indicated in Table 21-1, they are sometimes dividing into HDL2 and HDL3 density groups. In addition, there is a pre-HDL with lower phospholipid content and discoid forms low in cholesterol. Models of a reconstituted lipoprotein disc contain two molecules of apoA-I and ~160 phosphatidylcholines that form a bilayer core.[10–10b]

Each apolipoprotein has one or more distinct functions. The apoB proteins probably stabilize the lipoprotein micelles. In addition, apoB-100 is essential to recognition of LDL by its receptors. The 79-residue apoC-II has a specific function of activating the lipoprotein lipase that hydrolyses the triacylglycerols of chylomicrons and VLDL. Lack of either C-II or the lipase results in a very high level of triacylglycerols in the blood.[11]

The large apolipoprotein B-100 is synthesized in the liver and is a principal component of VLDL, IDL, and LDL. It is the sole protein in LDL, accounting for nearly 20% of the mass of LDL particles. Partly because of its insolubility in water, its detailed structure is uncertain. If it were all coiled into an α helix, it would be 680 nm long and could encircle the LDL particle nearly 10 times! While the true structure of apoB-100 is unknown, it is thought to be extended and to span at least a hemisphere of the LDL surface.[12] It consists of at least five domains. Sixteen cysteines are present in the first 25 residues at the N terminus, forming a crosslinked high-cysteine region. There are also 16 *N*-glycosylated sites. Domain IV (residues 3071–4011) is thought to contain the site that binds to its specific receptor, the LDL receptor.[12] Heterogeneity in the amide I band of the infrared absorption spectrum (Fig. 23-3) suggests that about 24% is α helix, 23% β sheet, and that a large fraction consists of turns, and unordered and extended peptide structures.[13]

In intestinal epithelial cells the same apoB gene that is used to synthesize apoB-100 in the liver is used to make the shorter **apoB-48** (48%) protein. This is accomplished in an unusual way that involves "editing" of the mRNA that is formed. Codon 2153 in the mRNA for the protein is CAA, encoding glutamine. However, the cytosine of the triplet is acted on by a deaminase, an editing enzyme, to form UAA, a chain termination codon.[14,15] A third form of apoB is found in **lipoprotein(a)** (Lpa). This LDL-like particle contains apoB-100 to which is covalently attached by a single disulfide linkage (probably to Cys 3734 of apoB-100) a second protein, **apo(a)**. The latter consists largely of a chain of from 11 to over 50 kringle domains resembling the 78-residue kringle-4 of plasminogen (see Fig. 7-30C)[16–19a] as well as a protease domain.[20] This additional chain may cause tighter binding to LDL receptors and may cause lipoprotein(a) to displace plasminogen from cell surface receptors.[21] The amount of Lp(a) varies over 1000-fold among individuals and is genetically determined. The number of kringle domains also varies.[17] Although the presence of high Lp(a) is associated with a high risk of atherosclerosis and stroke,[21a] many healthy 100-year olds also have high serum Lp(a).[22]

Apolipoprotein A-I is the primary protein component of HDL.[23–25b] Most of the 243 residues consist of a nearly continuous amphipathic α helix with kinks at regularly spaced proline residues.[26–28] Two disulfide-linked ApoA-I molecules may form a belt that encircles the discoid lipoprotein.[25b] ApoA-II is the second major HDL protein, but no clearly specialized function has been identified.[29,30] ApoA-I, II, and IV, apoC-I, II, and III, and apoE all have multiple repeats of 22 amino acids with sequences that suggest amphipathic helices. The 391-residue ApoA-IV has 13 tandem 22-residue repeats. Proline and glycine are present in intervening hinge regions.[23] This may enable these proteins to spread over and penetrate the surfaces of the lipoprotein micelles. Most of these proteins are encoded by a related multigene family.[7,30a]

The 299-residue apolipoprotein E plays a key role in metabolism of both triacylglycerols and cholesterol. Like apoB-100 it binds to cell surface receptors.[31–33a] Absence of functional apoE leads to elevated plasma triacylglycerol and cholesterol, a problem that is considered in Chapter 22, Section D. The N-terminal domain, from residues 23 to 164, forms a 6.5 nm-long four-helix bundle, which binds to the LDL receptors.[32] There are three common isoforms of alipoprotein E (apoE2, apoE3, and apoE4). ApoE3 is most common.[33b] The presence of apoE4 is associated with an increased risk of Alzheimer disease (Chapter 30).

The major lipoproteins of insect hemolymph, the **lipophorins**, transport diacylglycerols. The apolipophorins have molecular masses of ~250, 80, and sometimes 18 kDa.[34–37a] The three-dimensional structure of a small 166-residue lipophorin (apolipophorin-III) is that of a four-helix bundle. It has been suggested that it may partially unfold into an extended form, whose amphipathic helices may bind to a phospholipid surface of the lipid micelle of the lipophorin.[35] A similar behavior may be involved in binding of mammalian apolipoproteins. Four-helix lipid-binding proteins have also been isolated from plants.[38] See also Box 21-A. Specialized lipoproteins known as **lipovitellins**